Flight Vehicle System Identification: A Time-Domain Methodology

Second Edition

Ravindra V. Jategaonkar

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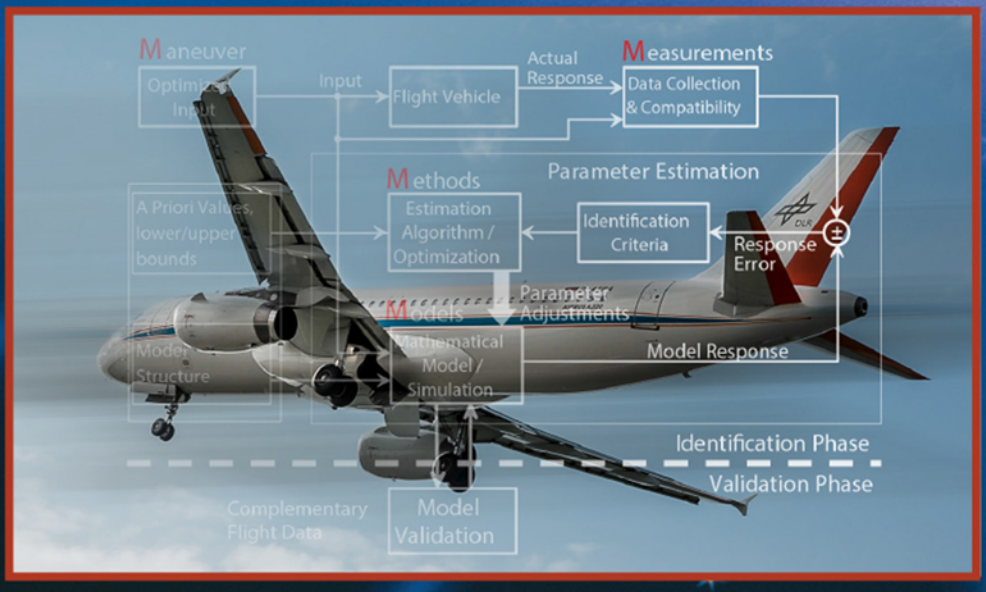

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Timothy C. Lieuwen, Editor-in-Chief
Volume 245

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Flight Vehicle System Identification:

A Time-Domain Methodology

Second Edition

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To Aai and Bapu, my parents

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PREFACE TO THE SECOND EDITION

It has been more than eight years since the first edition of this book provided readers with the theoretical background, concepts of coordinated approach of system identification, and certain well-researched guidelines to derive and validate high-fidelity models of flight vehicles from recorded flight data. During this period, I have received numerous emails commenting on the usefulness of the book in understanding the methodology, on the easy and understandable presentation of complex, multidisciplinary subjects, as well as making a few minor suggestions on the contents, mainly related to the scope of the basic software provided in Chapter 4 about output-error method, one of the most widely applied methods in the time domain methodology, and minor bugs in the filter-error method. I take this opportunity to thank them all and other readers of the first edition for their support, which counts for me the most.

During my subsequent technical consultations, I noticed that the basic version of the software provided with the first edition, although adequate to clearly understand the methodology and to follow the test cases presented in the book, was somewhat restricted in its scope of applications. Several extensions of the basic software to cater to individual needs or further advanced options were indicated in the first edition, but were left to the reader. The software used in this book has been programmed using the Matlab® computing environment. It has been developed using basic Matlab only, that is, without requiring any of the toolboxes. On the side of applications, use of the methodology to conventional applications such as verification and update of aerodynamic databases as well as flight-path reconstruction continues on a routine basis. More interestingly, advanced and challenging tasks such as extended flexible aircraft modeling from flight data, incorporating both the rigid-body dynamics and the structural dynamics, and other flight vehicles such as unmanned aerial vehicles have been demonstrated and established.

In consultation with David Arthur, acquisitions and development editor at AIAA, and the Progress in Astronautics and Aeronautics series editorial advisory board of AIAA, it was decided to publish the second edition to cover some of the points indicated herein. I extend my sincere thanks to them for their continued interest. Because many readers found the organization of the first edition to be useful, we continue to retain the same structure for the second edition as well. We have not added any new chapters; rather, the second half of Chapter 4, dealing with the output-error method, has been thoroughly modified to describe the extended software that now caters to all of the extensions that were mentioned in the first edition. Examples have been added to demonstrate the new options. Through these extensions, the extended software provided with the second edition is now much more flexible and suitable for advanced modeling tasks as well.

We also updated the contents of Chapter 10 on data compatibility check with additional examples. In Chapter 12 on selected advanced examples, we expanded

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on the case study on quasi-steady stall modeling with a different aircraft configuration, demonstrating efficacy of the simple model postulate for this unsteady aerodynamics phenomenon. Likewise, we expanded on the wake-vortex aircraft encounter model to include recent advances in the validating the aircraft interaction model. Furthermore, we have added a completely new case study on flexible aircraft modeling applying the time-domain methodology as discussed in the book. The model identified from flight data of a glider includes both the rigid-body dynamics and the structural dynamics. Consistent with the hitherto philosophy, practical aspects and limitations of the approach have been discussed in detail. Besides these main changes, we also made corrections in Chapter 2 to include a more detailed discussion on the practical aspects of smoothing noisy data and of numerical differentiation. To suit individual cases, more options are now provided to choose from for handling data with different noise characteristics.

While incorporating these updates, we were motivated by the comments from the readers and by the new applications pursued at our institute and other organizations. Nevertheless, the key aspects of the multidisciplinary task addressed in the second edition remain the same, namely, to analyze flight data to arrive at a model of minimum complexity, yet retain the salient dynamic effects, and have adequate accuracy for the intended purpose. Such models are mainly useful for applications to flight dynamics investigations, aerodynamic database validation and update, simulations, and control system design. As implied by the title, we continue to focus on the time-domain approach for this complex task. I hope that the readers will find, as before, the various topics covered in the book inspirational in embarking on the difficult task of flight vehicle modeling from flight data and also find the extended software encouraging and suitable to address more complex, challenging tasks. If you locate any errors in the text and software, or have any other comments or questions, please send your suggestions and queries to me (ravindra.jategaonkar@dlr.de).

I would like to sincerely thank Joaquim Neto Dias, IPEV, São José dos Campos, Brazil, for providing recent results of quasi-steady stall modeling from flight data. I extend my special thanks to Bruno Giordano de Oliveira Silva, IPEV, São José dos Campos, Brazil, for providing the case study on flexible aircraft modeling. I once again extend my thanks to my colleague Dietrich Fischenberg for updating the material on Aircraft Interaction Model, which is part of the case study on modeling of wake-vortex encounter. I am also thankful to Professor Stefan Levedag for the infrastructural support and for providing freedom of research granted at the Institute.

Ravindra V. Jategaonkar
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January 2015

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PREFACE TO THE FIRST EDITION

The objective of this book is to provide a consolidated account of flight vehicle system identification that has evolved during the recent decades, focusing particularly on nonlinear systems and the time-domain approach. It also aims to share the practical experience gained on aerodynamic modeling from flight data to a large number of flight vehicles because experience and engineering judgment are critical to generate good results. Effective system identification becomes possible based on a coordinated approach and by following certain well-researched guidelines. This book attempts to provide not only details of such a systematic approach, summarizing the general underlying concepts, methodologies, and computational procedures, and present examples of practical applications, but also to show the pitfalls of these methods. It gives practical tips on how to overcome the problems one is likely to face in developing nonlinear, high-fidelity models and analyzing flight data from complex flight vehicles, for example, intermediate divergence of optimization algorithms or estimation subject to bounds, or application of filter-error method to data with turbulence, which are generally not covered in theoretical books.

The layout of the book and the material presented here is partly based on the short courses on "Flight Vehicle System Identification in Time Domain" delivered as a part of the AIAA Professional Development program and at other educational and research organizations during the last few years, partly on the personal notes of discussions and experience gathered over two and half decades, and partly on several technical papers published jointly with my colleagues, including several guest scientists whom I had the pleasure of guiding. If you locate any errors in the text and software, or have any other comments or questions, please send your suggestions and queries to me (jategaonkar@dlr.de).

With great pleasure, I would like to acknowledge several individuals who helped me during the various phases of writing this book. First, I sincerely thank Professor Peter Hamel for his support and for the use of materials from some of his and our joint papers. My personal discussions with him have shaped the book layout to some extent. Next, I greatly appreciate the help provided by my colleague Wulf Mönnich in the tedious job of reading the draft manuscript and for making many helpful suggestions. It has helped me directly and indirectly in the thought process reflected in the book. I would also like to acknowledge my former colleague Dr. Ermin Plaetschke for his comments on the draft version, for the use of material from our joint papers and other notes. I am thankful to my colleague Dietrich Fischenberg for providing the case study on modeling of wake-vortex encounter and some material on other examples. Likewise, I extend my thanks to Dr. Wilhelm Gockel for the consent to report on the case study pertaining to reusable launch vehicle demonstrator. Help from Dr. Wolfgang von Grünhagen on rotorcraft example is appreciated. I am also thankful to Professor Stefan Levedag and Dr. Frank Thielecke for the

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facilities granted at the Institute. Interest of other present and former colleagues of the Institute is appreciated, particularly that of Dr. Karl Doherr. It has been a pleasure to work at the DLR Institute of Flight Systems for more than two decades. I also recall my past association with the former Systems Engineering Department of the National Aeronautical Laboratory, Bangalore, India.

I would also like to extend my special appreciation to my wife, Padma, for her patience throughout the extended period of writing this book. Without her understanding and support, this book would not have been possible. I would also like to mention here our daughters, Smita and Swati, for their continued interest.

Finally, I would like to acknowledge the interest of AIAA in publishing this book under the Progress Series. In particular, I would like to extend my appreciation to Dr. Peter Zipfel, member of the editorial board, for a discussion that led to embarking on this book project. I would like to acknowledge Rodger Williams, AIAA Publications Development for his help and encouragement. I extend my thanks to Alex McCray, Managing Editor, AIAA Books, for overseeing the book production; to Janice Saylor, Marketing Strategist, AIAA, for the front and back cover design and marketing; and Nick Barber, Books Manager, Techset Composition, Ltd., for copyediting, text composition, and incorporating text amendments efficiently. I am sure many other AIAA staff are involved down the line, whom I would like to thank as well.

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CHAPTER 1

Introduction

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System identification, as it is termed today, is a scientific discipline that provides answers to the age-old inverse problem of obtaining a description in some suitable form for a system, given its behavior as a set of observations. The inverse problem and, hence, system identification has been a fundamental part of obtaining knowledge of any physical system that is observed and, as such, has far-ranging areas of applications. It enables us to not only know more about the principle underlying (that is, model formulation of) the process being observed but also know in adequate details (that is, analysis and parameter estimation). System identification is implicitly concerned with modeling from experimental data and covers applications in all possible areas, for example, biology, medicine, chemical processes, economics, geology, materials, civil and mechanical engineering, automobiles, and, of course, flight vehicles. In this book, we are concerned with the last mentioned subject, namely, the flight vehicle system identification. As a rule, a flight vehicle is a dynamic system, and because a state-space representation in the time domain is closer to physical reality than any of the frequency-domain transform techniques, we deal here mainly with the time-domain methodology.

The real processes are, in general, too complex, and the exact internal behavior is unknown; the mathematical model, no matter how complex, at best represents a replica of what we purport to underlie the system behavior. It might be possible to postulate a very comprehensive model encompassing all conceivable influences or based on theoretical formulations. A large size model, however, does not necessarily mean a good system representation. The principle of parsimony, also called principle of simplicity or Ockham's Razor named after an early 14th century English theologian and philosopher, needs to be applied to determine the "best" model [1, 2]. Translated from Latin, in essence, it implies:

> The number of entities should not be increased beyond what is necessary to explain anything.

In our case *entities* refer to the assumptions or parameters appearing in the mathematical model. Speaking in layman's terms, this implies "*as simple as possible, and as complex as necessary.*" A similar principle has also been attributed to

Albert Einstein who said, "Everything should be made as simple as possible, but not simpler." Ockham's Razor is a methodological principle that helps to minimize redundancies and inconsistencies in the model. Thus, the best model representation having fidelity within a specified tolerance is that which has the minimum number of parameters. Simpler models are easier to interpret and are necessary for practical reasons of testing, estimation, and cross-validation. The definition of "best" model is, however, not unique and depends strongly on the intended purpose of the derived model. Here, engineering knowledge and experience play important roles.

The process of system identification involves certain fundamental assumptions, namely, the following:

1. The true state of the dynamic system is deterministic.
2. Physical principles underlying the dynamic process can be modeled.
3. It is possible to carry out specific experiments.
4. Measurements of system inputs and outputs are available.

The first and the second assumptions are somewhat related to each other. They suggest that we deal with systems that can be described mathematically, and not with chaotic systems whose input-output behavior cannot be described through fixed rules. It is also implied that, under ideal conditions, data analysis from repeated experiments converges to the truth, that is, lead to the same model. The third assumption implies that specific experiments will be necessary to excite different modes of dynamic motion, which might call for designing suitable experiments. This might be possible in some cases, for example, as in the case of flight vehicles, or might not be in others (such as economic systems?). The fourth assumption, as already pointed out, is implicit to system identification. Either directly measured or derived quantities (for example, engine thrust computed from other measured variables) must be available.

1.1 WHAT IS SYSTEM IDENTIFICATION?

The subject of system identification is closely related to the other problems in dynamics. For a dynamic system described in state space, three quantities that mainly describe the system are the inputs u, the outputs z or y, and the system model functions f and g; see Fig. 1.1. Depending upon the availability of any two, three different types of problems are encountered in system theory, namely:

1. Simulation, which deals with finding the outputs y for given inputs u and given system functions f and g
2. Control problem, which aims at finding the control inputs u for given f and g, and y

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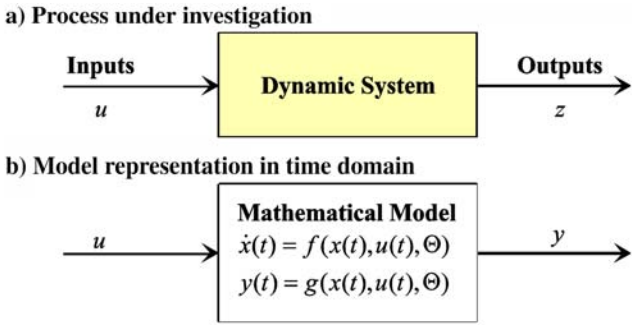

Fig. 1.1 Representation of dynamic system.

3. Identification problem, which attempts to find the system model f and g from the given measurements of inputs u and outputs z

Although the state variables x are not mentioned explicitly, they appear in the preceding description implicitly because the outputs are functions of internal system state variables. Likewise, the system functions f and g contain unknown parameters Θ.

System identification is the process of determining an adequate mathematical model, usually containing differential equations, with unknown parameters that have to be determined indirectly from measured data. The process includes not only the model postulation and determination of parameters, which is our ultimate goal, but also performing suitable experiments and gathering system inputs and responses. From the modeling aspect, basically, it amounts to determining the equations and unknown model parameters Θ such that the model response y matches adequately with the measured system response z; see Fig. 1.1. By saying "adequately," we introduce a notion that perfect fit might not be possible, which happens to be the case while dealing with real processes. The technical definition of system identification is provided by Zadeh [3]:

> System identification is the determination, on the basis of observation of input and output, of a system within a specified class of systems to which the system under test is equivalent.

Several critical aspects of system identification are encompassed by the preceding definition, namely, availability of input and output (that is, of experimental) data, postulating probable models (that is, class of systems) and choosing the best model from this specified class (that is, equivalency). Thus, the outcome of system identification will be heavily influenced by the models being considered in the process. It also implies that the mathematical model for the physical

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process might not be unique. In contrast to the preceding technical definition, Iliff provides a more philosophical definition [4]:

> Given the answer, what are the questions, that is, look at the results and try to figure out what situation caused those results.

That brings out the central approach behind the process of model building, namely, system identification constitutes an inverse problem.

Thus, system identification is mainly concerned with the determination of the mathematical model structure representing the dynamic system, which is in general unknown and not unique. As already mentioned, different model postulates might have to be investigated. For a specified model, the parameters within the particular model are quantified applying a numerical, usually statistical, procedure. This part of the overall model building process, which happens to be the core activity, is called parameter estimation; see Fig. 1.2. Parameter estimation needs to be followed by a step called model validation to assess model fidelity. If it turns out that the identified model does not meet the requirements, we will have to change the model structure and repeat the whole process. It might happen that additional model structures or even new experimental data might be necessary to arrive at a final model with sufficient fidelity. Thus, system identification is, in general, an iterative process, calling for substantial effort. The amount of effort spent on modeling depends on the required fidelity and end-use of the identified model.

If it happens that a structure is known from some a priori knowledge, or happens to be predefined (for example, analysis of data generated through

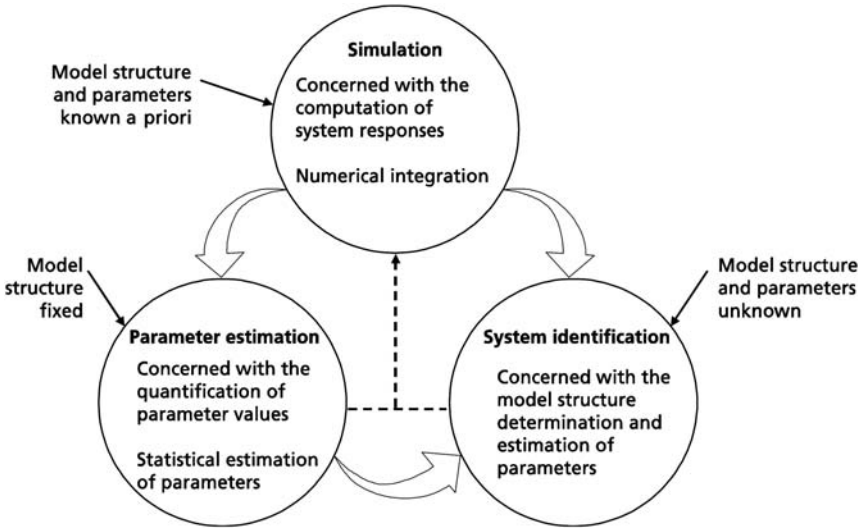

Fig. 1.2 Definitions of system identification, parameter estimation, and simulation.

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numerical simulation), then we deal with only a part of the system identification, called parameter estimation, to determine the numerical values of parameters; see Fig. 1.2. In the literature this step of quantifying numerical values is loosely referred to as "parameter identification" (PID). In our opinion, it is more appropriate to refer to it as "parameter estimation" because it is a quantification procedure. In case it is necessary to change the model structure, we then deal with system identification, which automatically encompasses parameter estimation.

As already pointed out, parameter estimates are determined by matching computed system responses y with the measured system responses z. Thus, having postulated a model, we need a numerical procedure, called simulation, to compute the system responses y for given inputs u. In this task the model structure as well as the numerical values of the parameters are known a priori and held fixed; see Fig. 1.2. Thus, it is now apparent that system identification, parameter estimation, and simulation are three interrelated aspects. From the specific aspect of flight vehicle modeling from flight data, simulation and parameter estimation are integral parts of the overall system identification. Simulations are also widely used in the other disciplines related to flight vehicles, namely, computational fluid dynamics and structural mechanics. Their scope, complexity, and the purpose, as well as the associated model uncertainty, vary significantly compared to flight mechanical simulations. We will address this particular aspect in a little more detail in Sec. 1.3.

1.2 MODEL CHARACTERIZATION

A suitable formulation of a model for the process being investigated is an aspect of paramount importance in system identification. Models can be broadly classified into two types characterized by the process of deriving the model [5].

1. Phenomenological models
2. Behavioral models

Typically, the phenomenological models are derived from basic principles and a theoretical formulation of the phenomenon involving the physics of the process. The theoretical formulation generally results from the Newtonian mechanics and necessarily leads to parameters having physical interpretations. This approach usually leads to a system representation in the form of ordinary or partial differential equations. It is based on a detailed description of each part of the subprocess and automatically leads to a predefined model structure based on the interactions in the subprocesses, providing a detailed representation of the real process in terms of model structure and system parameters. Such knowledge-based models might turn out to be highly complex. In some cases, simplifying assumptions are made to keep the theory tractable or in order to apply well-established analytical methods, or to arrive at lower-order models.

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Phenomenological models being complex might call for a high level of a priori information and effort in some applications. In some other cases it might not even be possible to define the underlying physics. In yet other cases it might be adequate to approximate the observed behavior without specific intent on duplicating the internal system behavior. In all such cases behavioral models provide an alternative form to describe the overall cause-effect relationship and are comparatively easier to derive. Models of this class can be developed using neural network techniques, which are characterized by the following: 1) input-output matching without any specific relevance to the internal behavior of the actual process, 2) reproduction of the overall system response, and 3) the so-called system parameters (or weights) are just numerical values without any physical meaning which the engineers mostly look for [6].

The two aforementioned models differ in terms of imbedded a priori knowledge, system parameters, computation of system response, that is, simulation, and as a result the range of validity. Table 1.1 summarizes these important characteristics and differences between the two types of model formulations [5].

In the case of flight vehicle system identification, we will be mostly dealing with phenomenological models, which can be further classified into two categories:

1. Parametric models
2. Nonparametric models

The state-space or transfer function models belong to the category of parametric models, which assume some suitable structure and order for the model. State-space models could be linear or nonlinear, continuous or discontinuous, time-invariant or time-variant, and deterministic or stochastic. Transfer function representations are mostly based on linear models. Impulse or step response, frequency response or power spectral density models belong to the second category of nonparametric models, which are not based on any specific model structure or order.

According to yet another classification, we can also characterize the models into the following three types:

1. White-box models
2. Black-box models
3. Gray-box models

The so-called "white-box" models are akin to the phenomenological models whereas "black-box" models are akin to the behavioral models.

The gray-box models are, as the name implies, a combination of the aforementioned two variants. Various degrees of combinations are possible. During the early stages of flight vehicle system identification and until recently, white-box models were mostly preferred. However, for specific operating conditions or

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TABLE 1.1 IMPORTANT PROPERTIES OF TWO TYPES OF MODELS

	Phenomenological Models	Behavioral Models
Parameters	Physical meaning	No concrete interpretation
Simulation	Complex and difficult	Quick and easy
A Priori Information	Included	Not necessary
Validity	Large	Restricted

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combinations, it might not be possible or if possible then involving substantial effort, to derive white-box models satisfying desired fidelity requirements. In other cases, safety critical issues might be based on white-box models whereas further improvements could be obtained through appended models based on a black-box approach. In such cases the gray-box models play a useful role. Model cascading can be expanded to include a combination of incremental models based on any of the preceding concepts. For simulation purposes, it is basically immaterial in which form the model is available, as long as the simulation replicates the input-output behavior of the actual process.

1.3 INTERDISCIPLINARY FLIGHT VEHICLE MODELING

As pointed out in Sec. 1.1, the scope and the basic purpose of simulation might differ significantly in the three interrelated disciplines: flight mechanics (FM), computational fluid dynamics (CFD), and structural mechanics (SM). For example, the flight mechanical simulations based on forces and moments, and structural applications involving generalized forces and moments, usually pertain to computing system responses (motion variables or structural deflections) to given inputs or disturbances. On the other hand, CFD simulations usually comprise computing pressure distributions, which are used in the aircraft design and development phase for configuration optimization. In contrast to the purely flight mechanical models characterized adequately by low-order dynamics or to purely structural models characterized by higher-order dynamics, the aeroservoelastic (ASE) models cover both the low- as well as high-frequency range. An interdisciplinary flight vehicle modeling approach shown in Fig. 1.3 is necessary to arrive at appropriate models for aeroservoelastic applications [7]. Advanced software tools catering to such interdisciplinary modeling and simulation have been developed [8]. In such applications, the model uncertainties play a dominant role [9].

The ASE dynamics include coupling due to structural, control, sensor, aero, and actuator dynamics. Because accurate models can be obtained for structural dynamics through ground vibration tests or through finite element method (FEM) computations, for the control dynamics through the design knowledge, and for the sensor dynamics through laboratory calibrations, the ASE modeling

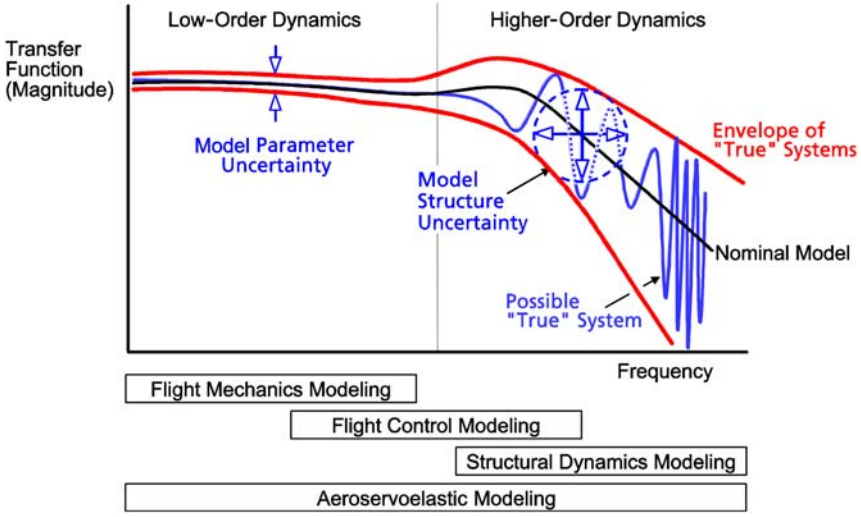

Fig. 1.3 Interdisciplinary flight vehicle modeling [7].

uncertainties are predominantly dictated by the aerodynamic and actuator models [10].

Determination of actuator models from test data, mostly ground tests, is generally restricted to equivalent models. Such models, also validated on flight data, are usually low-order models, typically second to fourth order. They are obtained either in the time domain through state-space representation by matching time histories of displacements or in the frequency domain by matching transfer function computed from the measured input-output data. However, for some applications it might be necessary to incorporate higher-order actuator models based on the physical properties of the system (main ram, servovalve, ram feedback, etc.) and its compliance with the structure [10].

The aerodynamic models must include both the rigid-body modes as well as the unsteady aerodynamics. The representative examples in this book will clearly demonstrate that the modern methods of system identification provide adequate tools to obtain high-fidelity flight-mechanics models. Hence, it is generally desirable to include such flight derived and validated models in terms of aerodynamic derivatives instead of those based on the generalized aerodynamic forces. They are mostly used to investigate stability and control aspects, handling qualities investigations or control system design, and invariably for high-fidelity flight simulators. Such simulations are usually performed by solving an initial-value problem applying numerical integration procedures. The computational requirements are relatively low. Even for large transport aircraft, the influence of structural deformations can usually be described in a quasi-steady manner through a dynamic pressure dependent flex factor, provided the rigid-body and

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structural frequencies are sufficiently separated [11]. Thereby, the generalized equations of motion are usually avoided, which are otherwise necessary to model the structural modes [12]. The structural models, particularly in the case of a large flexible airplane, can influence the accuracy of the flight mechanics models, that is, of the stability and control derivatives. The structural effects on the stability and control derivatives can be removed through filtering of data, or by applying pseudostatic or dynamic structural modeling techniques [13]. The advanced parameter estimation methods discussed in this book and applied widely to estimate flight mechanical models are also being used to derive flight validated models for aeroelastic and flutter analysis [14].

The CFD and ASE prediction and simulation codes are being applied in the advanced research and industrial applications [15, 16]. Application of the CFD and computational unsteady aerodynamics (CUA) methods to predict and validate nonlinear and unsteady aerodynamic responses and aircraft motion time histories was initiated and postulated by Tobak and colleagues [17–19]. There is no doubt that the CFD has revolutionized the aircraft design process and is capable of providing solutions even for complete configurations [20]. However, applicability of such solutions in the subsequent aircraft applications or flight vehicle life cycle is plagued with two restrictions: 1) enormous computational requirements compared to the classical FM models and 2) validation based on experimental data.

Although forces and moments required for the FM models can be obtained by integration of the pressure distributions provided by the CFD solutions, the enormous computational burden makes them unsuitable for flight mechanical investigations. Looking at the tremendous advancements in the computer technology, workstations with significantly more computational power will become available in the future. However, the gap between the computational requirements of FM and CFD models is drastic. One possible approach would be to provide alternative model representations to the available CFD solutions, or approximate those, using system identification methods such as time series modeling, input–output subspace matching, or other similar approaches [21]. Integrating these related technologies is a challenge of the future, and the potential of system identification is likely to pave way in the task. The other aspect of validation based on experimental data is equally involved [22] and calls for measurements of flowfield velocity and other surface variables such as pressures [23]. Today, the validation of CFD results is mostly attempted by comparing them with wind-tunnel measurements; CFD validation from flight-test data is attempted only in a very few cases [24–29]. On the other hand, as it will be demonstrated in this book, validation of FM models from data gathered during flight testing is a routine and well-established procedure. We will be dealing here with the FM models derived and validated from flight data.

1.4 WHY SYSTEM IDENTIFICATION?

Having discussed the meaning of system identification and the classification of models, we now turn our attention to another important aspect, namely, why

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and where do we need system identification. In the flight vehicle development, it is a necessary step because system identification leads to adequately accurate and validated mathematical models of the flight vehicle, which are required to do the following:

- Understand the cause–effect relationship that underlies a physical phenomenon
- Investigate system performance and characteristics
- Verify wind-tunnel and analytical predictions
- Develop high-fidelity aerodynamic databases for flight simulators meeting FAA fidelity requirements
- Support flight envelope expansion during prototype testing
- Derive high-fidelity and high-bandwidth models for in-flight simulators
- Design flight control laws including stability augmentation systems
- Reconstruct the flight-path trajectory, including wind estimation and incidence analysis
- Perform fault-diagnosis and adaptive control or reconfiguration
- Analyze handling qualities specification compliance

Although each of the aforementioned reasons is self-explanatory, we take a look at the two most important arguments. First, system identification yields a complete breakdown of the various components contributing to the observed response and thereby provides an overall understanding of the flight vehicle's dynamics. Second, system identification yields an accurate and comprehensive database for flight simulators and off-line digital simulations, which are extensively used for pilot training and to minimize the risk during experimental testing, which is very costly. The uncertainties in the parameters (aerodynamic database) affect the simulation results and predictions based thereupon.

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1.5 PARAMETER ESTIMATION IN FLIGHT MECHANICS

A flight vehicle is in general free to move in any direction and can be described by coupled equations of motion, which are derived from Newtonian mechanics, usually considering the flight vehicle as a rigid body. These equations involve the fundamental assumption that the forces and moments acting on the flight vehicle can be synthesized. Validity and utility of the mathematical models depend to a large degree on the adequacy and accuracy with which these external forces and moments acting on the flight vehicle can be modeled. The various forces and moments acting on a flight vehicle can be broadly classified as follows: 1) aerodynamic, 2) inertial, 3) gravitational, and 4) propulsive. During the ground roll, additional forces and moments due to landing gear are

encountered. The central issue in system identification of flight vehicles, be it a conventional transport aircraft, a highly augmented unstable aircraft flying at high angles of attack, a rotorcraft, or a missile, is to postulate appropriate models for the aerodynamic forces. The aerodynamic forces and moments acting on the flight vehicle cannot be measured directly. However, aerodynamic modeling followed by parameter estimation allows determination of specific aerodynamic characteristics (lift, drag, and side force coefficients, and rolling, pitching, and yawing moment coefficients in terms of stability and control derivatives) from the related measurements such as accelerations, angular rates, and flow angles.

The aerodynamic modeling, which provides a means to obtain relationships between the three forces X, Y, Z along the three Cartesian coordinates and the moments L, M, N about these axes as functions of the linear translational motion variables u, v, w, rotational rates p, q, r and control surface deflections, was introduced by Bryan in the early 20th century [30]. It is more convenient to express the aerodynamic reactions in terms of nondimensional coefficients because forces and moments depend on dynamic pressure. Different axes systems are defined to facilitate aerodynamic modeling. For example, in the body-fixed system the three force coefficients and three moment coefficients are given by C_x, C_y, C_z, C_ℓ, C_m, C_n. Typically, all of the coefficients depend on the flight condition, for example,

$$C_m = C_m(u,\ v,\ w,\ p,\ q,\ r,\ \delta_e, \ldots\ldots) \tag{1.1}$$

which can be expanded as

$$C_m = \frac{\partial C_m}{\partial u}\, u + \cdots\cdots + \frac{\partial C_m}{\partial q}\, q + \cdots\cdots + \frac{\partial C_m}{\partial \delta_e}\, \delta_e + \cdots\cdots \tag{1.2}$$

The partial derivatives with respect to the motion variables and control inputs, appearing as coefficients in the preceding Taylor series expansion, are defined as the stability and control derivatives. This model postulate is amenable to parameter estimation and marks the beginning of the evolution of flight vehicle system identification. The developments over the last nine decades have led to three different but complementary techniques of determining aerodynamic coefficients: 1) analytical methods, 2) wind-tunnel methods, and 3) flight-test methods.

The first two techniques are employed to generate basic information about the flight mechanical parameters. The analytical estimations can be based on certain simplified assumptions for a complex phenomenon. Nevertheless, as elaborated in Sec. 1.3, computational fluid dynamics has in recent years positively influenced the analytical scenario by providing numerical solutions of complete configurations via sophisticated and advanced Euler and Navier–Stokes flow solvers [31, 32]. Experimental methods are essential to corroborate the analytical predictions. Wind-tunnel techniques, which are comparatively less expensive, have provided in the past a huge amount of data on innumerable flight vehicle

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configurations and are, as a rule, a basis for any new flight vehicle design. These predictions are, however, associated with model scaling, Reynolds number, dynamic derivatives, cross coupling, and aeroservoelasticity effects. Moreover, these measurements could also be affected by tunnel unsteadiness or model support vibrations and interferences. Furthermore, some of the operational test conditions cannot be adequately tested in wind tunnels. Determination of aerodynamic derivatives from flight measurements is, therefore, important and necessary to reduce limitations and uncertainties of the aforementioned two methods [33]. Such an approach, although highly desirable, could turn out to be expensive if flight tests are to be performed at all flight conditions.

Because we are dealing with real processes and measured experimental data, certain complications arise due to the following reasons:

1. Presence of sensor errors and measurement noise
2. Modeling errors
3. Atmospheric turbulence

A block schematic of a system representation incorporating these practical issues is shown in Fig. 1.4. Determination of parameters in the presence of measurement noise requires an estimation procedure based on some statistical criterion; loosely speaking, we have to average out the noise effects on the parameter estimates. The measured data are likely to contain errors due to sensor calibration. Depending upon the location, the sensor can also measure contributions due to other motion variables, besides the primary variable itself. The signal processing and onboard recording system is likely to introduce time delays. Thus, appropriate sensor models will have to be included in the overall system representation.

Incorporation of the aerodynamic model in the equations of motion needs additional information in terms of aircraft mass characteristics (mass, moments

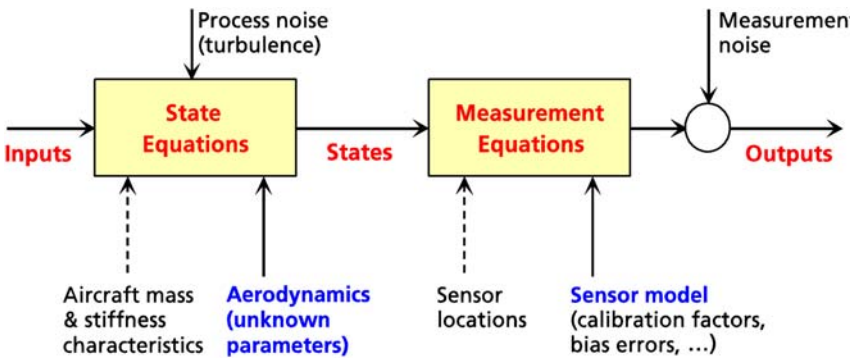

Fig. 1.4 Block schematic of system model for aircraft parameter estimation.

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of inertia, center-of-gravity position). Certain modeling errors are inevitable because the postulated aerodynamic model is an approximation of complex flow phenomena including interference effects. These errors are treated in the estimation process simply as noise. Rigorously, this is not justifiable because modeling errors are deterministic. But, in the absence of any better alternative, this is probably the best approach. The last aspect, atmospheric turbulence, also called process noise, is likewise unavoidable. The atmospheric turbulence during the flight test can be derived indirectly from the measurement of flow variables and flight-path velocity vector and used as additional model inputs. In the absence of such information, the presence of turbulence makes the dynamic system stochastic and significantly complicates the computation of the states.

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1.6 ESTIMATION TECHNIQUES OF THE PAST

The importance of obtaining flight-derived aircraft parameters was recognized early during the aircraft evolution. A couple of years after the introduction of the classical stability approach by Bryan, Glauert's work in 1919 on the analysis of phugoid motion and that of Norton during 1919–1923 on estimating a number of derivatives such as L_p, Y_v, L_v, N_v, and M_w marks the beginning of the experimental investigation of dynamic stability from actual flight data [34–36]. The interest in the dynamic behavior of aircraft grew steadily over the period. During the late 1940s and early 1950s, several techniques were introduced, such as steady-state oscillatory excitations by Milliken [37, 38], pulse-input methods incorporating Fourier analysis by Seamans et al. [39], weighted least-squares [40], and response curve fitting by Shinbrot [41]. An excellent account of dynamic stability and control research during this early period is found in [38] and a survey of methods for determining stability and control derivatives from dynamic flight measurements in [42]. During the 1950s and 1960s, the time-vector method was one of the commonly applied graphical procedures of determining aerodynamic derivatives from free oscillations initiated by an abrupt pulse [43–48]. The time-vector decomposition had to be carried out separately for each force and moment equation. The analog matching techniques were also applied towards the end of this era [49, 50].

The techniques of the 1940s to early 1960s were mostly frequency response methods and were either limited to estimation of incomplete sets of coefficients, to simple motions, or restricted due to other reasons [51]. The aforementioned sketch of the historical background is, by no means, exhaustive. Nevertheless, they represent the important milestones in the history of aircraft parameter estimation prior to the advent of digital computers. A brief account of some of these early methods, which have practically become techniques of the past, is found in [7] and [52].

Although Shinbrot had introduced the response curve fitting method [41], which is equivalent to today's output-error method, it was at that time found to

be impracticable due to the lack of adequate (digital) computational means. During the early 1960s, analog matching was a popular technique of updating and validating wind-tunnel predictions of stability derivatives based on flight tests [49, 50]. Briefly stated, the mathematical model, in several instances the decoupled equations of aircraft motion, was programmed on an analog computer, incorporating theoretical or wind-tunnel predictions of the stability derivatives as first approximations. The flight recorded control inputs, duplicated through function generators using diodes, amplifiers, etc., were fed into the simulation. Comparing qualitatively the simulated response with the actual aircraft response, the stability parameters were manually tuned to reduce the differences. The analog matching technique, although appealing, had several shortcomings, namely: 1) the success or failure depended on the operator's ingenuity in adjusting the proper parameter, 2) it was time consuming, and 3) it was limited to a small number of primary derivatives. Furthermore, the quality of the flight data significantly affected the tuning procedure and involved qualitative judgment of the operator; these two aspects will be involved in the modern methods of system identification to be discussed next, but they provide a more systematic approach to tackle these problems and minimize the resulting uncertainties.

1.7 MODERN METHODS OF AIRCRAFT PARAMETER ESTIMATION

The modern era of system identification is marked by the implementation of the maximum-likelihood method, which we will discuss in Chapters 4 and 5, on a digital computer by Åström and Bohlin in 1965, and applying it to an industrial plant represented by difference equations [53]. The automatic data processing capability provided by digital computers dramatically changed the focus of flight data analysis from frequency-domain methods to time-domain methods. It became possible to obtain a significantly larger number of stability and control derivatives from a single flight test. A coordinated approach based on flight-test techniques, flight-test instrumentation, and methods of data analysis gradually evolved for flight vehicle system identification.

- Instrumentation and filters, which cover the entire flight data acquisition process including adequate instrumentation and airborne or ground-based recording equipment, are needed to gather experimental data. Effects of all kinds of data quality have to be accounted for.
- Flight-test techniques, which are related to the selected flight vehicle maneuvering procedures, are required. The input signals have to be optimized in their spectral composition in order to excite all response modes from which parameters are to be estimated.
- Analysis of flight data, which includes the mathematical model of the flight vehicle and an estimation criterion that devises some suitable computational algorithm to adjust some kind of starting values or a priori estimates of the

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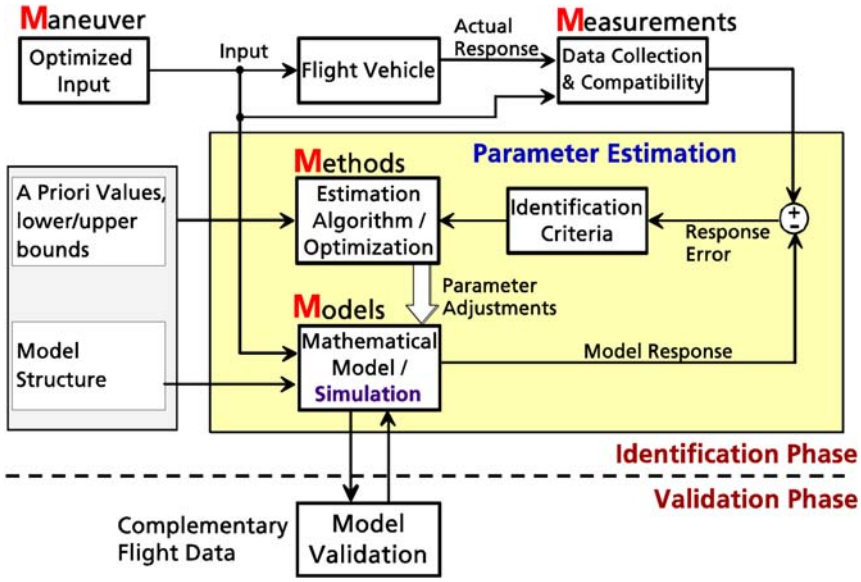

Fig. 1.5 Quad-M basics of flight vehicle system identification.

unknown parameters until a set of best parameter estimates is obtained that minimizes the response error, is also necessary.

Corresponding to these strongly interdependent topics, four important aspects of the art and science of system identification have to be carefully treated, namely, Maneuver, Measurements, Methods, and Models; see Fig. 1.5.

- Design of the control input shape in order to excite different modes of the vehicle dynamic motion
- Selection of instrumentation and filters for high accuracy measurements
- Quality of data analysis by selecting the most suitable time or frequency-domain identification method
- Type of flight vehicle under investigation in order to define the structure of a possible mathematical model

These **Quad-M** requirements must be carefully investigated for each flight vehicle and are the key to successful flight vehicle system identification [7]. It leads to an interdisciplinary task, covering the fields of statistical theory, control theory, numerical techniques, sensor and instrumentations, signal processing, flight-test techniques, and flight mechanics. To efficiently apply this methodology, appreciation of basic aspects of each of them is useful in terms of advantages,

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limitations, and likely problems. The Quad-M requirements fall within the framework of the contemporarily and widely accepted definition provided by Zadeh [3]. A systematic treatment of these key issues is followed in all of the applications presented in this book, focusing mainly on methods and modeling aspects.

1.8 GENERAL ASPECTS

The subject of statistical estimation theory being too vast, the number of flight vehicle applications too diversified, and the published literature too abundant, it is necessary to define the scope, the purpose, and overall organization of the book.

1.8.1 THE AIM

The present book aims at providing a consolidated account of state of the art of time-domain system identification with underlying mathematical principles from an engineer's view. It also aims at sharing the experience gained on aerodynamic modeling from flight-test data applying time-domain methods on a large number of flight vehicles because experience and engineering judgment are critical to generate good results. The pursuit is to expose the art and science of system identification by providing adequate theory and focusing on explanation of algorithmic steps and intricacies involved in the applications of the various methods. We aim at raising some of the issues that may deviate from procedures advocated in the past, and this might lead to some discussion, but then the reported schemes have worked in practice, providing an acceptable solution in some of the most demanding tasks. We will also bring out some topics of future research. We aim at addressing a broad spectrum of modeling cases with different degrees of complexities, in each case emphasizing practical issues.

1.8.2 THE LITERATURE

Flight vehicle system identification has been pursued at almost every organization dealing with the subject of atmospheric flight. In particular the major contributions to theory and applications have resulted from research activities at several organizations, like NASA Dryden and Langley Research Centers (RC), U.S. Army Aeroflightdynamics Directorate at NASA Ames RC, Calspan; Institute of Flight Systems, DLR, Germany; Delft University of Technology (TU-Delft) and National Aerospace Laboratory (NLR), The Netherlands; Defence Research Agency (DRA), England; National Research Council (NRC), Canada; and several other academic institutions. Pioneering contributions were made by individuals such as Taylor and Iliff [54], Mehra [55], and Gerlach[56]. Since more than four decades, the aerospace industry has been using this methodology for model validation, handling qualities evaluations, control law design, and flight vehicle design.

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The symbiosis of these international efforts has culminated into the present state of maturity. This has become possible mainly because of dissemination of information through published literature from various sources, in which American Institute of Aeronautics and Astronautics (AIAA) journal publications have played a major role. Broad overviews pertaining to flight vehicle system identification are found in the survey papers published by AIAA: 1) Iliff with 76 references summarizing contributions to flight vehicle system identification up to 1980 [52] and 2) Hamel and Jategaonkar having 183 references presenting the state of the art up to 1995 [7]. Other overview contributions are by 1) Klein giving an account of system identification methodology [57], 2) Padfield collecting 13 contributions in rotorcraft applications [58], and 3) Hamel and Kaletka surveying the progress in rotorcraft system identification up to 1997 [59]. Besides the survey papers, on every facet of aerodynamic modeling there are excellent technical papers, mostly in the AIAA *Journal of Aircraft*, including the two special focus issues on "Flight Vehicle System Identification – Engineering Utility" published in 2004 and 2005 [60, 61]. This was followed by the unique role played by AIAA in 2006 by publishing simultaneously three books related to flight vehicle system identification, two under their Education Series [62, 63] and one under the Progress in Astronautics and Aeronautics series [64]. They cover engineering approaches followed by practicing researchers to flight vehicle system identification, including both the time-domain and the frequency-domain methods. More recently the second edition of the book on Rotorcraft System Identification has also been published by AIAA [65].

The former Advisory Group for Aerospace Research and Development (AGARD), and to some extent the successor organization Research and Technology Organization (RTO), effectively stimulated and fostered international scientific and technical interchange, collaboration, and documentation in the field of flight vehicle system identification. Notable among the various publications are the three AGARDOGraphs (AG): one on estimation theory by Maine and Iliff [66] and two on aircraft applications, one by Maine and Iliff [67] and the other by Mulder et al. [68]. An excellent repertoire on rotorcraft applications is provided by the Advisory Report (AR) edited by Hamel [69]. Likewise, the AG compiled by Stoliker is a significant volume, providing a consolidated account on flight testing [70]. An exhaustive treatment of the various aspects of flight-test instrumentation for all possible applications is found in the 19 volumes of [71]. Noteworthy is also the unique RTO effort in bringing together the experts from four streams, namely, dynamics, system identification, control, and handling qualities, addressing the land, air, sea, and space vehicle applications to highlight challenges faced and the similarity in the employed expertise [72].

To a smaller extent, contributions come from the International Federation on Automatic Control (IFAC) symposiums, whose focus has been mainly on the theory and applications in areas other than flight vehicles. Apart from the aforementioned literature related to flight vehicle applications, the basic estimation theory is covered in textbooks such as those by Walter and Pronzato [5],

Eykhoff [73], Goodwin and Payne [74], Sorenson [75], Ljung and Söderström [76], and others.

We include at the end of each chapter a fairly exhaustive list of references pertinent to the topics discussed. These references help to trace the development and also to show the widespread applicability of this methodology. We have also tried to provide some historical references, which help us to appreciate the engineering approach and deep understanding of the physical phenomenon being analyzed by the early pioneers. If for any reason one or the other publication is missed out, the omission is unintentional and without any bias. In this book, it is not our claim that the references to the published literature provide a complete bibliography.

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1.8.3 THE LAYOUT

In Chapter 1, we have provided a general background on system identification, including various definitions, the purpose and the need. It has been brought out that system identification, parameter estimation, and simulation are three interrelated aspects. After a brief account of the historical background, the Quad-M basics of the modern approach to flight vehicle system identification have been introduced. The general layout of the book is summarized in Fig. 1.6, which shows at a glance that the complete material covered in this book can be broadly split into three important parts shown on the left-hand side, namely, experiment, methods, and applications. Some overlap of methods and models is unavoidable. The references to the four Ms of the Quad-M basics and to model validation have also been indicated in the same figure on the right-hand side.

In Chapter 2, we deal with the first of the four Ms, namely, maneuvers leading to data gathering. A brief introduction to flight testing relevant to system identification is provided. Typical flight maneuvers and commonly applied inputs to excite dynamic motion are presented. Although design of optimal inputs has been one of the subjects of extensive research, we approach this aspect more pragmatically from the practical implementation viewpoint. A brief account of flight measurements required for parameter estimation is presented, followed by a detailed discussion on the practical aspects of smoothing as well as numerical differentiation of noisy data and on the proper choice to handle data with different noise characteristics. This chapter covers a part of the second M for measurement; the other part is covered later in the book for reasons mentioned subsequently.

Chapter 3 introduces various mathematical model formulations that form the basis for the models analyzed in the following chapters. Several model extensions are proposed to account for errors in the measurements, to analyze multiple experiments simultaneously, and to estimate time delays. Commonly applied numerical integration procedures are briefly presented, making a recommendation on the suitable choice.

In Chapter 4, we introduce the most widely used maximum-likelihood principle, extend the same to general nonlinear dynamic systems, and develop the output-error method. Three gradient-based algorithms, namely, Gauss–Newton,

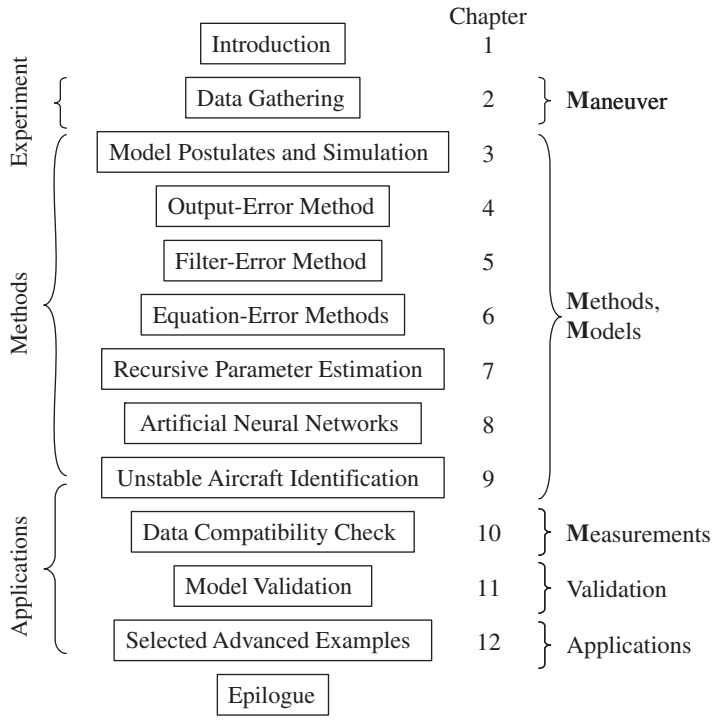

Fig. 1.6 Book layout.

quasi-linearization, and Levenberg–Marquardt, are presented, in each case discussing the algorithmic details. Solutions necessary to overcome convergence problems in some of these methods are presented. This is followed by introducing more recent techniques, for example, constrained optimization based on the bounded-variable Gauss–Newton method or the interior-point method. A brief summary of direct search optimization methods is provided. Details of efficient implementation to handle large-scale systems are elaborated. The techniques discussed here are applied to two test cases.

Chapter 5 presents the theory and development of the most complex and most general aircraft parameter estimation problem, namely, estimation accounting for atmospheric turbulence. The presence of process noise makes the system stochastic, requiring state estimation techniques to be incorporated in parameter estimation. Such techniques are called filter-error methods. Extensions to nonlinear systems are presented, giving step-by-step computational procedures for linear as well as nonlinear systems. The filter-error methods are applied to typical examples, highlighting the need, advantages, and limitations of these methods.

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The methods presented in Chapters 4 and 5 belong to the class of output-error methods and represent the two central topics of this book. In Chapter 6, we cover the methods belonging to the other class, namely, equation-error methods. We present the ordinary least-squares, weighted least-squares, and total least-squares methods; we discuss statistical properties of the estimates, practical aspects in the presence of noise, and demonstrate extensions to state-space models under certain assumptions. Nonlinear and multiple-output regression techniques are covered. We take a pragmatic look at the process of model-structure determination. Finally, we apply the one-shot and the iterative least-squares methods to linear and nonlinear models and the total least squares method to linear model.

The methods presented in Chapters 4 to 6 evaluate the complete set of data together at one time and are suitable for off-line analysis. In Chapter 7, we cover some basic applications of recursive estimation methods that are suitable for online implementations. Five different techniques, namely, the recursive least-squares method, the discrete Fourier transform method, the extended Kalman filter, the unscented Kalman filter in both simplified and general form, and the extended forgetting factor recursive least squares are presented. They are applied to flight data, and a comparative study based on the results is presented.

In Chapter 8, we cover a related approach to overall system characterization by mapping the input–output subspace. The basic concepts of neural processing are elaborated, concentrating on the feedforward neural networks that lead to black-box models, without explicit knowledge about the internal behavior and physics of the phenomenon. Various training algorithms are presented. These techniques are applied to three different examples of increasing complexity. The advantages, limitations, and issues related to practical applications of such networks in providing a general function approximation of any nonlinear phenomenon are brought out.

Chapter 9 addresses exclusively unstable aircraft identification. To start with, the problems encountered in the open-loop identification of inherently unstable aircraft are discussed. It is demonstrated that the difficulties encountered are practical issues of applying system identification and not of estimation methodology being applicable or not. This is followed by a summary of estimation methods applicable to unstable aircraft, highlighting that mostly the same estimation algorithms covered in the preceding chapters or some variants of them are applicable using measured data within the simulation and gradient computation. It is demonstrated that the major problem is due to data collinearity resulting from highly correlated control inputs and aircraft motion variables, which can be eliminated efficiently through separate surface excitations. A detailed comparative study is presented to evaluate different algorithms, and finally three of them are applied to flight data from an unstable aircraft.

Chapter 10 covers the various aspects of data compatibility checking, which ensures that recorded data are consistent and error free. Techniques applied to estimate the aircraft states and/or instrument errors are presented. It is argued that the output–error method developed in Chapter 4 is often adequate for this

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purpose. It is applied to typical flight data gathered from longitudinal and lateral-directional motions. Calibrations of a five-hole probe for flow angles and of static pressure ports are presented in some details. According to the Quad-M basics presented earlier in the current chapter, the logical sequence of presentation would have been the data gathering covered in Chapter 2 pertaining to the first M for maneuver, followed by the second M for measurements, and then the other two Ms, namely models and methods, which are covered in Chapters 3 through 9. Maneuvers and measurements are prerequisites for the off-line data analysis covering models and methods. We have, however, presented the discussion on data compatibility checking corresponding to the second M for measurements in Chapter 10, mainly because verification of data and estimation of instrument error requires algorithms that are part of the methods covered in Chapters 4 to 6.

Chapter 11 deals with model validation aspects, covering different techniques applied to assess the accuracy of estimated parameters and the fidelity of the identified models. Besides the statistical accuracy, residual analysis and inverse-simulation techniques are presented. Model predictive capability is the decisive factor in judging the fidelity; some practical issues in this connection are covered.

Chapter 12 provides an account of a wide variety of aerodynamic modeling and parameter estimation problems and analyzes flight data from different types of flight vehicles. A total of 12 different cases covering nonlinear models, primary and secondary aerodynamic effects, control surface malfunction, unsteady aerodynamics, symmetric and asymmetric quasi-steady stall, verification and update of high-fidelity databases, flexible aircraft modeling, wake vortex encounters, and rotorcraft are presented. The analyzed flight vehicles range from fixed-wing aircraft (high-wing and low-wing configurations), two propeller aircraft, unstable aircraft, glider to reusable launch vehicle demonstrator, and rotorcraft vehicles.

Finally, we conclude with an epilogue, summarizing the adopted system identification philosophy, which has worked well in practice. We attempt to indicate probable future directions of flight vehicle system identification.

The chapters are mostly self-contained. Although reading the chapters in a sequence as presented in the book will help to gain a systematic insight into the complete complex of flight vehicle system identification, some of the specific topics, like the filter-error method covered in Chapter 5 or unstable aircraft identification dealt with in Chapter 9, can be handled separately. Certain familiarity with matrix algebra, statistical theory, and system theory will be helpful whereas basic knowledge of flight mechanics in terms of equations of motion and modeling of forces and moments acting on flight vehicles is presumed [77–79].

1.8.4 SOFTWARE AND FLIGHT DATA

The book includes software programs that can be downloaded from the AIAA website (for details see Supporting Materials). The programs are mostly sorted

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by chapter and implement most of the algorithms described in the book. The software for the filter-error method capable of handling general nonlinear systems is provided for the first time. An extended and more flexible version of the software for the output-error method that is suitable for more advanced applications is included. The flight data analyzed in the test cases are supplemented to the book in the form of data files containing time histories of measured aircraft inputs and responses. The main aim of providing software is to help the reader better trace and correlate theory, computational steps, and results presented here. This will be possible through the algorithmic implementations elaborated in the text and comments provided in the programs, including equation numbers as they appear in the individual chapters. To facilitate this purpose, we have included the description of software in the individual chapters to allow treatment of a particular topic at one stretch. This differs from the usual practice of providing the software description as an appendix at the end; it leads to a better continuity and awakening reader's specific interests. We do not wish to provide a frozen software, that is, only the executable version, with little options to modify or extend the same to suit individual needs. Rather, we wish to provide the reader an opportunity to get himself or herself involved in the algorithmic details and programming. A deeper understanding of the estimation algorithms has proved helpful in appreciating the advantages and limitations of the methods.

The software is developed under the Matlab® computing environment [80]. It has been applied to test cases presented at the end of the chapters, describing the used models and data files analyzed. These fairly simple test cases enable familiarity with the software itself and also aid in tracing the results presented. The software has been developed using the basic Matlab® only. Additional toolboxes would lead to different ways of implementing the algorithms and possibly to a much simpler programming. For example, 1) availability of the Control Toolbox would allow solving the Riccati equation in Sec. 5.4.1, using the function CARE; we have, however, provided a simple basic version to do this job, and 2) definition of a state-space model is through user-defined functions. Availability of Matlab/Simulink® would change not only the interface, but also the loops over the data points. Such modifications are left to the reader. The software has been tested with six Matlab® versions: 1) Version 6.0.0, Release 12; 2) Version 6.5.1, Release 13 (Service Pack 1); 3) Version 7.0.0, Release 14; 4) Version 7.1.0.246, Release 14, Service Pack 3; 5) Version 7.5.0.342 (R2007b); and 6) Version 8.1.0.604 (R2013a). Matlab® itself is not a part of the book or of the software provided, but a product of The MathWorks, Inc. [80].

REFERENCES

[1] Boehner, P. (ed. & trans.), *William of Ockham: Philosophical Writings*, Hackett Publishing Co., Inc., Indianapolis , IN, 1990.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[2] Maurer, A. A., "Method in Ockham's Nominalism," *The Monist*, Vol. 61, No. 3, 1978, pp. 426–443.

[3] Zadeh, L. A., "From Circuit Theory to System Theory," *Proceedings of the IRE*, Vol. 50, May 1962, pp. 856–865.

[4] Iliff, K. W., "He Has a Question for Any Answer," Interview by M. McCall, Antelope Valley Press, CA, Nov. 1, 1994.

[5] Walter, É., and Pronzato, L., *Identification of Parametric Models*, Springer, Berlin, 1997.

[6] Sjöberg, J., Zhang, Q., Ljung, L., Benveniste, A., Deylon, B., Glorennec, P.-Y., Hjalmarsson, H., and Juditsky, A., "Nonlinear Black-Box Modelling in System Identification: A Unified Overview," *Automatica*, Vol. 31, No. 12, 1995, pp. 1691–1724.

[7] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[8] Gupta, K. K., "STARS – An Integrated, Multidisciplinary, Finite-Element, Structural, Fluids, Aeroelastic, and Aeroservoelastic Analysis Computer Program," NASA TM 4785, May 1997.

[9] Livne, E., "Future of Airplane Aeroelasticity," *Journal of Aircraft*, Vol. 40, No. 6, 2003, pp. 1066–1092.

[10] Brenner, M. J., "Actuator and Aerodynamic Modeling for High-Angle-of-Attack Aeroservoelasticity," NASA TM-4493, June 1993.

[11] Hamel, P. G., and Jategaonkar, R. V., "The Role of System Identification for Flight Vehicle Applications: Revisited," RTO-MP-11, Paper 2, March 1999.

[12] Waszak, M. R., and Schmidt, D. K., "Flight Dynamics of Aeroelastic Vehicles," *Journal of Aircraft*, Vol. 15, No. 6, 1988, pp. 563–571.

[13] Iliff, K. W., "Aircraft Identification Experience," AGARD, LS-104, Paper 6, Nov. 1979.

[14] Perangelo, H. J., and Waisanen, P. R., "Application of Advanced Parameter Identification Methods for Flight Flutter Data Analysis with Comparison to Current Techniques," AGARD, CP-373, Paper 5, April 1984.

[15] Meijer, J. J., Hounjet, M. H. L., Eussen, B. J. G., and Prananta, B. B., "NLR-TUDelft Experience in Unsteady Aerodynamics and Aeroelastic Simulation Applications," *Numerical Unsteady Aerodynamics and Aeroelastic Simulation*, AGARD, R-822, Paper 11, March 1998.

[16] Vos, J. B., Rizzi, A., Darracq, D., and Hirschel, E. H., "Navier-Stokes Solvers in European Aircraft Design," *Progress in Aerospace Sciences*, Vol. 38, 2002, pp. 601–697.

[17] Tobak, M., and L. B., "Aerodynamic Mathematical Modeling – Basic Concepts," AGARD, LS-114, Paper 1, May 1981.

[18] Tobak, M., and Chapman, G. T., "Nonlinear Problems in Flight Dynamics Involving Aerodynamic Bifurcation," AGARD, CP-386, Paper 25, Nov. 1985.

[19] Hui, W. H., and Tobak, M., "Bifurcation Theory Applied to Aircraft Motions," AGARD, CP-386, Paper 26, Nov. 1985.

[20] Tinoco, E. N., "An Assessment of CFD Prediction of Drag and Other Longitudinal Characteristics," AIAA Paper 2001-1002, Jan. 2001.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[21] Cowan, T. J., Arena, A. S., Jr., and Gupta, K. K., "Accelerating Computational Fluid Dynamics Based Aeroelastic Predictions Using System Identification," *Journal of Aircraft*, Vol. 38, No. 1, 2001, pp. 81–87.

[22] "Verification and Validation Data for Computational Unsteady Aerodynamics," RTO-TR-26, Oct. 2000.

[23] Marvin, J. G., "Perspective on Computational Fluid Dynamics Validation," *AIAA Journal*, Vol. 33, No. 10, 1995, pp. 1778–1787.

[24] Hawkins, R. W., and Dilley, A. D., "CFD Comparisons with Wind Tunnel and Flight Data for the X-15," AIAA Paper 1992-5047, 1992.

[25] Roberts, A. T., and Henderson, T. L., "Baseline CFD and Flight Test Comparison for SOFIA," AIAA Paper 1999-0533, Jan. 1999.

[26] Shivananda, T., McKeel, S., Salita, M., and Zabrensky, E., "Space Launch Vehicle Aerodynamics – Comparison of Engineering and CFD Predictions with Wind Tunnel Data," AIAA Paper 2001-0258, Jan. 2001.

[27] Hübner, A.-R., and Loeser, T., "Methods for Determination of the Unsteady Aerodynamic Derivatives for Transport Aircraft Configurations," RTO-MP-AVT-123, Paper 12, March 2005.

[28] Klausmeyer, S. M., "Drag, Lift, and Pitching Moment Estimates of Transonic Aircraft Using the Navier-Stokes Equations," AIAA Paper 2004-0533, Jan. 2004.

[29] Babcock, D. A., and Arena, A. S., Jr., "Estimating Aircraft Stability Derivatives Through Finite Element Analysis," AIAA Paper 2004-5174, Aug. 2004.

[30] Bryan, G. H., *Stability in Aviation: An Introduction to Dynamical Stability as Applied to the Motions of Aeroplanes*, Macmillan, London, 1911.

[31] Kroll, N., Radespiel, R., and Rossow, C.-C., "Accurate and Efficient Flow Solvers for 3D Applications on Structured Meshes," VKI Lecture Series 1994-05, Brüssel, Belgium, March 1994.

[32] Slooff, J. W., and Schmidt, W. (eds.), "Computational Aerodynamics Based on the Euler Equations," AGARD, AG-325, Sept. 1994.

[33] Hamel, P. G., "Determination of Aircraft Dynamic Stability and Control Parameters from Flight Testing," AGARD, LS-114, Paper 10, May 1981.

[34] Glauert, H., "Analysis of Phugoids Obtained by a Recording Airspeed Indicator," Aeronautical Research Council, A.R.C. R&M No. 576, Jan. 1919.

[35] Norton, F. H., "The Measurement of the Damping in Roll on a JN4h in Flight," NACA Report 167, 1923.

[36] Norton, F. H., "A Study of Longitudinal Dynamic Stability in Flight," NACA Report 170, 1923.

[37] Milliken, W. F., Jr., "Progress in Dynamic Stability and Control Research," *Journal of the Aeronautical Sciences*, Vol. 14, No. 9, 1947, pp. 493–519.

[38] Milliken, W. F., Jr., "Dynamic Stability and Control Research," *Proceedings of the 3rd Anglo-American Aeronautical Conference*, Brighton, England, U.K., 1951, pp. 447–524.

[39] Seamans, R. C., Jr., Blasingame, B. P., and Clementson, G. C., "The Pulse Method for the Determination of Aircraft Dynamic Performance," *Journal of the Aeronautical Sciences*, Vol. 17, No. 1, 1950, pp. 22–38.

[40] Shinbrot, M., "On the Analysis of Linear and Nonlinear Dynamical Systems from Transient Response Data," NACA TN-3288, Dec. 1954.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[41] Shinbrot, M., "A Least-Squares Curve Fitting Method with Applications to the Calculation of Stability Coefficients from Transient Response Data," NACA TN 2341, April 1951.

[42] Greenberg, H., "A Survey of Methods for Determining Stability Parameters of an Airplane from Dynamic Flight Measurements," NACA TN-2340, April 1951.

[43] Mueller, R. K., "The Graphical Solution of Stability Problems," *Journal of the Aeronautical Sciences*, Vol. 4, June 1937, pp. 324–331.

[44] Breuhaus, W. O., "Resumé of the Time-Vector Method as a Means for Analyzing Aircraft Stability Problems," Wright Air Development Center, U.S. Air Force, WADC TR-52-299, Nov. 1952.

[45] Doetsch, K. H., "The Time Vector Method for Stability Investigations," Royal Aircraft Establishment, RAE Report, Aero 2495, Aug. 1953; also Aeronautical Research Council, ARC R&M No. 2945, 1957.

[46] Larrabee, E. E., "Application of the Time-Vector Method to the Analysis of Flight Test Lateral Oscillation Data," Cornell Aero. Lab, Inc., Rept. FRM No. 189, Sept. 1953.

[47] Sternfield, L., "A Vector Method Approach to the Dynamic Lateral Stability of Aircraft," *Journal of the Aeronautical Sciences*, Vol. 21, No. 4, 1954, pp. 251–256.

[48] Wolowicz, C. H., and Holleman, E. C., "Stability Derivative Determination from Flight Data," AGARD, Rept. 224, Oct. 1958.

[49] Rampy, J. M., and Berry, D. T., "Determination of Stability Derivatives from Flight Test Data by Means of High Speed Repetitive Operation Analog Matching," Air Force Flight Test Center, Edwards Air Force Base, FTC-TDR-64-8, Edwards, CA, May 1964.

[50] Wolowicz, C. H., "Considerations in the Determination of Stability and Control Derivatives and Dynamic Characteristics from Flight Data," AGARD, Rept. 549, Part 1, 1966.

[51] Rynaski, E. G., "Application of Advanced Identification Techniques to Nonlinear Equations of Motion," *Proceedings of the Stall/Post-Stall/Spin Symposium*, Wright-Patterson AFB, OH, Dec. 1971, pp. 0.1–0.18.

[52] Iliff, K. W., "Parameter Estimation for Flight Vehicles," *Journal of Guidance, Control, and Dynamics*, Vol. 12, No. 5, 1989, pp. 609–622.

[53] Åström, K. J., and Bohlin, T., "Numerical Identification of Linear Dynamic Systems from Normal Operating Records," *Proceedings of the 2nd IFAC Symposium on Theory of Self-Adaptive Control Systems*, edited by P. H. Hammond, Plenum Press, New York, 1966, pp. 96–111.

[54] Taylor, L. W., Jr., and Iliff, K. W., "A Modified Newton-Raphson Method for Determining Stability Derivatives from Flight Data," *Computing Methods in Optimization Problems*, edited by L. A. Zadeh, L. W. Neustadt, and A. V. Balakrishnan, Academic Press, New York, 1969, pp. 353–364.

[55] Mehra, R. K., "Identification of Stochastic Linear Dynamic Systems Using Kalman Filter Representation," *AIAA Journal*, Vol. 9, No. 1, 1971, pp. 28–31.

[56] Gerlach, O. H., "Determination of Performance, Stability and Control Characteristics from Measurements in Non-Steady Manoeuvres," AGARD, CP-17, Sept. 1966, pp. 499–523.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[57] Klein, V., "Estimation of Aircraft Aerodynamic Parameters from Flight Data," *Progress in Aerospace Sciences*, Vol. 26, No. 1, 1989, pp. 1–77.

[58] Padfield, G. D., (guest ed.), "Applications of System Identification in Rotorcraft Flight Dynamics," *Vertica*, Vol. 13, No. 3, 1989, pp. 207–472.

[59] Hamel, P. G., and Kaletka, J., "Advances in Rotorcraft System Identification," *Progress in Aerospace Sciences*, Vol. 33, No. 3/4, 1997, pp. 259–284.

[60] Jategaonkar, R. V. (guest ed.), "Special Section: Flight Vehicle System ID – Part 1," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 681–764.

[61] Jategaonkar, R. V. (guest ed.), "Special Section: Flight Vehicle System ID – Part 2," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 11–92.

[62] Klein, V., and Morelli, E. A., *Aircraft System Identification: Theory and Practice*, AIAA Education Series, AIAA, Reston, VA, Aug. 2006.

[63] Tischler, M. B., and Remple, R. K., *Aircraft and Rotorcraft System Identification: Engineering Methods with Flight-Test Examples*, AIAA Education Series, AIAA, Reston, VA, Aug. 2006.

[64] Jategaonkar, R. V., *Flight Vehicle System Identification: A Time Domain Methodology*, AIAA Progress in Astronautics and Aeronautics, Reston, VA, Aug. 2006.

[65] Tischler, M. B., and Remple, R. K., *Aircraft and Rotorcraft System Identification: Engineering Methods with Flight-Test Examples*, 2nd ed., AIAA Education Series, AIAA, Reston, VA, Aug. 2012.

[66] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[67] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems – Applications to Aircraft. Part 1: The Output Error Approach," AGARD, AG-300, Vol. 3, Part 1, Dec. 1986.

[68] Mulder, J. A., Sridhar, J. K., and Breeman, J. H., "Identification of Dynamic Systems – Applications to Aircraft, Part 2: Nonlinear Analysis and Manoeuvre Design," AGARD, AG-300, Vol. 3, Part 2, May 1994.

[69] Hamel, P. G. (ed.), "Rotorcraft System Identification," AGARD, AR-280, Sept. 1991.

[70] Stoliker, F. N. (ed.), "Introduction to Flight Test Engineering," AGARD, AG-300, Vol. 14, Sept. 1995.

[71] AGARD Flight Test Instrumentation Series, AGARD AG-160, Vol. 1–19, 1971–1991.

[72] A'Harrah, R. C., "Technical, Evaluation Report," "Challenges in, Dynamics, System, Identification, Control and Handling Qualities for, Land, Air, Sea and Space Vehicles," RTO-MP-095, Jan. 2003, pp. T-1–T-4.

[73] Eykhoff, P., *System Identification, Parameter and State Estimation*, Wiley, New York, 1974.

[74] Goodwin, G. C., and Payne, R. L., *Dynamic System Identification: Experiment Design and Data Analysis*, Academic Press, New York, 1977.

[75] Sorenson, H. W., *Parameter Estimation – Principles and Problems*, Marcel Dekker, New York, 1980.

[76] Ljung, L., and Söderström, T. S., *Theory and Practice of Recursive Identification*, MIT Press, Cambridge, MA, 1983.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[77] Etkin, B., *Dynamics of Atmospheric Flight*, Wiley, New York, 1972.

[78] McRuer, D., Ashkenas, I., and Graham, D., *Aircraft Dynamics and Automatic Control*, Princeton Univ. Press, Princeton, NJ, 1973.

[79] Schmidt, D. K., *Modern Flight Dynamics*, McGraw-Hill, New York, 2010.

[80] "MATLAB, The Language of, Technical Computing: Computation, Visualization, Programming, Mathematics Version, 7, MATLAB 7.1, (Release 14SP3)," The MathWorks, Inc., Natick, MA, Sept. 2005.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

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CHAPTER 2

Data Gathering

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2.1 INTRODUCTION

The process of performing experiments and recording system inputs and outputs constitutes data gathering. Besides the ability to efficiently extract unknown system parameters applying advanced estimation techniques, which we will cover in the following chapters, data gathering is the other crucial aspect of flight vehicle system identification because of this basic rule: *"If it is not in the data, it cannot be modeled."* This rule applies to all exercises that attempt parameter estimation from experimental data. This is true irrespective of the type of flight vehicle we might attempt to model. Gathered data basically limits, both in terms of scope and accuracy, the model development and parameter estimation. Although data-acquisition-related aspects like parameters (aircraft motion variables and control surface deflections) to be recorded, quality of sensors in terms of accuracy and noise, sampling rate, signal conditioners and data recording equipment play a role in the overall process of data analysis, the most important aspects of data gathering are to do the following:

1. Define the scope of the flight testing
2. Define a suitable sequence of flight maneuvers to be performed at each test point
3. Choose an adequate form of the inputs to excite the aircraft motion in some optimum sense

The last mentioned topic is commonly called in the literature as "optimal input design."

Accuracy and reliability of parameter estimates, obtained applying either the recent modern methods like maximum likelihood or the methods of the past elaborated in Secs. 1.7 and 1.6 respectively, depend heavily on the amount of information available in the vehicle response, which can be maximized through

suitable excitations. This fact was recognized early during the evolution stages as evident from Milliken's statement in 1951 [1]:

> It would appear that the optimum input in a given case is that which best excites the frequency range of interest, and hence the harmonic content of the input should be examined before the test to ensure that it is suitable.

Throughout the history of flight vehicle system identification, this has been, in essence, the guiding principle in designing a proper flight-test maneuver for extracting aerodynamic parameters [2]. The early development in the mid-1940s and early 1950s led to heuristic input signals such as continuous sinusoidal input or pulse input. These designs were mainly governed by the method of analysis applied to extract stability and control derivatives from flight data. In the general field of system identification, theoretical developments on input design started during the 1960s [3–5]. Around the same time, optimal design for flight vehicle system identification started with the work of Gerlach [6, 7]. During the early 1970s, the subject of optimal input designs for aircraft parameter estimation has been extensively researched by Mehra, Gupta, Mulder, Koehler, Marchand, and Plaetschke et al., both from theoretically motivated design as well as more practical oriented approaches [8–13]. They led to inputs based on some suitable combinations of harmonic signals or of multiple pulses. An overview on optimal input and maneuver design is provided by Mulder et al. [14].

Although a lot of effort has been and continues to be spent on optimal input design [15–17], the methodology is plagued by a fundamental problem, namely, the design is based on an a priori model of the flight vehicle. Better a priori models lead to better designs; in other words, for a good input design a good model must be known ahead of flight testing. However, in such a case the estimation from flight data would no longer be needed. There is no exact solution to this paradoxical situation; the only way out is to approach the problem pragmatically and attempt to design an input that is sufficiently broadband in frequency content to cover uncertainties in the a priori model and at the same time easy to adapt to changing flight conditions.

In this chapter we discuss the scope and types of flight testing, providing explanation of maneuvers in a nutshell. The rigorous approaches of optimal input design based on theoretical formulations are briefly summarized without going deeply into mathematical details. A detailed explanation is, however, provided of the basic ideas behind a simple, yet efficient, engineering approach to design a multistep input. Simple software programs are provided to trace the discussion on multiple step inputs. Based on the basic knowledge about the natural frequencies of various modes of motion, typical empirical rules to arrive at time steps for multistep inputs are elaborated. From our experience of modeling different types of flight vehicles, we aim at suggesting a general procedure for performing dynamic flight maneuvers for system identification purposes. Finally, we take a look at other related aspects of data acquisition like a set of measurements required for estimation of stability and control derivatives, sensors, sampling rate, and signal preprocessing.

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2.2 FLIGHT TESTING AND MANEUVERS

Pertinent to our specific interest, flight testing can be broadly classified into two categories, namely, the following:

1. Flight testing for performance evaluation
2. Flight testing for system identification

The flight testing for aircraft certification usually falls in the first category whereas that for the aerodynamic database development is in the second group. Usually, a large number of specific maneuvers is required in each case. A part of these maneuvers is common to both whereas the rest is characteristically different. Flight maneuvers used in the proof-of-match to demonstrate the fidelity of aerodynamic database for flight simulators are defined in the Acceptance Test Guide (ATG). As defined by FAA (Federal Aviation Administration) and JAA (Joint Aviation Authorities), roughly 100 to 120 test cases, covering various configurations and maneuvers including special test cases for military aircraft, are usually needed for this purpose [18, 19]. Although special care is exercised in performing these maneuvers in terms of exactness and to minimize external disturbances, basically these maneuvers fall within the two classes already mentioned. Flight testing and its broad spectrum is a vast subject in itself; a detailed account is found, for example, in [20–22]. Here, it is not our aim to give a consolidated account of these two types of testing; rather, we will take a cursory look at them to understand the basic procedures and differences, providing a brief summary of common flight maneuvers.

2.2.1 FLIGHT TESTING FOR PERFORMANCE EVALUATION

Flight testing for performance evaluation is aimed at determining the ability to fulfill the mission in terms of range, fuel consumption, achievable maximum speed, rate of climb, altitude, etc. These tests are usually longer duration tests, requiring specific maneuvers, for example:

1. Acceleration and deceleration
2. Pushover–pull-up
3. Wind-up turn
4. Climb/sawtooth climb
5. Bank-to-bank roll
6. Steady sideslip
7. Landings and takeoffs

A level flight acceleration-deceleration maneuver is performed as follows: starting from a trimmed horizontal level flight, the aircraft is decelerated by

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changing the power lever to idle position and maintaining altitude using the pitch control. Once the minimum speed (just above the stall speed) is reached, the power lever is moved from idle to full throttle position, accelerating the aircraft. Once the maximum speed is reached, the power setting is changed to idle, decelerating up to the initial trim speed, at which the test is terminated. In some cases the test is performed for smaller speed variations about the trim condition. The acceleration-deceleration maneuver covers the large range of speed and is usually performed at different altitudes. It allows determination of acceleration capability and longitudinal speed stability; hence, it is also called as longitudinal static stability or speed stability test. The speed variations from minimum to maximum results in a large variation in the angle of attack, and as such the acceleration-deceleration maneuver is used for system identification purposes as well. For such purposes, during the acceleration phase elevator inputs, usually doublet, which we will discuss in the next section, are applied at regular interval to increase the information content in the data.

Pushover–pull-ups, also called roller-coaster, are primarily performed to determine lift and drag characteristics, longitudinal stability, and the elevator trim requirements. The maneuver is initiated from a trimmed level flight by slowly pushing the stick, reducing the angle of attack continuously, which leads to a slow diving motion. Once the maximum speed is reached, the stick is pulled slowly to result in increasing angle of attack, which leads to a slow climbing motion. Once the minimum speed (just above the stall speed) is reached, the stick is once again pushed slowly to return to the starting trim condition. In contrast to the acceleration-deceleration maneuver, pushover–pull-up maneuvers are performed at constant thrust. This maneuver results in a significant variation in the vertical acceleration, typically covering the range of 0 to 2 g and as such is used mostly to determine the drag polars. The maneuver is performed in about 40 to 50 s.

Wind-up turns are used to determine the gradient of "stick force per g," which is a design and certification criterion. It is a measure of the stick force required to change the load factor; it is an indicator of the compromise between the two conflicting requirements of longitudinal stability and maneuverability. A higher slope of stick force vs load factor curve implies higher apparent stability and less maneuverability because a larger pilot force will be necessary to move away from the trim and vice versa. The maneuver starts by initiating a level turn and continuously increasing the bank angle. Increasing bank angle leads to a nose-down motion at increasing speed. Through pitch control the angle of attack is increased to maintain the speed constant. The throttle position is kept fixed. The process of simultaneously increasing the bank angle and angle of attack results in a descending spiraling and wind-up motion with decreasing turn radius, resulting in an increasing load factor. The maneuver is terminated when a g-limit is reached. This is one of the demanding maneuvers for the pilot, particularly for fighter aircraft with higher g-limits. The duration of a wind-up turn is roughly 60 to 80 s and repeated at different nominal speeds. Besides the primary purpose of finding the

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gradient of "stick force per *g*," this maneuver also aids in determination of drag and lift coefficients at different angles of attack for a constant Mach number and also in the system identification.

The main aim of the climb/sawtooth climb is to determine the best climb rate at different altitudes to arrive at a best climb schedule to reach a desired altitude in a minimum amount of time. Starting from a level flight, repeated climbs and dives are performed to result in predefined variations in the altitude by applying maximum and idle power alternatively. The airspeed is maintained constant using pitch control. This maneuver is rarely used in system identification; however, it is a part of the proof-of-match process that is necessary to demonstrate the model fidelity.

Determination of the maximum roll capability is important for fighter aircraft, as it mainly governs the ability to turn quickly and change direction of flight. In the case of transport aircraft, adequate roll performance is necessary to quickly compensate for disturbances such as turbulence or asymmetric configurations. Determination of aerodynamic coupling between roll and yaw axes and ability to compensate adverse yaw following an aileron input is part of this test. This test is performed at several trim conditions. The bank-to-bank roll, also called rapid roll, tests are widely used for parameter estimation purposes.

The primary aim of steady sideslip maneuvers is to determine the gradient of rudder deflection and aileron deflection per degree of sideslip. These maneuvers are particularly of interest at low speeds for the landing configuration because larger side slipping might be necessary to compensate for strong crosswind during landing phases. Starting from a horizontal level flight, the maneuver is initiated by applying and holding a constant pedal force, leading to a gradual increase in the angle of sideslip. Constant heading flight is maintained in a banked position by applying ailerons. These maneuvers are widely used in system identification.

The landings and takeoffs are critical flight phases. A careful planning is necessary during these tests because the performance is greatly affected by wind gradients, particularly during landing phases. The ground effect also affects the aircraft performance at low altitudes. Besides the flybys at different low altitudes over the runway, landing maneuvers are also used in the estimation of ground effect. Landings under different external weather conditions and one-engine out on takeoff are an important part of the data gathering for simulator validation.

Besides the preceding elaborated performance evaluation specific tests, other stability-related maneuvers enabling frequency and damping determination are also performed as a part of overall flight-test program. They are primarily used for handling qualities evaluations and for estimation of aerodynamic derivatives. Such maneuvers and those commonly used for system identification purpose, for example, steady heading sideslips, are elaborated in the next section.

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2.2.2 FLIGHT TESTING FOR SYSTEM IDENTIFICATION

Estimation of stability and control derivatives is carried out primarily from the dynamic response of an aircraft to specific control inputs. In general, dynamic motion is excited by applying control pulse, step, multistep, or harmonic inputs. A variety of maneuvers is usually necessary to excite dynamic motion about different axes using independent inputs on every control. Dynamic maneuvers allow extraction of a larger number of static as well as dynamic parameters from a single test. Combining several stationary maneuvers at different trim conditions principally allows determination of some of the aerodynamic derivatives, for example, slope of the lift coefficient or the longitudinal static stability parameter. Compared to dynamic maneuvers, this approach is less efficient because a larger number of test points will be necessary and the uncertainties in the estimates might be larger. Usually, the best choice is to combine the dynamic maneuvers and different stationary conditions to extract a single set of aerodynamic derivatives.

A set of common flight maneuvers for system identification consists of the following:

1. Short-period maneuver
2. Phugoid maneuver
3. Pushover–pull-up
4. Level turn
5. Thrust variation
6. Bank-to-bank roll
7. Dutch roll maneuver
8. Steady heading steady sideslip

Besides these maneuvers, as already mentioned in Sec. 2.2.1, acceleration-deceleration is also used for the purpose at hand. The main source for estimation of ground effect is the flybys at different altitudes over the runway and the landings. Generally, for control applications, for safety purposes, and for passenger comfort, excitation of the eigenmodes is suppressed. However, for parameter estimation purposes, it is essential that the dynamic response exhibits frequencies and damping of the oscillatory modes. In general, it is recommended to start each maneuver from a trimmed level flight and allow about 5 to 10 s of steady flight before applying specific control inputs and, depending upon the mode of motion, allow sufficient time after the input to allow the aircraft to oscillate.

The short-period motion, which is a fast responding longitudinal mode, provides the most information to enable estimation of derivatives pertaining to the vertical and pitching motion. The short-period maneuver is initiated from a horizontal level flight by applying an elevator or other longitudinal control input,

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which is usually a doublet or multistep input. Doublet input is a two-sided pulse input whereas multistep input consists of a suitable combination of more than two pulses. The time step of such inputs is to be chosen such that the short-period mode is excited well. We will deal with the aspects of arriving at a suitable form of multistep input and choosing proper time steps for such input in the next section. Input amplitude should be such as to result in about $\pm$3- to 4-deg variation in angle of attack about the trimmed value, or in terms of the load factor the variation should be typically $\pm$0.4 to 0.5 g. A typical duration of such a maneuver is about 15–20 s. It is performed at different trim speeds. Larger amplitude maneuvers might be necessary to enable estimation of nonlinearities. While applying a multistep elevator input, pilots have to pay attention because in special cases they tend to maintain the angle of attack, particularly during approach; this practice is in contrast to the basic principle of exciting the longitudinal mode resulting in variation in the angle of attack.

The phugoid mode is a long-duration mode, which can be excited either through an elevator pulse or thrust variation. Typically, one full cycle of phugoid motion is required for parameter estimation purposes. For the validation of databases for simulators, the model accuracy has to be demonstrated for three full cycles.

As already mentioned in Sec. 2.2.1, the pushover–pull-ups result in variation in the vertical acceleration and are useful to determine lift and drag characteristics. A level-turn maneuver is initiated from a horizontal level flight by banking the aircraft smoothly into 30- (45- or 60-) deg bank at a rate of 10 deg/s, holding the bank attitude steady for several seconds, then rolling the aircraft to the opposite direction, again holding steady the bank angle for several seconds, and finally returning to the wings-level conditions. Altitude is maintained during the maneuvers through additional power. It results in an S-type flight path in the horizontal plane, the heading at the start and at the end of the maneuver roughly the same. As will be elaborated in Sec. 12.2, this maneuver combined with an elevator doublet input in the banked position is useful to separate out the pitch rate and rate of change of angle-of-attack components of longitudinal motion.

Dynamic thrust input maneuvers are performed applying a longer-duration doublet input to throttle. They are useful to determine the dynamic effects due to thrust variations on the longitudinal motion, for example, on the downwash effect on the horizontal tail. They also aid determination of the effect of the thrust coefficient on the various aerodynamic derivatives, particularly for propeller aircraft where such effects are dominant. The dependencies of the aerodynamic derivatives on the thrust coefficient can also be determined by performing other longitudinal and lateral-directional dynamic maneuvers starting from trims with different thrust levels.

To enable estimation of lateral-directional derivatives, bank-to-bank roll and Dutch roll maneuvers are necessary. A bank-to-bank maneuver is performed applying a series of aileron pulses. Starting from a level flight, sharp aileron input is applied, holding it for some time to roll the aircraft from wings level to

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at least 30-deg bank on one side; this is followed by rapidly changing the input and going smoothly through wings level to the opposite bank angle, followed by returning to the wings-level conditions. The test is repeated for different levels of bank angles (30, 45, and 60 deg). The bank-to-bank maneuver is also sometimes called rapid-roll maneuver because the abrupt changes in aileron pulses result in rapid variation in roll rate and acceleration. A typical duration of such a maneuver is 30 to 40 s. The doublet or multistep aileron input with short time steps might not be quite adequate because sufficient time should be allowed for bank-angle buildup; rolling motion corresponds to a first-order system.

In many aircraft, ailerons are coupled to roll spoilers, which are additional flaps on the upper surface of the wings. They are drawn out either on the left or the right wing corresponding to the upper aileron deflection, that is, on the wing going down, and help to improve the overall roll efficiency without being too sensitive at small aileron deflections. For aileron deflections less than a threshold value, the spoilers are inactive; they are deflected proportional to aileron deflection larger than the threshold. For a given aircraft, the threshold aileron deflection for roll spoiler activation is a fixed value, which usually ranges from 3 to 5 deg. Because of this basic construction, ailerons and roll spoilers are deflected proportionately, except for small aileron deflections. In case we are interested in estimating from flight data both aileron and roll-spoiler effectiveness, then it is necessary to perform the bank-to-bank maneuver with small aileron deflection (smaller than the threshold values) during which roll spoilers are inactive, followed by larger aileron input in which case both control surfaces would contribute to the rolling motion. A similar situation arises for aircraft having aileron-to-rudder interconnect. In this case, automatic rudder deflection following an aileron input is detrimental to extraction of independent derivatives, although both surfaces move. Basically, it amounts to the fundamental aspect of generating independent inputs on every control; otherwise, it would not be possible to separate out the effects due to multiple surfaces moving proportionately.

The Dutch roll maneuver provides maximum information on the frequency and damping of this oscillatory mode. It is excited by applying rudder inputs. Usually, several cycles of oscillations are recorded; the duration of the total maneuver could be typically 30–35 s or even longer. The resulting maximum peak-to-peak variation in the angle of sideslip is typically of the order of ± 4 deg, or 0.1-g lateral acceleration. The Dutch roll mode of the basic aircraft is usually lightly damped and can be excited very well with a simple doublet input. Many aircraft are equipped with a yaw damper, which is turned off during maneuvers performed for system identification purposes, because we are mainly interested in the aerodynamic characteristics of the bare airframe. It is required to perform the Dutch roll and bank-to-bank maneuvers at different trim speeds because most of the lateral-directional derivatives depend on the angle of attack.

Both aileron and rudder input maneuvers are necessary not only to allow estimation of the respective control surface effectiveness, but also to improve

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estimation of other derivatives due to roll and yaw rates. The aileron input maneuver yields significantly larger roll rates, which are characteristically different in shape than compared to those from the rudder input. In general, it is our experience that better estimation results are obtained by performing the aileron and rudder input maneuvers separately starting with a bank-to-bank maneuver followed by a Dutch roll. This sequence is easy for pilots to execute manually. In some cases we have also applied computerized aileron and rudder inputs simultaneously. In such cases, the starting points of the two inputs should be time skewed, that is, displaced in time. If the starting point is time synchronized, the duration of the two inputs should be different, or an independent maneuver with a single control surface input would be desirable to improve separation of the components due to these two control surfaces.

The steady heading steady sideslip (SHSS) maneuver provides additional information on the directional stability, lateral-directional control, and cross coupling effects. It is usually flown in both the positive and negative sideslipping conditions to allow detection of any asymmetric effects. Such tests are particularly important for landing configurations. SHSS are performed in different variations to cover a large range in angle of sideslips in both directions and at different angles of attack and thrust levels. For example, starting from a horizontal level flight, SHSS is performed by applying abrupt pedal step and maintaining the sideslip angle at some chosen level for roughly 10–15 s, followed by releasing the pedal to neutral position allowing few seconds for dynamic recovery response. This is followed by repeating the procedure in the other direction of side slipping. In each case, the track angle is maintained by adjusting the bank angle applying aileron input. The side slipping conditions should be achieved rapidly, ideally as shown schematically in Fig. 2.1a. However, in practice some overshoot is unavoidable; strong excitation of the Dutch roll is to be suppressed by regulating the pedal input. For larger sideslips, the procedure is similar. The alternative is to perform building up the larger sideslip angles in two steps before releasing the controls to neutral. Yet another variation is to change abruptly from one steady sideslip in one direction to the other direction, without a stabilizing phase for no side slipping conditions, as shown in the two plots on the right side of Fig. 2.1a. If SHSS are carried out for performance evaluations, then the overall procedure is similar, but for maintaining the steady side slipping conditions for a considerably longer time than indicated earlier.

Besides SHSS with abrupt changes in asymmetric trim conditions, a quasi-steady type of maneuver, called wings-level steady sideslip, with gradual variation in side slipping is also useful. Starting from horizontal level flight, rudder is applied gradually to increase angle of sideslip from zero to some chosen value (say 5 or 10 deg), then gradually releasing the pedal pressure to reduce the side slipping, crossing over through zero to angle of sideslip in the opposite direction, finally returning to neutral; see Fig. 2.1b. Throughout the maneuver, the wings-level conditions are maintained applying lateral-stick cross-control. In

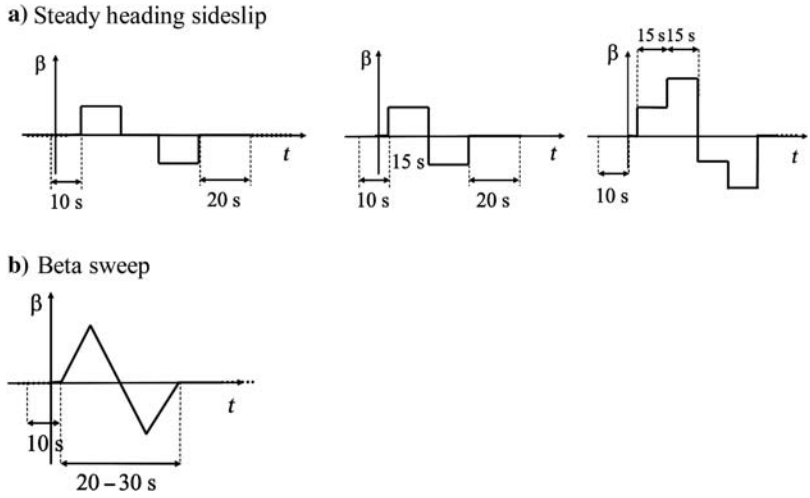

Fig. 2.1 Schematic of steady heading sideslip and sweep input maneuvers.

steady heading sideslip as well as wings-level steady sideslip maneuvers, the variations in speed are compensated through the longitudinal control.

Precaution and constant monitoring is necessary during large constant sideslipping conditions due to large aerodynamic loads acting on the vertical tail and to make sure that the structural limits are not exceeded. The piloting techniques to perform SHSS are demanding because considerable power is necessary for rudder deflections required to achieve asymmetric trim conditions. New safety recommendations have been defined to avoid structural damages resulting from large rudder deflections for transport-category airplanes [23].

Depending upon the aircraft, if there are additional controls available, for example direct-lift-control flaps or all-moving trimmable horizontal tail, then maneuvers with appropriate inputs (either multistep or longer duration pulses) will be necessary. In the case of unstable aircraft, the flight control laws prevent dynamic excitation of the aircraft motion and result in highly correlated control surface deflections and aircraft motion. Special techniques are necessary to generate flight data suitable for parameter estimation; we defer this specific case to Chapter 9, dealing exclusively with unstable aircraft identification.

The frequency sweep test techniques, although rarely used for fixed-wing aircraft, are used more routinely in the field of rotorcraft identification [24]. These techniques are useful and necessary not only for the next generation specification requirements [25] but also are an integral part of the interdisciplinary modeling aspects. Critical flight incidences have, however, occurred while sweep testing due to, for example, exceeding the aeroservoelastic range or the flight permissible maximum loads [26]. Proper coordination is, hence, necessary through careful

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preparation, buildup, real-time monitoring, and analysis to prevent possible structural damage and to avoid any increase in the risk factor.

The central idea behind system identification maneuvers described here is to excite pertinent modes of aircraft motion independently and sufficiently. In general, while exciting a particular mode, excitation of the other modes is to be minimized. The amplitudes are usually based on heuristic considerations, which have been discussed. Thus, the task now boils down to finding suitable shapes and durations of inputs that determine the frequency content.

2.3 OPTIMAL INPUT DESIGN

There are two approaches that have been applied in the past to design optimal inputs. The first one, based on the estimation error criterion, is more rigorous, involved, and tends to be theoretical. The second one is an engineering approach, based on the spectral behavior of the model.

2.3.1 INPUT DESIGN BY ESTIMATION ERROR ANALYSIS

As will be described in Chapter 4, statistical properties of the estimates are most conveniently expressed in terms of the bias and covariances of the estimates. It will be shown in the same chapter that the maximum likelihood estimation is based on the maximization $p(z|\Theta)$, the conditional probability density function of measurements z for a given parameter vector Θ, and that the Fisher information matrix, which is an indicator of the information content in the data being analyzed, is given by

$$\mathcal{F}_{ij} = E\left\{-\frac{\partial^2 L(z|\Theta)}{\partial\Theta_i \partial\Theta_j}\right\} \tag{2.1}$$

where $L(z|\Theta) = \ell n(p(z|\Theta))$ and $E\{\}$ denotes the expected value. It will also be shown that the maximum likelihood estimation is bias free and efficient in a statistical sense. For further development of the input signals, we simply accept these properties at this point. Under such conditions, the inverse of the Fisher information matrix given by Eq. (2.1) provides a good approximation to P, the parameter error covariance matrix [27, 28]. Without going into the exact mathematical derivation which will follow in the subsequent chapters, we also draw on the expression for the information matrix $\mathcal{F}$ in terms of $\partial y/\partial\Theta$, the gradient of the system responses with respect to the parameters.

$$P \approx \mathcal{F}^{-1} \approx \left\{\sum_{k=1}^{N}\left[\frac{\partial y(t_k)}{\partial\Theta}\right]^T R^{-1}\left[\frac{\partial y(t_k)}{\partial\Theta}\right]\right\}^{-1} \tag{2.2}$$

Thus, the parameter error covariance matrix P depends upon the response sensitivities to the model parameters $\partial y/\partial\Theta$, the data points being analyzed N, and a

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weighting matrix R, which depends on the measurement noise. The system responses y and its gradients $\partial y/\partial\Theta$ are obtained from the a priori model used for designing the inputs. At this stage the parameters Θ are known and kept fixed at their a priori values. Thus, the information content in $\mathcal{F}$ will be mainly determined by the response gradients that depend on the input excitation. Proper tuning of the input shapes will result in an optimal excitation of the modes of a system defined by the a priori model. In other words, it would lead to maximization of the information matrix. In turn, this is the same as minimizing the error covariance matrix P, which suggests that the parameters that will be subsequently estimated from such inputs will have lowest statistical errors and hence will be more accurate. This is the basic idea behind designing the optimal inputs based on the theoretical approach.

Different measures, all based on the Fisher information matrix, can be chosen as criteria to design optimal inputs [11, 28]. The first criterion is

$$\min_{u(t)}\{\det(P)\} \Longrightarrow u_{\text{opt}}(t) \tag{2.3}$$

which is the same as maximizing the determinant of the Fisher information matrix $\mathcal{F}$. The criterion of Eq. (2.3) is the overall measure in terms of the volume of the estimation error ellipsoid, which is proportional to the determinant of P. Input signals that maximize the determinant of $\mathcal{F}$ (or equivalently minimize the determinant of P) are commonly called in the literature as D-optimal, where "D" stands for "determinant." It can be shown that the optimum is invariant to scaling of the system states. Let us take a look at the physical significance of D-optimality. The columns of the information matrix $\mathcal{F}$ represent contributions of the individual parameters. If the columns are independent, the determinant of the matrix will have a maximum value subject to the given constraints in terms of the postulated model. On the other hand, if parameters are correlated, the determinant will be smaller; in the worst case, linear dependency of two parameters results in a zero determinant, indicating that the two parameters cannot be estimated independently. Thus, D-optimality minimizes the redundancy and leads to better identifiable parameters.

The other criteria are the sum or product of the diagonal elements of the matrix P:

$$\min_{u(t)}\{tr(P)\} \Longrightarrow u_{\text{opt}}(t) \tag{2.4}$$

and

$$\min_{u(t)}\left\{\prod_i P_{ii}\right\} \Longrightarrow u_{\text{opt}}(t) \tag{2.5}$$

The diagonal elements of P represent the variances of the estimates and are indicators of the accuracy of the estimates. As it will be discussed in Chapters 4 and 11,

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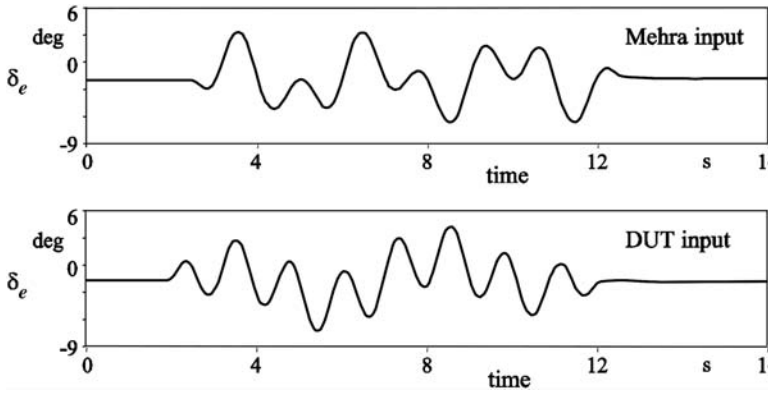

Fig. 2.2 Time histories of typical harmonic inputs.

the standard deviations are given by the square root of the variances. Thus, the last two criteria attempt to minimize the standard deviations or, in other words, increase the accuracy of the estimates. Input signals resulting from Eq. (2.4), that is, by minimizing the sum of the variances of the parameter estimation errors, are commonly called in the literature as A-optimal, where "A" stands for "average." In contrast to the D-optimal design, the A-optimality criterion of Eq. (2.4) or (2.5) is strongly dependent on the units and is not scale invariant. This aspect needs to be carefully considered while designing A-optimal inputs.

Both time-domain and frequency-domain methods have been applied to solve the preceding optimal design problem. To apply any of these criteria, it is necessary to define an input space (in terms of forms and amplitudes) to be searched for the optimal input. Different researchers have followed different approaches. The best known two designs shown in Fig. 2.2 are the following [13]: 1) Mehra input, based on optimization of harmonic signals selecting integer multiples of the basic frequency, where the basic frequency corresponds to the observation time and 2) Delft University of Technology (DUT) input, based on sum of sine functions. Both of these methods and other similar approaches lead to continuous and smoothly varying inputs. These signals are tuned to the a priori model used in their design, but they are not quite suitable to be flown manually. In all of these cases, an onboard computer implementation is necessary to generate such harmonic inputs.

2.3.2 DESIGN OF MULTISTEP INPUT SIGNALS

The design of multistep input signals is a two-step procedure: first, investigating the range of frequencies needed for accurate estimation of parameters, and second, designing suitable multiple step input to cover the desired frequency

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range. As in the preceding case, the design procedure requires an a priori model of the aircraft.

Investigation of the range of frequencies for accurate estimation of parameters is based on synthesizing the contributions due to each of the parameters appearing in each of the force and moment equations. This is conveniently done using the Bode diagram, which helps to determine which of the frequencies must be included in the input signal to enable extraction of particular derivatives. The following procedure developed by Marchand [10, 29] is briefly illustrated here considering the following linearized model pertaining to the longitudinal motion:

$$\begin{bmatrix} \dot{u} \\ \dot{\alpha} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_\alpha & X_q & -g \\ Z_u/U_0 & Z_\alpha/U_0 & 1 & 0 \\ M_u & M_\alpha & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ \alpha \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_{\delta e} \\ Z_{\delta e} \\ M_{\delta e} \\ 0 \end{bmatrix} \delta_e \tag{2.6}$$

where U_0 is nominal trim speed, u the component of true airspeed along the x axis, α the angle of attack, q the pitch rate, θ the pitch attitude, δ_e the elevator input, and ($X_{()}$, $Z_{()}$, $M_{()}$) the dimensional derivatives. For the case considered here for demonstration, the a priori values of the dimensional derivatives for a forward speed of 98 m/s lead to system state and control matrices in Eq. (2.6), which can be denoted as A and B, given by

$$A = \begin{bmatrix} -0.0091 & 9.43 & 0 & -9.80665 \\ -0.0022 & -0.867 & 1 & 0 \\ 0 & -3.49 & -2.04 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$B = [0 \quad -0.11 \quad -5.09 \quad 0]^T$$

Now, for each equation in Eq. (2.6), the frequency response magnitudes of the various terms in that equation are plotted as a function of the input signal frequency. To demonstrate the procedure, we consider here the pitching moment equation. We compute the frequency response magnitudes for each of the terms, namely, M_u, M_α, M_q, and $M_{\delta e}$ as well as for the total pitch acceleration $\dot{q}$ with respect to the elevator input δ_e. In other words, it amounts to computing the magnitudes:

$$\left|\frac{\tilde{\dot{q}}(\omega)}{\tilde{\delta}_e(\omega)}\right|, \quad \left|\frac{M_u\tilde{u}(\omega)}{\tilde{\delta}_e(\omega)}\right|, \quad \left|\frac{M_\alpha\tilde{\alpha}(\omega)}{\tilde{\delta}_e(\omega)}\right|, \quad \left|\frac{M_q\tilde{q}(\omega)}{\tilde{\delta}_e(\omega)}\right|, \quad \left|\frac{M_{\delta e}\tilde{\delta}_e(\omega)}{\tilde{\delta}_e(\omega)}\right| \tag{2.7}$$

where $\sim$ denotes the Fourier transform. The individual components for $\dot{q}$, α, q, and δ_e can be computed from the output equation $y = C[u \;\; \alpha \;\; q \;\; \theta]^T + D\delta_e$

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by defining the observation matrices C and D appropriately as follows: 1) $C_{\dot{q}} = [0\ \ -3.49\ \ -2.04\ 0]$, $D_{\dot{q}} = [-5.09]$; 2) $C_{\alpha} = [0\ \ -3.49\ 0\ 0]$, $D_{\alpha} = [0]$; 3) $C_{q} = [0\ 0\ \ -2.04\ 0]$, $D_{q} = [0]$; and 4) $C_{\delta e} = [0\ 0\ 0\ 0]$, $D_{\delta e} = [-5.09]$, where the subscripts denote matrices corresponding to the respective variables. These components are computed using the program "/FVSysID2/chapter02/FreqCompInput.m" and plotted in Fig. 2.3. The contribution due to the term M_u is not shown because the a priori value of this derivative is zero. The magnitude of the control term corresponding to $M_{\delta e}$ with respect to δ_e is largest compared to the other terms in the higher frequency region. This suggests that the control effectiveness is best identifiable at such frequencies. Identifiability of other parameters depends on the frequency range as will be discussed next. The preceding elaborated procedure evaluates the contributions due to control surface deflections, flow angles, velocity components, and angular rates, but does not include the indirect influence of attitude angles, which are used in the output-error method that we will study in detail subsequently. In Sec. 2.5, it has been demonstrated that attitude angles are also useful in the estimation of aerodynamic models.

The procedure is repeated for the X and Z equations for the longitudinal and vertical motion. From such plots, identifiability of the derivatives can be ascertained. At any given frequency, a large magnitude of any particular term compared to the other contributions suggests a dominant influence of that

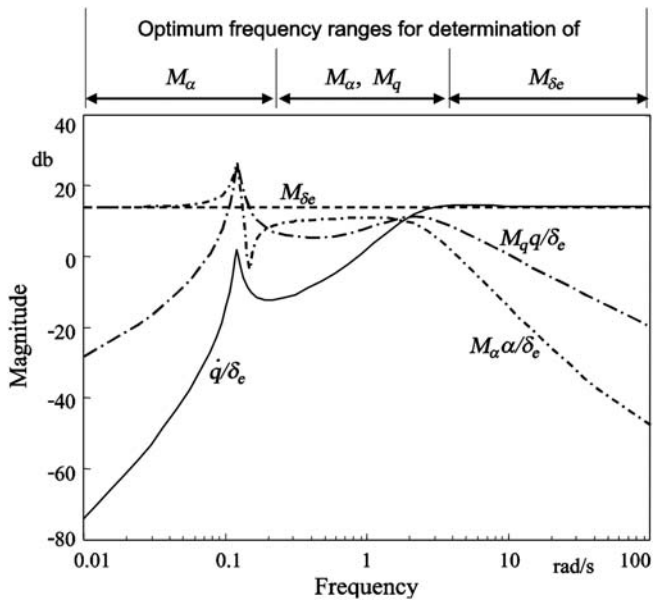

Fig. 2.3 Bode magnitude plot of the pitching moment equation terms.

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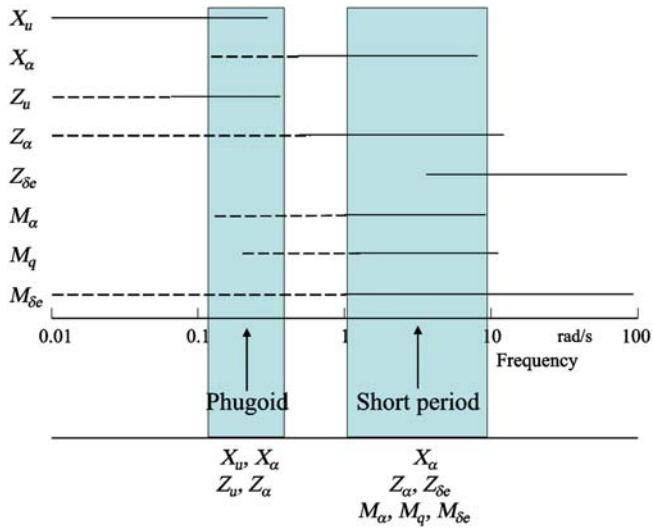

Fig. 2.4 Regions of identifiability.

derivative, which indicates good information content necessary for estimation of the parameter. Conversely, the derivative cannot be accurately estimated if the contribution is small. It is now necessary to specify a measure to decide upon relative contributions. As a rule of thumb, a derivative is considered identifiable when its term has a magnitude of at least 10% of the largest term's magnitude [13].

From frequency response magnitude plots for the X and Z forces and for the pitching moment, the ranges of frequencies are determined in which contributions due to each derivative are appreciable in the sense just described. These ranges are plotted by horizontal lines in Fig. 2.4, which shows that all of the derivatives can be identified from excitation in the two regions marked in gray. The solid lines indicate regions where the derivatives are directly identifiable and dashed line the regions in which they can be estimated only as the ratio of two parameters. The regions surround the natural frequencies of the phugoid and the short-period mode. This is not really surprising because from our basic intuition we know that the system excited at its natural mode will exhibit dominant dynamic motion. Excitation of the long-period phugoid mode is easily obtained through a longer duration pulse input, but the excitation of the fast responding short-period motion is more critical. The preceding analysis combined with additional issues provides a good basis for designing elevator input signals for longitudinal motion, namely: 1) the optimum range for frequencies covers a range below and above the natural frequency of the short-period mode, 2) the eigenfrequencies of the modes based on the a priori model are subjected to uncertainties, and 3) the eigenfrequency changes with the flight

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condition. These considerations imply that besides the eigenfrequency it is also necessary to excite the frequencies around the eigenfrequency as well. Based on these considerations, a bandwidth of 1:10 is recommended for the design of a multistep input signal. Thus, the task now is to design an input signal covering this required frequency range.

A multistep input signal of arbitrary shape can be synthesized by a suitable sequential combination of pulse inputs. For an input signal consisting of an arbitrary number of equidistant time steps, in any combination of input levels, as derived in Appendix A, the power (or energy) spectrum is given by [30]

$$\mathrm{E}(\omega) = 2\Delta t^2 \, \frac{1-\cos\Omega}{\Omega^2} \left[\sum_{i=1}^{N} V_i^2 + 2 \sum_{j=1}^{N-1} \cos j\Omega \sum_{i=1}^{N-j} V_i \, V_{i+j} \right] \tag{2.8}$$

where $\Omega = \omega\Delta t$ is the normalized frequency, $T = N\Delta t$ the total duration of the input consisting of N impulses each of duration Δt, and V_i the amplitude for the current impulse ($t_{i-1} < t < t_i$). Equation (2.8) has been coded in the function "/FVSysID2/chapter02/EnergySpectrum.m," which is used in the following to compute the spectra of various input signals. It is also possible to compute the spectra by fast Fourier transform (FFT) using the function "InputSigPSD.m," which, however, requires the signal processing toolbox of Matlab®.

The simplest input to excite the oscillatory motion is a pulse input. After the input over a specified time, the control is released to allow the aircraft to oscillate freely without pilot inputs. Equation (2.8) leads to the energy spectra shown in Fig. 2.5 for such an input corresponding to different durations of Δt, but with the same input amplitude. The spectra in Fig. 2.5 are computed and plotted using the program "/FVSysID2/chapter02/EnergyImpulse.m." From this figure, we observe that a pulse with $\Delta t = 0.4$ s has energy spread over a larger range of frequencies, but the magnitude is very small, which is not enough to excite the system adequately. Increasing the duration leads to larger energy content in the signal, but at the same time decreasing the spread to lower frequencies. For $\Delta t = 1.2$ s, it has large energy, but unfortunately concentrated at low frequencies with rapid decay. The pulse input is asymmetric about the starting trim values, which leads to nonzero energy at zero frequency. The asymmetric excitation can lead to changes in the stationary conditions, in many cases deviating significantly compared to the starting trim conditions from which the maneuver

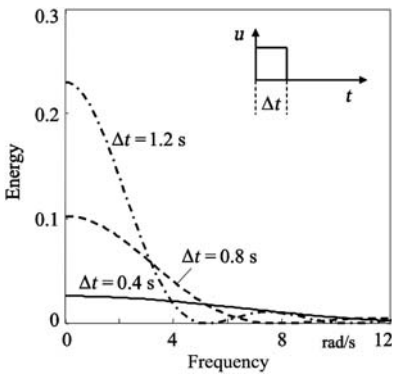

Fig. 2.5 Energy spectra of pulse inputs.

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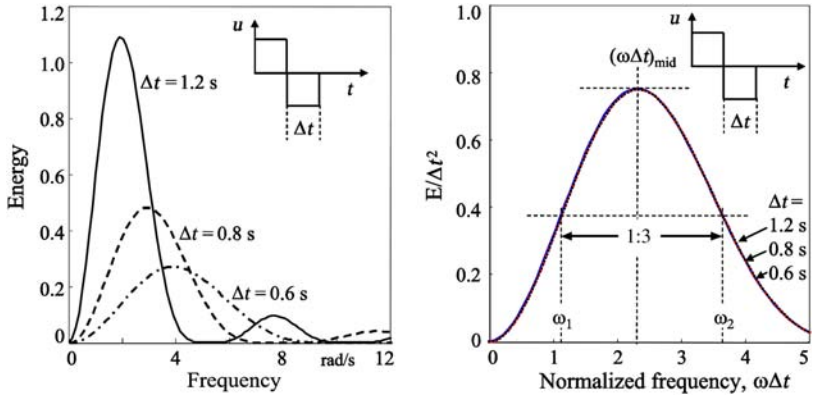

Fig. 2.6 Energy spectra of doublet inputs.

might have been initiated. This in itself is not a serious drawback, provided we can account for such changes in the identified model, which may turn out to be flight condition dependent or nonlinear. If linear models are to be identified, it may lead to violation of the range of validity. The major limitation of this input is that it is less suitable for exciting the rapid eigenmodes at higher frequencies, like short period or Dutch roll.

A more commonly used input to excite the dynamic motion is a doublet: control moved abruptly in one direction, held fixed for certain time Δt, then moved abruptly to other direction, held fixed for same specified time before releasing it to neutral. The doublet input, which is a two-sided pulse, results in a symmetrical signal, having energy concentrated at a frequency that varies with Δt. Figure 2.6 shows the energy spectra for three different time steps $\Delta t = 0.6$, 0.8, and 1.2 s, which are computed and plotted using the program "/FVSysID2/chapter02/EnergyDoublet.m." The drop in the energy content around the peak is relatively fast. As observed on the right side of Fig. 2.6, the normalized frequency $\omega\Delta t$ for the peak is roughly 2.3, that is,

$$\omega_n \Delta t_{\text{DBLT}} \approx 2.3 \tag{2.9}$$

where ω_n is the short period frequency. Equation (2.9) can be expressed in terms of the period of oscillation, yielding an estimate of the time step for the doublet given by

$$\Delta t_{\text{DBLT}} \approx \frac{2.3}{\omega_n} \approx \frac{2\pi}{2.7\ \omega_n} \approx \frac{1}{2.7} \cdot \text{ period of oscillation} \tag{2.10}$$

Although Eq. (2.10) shows a factor of 1/2.7, in practice, a simpler factor of 1/2 is often used, which corresponds to choosing the total length of the doublet ($=2\Delta t$) equal to the period of the eigenmotion. Through this very simple rule of

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thumb, the corresponding mode based on the a priori model can be easily and adequately excited. It is observed from Fig. 2.6 that the doublet input has a bandwidth of roughly 1:3, which is the ratio of the normalized frequencies ω_2 ($\approx$3.63) to ω_1 ($\approx$1.1, cf. Fig. 2.6). In this range the corresponding spectrum is at least half of its maximum value. Because of its simplicity, the doublet input is widely used for estimation of stability and control derivatives. A combination of doublet and pulse inputs is also quite useful in some cases; a somewhat smaller step size than given by Eq. (2.10) leads to a slightly higher central frequency of the doublet, which is followed by a pulse to cover the lower frequency range.

Extending the logic that progressing from single-step input to two-step input (doublet) leads to a spread of the power spectrum, a much broader band signal can be achieved through a multistep input. Without going into exact details, a multistep input can be designed by minimizing the deviation of the computed spectrum for a design from the desired spectrum suitably defined from the results of frequency response magnitude plots (Fig. 2.3). In such cases it is desirable to minimize the variations in the energy contents in the range of frequencies covered, with drop on the two sides being smooth and rapid, and ideally zero or very small energy at zero frequency. The parameters to be adjusted during the minimization are the number of time steps, N in Eq. (2.8), and amplitudes at each step. Design of such a multistep input was performed by Koehler, resulting in the widely known 3-2-1-1 input [12]. The 3-2-1-1 input is $7\Delta t$ long and consists of alternating positive and negative, equal amplitude, steps of relative duration 3, 2, 1, and 1. The energy spectrum of such a 3-2-1-1 input, computed from Eq. (2.8) using the program "/FVSysID2/chapter02/EnergyImpDbtS3211M3211.m," is shown in Fig. 2.7.

The time step Δt for the 3-2-1-1 input is usually chosen such that the natural frequency of the mode being excited lies in the center or in the upper third of the

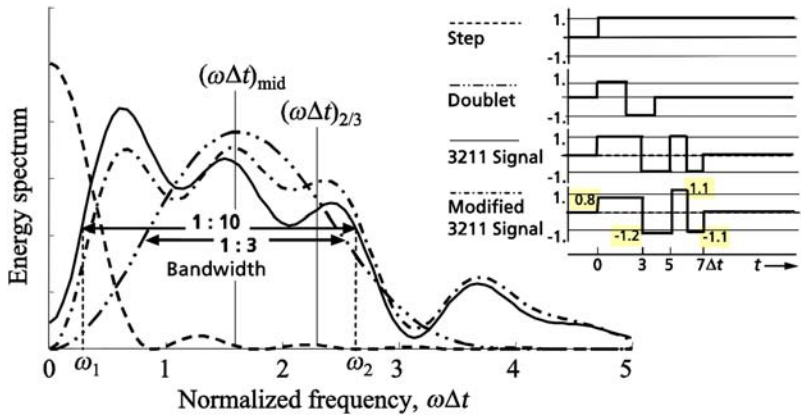

Fig. 2.7 Frequency-domain comparison of standard inputs.

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input spectrum. As seen from Fig. 2.7, the normalized frequencies $\omega\Delta t$ for these two conditions are roughly $(\omega\Delta t)_{\text{mid}} = 1.6$ and $(\omega\Delta t)_{2/3} = 2.1$, respectively. This leads to an estimate for Δt_{3211} given by

$$\Delta t_{3211} \approx \frac{1.6}{\omega_n} \approx \frac{\pi}{2\,\omega_n} \approx \frac{1}{4} \cdot \text{ period of oscillation} \tag{2.11}$$

or

$$\Delta t_{3211} \approx \frac{2.1}{\omega_n} \approx \frac{1}{3} \cdot \text{ period of oscillation} \tag{2.12}$$

Thus, the time step Δt_{3211} is somewhat smaller than Δt_{DBLT}. As a rule of thumb, a time step Δt_{3211} given by $\Delta t \approx 0.3/f_C$ is found to be quite effective in several aircraft cases, where f_C in hertz is the frequency of the mode to be excited. This matches well with the range of 0.25 to 0.35 based on an empirical rule and numerical studies [31].

The spectra for various signals are computed with suitable time steps using the program "/FVSysID2/chapter02/EnergyImpDbtS3211M3211.m"; for comparison purposes, they are plotted in Fig. 2.7 as a function of normalized frequency of the 3-2-1-1 signal. Assuming a time step of $\Delta t_{3211} = 1$ s for the 3-2-1-1 signal, the equivalent time step for the doublet is computed from Eqs. (2.11) and (2.9), which turns out to be roughly 1.44 s ($\Delta t_{\text{DBLT}} \approx 2.3/\omega_n$; $\omega_n \approx 1.6/\Delta t_{3211}$). It is observed from Fig. 2.7 that the 3-2-1-1 input has a bandwidth of roughly 1:10, ($=\omega_2/\omega_1 \approx 2.7/0.3$) having at least half of its maximum energy. Thus, the 3-2-1-1 input has a much wider spectrum compared to that of the impulse or doublet inputs. For reference purposes, another equivalent program based on FFT, "PSDImpDbtS3211M3211.m," is provided to compute the various spectra.

Because the natural frequency of the mode is flight condition dependent, strictly speaking the optimal Δt varies from flight condition to flight condition. However, the time variations and step reversals in the 3-2-1-1 signal result in a broad frequency content, and in spite of some deviations from the optimal value of Δt, the input is adequate to excite a band around the natural frequency. On the other hand, tuning of Δt for a doublet input is a little more critical because it has a smaller spread of frequency content around the central frequency. It is primarily due to this reason that the 3-2-1-1 signal is superior to the doublet input.

The main advantage of the 3-2-1-1 input lies in its simplicity and ability to execute it manually. It can be fairly easily remembered without any external aid. Different procedures have been adopted in practice to apply the 3-2-1-1 input manually, matching the desired Δt or for repeatability. For example, just counting "twenty-one, twenty-two, twenty-three" to keep the stick pushed, then "twenty-one, twenty-two" to keep it pulled, then "twenty-one" to pushed, and finally once again "twenty-one" to pulled position, and each time changing the stick position abruptly has produced remarkably good quality 3-2-1-1 inputs.

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Another procedure is to provide audio-cuing, consisting of series of beeps. Optical cuing has also been used in a few cases, by providing on the display the multistep input form which the pilot tries to track.

Two minor aspects of 3-2-1-1 inputs are as follows:

1. It is asymmetric about the trim deflection (with four time steps in one direction and three time steps in the other). As a consequence, it has nonzero energy at zero frequency.
2. With the first step being of larger duration, namely, three units of Δt, it may lead to motions far from the initial trim condition, before the following steps are applied.

These undesirable effects can be minimized by modifying the input amplitudes of the 3-2-1-1 input; such a modified 3-2-1-1 input is shown in Fig. 2.7. It has much reduced energy at zero frequency, and the variations in the spectrum are much less. Precise manual application of the modified 3-2-1-1 is, however, difficult and calls for onboard computer implementation. Another way is to apply 3-2-1-1 twice, and once starting by pulling up the aircraft and second time by pushing the stick.

Another variation of the 3-2-1-1 input is the 1-1-2-3 input obtained by time twisting the steps. The 1-1-2-3 input first excites the higher frequencies at the trimmed condition and prevents the vehicle from going far away from the trim condition before the larger duration time steps are applied. Like the 3-2-1-1, the 1-1-2-3 input can be easily applied manually with little practice. The 1-1-2-3 has the same energy spectrum as the 3-2-1-1 input. In some cases a 1-2-1 input, which is a combination of a normal doublet followed by an inverted doublet, is used. By using a time step for the 1-2-1 signal that is half of that of the doublet, it becomes possible to shift the spectrum to higher frequencies. This can be easily verified using the sample program "/FVSysID2/chapter02/EnergyDblt121.m." Much more complex multistep maneuvers have also been designed in some cases, which of course cannot be flown manually [15, 17]. In general, for more complex multistep inputs, the time step has to be chosen carefully; otherwise, there will be pronounced variations with valleys in the input spectrum.

Following a similar procedure based on analyzing the frequency response magnitude plots and approximations given in Eqs. (2.10) or (2.11), we can also design a 3-2-1-1 input for Dutch roll excitation. However, the Dutch roll is lightly damped compared to the short period, and hence a doublet input is found adequate to excite the Dutch roll adequately. Furthermore, although based on the a priori knowledge the time step of a doublet can be optimized, mostly the duration as well as the exact shape is of secondary importance. This is evident from the typical Dutch roll responses shown in Fig. 2.8 for clean configuration at four different speeds. The durations of the oscillations as measured from the recorded data and the actual time steps of the inputs are also provided in

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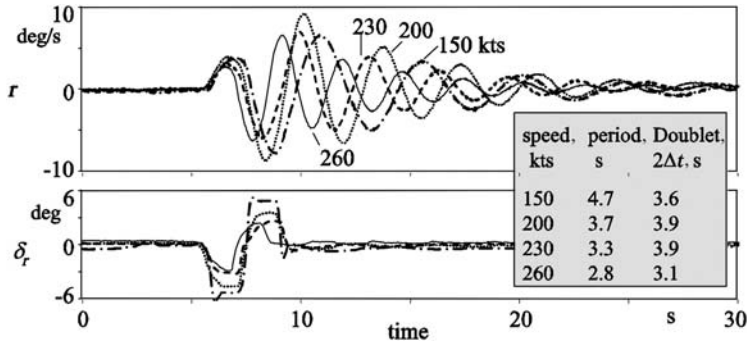

Fig. 2.8 Typical Dutch roll responses.

the same figure. Based on the very good results obtained for several aircraft cases, we recommend a doublet input on the rudder to excite the Dutch roll motion and 3-2-1-1 or 1-1-2-3 on the elevator to excite the short-period motion.

A detailed investigation based on flight data [32], comparing five different types of inputs, has shown that the simpler 3-2-1-1 or doublet inputs yield comparable and in many cases even better estimates compared to the optimal harmonic inputs such as Mehra or DUT elaborated in Sec. 2.3.1. This is mainly because the multistep inputs are richer in frequency contents than the harmonic inputs. For the multistep inputs, if applied automatically using an onboard computer, it becomes possible to change the input levels within one sampling time. Such abrupt changes can lead to rate saturation in the electrohydraulic actuators or can result in much higher load factors at off center-of-gravity locations, for example, in the cockpit, than at the center of gravity. They can also lead to excitation of much higher frequencies and of aeroelastic modes. The optimal harmonic inputs are limited in frequency range and tuned to the desired rigid-body frequencies. This is particularly of importance when the rigid-body and structural frequencies are close to each other, as tends to be the case for very large aircraft. To avoid such undesirable excitations from multistep inputs, the slopes of the input changes should be limited. On the other hand, if they are applied manually, the inputs do not have extreme sharp edges because the pilot acts as a filter. This does not affect the estimation of the stability and control derivatives because, as already mentioned, the exact shape and time step are not critical.

Thus, based on our discussion in Secs. 2.2.2 and 2.3.2, a variety of maneuvers is usually necessary to excite dynamic motions about different axes. We recommend independent control inputs applied one at a time and prefer those that can be flown manually by the pilot. The excitation level should be adjusted to the flight condition being tested; at lower dynamic pressures larger amplitudes are necessary, whereas at higher dynamic pressures lower amplitudes can be applied. We have performed a large number of flight tests, first with a small

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amplitude input followed by a larger amplitude input, but still within the range of the linear model. The large amplitude maneuvers, in which nonlinear aerodynamic effects are encountered, are performed separately. Figure 2.9 summarizes schematically the most important system identification maneuvers.

The various multistep inputs just elaborated belong to the class of binary signals, which consist of a series of optimally spaced pulses, yielding desired frequency spectrum. It is also possible to realize an arbitrary power spectral density spread across a specified bandwidth through superposition of phased harmonic signals [33–36]. Such inputs belong to the class of multisine signals. One such input that is used in some applications is called the Schroeder-phased multisine signal. By introducing orthogonalization, simultaneous multiaxis optimal excitation signals can be generated. Such input sequences are preferred in modal estimation of structural systems and for estimation of aerodynamic derivatives of hypersonic flight vehicles because the total test time is usually very short and the flight condition (Mach number) changes rapidly. Furthermore, such input signals have a lot of energy, keeping the input amplitude levels low. In the case of helicopters too, they provide a broadband excitation with improved low-frequency contents [37]. As already mentioned in Sec. 2.3.1, an onboard computer implementation is necessary to apply such multisine inputs. Other signals in the class of frequency sweeps, such as chirp or Fresnel chirp input signal, are also possible.

Although the control inputs such as impulse, doublet, and 3211 are sometimes used for ASE applications and for flight flutter investigation, it is more common to adopt the frequency sweep testing in this case [38]. In general, the frequency sweeps might not be time optimal, but they have a broader bandwidth. The

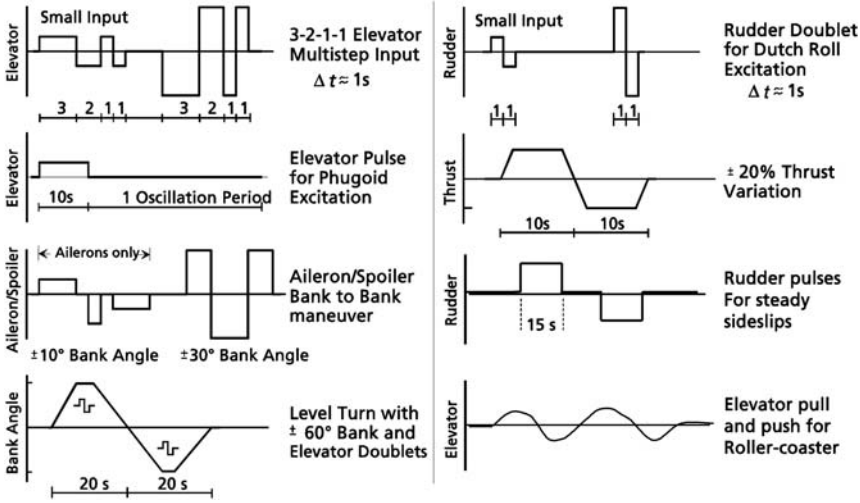

Fig. 2.9 Representative manual control inputs.

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frequency sweep testing has been applied, for example, to X-29, EAP aircraft [39, 40] and to identify several structural modes of an XV-15 tilt-rotor aircraft [41].

2.4 SCOPE OF FLIGHT TESTING

Flight testing is costly; hence, it is necessary to limit the number of flight tests to a minimum and also to optimize the test procedure and maneuver sequence within each flight. As pointed out in Sec. 1.4, the goals of system identification could be different, each of which poses different requirements on the range of validity and fidelity of the model to be identified. The scope of the flight testing for system identification depends mainly on these requirements, that is, on the specific goal for which the model is intended. To some extent, it also depends upon the type of flight vehicle being investigated. The larger the number of configurations to be modeled, the larger is the number of tests to be performed. Thus, the following important aspects need to be considered while organizing a flight-test program:

1. Define the specific goal of flight testing.
2. Select the appropriate configurations to be flown.
3. Choose the trim conditions to be tested.
4. Define maneuvers to be performed and the input signals to be applied.

Extensive flight testing is required for generating aerodynamic databases for flight simulators meeting the FAA Level-D fidelity requirements. Estimation of a so-called global model valid over the complete operational envelope needs a comprehensive flight-test program covering all possible configurations. Because the flight vehicle response characteristics depend on flight conditions (angle of attack, sideslip angle, Mach number), dynamic maneuvers will have to be performed at various trim conditions. Furthermore, depending upon the aircraft, special configurations and effects will have to be investigated, which may call for additional tests. As already mentioned in Sec. 2.2, the data for the simulator certification might have to be gathered separately.

As summarized in Fig. 2.9, a variety of maneuvers is usually necessary to allow collection of data necessary to excite significantly all modes that we will be analyzing. The maneuvers are usually performed in blocks, repeated at different trim conditions defined by Mach number, angle of attack, altitude, configuration, which will cover flaps, thrust levels (particularly for propeller aircraft), landing gear, center of gravity, etc. The general procedure consists of the following:

1. Fly to the desired level and speed.
2. Select the configuration and trim the aircraft at a reference flight condition.
3. Apply the sequence of inputs; choose the amplitudes such that operational limits are not exceeded.
4. Repeat the sequence multiple times at the same trim condition.

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Between maneuvers, the aircraft is trimmed to horizontal level flight. The same input maneuver is repeated a number of times to enable comparison of estimates, to minimize the influence of external disturbances, and to determine the amount of variation in the measurements. If possible, the same maneuver should be performed three or more times. If the maneuver is performed twice, we face the common dilemma of experimental investigations; namely, if the estimates from the two repeat runs deviate significantly, it becomes difficult to choose or discard one or the other. External disturbances are to be minimized, which means that maneuvers should be performed in calm air. Atmospheric disturbances due to thermal activities are less early in the morning. The reference point for the estimates obtained from the dynamic maneuvers is usually the starting trim condition. Another possibility is to treat reference point as the average of the initial trim and flight condition to which the flight vehicle returns after the dynamic maneuver; ideally, both should the same. In a few cases, average over the complete maneuver has also been considered.

The reference flight conditions are defined in terms of altitude and calibrated airspeed. Figure 2.10 shows the flight envelope for the Transall C-160 military transport aircraft. As an example, various test flight conditions are shown spread over four altitude levels and several speeds at each level. The equidistant stapling of trim points over altitude and speed, shown in the figure, might not be optimal and needs to be adapted to each case appropriately. For example, fewer altitude levels would suffice or more number of points at lower speeds might provide more data at operationally critical flight conditions. Likewise, trim points at several thrust levels should be part of the testing although not

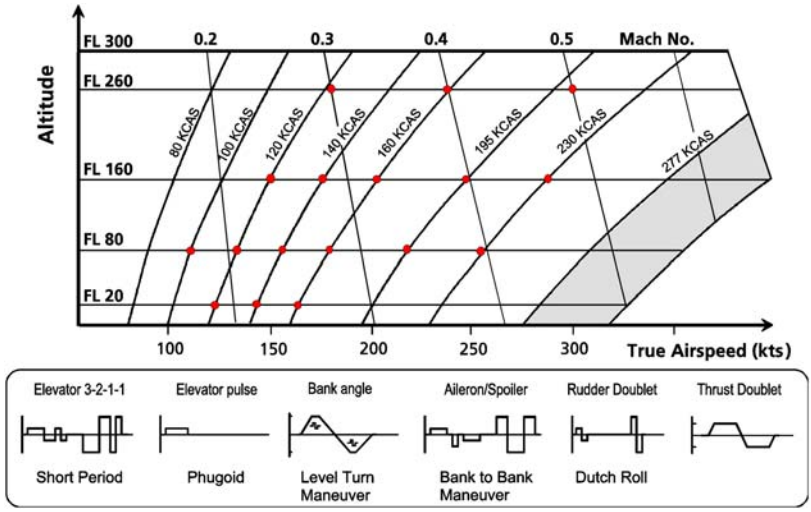

Fig. 2.10 Typical flight-test program for system identification.

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explicitly shown in the figure. Different altitude levels allow separation of the Mach number and dynamic pressure effects in the model because the same trim speed at two different altitudes results in two different Mach numbers. At each point the sequence of maneuvers to be performed is shown at the bottom of the same figure. Usually, it is efficient to perform all of the maneuvers at different trim conditions at one flight level, including the repeat runs and then change the flight level to follow the same procedure. During the actual flight testing, the flight levels might have to be adapted to the prevailing weather conditions.

In some cases, at high Mach numbers it might not be possible to fly level; we may have to then perform the tests in descending flight. Special flight conditions and those pertaining to flight at low altitudes above ground, for example, flybys, takeoffs, and landings, are not marked in Fig. 2.10. Similarly, tests at all of the altitude levels and speeds will not be possible for each configuration. The test points will have to be chosen accordingly, for example, dynamic tests with landing gear extended will be possible only up to certain altitudes and speeds. It is also not meaningful to perform such tests with landing gear extended at high altitudes, even if possible, because such test conditions are not realistic.

Development of a global model valid over the complete operational envelope requires flight testing for different landing flaps. The common approach is to perform dynamic maneuvers at each landing flap and identify models for each configuration separately. Interpolation over the flaps is incorporated in the final database to cater to the flap change dynamics, which is one of the tests to be covered in validating the models for flight simulators. If the models identified for the individual flap settings are accurate enough, usually the preceding procedure of interpolation is adequate to meet the fidelity requirements. If not, then some model adjustments will be necessarily late in the database development, which is a drawback of this approach. On the other hand, the preceding approach yields more accurate models for each flap configuration, covering significantly larger flight durations than the transition from one flap setting to another. The other approach of identifying the global model from simultaneous evaluation of flight maneuvers at different flap settings is not considered efficient although it might theoretically be possible, given sufficient computing power and computer memory for data processing. Irrespective of the approach we adopt, it is important to note that the input sequence does not include specific dynamic inputs to the landing flaps because they are rate limited. Instead, various flaps are chosen as reference configuration, performing the dynamic tests at each of them.

In a few specific cases, we deviate from this basic philosophy of performing various flight maneuvers at constant flap setting as a chosen reference configuration. If the aircraft is equipped with fast responding direct-lift flaps, which are modified parts of the landing flaps, then we apply the dynamic inputs to the flaps as well. The direct lift flaps provide an independent vertical control necessary in the aircraft that serve as in-flight simulators. Similarly, in the case of glider the flaps are purely mechanically actuated, limited only by ergonomics; the flaps can be deflected dynamically with multistep inputs to provide a direct generation of

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lift on the wing, which is expected to be more efficient to excite the wing symmetric bending modes than the indirect effect obtained through elevator deflection. We turn to these two cases in Chapter 12 to discuss in more details the relevance and advantages of dynamic flap inputs.

In the foregoing discussion we recommended exciting one mode at a time through appropriate input; this results from the practical aspects of performing flight maneuvers. In practice, it is difficult to excite optimally all of the modes simultaneously. Thus, flight testing will generate a large set of flight data with several dynamic maneuvers at a chosen trim point, which are repeated the same at different points over the flight envelope. However, estimation of the global model will require combining multiple maneuvers to estimate a set of aerodynamic parameters common to different modes and account for nonlinearities and dependencies on flight conditions. Even when we restrict ourselves to single trim point, it will be necessary to analyze data from several maneuvers. This requirement calls for special capabilities in processing flight data. We turn our attention to this aspect in Sec. 3.3.

2.5 FLIGHT-TEST INSTRUMENTATION AND MEASUREMENTS

Accuracy of parameter estimates is directly dependent on the quality of flight measured data, and hence high accuracy measurements of control inputs and of motion variables are a prerequisite for successful application of the modern methods of flight vehicle system identification. Classical information on flight-test instrumentation for aircraft parameter estimation is provided by Wolowicz [42] and also found in other references, for example, [43]. During the 1970s, the various aspects of flight-test instrumentation have been investigated in detail [44]. Dedicated flight-test instrumentation systems have been developed in several applications of flight vehicle system identification, mainly because 1) several signals from the basic aircraft systems can be tapped, that is, no additional sensors are necessary (for example, engine parameters, static pressure, rate gyros, etc.); and 2) control surfaces to be instrumented are flight vehicle dependent. Although more recently flight instrumentation systems based on commercially available sensors and standard signal processors have been developed to simplify this task, flight instrumentation is still a laborious and time-consuming job. Instead of providing a detailed description of a particular setup, we take a look at the basic requirements in terms of the necessary signals and their importance in the estimation of parameters.

A typical set of measurements required for aerodynamic model extraction from flight data consists of the following:

1. Control surface deflections
2. Linear accelerations
3. Angular rates

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4. Attitude angles
5. Air data
6. Static pressure
7. Engine parameters
8. Pilot forces and inceptor positions

If the scope of modeling is extended to validation of CFD simulations, additional measurements of flowfield velocity, other surface variables such as pressures, and temperatures, and of loads will become necessary [45].

Control surface deflections are inputs to the aerodynamic model being identified; inaccuracies in their measurements directly affect estimation results. The deflections are usually measured using potentiometers or LVDT (linear variable differential transducer), mounted directly at the respective hinges. Besides the primary controls (elevator, ailerons, and rudder), other control surfaces like flaps, speed brakes, spoilers, etc. need to be measured.

Accelerations and angular rates along and about the three axes provide very good information about aerodynamic effects. The inertial navigation system (INS) or sometimes called IMU (inertial measuring unit) is an integral unit providing these measurements and those of attitude angles. Some INS incorporate low-pass filters, mainly because such systems are used for long-term navigation purposes whereas short-term dynamic response is not of particular importance. For aerodynamic modeling purposes, however, dynamic response is of primary importance; hence, it needs to be ascertained that raw data from the INS are available or at least the cutoff frequency of the filter included is high enough. A dedicated tri-axis accelerometer is usually recommended. Accelerations are also measured at several other locations such as cockpit or the wing tips, horizontal tail, rudder tip, and engines, which are useful for other investigations such as flutter. For rigid-body aerodynamic model identification, a single of set of acceleration is sufficient.

The measurements of attitude angles are mostly provided by INS, usually through internal integration of measured angular rates; in rare cases they are obtained using magnetometers. These measurements are primarily useful in the data compatibility check, which we will discuss in Chapter 10 to ascertain quality of the recorded data and to determine biases in the measurements of the angular rates. Usually for aerodynamic model identification, the attitude angles are considered to be unnecessary or of secondary importance because aerodynamic forces and moments do not depend on the aircraft attitude relative to Earth axes. They are not required for the least-squares method discussed in Chapter 6, which also includes preprocessing of data to compute aerodynamic coefficients from the measured translational accelerations and angular rates; see Sec. 6.9. On the other hand, the attitude angles are useful in the output-error method discussed in Chapter 4, as they directly aid the determination of aerodynamic zero terms and of the other derivatives indirectly because they contain

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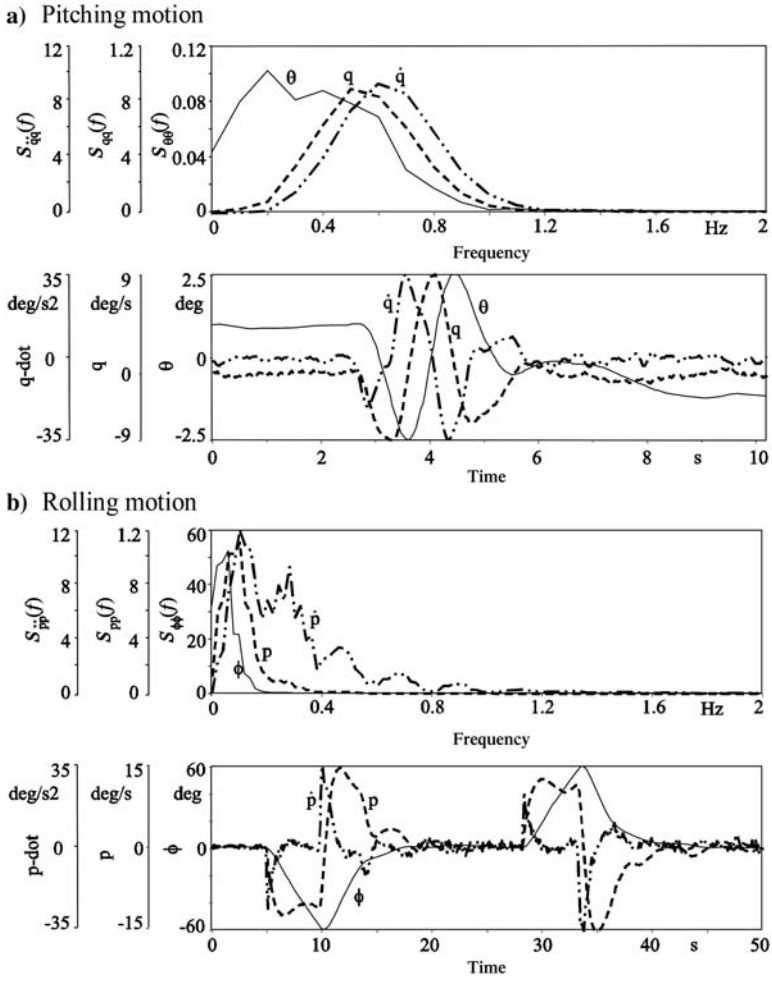

Fig. 2.11 Time histories and power spectra.

information at lower frequencies; see Fig. 2.11 showing typical plots of time histories and power spectra for attitude angle, angular rate, and angular acceleration related to pitching and rolling motion.

The angular accelerations are generally not measured directly. In many exercises related to system identification, they are derived by numerical differentiation of the measured angular rates. It is our experience that inclusion of angular accelerations, besides the angular rates and attitude angles, improves accuracy of parameter estimates and overall convergence of the optimization methods significantly. The contributions due to control surface deflections appear directly in the

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accelerations, and hence they come out much better from the acceleration signals. As seen from Fig. 2.11, these signals contain frequencies higher than those from the corresponding attitude angles and angular rates. Their inclusion leads to a model based on a wider range of frequencies. Based on these considerations, although angular accelerations may not be measured using dedicated sensors and aerodynamic forces and moments do not depend on the attitude angles, all three variables (angular accelerations, angular rates, and attitude angles) are considered useful in estimating aerodynamic derivatives.

Air data measurements, consisting of angle of attack, angle of sideslip, and airspeed, are also very useful in the parameter estimation. The flow angles are commonly measured using mechanical vanes and the airspeed by stagnation of the flow using a pitot tube or by propeller. Swiveling probes minimize the alignment errors. These variables are also measured quite often using a five-hole probe, measuring the difference and total pressures. In specific cases, a more advanced flush air data sensing system avoiding an intrusive boom is necessary to measure these parameters. If possible, the flow angles should be measured at multiple locations, typically in front of the aircraft or near pilot station and wing tips. A static pressure measurement is necessary to derive the information on the dynamic pressure, which is the difference between the total and static pressure and also for thrust computations. The static pressure varies along the length of the aircraft, and these measurements usually contain position errors. Availability of additional measurements such as altitude above ground measured using a radio altimeter, the ground speed, and the geodetical positions might be necessary in specific cases.

The engine parameters are necessary to correctly account for the thrust, which is necessary to enable estimation of drag. During aerodynamic model identification, the engine-manufacturer supplied data are used to compute the thrust, which is then treated as known input. In other words, we do not estimate the engine-thrust model. The identified drag model will, of course, depend upon the thrust model. Validating and updating engine thrust (performance) model from flight tests is possible, but it is not the subject addressed here [46, 47]. The exact set of required engine parameters depends on the aircraft (jet or propeller engines). For example, measurements of power lever and condition lever positions, propeller rpm, low-/high-pressure rpm, torque, and turbine gas temperature are necessary for propeller engines whereas those of low-pressure engine rpm are necessary for jet engines. The engine-thrust model also needs other variables such as Mach number, airspeed, and static pressure or altitude.

The pilot inputs (stick, wheel, and pedal positions) and pilot forces are necessary only for the modeling of the flight control system, particularly in the case of reversible controls, driven through mechanical rods and linkages. For estimation of rigid-body aerodynamics, the measured control surface deflections are treated as inputs to the model.

Besides the foregoing aspect of measurements required for extracting aerodynamic derivatives, other aspects related to data recording also affect data

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analysis. We simply mention them here without going into any details. They are as follows:

1. The sampling rate is the rate at which data are sampled and recorded. The minimum sampling rate is given by the Nyquist frequency, which should be twice the frequency of interest. In general, the data are sampled at a much higher frequency. A sampling frequency of 20–25 Hz is usually sufficient for rigid-body aerodynamic model estimation. For an extended model comprising rigid-body and structural modes, a higher sampling rate will be necessary.
2. The anti-aliasing filter for all measurements should have the same cutoff frequency, so that all signals would have the same time delays introduced by such filters.
3. Recording of raw data is usually preferable.
4. Ideally, all data channels should be recorded at the same sampling rate. Some of the data channels tapped from the basic aircraft systems may be available at lower sampling rates, for example, slowly varying parameters like altitude. But, they are less critical in the estimation of aerodynamic parameters. More critical measurements like translational accelerations, angular rates, and control surface deflections should be sampled at higher and uniform rates.
5. Ideally, all data channels should be time synchronized. This might not be possible due to serial data recording or the onboard system might introduce time delays. It is usually sufficient to know which data channel is recorded fastest; the others can be time synchronized through data compatibility checking.
6. The signal-to-noise ratio of 10:1 is desirable.
7. All sensors should be calibrated in laboratory and in situ as far as possible.
8. Data reduction should be avoided at the time of recording; if necessary, data reduction can be done while reading the data stored with higher sampling rate by skipping every point or the other during the off-line analysis. Such a procedure is a better one because once the information is lost due to data compression in the recording, it can never be recovered.

Finally, on-site preliminary verification of recorded data is recommended to ensure the adequacy of the data gathered for off-line analysis.

2.5.1 DIGITAL FILTER

Because recording of raw data is recommended, in a few cases it might be necessary to suppress high-frequency noise before using the signals. The high-frequency noise components are usually unrelated to the signal of interest. While generating the angular accelerations by numerical differentiation of measured angular rates,

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it is preferable to filter the measured signals for this specific purpose only. Digital filters based on a linear combination of input values are widely used for this purpose. They have the advantage that due to symmetry, they do not introduce any lag in the filtered values. In our applications, we have used a low-pass digital filter, developed by Spencer, based on 15 points, the current value, and preceding as well as following seven data points [48–50]. It belongs to a class of weighted moving average filter, implying that the averaging window moves point by point through the time series, whereby the number of data points for averaging remains the same. Weighted averaging leads to a smoother signal. Such a 15-point Spencer filter is given by

$$y_n = \frac{1}{320}\left[-3\,u_{n-7} - 6\,u_{n-6} - 5\,u_{n-5} + 3\,u_{n-4} + 21\,u_{n-3} + 46\,u_{n-2} + 67\,u_{n-1} + 74\,u_n + 67\,u_{n+1} + 46\,u_{n+2} + 21\,u_{n+3} + 3\,u_{n+4} - 5\,u_{n+5} - 6\,u_{n+6} - 3\,u_{n+7}\right] \tag{2.13}$$

where u is the noisy time series, y the filtered signal, n the discrete time index, u_{n-i} the ith point in the past, and u_{n+i} the ith point in the future from the nth discrete point being processed.

The magnitude response of a discrete filter, such as the one given by Eq. (2.13), can be computed as

$$H(\omega) = C(0) + 2\sum_{j=1}^{m} C(j)\cos(j\,\omega) \tag{2.14}$$

where $m = (N-1)/2$, N the number of data points for weighted averaging, that is, the window size, and C the filter coefficients, also called weights. For the Spencer's 15-point moving average filter given by Eq. (2.13), N is 15, and the filter weights $C = [-3\ \ -6\ \ -5\ \ 3\ \ 21\ \ 46\ \ 67\ \ 74\ \ 67\ \ 46\ \ 21\ \ 3\ \ -5\ \ -6\ \ -3]$, with $C(0) = 74$ at the window midpoint. The magnitude response plot for the 15-point Spencer filter is shown in Fig. 2.12, which shows that it has negligible oscillatory behavior at high frequencies and that the separation between low and high frequencies is well defined and sharp. The magnitude responses are plotted using the function "/FVSysID2/chapter02/filter_Sp_Hen_FreqResp.m."

Another commonly found Spencer filter with 21 data points is given by

$$y_n = \frac{1}{350}\left[-u_{n-10} - 3\,u_{n-9} - 5\,u_{n-8} - 5\,u_{n-7} - 2\,u_{n-6} + 6\,u_{n-5} + 18\,u_{n-4} + 33\,u_{n-3} + 47\,u_{n-2} + 57\,u_{n-1} + 60\,u_n + 57\,u_{n+1} + 47\,u_{n+2} + 33\,u_{n+3} + 18\,u_{n+4} + 6\,u_{n+5} - 2\,u_{n+6} - 5\,u_{n+7} - 5\,u_{n+8} - 3\,u_{n+9} - u_{n+10}\right] \tag{2.15}$$

Figure 2.12 shows the magnitude response of this filter as well. It can be shown that the Spencer filter leaves the local quadratic polynomials untouched and,

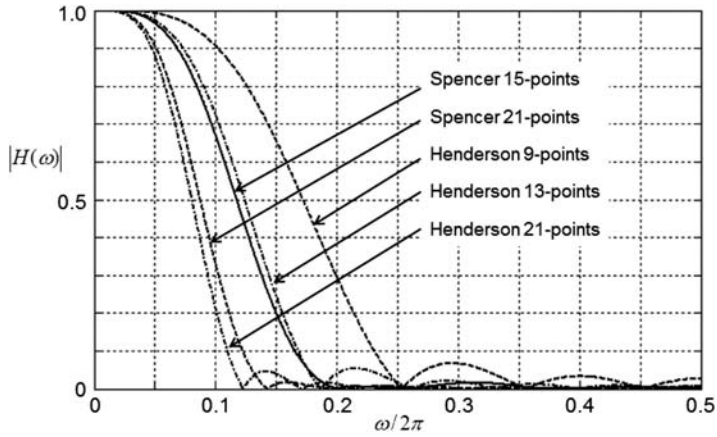

Fig. 2.12 Magnitude responses of Spencer and Henderson filters.

due to symmetry, the cubic polynomials as well. This implies that they can match peaks and troughs well. In all of our applications we have used the 15-point Spencer filter given by Eq. (2.13).

Yet another filter commonly applied in the trend estimation is due to Henderson, mostly based on 7, 9, 13, or 23 points [51–53]. The general formula to compute weights of Henderson filter is given by

$$C_j = \frac{315[(m+1)^2 - j^2][(m+2)^2 - j^2][(m+3)^2 - j^2][3(m+2)^2 - 11j^2 - 16]}{8(m+2)[(m+2)^2 - 1][4(m+2)^2 - 1][4(m+2)^2 - 9][4(m+2)^2 - 25]} \tag{2.16}$$

for $j = -m$ to m, where $m = (N-1)/2$, N being the window size (filter length), that is, number of data points. We consider here only symmetrical filter, that is, N must be an odd number. Henderson's filter is designed to follow local cubic polynomial and can fit different curvatures in the noisy time series, whereby smoothing is obtained through weighted moving averaging process [53]. The preceding formula yields weights for the 7-, 9-, and 13-point Henderson filters as (−0.0587, 0.0587, 0.2937, 0.4126, 0.2937, 0.0587, −0.0587), (−0.0407, −0.0099, 0.1185, 0.2666, 0.3311, 0.2666, 0.1185, −0.0099, −0.0407), and (−0.0193, −0.0279, 0, 0.0655, 0.1474, 0.2143, 0.2401, 0.2143, 0.1474, 0.0655, 0, −0.0279, −0.0193), respectively. They are computed using the function "/FVSysID2/chapter02/HendersonCoeff.m." The magnitude responses of these Henderson filters with different filter lengths are shown in Fig. 2.12. It is observed that 13-point Henderson filter has characteristics similar to 15-point Spencer filter, but with small oscillations around zero level at higher frequencies. The

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same is true for the two filters with 21 points. The 9-point Henderson filter has a wider bandpass, but at the same time somewhat larger oscillations at higher frequencies than those with more data points. Thus, there is tradeoff between the required range of bandpass and the oscillations around zero at higher frequencies that can be tolerated, which might have to be investigated in the case of very noisy data.

The preceding equations can be correctly implemented for a truncated time series for all data points except for few data points, namely, $(N-1)/2$ points, from the two ends; for example, for the 15-point Spencer filter for the first and the last seven points from the two ends of the time series. To smooth the two ends of a time series, a filter based on fewer points is applied. Because of the lack of sufficient information, the first two and the last two data points cannot be filtered accurately. This needs to be kept in mind while using preceding procedure. In the case of 15-point Spencer filter given by Eq. (2.13), for the third to seventh data points from the two ends the following filter using two data points can be incorporated:

$$y_n = \frac{1}{96}[7\,u_{n-2} + 24\,u_{n-1} + 34\,u_n + 24\,u_{n+1} + 7\,u_{n+2}] \tag{2.17}$$

Equations (2.13) and (2.17) have been coded for a single time segment in the function "/FVSysID2/chapter02/smooth.m" and for multiple time segments in "smoothMulTS.m," which is used in the following chapters. Similarly the Henderson filter can be coded based on Eq. (2.16), if the need arises.

2.5.2 NUMERICAL DIFFERENTIATION

As already pointed out earlier in this section, angular accelerations are obtained by numerical differentiation of the measured angular rates. Time derivatives of other measurements may also be needed in specific cases. The simplest way to obtain such time derivatives is to use the one-sided forward or two-sided central difference formulas. We will deal in more detail with such formulas in Sec. 3.5.3 in connection with approximating the system functions. Such simple formulas are not quite suitable for numerical differentiation of measured time histories because they tend to amplify the noise. In other words, small errors in measured data can induce large deviations in the numerically approximated derivative. In spite of high quality sensors, the presence of measurement noise is unavoidable. In general, there are two ways to go about doing differentiation of noisy data: 1) filter the raw data first using suitable filter such as a 15-point Spencer or 9-point Henderson filter discussed in the foregoing section and then differentiate the smoothed data using one of the differentiator having near-ideal response characteristics and 2) choose numerical differentiator incorporating a suitable low-pass digital filter that suppresses the noisy high frequencies. We study both the approaches.

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Without going into mathematical details, the frequency response of a differentiating smoother given by $i\Omega/\Delta t$, where $\Omega = \omega\Delta t$, having no real part, can be constructed by the following digital filter of arbitrary order given by [54]

$$y(k) = \frac{1}{\Delta t} \sum_{i=1}^{N} C_i[x(k+i) - x(k-i)] \tag{2.18}$$

where k is the discrete time point, x the measured signal, y the time derivative of the measured signal, Δt the sampling time, N the order of the filter incorporated within a differentiator, and C_i the corresponding coefficients. The coefficients C_i are obtained from the equation:

$$A\ C = b \tag{2.19}$$

where the vector b and the elements of n–by–n matrix A are given by

$$b^T = [1/2 \quad 0 \quad 0 \quad 0 \quad \cdots]; \quad a_{ij} = (-1)^{i+1}\ j^{2i-1} \tag{2.20}$$

Equations (2.19) and (2.20) yield the following coefficients for differentiator from first to fourth order, respectively: (0.5), (0.666666667, −0.08333333), (0.75, −0.15, 0.016666667), and (0.8, −0.2, 0.038095238, −0.003571428). These coefficients and those for the higher-order smoother differentiator can be generated using the program "/FVSysID2/chapter02/numDiffCoeff.m." Utility functions "ndiff_Filter02," "ndiff_Filter04," "ndiff_Filter08," and "ndiff_Filter12" for differentiation, catering to multiple time segments, have been developed based on second-, fourth-, eighth-, and twelfth-order formulas. They have the magnitude responses shown in Fig. 2.13, whereby the magnitude response of a differentiator is given by

$$H(\omega) = \frac{i}{\Delta t}\ 2\sum_{j=1}^{N} C(j)\ \sin(j\omega) \tag{2.21}$$

and computed using the function "/FVSysID2/chapter02/nDiff_FreqResp.m." The higher the order (that is, number of data points used to compute the derivative), then the more closer is the differentiator to the ideal response of $H(\omega) = i\omega$. These four differentiators and those discussed hereafter can be easily verified on a simple test case using "ndiffVerify.m." It is fairly obvious that for $N = 1$, Eq. (2.18) corresponds to the standard central difference formula. These formulas with more data points are, in general, termed as N-point central differences. Other ways to arrive at these centrally pivoted algorithms for smoothing and differentiation are possible [55]. We have used eighth- and twelfth-order differentiator formulas in several applications. We will be using the differentiator with eighth-order central differences ("ndiff_Filter08.m") preceded by "smoothMulTS.m" based on the 15-point Spencer filter in some of the test cases analyzed in this book. From Eq. (2.18), we note that these formulas are based on the use of past and future data about a point. The number of data points used to compute the

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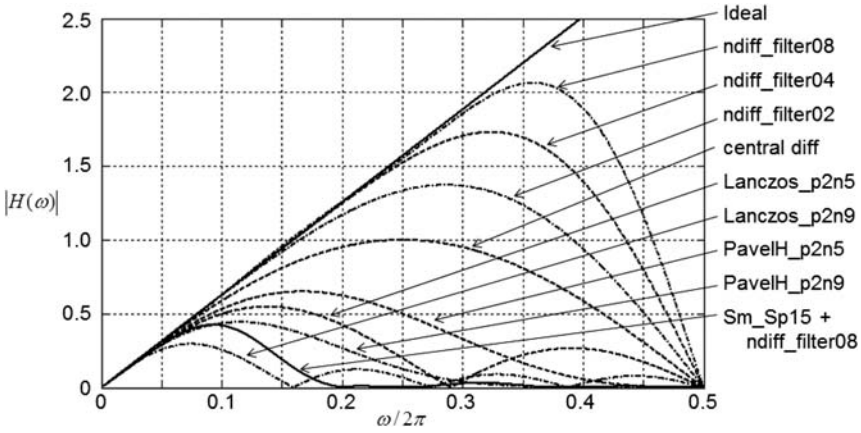

Fig. 2.13 Magnitude responses of various differentiators.

derivative at any point is twice the order plus one for the data point being evaluated. As these differentiators do not suppress sufficiently the high-frequency noise, as already pointed out, they are usually preceded by smoothing the raw data using a suitable filter.

Another approach based on first fitting a polynomial to a few data points by the least-squares method and then analytically differentiating the approximated polynomial is also frequently used to compute the first derivative. Fitting a polynomial automatically leads to smoothing out the noisy data. The method, called Lanczos differentiator, is once again centrally pivoted and needs an equal number of data points on either side of the data point at which the derivative is being computed. It can be shown that for five data points the formula for the first derivative at the midpoint, fitting a parabola by least-squares minimization, is given by [56, 57]

$$y(k) = \frac{2x(k+2) + x(k+1) - x(k-1) - 2x(k-2)}{10\,\Delta t} \tag{2.22}$$

where, as before in Eq. (2.18), k is the index of the data point, y the first derivative of the measured signal x, $x(k+1)$, $x(k+2)$ the two point in the future, and $x(k-1)$, $x(k-2)$ the two point in the past from the kth point.

It is also possible to use a greater number of data points in the Lanczos differentiator. The general formula for the Lanczos differentiator based on a second-order polynomial using N data points is given by [58]

$$y(k) = \frac{3}{\Delta t}\sum_{i=1}^{m} i\frac{x(k+i) - x(k-i)}{m(m+1)(2m+1)}, \quad m = \frac{N-1}{2} \tag{2.23}$$

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where N must be an odd number. It can be easily verified that for $N = 5$, Eq. (2.23) leads to the formula in Eq. (2.22).

Note that the coefficients for this Lanczos differentiator with five data points are (0.1, 0.2), which are different from (0.666666667, −0.083333333) for the second-order differentiator with similar structure given by Eqs. (2.18–2.20). For the case with nine data points, the coefficients for the Lanczos differentiator turn out to be (0.01666666666667, 0.03333333333333, 0.050, 0.06666666666667), which are different from (0.8, −0.2, 0.038095238, −0.003571428) for the fourth-order formula given by Eqs. (2.18–2.20). As such, these differentiator filters show different magnitude response and noise suppression characteristics. The Lanczos differentiator based on a second-order polynomial using five data points given by Eq. (2.22) is coded in a function "ndiff_Lanczos_p2n5.m" and for nine data points in "ndiff_Lanczos_p2n9.m." These differentiator filters based on a second-order polynomial approximation are termed as *low-noise Lanczos differentiators* because they are less sensitive to noise due to the low gain at high frequencies. At low frequencies their characteristics approach the ideal response. In other words, they have a good suppression of high frequencies. The limitations are that the linear frequency range is somewhat limited and that they lead to wavy nonzero response at higher frequencies as seen in Fig. 2.13. The Lanczos formulas can also be derived for higher-order polynomials [58, 59]. They are usually termed as *super Lanczos low-noise differentiators*, which have a wider linear range, but the high-frequency suppression is poorer than that of the low-noise Lanczos differentiator just discussed. We do not pursue further the super Lanczos low-noise differentiators, but their formulas can be found in [58–60].

A more general polynomial fitting approach is commonly termed in the field of digital signal processing as Savitzky–Golay filter [61–64]. Such a general-purpose, more flexible version of Savitzky–Golay digital differentiators is available as a Matlab® function "sgolay" found in the Signal Processing Toolbox [65]. The same function can also be readily used to obtain higher-order derivatives or just for smoothing the noisy data, in which case it provides an alternative to Spencer digital filter given by Eq. (2.13). It can be verified that the coefficients given by Eqs. (2.22) for a second-order polynomial with five data points, and those by Eq. (2.23) for nine data points, are exactly the same as those provided by the function sgolay for the same order of the polynomial and the same number of data points; the second column of the differentiation filter in sgolay provides the coefficients for the first derivative. For our specific application involving data analysis with multiple time segments, we provide two functions, namely, "ndiff_SGolay_p2n5" and "ndiff_SGolay_p2n9" based on the function sgolay for the preceding choice of polynomial order and data points. They are numerically equivalent to "ndiff_Lanczos_p2n5.m" and "ndiff_Lanczos_p2n9.m," respectively.

For all of the differentiators, as in the case of digital filter for smoothing discussed in the preceding section, for a truncated time series an adequate procedure needs to be developed for a few data points at the two ends. This aspect has been

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suitably taken care of in the various functions for numerical differentiation provided herewith. As the algorithms elaborated here are based on the use of past and future data, they are noncausal and suitable for off-line data analysis only. For real-time application, a different class of filters needs to be developed. We do not go into details of such algorithms. For a simple test of differentiating a data with sine wave without any noise in "ndiffVerify.m," all the differentiators yield, as expected, very similar results. With data gathered from different aircraft, which showed noise characteristics commonly found with typical sensors and data acquisition systems, different noise-suppressing differentiators discussed hitherto provided comparable and acceptable results. In general, smoothing the data first by Spencer filter "smoothMulTS.m" followed by "ndiff_Filter08.m" differentiator discussed in Eqs. (2.18–2.20) showed marginally better performance. This conclusion is also supported by the magnitude response of the combined 15-point Spencer filter followed by ndiff_Filter08 shown in Fig. 2.13 and enlarged in Fig. 2.14, which shows much better suppression of high frequencies and negligible oscillations around zero at higher frequencies. Figure 2.14 is generated using the function "/FVSysID2/chapter02/nDiff_FreqResp2.m." If larger linear range is desired, we will have to go for the nine-point Henderson filter to smooth out the noise first and then apply the ndiff_Filter08 to differentiate the filtered data.

In a few cases, particularly with unmanned aerial vehicle with a very high level of noise with wider frequency spread, the choice of differentiator was more involved. The differences in the performance in such a case are mostly observed at the peaks of the rapidly changing responses in terms of peak heights and width and around the zero level in the form of oscillations. For smoothing out the noise, longer window length is preferable; the longer the length, the better the

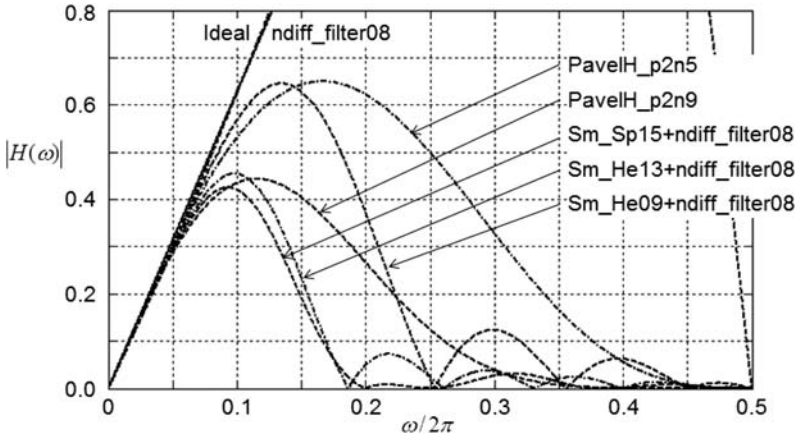

Fig. 2.14 Magnitude responses of smooth noise-robust differentiators.

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smoothing. However, for a fast responding dynamic system, longer window length may distort the actual fast system response by smoothing out the desired signal itself. As smoothing and noise-suppressing differentiators may, in general, reduce the information content, it is a tradeoff between noise suppression, mostly determined by the window length, that is, the data points, and acceptable reduction in the information, which needs to be carefully considered. Optimal choice of differentiator depends on the noise contained in the data and may have to be investigated in each case.

The just-discussed differentiator filters might show poor performance in specific cases because the least-squares polynomial approximation does not guarantee complete noise suppression of high frequencies. A smooth noise-robust differentiator developed recently by Holoborodko guarantees suppression of high frequencies and yet is precise on low frequencies and retains exactness on the polynomial [58]. The general formula for such a smooth noise-robust differentiator is given by

$$y(k) \approx \frac{1}{\Delta t} \sum_{i=1}^{M} c_i (x(k+i) - x(k-i))$$
$$c_i = \frac{1}{2^{2m+1}} \left[\binom{2m}{m-k+1} - \binom{2m}{m-k-1} \right], \quad m = \frac{N-3}{2}; \quad M = \frac{N-1}{2} \tag{2.24}$$

where, as before, k is the index of the data point, y the first derivative of the measured signal x, and N the number of data points used in the filter differentiator; it must be an odd number. The terms inside the two brackets within the square bracket are the binomial coefficients.

We note that the structure of this smooth noise-robust differentiator is basically the same as that of the other differentiators, with coefficients as defined in Eq. (2.24). In the present case, they turn out to be (0.25, 0.125) for differentiator with five data points and (0.109375, 0.109375, 0.046875, 0.0078125) for differentiator with nine data points. Functions for differentiators based on the preceding formula using five and nine data are provided as utility functions "ndiff_PavelH_p2n5.m" and "ndiff_PavelH_p2n9.m," respectively, and can be used for multiple time segments analyzed simultaneously. The magnitude responses of these differentiators are shown in Figs. 2.13 and 2.14. They appear to have a wider linear range than the Lanczos differentiators and suppress the high frequencies better without showing wavy characteristics. Similar characteristics were observed for the combined 15-point Spencer filter followed differentiator using 9-point or 17-point central differences.

From Fig. 2.14, it is observed that the smooth noise-robust differentiator given by Eq. (2.24) has no nonzero wavy response at higher frequencies, but the separation between the low and high frequencies is gradual compared to the procedure recommended and followed in this book, namely, filter the data

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first using 15-point Spencer filter followed by differentiation using 9-point or 17-point central differences. If a wider linear range is desired, then the same two-step procedure using the 9-point Henderson filter can be followed, but at the cost of a small nonzero wavy response at higher frequencies. The optimal choice in a particular case will depend upon the noise characteristics. As pointed out in [58], however, the smooth noise-robust differentiator given by Eq. (2.24) is suitable in special cases when a certain range of frequencies is corrupted with noise whereas Savitzky–Golay is optimal for Gaussian noise suppression. In connection with flight data analysis, we have not investigated the performance of this smooth noise-robust differentiator filter in detail and hence refrain from comparing its performance with other formulas, mainly because the majority of the test cases showed noise frequencies for which the two-step procedure of "smoothMulTS.m" followed by "ndiff_Filter08.m" was performed adequately.

In spite of great care in choosing sensors, their installation and calibration, and in recording the data, instrument errors such as bias and time delays are unavoidable. In some cases verification of scale factors is necessary, particularly for flow angles, because they can be configuration dependent and realistic conditions might not be duplicated in a laboratory. Such a step dealing with data checking, estimating instrument errors, and generating refined measurements is part of the overall data gathering process. We defer the discussion of this step, called data compatibility checking, to Chapter 10 because it requires some of the estimation algorithms that we will discuss in the following chapters.

2.6 CONCLUDING REMARKS

In this chapter we have discussed an important aspect of data gathering for system identification purposes. In terms of the Quad-M basics introduced in the preceding chapter, we covered in detail the aspects pertaining to the first M for maneuvers and a part of the second M for measurements. In essence, optimal input design boils down to two simple basic principles: 1) inputs must be such as to excite significantly a range of frequencies around the eigenmodes that are to be modeled and 2) independent input on every control is necessary. Two approaches to design optimal inputs have been discussed; the one based on the estimation error criterion is more rigorous, and the other is the more practical one based on a simple engineering process of synthesizing the contributions due to each of the parameters appearing in each of the force and moment equations. Because both designs are based on the a priori model involving parameter uncertainties, it is necessary to provide for a broader band input signal and one that can be easily adapted to changing flight conditions.

It is our experience that from the basic a priori knowledge about the modes of motion, which we mostly possess, adequate inputs can be designed for

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conventional aircraft without detailed investigations by following a few simple guidelines:

1. The multistep 3-2-1-1 input is preferable for short-period mode whereas a simple doublet is quite sufficient for Dutch roll; rolling motion is adequately excited by a bank-to-bank maneuver.
2. The exact shape of the doublet or of the multistep inputs and the optimal time step for these inputs are not critical.
3. The duration of time steps is obtained from simple empirical rules suggested here.

To further improve upon the estimation results, it is preferable to apply one control input at a time during dynamic maneuvers and to minimize excitation of the unmodeled modes. We believe that the gains obtained through a more detailed design of input signals are marginal; furthermore, the improvements in the statistical accuracies as predicted by the error covariance matrix are associated with certain limitations with which we will deal in one of the following chapters. Thus, we have followed the simple philosophy in many applications, namely, stick to the basics just elaborated, and use a combination of different system identification maneuvers briefly discussed in this chapter.

In special cases such as hypersonic flight vehicles, however, more advanced experiment design will be required because the duration of the total test time is usually short and the flight condition (Mach number) changes rapidly. In such cases it is desirable to have input signals with a lot of energy, keeping the input amplitude levels low. Multi-axis orthogonal phase-optimized sweeps having a wide frequency band are more suitable in these applications. Similarly, inputs other than the multistep signals might be more suitable for aeroservoelastic applications.

The scope of the flight testing depends on the desired range of validity and fidelity of the model; hence, defining the goal of system identification prior to data gathering is necessary to optimize the scope of flight testing. The larger the number of configurations to be modeled and the range of the desired validity, the larger is the number of tests to be performed. A brief summary of various flight maneuvers has been provided to gain insight into experimental techniques. It is necessary to point out here that the primary goal of flight testing is to certify flight worthiness and safety; however, conforming to the subject of this book we have focused here only a very specific part, namely, that which allows more accurate validation and update of aerodynamic database.

A brief summary of typical measurements required for parameter estimation has also been provided. It has been argued that besides the control surface deflections, translational accelerations, angular rates, and flow angles that are necessary, the angular accelerations and attitudes angles are useful in extracting aerodynamic derivatives, yielding a model covering a broader frequency range. Practical aspects of data smoothing using digital filters and of differentiating noisy signals have been discussed in some depth.

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REFERENCES

[1] Milliken, W. F. Jr., "Dynamic Stability and Control Research," *Proceedings of the 3rd Anglo-American Aeronautical Conference*, Royal Aeronautical Society, London, England, U.K., 1952, pp. 447–524.

[2] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[3] Levin, M. J., "Optimal Estimation of Impulse Response in the Presence of Noise," *IRE Transactions on Circuit Theory*, Vol. CT-7, No. 1, 1960, pp. 50–56.

[4] Litmann, S., and Huggins, W. H., "Growing Exponentials as a Probing Signal for System Identification," *Proceedings of the IEEE*, Vol. 51, June 1963, pp. 917–923.

[5] Levadi, V. S., "Design of Input Signals for Parameter Estimation," *IEEE Transactions on Automatic Control*, Vol. AC-11, No. 2, 1966, pp. 205–211.

[6] Gerlach, O. H., "Analysis of a Possible Method for the Measurement of Performance and Stability and Control Characteristics of an Aircraft in Non-Steady Symmetric Flights," Dept. of Aerospace Engineering, Delft Univ. of Technology, Rept. VTH-117, Delft, The Netherlands, Nov. 1964 (in Dutch with Summary in English).

[7] Gerlach, O. H., "Determination of Performance, Stability and Control Characteristics from Measurements in Non-Steady Manoeuvres," AGARD, CP-17, Sept. 1966, pp. 499–523.

[8] Mehra, R. K., "Optimal Inputs for Linear System Identification," *IEEE Transactions on Automatic Control*, Vol. 19, No. 3, 1974, pp. 192–200.

[9] Stepner, D. E., and Mehra, R. K., "Maximum Likelihood Identification and Optimal Input Design for Identifying Aircraft Stability and Control Derivatives," NASA CR-2200, March 1973.

[10] Marchand, M., "Untersuchung der Bestimmbarkeit von Flugzeugderivativen aus Flugversuchen," DFVLR IB 154-74/32, Dec. 1974 (in German).

[11] Gupta, N. K., and Hall, W. E. Jr., "Input Design for Identification of Aircraft Stability and Control Derivatives," NASA CR-2493, Feb. 1975.

[12] Koehler, R., and Wilhelm, K., "Auslegung von Eingangssignalen für die Kennwertermittlung," DFVLR-IB 154-77/40, Dec. 1977 (in German).

[13] Plaetschke, E., and Schulz, G., "Practical Input Signal Design," AGARD, LS-104, Paper 3, Nov. 1979.

[14] Mulder, J. A., Sridhar, J. K., and Breeman, J. H., "Identification of Dynamic Systems – Applications to Aircraft, Part 2: Nonlinear Analysis and Manoeuvre Design," AGARD, AG-300, Vol. 3, Part 2, May 1994.

[15] Proskawetz, K.-O., "Optimierung Stufenförmiger Eingangssignale im Frequenzbereich für die Parameter Identifizierung," *ZFW*, Vol. 9, No. 6, 1985, pp. 362–370 (in German).

[16] Gates, R. J., Bowers, A. H., and Howard, R. M., "A Comparison of Flight Input Techniques for Parameter Estimation of Highly-Augmented Aircraft," AIAA Paper 96-3363, Aug. 1996.

[17] Morelli, E. A., "Flight Test of Optimal Inputs and Comparison with Conventional Inputs," *Journal of Aircraft*, Vol. 36, No. 2, 1999, pp. 389–397.

[18] "Airplane Simulator Qualification," Federal Aviation Administration, FAA Advisory Circular, AC 120-40C, Draft Version, July 1995.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[19] "Joint Aviation Requirements – Aeroplane Flight Simulators," Westward Digital, Ltd., JAR-STD 1A, Cheltenham, England, April 1997.

[20] Kimberlin, R. D., *Flight Testing of Fixed-Wing Aircraft*, AIAA, Reston, VA, Sept. 2003.

[21] Lee, R. E., Jr., "Handling Qualities," Vol. 14, AGARD-AR-300, Paper No. 15, Sept. 1995.

[22] Gallagher, G. L., Higgins, L. B., Khinoo, L. A., and Pierce, P. W., "Fixed Wing Performance," Naval Test Pilot School Flight Test Manual, Rept. USNTPS-FTM-No. 108, Sept. 1992.

[23] "Safety Recommendation," National Transportation Safety Board, No. A-02-01 and -02, Washington, D.C., Feb. 8, 2002.

[24] Tischler, M. B., Fletcher, J. W., Diekmann, V. L., Williams, R. A., and Cason, R. W., "Demonstration of Frequency-Sweep Testing Techniques Using a Bell 214-ST Helicopter," NASA TM-89422, April 1987.

[25] Kolwey, H., "Rotorcraft Frequency Domain Flight Testing," American Helicopter Society T&E Committee, Tech Memo RW-51, Alexandria, VA, March 1994.

[26] Ham, J. A., "Frequency Domain Flight Testing and Analysis of an OH-58D Helicopter," *Journal of the AHS*, Vol. 37, No. 4, Oct. 1992, pp. 16–24.

[27] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[28] Goodwin, G. C., and Payne, R. L., *Dynamic System Identification: Experiment Design and Data Analysis*, Academic Press, New York, 1977.

[29] Marchand, M., "Untersuchung der Bestimmbarkeit der Flugmechanischen Derivative des CCV-Versuchsträgers F-104 G," DFVLR IB 154-77/12, March 1977 (in German).

[30] Blackman, R. B., and Tukey, J. W., *The Measurement of Power Spectra*, Dover, New York, 1958.

[31] Paris, A. C., and Bonner, M., "Nonlinear Model Development from Flight-Test Data for F/A-18E Super Hornet," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 692–702.

[32] Plaetschke, E., Mulder, J. A., and Breeman, J. H., "Results of Beaver Aircraft Parameter Estimation," DFVLR-FB 83-10, Feb. 1983.

[33] Schroeder, M. R., "Synthesis of Low-Peak-Factor Signals and Binary Sequences with Low Autocorrelation," *IEEE Transactions on Information Theory*, Vol. 16, No. 1, 1970, pp. 85–89.

[34] Schoukens, J., Pintelon, R., Van der Ouderaa, E., and Renneboog, J., "Survey of Excitation Signals for FFT Based Signal Analyzers," *IEEE Transactions on Instrumentation and Measurement*, Vol. 37, No. 2, 1988, pp. 342–352.

[35] Roberto, J., and Arruda, F., "Multisine Multiexcitation in Frequency Response Fucntion Estimation," *AIAA Journal*, Vol. 31, No. 1, 1993, pp. 215, 216.

[36] O'Neill, C. R., and Arena, A. S., Jr., "Time-Domain Training Signals Comparison for Computational Fluid Dynamics Based Aerodynamic Identification," *Journal of Aircraft*, Vol. 42, No. 2, 2005, pp. 421–428.

[37] Young, P., and Patton, R. J., "Comparison of Test Signals for Aircraft Frequency Domain Identification," *Journal of Guidance, Control, and Dynamics*, Vol. 13, No. 3, 1990, pp. 430–438.

[38] Brenner, M. J., Lind, R. C., and Voracek, D. F., "Overview of Recent Flight Flutter Testing Research at NASA Dryden," NASA TM-4792, April 1997.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[39] Clarke, R., Burken, J. J., Bosworth, J. T., and Bauer, J. E., "X-29 Flight Control System: Lessons Learned," *International Journal of Control*, Vol. 59, No. 1, Jan. 1994, pp. 199–219.

[40] Caldwell, B. D., "The FCS-Structural Coupling Problem and Its Solution," AGARD, CP-560, Paper 16, Jan. 1995.

[41] Acree, C. W., Jr., and Tischler, M. B., "Identification of XV-15 Aeroelastic Modes Using Frequency Sweeps," *Journal of Aircraft*, Vol. 26, No. 7, 1989, pp. 667–674.

[42] Wolowicz, C. H., "Considerations in the Determination of Stability and Control Derivatives and Dynamic Characteristics from Flight Data," AGARD, AR-549, Part 1, 1966.

[43] Borek, R. W., and Pool, A. (eds.), "Basic Principles of Flight Test Instrumentation Engineering," AGARD, AG-160, Vol. 1, No. 2, March 1994.

[44] Breeman, J. H., van Woerkom, K., Jonkers, H. L., and Mulder, J. A., "Aspects of Flight Test Instrumentation," AGARD, LS-104, Paper 4, Nov. 1979.

[45] Marvin, J. G., "Perspective on Computational Fluid Dynamics Validation," *AIAA Journal*, Vol. 33, No. 10, 1995, pp. 1778–1787.

[46] Rooney, E. C., and Wilt, C. E., "Development of in-Flight Thrust Measurement Procedures for an Afterburning Turbofan Engine," AIAA Paper 1985-1405, July 1985.

[47] Niewald, P. W., and Parker, S. L., "Flight-Test Techniques Employed to Successfully Verify F/A-18E in-Flight Lift and Drag," *Journal of Aircraft*, Vol. 37, No. 2, 2000, pp. 194–200.

[48] Spencer, J., "On the Graduation of the Rates of Sickness and Mortality Presented by the Experience of the Manchester Unity of Oddfellows During the Period 1893-97," *Journal of the Institute of Actuaries*, Vol. 38, 1904, pp. 334–343.

[49] Spencer, J., "Some Illustrations of the Employment of Summation Formulas in the Graduation of Mortality Tables," *Journal of the Institute of Actuaries*, Vol. 41, 1907, pp. 361–408.

[50] Kendall, M., Stuart, A., and Ord, J. K., *The Advanced Theory of Statistics, Volume 3: Design and Analysis, and Time-Series*, Charles Griffin & Company, Ltd., London, 1983.

[51] Henderson, R., "Note on Graduation by Adjusted Average," *Transactions of the Actuarial Society of America*, Vol. 17, 1916, pp. 43–48.

[52] Kenny, P. B., and Durbin, J., "Local Trend Estimation and Seasonal Adjustment of Economic and Social Time Series," *Journal of the Royal Statistical Society*; Series A (Statistics in Society), Vol. 145, 1982, pp. 1–41.

[53] Chamberlin, G., "Fitting Trends to Time Series Data," *Economic Trends*, Vol. 633, Aug. 2006, pp. 64–70.

[54] Koehler, R., "Experimentelle Aspekte," *Lecture Notes: Praktische Einführung in die Systemidentifizierung*, Carl-Cranz-Gesellschaft, Oberpfaffenhofen, Germany, 11–13 June 1985, Sec. 9 (in German).

[55] Forsythe, W., "Digital Algorithms for Prediction, Differentiation and Integration," *Transactions of the Institute of Measurement and Control*, Vol. 1, No. 1, Jan.–March 1979, pp. 46–56.

[56] Lanczos, C., *Applied Analysis*, Prentice–Hall, Englewood Cliffs, NJ, 1956.

[57] Wood, G. A., "Data Smoothing and Differentiation Procedures in Biomechanics," *Exercise and Sport Sciences*, Vol. 10, 1982, pp. 308–362.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[58] Holoborodko, Pavel, "Low-Noise Lanczos Differentiators," http://www.holoborodko.com/pavel/numerical-methods/numerical-derivative/lanczos-low-noise-differentiators/.

[59] Hamming, R., *Digital Filters*, 3rd ed., Dover, New York, 1998, pp. 133–137.

[60] Lyons, R. G., *Understanding Digital Signal Processing*, 3rd ed., Prentice–Hall, Upper Saddle River, NJ, 2011, pp. 366, 367.

[61] Savitzky, A., and Golay, M. J. E., "Smoothing and Differentiation of Data by Simplified Least Squares Procedures," *Analytical Chemistry*, Vol. 36, No. 8, July 1964, pp. 1627–1639.

[62] Steinier, J., Termonia, Y., and Deltour, J., "Comments on Smoothing and Differentiation of Data by Simplified Least Square Procedure," *Analytical Chemistry*, Vol. 44, No. 11, Sept. 1972, pp. 1906–1909.

[63] Madden, H. H., "Comments on the Savitzky-Golay Convolution Method for Least-Squares Fit Smoothing and Differentiation of Digital Data," *Analytical Chemistry*, Vol. 50, No. 9, Aug. 1978, pp. 1383–1386.

[64] Orfanidis, S. J., *Introduction to Signal Processing*, Prentice–Hall, Englewood Cliffs, NJ, 1996.

[65] "MATLAB, "The Language of Technical Computing," ver. 6.5.1, Release 13 (Service Pack 1), The MathWorks, Inc., Natick, MA, 2003.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

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CHAPTER 3

Model Postulates and Simulation

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3.1 INTRODUCTION

Dynamical systems can be described by differential equations, whose order depends upon the process complexity, coupling between the subsystems, and on the degree of accuracy required for the specific application. The state-space representation transforms higher-order differential equations into a set of coupled first-order equations. The two important issues that characterize the postulated model are as follows: 1) choice of the state variables and 2) input–output and internal system behavior. The set of states that can be chosen to represent a system is never unique, but will depend upon the pertinent physical characteristics of the system being modeled. Having selected appropriate states, the internal system behavior is then characterized by the system parameters. Such models in terms of states and parameters for real-world processes are mostly nonlinear. Linear system models are simplified representations of nonlinear processes. They are obtained through linearization about a predefined operating point and hence are valid for small variations around the point of linearization.

Having postulated a model, it becomes possible to investigate the time propagation of states through simulation, which is usually performed by solving an initial-value problem applying numerical integration procedures. The model responses are composed of different modes, some of which may be fast decaying and others may be slow requiring longer oscillation time. For stiff systems, that is, when two modes characterized by the smallest and the largest eigenvalues differ greatly, special integration algorithms are necessary. Fidelity of simulated model responses is determined by comparing them with the measured system outputs. Because measurements are likely to be corrupted by deterministic errors such as scale factor or bias as well as by stochastic noise, additional sensor models might be necessary. Such recorded measurements are usually available at discrete time points although the actual process evolves in continuous time.

For aircraft parameter estimation, state-space models describing the aircraft motion are generally necessary. In the field of flight mechanics, such models are invariably derived from the Newtonian mechanics; they lead to

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well-formulated kinematic equations pertaining to the equation of aircraft motion with translational and rotational degrees of freedom [1, 2]. The instantaneous aerodynamic forces and moments acting on a vehicle characterize the internal and external behavior. Because the kinematic equations are well defined, the efficacy of aircraft parameter estimation depends primarily on aerodynamic model postulates. Modeling of aerodynamic forces and moments is much more complex. The kinematic equations and aerodynamic models valid over the entire flight envelope are invariably nonlinear. We often make the assumption of rigid-body aircraft to simplify the model formulation. This has mostly been adequate in the past for the purpose of aircraft parameter estimation. For flight mechanic simulations, even for large transport aircraft, the influence of structural deformations can be described in a quasi-steady manner through a dynamic pressure-dependent flex factor, provided the rigid-body and structural frequencies are sufficiently separated. Thereby, the complex generalized equations of motion are usually avoided. More recently, the focus has been on developing and validating from flight data the integrated model of flexible aircraft. The order of such integrated models that characterize rigid-body and structural dynamics is naturally much higher. Usually, only salient dynamic effects due to structural modes closer to the rigid-body frequency are retained in the model to facilitate simulation, flight mechanics investigations, and control system design.

In this chapter, we mainly concentrate on the general formulation of the state-space models, which are amenable to time-domain methods that we will discuss in great detail in the following chapters. It is not the goal of this chapter to provide a treatment on aircraft equations of motion; these equations are found in any standard text book on flight mechanics. They are also provided for the examples covered in this book at respective places. Specifically, we look here at the various forms including linear and nonlinear models and at the extensions necessary to deal with practical issues such as mixed continuous/discrete system representation, initial conditions, deterministic bias errors in the measurements, and simultaneous evaluation data recorded from multiple experiments. A special emphasis is placed on the treatment of time delays. Those can result either from the instrumentation system or can be due to the internal behavior of the system. Finally, commonly applied numerical integration techniques are briefly discussed. Based on typical results, some general recommendations are made for the adequate choice of integration method to estimate aerodynamic derivatives from measured flight data.

3.2 MODEL DESCRIPTION

In a general case, the mathematical model of a process in state space is given by [3–5]

$$\dot{x}(t) = f[x(t), u(t), \beta] + F(\lambda)\, w(t), \qquad x(t_0) = x_0 \tag{3.1}$$

$$y(t) = g[x(t), u(t), \beta] \tag{3.2}$$

where x is the $n_x \times 1$ column vector of state variables, u the $n_u \times 1$ control input vector, y the $n_y \times 1$ system output vector, and β the $n_q \times 1$ vector of system parameters. The n_x and n_y dimensional system functions f and g are general nonlinear real valued functions. These system functions are assumed to have sufficient differentiability to be able to invoke Taylor series expansion. Besides the deterministic control input u, the system is also excited by stochastic input, called process noise $w(t)$, a $n_w \times 1$ column vector, which is usually nonmeasurable. In our investigations, we assume $n_w = n_x$. The process noise is usually assumed to be a zero-mean white Gaussian noise with an identity power spectral density. The matrix F represents the additive process (state) noise distribution matrix.

Because it is not possible to measure the system parameters β, they have to be estimated from the discrete measurements $z(t_k)$ of the system outputs $y(t_k)$. Because measurements are invariably corrupted by noise, the output equation can be formulated as

$$z(t_k) = y(t_k) + G\,v(t_k) \tag{3.3}$$

where k is the discrete time index and $v(t_k)$ the $n_v \times 1$ measurement noise vector. Throughout our discussion, we consider $n_v = n_y$. The measurement noise is assumed to be characterized by a sequence of independent Gaussian random variables with zero mean and identity covariance. The matrix G represents the additive measurement noise distribution matrix.

It is assumed that besides the system parameters β and the elements λ of process noise distribution matrix F, the initial conditions x_0 are also unknown. Accordingly, the unknown parameter vector is given by

$$\Theta = [\beta^T \quad \lambda^T \quad x_0^T]^T \tag{3.4}$$

The measurement noise distribution matrix G is also unknown. However, estimation of G, or alternatively the covariance matrix R $(=GG^T)$, is treated separately. This will be elaborated in more details in Chapters 4 and 5. For the present, it suffices to deal with the unknown parameter vector of Eq. (3.4). In our particular case of flight vehicle system identification, the system parameters β correspond to the stability and control derivatives or other parameters modeling aerodynamic forces and moments.

3.3 EXTENSIONS OF THE MATHEMATICAL MODELS

In the preceding section, it was assumed that measurements of control variables u are error free and that the output measurements y are corrupted by Gaussian noise only. In practice, however, the measurements of both of these variables contain systematic errors. These measurement errors are unavoidable, and we have to account for them in the parameter estimation through the postulated model. Let us denote these so-called zero shifts (also termed measurement biases or offsets) in the control variables u and output variables z as Δu and

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Δz, respectively. The zero shifts are treated as constant over the period of observation and are usually unknown. They represent, of course, very simple sensor models; in reality, the sensor models can be much more complex.

Considering the zero shifts in the control and output variables, the general system representation of Eqs. (3.1–3.3) can be rewritten as

$$\dot{x}(t) = f[x(t), u(t) - \Delta u, \beta] + F(\lambda)\, w(t), \quad x(t_0) = x_0 \tag{3.5}$$

$$y(t) = g[x(t), u(t) - \Delta u, \beta] \tag{3.6}$$

$$z(t_k) = y(t_k) + \Delta z + G v(t_k) \tag{3.7}$$

The unknown parameter vector in this case is given by

$$\Theta = [\beta^T \;\; \lambda^T \;\; x_0^T \;\; \Delta u^T \;\; \Delta z^T]^T \tag{3.8}$$

Besides the n_q number of unknown system parameters β and λ, we have $(n_x + n_u + n_y)$ initial conditions and zero shifts. In practice, it might not always be possible to estimate all of the initial conditions x_0 and the zero shifts Δu and Δz independently because of high correlation. For nonlinear systems, it is essential to estimate the initial condition x_0. The linear dependence of zero shifts can be determined from the estimation error correlation matrix, which we will study in Sec. 4.17. Theoretical investigation of parameters' identifiability is difficult for nonlinear systems. However, a more detailed treatment of this important issue is presented in Sec. 3.5.1 considering linear system models.

Yet another extension of the model represented in Eqs. (3.5–3.7) results from the practical limitation of performing experiments. A large number of experiments might have to be carried out separately to enable estimation of a single set of system parameters. A typical example pertains to estimation of derivatives pertaining to the lateral-directional motion. Even when we restrict our attention to a simple model valid for a single trim point, the lateral-directional motion consisting of rolling, yawing, and sideslipping requires aileron and rudder control inputs. Although it would be possible to apply aileron as well as rudder inputs simultaneously, it is preferable to carry out these maneuvers separately. In general, to aid identifiability of parameters, in many cases only one control input is varied keeping the other controls fixed, and if it is not possible to keep the other controls fixed, at least minimize the variations in them. Furthermore, many of the stability and control derivatives are functions of angle-of-attack or angle-of-sideslip. To estimate such dependencies and other nonlinear effects, it will be necessary to perform flight experiments at different trim points and excite the dynamic motion about each axis separately.

Thus, a capability to process simultaneously multiple experiments (flight maneuvers) containing different information to estimate a single set of parameters is a necessity that arises out of practice. The initial conditions on the state variables, that is, trim conditions, vary from maneuver to maneuver. Theoretically, the zero shifts Δu and Δz should be independent of the experiments because

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they represent systematic errors in the sensors; as long as the same sensors and recording system are used, these errors should be constant. However, in reality due to temperature effects, vibrations, differences in laboratory and flight calibrations, they vary from experiment to experiment, albeit the variations might be small. Therefore, in a most general case, we have to account for the initial conditions and systematic errors separately for each experiment. In this case the system representation of Eqs. (3.5–3.8) can be extended as follows:

$$\dot{x}(t) = f[x(t), u(t) - \Delta u(\gamma_l), \beta] \quad x(t_0) = x_0(\alpha_l) \qquad l = 1, 2, \ldots, n_E \tag{3.9}$$

$$y(t) = g[x(t), u(t) - \Delta u(\gamma_l), \beta] \tag{3.10}$$

$$z(t_k) = y(t_k) + \Delta z(\delta_l) + Gv(t_k) \tag{3.11}$$

where α_l, γ_l, and δ_l represent respectively the unknown components of x_0, Δu, and Δz for the lth experiment, and n_E is the number of experiments (time segments) to be analyzed simultaneously.

The complete unknown parameter vector in this case is given by

$$\Theta = [\beta^T \ \alpha_1^T \ \alpha_2^T \ldots \alpha_{nE}^T \ \ \gamma_1^T \ \gamma_2^T \ldots \gamma_{nE}^T \ \ \delta_1^T \ \delta_2^T \ldots \delta_{nE}^T]^T \tag{3.12}$$

We notice here that the process noise matrix $F(\lambda)$ has been dropped from the system representation in Eq. (3.9). Theoretically, it would be possible to include λ separately for each of the n_E experiments. However, treatment of separate distribution matrices for multiple experiments leads to some algorithmic difficulties that we will address in more detail in Chapter 5 dealing with estimation accounting for process noise. If we assume that the process noise distribution matrix F remains the same, then the elements λ can be easily included in the vector of unknown parameters of Eq. (3.12), just like the system parameter β. This is considered as a viable approach because in many of the cases, we analyze multiple experiments carried out under similar atmospheric conditions. Moreover, the estimates of system parameters are relatively insensitive to exact values of the process noise distribution matrix.

The necessity of the preceding capability to evaluate multiple experiments simultaneously was briefly discussed in Sec. 2.4. This capability relaxes the demands on performing flight testing in any particular sequence. Concatenation of arbitrary time segments, however, at first glance will show discrete jumps in the time scale. But these jumps are only in the graphical plots of time histories; they do not affect the computational results. The numerical procedure based on the preceding extended model formulation treats each time segment separately, enabling estimation of initial trim condition separately and a single set of aerodynamic parameters common to all of the maneuvers analyzed.

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3.4 RETARDED SYSTEMS

Modeling of physical phenomena requires in many cases accounting for time delays. Those can result either from measurement and recording systems or in our specific case of aerodynamic modeling as transit time effects. First, let us consider a case of calibrating a five-hole probe for flow angles mounted on a nose boom, in particular the differential pressure for the angle of attack, although any arbitrary variable would have served the purpose here of explaining model formulation. For the recorded variable corresponding to the differential pressure $p_{d\alpha}$, let us consider a typical measurement equation of the form:

$$p_{d\alpha m}(t) = K_\alpha \; p_{\text{dyn}}(t) \; \alpha_{\text{nb}}(t) + \Delta p_{d\alpha} \tag{3.13}$$

where the subscript m refers to the measurements, that is, $p_{d\alpha m}$ is the measured differential pressure, α_{nb} the computed angle of attack at the nose boom (model output), p_{dyn} the computed dynamic pressure, K_α the unknown scale factor, and $\Delta p_{d\alpha}$ the unknown bias in the measurement. Note that the variables α_{nb} and p_{dyn} appearing on the right-hand side of Eq. (3.13) are obtained from the model (state) variables estimated through flight-path reconstruction technique, which will be discussed in Chapter 10.

To account for the time delays, we reformulate Eq. (3.13) as follows:

$$p_{d\alpha m}(t) = p_{d\alpha C}(t - \tau_\alpha) + \Delta p_{d\alpha} \tag{3.14}$$

where τ_α denotes the time delay in the measured variable and $p_{d\alpha C}$ is the computed quantity, which is basically the same as that given by the first term on the right-hand side of Eq. (3.13), namely, $p_{d\alpha C}(t) = K_\alpha \; p_{\text{dyn}}(t)\alpha_{\text{nb}}(t)$. Equation (3.14) implies that current measurement $p_{d\alpha m}$ (on the left-hand side) at time t is a function of the past variable $p_{d\alpha C}$ at time $t - \tau_\alpha$. In other words, the computed variable is to be time delayed to match the measured data, which is the same as saying that the measurements of $p_{d\alpha}$ contains time delay. As postulated in Eq. (3.14), the time delays τ always come out positive, which is consistent with the physical interpretation of the phenomenon. For causal systems, negative time delays are not physically meaningful because that would then imply anticipating the future system characteristics ahead of time.

The second example that we consider pertains to modeling of transit time effect resulting from internal system behavior. As a specific case pertaining to aircraft motion, multipoint aerodynamic models considering, for example, the wing-body combination and horizontal tail separately include such transit time effects to model the time required for flow variations generated at the wing to reach the tail [1, 6]. Such a transit time is a function of the forward speed and hence variable. This aerodynamic phenomenon is commonly termed as downwash lag effect. From flight mechanics we know that the angle of attack at the horizontal tail α_H can be modeled as [1, 6, 7]

$$\alpha_H = \alpha + i_H - \varepsilon_H + \alpha_{\text{dyn}} \tag{3.15}$$

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where α is the angle of attack at the wing, i_H the horizontal-tail trim angle, ε_H the downwash angle at the tail, and α_{dyn} the dynamic angle of attack. The downwash and the lag effect can be modeled as

$$\varepsilon_H = \frac{\partial \varepsilon_H}{\partial \alpha}\,\alpha(t-\tau) + \frac{\partial \varepsilon_H}{\partial C_T}\,C_T(t-\tau) \tag{3.16}$$

where $\partial\varepsilon_H/\partial\alpha$ and $\partial\varepsilon_H/\partial C_T$ denote the unknown downwash parameters, C_T the thrust coefficient, and $\tau = r_H/V$ the transit time, where r_H is the tail length (that is, the horizontal distance between the neutral points of wing and horizontal tail) and V the airspeed. Equation (3.16) models the downwash generated due to angle-of-attack variations and also due to the thrust changes for wing-mounted engines. Other wing-mounted control devices such as direct-lift-control flaps or speed brakes also contribute to downwash and might have to be included in the model of Eq. (3.16). An example of accounting for transit time delay in the estimation from flight data using such a model is presented in Chapter 12.

Having brought out the need to account for time delays, we now turn our attention to possible ways to account for them in the estimation procedure. In general, there are three different ways:

1. Data preprocessing
2. First-order lag or Padé approximation
3. Delay array

The first option, called data preprocessing, is a fairly simple and commonly applied procedure, which is a part of the flight data reading process. Being a data preprocessing step, no changes to the estimation algorithm are necessary; the only program changes that might be necessary are for the part that reads the flight data; see Fig. 3.1. It enables time shifting variables through a prespecified fixed value in seconds, usually multiples of the sampling time, although through linear interpolation even time shifts of partial sampling time can be done. Time shifting the data in both directions, that is, for positive and negatives values, is theoretically feasible. If the recorded length is long enough, selection of a portion of time segment for further analysis after time shifting does not pose any problem. If recorded data are short

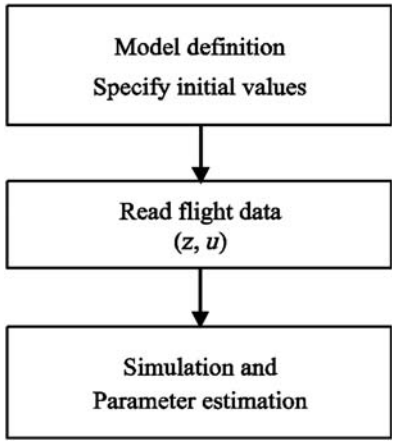

Fig. 3.1 General flowchart of data processing.

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in length, for example, just the maneuver part, then at respective ends time shifting might not be possible; in such a case the time segment for data analysis will have to be reduced by a few data points. However, this is not a major limitation because usually longer records are available. The approach is necessarily limited to directly measured variables and can be used to time shift the measurements of the specified output, control, and state variables prior to estimation. The time delays in various channels can be determined from flight-path reconstruction techniques discussed in Chapter 10 or through elaborative data recording system with an option to time stamp each channel. In any case, a time reference is needed to perform the time synchronization and also for estimation using other methods. This is provided by the channel that is recorded fastest with a minimum of time delay. All other channels are shifted relative to the time frame of this best available signal, which in many cases happens to be linear accelerations measured by dedicated accelerometers and stored directly.

The second option to account for the time delays is through an approximation by a first-order lag [8]. To illustrate the approach, let us consider the following equation for forward speed u:

$$\dot{u} = X_0 + X_u u + X_w w - qw + rv - g\sin\theta + \frac{F_e}{m}\cos\sigma_T \tag{3.17}$$

To introduce now time delay in the state variable u, we introduce an additional state variable u_ℓ and model the same as follows:

$$\dot{u}_\ell = (u - u_\ell)/\tau_u \tag{3.18}$$

It is apparent that Eq. (3.18) models a first-order lag effect. Thus, the new state variable u_ℓ is time delayed through τ_u. Having extended the state model, we now compare the time delayed u_ℓ obtained through integration of the state equation, and not the u, with the measured variable of forward speed. The preceding approach is simple and does not require any sophisticated estimation program capabilities. Although this approach is adopted by some analysts, it is not the most efficient one because 1) an additional first-order differential equation of the form of Eq. (3.18) is required for each variable to be time shifted, leading to larger computational overhead; computational time is directly proportional to the number of state variables that have to be integrated and 2) some dynamic effects, albeit small, are introduced due to the approximation. For a complex aerodynamic model, the increase in computational overhead due to additional state equations for time delays might not be too large because, as it will be demonstrated in Sec. 4.18, the major burden is usually to compute the right-hand sides of the state equations containing forces and moments.

The third option uses an array to generate time delay during the estimation procedure in any specified variable. It is much more complex, but provides a more accurate representation of time shift phenomenon. Separate work space arrays are required for each variable to be time shifted. Because it is a part of the computational procedure, time delays can be treated as unknown parameters

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and estimated. It is much more flexible than the other two options because it enables identification of time delays in any arbitrary variable (directly measured or computed in the model). However, this option requires an estimation program capable of handling general nonlinear systems and a special procedure (subroutine/function) to generate the delay [9, 10]. The system model of Eqs. (3.9–3.11) can be modified to account for time delays as follows:

$$\dot{x}(t) = f[x(t), X(t, \tau), u(t) - \Delta u(\gamma_l), U(t, \tau), \beta] \quad x(t_0) = x_0(\alpha_l) \quad l = 1, 2, \ldots, n_E \tag{3.19}$$

$$y(t) = g[x(t), X(t, \tau), u(t) - \Delta u(\gamma_l), U(t, \tau), \beta] \tag{3.20}$$

$$z(t_k) = y(t_k) + \Delta z(\delta_l) + Gv(t_k) \tag{3.21}$$

where x, u, y, and z are as before the state, input, output, and measured variables, β the unknown system parameters, and Δu and Δz the zero shifts. The terms $X(t, \tau)$ and $U(t, \tau)$ denote the time-delayed state and input variables:

$$[X(t, \tau)]_{ij} = x_i(t - \tau_j) \quad \text{and} \quad [U(t, \tau)]_{ij} = u_i(t - \tau_j) - \Delta u(\gamma_l) \tag{3.22}$$

The complete unknown parameter vector in this case is given by

$$\Theta = \left[\beta^T \;\; \tau^T \;\; \alpha_1^T \; \alpha_2^T \ldots \alpha_{nE}^T \;\; \gamma_1^T \; \gamma_2^T \ldots \gamma_{nE}^T \;\; \delta_1^T \; \delta_2^T \ldots \delta_{nE}^T\right]^T \tag{3.23}$$

where τ denotes the vector of time delays. It is obvious that time delays can be included in any arbitrarily selected variables. The presence of time delays affects the simulation and consequently parameter estimation as well. They must be accounted for to obtain accurate and reliable estimates [9, 11].

In the extended version of the software, which is a part of the book, we provide a utility function "timeDelay.m," which is based on the delay matrix appearing in Eq. (3.22). The functional call is of the form:

$$[rTz,\ xWS,\ tNewX,\ iNewX] = timeDelay(rTz,\ tDelay,\ xWS,\ tNewX,\ iNewX,\ nTdMx)$$

where *rTz* is the variable to be shifted; *tDelay* the time delay; *xWS*, *tNewX*, *iNewX* the auxiliary work space arrays and indices; and *nTdMx* the maximum time delay as a multiple of the sampling time. These work space arrays need to be defined as global arrays in the calling function and have to be defined uniquely for each signal to be time shifted. In other words, we need to call the preceding function with its own set of work space arrays for each signal to be time shifted. The use of this function is demonstrated in Sec. 4.20.1.3 on a simple test case and also in Chapter 10 for a case with more than one time-delayed variables.

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3.5 LINEARIZED MODELS

We are primarily concerned in this book with general nonlinear systems as represented in the preceding sections. In a few specific cases, however, we can consider simplified linear system representations. As already pointed out in Sec. 3.1, such linear models are valid over small variations about the operating point. The system equations (3.1) and (3.2) can be linearized about some suitable operating point, leading to

$$\dot{x}(t) = Ax(t) + Bu(t) + Fw(t), \quad x(t_0) = x_0 \tag{3.24}$$

$$y(t) = Cx(t) + Du(t) \tag{3.25}$$

where the system matrices A, B, C, and D are respectively denoted by

$$A = \frac{\partial f[x, u, \beta]}{\partial x}, \quad B = \frac{\partial f[x, u, \beta]}{\partial u}$$
$$C = \frac{\partial g[x, u, \beta]}{\partial x}, \quad D = \frac{\partial g[x, u, \beta]}{\partial u} \tag{3.26}$$

The unknown system parameters β appearing in Eqs. (3.1) and (3.2) now appear as elements of these systems matrices, either directly or as their linear effects. The linearization point is usually chosen to be the trim point or estimated initial conditions. The measurement equation (3.3) is the same in this case and hence not repeated here. The state, input, and output variables appearing in Eqs. (3.24) and (3.25) are the perturbations around the trim conditions.

The formulation of Eqs. (3.5) and (3.6) accounting for the measurement biases, Δu and Δz, leads in the present case to

$$\dot{x}(t) = Ax(t) + B[u(t) - \Delta u] + Fw(t), \quad x(t_0) = x_0 \tag{3.27}$$

$$y(t) = Cx(t) + D[u(t) - \Delta u] + \Delta z \tag{3.28}$$

For convenience, we have now equivalently included the measurement bias in Eq. (3.28) for y instead of that for z. Once again, the system of Eqs. (3.27–3.28) contain $(n_x + n_u + n_y)$ constant parameters consisting of biases and initial conditions. In contrast to the nonlinear systems discussed in Sec. 3.3, linear model postulates are directly amenable to theoretical analysis using linear system theory.

3.5.1 IDENTIFIABILITY OF AERODYNAMIC DERIVATIVES AND CONSTANT PARAMETERS

Identifiability of parameters is closely related to the concepts of observability and controllability [12]. In our case, we have to consider this issue separately for the system parameters (aerodynamic derivatives) and for the constant terms (initial conditions and biases). Identifiability of aerodynamic derivatives is determined by the information content in the data being analyzed. Unique component due to each derivative must be available in the response data to enable its estimation

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accurately. We have covered in Chapter 2 some of the important aspects of performing flight maneuver for this purpose. Concatenation of different flight maneuvers, as discussed therein and in Sec. 3.3, may be necessary in most of the cases. Thus, the first part of this general issue is directly linked to the experiment.

With regard to the identifiability of constant parameters comprising measurement biases in the output and input variables and initial conditions on the state variables, we treat linear and nonlinear models differently. In Sec. 3.3, it has already been pointed out that in the case of nonlinear models, such as those given by Eqs. (3.5–3.8), initial conditions have to be estimated; we will also address this issue in Sec. 3.7 once again. Regarding the other bias parameters, because of correlation, it might not be possible to estimate all of them, and the theoretical analysis being difficult we choose to make use of the estimation error correlation matrix to decide upon the appropriate choice of biases that can be estimated. On the other hand, linear models can be analyzed readily applying the conventional observability analysis to determine the exact number of constant parameters that can be theoretically estimated. In Appendix B, it has been demonstrated that for a linear model represented in Eqs. (3.27) and (3.28), a maximum of $(n_x + n_y)$ terms can be determined independently and not $(n_x + n_u + n_y)$ as ideally desired.

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3.5.2 MODELS WITH LUMPED BIAS PARAMETERS

Besides the just-discussed limitation of being able to estimate only a subset of constant zero shifts and initial conditions, the linear model extension of Eqs. (3.27) and (3.28) is plagued with yet another undesirable effect. It leads to a model containing terms $B \cdot \Delta u$ and $D \cdot \Delta u$. Thus, the model is still linear in the state and control variables, but not in the parameters being estimated. This affects the convergence of the parameter estimation through optimization of some suitably defined cost function.

To overcome both these difficulties, we reformulate Eqs. (3.27) and (3.28) using the transformation $x^* = x - x_0$. It can be easily verified that the substitution of $x = x^* + x_0$ in these equations followed by simple manipulation, and agreeing, for convenience, to still label x^*, the transformed x, as x, leads to an equivalent system representation of

$$\dot{x}(t) = Ax(t) + Bu(t) + b_x + Fw(t), \quad x(t_0) = 0 \tag{3.29}$$

$$y(t) = Cx(t) + Du(t) + b_y \tag{3.30}$$

The constant terms x_0, Δu, and Δz now appear equivalently as lumped parameters $b_x(=Ax_0 - B\Delta u)$ and $b_y(=Cx_0 - D\Delta u + \Delta z)$. The transformed system is now linear in all of the parameters. The initial conditions for simulation reduce to zero and, hence, are no more unknown and as such not to be estimated. We now have exactly $(n_x + n_y)$ constant bias terms. Other transformations, such as $x^* = x - A^{-1} B\Delta u$, are also possible, but the one just elaborated is found to be

more efficient. Based on the aspects presented in the preceding and this section, whenever we wish to estimate parameters of a linear system, we will formulate our equations according to Eqs. (3.29) and (3.30).

3.5.3 NUMERICAL APPROXIMATION OF SYSTEM MATRICES

Computing the derivatives of a function analytically is the most accurate approach, but not that convenient, particularly when dealing with complex and nonlinear systems. Numerical approximations are more convenient in such applications and found to work quite satisfactorily, provided we pay attention to the limitations of these approaches. Such procedures to numerically approximate functional derivatives or response gradients are commonly used to derive system matrices or in optimization algorithms.

The elements of the system state and control input matrices, A and B, are approximated using the central difference formula given by

$$A_{ij} \approx \frac{f_i[x + \delta x_j\, e^j,\, u,\, \beta] - f_i[x - \delta x_j e^j,\, u,\, \beta]}{2\delta x_j}, \quad j = 1, 2, \ldots, n_x \tag{3.31}$$

and

$$B_{ij} \approx \frac{f_i[x,\, u + \delta u_j e^j,\, \beta] - f_i[x,\, u - \delta u_j e^j,\, \beta]}{2\delta u_j}, \quad j = 1, 2, \ldots, n_u \tag{3.32}$$

where e^j is a column vector with one in the jth row and zeros elsewhere and δx_j and δu_j are small perturbations in each of the n_x states and n_u control variables. The matrices C and D are similarly approximated from the observation function g. The truncation error of the centered formulas of Eqs. (3.31) and (3.32) is of the order of $\mathcal{O}(h^2)$, where h is the perturbation. A proper choice of step size is critical to obtain valid and accurate approximations because on one hand these increments have to be small enough to give valid approximation to the derivative, whereas on the other if they are too small then roundoff errors may adversely affect the approximation. Some tradeoff between roundoff and truncation error is usually necessary. We will address this aspect in some more details in Sec. 4.8.

The higher-order centered formula having truncation error of the order of $\mathcal{O}(h^4)$ is given by [13]

$$A_{ij} \approx \frac{-f_i[x + 2\delta x_j e^j] + 8 f_i[x + \delta x_j e^j] - 8 f_i[x - \delta x_j e^j] + f_i[x - 2\delta x_j e^j]}{12\delta x_j} \tag{3.33}$$

For notational simplicity, we have dropped the arguments u and β in Eq. (3.33). The advantage of the preceding fourth-order formula is that the truncation error goes to zero faster than that of the second-order formula. Hence, larger step sizes

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are possible. The main disadvantage is that it requires four function evaluations. For a detailed error analysis of the numerical approximation, the reader is referred to any standard textbook on numerical computations [13].

3.6 PSEUDO CONTROL INPUTS

During the 1970s, a good number of estimation programs were developed based on the maximum likelihood estimation that we will address in the next two chapters. Although these programs provided capabilities to estimate linear stability and control derivatives, analysis of flight data at high angles of attack and at extreme flight conditions required capabilities to handle nonlinear terms such as w^2, V^2, α^2, $\alpha\delta_e$, qu, or $|\delta_a|$. The approach that was adopted then was that of the pseudo control inputs [14–16]. In this case, the control input vector u was augmented by additional terms, namely, with those corresponding to the nonlinear terms, but now computed prior to estimation using the measured variables. These computed nonlinear terms are then treated as additional pseudo-inputs and lead to a system that is linear in these derived inputs.

The approach of pseudo control inputs can also be used to reduce the model size and yet account for the effects of other motion variables that are not treated as state variables. For example, in the decoupled model for longitudinal motion with the forward speed, vertical speed, pitch rate, pitch attitude and pressure altitude, (u, w, q, θ, h) as state variables, and elevator deflection δ_e as input, we can treat the variables pertaining to the lateral-directional motion as pseudo inputs. The vice versa is, of course, true whereby we can analyze the lateral-directional motion considering the typical variables of longitudinal motion, say, angle of attack, as pseudo inputs. The approach of pseudo control inputs, also applicable to linear models, relaxes the demands on the performing flight maneuvers with minimum variations of the other mode of motion.

To illustrate the basic concept of pseudo control inputs, we consider the following state equation:

$$\dot{u} = X_0 + X_u u + X_w w + X_{w^2} w^2 - qw + rv - g\sin\theta + \frac{F_e}{m}\cos\sigma_T \tag{3.34}$$

For the purpose at hand, without explaining each variable, it would suffice to make a note that Eq. (3.34) pertains to forward speed u in terms of the dimensional derivatives $X_{()}$ and involves nonlinear terms w^2, qw, rv, and $g\sin\theta$. If we compute these nonlinear terms prior to estimation using the measured values of w, q, r, v, and θ, and denote them as σ_1, σ_2, σ_3, and σ_4 respectively, then Eq. (3.34) can be rewritten as

$$\dot{u} = X_0 + X_u u + X_w w + X_{w^2}\sigma_1 - \sigma_2 + \sigma_3 - \sigma_4 + \frac{F_e}{m}\cos\sigma_T \tag{3.35}$$

Equation (3.35) is clearly linear in all of the variables.

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In this approach the basic linear representation is retained; therefore, the estimation programs capable of handling linear models can be applied without any modifications. However, the measurement errors in the variables from which the nonlinear terms are computed propagate during estimation. The approach, therefore, yields biased estimates in the presence of noise and measurement errors. Moreover, it is limited to nonlinearities in variables for which the measurements are available. We do not pursue this approach anymore because we will arrive at estimation techniques that can handle general nonlinear systems more efficiently.

An alternative approach based on the so-called state vector augmentation is also possible to transform a nonlinear model into an equivalent linear model to which an estimation program for linear systems could be applied [17]. It is applicable when the nonlinearities are in terms of the state variables or when measured state variables contain significant noise and modeling errors, making the use of pseudo controls questionable. In this approach the state vector is augmented through the nonlinear terms. Although the approach is more justified, it often involves modifications of the estimation program. These changes are required each time to incorporate the new sensitivity equations resulting from different nonlinear terms considered. The approach has been rarely used in practice, mainly because it is not flexible enough to investigate different nonlinear model postulates.

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3.7 TREATMENT OF INITIAL CONDITIONS

It has been clearly pointed out in Sec. 3.5 that for linear systems the model representation in terms of lumped bias parameters, Eqs. (3.29) and (3.30), is preferable, which reduces the initial conditions to be used in the simulation to zero. On the other hand, as mentioned in Sec. 3.3, for nonlinear models it is necessary to specify and/or estimate the initial conditions explicitly. Accordingly, they are treated as part of the unknown parameter vector. Generally, the convergence of x_0 has poor asymptotic properties because the information content is concentrated only at the beginning of the maneuver [2]. Loosely speaking, increasing the length of the segment does not necessarily add to the information useful for estimation of the initial conditions, at least for fast responding modes such as short period and Dutch roll. For slow motion modes, such as Phugoid, longer duration records will be necessary.

For large size models, for example, six-degrees-of-freedom aircraft motion with 10 states or combined rigid-body and dynamic models for reversible flight controls [18], requiring concatenation of multiple maneuvers leads to a large number of initial conditions, namely, $(n_E \cdot n_x)$. This can far exceed the number of aerodynamic derivatives in which we are primarily interested. Although identifiability of this large number of initial conditions is principally no problem, because each maneuver is treated separately and provides unique uncorrelated

information, it increases the computational time disproportionately. To reduce the computational burden, a pragmatic approach would be to define the initial conditions appropriately and keep them fixed. In practice, provided measurements of adequate variables are available, the initial conditions are set to the first data point or average of first few data points to reduce the effects of noise and other errors. Keeping appropriately defined initial conditions fixed during the first few iterations of the optimization procedure, and estimating the same in the last few iterations helps to speed up the process, the reduction in computational time can be significant for large-scale systems.

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3.8 SIMULATION

As defined in Sec. 1.1, simulation is the process of reproducing (in technical terminology computing) numerically the system response to a given input for a prespecified system model. At this stage it is irrelevant to think over the model quality. It is always possible to carry out a simulation of a physically existing process, or even of an unreal (anticipated) process. The model updates, fidelity, and adequacy are addressed in the subsequent chapters on parameter estimation and model validation. Computation of system responses calls for integrating the state equations, which is usually performed applying suitable numerical procedures. An adequate choice of the integration method is important for efficient and reliable parameter estimation.

In many of the engineering and scientific problems, it is convenient to solve the initial-value problem through numerical approximations because the exact solution might be theoretically possible but very complex or the explicit solution might not exist. Therefore, we focus in this section on numerically integrating first-order ordinary differential equations. Starting from the specified initial conditions and neglecting the process noise, numerical integration over one sampling period yields state variables at each successive discrete point given by

$$x(t + \Delta t) = x(t) + \int_{t}^{t+\Delta t} f[x(t), u(t), \beta]\, \mathrm{d}t \tag{3.36}$$

In our case, for stable aircraft a solution to Eq. (3.36) always exists. For unstable aircraft, integration of an open-loop plant may numerically diverge. How to deal with numerical divergence will be considered in Chapter 9 dealing specifically with unstable aircraft. Equation (3.36) applies to deterministic systems only. For a stochastic system of Eq. (3.1) or (3.5), with process noise $w(t)$ being nonmeasurable we need a state estimator consisting of a prediction step that is the same as Eq. (3.36) and a correction step to improve the predicted states based on the process noise distribution matrix F and measurements z. We will address this problem in Chapters 5 and 7 and study different approaches of steady-state and time-varying filters. The state estimator includes in the first step of prediction the numerical

integration methods that we apply to the deterministic system and that we will study in this section. It is also relevant to make a specific mention that for the purpose of parameter estimation here we are concerned only with off-line simulation techniques. Real-time simulation, as necessary, for example, in the flight simulators, calls for different implementation, although the basic algorithms might be similar.

3.8.1 NUMERICAL INTEGRATION METHODS

The numerical integration methods can be classified into four general categories [13, 19, 20]:

1. Taylor-series methods
2. Runge–Kutta methods
3. Multistep methods
4. Extrapolation methods

The multistep Adams–Bashforth–Moulton method is a two-step predictor-corrector procedure for higher accuracy requirements, but not suitable for our application because it is not self-starting. The extrapolation methods are particularly suitable for integrating over a large interval (time step) and when high precision is necessary. In the exercise that we are mainly concerned with, namely, estimation of aerodynamic model from flight data, the sampling times are typically 20 to 100 ms corresponding to sampling frequencies of 50 and 10 Hz. The accuracy requirements are moderate. A comparative study has demonstrated that in such cases the advantages of the extrapolation methods are not apparent, besides being slower than the conventional Runge–Kutta methods [21]. Accordingly, we will briefly cover the simplest of the Taylor-series method, namely, Euler's method, to understand the principle of numerical integration and then most commonly applied Runge–Kutta formulas.

The numerical techniques provide a large number of integration formulas. Table 3.1 gives those that have found application on a routine basis in the aircraft parameter estimation. The five algorithms provide the solution to the initial value problem for systems with first-order ordinary differential equations, including retarded systems, that is, those with time delays. Because integration is across two data points, different procedures have been adopted while using control input u for function evaluations. The simplest one is to use the same control input corresponding to the start of or that at the end of the interval for all of the function evaluations, that is, one at the beginning, between, and at the end of the interval. Another approach is to use an average of the inputs at the two discrete points, as denoted by $\overline{u}$ in Table 3.1. To achieve more accurate numerical results, it is also possible to use $u(t_k)$ and $u(t_{k+1})$ for function evaluations at the two ends of the interval and appropriately interpolated values (one-half, one-third, or two-thirds) between $u(t_k)$ and $u(t_{k+1})$ for the intermediate steps.

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TABLE 3.1 COMMONLY APPLIED NUMERICAL INTEGRATION FORMULAS

	Method	Function Evaluations	Solution
1	Euler method 1st order	$f_1 = f(x, u, \beta)$	$x(k+1) = x(k) + f_1 \Delta t$
2	Heun method 2nd order	$f_1 = f(x, u, \beta)$ $f_2 = f(x + f_1 \Delta t, \bar{u}, \beta)$	$x(k+1) = x(k) + (f_1 + f_2)\Delta t/2$
3	Runge–Kutta 2nd order	$f_1 = f(x, u, \beta)$ $f_2 = f\left(x + f_1 \frac{\Delta t}{2}, \bar{u}, \beta\right)$	$x(k+1) = x(k) + f_2 \Delta t$
4	Runge–Kutta 3rd order	$f_1 = f(x, u, \beta)$ $f_2 = f\left(x + f_1 \frac{\Delta t}{3}, \bar{u}, \beta\right)$ $f_3 = f\left(x + 2 f_2 \frac{\Delta t}{3}, \bar{u}, \beta\right)$	$x(k+1) = x(k) + [f_1 + 3f_3]\Delta t/4$
5	Runge–Kutta 4th order	$f_1 = f(x, u, \beta)$ $f_2 = f\left(x + f_1 \frac{\Delta t}{2}, \bar{u}, \beta\right)$ $f_3 = f\left(x + f_2 \frac{\Delta t}{2}, \bar{u}, \beta\right)$ $f_4 = f(x + f_3 \Delta t, \bar{u}, \beta)$	$x(k+1) = x(k) + [f_1 + 2f_2 + 2f_3 + f_4]\frac{\Delta t}{6}$

We will follow this procedure of interpolating the control inputs in the software that will be developed in Chapter 4.

The five methods listed in Table 3.1 are characterized by the number of function calls required to compute the solution at $x(t_{k+1})$ starting from the known (given) solution $x(t_k)$, where k is the discrete time point index. The Euler integration based on forward step requires one evaluation of the state derivative function f for each time point whereas the Heun and Runge–Kutta second-, third-, and fourth-order formulas require two, two, three, and four evaluations, respectively. These methods differ not only in the number of function calls, but in the accuracy of the solution. The errors at the end of the interval for these methods are of the order of $\mathcal{O}(h^2)$, $\mathcal{O}(h^3)$, $\mathcal{O}(h^3)$, $\mathcal{O}(h^4)$, and $\mathcal{O}(h^5)$, respectively, where h $(=\Delta t)$ is the integration interval.

The Euler method is the simplest one based on the gradient information at a single point (at the beginning of the integration interval); see Table 3.1. At each point just one function evaluation is necessary. It has limited accuracy and usage in parameter estimation because the errors get propagated as the process progresses in time. The error accumulation also depends upon the number of

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data points and on the integration interval; the smaller the sampling time, the slower is the error propagation and vice versa. Euler integration is rarely used particularly for longer duration flight maneuvers and for sampling times greater than 20 ms because of adverse error propagation.

The Runge–Kutta formulas with constant step size provided in Table 3.1 are most widely applied in the off-line simulation. They are adequate for moderate accuracy requirements, typical of rigid-body parameter estimation. Higher-order methods, possibly with step size control, will be useful for high-bandwidth models, such as those for helicopter rigid-body mode extended with rotor degrees of freedom. It requires five or more function evaluations per time point depending upon the accuracy required. The computational load is directly proportional to the number of function calls, and, hence, it is obvious that the Euler integration is the fastest and fourth or higher-order formula the slowest. On the other hand, higher-order integration formulas are more accurate. The higher the order of the formula, the more accurate is the approximation, but requires more function evaluations and computational time. Depending upon the complexity of the problem, we might have to make a tradeoff between speed and accuracy.

The choice of the integration method depends on the type of the system being investigated. Although the fourth Runge–Kutta method is generally recommended for rigid-body aircraft dynamics, this choice may be conservative [22]. As a typical example, estimation of derivatives pertaining to the longitudinal motion from a two-point aerodynamic model accounting for wing-body and tail separately is considered. As seen from Table 3.2, the second- and third-order methods yield equally acceptable results compared to the more time-consuming fourth- and fifth-order Runge–Kutta formulas. The Euler integration, although it appears to converge to almost the same minimum, which might just be a coincidence, does show some changes in the numerical values; the convergence was affected as evident from the increase in the number of iterations. Moreover, the performance deteriorated for sampling times greater than 20 ms, and propagation of errors was unacceptable, particularly when the flight maneuver was of longer duration. For systems characterized by higher-order dynamics, integration methods with step size control will be necessary to ensure that the errors are low and that they do not propagate adversely, which can affect the accuracy of the estimates and convergence of the optimization method.

For stiff systems, that is, for those where all eigenvalues of the linearized system matrix have a negative real part and the largest and the smallest differ a lot, special integration methods are necessary because the system is characterized by a fast part (corresponding to a large eigenvalue) and also by a slowly decaying part (corresponding to a small eigenvalue). Typically, we encounter such systems, for example, in the modeling of elasto-viscoplastic deformations of metallic material [22]. The Runge–Kutta algorithms in such cases encounter numerical problems and lead to unreliable results. Several algorithms that overcome these problems are available; Gear's method, also called backward differentiation formula, is one of them that is widely used [23, 24]. The primary limitation of

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TABLE 3.2 ESTIMATES OF AERODYNAMIC DERIVATIVES APPLYING DIFFERENT INTEGRATION METHODS

Method Parameter	Euler	2nd-order Runge–Kutta	3rd-order Runge–Kutta	4th-order Runge–Kutta	5th-order Runge–Kutta–Fehlberg
C_{L0}	0.0682	0.0770	0.0775	0.0775	0.0775
C_{D0}	0.0232	0.0296	0.0297	0.0297	0.0297
C_{m0}	−0.1732	−0.1731	−0.1732	−0.1732	−0.1732
$C_{L\alpha}$	5.536	5.443	5.437	5.438	5.437
e	0.646	1.038	1.049	1.048	1.048
$C_{L\delta e}$	1.436	1.427	1.433	1.434	1.433
C_{mqWB}	10.362	11.39	11.23	11.23	11.23
Det(R)*	$2.32^{*}10^{-9}$	$1.25^{*}10^{-9}$	$1.25^{*}10^{-9}$	$1.25^{*}10^{-9}$	$1.25^{*}10^{-9}$
Iterations	7	5	5	5	5

*Cost function defined as determinant of the residual covariance matrix R.

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such methods is that they are not applicable to systems with time delays. In general, special algorithms such as Gear's method for stiff systems work inefficiently when applied to nonstiff systems.

For estimation of rigid-body aerodynamic models, the second-order Runge–Kutta algorithm is usually adequate and hence recommended during the initial iterations of the iterative estimation algorithms, switching over to the fourth-order Runge–Kutta only during final iterations. Such sophistications lead to speeding up the overall estimation procedure for large-scale systems without affecting the final results. Here, we restrict ourselves to the use of a single algorithm throughout the iterative estimation procedure. Three utility functions "ruku2," "ruku3," and "ruku4" based on the second-, third-, and fourth-order Runge–Kutta formulas are provided in the software. In these functions for integration, apart from the formulas given in Table 3.1, a global flag "rk_IntStp" is set for intermediate step of the algorithm, and also the current time is assigned to global variable "tCur." These two quantities are required to properly account for the time when the time delay function discussed in Sec. 3.4 is called.

3.8.2 INTEGRATION OF LINEAR SYSTEMS

In the case of linear systems, as represented in Eq. (3.29), there are two approaches to integrate the state equation. They are as follows: 1) discretize the state equation and solve for x using the state transition matrix and 2) apply one of the numerical

integration methods covered in Sec. 3.8.1. Using the discrete-time theory of linear dynamic systems, it can be shown that discretization of Eq. (3.29) leads to

$$x(t_{k+1}) = \Phi\, x(t_k) + \Psi\, B\, \overline{u}(t_k) + \Psi\, b_x \tag{3.37}$$

where $\Phi = e^{A\Delta t}$ is the state transition matrix (also called matrix exponential), and its integral will then be given by $\Psi = \int_0^{\Delta t} e^{A\tau} \mathrm{d}\tau$, where $\Delta t (= t_k - t_{k-1})$ is the sampling time interval. For numerical computational purpose, the exponential of a matrix can be computed in many ways [25]. Here, we approximate Φ through the Taylor-series expansion of $e^{A\Delta t}$ given by

$$\Phi = e^{A\Delta t} \approx \mathrm{I} + A\,\Delta t + A^2\,\frac{\Delta t^2}{2!} + \cdots \tag{3.38}$$

and the integral of Eq. (3.38) is given by

$$\Psi = \int_0^{\Delta t} e^{A\tau} \mathrm{d}\tau \approx \mathrm{I}\,\Delta t + A\,\frac{\Delta t^2}{2!} + A^2\,\frac{\Delta t^3}{3!} + \cdots \tag{3.39}$$

Typically, 8 to 10 terms of the Taylor series in Eqs. (3.38) and (3.39) are found to be adequate for the accurate computational purposes.

Although the well-established state transition matrix approach was widely used in the parameter estimation programs capable of handling linear systems only, during the last few decades more sophisticated programs have been developed capable of handling nonlinear models. Such programs invariably integrate the state equations using numerical integration methods. Because the same estimation program is applied to linear models as well, they will then be integrated using the chosen numerical integration method. Detailed investigation performed to evaluate the differences between the two approaches clearly showed that they are equivalent and yield the same estimates within the numerical accuracy of number representation. In general, it is preferable to have a single estimation program capable of handling both types of models, and from this view point there is little choice but to go for the methods covered in Sec. 3.8.1.

3.9 CONCLUDING REMARKS

In this chapter we have postulated general state-space models and discussed several extensions to account for practical requirements of catering to 1) multiple experiments, 2) systematic errors in the measurements of control inputs and system outputs, and 3) time delays either in the direct output measurements or in the internal variables of the system. Issues related to identifiability of initial conditions and constant bias terms have been elucidated. The two approaches that were adopted in the past to account for nonlinear terms within the framework of linear estimation have been brought out, emphasizing the advanced

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estimation techniques that have evolved over the last few decades and, as will be covered in this book, that are more flexible and capable of handling the nonlinear systems directly. The equations of aircraft motion and aerodynamic model can be put within the framework of models postulated in this chapter. Hence, while developing the estimation methods in the next few chapters, model postulates as described in this chapter will form the basis to describe the flight vehicle, or dynamic system in general. Based on a brief discussion, it has been pointed out that multiple experiment analysis may involve estimation of a large number of initial conditions, far exceeding the actual number of aerodynamic derivatives in which we are mainly interested, and a pragmatic approach has been suggested to reduce the computational burden. Finally, different approaches to integrate both linear and nonlinear system have been presented. In general, the classical Runge–Kutta fourth-order formula is commonly used for parameter estimation from flight data. However, substantiated by typical results obtained applying the output-error method that will be studied in the next chapter, it is argued that even second- or third-order formulas may be sufficient in many cases. It leads to speeding up the overall process, which will be particularly useful while analyzing large-scale systems. Aspects related to integration of stiff systems and limitations of such algorithms have been brought out briefly.

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REFERENCES

[1] Etkin, B., *Dynamics of Atmospheric Flight*, Wiley, New York, 1972.

[2] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems – Applications to Aircraft Part 1: The Output Error Approach," AGARD, AG-300, Vol. 3, Part 1, Dec. 1986.

[3] Jategaonkar, R. V., and Plaetschke, E., "Maximum Likelihood Parameter Estimation from Flight Data for General Nonlinear Systems," DFVLR-FB 83-14, March 1983.

[4] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[5] Jategaonkar, R. V., and Plaetschke, E., "Algorithms for Aircraft Parameter Estimation Accounting for Process and Measurement Noise," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 360–372.

[6] Mönnich, W., "Ein 2-Punkt-Aerodynamikmodell für die Identifizierung," *Proceedings of the Symposium on "Systemidentifikation in der Fahrzeugdynamik,"* DFVLR-Mitt. 87-22, Paper No. 3.1, Nov. 1987 (in German).

[7] Jategaonkar, R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DFVLR-FB 90-40, July 1990.

[8] Mackie, D. B., "A Comparison of Parameter Estimation Results from Flight Test Data Using Linear and Nonlinear Maximum Likelihood Methods," DFVLR-FB 84-06, Dec. 1983.

[9] Plaetschke, E., "Ein FORTRAN-Programm zur Maximum-Likelihood-Parameterschätzung in nichtlinearen retardierten Systemen der Flugmechanik – Benutzeranleitung," DFVLR-Mitt. 86-08, Feb. 1986 (in German).

[10] Weiß, S., "NLHP1L: Ein Programm zur Maximum-Likelihood-Parameter-Schätzung für Nichtlineare Systeme – Benutzeranleitung," DFVLR IB 111-87/29, July 1987 (in German).

[11] Blackwell, J., and Feik, R. A., "Identification of Time Delays in Flight Measurements," *Journal of Guidance, Control, and Dynamics*, Vol. 14, No. 1, 1991, pp. 132–139.

[12] Gelb, A., *Applied Optimal Estimation*, MIT Press, Cambridge, MA, 1974.

[13] Mathews, J. H., *Numerical Methods for Mathematics, Science, and Engineering*, 2nd ed., Prentice–Hall, Englewood Cliffs, NJ, 1992.

[14] McBrinn, D. E., and Brassell, B. B., "Aerodynamic Parameter Estimation for the A-7 Airplane at High Angles of Attack," *Proceedings of the AIAA 3rd Atmospheric Flight Mechanics Testing Conference*, AIAA, New York, 1976, pp. 108–117.

[15] Maine, R. E., "Aerodynamic Derivatives for an Oblique Wing Aircraft Estimated from Flight Data by Using Maximum Likelihood Technique," NASA TP 1336, Oct. 1978.

[16] Ross, A. J., "Identification Experience in Extreme Flight," AGARD, LS-104, Paper No. 8, Nov. 1979.

[17] Iliff, K. W., "Maximum Likelihood Estimation of Lift and Drag from Dynamic Aircraft Maneuvers," *Journal of Aircraft*, Vol. 14, No. 12, 1977, pp. 1175–1181.

[18] Jategaonkar, R. V., and Mönnich, W., "Identification of Do-328 Aerodynamic Database for a Level D Flight Simulator," AIAA Paper 97-3729, Aug. 1997.

[19] Forsythe, G. E., Malcolm, M. A., and Moler, C. B., *Computer Methods for Mathematical Computations*, Prentice–Hall, Englewood Cliffs, NJ, 1977.

[20] Hairer, E., Nørsett, S. P., and Wanner, G., *Solving Ordinary Differential Equations I – Nonstiff Problems*, Springer-Verlag, Berlin, 1987.

[21] Jategaonkar, R. V., and Plaetschke, E., "Non-Linear Parameter Estimation from Flight Test Data Using Minimum Search Methods," DFVLR-FB 83-15, March 1983.

[22] Jategaonkar, R. V., and Thielecke, F., "Aircraft Parameter Estimation – A Tool for Development of Aerodynamic Databases," *Sadhana*, Indian Academy of Sciences, Vol. 25, Part 2, April 2000, pp. 119–135.

[23] Gear, C. W., *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Inc., Englewood Cliffs, NJ, 1971.

[24] Radhakrishnan, K., and Hindmarsh, A. C., "Description and Use of LSODE, the Livermore Solver for Ordinary Differential Equations," NASA RP 1327, Dec. 1993.

[25] Moler, C., and van Loan, C., "Nineteen Dubious Ways to Compute the Exponential of a Matrix, Twenty-Five Years Later," *SIAM Review*, Vol. 45, No. 1, 2003, pp. 3–49.

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CHAPTER 4

Output-Error Method

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4.1 INTRODUCTION

Having dealt in the preceding chapter with the formulation of nonlinear state-space model postulates to define the cause-effect relationship purported to underlie the physical phenomenon, and having elaborated on the simulation (that is, computation of system responses) using numerical procedures, we now turn our attention to the first of the two central methods of aircraft parameter estimation, namely, the output-error method. The other method called filter-error method will be discussed in the next chapter. Both of these methods belong to a general class of output error, also called response-curve-fitting, methods. The class of estimation methods called the equation-error methods will be considered separately. In this class of output-error methods, model parameters are adjusted iteratively to minimize the error between the measured variables (system output) and the estimated (model predicted) responses. The method, however, leads to a nonlinear optimization problem, in which the computational burden is relatively high. The method of weighted least squares, the simplest among this class, accounts for measurement noise. However, it assumes a priori specification of the weighting matrix. Based on probability theory, a more profound formulation called the maximum-likelihood principle was provided by Fisher [1–3]. It can handle both process and measurement noise and has several desirable statistical properties of a "good estimator." This chapter considers the case where we assume that the process noise is negligible and that the measurements are corrupted by additive measurement noise only. It leads to the popularly called "output-error method" (OEM). The output-error as well as filter-error methods represent a natural formulation for dynamic systems, be it a linear or nonlinear. In the majority of practical applications, the dynamic process being investigated evolves continuously in time whereas the observations are recorded at discrete time points. Accordingly, we consider here the mixed continuous-/discrete-time formulation.

Since its introduction in the 1960s, the output-error method is the most widely applied time-domain method to estimate aircraft parameters from flight data.

This is quite evident from the numerous applications reported in the literature; Refs. [4–9] provide a good survey of the flight vehicle applications. Initially, during the 1960s to 1970s, the scope of the output-error method was limited to linear systems. The limitations were not related to the applicability of the maximum likelihood principle to nonlinear systems; rather, they were practical difficulties related to handling general nonlinear model structures and to compute the sensitivity matrix required in the optimization of the cost function. In the 1980s, these difficulties were solved through a numerical approach, which has paved the way to analyze general nonlinear systems of arbitrary complexity [6].

In this chapter, we begin with the basic formulation of the maximum likelihood principle, the assumptions that are made, and study briefly its statistical properties. This is followed by a detailed study of the cost function and various optimization methods. For each aspect, we indicate the pros and cons and attempt to make recommendations based on our experience of modeling different types of flight vehicles [6, 7]. We study the performance of the output-error method on two sample examples, using estimation software that we also supply with the book to follow the discussion and to gain more insight into the algorithmic steps and numerical aspects. The data analyzed are also supplied as sample data to check the results and possible adaptation of the software to individual needs.

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4.2 PRINCIPLE OF MAXIMUM LIKELIHOOD ESTIMATION

In the general theory of estimation founded by Fisher [1–3], it is assumed as a working hypothesis that the parent population has a known determinate mathematical form but for the numerical value of some parameter vector Θ (a column vector). Parent population is a set of all possible measurements as the number of samples N goes to ∞. It is required to deduce the unknown value of the parameter vector Θ using a set of observations $(z_1, z_2, \ldots, z_N)$.

Considering the given samples of N random observations $(z_1, z_2, \ldots, z_N)$ to be independent and taken from the same population, the likelihood function can be defined as

$$p(z|\Theta) = p(z_1|\Theta) \cdot p(z_2|\Theta) \cdot \cdots \cdot p(z_N|\Theta) = \prod_{k=1}^{N} p(z_k|\Theta) \tag{4.1}$$

where $p(z|\Theta)$ is the probability of z given Θ. The maximum likelihood (ML) method, introduced by Fisher as a general estimation procedure, amounts to selecting that value of Θ within the admissible range such that $p(z|\Theta)$ is maximized. The likelihood function represents probability density, but the probability density of observed variables and not of the parameters. The unknown parameters are assumed not to depend on chance [10, 11].

Because of the exponential nature of many density functions [12], the logarithm of the likelihood function, which has the same optimal solution, is generally preferred. The maximum likelihood estimate is thus obtained as

$$\widehat{\Theta}_{\mathrm{ML}} = \arg\left\{ \min_{\Theta} \ell n\, p(z|\Theta) \right\} \tag{4.2}$$

Assuming that $p(z|\Theta)$ is a twice differentiable function, practical application of the maximum likelihood method requires solving the likelihood equation:

$$\frac{\partial\, \ell n\, p(z|\Theta)}{\partial \Theta} = 0 \tag{4.3}$$

Equation (4.3), a column vector, is a set of nonlinear equations that can be solved only by successive approximation. Linear expansion of the likelihood equation about a first approximation Θ_0 of Θ yields

$$\frac{\partial\, \ell n\, p(z|\Theta_1)}{\partial \Theta} \approx \frac{\partial\, \ell n\, p(z|\Theta_0)}{\partial \Theta} + \frac{\partial^2\, \ell n\, p(z|\Theta_0)}{\partial \Theta^2}\, \Delta\Theta \tag{4.4}$$

where $\Theta_1 = \Theta_0 + \Delta\Theta$ is the improved approximation to Θ. Equating Eq. (4.4) to zero yields a linear system of equations:

$$\frac{\partial^2\, \ell n\, p(z|\Theta_0)}{\partial \Theta^2}\, \Delta\Theta = -\frac{\partial\, \ell n\, p(z|\Theta_0)}{\partial \Theta} \tag{4.5}$$

which can be solved by any procedure of linear algebra to yield the parameter improvement vector $\Delta\Theta$. The expected value of the matrix of second gradient $[\partial^2 \ell n\ p(z|\Theta)/\partial \Theta^2]$ is called the Fisher information matrix.

To apply the maximum likelihood function to the dynamical system that we have considered in Chapter 3, it is now required to write the expression for $p(z|\Theta)$, that is, the conditional probability density function. Although the method is applicable to any form of the density function distribution, for mathematical tractability we consider the Gaussian (normal) distribution, which is completely determined by the first and second moments, that is, the mean and the covariance matrix. It is the most widely used assumption in practical cases. For such a case, and further assuming the error $v(t_k) = z(t_k) - y(t_k)$ at different time points t_k to be statistically independent, that is,

$$E\{v(t_k)\, v^T(t_\ell)\} = R\, \delta_{k\ell} \tag{4.6}$$

where $\delta_{k\ell}$ is the Kronecker delta symbol ($\delta_{k\ell} = 1$ for $k = \ell$ and $= 0$ for $k \neq \ell$), it is shown in Appendix C that the likelihood function $p(z(t_1), \ldots, z(t_N)|\Theta, R)$ of n_y-dimensional measurement vector at N discrete time points for a given

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parameter vector Θ and given measurement error covariance matrix R is [13, 14]

$$p(z(t_1),\ldots,z(t_N)|\Theta,R)=\prod_{k=1}^{N}p(z(t_k)|\Theta,R)$$

$$=\{(2\pi)^{n_y}|R|\}^{-N/2}\exp\left[-\frac{1}{2}\sum_{k=1}^{N}[z(t_k)-y(t_k)]^T\,R^{-1}\,[z(t_k)-y(t_k)]\right] \quad (4.7)$$

The reader may note that we have implicitly extended here the basic definition of Eq. (4.1) to include the conditioning on R, the covariance matrix of the residuals. For notational simplicity we denote

$$p(z|\Theta,R)=p(z(t_1),\ldots,z(t_N)|\Theta,R) \quad (4.8)$$

We pay specific attention here to the commonly misunderstood concept and interpretation of the exact meaning of the maximum likelihood function. The likelihood function $p(z|\Theta,R)$ does not represent the probability distribution of the unknown parameters Θ, but of the measurements z. Although looking at the observations some parameters may appear more plausible and some less, the unknown parameters are not random variables and do not have probability density. Maximum likelihood estimation means that the Θ vector is searched, which maximizes the function $p(z|\Theta,R)$. Such a vector is "the most plausible" because it gives the highest probability to the measurements.

From Eq. (4.7), we can immediately compute the maximum likelihood estimates of the unknown parameters Θ and of the measurement noise covariance matrix R by setting the first derivatives with respect to each of them to zero. However, an equivalent and more practical solution to the optimization problem is the minimization of $L(z|\Theta,R)$, the negative logarithm of the likelihood function:

$$L(z|\Theta,R)=\frac{1}{2}\sum_{k=1}^{N}[z(t_k)-y(t_k)]^T\,R^{-1}[z(t_k)-y(t_k)]$$

$$+\frac{N}{2}\ell n(\det(R))+\frac{Nn_y}{2}\ell n(2\pi) \quad (4.9)$$

We prefer to deal with Eq. (4.9) instead of (4.7) because, as already mentioned, the natural logarithm of the normal density function is a simpler function than the density function itself. The differentiation result $\partial(\ell n\ p)/\partial\Theta=1/p\cdot\partial p/\partial\Theta$ guarantees that the derivative of log of p is zero, when $\partial p/\partial\Theta$ is zero, and thus yields the same results.

4.3 PROPERTIES OF MAXIMUM LIKELIHOOD ESTIMATES

Solving the likelihood equation yields parameter estimates $\widehat{\Theta}_{\mathrm{ML}}$ that lend the largest probability to the observations $(z_1, z_2, \ldots, z_N)$, which are assumed to be

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statistically independent. Although the maximum likelihood principle is intuitively appealing, we will study here briefly the statistical properties of the estimates. For a more detailed treatment of these statistical properties, the reader can refer to Appendix D or any one of the standard books on statistics [10–17].

The concepts of efficiency and consistency of the maximum likelihood estimates were introduced by Fisher [3]. Based on the assumption of independent observations, Wald established the asymptotic consistency and unbiasedness of the maximum likelihood estimates [15]. The asymptotic normality was proved by Cramer [16]. We recapture here only the essentials of a "good" estimator [18].

1. Unbiased estimates:
   The maximum likelihood estimates are asymptotically unbiased, that is,

$$\lim_{N\to\infty} E(\widehat{\Theta}_{\mathrm{ML}}) = \Theta$$

   where Θ represents the true values of the parameters.

2. Asymptotic consistency:
   Maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ are asymptotically consistent, that is, $\widehat{\Theta}_{\mathrm{ML}}$ converges in probability to the true value Θ.

3. Asymptotic normality:
   The maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ obtained from different sets of data samples are normally distributed around the true value Θ, that is,

$$\sqrt{N}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \to r_1 \quad \sim \mathcal{N}(0, \overline{\mathcal{F}}^{-1})$$

   where r_1 the random variables, $\overline{\mathcal{F}}$ the average Fisher information matrix per sample, and $\mathcal{N}(0, \overline{\mathcal{F}}^{-1})$ the normal (Gaussian) distribution with zero mean and variance $\overline{\mathcal{F}}^{-1}$.

4. Asymptotic efficiency:
   The maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ are asymptotically efficient in the sense that they attain the Cramér–Rao lower bounds, which is a measure of achievable statistical accuracy.

The property of consistency implies that the maximum likelihood estimates converge in probability to the true values. The property of asymptotic normality implies that the estimates obtained from different sets of data samples corresponding to different experiments are clustered around the true value with a normal distribution. The property of asymptotic efficiency is of practical significance. It implies that the maximum likelihood estimator makes efficient use of the available data. The Cramér–Rao lower bound indicates the theoretically maximum achievable accuracy of the estimates; see Appendix D.

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4.4 MAXIMUM LIKELIHOOD FUNCTION FOR ESTIMATION OF PARAMETERS IN DYNAMICAL SYSTEMS

The mathematical model of the dynamical system, whose parameters are to be estimated, is assumed to be described by the following general nonlinear system representation:

$$\dot{x}(t) = f[x(t), u(t), \beta], \quad x(t_0) = x_0 \tag{4.10}$$

$$y(t) = g[x(t), u(t), \beta] \tag{4.11}$$

$$z(t_k) = y(t_k) + G\,\nu(t_k) \tag{4.12}$$

This is the same set of equations that we had postulated in Chapter 3 to represent a dynamical system in state space. All of the variables have already been defined there. To recollect, the nonlinear functional relationships f and g are assumed to be real valued. They may be nonlinear in the state variables x and/or in the control variables u, as well as in the parameters to be estimated [18–21]. This class of nonlinear system description is in general quite adequate for various estimation problems encountered in flight vehicle modeling from flight data [4–6].

To apply the likelihood function to estimate the parameters of a dynamical system represented in Eqs. (4.10–4.12), we make the following assumptions:

1. The input sequence $(u(t_k), k = 1, 2, \ldots, N)$ is exogenous, that is, generated independent of the system output.
2. The measurement errors $\nu(t_k) = z(t_k) - y(t_k)$ at different discrete time points are statistically independent. They are assumed to be distributed with zero mean and covariance matrix R, that is,

$$E\{\nu(t_k)\} = 0$$

$$E\{\nu(t_k)\nu^T(t_\ell)\} = R\delta_{k\ell}$$

3. The system is corrupted by measurement noise only.
4. Control inputs $u(t_k)$ are sufficiently and adequately (that is, in magnitude and frequencies) varied to excite directly or indirectly the various modes of the dynamical system being analyzed.

By definition, exogenous inputs are those that affect the system, but are not affected by the system itself. It is the key to success of parameter estimation and also helps to ensure that the residuals are independent at different time points. For stable aircraft, it is usually possible to apply control inputs about all axes independent of the aircraft response. However, in the case of unstable aircraft, due to feedback, this assumption might not be met, and hence special treatment is necessary. We will discuss problems and solutions of unstable aircraft in a separate chapter later.

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The second assumption allows applying directly the likelihood function that we already discussed. The third assumption implies that we treat in this chapter the dynamic system as deterministic. Note that we have dropped the process noise w and the process noise distribution matrix F in Eq. (4.10). For such cases, the state variables are obtained through simple integration of the state equations. The case of process noise will be dealt with in the next chapter. Through the fourth assumption, we ensure identifiability of the parameters because the basic rule of *"If it is not in the data, it can not be identified"* applies to all exercises that attempt parameter estimation from experimental data. This is true irrespective of the type of flight vehicle we might attempt to model.

In the preceding discussion of Secs. 4.2 and 4.3, the likelihood function $p(z|\Theta, R)$ included the arguments z, Θ, and R. Strictly speaking, mathematically it would be more appropriate to write the likelihood function for the given data as $p(z|\Theta, R, u)$, where we have now included the argument u for the system inputs. However, the argument u is dropped in the further discussions without loss of generality, because 1) identical control inputs are used to excite the system as well as the postulated model (see Fig. 4.1) and 2) the control inputs are assumed to be exogenous, noise free, and known a priori.

From Sec. 4.2, we know that maximum likelihood estimates are obtained by minimization of Eq. (4.9), the negative logarithm $L(z|\Theta, R)$ of the likelihood function. For convenience, we rename $L(z|\Theta, R)$ as J, the cost function to be minimized

$$J(\Theta, R) = L(z|\Theta, R) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - y(t_k)]^T R^{-1}[z(t_k) - y(t_k)] + \frac{N}{2}\ell n(\det(R)) + \frac{Nn_y}{2}\ell n(2\pi) \tag{4.13}$$

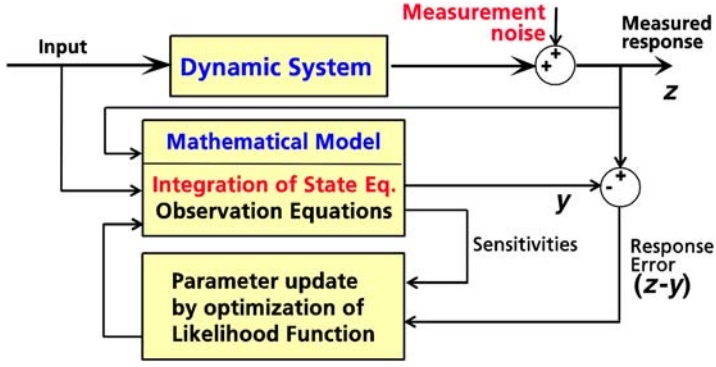

Fig. 4.1 Block schematic of output-error method.

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subject to the system model postulated in Eqs. (4.10–4.12). Figure 4.1 shows a block schematic of the output-error method. The details of each of the blocks in this figure will be discussed hereafter. Note that the system responses y are functions of the parameters Θ, and hence the exact notation of the preceding equation should include $y(\Theta)$. However, for the sake of brevity, this dependence is not shown explicitly in any of the equations that we deal with.

4.5 BASICS OF COST FUNCTION OPTIMIZATION

Before we turn to actual methods of optimization, we further analyze the likelihood cost function $J(\Theta, R)$ to gain more insight [22, 23]. For a postulated model and data set being analyzed, the number of observation variables n_y and the number of data points N are fixed. Thus, the last term in Eq. (4.13) is a constant and hence neglected in the optimization without affecting the results. For the first two terms of Eq. (4.13), there are two cases that are of interest depending upon whether the measurement error covariance matrix R is known or not.

4.5.1 KNOWN MEASUREMENT NOISE COVARIANCE MATRIX

In the first case, we assume that the measurement noise covariance matrix R is known; at this stage it is immaterial as to how we might arrive at the knowledge of R. In such a case, because R is assumed to be known, the second term in Eq. (4.13), namely, $N\,\ell n(\det(R))/2$, is constant. The cost function then reduces to

$$J(\Theta) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - y(t_k)]^T R^{-1}[z(t_k) - y(t_k)] \tag{4.14}$$

To understand the nature of Eq. (4.14), let us consider a simpler case of having just two observation variables and assume R to be a diagonal matrix. In such a case, multiplying out the terms in the summation on the right-hand side leads to $J(\Theta) = [r_{11}e_1^2(t_k) + r_{22}\,e_2^2(t_k)]$, where e_1 and e_2 denote the response errors in the two observation variables and r_{11} and r_{22} the diagonal elements of R representing the variances. Thus, the cost function $J(\Theta)$ in Eq. (4.14) is nothing but the weighted sum of squares of the response error and is quadratic in nature. We can apply any one of the optimization methods we will discuss in the following sections.

4.5.2 UNKNOWN MEASUREMENT NOISE COVARIANCE MATRIX

In the other case of unknown R, optimization of Eq. (4.13) is little more involved. The brute-force method would be to include the elements of the covariance matrix to the unknown parameters vector Θ and then apply one of the optimization methods. This approach is, however, never practiced because there is no closed-form solution to this minimization problem. During the optimization, the estimates of the system parameters depend on the measurement noise covariance

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matrix and vice versa. Any attempt in this direction is bound to fail. To overcome this difficulty, we use the relaxation strategy in which optimization of the likelihood function, Eq. (4.13), is carried out in two steps.

In the first step, it can be shown that for any given value of the parameter vector Θ, the maximum likelihood estimate of R is given by

$$R = \frac{1}{N} \sum_{k=1}^{N} [z(t_k) - y(t_k)] \; [z(t_k) - y(t_k)]^T \tag{4.15}$$

Equation (4.15) is obtained by partially differentiating Eq. (4.13) with respect to R and setting it to zero. A detailed derivation of the preceding expression for R is provided in Appendix E.

Having obtained a maximum likelihood estimate of R, substitution of Eq. (4.15) in (4.13) yields

$$J(\Theta) = \frac{1}{2} n_y N + \frac{N}{2} \ell n(\det(R)) + \frac{N n_y}{2} \ell n(2\pi) \tag{4.16}$$

Once again, recall that n_y and N are fixed for a postulated model and a data set being analyzed. Hence, in addition to the last term, the first term of Eq. (4.16) also turns out to be a constant and can be neglected without affecting the minimization results. Thus, the cost function reduces to

$$J(\Theta) = \det(R) \tag{4.17}$$

Determination of the parameter vector Θ, which minimizes $\det(R)$, or equivalently the function $L(z|\Theta)$, is an optimization problem that can be solved by applying different methods.

Thus, the relaxation procedure can be summarized as follows:

1. Choose suitable initial values for Θ.
2. Compute system outputs y and the residuals $(z - y)$; estimate the measurement noise covariance matrix R.
3. Minimize $J(\Theta)$ with respect to Θ applying one of the nonlinear optimization methods.
4. Iterate on step 2, and check for convergence.

Mathematical proof of a global convergence of the relaxation procedures is difficult, but it is convenient to use and mostly works well in practice. From the abundant examples pertaining to flight vehicles of diversified complexity reported in the literature, it is apparent that this is the only pragmatic approach to nonlinear optimization.

In general, the covariance matrix of the output errors is unknown and, hence, has to be estimated. In such a case, as just shown, the cost function to be minimized reduces to the determinant of the covariance matrix of the output errors. Because the covariance matrix is usually assumed to be diagonal, a simplification

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that has found widespread use in practice, the cost function then represents nothing but the product of the variances. Any optimization method, direct search or gradient based, can be applied to obtain the estimates of Θ. In any case, it is necessary to compute the system states x and responses y to obtain the cost function. Furthermore, depending upon the optimization algorithm, response gradients might be required.

4.6 GAUSS–NEWTON ALGORITHM

The Gauss–Newton method of nonlinear optimization belongs to a class of second-order algorithms and is one of the most widely applied method in flight vehicle system identification in the time domain. It is a modification of the Newton, also called Newton–Raphson, method. Therefore, we first study this basis method before discussing the modifications and its implications. A necessary condition for minimization of the likelihood function with respect to the unknown parameters is given by

$$\frac{\partial J(\Theta)}{\partial \Theta} = 0 \tag{4.18}$$

The Taylor-series expansion of $\partial J/\partial\Theta$ about the ith value of the parameter vector Θ, truncated after two terms, is given by

$$\left(\frac{\partial J}{\partial \Theta}\right)_{i+1} \approx \left(\frac{\partial J}{\partial \Theta}\right)_i + \left(\frac{\partial^2 J}{\partial \Theta^2}\right)_i \Delta\Theta \tag{4.19}$$

where $\Delta\Theta = \Theta_{i+1} - \Theta_i$ is the parameter change and $(\partial^2 J/\partial\Theta^2)_i$ is the second gradient of the cost function with respect to Θ at the ith iteration. In optimization theory, the matrix of second gradients is termed as Hessian. Now, using the necessary condition of Eq. (4.18) for the minimum, we set the right-hand side of Eq. (4.19) to zero and solve for $\Delta\Theta$:

$$\Delta\Theta = -\left[\left(\frac{\partial^2 J}{\partial \Theta^2}\right)_i\right]^{-1}\left(\frac{\partial J}{\partial \Theta}\right)_i \tag{4.20}$$

The preceding change in Θ on the $i+1$th iteration makes local $(\partial J/\partial\Theta)_{i+1}$ approximately zero. Thus, starting from an initial guess value Θ_0, $\Theta_{i+1} = \Theta_i + \Delta\Theta$ provides an iterative solution to find the minimum of a function. Such an algorithm is commonly known as the Newton–Raphson method [24, 25].

To demonstrate the preceding optimization procedure, we consider a simple one-dimensional case. First, we apply the preceding procedure to find θ that minimizes a quadratic cost function $J(\theta) = 8\theta^2$. Let the starting value of θ be 4. The first and second gradients, $\partial J/\partial\theta$ and $\partial^2 J/\partial\theta^2$, of this cost function evaluated at $\theta_0 = 4$ are given by $16\theta_0 = 16 \times 4$ and 16 respectively. Equation (4.20) then yields an incremental value of $\Delta\theta = -(16)^{-1} \times 16 \times 4 = -4$. Thus, after the

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first iteration the iterative update leads to $\theta_1 = \theta_0 + \Delta\theta = 4 - 4 = 0$. For the quadratic cost function, this solution happens also to be the minimum and does not require any further iteration. This is always the case for any arbitrary quadratic function because the second gradient $\partial^2 J/\partial\theta^2$ is constant, which allows, as illustrated schematically in Fig. 4.2a, determination of the minimum in a single step irrespective of the starting value θ_0. This simple test case can be run using the program "/FVSysID2/chapter04/ExQuadFun.m."

In the second example, we consider optimization of a nonquadratic function $J(\theta) = 4\theta^2 + 0.4\theta^3$. The first and the second gradients are given by $\partial J/\partial\theta = 8\theta + 1.2\theta^2$ and $\partial^2 J/\partial\theta^2 = 8 + 2.4\theta$, respectively. The second gradient in this case is not a constant, but is a function of θ. The optimization procedure yields step by step an improved solution given by Eq. (4.20) based on the local quadratic approximation of the nonquadratic cost function. For example, starting from $\theta_0 = 4$, the first iteration yields the increment $\Delta\theta_1 = -(8 + 2.4 \times 4)^{-1}(8 \times 4 + 1.2 \times 4^2) = -2.9091$, and the updated parameter $\theta_1 = \theta_0 + \Delta\theta_1 = 4 - 2.9091 = 1.0909$. Now, starting from the updated value θ_1, the second iteration leads to $\Delta\theta_2 = -(8 + 2.4 \times 1.0909)^{-1}(8 \times 1.0909 + 1.2 \times 1.0909^2) = -0.9564$, yielding the updated parameter value of $\theta_2 = \theta_1 + \Delta\theta_2 = 1.0909 - 0.9564 = 0.1345$. The iterative procedure continues until the minimum is achieved after the fourth iteration. As already pointed out, in each iteration the first and second gradients change and have to be computed for the local value of θ. The optimization procedure is schematically illustrated in Fig. 4.2b. This test case is run using the program "/FVSysID2/chapter04/ExNonQuadFun.m." Thus, as just illustrated, geometrically interpreted, the basic idea behind the Newton–Raphson method is to construct, at each iteration, a quadratic approximation to the cost function and then take a step in the direction of the

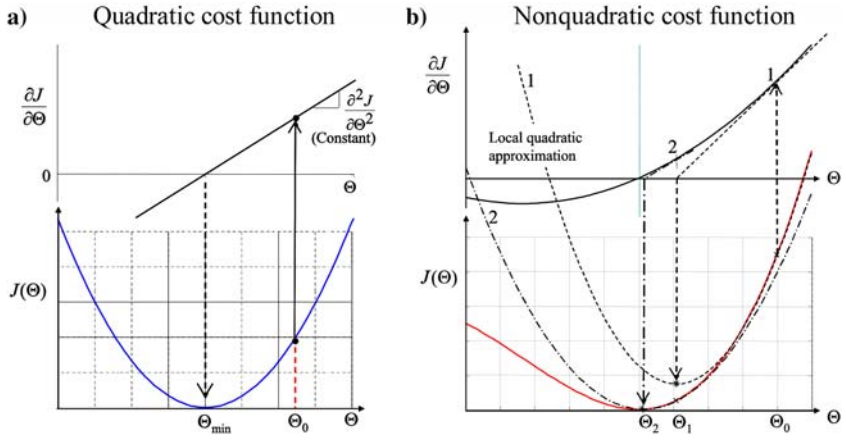

Fig. 4.2 Schematic of optimization procedure.

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minimum of that approximated quadratic function. Because the step is directly towards the local minimum, the method is much more efficient than the gradient-based methods that merely step in the local downhill direction.

In general, away from the minimum of the cost function, the algorithm takes larger incremental steps $\Delta\theta$ corresponding to larger values of the gradient whereas they become smaller and smaller as the minimum is approached; the number of steps required finding the minimum in a specific case will depend upon the shape of the cost function. If the cost function has multiple minima depending upon the initial starting values, the search procedure might find a local extremum.

Now, in our case of parameter estimation from flight data, having determined the residual covariance matrix using Eq. (4.15), we now minimize the quadratic cost function $J(\Theta)$ given by Eq. (4.14) resulting from the likelihood function, Eq. (4.13). The partial differentiation of Eq. (4.14) results in

$$\frac{\partial J}{\partial \Theta} = -\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}[z(t_k) - y(t_k)] \tag{4.21}$$

and the partial differentiation of $\partial J/\partial\Theta$ in Eq. (4.21) with respect to Θ yields

$$\frac{\partial^2 J}{\partial \Theta^2} = \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \frac{\partial y(t_k)}{\partial \Theta} - \sum_{k=1}^{N} \left[\frac{\partial^2 y(t_k)}{\partial \Theta^2}\right]^T R^{-1}[z(t_k) - y(t_k)] \tag{4.22}$$

The computation of the first gradient $\partial J/\partial\Theta$ from Eq. (4.21) is relatively straightforward. It requires just the measured data z, the corresponding computed variables y, and the response gradient $\partial y/\partial\Theta$. Recall that the estimate of the weighting matrix R, given by Eq. (4.15), is already obtained for the current Θ.

Computation of the second gradient $\partial^2 J/\partial\Theta^2$ given by Eq. (4.22) is more complex and time consuming, as it requires the second gradient of the response $\partial^2 y/\partial\Theta^2$. However, the second term on the right-hand side of Eq. (4.22) includes a term $[z(t_k) - y(t_k)]$, and it is found that this contribution goes to zero as the process converges. In an ideal case, the residuals $[z(t_k) - y(t_k)]$ should just be the random measurement error at each time point. Based on our assumption of zero mean and independent noise, the second term tends to cancel out when summed over a sufficient length of data points. Based on this practical consideration, we neglect the second term in Eq. (4.22) and approximate the second gradient $\partial^2 J/\partial\Theta^2$ as

$$\frac{\partial^2 J}{\partial \Theta^2} \approx \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta}\right] \tag{4.23}$$

The main advantage of the preceding simplification is the saving in the computational burden without affecting much the rapid, quadratic convergence. It is also associated with another minor advantage, namely, the second gradient given by Eq. (4.23) always remains positive definite, because it is the sum of quadratic

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terms. The preceding simplification was suggested by Balakrishnan [26] and leads to a more tractable algorithm called modified Newton–Raphson method, which is found to work well in most of the practical applications. In the literature this algorithm is also sometimes called the Newton–Balakrishnan algorithm or also the Gauss–Newton method, the terminology that we will follow in this book.

4.7 METHOD OF QUASI-LINEARIZATION

In this section we follow a slightly different approach to derive an expression to compute the parameter update. We already know from Eq. (4.18) that $\partial J/\partial\Theta$ must vanish at the minimum. Accordingly, setting Eq. (4.21) to zero yields

$$\frac{\partial J}{\partial \Theta} = -\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}[z(t_k) - y(t_k)] = 0 \tag{4.24}$$

Now, applying the quasi-linearization [27, 28], that is, a first-order approximation, to the system responses around some nominal value Θ_0, we obtain

$$y(\Theta) = y(\Theta_0 + \Delta\Theta) \approx y(\Theta_0) + \frac{\partial y}{\partial \Theta}\,\Delta\Theta \tag{4.25}$$

Substituting the linearized $y(\Theta)$ in Eq. (4.24) leads to

$$\frac{\partial J}{\partial \Theta} = -\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \left[z(t_k) - \left\{y(t_k) + \frac{\partial y(t_k)}{\partial \Theta}\,\Delta\Theta\right\}\right] = 0 \tag{4.26}$$

Note that for convenience, we have dropped the argument Θ_0 from y in the preceding equation. Equation (4.26) can be rewritten as

$$\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}\,[z(t_k) - y(t_k)] - \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta}\right] \Delta\Theta = 0 \tag{4.27}$$

which leads to a system of linear equations:

$$\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta}\right] \Delta\Theta = \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}[z(t_k) - y(t_k)] \tag{4.28}$$

Equation (4.28) derived applying quasi-linearization is exactly the same as Eqs. (4.21) and (4.23) derived for the Gauss–Newton method. Thus, the Gauss–Newton and quasi-linearization methods are equivalent.

The Gauss–Newton and quasi-linearization methods lead to a system of linear equations, which can be represented in a general form as

$$\Theta_{i+1} = \Theta_i + \Delta\Theta \quad \text{and} \quad \mathcal{F}\Delta\Theta = -\mathcal{G} \tag{4.29}$$

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where i is the iteration index and $\mathcal{F}$ and $\mathcal{G}$ are given by

$$\mathcal{F} = \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]$$
$$\mathcal{G} = -\sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1} [z(t_k) - y(t_k)] \tag{4.30}$$

$\mathcal{F}$ is the information matrix (also called Hessian), $\mathcal{G}$ is the gradient vector, and $\Delta\Theta$ is the parameter change vector. Once the gradient vector $\mathcal{G}$ and the matrix of second gradients $\mathcal{F}$ are computed, the parameter updates are given by Eq. (4.29). Thus, we now have both of the steps of the relaxation algorithm, which can be carried out successively, namely, compute the measurement noise covariance matrix using Eq. (4.15) and then apply the Gauss–Newton method given by Eqs. (4.29) and (4.30) to update the parameters. Equation (4.29), $\mathcal{F}\Delta\Theta = -\mathcal{G}$, can be solved for $\Delta\Theta$ by any standard algorithm from linear algebra. Because of inaccurate numerical results, direct matrix inversion methods are rarely used. It is obvious from Eq. (4.30) that the information matrix $\mathcal{F}$ is symmetric, positive-definite. For such cases the Cholesky factorization is an efficient procedure. Singular value decomposition is yet another approach that is also efficient. Details of these techniques can be found in any standard text book on numerical methods.

4.8 SYSTEM RESPONSE AND SENSITIVITY COEFFICIENTS

The iterative update of the parameter vector Θ by Gauss–Newton or quasi-linearization method using Eqs. (4.29) and (4.30) requires the following:

1. Computation of the system responses $y(t_k)$
2. Computation of the response gradients $\partial y(t_k)/\partial\Theta$

Efficient implementation of these computational aspects, together with flexibility to handle conveniently different model structures, is important for parameter estimation in nonlinear systems [20, 21, 29].

The model predicted response $y(t)$, Eq. (4.11), is a function of the parameters being estimated and the state variables $x(t)$. According to our assumption of the system being corrupted by measurement noise only and that the measured control inputs contain no errors and noise, we are dealing in this chapter with deterministic systems. For such a case we can simply integrate the state equations, Eq. (4.10), using one of the numerical integration procedures elaborated in Sec. 3.8.

Computation of the response gradients $\partial y(t_k)/\partial\Theta$, also called the sensitivity matrix, involves some critical issues that directly affect the capability to handle

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different nonlinear models. In the conventional approach, the sensitivity matrix $\partial y/\partial \Theta$ is obtained by solving the sensitivity equations, which are obtained analytically by partial differentiation of the system equations (4.10) and (4.11):

$$\frac{\partial \dot{x}}{\partial \Theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \Theta} + \frac{\partial f}{\partial \Theta} \tag{4.31}$$

$$\frac{\partial y}{\partial \Theta} = \frac{\partial g}{\partial x}\frac{\partial x}{\partial \Theta} + \frac{\partial g}{\partial \Theta} \tag{4.32}$$

In the case of linear systems, the method of transition matrices for the solution of Eq. (4.31) and simple matrix multiplications provides an elegant procedure. However, if the system model is nonlinear, as in Eqs. (4.10) and (4.11), any changes in the model structure entail rederivation of the sensitivity equations of the form given in Eqs. (4.31) and (4.32). As a consequence, changes in the parameter estimation software are necessary. This additional programming effort becomes cumbersome especially when alternative model structures are to be tried out to fit the experimental data. Furthermore, in case of systems with discontinuous nonlinearities numerical problems might be encountered due to the difficulty of defining the derivatives analytically. These are the practical difficulties arising out of the model postulate, which is nonlinear, and the need to have flexibility of handling different model structures conveniently without software changes [29].

The need for explicit derivation of the sensitivity coefficients, Eq. (4.31) and (4.32), can be eliminated by approximating the sensitivity coefficients by numerical differences. The forward difference approximation yields each element of the response gradient matrix:

$$\left[\frac{\partial y(t_k)}{\partial \Theta}\right]_{ij} \approx \frac{y_i^p(t_k) - y_i(t_k)}{\delta \Theta_j}; \quad i = 1, \ldots, n_y; \quad j = 1, \ldots, n_q$$

$$\approx \frac{g_i(x^p(t_k), u(t_k), \Theta + \delta\Theta_j e^j) - g_i(x(t_k), u(t_k), \Theta)}{\delta \Theta_j} \tag{4.33}$$

where n_y is the number of output variables, n_q the total number of unknown parameters, superscript p the perturbed variables, $\delta\Theta_j$ a small perturbation in the jth component of Θ, e^j a column vector with one in the jth row and zeros elsewhere, $y^p(t_k)$ the perturbed response variables corresponding to the perturbation $(\Theta + \delta\Theta_j\, e^j)$, and $y(t_k)$ the unperturbed responses corresponding to Θ. The perturbed responses $y^p(t_k)$ at each time point are obtained from the perturbed system equations. These equations have a form similar to those of Eqs. (4.10) and (4.11), except that each time the parameter vector Θ is replaced through the perturbed parameter vector Θ_p. The reader can easily write these equations. The perturbed responses $y^p(t_k)$ are also functions of the perturbed states $x^p(t_k)$, which are once again obtained by numerically integrating the perturbed state equations, which are basically the same state equations, except for a different

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set of parameter values. Thus, we use the same model and thereby avoid the changes to the estimation software.

Thus, the finite difference approximation of the response gradients provides a more flexible approach to handle nonlinear models. The sensitivity coefficients in Eq. (4.33) were approximated by forward differences. A more accurate method would be based on the two-sided, central difference formula. It, however, requires two evaluations of the perturbed states and of the response variables at each time point and needs twice the computational time. From our experience, the one-sided forward difference approximation works fairly well in most of the cases. For large-scale problems (that is, large numbers of outputs n_y, parameters n_q, and data points N), near the minimum of the cost function a better performance can be obtained through central differencing [30]. The other alternatives that also provided good results in such cases were forward difference approximation in conjunction with the line search that we will discuss in Sec. 4.10.2 to overcome stalling of the optimization or to apply the Levenberg–Marquardt method we will address in Sec. 4.13.

The choice of the parameter perturbation will affect the accuracy of the approximated sensitivity coefficients. Some care is necessary in choosing the perturbation size; it should be small enough to give a valid approximation of the true gradients, but too small a value might pose difficulties due to machine accuracy and roundoff errors. We have already discussed this aspect of tradeoff in Sec. 3.5.3 in connection with approximating system matrices. Because the parameters might vary in the order of their magnitudes, a relative parameter perturbation is more appropriate. Typically,

$$\delta\Theta_j = 10^{-6}\,\Theta_j \tag{4.34}$$

is found to be a reasonable choice. In some cases, larger perturbation might be required, for example, when the particular parameter appears as a discrete nonlinear term, or when the effect (changes) in the system response because that parameter is small. If a particular parameter happens to be zero, either as initially specified or during iteration, then deviating from Eq. (4.34) it is necessary to specify a small perturbation; otherwise, Eq. (4.34) would yield zero perturbation, and the computed gradient would be zero. A small absolute value, say, $\delta\Theta_j = 10^{-6}$, is usually adequate to overcome this numerical peculiarity.

The procedure elaborated already is a classical approach in which the parameters are varied one at a time [18, 20, 21, 29]. Thus, for n_q number of unknown parameters, at each time point we have to solve n_q times the perturbed state equations. More recently, in connection with the stochastic optimization a procedure called "simultaneous perturbations" has been investigated [31], requiring just two evaluations at each discrete time point. For large n_q, such a method will be much faster. We have not investigated such a concept of simultaneous perturbation for dynamic systems and in the framework of Gauss–Newton method, and hence do not go into any further discussion of it.

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4.9 AUTOMATIC GRADIENT COMPUTATION

As elaborated in Sec. 4.8, gradients are usually approximated using either one-sided (forward or backward) or two-sided central differencing. The latter is more accurate, but requires twice the computational time. Basically, in both the cases proper choice of step size is critical to obtain valid and accurate approximations. If the step size is too small, the roundoff errors (resulting from subtraction of two almost equal numbers) may dominate; on the other hand, too large a step can lead to nonnegligible truncation errors (resulting from neglected higher-order terms of Taylor series) yielding an erroneous gradient. Some tradeoff is usually necessary between these errors. We had resorted to the finite difference approximation due to various reasons, for example:

1. Analytical differentiation of the postulated model can be very tedious for complex models.
2. Flexibility is needed to consider different nonlinear model structures without programming changes each time.
3. In some cases a program may be available just for computing the desired function.

However, if we can find a technique to do all of these jobs automatically and also generate program code equivalent to an analytic derivative without actually going through the steps of deriving the mathematical expressions, we will not only achieve the desired goal, but also improve the accuracy by eliminating the errors introduced by finite differencing. This becomes possible through an upcoming technology in the field of mathematical computations, called automatic differentiation (AD) [32, 33]. We cover here just the basic fundamentals of this recent approach to indicate new possibilities.

The concept of automatic differentiation is based on the fact that any function, no matter in which programming language, is executed as a sequence of elementary operations such as additions, multiplications, and intrinsic functions. Such basic operations and functions are limited in number, and their analytic derivatives are known. By repeated application of chain and product rules, any function can be differentiated. To illustrate the procedure, we quote here a simple example from [33] considering a function $y = (x + x^2)^2$. Figure 4.3a gives a possible way of programming the preceding function and Fig. 4.3b that for the automatically generated gradient, where the value of dx is arbitrary but not zero.

In simple terms, given a program code to evaluate a function in terms of the dependent and independent variables, the AD processor first synthesizes (breaks down) the code in terms of elementary operations. Based on this code synthesis, a program code is constructed to compute the derivative of the dependent variables with respect to the independent variables. Because the derivatives are computed now using elementary and intrinsic functions, the approximation errors vanish,

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Fig. 4.3 Sample example of automatic differentiation by chain rule.

a) Function evaluation

```
Function y = f(x)
z=x*x
w=x+z
y=w*w
end
```

b) Function and gradient evaluation

```
Function (y, dy) = fdot(x, dx)
z=x*x
dz = 2*x*dx
w = x+ z
dw = dx + dz
y= w*w
dy = 2 * w * dw
end
```

giving derivatives to machine precision. The approach being analytical, it does not involve perturbations, and hence roundoff or truncation errors are eliminated.

Thus, automatic differentiation provides an option to overcome numerical difficulties that might arise from finite difference approximations. Automatic differentiation will perform better, particularly in those optimization cases where the algorithm is sensitive to gradients. Applications of this new approach have been reported in few fields based on static models (that is, systems not incorporating state equations). Preliminary investigations suggest that the approach can be applied to dynamic systems with state and observation equations. Utility in terms of estimating aerodynamic derivatives and catering to special cases where finite differences posed severe problems will have to be established through applying the new approach to a large number of cases. The only minor issue appears to be that of using a generated code that looks like a black box and thereby tends to loose engineer's insight into a program code he or she might use.

4.10 STEP-SIZE CONTROL

The unconstrained Gauss–Newton method, described in Sec. 4.6, takes a full step of parameter update at each iteration. Generally, the Gauss–Newton method performs very well. However, as presented in Secs. 4.6 and 4.7, the update is based on the assumption of local linearity and a quadratic cost function. Furthermore, an approximation to the second gradient is incorporated to reduce the numerical burden, and the response gradients are approximated through numerical approximations. Because of these limitations, the Gauss–Newton method might perform poorly, for example, due to initial values far away from optimum or due to numerical errors introduced through the finite difference approximation of gradients near the optimum, and it might show intermediate local divergence or stalling. In such cases it is necessary to control the step size carefully once the direction of the correction vector has been established. Three options are possible to overcome these typical numerical difficulties:

1. Heuristic approach of parameter halving
2. Line search
3. Dominant directions approach

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4.10.1 HEURISTIC APPROACH

In the first approach based on heuristic considerations, if the full step of the Gauss–Newton algorithm, Eq. (4.29), diverges at any iteration, the parameter update $\Delta\Theta$ is successively reduced by halving each time until reduction in the cost function compared to the prior iteration is achieved. It amounts to a successive reduction of the full step through the factors of (2, 4, 8, . . .). During these halving steps, just the cost-function evaluation is necessary, without computing the gradient. Halving of the parameter increments is generally performed a maximum of 10 times, which amounts to a factor of ($2^{10} = 1024$), and thereby at the end of 10 halving steps we practically reach the parameters of the preceding iteration. It implies that under given conditions the current iteration does not lead to any further improvement in the optimization. In that case, the optimization procedure is terminated. Usually few halving steps are sufficient to overcome the intermediate divergence. This heuristic procedure is quite simple and in many cases found to be adequate.

4.10.2 LINE SEARCH

A more systematic approach and one that guarantees efficient performance is based on introducing a line search in the Gauss–Newton method. The update to Eq. (4.29) in such a case becomes

$$\Theta_{i+1} = \Theta_i + \alpha_i \Delta\Theta \quad \text{with} \quad \Delta\Theta = -\mathcal{F}^{-1}\mathcal{G} \tag{4.35}$$

where α_i is determined by a line search algorithm in each iteration. Line searches, also called one-dimensional searches, are the basic optimization procedures of finding a minimum of a cost function along the specified search direction. In the present case it amounts to reducing or increasing $\Delta\Theta$ of the basic Gauss–Newton method optimally to obtain maximum reduction in the cost function at each iteration.

Because just a few function evaluations are necessary for line search, the additional computational overhead is marginal. Such damping strategies help to widen the convergence region and to overcome other numerical problems [18, 34, 35]. Any standard line search algorithm can be applied here; a quadratic line search procedure has shown good performance in many practical problems. Line search algorithm is also necessary for the bounded-variable Gauss–Newton method, which will be discussed in the next section.

4.10.3 DOMINANT DIRECTIONS

The third approach to overcome intermediate divergence or stalling is based on evaluating the eigenvalues of the information matrix $\mathcal{F}$ and determining the dominant search directions by checking the smallest and the largest eigenvalues. To understand the basic philosophy behind this approach, we begin by analyzing

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the various contributions to the parameter step $\Delta\Theta$. Recall that the parameter step is given by

$$\Delta\Theta = -\mathcal{F}^{-1}\mathcal{G} \tag{4.36}$$

where $\mathcal{F}$ is the information (Hessian) matrix and $\mathcal{G}$ is the gradient vector. If $\mathcal{F}$ is ill-conditioned, the Gauss–Newton method can result in an extremely large step and divergence. We now attempt to detect the ill-conditioning through singular value decomposition (SVD). We know that $\mathcal{F}$ is a $(n_q \times n_q)$ real, symmetric matrix. In this case we can write $\mathcal{F}$ as follows:

$$\mathcal{F} = T\Lambda T^{-1} \tag{4.37}$$

where Λ is the diagonal matrix of eigenvalues of $\mathcal{F}$ and T is the matrix of eigenvectors. By inverting the preceding equation and using the matrix algebra results $(AB)^{-1} = B^{-1}A^{-1}$ and $T^{-1} = T^T$ for orthogonal matrices, we obtain

$$\mathcal{F}^{-1} = \left(T\Lambda T^{-1}\right)^{-1} = \left(T^{-1}\right)^{-1}\Lambda^{-1}T^{-1} = T\Lambda^{-1}T^T \tag{4.38}$$

Thus, the parameter improvement $\Delta\Theta$ is given by

$$\Delta\Theta = -T\Lambda^{-1}T^T\mathcal{G} \tag{4.39}$$

Equation (4.38) can be rewritten in SVD form as

$$\mathcal{F}^{-1} = \frac{1}{\lambda_1}t_1t_1^T + \frac{1}{\lambda_2}t_2t_2^T + \cdots + \frac{1}{\lambda_q}t_qt_q^T \tag{4.40}$$

where the matrix of eigenvectors is partitioned into n_q column vectors as $T = [t_1, t_2, \ldots, t_q]$ corresponding to the $(\lambda_1, \lambda_2, \ldots, \lambda_q)$ eigenvalues.

Substituting Eq. (4.40) into Eq. (4.36) leads to

$$\Delta\Theta = -\frac{t_1^T\mathcal{G}}{\lambda_1}t_1 - \frac{t_2^T\mathcal{G}}{\lambda_2}t_2 - \cdots - \frac{t_q^T\mathcal{G}}{\lambda_q}t_q \tag{4.41}$$

which is a well-known result in the optimization theory, saying that the parameter step $\Delta\Theta$ is a sum of n_q steps of magnitude $[t_i^T\mathcal{G}/\lambda_i]$ in the directions of the eigenvectors t_i, $i = 1, 2, \ldots, n_q$ of the information matrix.

For an ill-conditioned information matrix $\mathcal{F}$, at least one of its eigenvalues is very small. Let us assume that there are r such eigenvalues that might be considered very small. Because the step sizes in Eq. (4.41) are inversely proportional to the value of the eigenvalues, it can lead to very large step sizes along these r eigenvectors and in worst cases to divergence. The intuitive idea is to neglect these so-called bad directions and consider only the remaining $(n_q - r)$ dominant

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search directions. Mathematically, Λ and T are partitioned as

$$\Lambda = \begin{bmatrix} \Lambda_d & | & 0 \\ - & - & - \\ 0 & | & \Lambda_r \end{bmatrix} \quad \text{and} \quad T = [T_d \quad T_r] \tag{4.42}$$

such that the dominant eigenvalues greater than ε (a small positive number) are grouped together as Λ_d and the corresponding eigenvectors in the left partition T_d. Substituting Eq. (4.42) into (4.39) and simple manipulation leads to

$$\Delta\Theta = -T_d\Lambda_d^{-1}T_d^T\mathcal{G} - T_r\Lambda_r^{-1}T_r^T\mathcal{G} \tag{4.43}$$

and neglecting the r small eigenvalues gives the step in the dominant search direction as

$$\Delta\Theta \approx -T_d\Lambda_d^{-1}T_d^T\mathcal{G} \tag{4.44}$$

Let us discuss some practical issues and performance of the preceding modified form of the Gauss–Newton method. It requires specifying ε appropriately. This can depend upon the model size, modeling errors, and the noise in the data analyzed. In several practical cases of flight vehicle parameter estimation, it turned out that the preceding modification worked only in a third of the cases of an ill-conditioned matrix. This is not totally surprising because neglecting large $[t_i^T\mathcal{G}/\lambda_i]$ steps does not necessarily mean that new directions are used in the optimization. All of the remaining so-called dominant directions, in many cases a major number of them, are still the Gauss–Newton directions. A better way could be to generate a new set of search directions, as suggested in [35], possibly interpolating between Gauss–Newton and the Newton direction. The approach is again based on SVD and dominant directions. This is left to the reader to pursue, but with a remark that the best strategy would be one that is simple and straightforward like the heuristic approach or line search presented earlier.

4.11 BOUNDED-VARIABLE GAUSS–NEWTON METHOD

The Gauss–Newton method discussed in Sec. 4.6 provides an unconstrained solution to a nonlinear optimization problem. Parameter estimation subject to simple bounds might, however, be relevant in some cases. Typical applications related to flight vehicle system identification are the following [36]:

1. Parameters that describe the physical effects, in the present case aerodynamic effects, are often constrained to lie in a certain range, for example, the Oswald's factor [37] characterizing the increase in drag over ideal condition caused by nonelliptical lift distribution and interference is typically limited to less than one.

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2. Time delays in the measurement variables are by definition always positive and hence greater than zero.
3. Estimation of highly nonlinear model parameters such as friction, which can lead to numerical difficulties due to different reasons like poor guess of initial values [38], is needed.
4. Attempt to estimate parameters with almost zero sensitivity, resulting either from the use of inappropriate maneuvers or observation variables in the estimation procedure can lead to divergence.

In this section we address the issues pertaining to extending the Gauss–Newton method to account for simple bounds.

The linearly constrained optimization problem in which the constraints are simple bounds on the variables is formulated as

$$\min_{\Theta} J(\Theta) \quad \text{subject to } \Theta_{\min} \leq \Theta \leq \Theta_{\max} \tag{4.45}$$

where $\Theta_{\min}$ and $\Theta_{\max}$ are the lower and upper bounds on the parameter respectively.

Optimization theory provides several approaches to solve this problem, such as the following:

1. Transformation techniques
2. Penalty function methods
3. Barrier function or Lagrangian approach
4. Active set strategy

The transformation technique is based on transforming the independent variables such that the constraints are satisfied automatically. Once this is done, any standard unconstrained optimization method can be applied [39]. Although the concept is simple to understand, it is not always possible to transform all constraints, and the form of the transformation equation might not be simple to find. The penalty function and Lagrangian approaches are more general and suitable for complex nonlinear inequality constraints. They are also more complex. In the present case we are considering a simpler problem of linear constraints. Several algorithms and software programs, for example, limited memory BFGS [40], Extrem [41], quasi-Newton [42], and bounded-variable Least Squares [43], provide solutions to this problem. However, as already pointed out, the Gauss–Newton method is preferred here. The active set strategy is conceptually very appealing and can be readily extended to the Gauss–Newton method [36].

Starting from the initially specified parameter values Θ_0, an active set IA containing the indices of the variables hitting the bounds is formed and updated for every iteration. A variable is called a free variable if it is within

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the permissible bounds and hence not in the active set. The Gauss–Newton search directions for the free variables are computed as follows:

$$\Delta\Theta_{\text{free}} = -\mathcal{F}_{\text{free}}^{-1}\,\mathcal{G}_{\text{free}} \tag{4.46}$$

where the information matrix $\mathcal{F}_{\text{free}}$ and the gradient vector $\mathcal{G}_{\text{free}}$ are computed using Eq. (4.30) for the free variables. The parameter updates resulting from Eq. (4.46) are checked for the specified bounds, and any violation of the constraints leads to modification of the active set IA. For such parameters the values are set to the respective bounds and the search directions of Eq. (4.46) to zero. For the remaining free parameters a new set of values is computed using a line search, which is necessary because changes in the number of free parameters change the search directions. As already addressed briefly in Sec. 4.10.2, we have used the quadratic line search procedure in our applications. It is interesting to point out that because the bounded-variable Gauss–Newton method includes the line search as an integral part of the algorithm, it automatically eliminates the problems of poor performance due to intermediate divergence.

An important aspect of the active set strategy is to develop a procedure of finding variables leaving or entering the feasible region, and accordingly to alter the active set IA in each iteration as the optimization progresses. The active set is changed whenever a free variable hits its bounds during iteration. Furthermore, if the Kuhn–Tucker optimality conditions [39]

$$\mathcal{G}_i < 0, \quad \text{for } \Theta_i = \Theta_{i_\max}$$

or (4.47)

$$\mathcal{G}_i > 0, \quad \text{for } \Theta_i = \Theta_{i_\min}$$

are not satisfied for any of the variables in the active set, then those variables are dropped from the active set and made free; $\mathcal{G}_i$, Θ_i, $\Theta_{i_\min}$, and $\Theta_{i_\min}$ are respectively the components of the gradient vector given by Eq. (4.30), the current parameter value, and its lower and upper bounds. In other words, conditions checked in Eq. (4.47) guarantee that the gradients for the variables hitting the bounds are such that they point outwards of the feasible region, implying that any further minimization of the cost function would be possible only when the particular parameters are not constrained within the specified limits.

The computational overhead to implement the active set strategy in an existing unconstrained Gauss–Newton method is minor; it is just required to check for the variables that hit the bounds and for the optimality conditions of Eq. (4.47) to enter or drop parameters from the active set. The advantages of this particular approach are twofold: 1) it provides an efficient solution retaining the desirable properties of the Gauss–Newton method, namely, the quadratic convergence property and 2) statistical information regarding the accuracy of the estimates, which we will address in Sec. 4.17, is readily available.

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4.12 CONSTRAINED GAUSS–NEWTON METHOD USING INTERIOR-POINT ALGORITHM

Yet another approach to constrained optimization that has attracted quite some attention in the area of nonlinear programming is based on the interior-point algorithm [44–46]. Without going into the historical background on the development of such techniques, we straightaway address the applicability of such an algorithm within the framework of the maximum likelihood estimation that we have so far considered. To restate the problem, the maximum likelihood estimates of the unknown parameters and of the unknown noise covariance matrix are obtained by minimizing the cost function:

$$J(\Theta, R) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - y(t_k)]^T R^{-1}[z(t_k) - y(t_k)] + \frac{N}{2}\ell n|R| \tag{4.48}$$

where Θ is the n_q-dimensional vector of unknown parameters. Minimization of Eq. (4.48) yields the unconstrained estimates of Θ. In the present case, the parameters to be estimated are subjected to inequality constraints:

$$\Theta_{\min} \leq c(\Theta) \leq \Theta_{\max} \tag{4.49}$$

where $c(\Theta)$ are the inequality constraints. Through simple manipulation, the two-sided constraints of Eq. (4.49) can be transformed into the form:

$$\begin{aligned} c(\Theta) - \Theta_{\max} &\leq 0 \\ \Theta_{\min} - c(\Theta) &\leq 0 \end{aligned} \tag{4.50}$$

For simplicity of notation and for derivation of the algorithm, we express the constraints in Eq. (4.50) in the general form given by

$$c_i(\Theta) \leq 0, \quad i = 1, \ldots, p \tag{4.51}$$

where p is the total number of constraints and c can be of any form, but is assumed to be twice continuously differentiable. Thus, it is required to optimize the cost function of Eq. (4.48) subject to the constraints specified in Eq. (4.51). To apply the interior-point approach, we first reformulate the problem using the slack variables s_i to each of the constraints of Eq. (4.51).

$$\underset{\Theta,R}{minimize}\, J(\Theta, R) \quad \text{subject to } c(\Theta) + s = 0, \quad \text{and} \quad s \geq 0 \tag{4.52}$$

where c and s are p-dimensional vectors. The inequality constraints in Eq. (4.52) are eliminated by introducing a barrier parameter $\mu(>0)$, which results in the following problem:

$$\underset{\Theta,R}{minimize}\left\{J(\Theta, R) - \mu\sum_{i=1}^{p}\log s_i\right\} \quad \text{subject to } c(\Theta) + s = 0 \tag{4.53}$$

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where $J(\Theta, R)$ is given by Eq. (4.48). Adopting the Lagrangian approach, we can rewrite Eq. (4.53) as

$$\tilde{J}(\Theta, R, s, \lambda, \mu) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - y(t_k)]^T R^{-1}[z(t_k) - y(t_k)] + \frac{N}{2}\ell n|R| - \mu\sum_{i=1}^{p}\log s_i + \lambda^T(c(\Theta) + s) \tag{4.54}$$

where λ is the vector of Lagrange multipliers.

The first-order condition for a minimum is that the first derivatives of $\tilde{J}$ with respect to the respective variables, namely, Θ, R, s, and λ, vanish. As in the case of the classical unconstrained output-error method described in Sec. 4.5, we apply the two-step relaxation strategy, estimating R first and then the other parameters. Following the development presented in Sec. 4.5.2 and Appendix E, it can be shown that the estimate of the covariance matrix of residuals R is given by

$$R = \frac{1}{N}\sum_{k=1}^{N}[z(t_k) - y(t_k)][z(t_k) - y(t_k)]^T \tag{4.55}$$

which is the same expression as given by Eq. (4.15).

For a known R given by Eq. (4.55) during the particular iteration, equating $\partial\tilde{J}/\partial\Theta$, the partial differentiation of Eq. (4.54) with respect to Θ, to zero yields

$$-\sum_{k=1}^{N}\left(\frac{\partial y(t_k)}{\partial\Theta}\right)^T R^{-1}[z(t_k) - y(t_k)] + \left(\frac{\partial c}{\partial\Theta}\right)^T\lambda = 0 \tag{4.56}$$

Similarly, partially differentiating Eq. (4.54) with respect to s and λ and equating to zero leads respectively to

$$-\mu e + S\lambda = 0 \tag{4.57}$$

and

$$c(\Theta) + s = 0 \tag{4.58}$$

where S is a diagonal matrix with elements s_i and e is a vector of ones. It can be shown that Eqs. (4.56–4.58) represent the so-called Karush–Kuhn–Tucker conditions for the constrained minimum.

As in the case of the standard unconstrained Gauss–Newton method, we start from a valid (interior) point Θ_0, s_0, λ_0. Without going into the detailed mathematical derivation of individual steps, we state here the final expression for $\Delta\Theta$,

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$\Delta s, \Delta\lambda$ updates. For a prespecified μ, the Gauss–Newton step is computed according to

$$\left\{\sum_{k=1}^{N}\left(\frac{\partial y(t_k)}{\partial\Theta}\right)^T R^{-1}\left(\frac{\partial y(t_k)}{\partial\Theta}\right)+\left(\frac{\partial c}{\partial\Theta}\right)^T S^{-1}\Lambda\left(\frac{\partial c}{\partial\Theta}\right)\right\}\Delta\Theta$$
$$=\sum_{k=1}^{N}\left(\frac{\partial y(t_k)}{\partial\Theta}\right)^T R^{-1}[z(t_k)-y(t_k)]-\left(\frac{\partial c}{\partial\Theta}\right)^T S^{-1}\Lambda[s+c(\Theta)+\mu\Lambda^{-1}e] \quad (4.59)$$

$$\Delta s=-\left(\frac{\partial c}{\partial\Theta}\right)\Delta\Theta-c(\Theta)-s \quad (4.60)$$

$$\Delta\lambda=S^{-1}[\mu e-S\lambda-\Lambda\Delta s] \quad (4.61)$$

where Λ is a diagonal matrix with elements λ_i.

The matrix on the left-hand side of Eq. (4.59) is symmetric and positive definite. Any standard numerical procedure like Cholesky factorization can be used to solve Eq. (4.59) for the parameter update $\Delta\Theta$. The first term in the flower bracket on the left-hand side of Eq. (4.59) is the same as that appearing in Eq. (4.28) or (4.30) applying the unconstrained Gauss–Newton method. The full Gauss–Newton step, resulting from Eqs. (4.59–4.61), does not guarantee that the updated parameters satisfy all of the constraints. Therefore, a suitable damping strategy needs to be applied. The common approach is

$$\Theta_{i+1}=\Theta_i+\alpha_P\,\Delta\Theta_i \quad (4.62)$$
$$s_{i+1}=s_i+\alpha_P\,\Delta s_i \quad (4.63)$$
$$\lambda_{i+1}=\lambda_i+\alpha_D\,\Delta\lambda_i \quad (4.64)$$

where i is the iteration index and $(\alpha_P,\ \alpha_D)$ are the damping coefficients. These coefficients guarantee that the nonnegative variables remain nonnegative and prevent the estimates from approaching the boundaries too closely [44, 45]. They are chosen based on the standard ratio test.

$$\alpha_P=\min\left\{0.9995\min_{(j:\Delta z_j<0)}\{-s_j/\Delta s_j\},\ 1\right\} \quad (4.65)$$

$$\alpha_D=\min\left\{0.9995\min_{(j:\Delta y_j<0)}\{-\Lambda_j/\Delta\lambda_j\},\ 1\right\} \quad (4.66)$$

The last aspect of the interior point algorithm pertains to the choice of the barrier parameter μ. In general, the choice is model and data dependent. However, the

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following procedure was also found quite adequate for a number of examples of flight data analysis [45, 46].

$$\mu = \frac{\lambda^T s}{(n_q + p)^2} \tag{4.67}$$

It is fairly obvious that the interior point algorithm with μ equal to zero reduces to the unconstrained method. For $\mu > 0$, the algorithm ensures that the estimated parameters remain within the specified constraints.

For linear constraints specified as simple upper and lower bounds, the two approaches to account for the limits, namely, the bounded-variable Gauss–Newton method described in Sec. 4.11 and the interior-point algorithm described in this section, were found to be comparable in terms of convergence, numerical values, and accuracies. The more sophisticated formulation based on the interior-point will be efficient when more complex constraints have to be satisfied.

4.13 LEVENBERG–MARQUARDT METHOD

The Levenberg–Marquardt method is yet another approach to overcome the numerical problems discussed in the preceding sections [47, 48]. It combines the best features of the standard, unconstrained Gauss–Newton method and the steepest descent technique that moves in the gradient direction. Because of the optimal combination of two search directions, the Levenberg–Marquardt method has a wider convergence region. The update formula in such a case is given by

$$\Theta_{i+1} = \Theta_i + \Delta\Theta \quad \text{with } (\mathcal{F} + \lambda I)\Delta\Theta = -\mathcal{G} \tag{4.68}$$

where λ is the Levenberg–Marquardt (LM) parameter, which can be smoothly varied to control whether the update search direction is more like steepest descent or along the Gauss–Newton direction. It is easily seen from Eq. (4.68) that for $\lambda \to \infty$ the method becomes steepest descent, and as $\lambda \to 0$, it becomes like Gauss–Newton.

The computation of $\mathcal{F}$ and $\mathcal{G}$ is exactly the same as in the preceding case of the Gauss–Newton method, given by Eq. (4.30). It now remains to find proper value for λ that performs optimum interpolation between the two methods. Many methods exist for computing the LM parameter λ. The control strategy described in [47] is simple and found to work well in practical cases.

1. Choose a reduction factor $\nu > 1$, typically $\nu = 10$; choose $\lambda_0 = 0.001$ as initial starting value.
2. Perform one step of the Gauss–Newton method, and increment the iteration counter i by one; let λ^{i-1} denote the value from the preceding iteration.
3. Solve the system of linear equations, Eq. (4.68), for two values of the LM parameter λ^{i-1} and (λ^{i-1}/ν).

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4. Now, for the two solutions of $\Delta\Theta$ from step 3, perform the parameter update, and compute the respective cost functions $L_i = L(\Theta(\lambda^{i-1}))$ and $L_i^{\nu} = L(\Theta(\lambda^{i-1}/\nu))$.
5. Comparison of L_i and L_i^{ν} leads to the following three possible cases:
   a. If $L_i^{(\nu)} \leq L_{i-1}$, then set $\lambda^{(i)} = \lambda^{(i-1)}/\nu$ and $\Delta\Theta = \Delta\Theta(\lambda^{i-1}/\nu)$. (That is, smaller λ results in reduced cost function; hence, accept the new point and reduce the LM parameter for the next iteration.)
   b. If $L_i^{(\nu)} > L_{i-1}$ and $L_i \leq L_{i-1}$, then set$\lambda^{(i)} = \lambda^{(i-1)}$ and $\Delta\Theta = \Delta\Theta(\lambda^{i-1})$. (That is, accept the new point, but retain the LM parameter from preceding iteration.)
   c. Otherwise, $L_i^{(\nu)} > L_{i-1}$ and $L_i > L_{i-1}$, then set $\lambda^{(i-1)} = \lambda^{(i-1)}\nu$, (i.e., increase λ), and repeat the steps 2 through 5, until an acceptable new point is found.

In simple words, the control strategy just elaborated evaluates the cost function values (among themselves and compared to the previous lowest cost) for two LM parameters. The one corresponding to the most reduction is accepted, and the program proceeds to the next iteration. In case both of the new updates have costs that are higher than the cost in the previous iteration, the factor λ is increased, and the cycle repeated. Adaptation of the LM parameter generally guarantees convergence to the optimum.

The choice of the reduction factor ν is arbitrary, and $\nu = 10$ has been found to be a good choice for many test cases pertaining to aircraft parameter estimation. Because the properties of the gradient optimization methods are not scale invariant, the starting value for the LM parameter λ depends on the order of magnitude of the system (information) matrix. A scaling, widely used in linear least-squares problems, is given by

$$\mathcal{F}^* = (f_{ij}^*) = \left(\frac{f_{ij}}{\sqrt{f_{ii}}\,\sqrt{f_{jj}}}\right) \tag{4.69}$$

$$\mathcal{G}^* = (g_i^*) = \left(\frac{g_i}{\sqrt{f_{ii}}}\right) \tag{4.70}$$

This leads to a scaled system of linear equations:

$$\mathcal{F}^*\Delta\Theta^* = -\mathcal{G}^* \tag{4.71}$$

which can be solved by any standard procedures of linear algebra. The solution for the transformed $\Delta\Theta^*$ is then scaled back as

$$\Delta\Theta_i = \frac{\Delta\Theta_i^*}{\sqrt{f_{ii}}} \tag{4.72}$$

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The transformed matrix $\mathcal{F}^*$ is a simple correlation matrix, whose diagonal elements are one. For such a scaling, a starting value of $\lambda_0 = 0.001$ is recommended, which proved adequate for several test cases investigated [30, 49].

The Levenberg–Marquardt method requires solving Eq. (4.68) twice per iteration compared to once in the Gauss–Newton method. However, this overhead is just a minor part of the total overhead; as elaborated in Sec. 4.18, a large portion being spent on computing the information matrix, that is, the second gradient of the cost function, which is done only once in both the cases. The adaptation of the LM parameter ensures that the convergence is monotonic and guarantees convergence from iteration to iteration, and as such the procedures described in Sec. 4.10 to overcome intermediate divergence are not relevant here. The Levenberg–Marquardt method works well for starting points far from the optimum. It is also useful to overcome the numerical problems due to one-sided approximation of gradients near the optimum.

4.14 DIRECT SEARCH METHODS

Optimization methods that require just function evaluations and not gradients belong to the class of direct search methods. They are, in general, much slower compared to gradient-based methods [29, 50, 51] but are somewhat more robust, particularly starting from a point that is far from the optimum. Moreover, standard program codes are readily available. Because the literature on optimization theory is abundant with the details of these methods, only a very brief verbal description is provided here.

4.14.1 SIMPLEX AND SUBPLEX METHODS

The downhill simplex method developed by Nelder and Mead and the subspace searching method developed by Rowan are based on the principles of reflection, expansion, contraction, and shrinkage (massive contraction) of an initially specified q-dimensional convex hull, a geometrical figure with $n_q + 1$ vertices called simplex [52, 53].

In the Nelder and Mead simplex method, a simplex moves through the q-dimensional space of the cost function, changing shape and size, and automatically shrinks when the minimum is enveloped by the simplex [52]. The simplex method is widely used in the optimization of cost functions in several applications, but rarely for flight vehicle parameter estimation from flight data. Nevertheless, because of its robustness with respect to the initial guess of the unknown parameters, it could be useful in a few cases, for example, estimation of parameters with discrete nonlinearities, friction parameters, etc.

The subplex method is a generalization of the simplex method of Nelder and Mead. It decomposes the higher-dimensional problem into smaller-dimensional subspaces in which the simplex method can search efficiently [53]. The task is

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twofold: first, to determine the improved set of subspaces, and second to apply the simplex method. The concept of orthogonality used for the subspaces is similar to the concept of primary and secondary search directions used in Jacob's heuristic minimum search method called "Extrem"; see Sec. 4.14.3.

When the number of parameters to be estimated is large, the classical simplex method of Nelder and Mead rapidly becomes inefficient. A restart procedure helps in some cases to improve the performance somewhat. On the other hand, the computational overhead for the subspace searching method increases roughly linearly with the problem dimension. From this viewpoint, it is considered as the most promising of the simplex-based methods.

4.14.2 POWELL'S METHOD

Powell's method is a widely used direct search method [54]. It can be shown to be a method of conjugate directions that minimizes the cost function in a finite number of cost function evaluations. The method is an iterative procedure and involves computing the search directions at each iteration and finding the minimum along these search directions applying a line search method.

4.14.3 JACOB'S METHOD

The Jacob's method is based on the choice of optimal search directions, determination of the optimum along a line and defining the step size for each search [41]. The Gram–Schmidt orthogonalization procedure is used to select the orthogonal search directions, in contrast to the conjugate directions used in Powell's method. The search directions are adapted to the local properties of the parameter space being searched. Basically, the procedure is very similar to the Gauss–Newton method. A program called "Extrem," which is available in the literature, caters to constrained optimization. For example, simple bounds on the parameters being estimated can be accounted for.

Having covered in Secs. 4.4–4.14 several methods to optimize the cost function, the choice of a particular method might appear to be a difficult one. However, based on comparative studies performed in the past [29, 50, 51], it can be concluded that the direct search methods covered in Sec. 4.14 are extremely slow compared to the gradient-based Gauss–Newton, its variation accounting for bounds or the Levenberg–Marquardt method. The last mentioned methods are far superior in performance, mainly because they make use of the first and second gradients to predict the minimum and directly step in the best local direction. Any one of these will perform equally well on the chosen problem. We have widely used the Gauss–Newton and bounded-variable Gauss–Newton method, incorporating either line search or the heuristic approach of parameter halving to overcome any intermediate divergence or stalling of the optimization procedure. Nevertheless, in certain rare cases, like estimation of friction parameters, the simplex or subplex may give better starting values for the parameters.

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It is also possible to switch from one method to another as the optimization progresses.

4.15 REGRESSION STARTUP PROCEDURE

In Sec. 4.10, it has been pointed out that the Gauss–Newton method might perform poorly for initial values far away from optimum, and we had discussed some techniques to overcome this problem. Yet another way would be to generate better starting values using some other technique that does not need initial guess values. This is possible by using a procedure belonging to the broad class of least-squares methods that we call regression startup method. It represents a mixed least-squares and output-error approach. To illustrate the method, first consider the linearized system representation:

$$\dot{x}(t) = A(\beta)x(t) + B(\beta)u(t) \tag{4.73}$$

$$y(t) = C(\beta)x(t) + D(\beta)u(t) \tag{4.74}$$

Now, assuming that measurements of all of the state variables are available and denoting them as x_m, the system equations are reformulated as

$$\dot{x}(t) = A_F x(t) + [B(\beta); \quad A_E(\beta)] \begin{bmatrix} u(t) \\ x_m(t) \end{bmatrix} \tag{4.75}$$

$$y(t) = C_F x(t) + [D(\beta); \quad C_E(\beta)] \begin{bmatrix} u(t) \\ x_m(t) \end{bmatrix} \tag{4.76}$$

The matrices A and C are resolved each into two, namely, (A_F, A_E) and (C_F, C_E). The matrices with the subscript F contain the possibly fixed coefficients, if any, and those with the index E contain the parameters to be estimated.

In general, the fixed parameters appearing in matrices A_F and C_F correspond to those resulting from kinematic terms or derivatives fixed using the a priori values whereas those appearing in A_E and C_E correspond to stability and control derivatives, which are unknown. This reformulation makes the state variables x and thereby the observation variables y and hence the error $z(t_k) - y(t_k)$ linear in the parameters being estimated. For such cases, the cost function being quadratic, the least-squares method will yield estimates in a single iteration. A detailed treatment of the least-squares technique will be dealt with in Chapter 6. The preceding formulation has the advantage that the initial values of the unknown parameters are not required. The optimization procedure can be started with zero values of all of the parameters. Note that the preceding formulation retains state equations that have to be integrated, but it incorporates measured states as control inputs in conjunction with unknown parameters, thus leading to a combined output-error and least-squares approach. The optimization of the cost function is carried out using the same iterative algorithm that we have studied earlier (Sec. 4.7).

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Applying regression analysis to the nonlinear model postulated in Eqs. (4.10) to (4.12) is not straightforward. If the model is linear in parameters, the optimization algorithm is a one-shot procedure whereas for a model nonlinear in parameters, the iterative optimization is necessary. In such a case some reasonable start values might be required. Treatment of unknown initial conditions is a little trickier. However, based on practical considerations, an acceptable solution becomes feasible. Assuming that the measurements of the state variables x_m are available; Eqs. (4.10) to (4.12) are reformulated as follows:

$$\dot{x}(t) = f[x(t), x_m(t), u(t) - \Delta u(b_{u,l}), (\beta_{xF}, \beta_{xE})], \quad x(t_{0,l}) = x_m(t_{0,l}) \tag{4.77}$$

$$y(t) = g[x(t), x_m(t), u(t) - \Delta u(b_{u,l}), (\beta_{xF}, \beta_{xE}), (\beta_{yC}, \beta_{yE})] + \Delta z(b_{y,l}) \tag{4.78}$$

where the unknown parameters β_x and β_y are subdivided into two components (β_{xF}, β_{xE}) and (β_{yF}, β_{yE}) corresponding to those parameters that are constants and those that are to be estimated. The parameters to be estimated are multiplied by the measured states x_m or by measured control inputs u whereas the constant parameters are multiplied by the integrated states. The initial values $x(t_0)$ are set to the measured values at time t_0 and held fixed. Furthermore, the zero shifts in the control variables Δu, which represent the systematic errors, are assumed to be small and are, hence, either set to zero or kept fixed at the value specified. Estimation of the zero shifts in the observation variables Δz poses no difficulties because the model is linear in them.

Thus, associating the unknown parameters with the measured states leads to a mixed output-error and least-squares approach, which can be used to generate good initial guess values for the unknown parameters, both for linear and nonlinear model postulates. As will be discussed in Sec. 9.5, such a mixed least-squares and output-error formulation is also useful in estimating parameters of unstable aircraft without facing the numerical problems otherwise encountered in applying the classical output-error method. In a few cases the preceding general approach has been used without subdividing the matrices and using the measured states throughout. Such a procedure is much simpler to implement and found to work equally well.

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4.16 ESTIMATION ACCOUNTING FOR A PRIORI INFORMATION

In some cases a priori information about certain aerodynamic derivatives might be available from wind-tunnel tests or from estimation results of other flight-test investigations. The Bayesian estimator enables accounting for a priori information in a probabilistic way by treating the parameter vector Θ $(n_q \times 1)$ as a random vector [55]. In such a case, a priori knowledge about the probability density functions of the measurements and of the parameters is necessary. The Bayesian estimator represents the most general type of parameter estimation methodology, but it also needs the maximum amount of a priori information.

The conditional probability density function $p(\Theta|z)$, which represents the a posteriori probability density function of the parameters Θ, given the measurements z, can be expressed using Bayes' rule as [55]

$$p(\Theta|z) = \frac{p(\Theta)p(z|\Theta)}{p(z)} \tag{4.79}$$

For convenience, both $p(\Theta)$ and $p(z|\Theta)$ are assumed to be Gaussian. In this case, they can be expressed as

$$p(z|\Theta) = \{(2\pi)^{n_y}|R|\}^{-N/2}\exp\left(-\frac{1}{2}\sum_{k=1}^{N}[z(t_k)-y(t_k)]^T R^{-1}[z(t_k)-y(t_k)]\right) \tag{4.80}$$

and

$$p(\Theta) = \{(2\pi)^{n_q}|R_2|\}^{-1/2}\exp\left(-\frac{1}{2}(\Theta-\Theta^*)^T R_2^{-1}(\Theta-\Theta^*)\right) \tag{4.81}$$

where R is the measurement noise covariance matrix, Θ^* is the vector of a priori values of the derivatives, and R_2 is the error covariance matrix of the a priori values. It follows that the a priori values and the associated uncertainty are represented as

$$E\{\Theta\} = \Theta^* \tag{4.82}$$

and

$$E\left\{(\Theta-\Theta^*)(\Theta-\Theta^*)^T\right\} = R_2 \tag{4.83}$$

In the estimation procedure, for a given set of measured data, $p(z)$ can be treated as a constant. Thus, maximization of $p(\Theta|z)$ in Eq. (4.79) is equivalent to maximization of J_{ap} given by

$$J_{\text{ap}} = (2\pi)^{-(n_q+n_yN)/2}|R|^{-N/2}|R_2|^{-1/2} \times\exp\left(-\frac{1}{2}\sum_{k=1}^{N}[z(t_k)-y(t_k)]^T R^{-1}[z(t_k)-y(t_k)] - \frac{1}{2}(\Theta-\Theta^*)^T R_2^{-1}(\Theta-\Theta^*)\right) \tag{4.84}$$

Setting the gradient with respect to Θ to zero and solving yields

$$\left\{\sum_k\left[\frac{\partial y(t_k)}{\partial\Theta}\right]^T R^{-1}\left[\frac{\partial y(t_k)}{\partial\Theta}\right] + R_2^{-1}\right\}\Delta\Theta = \sum_k\left[\frac{\partial y(t_k)}{\partial\Theta}\right]^T R^{-1}[z(t_k)-y(t_k)] - R_2^{-1}(\Theta-\Theta^*) \tag{4.85}$$

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The just-discussed estimator is commonly called in the literature as the maximum a posteriori probability estimator (MAP). Loosely speaking, it pulls the estimates toward the a priori values, depending upon the confidence in them. Although it represents a more general approach, the a priori information about the probability density function $p(\Theta)$ is often difficult to obtain. Relaxing the assumption of parameters being random variables leads to the maximum likelihood estimation that we had already discussed in some depth in Secs. 4.2 through 4.5. To recall once again, the maximum likelihood method assumes that Θ does not depend on chance and thereby requires a priori knowledge only about $p(z|\Theta)$. In such a case Eq. (4.85) simplifies to Eq. (4.28). In other words, maximum likelihood estimation is a special case of MAP, which, in turn, is an approximation of the Bayesian estimator.

Because it is difficult in practice to specify accurately the error covariance matrix of the a priori values, a weighting factor W is additionally introduced in conjunction with R_2. In such a case, it can be shown that Eq. (4.85) leads to

$$\left\{\sum_k \left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}\left[\frac{\partial y(t_k)}{\partial \Theta}\right] + W\,R_2^{-1}\right\}\Delta\Theta = \sum_k \left[\left[\frac{\partial y(t_k)}{\partial \Theta}\right]^T R^{-1}[z(t_k) - y(t_k)]\right] - W\,R_2^{-1}(\Theta - \Theta^*) \tag{4.86}$$

Introduction of the weighting factor W provides a convenient means to study the effect of a priori values on the estimates. It is obvious from Eq. (4.86) that $W = 0$ implies suppression of the a priori values and $W \to \infty$ implies suppression of the measurements. In the latter case, the algorithm will neglect the measurements completely, yielding the a priori values as the estimates. This is a hypothetical case because if we are 100% sure that the a priori values are perfect, then we do not have to worry about the estimation from experimental data. This is in contradiction to our hypothesis. We have gone through the whole exercise of statistically estimating the parameters because we want to improve upon the accuracy of the estimated parameters and the model structure from the experimental data. This approach could, however, help to avoid convergence problems encountered when the parameters are near linearly dependent.

In practice, if the a priori information is to be included in the estimation, it is usually done for system parameters (that is, aerodynamic derivatives) only. When this option is invoked, it is necessary to specify the a priori values and corresponding error covariances along with the values for parameters appearing in the state equations. In practice, specifying an appropriate error covariance matrix R_2 for the a priori values and of the weighting factor W is difficult. It is this aspect that restricts the utility of this option. In general, this option is rarely used in flight vehicle system identification.

4.17 STATISTICAL ACCURACY OF PARAMETER ESTIMATES

Having applied the statistical estimation methods, the first and the most natural question that comes to our mind is about the statistical accuracy of the estimates that we have obtained from the experimental data. In other words, we would like to gain some appreciation of the confidence that we might have in the estimates obtained by applying the maximum likelihood or Bayesian estimator. In some sense the issue of accuracy of the estimates belongs to the category of tests that are required to evaluate the model adequacy and validation. We will dwell on this aspect separately in Chapter 11 dealing with various procedures and practical issues of model validation. However, for the sake of completeness, we cover here only the mathematical basics of statistical evaluation of the estimates.

From Sec. 4.3 and Appendix D, we know that the maximum likelihood estimator is asymptotically efficient in the sense of achieving the Cramér–Rao lower bound. It follows from Eq. (D35) that the parameter error covariance matrix P is given by the inverse of the Fisher information matrix $\mathcal{F}$ [Eq. (4.30)].

$$P = \left\{ \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta} \right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta} \right] \right\}^{-1} \tag{4.87}$$

If a priori information, as discussed in Sec. 4.16, is included in the estimation, it is required to use the system matrix, that is, terms within the flower bracket on the left-hand side, of Eq. (4.86) while computing the preceding parameter error covariance matrix. The standard deviations and correlation coefficients of the estimates are given by

$$\sigma_{\Theta i} = \sqrt{p_{ii}} \tag{4.88}$$

$$\rho_{\Theta i \Theta j} = \frac{p_{ij}}{\sqrt{p_{ii} p_{jj}}} \tag{4.89}$$

The property of asymptotic efficiency is of practical significance. It implies that the ML estimator makes efficient use of the available data and that the Cramér–Rao lower bound indicates the theoretically maximum achievable accuracy of the estimates. Although Eqs. (4.88) and (4.89) provide a simple way to obtain theoretical information about the accuracy of the estimates and correlation among them, unfortunately, in practical applications this issue is a little more complex and not without ambiguities. We will discuss these implications separately in Chapter 11 dealing with model validation.

Because the information matrix $\mathcal{F}$ is required in the optimization of the cost function, the parameter error covariance matrix is automatically available without any extra computations. The standard deviations and correlation coefficients can be readily obtained. Thus, information on the statistical accuracy of the estimates is a byproduct of the Gauss–Newton or Levenberg–Marquardt methods. Other optimization methods we had just briefly mentioned, namely, Powell's, Jacob's,

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simplex, and subplex, do not provide this information directly, and this is one of the limitations of these methods, besides being very slow in convergence.

4.18 ALGORITHMIC IMPLEMENTATIONS

The computational demands for large-scale systems and huge amounts of measured data required for global model identification are significant and can become prohibitive; the job can run to hours of CPU time [56]. Therefore, special emphasis is placed on computational details to achieve reasonable turnaround time in such cases. It makes use of the special structure of the information matrix resulting from multiple time segments (maneuvers) being processed.

As already elaborated in the preceding sections, the iterative unconstrained or bounded-variable Gauss–Newton methods require solving a linear algebraic equation once and the Levenberg–Marquardt twice per iteration. However, depending upon the number of unknown parameters being estimated, the dimension of the information matrix could be large, and the main computational burden is not due to solving this algebraic equation required for matrix inversion; see Eq. (4.29). The major portion of the computational time is consumed in computing the $n_y \times n_q$ dimensional sensitivity coefficients $\partial y/\partial \Theta$ and the $n_q \times n_q$ dimensional information matrix $\mathcal{F}$ over N data points; see Eq. (4.30).

As a typical case, we consider an example pertaining to the six-degree-of-freedom (DOF) equations of motion, with number of states $n_x = 10$, number of inputs $n_u = 12$, number of outputs $n_y = 19$, and the number of parameters to be estimated $n_q = 30$. Throughout the following comparative investigations, we use the fourth-order Runge–Kutta integration method. We perform five different evaluations using different number of maneuvers analyzed simultaneously each time, namely, $n_E = 10$, 20, 30, 40, and 60, with $N = 13{,}424$; 25,453; 37,720; 51,144; and 75,724 data points, respectively. In each case the initial conditions are specified appropriately (as average of first few points of measured data, see Sec. 3.7) and held fixed, and no other bias parameters are considered. Thus, we have in each case the same number of unknown parameters $n_q = 30$, and the information matrix is 30×30. On a workstation with 3.4-GHz CPU, they required on an average 9.1, 17.4, 25.7, 34.8, and 52 s per iteration to compute the information matrix $\mathcal{F}$ and gradient vector $\mathcal{G}$ (time linearly increasing with the number of data points) whereas the matrix inversion took negligible time (less than 0.01 s).

Now, let us consider estimating the initial conditions for each of the n_E time segments analyzed simultaneously. Then, the total number of parameters n_q is given by $30 + n_x \times n_E$, where in the present case the number of states is $n_x = 10$. Thus, in the five cases it amounts to estimating 130, 230, 330, 430, and 630 parameters, from which each time 30 are the aerodynamic derivatives. On the same 3.4-GHz workstation they required roughly 20, 78, 214, 485, and 1518 s per iteration to compute the information matrix $\mathcal{F}$ and gradients $\mathcal{G}$. The

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information matrix $\mathcal{F}$ being symmetrical, we compute only the lower triangular part and then fill it up before inverting the same. Apart from this single consideration, no other special treatment is performed, implying that the buildup of $\mathcal{F}$ and $\mathcal{G}$ is done in a straightforward fashion using Eq. (4.30) for all n_q parameters. The matrix inversion required from 0.01 to 3 s is once again a negligible part of the overall computations even for the last case with 630×630 size matrix.

Thus, having substantiated our statement made earlier that the major portion of the computational time is consumed in computing the $n_q \times n_q$ information matrix, we now look for the ways to optimize computations to reduce the overall CPU time. If only aerodynamic parameters are estimated, keeping initial conditions and other bias parameters fixed, then no optimization is possible irrespective of how many n_E maneuvers are analyzed simultaneously. As just demonstrated, in such a case for a given model postulate the CPU time is linearly proportional to the number of data points. However, if initial conditions or bias parameters are also estimated, then it turns out that through a judicious implementation of the computations in the $\mathcal{F}$ matrix buildup pertaining to n_E time segments, significant reduction in the computational burden can be achieved. This becomes possible because concatenation of several time slices leads to a special block structure for the information matrix with respect to the initial conditions and other bias parameters. For illustration purposes, just three time segments and for each time segment the initial conditions and bias parameters of the observation equations are considered. The resulting matrix structure is shown schematically in Fig. 4.4. The nonzero elements appear in gray blocks

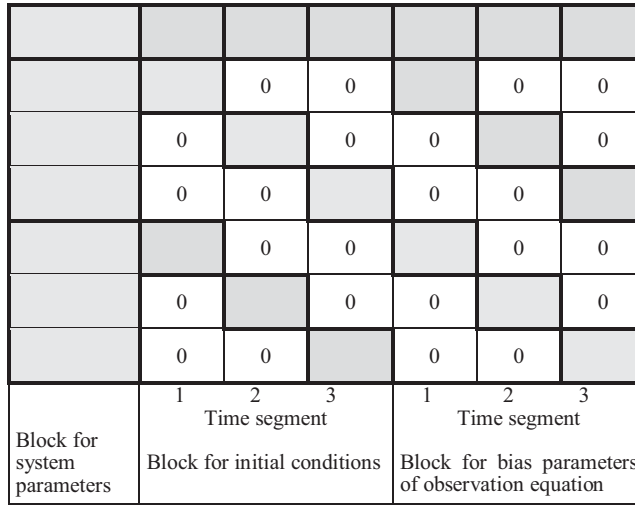

Fig. 4.4 Schematic of block structure of the information matrix for three time slices analyzed simultaneously.

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with bold borders. The sparse matrix structure can be used advantageously to reduce the numerical computations while computing the $n_q \times n_q$ dimensional information matrix [57]. Furthermore, a part of the computation is done judiciously just once, namely, $[\partial y/\partial \Theta]^T R^{-1}$, which is common to both $\mathcal{F}$ and $\mathcal{G}$.

The use of this optimization led to CPU-time requirements of 12, 23, 35, 48, and 70 s per iteration for the five cases. Compared to the CPU times of 20, 78, 214, 485, and 1518 s using straightforward approach, the reduction is significant. The larger the number of time segments, the larger is the ratio of time saving. Because it is usually preferable to estimate the initial conditions and bias parameters, and the use of a large number of time segments is not uncommon, as well as 10 or more iterations are typically needed for the convergence, a significant amount of reduction in the computational overhead can be achieved by making optimum use of the special block structure for multiple maneuver evaluations. This issue is not so relevant for small-size problems, but it can become critical to ensure reasonable turnaround time during model development process for large-size problems encountered during global model identification addressed in Sec. 12.8.

Yet another special implementation feature pertains to avoiding large-size matrices through judicious implementation of computational sequence. Several estimation programs are based on computing system responses and those for the perturbed parameters required for approximating gradients, in a sequential way. In this direct approach, one parameter is perturbed at a time, and simulation is performed at a stretch over the complete set of N data points. The process is repeated n_q times, storing the responses in separate data arrays each time. This so-called sequential processing, shown on the left-hand side of Fig. 4.5, needs $(n_q + 1)$ arrays, each of size $(N \times n_y)$. At the end of such $(n_q + 1)$ simulations, the gradients are computed from these responses using Eq. (4.33) and summed over N data point according to Eq. (4.30). Thus, depending upon the number of data points N, the program might need very large size arrays; a program capable of analyzing up to 80,000 data points and 60 observations, 30 states, 60 inputs, 1200 parameters has been used in practice; in a rare case up to 150,000 data points or more have also been processed at a time.

For such large-size problems, the following alternative implementation, called pseudo-parallel processing, leads to substantial reduction in the storage needed. As depicted on the right-hand side of Fig. 4.5, instead of performing the simulations at a stretch over N data points for each perturbation, we handle each data point separately. At each data point, we perturb each of the n_q parameters, compute the responses, compute the gradients according to Eq. (4.30), and incrementally build up the matrices $\mathcal{F}$ and $\mathcal{G}$. Having done this for all n_q parameters, then we jump to the next data point, restore the perturbed states for continuation, and repeat the process to process the next point, until all of the data points are processed. Restoration of appropriate states at each point needs saving perturbed states for each parameter, requiring array of the size $n_x \times n_q$. This pseudoparallel approach characterized by horizontal processing at each point needs just two

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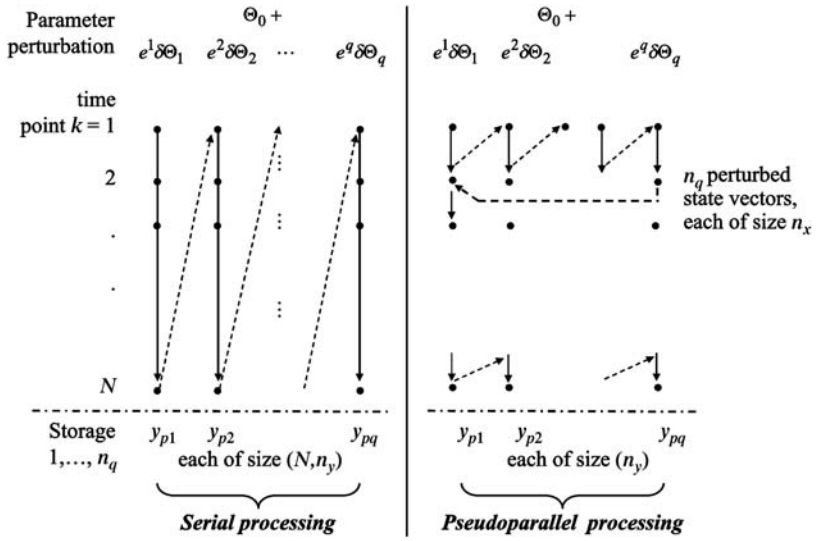

Fig. 4.5 Two different computational sequences for perturbed simulations.

arrays, one of size $(n_y \times n_q)$ and the other of $(n_x \times n_q)$. Compared to the direct implementation discussed earlier, requiring arrays of total size $(N \times n_y \times n_q)$, the storage requirements of the second approach are negligible. We have called this approach as pseudoparallel because, strictly speaking, the terminology of parallel processing has, in general, a different connotation, implying several simulations running at the same time whereas we still perform perturbations and simulations sequentially, but the order of performing the same over N data points and for n_q has been interchanged, leading to remarkable saving in the storage requirements.

4.19 OEM SOFTWARE

There are so very many ways to implement the output-error method; attempts to standardize the software program will be in vain because the model postulates vary from case to case. (Remember, we mostly deal with a general nonlinear system of arbitrary complexity.) Furthermore, the format of flight data to be analyzed will vary from organization to organization, more likely from project to project within the organization itself. Nevertheless, the basic parameter estimation algorithm remains the same. We provide in this book an extended version of software for general nonlinear systems. It caters to estimating not only the system parameters, time delays, and zero shifts, but also the initial conditions from multiple experiments. A more sophisticated estimation algorithm called

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bounded-variable Gauss–Newton algorithm, discussed in Sec. 4.11, is also provided. In the case of linear systems, the bias parameters of the state and observation equations, as discussed in Sec. 3.5.2, can be included and estimated along with the system parameters. The software is quite adequate to analyze the test cases that are considered in this and in few other chapters, as well as for advanced and more complex applications. At the same time, it allows us to understand more closely the various algorithmic steps that we have so far studied in this chapter. To conform to the current trends, the provided software runs under Matlab® [58]. In general, the current trend shows that in the future more and more flight vehicle system identification work will be carried out under the Matlab® operating environment.

The extended version of the software caters to estimation of the following: 1) system parameters common to all time segments being analyzed, 2) initial conditions separately for multiple time segments being analyzed, 3) time delays, and 4) bias parameters separately for multiple time segments being analyzed simultaneously. These features have been achieved by treating the complete set of parameters into three different sets, namely, 1) system parameters (such as, for example, aerodynamic derivatives) common to all time segments, 2) initial conditions x_0 for each of the n_E experiments, and 3) separate bias parameters for each of the n_E experiments. If desired, the bias parameters can also be clubbed in the first category and treated as common to all time segments. This choice is left to the analyst. Note: because of the preceding extensions, which lead to highly flexible software and expand the scope significantly, the user interface to define the model has changed compared to that in the first edition of the book. As such, it is imperative that the model definition functions have to be adapted to the new structure and user interface.

Other estimation software packages are available as Matlab® Toolboxes based on restricted circulation or at a license fee [59, 60]. It is also possible to include models developed in Matlab/Simulink® using the block-diagram functionality [60]. Matlab® and Simulink provide an easy-to-use computing and modeling description environment. We do not go into the details of such software tools, except to say that they are based on the basic theory that we have covered in this book. More sophisticated software tools in FORTRAN language have been developed in the past and extensively used for flight vehicle system identification, for example, [57, 61, 62] among others. Here too, we do not go into any further details of these software tools.

Figure 4.6 shows a flowchart of the computational procedure for the output-error method. The function names for various parts are indicated at the bottom of each block. The source codes (Matlab m-files) for output-error method are provided in the directory /FVSysID2/chapter04/ and a README providing the details of various functions and test cases covered here. The starting point is the main program called "ml_oem2" and provides the user interface to define the model and to choose the optimization and integration methods and other information.

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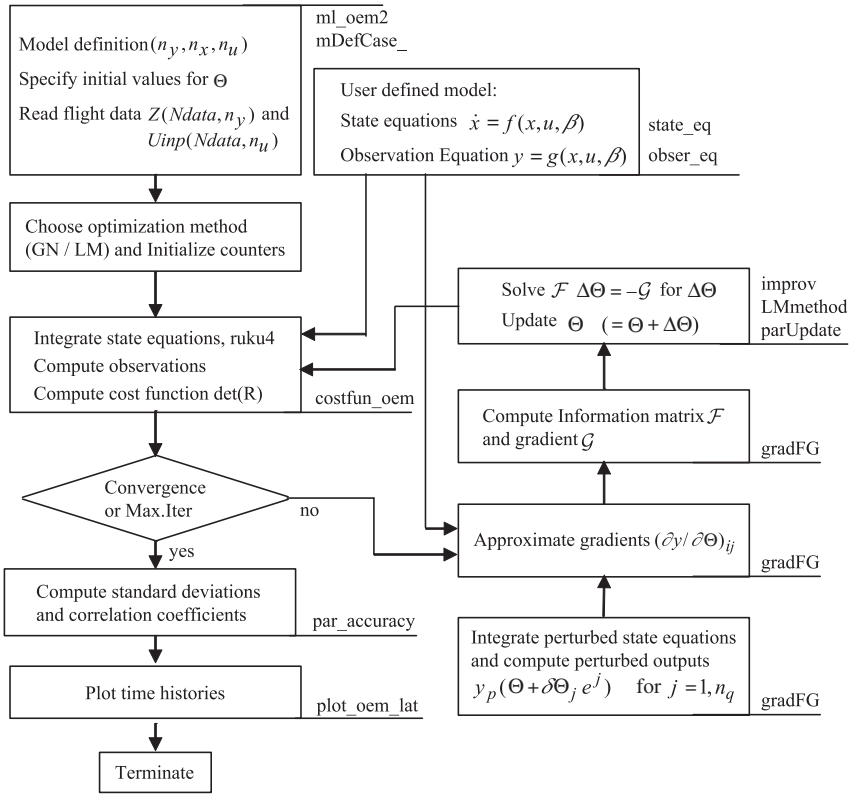

Fig. 4.6 Details of implemented output-error method.

test_case	index for the test case
iOptim	index for the optimization method 1: Gauss Newton; 2: Levenberg–Marquardt
integMethod	flag for integration method (ruku4, ruku3, ruku2)
niter_max	maximum number of iterations
tolR	relative change in the cost function for termination
par_step	parameter perturbation size for gradient approximations

Default values are defined for the maximum number of iterations niter_max, for the convergence criterion tolR and for the step size for parameter perturbations par_step. They can be suitably changed, if necessary.

A particular system model to be analyzed is denoted through the integer flag "test_case," which is to be uniquely associated with the state-space model and with the user-provided interface in the form of a function defining the model parameters, flight data, and other relevant details. Different state-space

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models, to be coded by the user, are invoked by defining the flags and strings in a function mDefCase** called from "ml_oem2," where** denotes the test case index:

state_eq	function name: to code the right-hand side of state equations
obser_eq	function name: to code the right-hand side of observation equations
Nx	number of state variables (n_x)
Ny	number of output variables (n_y)
Nu	number of input variables (n_u)
NparSys	number of system parameters
Nparam	total number of unknown parameters
NparID	total number of parameters to be estimated (free parameters) (n_q)
param	starting values for unknown parameters (Θ)
parFlag	flags for free and fixed parameters = 1: free parameters (to be estimated) = 0: fixed parameters (not to be estimated)
param_min	lower bounds on the parameters; $= -\infty$ for unconstrained optimization
param_max	upper bounds on the parameters $= \infty$ for unconstrained optimization
Nminmax	number of lower and upper bounds
x0	starting values for initial conditions on state variables
parFlagX0	flags for free and fixed initial conditions = 1: free parameters (to be estimated) = 0: fixed parameters (not to be estimated)
NX0ID	total number of free (to be estimated) initial conditions
bXpar	starting values for bias parameters for each time segment
parFlagBX	flags for free and fixed bias parameters for each time segment = 1: free parameters (to be estimated) = 0: fixed parameters (not to be estimated)
NbX	total number of (fixed and free) bias parameters per time segment
NBXID	number of free (to be estimated) bias parameters
iArtifStab	flag for artificial stabilization (=0 for OEM)
StabMat	artificial stabilization matrix
LineSrch	flag for line search option in the Gauss–Newton method (=0, No line search, =1, Line search)

The variable names used in the program "ml_oem2" and calling functions are shown in the preceding list on the left-hand side, followed by the description and the notation used to denote these variables in the text. The option of artificial stabilization is not relevant for the present chapter, but it is required in Chapter 9,

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which extends the basic OEM method to unstable aircraft. In all examples covered in this book, except for those in Chapter 9, the flag iArtifStab and the matrix StabMat are dummy variables, and we initialize them to zero and an empty matrix.

A couple of relevant indices such as: 1) total number of parameters to be estimated (free parameters) NparID, 2) the number of lower and upper bounds Nminmax, 3) total number of free initial conditions NX0ID, 4) total number of (fixed and free) bias parameters per time segment NbX, and 5) number of free (to be estimated) bias parameters NBXID can be readily computed from the size of the corresponding flags. The default values for lower and upper bounds on the parameters are $-\infty$ and ∞ respectively, implying unconstrained optimization. The flag LineSrch allows choosing between the two approaches of overcoming the intermediate divergence discussed in Secs. 4.10.1 and 4.10.2. Setting LineSrch equal to zero implies heuristic approach whereas setting it to one implies applying the line search option. The flag LineSrch is relevant only when the integer flag for optimization method iOptim is set to one, that is, when the Gauss–Newton method is to be applied.

It is obvious that model specification in terms of (Nx, Ny, Nu, Nparam) must match with those coded in the two user functions for the postulated model. It is also required to specify suitable starting values for the parameters Θ and the integer flags parFlag to indicate whether the particular parameter is free (to be estimated) or fixed (not to be estimated). Any arbitrary combination of free (to be estimated) and fixed system parameters is possible and can be suitably specified by parFlag. The same procedure of specifying stating values and the flags to keep them free or fixed is required for the initial conditions x_0 and the bias parameters bXpar. They are to be specified for n_E time segments being analyzed simultaneously. The corresponding flags parFlagX0 and parFlagBX can be specified in any arbitrary combination. This option to handle the three sets of parameters, as either unknowns or as known, or not identifiable from data analyzed, and hence to be kept fixed, makes the software flexible. For example, the function codes for the state and observation equations with larger number of parameters can be used without modifications to estimate a subset of them, depending upon the data available.

The flight data to be analyzed are also to be loaded in the function for model definition and require specification or assignment of the following information:

Ndata	number of data points
dt	sampling time
t	time vector [=0 to (Ndata-1)*dt]
Z(Ndata,Ny)	flight data for measured outputs (N, n_y)
Uinp(Ndata,Nu)	flight data for measured control inputs (N, n_u)

While loading the flight data in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu), it is obvious that the units of the variables must match with those used in the

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postulated model and programmed in the state and observation functions "state_eq" and "obser_eq," respectively.

Because we provide an option to analyze multiple experiments, it is required to define additionally the following time-segment related information:

Nzi	number of time segments to be analyzed simultaneously (n_E)
izhf	cumulative index at which the maneuvers end when concatenated.

For example, if we analyze three time segments, each having Nts1, Nts2, and Nts3 data points, respectively, then izhf is defined as [Nts1; Nts1 + Nts2; Nts1 + Nts2 + Nts3]. The total number of data points, Ndata, is just the sum of data points for each of the Nzi segments. In this chapter, we will be dealing with test cases analyzing a single maneuver at a time, but the option of multiple experiments will be invoked in examples covered in other chapters.

From Fig. 4.6, it can be seen that the function "costfun_oem" integrates the states and computes the observations and the cost function value; the function "gradFG" propagates the perturbed state equations, computes the perturbed system responses, and computes the gradient vector $\mathcal{G}$ and information matrix $\mathcal{F}$; the parameter updates $\Delta\Theta$ are computed in the function "improv" when the Gauss–Newton method is applied; otherwise, the function "LMmethod" for the Levenberg–Marquardt method is called. Depending upon whether lower and upper bounds are specified or not, as determined by Nminmax, either the unconstrained Gauss–Newton method or the bounded-variable Gauss–Newton method is applied. We prefer to use Cholesky factorization to solve for $\Delta\Theta$; other methods, if desired by users, can be easily invoked in these functions. The convergence checking is performed in "ml_oem2," so also the step-size control including halving of the parameter steps to overcome intermediate divergence. At the end, we compute the standard deviations and correlation coefficients in "par_accuracy" and finally make plots of the time histories and of estimates.

For each test case, a program is provided to plot time histories of the control variables and generate superimposed plots of measured and computed output variables that allow direct comparison. In a separate figure, the estimates plotted against iteration count show the convergence of the optimization method. The vertical bars are provided, showing the standard deviations. The plot programs are called "plots_TCnn_oem_ttt.m," where nn refers to the test case index and ttt to additional explanation usually giving details of model or aircraft analyzed.

4.20 EXAMPLES

Although we will cover some typical examples related to flight vehicles in a separate chapter, we consider here two simple examples to demonstrate the output-error method and the use of the software provided. The first example

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pertains to analysis of flight data recorded during an aileron and rudder input maneuver, performed under seemingly steady atmospheric conditions. To this example, we will apply the Gauss–Newton, bounded-variable Gauss–Newton, and Levenberg–Marquardt methods. The second example also pertains to the lateral-directional motion, analyzing simulated data that have been generated with a moderate to high level of atmospheric turbulence. The test cases are selected through the flag "test_case" and run using the program "ml_oem2." The flight data analyzed in these cases are supplied to the readers as sample flight data for verification and to try out other options and possibly to verify software that they might modify or develop themselves. These data files are found under the directory /FVSysID2/flt_data/. More complex test cases are provided in Chapter 10, which provide details of analyzing multiple time segments and estimating initial conditions and bias parameters for each time segment separately.

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4.20.1 LATERAL-DIRECTIONAL MOTION

The model pertaining to the lateral-directional motion is postulated as follows:

State equations:

$$\begin{aligned} \dot{p} &= L_p p + L_r r + L_{\delta a}\delta_a + L_{\delta r}\delta_r + L_\beta \beta + b_{x\dot{p}} \\ \dot{r} &= N_p p + N_r r + N_{\delta a}\delta_a + N_{\delta r}\delta_r + N_\beta \beta + b_{x\dot{r}} \end{aligned} \tag{4.90}$$

Observation equations:

$$\begin{aligned} \dot{p}_m &= L_p p + L_r r + L_{\delta a}\delta_a + L_{\delta r}\delta_r + L_\beta \beta + b_{y\dot{p}} \\ \dot{r}_m &= N_p p + N_r r + N_{\delta a}\delta_a + N_{\delta r}\delta_r + N_\beta \beta + b_{y\dot{r}} \\ a_{ym} &= Y_p p + Y_r r + Y_{\delta a}\delta_a + Y_{\delta r}\delta_r + Y_\beta \beta + b_{yay} \\ p_m &= p + b_{yp} \\ r_m &= r + b_{yr} \end{aligned} \tag{4.91}$$

where the subscript m denotes the measured variables; p the roll rate; r the yaw rate; a_y the lateral acceleration; $\dot{p}$ the roll acceleration; $\dot{r}$ the yaw acceleration; δ_a the aileron deflection; δ_r the rudder deflection; β the angle of sideslip; and $L_{(\,)}$, $N_{(\,)}$, and $Y_{(\,)}$ the dimensional aerodynamic derivatives. As discussed in Sec. 3.5.2, the postulated model contains bias terms for the state and observation equations $b_{x(\,)}$ and $b_{y(\,)}$, respectively. The readers might recall that the lumped bias parameters provide a convenient means to appropriately account for the initial conditions and systematic errors in the measurements. Besides the aileron and rudder inputs, as discussed in Sec. 3.6, the angle of sideslip β is treated as a pseudo control input. The unknown parameter vector Θ consisting of the

dimensional derivatives and the bias parameters is given by

$$\Theta^T = [L_p \;\; L_r \;\; L_{\delta a} \;\; L_{\delta r} \;\; L_\beta \;\; N_p \;\; N_r \;\; N_{\delta a} \;\; N_{\delta r} \;\; N_\beta \;\; Y_p \;\; Y_r \;\; Y_{\delta a} \;\; Y_{\delta r} \;\; Y_\beta \;\; b_{x\dot{p}} \;\; b_{x\dot{r}} \;\; b_{y\dot{p}} \;\; b_{y\dot{r}} \;\; b_{yay} \;\; b_{yp} \;\; b_{yr}] \tag{4.92}$$

Thus, the postulated model is defined as follows:

	No. of Variables		Function Name
States	2	p, r	"xdot_TC01_attas_lat"
Outputs	5	$\dot{p}$, $\dot{r}$, a_y, p, r	"obs_TC01_attas_lat"
Inputs	3	δ_a, δ_r, β	

The right-hand sides of the state equation, Eq. (4.90), are programmed in the function "xdot_TC01_attas_lat" and those of the observation equation, Eq. (4.91), in "obs_TC01_attas_lat." Thus, the model definition for this example provided by the function "/FVSysID2/chapter04/mDefCase01.m," which is called from the main program "/FVSysID2/chapter04/ml_oem2" is as follows:

```
test_case = 1;                          % index for the test case
state_eq  = "xdot_TC01_attas_lat";      % function for state equations
obser_eq  = "obs_TC01_attas_lat";       % function for observation equations
Nx        = 2;                          % number of states
Ny        = 5;                          % number of observation variables
Nu        = 3;                          % number of input (control) variables
NparSys   = 15;                         % number of system parameters
Nparam    = NparSys + Nx + Ny;          % total number of parameters
dt        = 0.04;                       % sampling time
```

The flight data analyzed were gathered from flight tests carried out with the research aircraft ATTAS [63]. The rolling and yawing motions were excited in a sequence through pilot-applied aileron and rudder inputs. The flight maneuvers were carried out under seemingly steady atmospheric conditions when the turbulence was considered negligible. The data were recorded with a sampling time of 40 ms, and the duration of the maneuver is 60 s. The resulting lateral-directional motion was recorded and analyzed as a single maneuver. The data to be analyzed are loaded from the data file "/FVSysID2/flt_data/fAttasAilRud1.mat." The number of time segments is given by Nzi = 1, and the izhf is set to the total number of data points. The output and input variables in data arrays Z(Ndata,Ny) and Uinp(Ndata,Nu) are appropriately assigned; a list of channels recorded is provided in the README. Suitable starting values are specified for the unknown parameters; they are mostly 50% off the expected values. The integer flags parFlag are set to one, indicating that all parameters are to be estimated. The lower and upper bounds are set to $-\infty$ and ∞

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respectively, leading to unconstrained optimization. All of these details can be easily traced from the model definition function "mDefCase01.m." In the main program, the flag integMethod is set to ruku4 for the fourth-order Runge–Kutta integration method.

4.20.1.1 GAUSS–NEWTON AND LEVENBERG–MARQUARDT METHODS

The results of parameter estimation applying the output-error method are summarized in Table 4.1, showing convergence obtained from the Gauss–Newton and Levenberg–Marquardt methods. By setting the flag iOptim = 1 (in ml_oem2.m) and LineSrch = 0 (in mDefCase01.m), we choose first the Gauss–Newton optimization and heuristic approach to cater to intermediate divergence. It is observed that the Gauss–Newton method shows intermediate divergence during the first iteration. As pointed out in Sec. 4.10.1, the heuristic approach of halving the parameter update proved quite sufficient to reduce the cost function through a single step. The subsequent convergence was smooth, and the process was terminated when the relative change in the cost function, that is, determinant of the covariance matrix R was less than tolR = 0.0001. The convergence is achieved in six iterations, and the cost function values are provided in the second column of Table 4.1.

In the second attempt, we continue applying the Gauss–Newton method, but choose the line search strategy to correct local intermediate divergence (selected by setting the integer flag LineSrch = 1 in mDefCase01.m). We start with the same initial parameter values and the cost function. It is observed that the line

TABLE 4.1 CONVERGENCE OF GAUSS–NEWTON WITH HEURISTIC APPROACH AND LINE SEARCH AND LEVENBERG–MARQUARDT METHODS FOR TEST CASE 1

Iteration	Gauss–Newton: Cost Function	Gauss–Newton with Line Search: Cost Function	Levenberg–Marquardt: Cost Function (LM Parameter)
0	1.3568×10^{-11}	1.3568×10^{-11}	1.3568×10^{-11}
1	3.6039×10^{-10} 6.7975×10^{-15}	2.2964×10^{-15}	3.5663×10^{-10} (λ: 1.0×10^{-2}) 3.2505×10^{-10} (λ: 1.0×10^{-1}) 4.0265×10^{-13} (λ: 1.0×10^{-1})
2	2.437×10^{-19}	2.1453×10^{-19}	8.5981×10^{-18} (λ: 1.0×10^{-2})
3	9.3597×10^{-20}	9.3548×10^{-20}	1.6038×10^{-19} (λ: 1.0×10^{-3})
4	9.1717×10^{-20}	9.1706×10^{-20}	9.2428×10^{-20} (λ: 1.0×10^{-4})
5	9.1702×10^{-20}	9.1702×10^{-20}	9.1707×10^{-20} (λ: 1.0×10^{-5})
6	9.1702×10^{-20}		9.1702×10^{-20} (λ: 1.0×10^{-6})

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search option automatically overcomes the intermediate divergence during the first iteration. For this simple case the convergence is achieved in five iterations. The cost function values during the optimization in this case are tabulated in the third column of Table 4.1.

In the case of the Levenberg–Marquardt method (selected by setting the integer flag iOptim = 2 in ml_oem2.m), we start once again with the same initial parameter values and cost function and the default value for the LM parameter $\lambda = 0.001$, as suggested in Sec. 4.13. During the first iteration, intermediate divergence is encountered; at this stage the algorithm adapts the LM parameter λ until a reduction in the cost function is achieved. In the present case, this required three steps, changing λ from 0.001 to 0.01; note that during the third step of LM-parameter adjustment, λ is not reduced as a different condition is satisfied here; see Sec. 4.13. The λ values are shown in brackets in the fourth column of Table 4.1. During the subsequent iterations, it turns out that the Gauss–Newton directions are the best choice and the optimization progresses smoothly; hence, λ is reduced continuously from iteration to iteration.

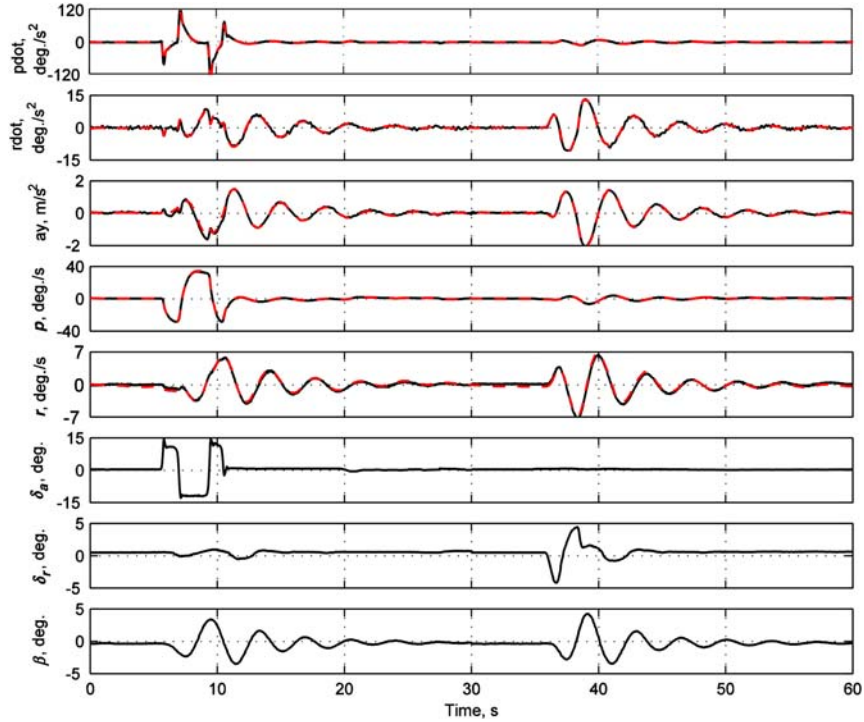

Fig. 4.7 Time histories of control inputs and a comparison of flight measured and model predicted outputs for test case 1: ——, flight measured; -----, model predicted.

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The time history match is shown in Fig. 4.7 only once because there was no qualitative difference between the three methods. The three plots from the bottom show the control inputs, namely, the angle of sideslip and the rudder and aileron deflections, respectively. The five plots from the top show the match between the flight measured responses (shown in continuous lines) and the model estimated responses (shown in dashed). The match between the two is excellent. The computations in the postulated model are carried out in radians and radians/s whereas the time histories are plotted in degrees and degrees/s.

It is observed that the Gauss–Newton method with line search option requires one iteration less for convergence. But in general, for more complex cases, it is not necessary that the Gauss–Newton method with line search option will perform better, requiring less iterations for convergence. In a few cases, we have also observed that the line search option requires the same number of iterations and rarely one or two iterations more. Thus, the choice of heuristic approach or line search is not quite obvious. In most of the cases, we have been using the heuristic approach, which is found to be quite adequate, even for much more complex problems. Convergence applying the Levenberg–Marquardt method is achieved in the same number of iterations as that by the Gauss–Newton method with heuristic approach for intermediate divergence, which is more or less a coincidence. Depending upon the starting values and the nature of the cost function, the two methods may progress towards the minimum differently and may require different number of iterations.

The convergence plots for the derivatives $L_{()}$ and $N_{()}$ obtained from the Gauss–Newton method are provided in Fig 4.8, showing the error bars, that is, standard deviations, given by Eq. (4.88). It is observed that all of the parameters are estimated with very low standard deviations and there was no correlation between the aerodynamic derivatives. Numerical values and the standard deviations are not provided here, but it can be easily verified that the three methods yield the same results within the numerical accuracy and roundoff errors.

4.20.1.2 BOUNDED-VARIABLE GAUSS–NEWTON METHOD

We continue with the same example to demonstrate the use of bounded-variable Gauss–Newton method discussed in Sec. 4.11. For this purpose we arbitrarily define lower and upper bounds on two parameters appearing in the model postulated in Eqs. (4.90) and (4.91); they are $(-6.0, -3.0)$ for the third parameter $L_{\delta a}$ and $(-5.0, -3.5)$ for the fifth parameter L_{β}. These limits serve the demonstration purpose only. Accordingly, in "mDefCase01.m," we set param_min(3) $= -6.0$, param_max(3) $= -3.0$, and param_min(5) $= -5.0$, param_max(5) $= -3.5$. The starting values of parameters for which lower and upper bounds are specified should lie within the range defined by the bounds. The rest of the settings in model definition function remain unchanged. Now, we rerun the case by starting ml_oem2.m. It is observed that starting from the same cost function value

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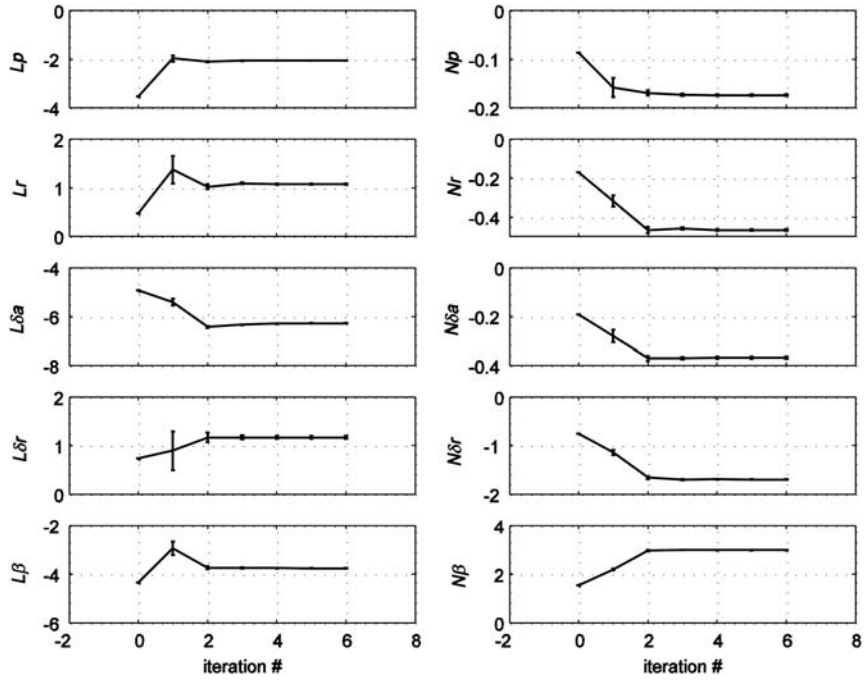

Fig. 4.8 Convergence plot of dimensional derivatives estimated from flight data for test case 1.

1.3568×10^{-11}, as in the preceding cases, the convergence is achieved within five iterations. The optimized cost function in this case is 1.3452×10^{-19}. At the end of the optimization, the log of final results shows the estimates of free parameters, their standard deviations, and the lower and upper bounds, if any are specified. If any parameter remains at any one of the bounds, this is indicated by "*" after the estimate in the printout. It helps to pinpoint that the results are obtained though constrained optimization. For parameters remaining at its bounds, one cannot compute the standard deviations.

The convergence plot of the various parameters is shown in Fig. 4.9. The lower and upper bounds are marked for the two parameters (see third and fifth plot on the left-hand side of Fig. 4.9). It is observed that parameter L_β hits the upper bound of -3.5 during the first iteration, but gets freed during the subsequent iterations. On the other hand, the parameter $L_{\delta a}$ hits the lower bound of -6.0 during the second iteration and remains at its bound until the convergence. As discussed in Sec. 4.11, the algorithm automatically caters to either keep the parameters fixed at its bounds or free them as the optimization progresses.

Comparing the results of optimization with those given in Table 4.1, we notice that the cost function value (1.3452×10^{-19}) in this case is somewhat larger than

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that obtained through unconstrained optimization (9.1702×10^{-20}). This behavior is typical of the constrained optimization. Any further reduction in the cost function would be possible only when the particular parameters are not constrained. Further, the number of iterations required for the convergence in the present case is the same as that for the unconstrained Gauss–Newton method with line search (third column, Table 4.1). In a couple of other more complex cases too, it was observed that the bounded-variable method required the same number of iterations. In a few problems, however, the constrained optimization by bounded-variable Gauss–Newton method required one or two additional iterations, particularly when the bounds were inappropriate. Thus, in general, some care is necessary while specifying the parameter bounds. In some cases cycling of active set, that is, parameters entering the active set on reaching the bounds followed immediately by an indication to leave the active set, can result, particularly as the minimum of the cost function is approached. The reason for this is twofold: 1) with the roundoff errors becoming dominant, the gradients computed using the numerical approximation are not sufficiently accurate; and 2) some parameters appearing in the model are approximately linearly dependent. This

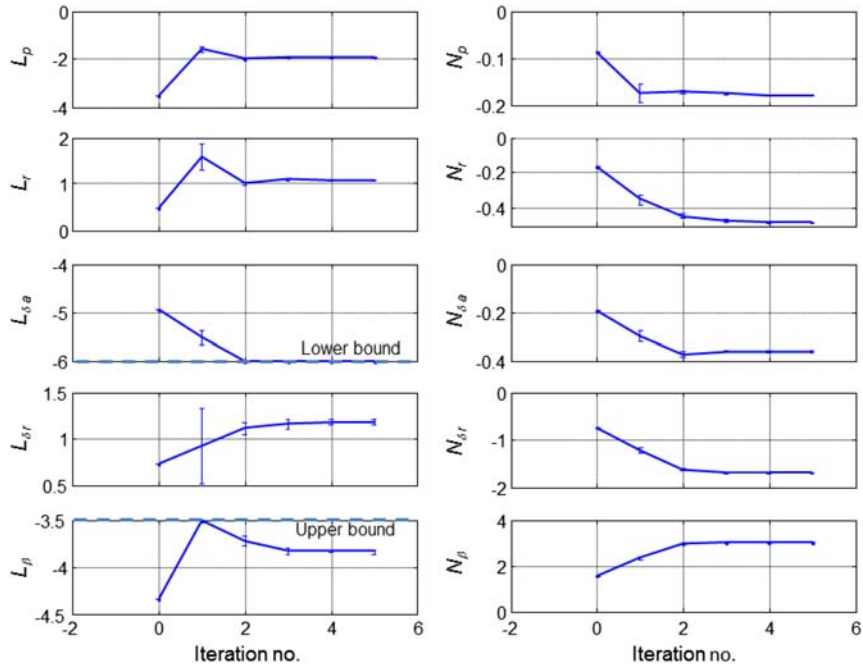

Fig. 4.9 Convergence plot of dimensional derivatives estimated applying bounded-variable Gauss–Newton method for test case 1.

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phenomenon was, however, not encountered in several examples of estimating nonlinear aerodynamic parameters with various degrees of complexity.

4.20.1.3 ESTIMATION OF TIME DELAY

To demonstrate the use of software to estimate time delay, we continue with the same example discussed hitherto. The model postulate for this case remains the same as in Eqs. (4.90–4.91). For demonstration purposes, we choose to estimate a time delay in the measured yaw rate. We define an additional parameter called τ_r, the time delay in yaw rate r, and append the parameter vector Θ in Eq. (4.92) with τ_r. We call this test case as test_case = 31. Thus, now the number of system parameters NparSys is set to 16 in the model definition mDefCase_TC31.m (instead of 15 in the first case). Because we wish to account for the time delay in the measured data, shifting of the integrated variable is to be carried out in the function for the observation variables, in the present case "obs_TC31_attas_lat.m." Herein, we call the utility function "timeDelay.m" to shift the computed yaw rate r, which is based on the delay matrix discussed in Sec. 3.4 as follows:

$$[rTz, xWS, tNewX, iNewX] = timeDelay(rTz, tDelay_r, xWS, tNewX, iNewX, nTdMx)$$

where rTz is the computed yaw rate to be time shifted by a parameter $tDelay_r$ being estimated and xWS, $tNewX$, $iNewX$ are the work space arrays associated with time delay. We assume that the maximum time delay is 40 times the sampling time and accordingly define $nTDMX = 40$. If this is not sufficient, the computations are terminated in the function timeDelay, with a warning message for the analyst. The exact details of calling the time delay function discussed in Sec. 3.4 can be traced in the function for the observations "obs_TC31_attas_lat.m." The time-delayed computed yaw rate is then matched with the measured data.

The state equations for test_case = 31 are provided in "xdot_TC31_attas_lat.m," which are exactly the same as those in "xdot_TC01_attas_lat.m" for test_case = 1. The observation equations in "obs_TC31_attas_lat.m" differ only slightly compared with "obs_TC01_attas_lat.m"; it includes changes to account for the call of the function "timeDelay.m." A major part of the model definition mDefCase_TC33.m remains the same, except for increasing number of system parameters NparSys by one, and setting starting value for τ_r to zero in the parameter vector param and setting the corresponding flag parFlag to one. Now choosing the index Test_case = 31 in the main program ml_oem2, we run this case. Because we start with the same values for the aerodynamic derivatives and the starting value for time delay is set to zero, the starting cost function values of 1.3568×10^{-11} is the same as before found in Table 4.1. We apply the Gauss–Newton method and note that convergence is very similar to that

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observed for the first example (test_case = 1), once again encountering intermediate divergence in the first iteration, which is corrected by the heuristic approach. The convergence is achieved in the same number of iterations. The cots function value is 9.0444×10^{-20}, which is slightly lower than 9.1702×10^{-20} for the first case. This is typical of parameter estimation; the more the number of parameters, the lower is the cost function and thereby better is the fit.

In this test case, it is also possible to set the time delay to zero and keep it fixed by setting the corresponding parFlag to zero in the model definition function "mDefCase_TC31.m." This yields exactly the same results given in Table 3.1 and nicely confirms that the procedure adopted to generate time delays does not affect the algorithm provided. The preceding simple example serves the only purpose of demonstrating estimation of time delay. In the present case, only one set of the global work spaces and indices *xWS*, *tNewX*, *iNewX*, *nTdMx* is necessary because we considered time delay in just one signal. However, if more than one signal is to be time shifted, recall from Sec. 3.4 that the calling procedure would be similar for each variable, and the work space variables have to be defined uniquely for each of them. A little more complex problem based on multiple time segments is presented later in Chapter 10.

4.20.2 APPLICATION OF OUTPUT-ERROR METHOD TO DATA WITH PROCESS NOISE

To evaluate the performance of more advanced estimation algorithms accounting for process noise that we will be addressing in the next chapter, aircraft responses pertaining to lateral-directional motion with an appreciable level of turbulence are generated through simulation [64]. The nominal values of the aerodynamic derivatives used correspond to those obtained from flight tests carried out in steady atmospheric conditions with the research aircraft de Havilland DHC-2. Equations (4.90) and (4.91), incorporating additional state and measurement noise, are used to generate the data. The angle of sideslip has been replaced through the lateral-velocity component v and the corresponding derivatives too. To provide realistic control inputs, the rudder and aileron excitations actually applied in a particular flight test are used to excite the postulated model. Independent process and measurement noise vectors are generated using pseudorandom noise generators. The state noise matrix was assumed to be diagonal. Note that state variables p and r only affected by the state noise and not the other control inputs δ_a, δ_r, and v. A total of 16 s of data with a sampling time of 0.05 s has been generated. These simulated responses are analyzed applying parameter estimation methods; see Fig. 4.10. Time histories of these data are provided as /FVSysID2/flt_data/y13aus_da1.asc.

To these data with process noise, we apply the output-error method. We use the model postulated in Eqs. (4.90) and (4.91) to estimate the unknown parameter vector of Eq. (4.91), except that, as already pointed out, the pseudo control

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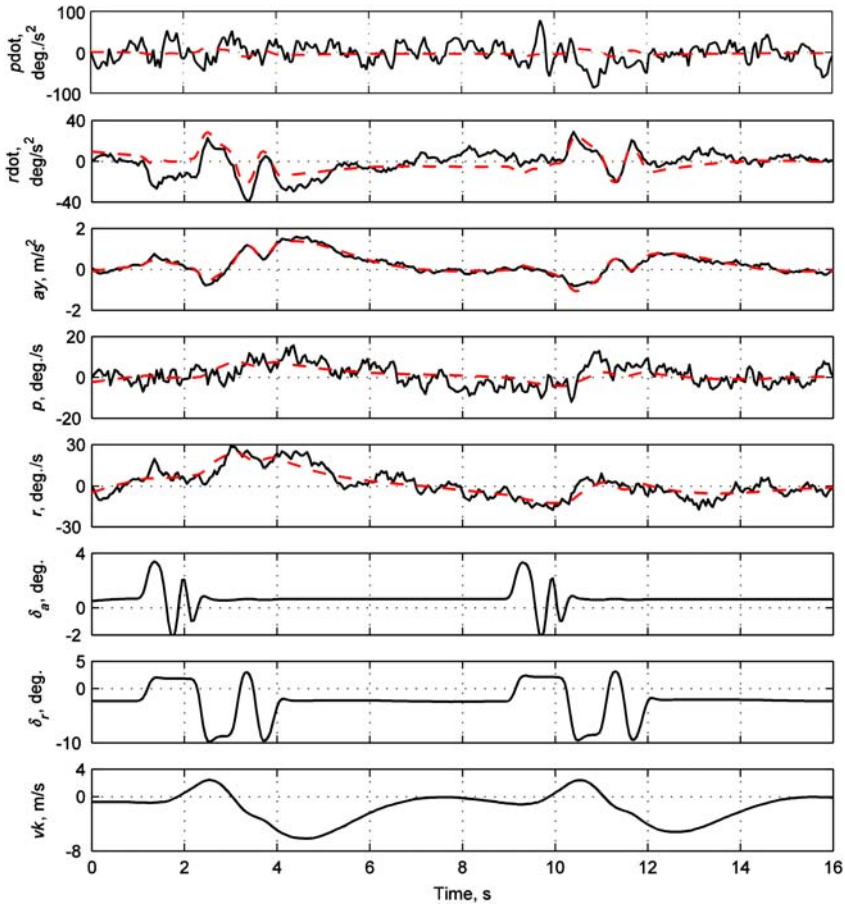

Fig. 4.10 Output-error method applied to data with moderate to high level of turbulence — test case 2: ——, flight measured; -----, model predicted.

input β has been replaced through v. This test case is designated as test_case = 2. The model definition is very similar to that of test_case = 1. The data to be analyzed, that is, the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu), are loaded from the file /FVSysID2/flt_data/y13aus_da1.asc. The model definition for this example is provided by the function "/FVSysID2/chapter04/mDefCase02.m," which is called from the main program "/FVSysID2/chapter04/ml_oem2" for test_case = 2. To apply the Gauss–Newton, we set the integer flag iOptim = 1.

In this case, it is observed that the output-error method has severe convergence problems. Almost at every iteration, intermediate divergence is encountered. The process terminates after 42 iterations, yielding parameter values

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that are not anywhere near the expected values. The estimated parameters are not provided here because we will get back to this case in Chapter 5, when we apply the filter-error method and make a comparison with the output-error method. The time history plots are provided in Fig. 4.10, showing the control inputs and the match for the output variables. The discrepancies in the match are clearly evident. Although this example is a simple one, it helps to demonstrate the limitations of the output-error method when we encounter process noise, that is, for flight data gathered under turbulent atmospheric conditions.

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4.21 CONCLUDING REMARKS

In this chapter we presented the general principle of maximum likelihood estimation, highlighting the exact physical interpretation, and showed that it can be applied to nonlinear dynamical systems of arbitrary complexity. It was shown that the maximum likelihood estimator has several desirable statistical properties, namely, unbiased estimates, asymptotic consistency, normality, and efficiency. The advantages of the method are that it makes optimal use of the measured data and as a byproduct provides, without extra computations, statistical information on the accuracy of the parameter estimates in terms of the standard deviations and correlation coefficients. The subtle issues associated with evaluation of these statistical accuracies are deferred to Chapter 11 dealing with model validation.

The output-error method is one of the most widely applied algorithms to estimate aircraft parameters. It considers measurement noise only, but this assumption simplifies the algorithm significantly. It is also considered acceptable from a practical viewpoint because we generally carry out flight tests specifically for parameter estimation purposes, and as such these flights can be carried out as well when the atmospheric conditions are stable. The relaxation strategy provides the best approach to optimize the likelihood function and overcomes convergence problems through the closed-form solution for the estimation of the measurement noise covariance matrix. We studied in some depth the three most commonly applied optimization methods, namely, unconstrained Gauss–Newton, bounded-variable Gauss–Newton, and Levenberg–Marquardt. Various options to overcome the problems of intermediate divergence in the unconstrained Gauss–Newton have been presented; these finer practical details are very important to enlarge the scope of application of these methods. Several other numerical aspects, like approximation of sensitivity coefficients, nonlinear model structures and structure of the sensitivity matrix for large-scale systems were discussed.

We also covered briefly the aspects of Bayesian estimation, the most general form of an estimator, and showed that a priori information about the aerodynamic derivatives can be accounted for in the output-error method. We further

argued that due to uncertainties in the error covariance matrix for such a priori values, the approach is rarely used in practice.

Finally, samples of flight data and extended software implemented under Matlab® have been provided to follow the various computational steps of the algorithm. It has been applied to the two examples that were presented to demonstrate the use of the software and to evaluate the performance of the output-error method. The extended software provides options to estimate system parameters and initial conditions and bias parameters of multiple time segments being analyzed simultaneously, thereby extending the scope of the application significantly and even to more complex problems.

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REFERENCES

[1] Fisher R. A., "On an Absolute Criterion for Fitting Frequency Curves," *Messenger of Mathematics*, Vol. 41, Macmillan, London, 1912, pp. 155–160.

[2] Fisher R. A., "On the Mathematical Foundations of Theoretical Statistics," *Philosophical Transactions of the Royal Society of London*, Vol. 222, 1922, pp. 309–368.

[3] Fisher R. A., "Theory of Statistical Estimations," *Proceedings of the Cambridge Philosophical Society*, Vol. 22, 1925, pp. 700–725.

[4] Iliff K. W., "Parameter Estimation for Flight Vehicles," *Journal of Guidance, Control, and Dynamics*, Vol. 12, No. 5, 1989, pp. 609–622.

[5] Klein V., "Estimation of Aircraft Aerodynamic Parameters from Flight Data," *Progress in Aerospace Sciences*, Vol. 26, No. 1, 1989, pp. 1–77.

[6] Hamel P. G., and Jategaonkar R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[7] Jategaonkar R. V., Fischenberg D., and von Gruenhagen W., "Aerodynamic Modeling and System Identification from Flight Data - Recent Applications at DLR," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 681–691.

[8] Wang K. C., and Iliff K. W., "Retrospective and Recent Examples of Aircraft Parameter Identification at NASA Dryden Flight Research Center," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 752–764.

[9] Morelli E. A., and Klein V., "Application of System Identification to Aircraft at NASA Langley Research Center," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 12–25.

[10] Kendall M. G., and Stuart A., *The Advanced Theory of Statistics*, 3rd ed., Griffin, London, 1969.

[11] Waerden B. L., *Mathematical Statistics*, Springer, New York, 1969.

[12] Sorenson H. W., *Parameter Estimation – Principles and Problems*, Marcel Dekker, New York, 1980.

[13] Davenport W. B., and Root W. L., *Random Signals and Noise*, McGraw–Hill, New York, 1958.

[14] Goodwin G. C., and Payne R. L., *Dynamic System Identification*, Academic Press, New York, 1977.

[15] Wald A., "Note on the Consistency of the Maximum Likelihood Estimates," *Annals of Mathematical Statistics*, Vol. 20, 1949, pp. 595–601.

[16] Cramér H., *Mathematical Methods of Statistics*, Princeton Univ. Press, Princeton, NJ, 1946.

[17] Eykhoff P., *System Identification – Parameter and State Estimation*, Wiley, New York, 1977.

[18] Maine R. E., and Iliff K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[19] Iliff K. W., "Maximum Likelihood Estimation of Lift and Drag from Dynamic Aircraft Maneuvers," *Journal of Aircraft*, Vol. 14, No. 12, 1977, pp. 1175–1181.

[20] Trankle T. L., Vincent J. H., and Franklin S. N., "System Identification of Nonlinear Aerodynamic Models," AGARD, AG-256, Paper 7, March 1982.

[21] Jategaonkar R. V., and Plaetschke E., "Maximum Likelihood Estimation from Flight Test Data for General Non-Linear Systems," DFVLR-FB 83-14, Germany, March 1983.

[22] Taylor L. W., and Iliff K. W., "System Identification Using a Modified Newton-Raphson Method – A Fortran Program," NASA TN D-6734, May 1972.

[23] Plaetschke E., "Kennwertermittlung mit Maximum-Likelihood-Verfahren," DFVLR-IB 154-74/20, 1974 (in German).

[24] Iliff K. W., and Taylor L. W., "Determination of Stability Derivatives from Flight Data Using a Newton-Raphson Minimization Technique," NASA TN D-6579, March 1972.

[25] Ortega J. M., and Rheinboldt W. C., *Iterative Solution of Nonlinear Equations in Several Variables*, Academic Press, New York, 1970.

[26] Balakrishnan A. V. (ed.), *Communication Theory*, McGraw–Hill, New York, 1968.

[27] Kumar K. S. P., and Sridhar R., "On the Identification of Control Systems by the Quasi-Linearization Method," *IEEE Transactions on Automatic Control*, Vol. AC-9, No. 2, 1964, pp. 151–154.

[28] Larson D. B., and Fleck J. T., "Identification of Parameters by the Method of Quasilinearization," Cornell Aeronautical Lab., CAL Rept. No. 164, Buffalo, NY, May 1968.

[29] Jategaonkar R. V., and Plaetschke E., "Maximum Likelihood Estimation of Parameters in Nonlinear Flight Mechanics Systems," *Proceedings of the IFAC Symposium on "Identification and System Parameter Estimation*," York, England, U.K., 1985, pp. 665–668.

[30] Jategaonkar R. V., and Thielecke F., "Aircraft Parameter Estimation – A Tool for Development of Aerodynamic Databases," *Sadhana*, Vol. 25, Part 2, Indian Academy of Sciences, Bangalore, India, April 2000, pp. 119–135.

[31] Spall J. C., "An Overview of the Simultaneous Perturbation Method for Efficient Optimization," *John Hopkins APL Technical Digest*, Vol. 19, No. 4, 1998, pp. 482–492.

[32] Bischoff C., Carle A., Khademi P., and Mauer A., "ADIFOR 2.0: Automatic Differentiation of Fortran-77 Programs," *IEEE Computational Science and Engineering*, Vol. 3, No. 3, 1996, pp. 18–32.

[33] Verma A., "An Introduction to Automatic Differentiation," *Current Science*, Vol. 78, No. 7, 2000, pp. 804–807

[34] Foster G. W., "The Identification of Aircraft Stability and Control Parameters in Turbulence," Royal Aircraft Establishment, RAE TR-83025, March 1983.

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[35] Gill P. E., Murray W., and Wright M. H., *Practical Optimization*, Academic Press, San Diego, 1981.

[36] Jategaonkar R. V., "Bounded-Variable Gauss-Newton Algorithm for Aircraft Parameter Estimation," *Journal of Aircraft*, Vol. 37, No. 4, 2000, pp. 742–744.

[37] Oswald W. B., "General Formulas and Charts for the Calculation of Airplane Performance," NACA Report No. 408, 1932.

[38] Weiss S., Gockel W., Monnich W., and Rohlf D., "Identification of Dornier-328 Reversible Flight Control Systems," AIAA Paper 98-4163, Aug. 1998.

[39] Rao S. S., *Engineering Optimization: Theory and Practice*, Wiley, New York, 1996.

[40] Byrd R. H., Lu P., Nocedal J., and Zhu C., "A Limited Memory Algorithm for Bound Constrained Optimization," *SIAM Journal of Scientific Computing*, Vol. 16, No. 5, 1995, pp. 1190–1208.

[41] Jacob H. G., "An Engineering Optimization Method with Application to STOL-Aircraft Approach and Landing Trajectories," NASA TN D-6978, Sept. 1972.

[42] Gill P. E., and Murray W., "Minimization Subject to Bounds on the Variables," National Physical Lab., Rept. NAC-72, Middlesex, England, U.K., Dec. 1976.

[43] Stark P. B., and Parker R. L., "Bounded Variable Least Squares: An Algorithm and Applications," *Journal of Computational Statistics*, Vol. 10, No. 2, 1995, pp. 129–141.

[44] Vanderbei R. J., and Shanno D. F., "An Interior-Point Algorithm for Nonconvex Nonlinear Programming," *Computational Optimization and Applications*, Vol. 13, Nos. 1–3, 1999, pp. 231–252.

[45] Handschin E., Langer M., and Kliokys E., "An Interior Point Method for State Estimation with Current Magnitude Measurements and Inequality Constraints," *Proceedings of the IEEE Power Industry Computer Application Conference*, Salt Lake City, UT, 1995, pp. 385–391.

[46] Lustig I. J., Marsten R. J., and Shanno D. F., "Computational Experience with a Primal-Dual Interior-Point Method for Linear Programming," *Linear Algebra and Its Applications*, Vol. 152, 1991, pp. 191–222.

[47] Marquardt D. W., "An Algorithm for Least Squares Estimation of Nonlinear Parameters," *SIAM Journal of Applied Mathematics*, Vol. 11, No. 2, 1963, pp. 431–441.

[48] Moré J. J., "The Levenberg-Marquardt Algorithm: Implementation and Theory," *Springer Lecture Notes in Mathematics*, No. 630, edited by G. A. Watson, Springer, Berlin, 1978, pp. 105–116.

[49] Pashilkar A. A., and Jategaonkar R. V., "Some Investigations Related to Gradients for Aircraft Parameter Estimation," DLR IB 111-98/44, Germany, Nov. 1998.

[50] Jategaonkar R. V., and Plaetschke E., "Non-Linear Parameter Estimation from Flight Test Data Using Minimum Search Methods," DFVLR-FB 83-15, Germany, March 1983.

[51] Murphy P. C., "A Methodology for Airplane Parameter Estimation and Confidence Interval Determination in Nonlinear Estimation Problems," NASA RP 1153, April 1986.

[52] Nelder J. A., and Mead R., "A Simplex Method for Function Minimization," *Computer Journal*, Vol. 7, No. 4, Jan. 1965, pp. 308–313.

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[53] Rowan T., "The Subplex Method for Unconstrained Optimization," Ph.D. Dissertation, Dept. of Computer Sciences, Univ. of Texas, Austin, TX, 1990, pp. 50–74.

[54] Powell M. J. D., "An Efficient Method for Finding the Minimum of a Function of Several Variables without Calculating Gradients," *Computer Journal*, Vol. 7, 1964, pp. 155–162.

[55] Wilks S. S., *Mathematical Statistics*, Wiley, New York, 1962.

[56] Jategaonkar R. V., and Mönnich W., "Identification of DO-328 Aerodynamic Database for a Level D Flight Simulator," AIAA Paper 97-3729, Aug. 1997.

[57] Jategaonkar R. V., and Thielecke F., "ESTIMA – An Integrated Software Tool for Nonlinear Parameter Estimation," *Journal of Aerospace Science and Technology*, Vol. 6, No. 8, 2002, pp. 565–578.

[58] "MATLAB, The Language of Technical Computing, Version 6.5.1, Release 13, (Service Pack 1)," The MathWorks, Inc., Natick, MA, 2003.

[59] Morelli E. A., "System Identification Programs for Aircraft (SIDPAC)," AIAA Paper 2002-4704, Aug. 2002.

[60] Weiss S., "PENSUM: Parameter Estimation of Nonlinear Systems Using Matlab, Version 1.0," DLR-IB 111-1999/32, Germany, Dec. 1999.

[61] Maine R. E., and Iliff K. W., "User's Manual for MMLE3, a General FORTRAN Program for Maximum Likelihood Parameter Estimation," NASA TP-1563, Nov. 1980.

[62] Murray J. E., and Maine R. E., "pEst Version 2.1 User's Manual," NASA TM-88280, Sept. 1987.

[63] Jategaonkar R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DLR-FB 90-40, Germany, July 1990.

[64] Jategaonkar R. V., and Plaetschke E., "Algorithms for Aircraft Parameter Estimation Accounting for Process and Measurement Noise," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 360–372.

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CHAPTER 5

Filter-Error Method

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5.1 INTRODUCTION

The maximum likelihood estimates of model parameters accounting for measurement noise only can be efficiently obtained for linear and general nonlinear systems. In Chapter 4, we discussed the output-error method for this purpose in quite some detail, presenting the most commonly applied optimization algorithms and several practical issues. Although the output-error method has been widely used in the past, and will continue to be used in the future as well, it is required to gather the data for estimation purposes from flight tests in a steady atmosphere. In the presence of atmospheric turbulence, the output-error method is found to yield poor estimation results, both in terms of convergence and estimates; we had also seen one such example in Sec. 4.20.2.

There are two ways to account for turbulence in parameter estimation, namely, 1) to measure the wind components, more appropriately said, derive them from other measured variables such as true airspeed, inertial speed, attitude angles and flow angles; and 2) to model generically or explicitly the turbulence mathematically and estimate the corresponding parameters. The first approach is a data preprocessing step that yields wind components along the three body-fixed coordinates; those can be treated as known inputs and accounted for in the estimation through minor modifications of the postulated models. The advantage is that the fairly simple output-error method can be applied directly. The approach, however, requires precise measurements of the said variables. Any inaccuracies in the measurements, for example, those resulting from calibration errors or time delays in the recorded flow angles, will affect the derived wind and consequently the estimates of the aerodynamic derivatives. Furthermore, accounting for the effects of turbulence on other motion variables, for example, angular rates, is more involved and generally not considered. Nevertheless, it is a viable approach. The second approach of mathematically accounting for turbulence is algorithmically more involved, but at the same time comparatively more flexible because it is not limited by the difficulties just mentioned. Here, we consider

the second approach only because it has been used more commonly in practice than the other.

The presence of process noise (turbulence) complicates parameter estimation considerably. Because the nonmeasurable process noise makes the system stochastic, as a consequence, it requires a suitable state estimator to propagate the states; for this reason, we avoid here the term "integrate the state equations." The state estimation is performed using either a Kalman filter or an extended Kalman filter, depending upon whether the postulated model is linear or nonlinear. There are two possible approaches to account for process noise in parameter estimation: 1) techniques belonging to the general class of output error and 2) the filtering approach. The second approach is an indirect approach that transforms the parameter estimation problem into a nonlinear state estimation problem by artificially defining the unknown parameters as additional state variables. We will discuss this approach having certain limitations, in Chapter 7. In this chapter we focus on the first approach, namely, the techniques called filter-error methods (FEM). These techniques are a logical extension of the output-error method. Such methods accounting for both process and measurement noise, although essentially more complex, are not only necessary to analyze flight-test data from flights in turbulent atmosphere, but can even yield improved estimation results for flight data in a seemingly smooth air.

Filter-error methods represent the most general stochastic approach to aircraft parameter estimation that was proposed by Balakrishnan [1]. With the pioneering work of Mehra and colleagues [2–4] and Iliff [5] during the early 1970s, these techniques provided capabilities to estimate aircraft parameters from flight in turbulent atmosphere. Highly successful applications of estimation methods to linear systems, combined with the fact that the linear filtering theory provides an optimal solution to state estimation in linear systems, had paved way to the use of filter-error methods for linear systems [6]. Several applications of aircraft parameter estimation have since been reported [7–9]. Although extension of the filter-error methods to nonlinear systems is a natural further step, such extensions and applications are particularly complicated. During the 1980s, Jategaonkar and Plaetschke demonstrated a workable approach to handle general model structures using a numerical approach and incorporating a mixed state estimator based on a prediction step with the nonlinear model and a correction step using a first-order linear approximation [10–12].

Besides the capability to account for process and measurement noise, the filter-error methods offer few more advantages; they lead, in general, to a more nearly linear optimization problem, which has fewer local minima and a faster convergence rate, because these methods are less sensitive to stochastic disturbances. Because of the basic formulation that we will address in this chapter, they contain a feedback proportional to the fit error. This feedback stabilizes numerically the filter-error algorithm and also helps to improve the convergence properties. The stabilizing property of the filter-error algorithm makes it suitable for open-loop identification of unstable aircraft. We will address this issue of

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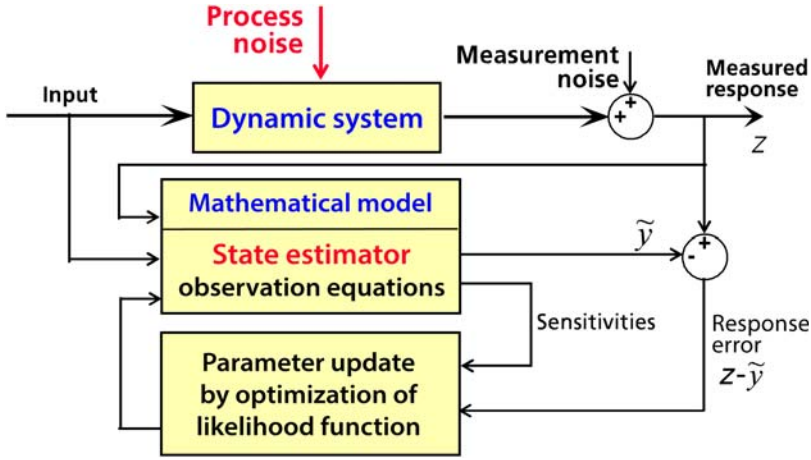

Fig. 5.1 Block schematic of filter-error method.

unstable aircraft in more detail separately, but the filter-error methods work for unstable aircraft; output-error methods may not.

Figure 5.1 shows a block schematic of the filter-error method. In this chapter, we begin with the theoretical development and computational aspects for linear systems, followed by a discussion of various possible formulations. We then move on to the ultimate step of implementation for general nonlinear systems. As in Chapter 4, we highlight numerical intricacies, practical aspects, and limitations, providing our recommendations in each case. We also take a critical look at the equivalence between the output-error and filter-error methods. We do not go into the details of the statistical properties of the estimates [13] because the technique is based on the principle of maximum likelihood, whose desirable statistical properties have already been discussed in Chapter 4. We study the performance of the filter-error method on two sample cases using estimation software capable of handling nonlinear systems, which we also supply with the book to follow and to gain more insight into the algorithmic steps and numerical aspects. The data analyzed are also supplied as sample data to verify the results. Yet another aircraft example is presented to bring out the practical utility of these techniques.

5.2 FILTER-ERROR METHOD FOR LINEAR SYSTEMS

Although it has been said that our main focus in this book is on nonlinear systems, we consider in this chapter first the linear system representation, mainly to highlight the mathematical details of a fairly complex algorithm, and thereby appreciate the difficulties encountered in handling nonlinear systems with process noise. In such a case, the dynamic system is assumed to be described by the

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following stochastic linear mathematical model:

$$\dot{x}(t) = A\,x(t) + Bu(t) + F\,w(t) + b_x, \quad x(t_0) = 0 \tag{5.1}$$

$$y(t) = C\,x(t) + Du(t) + b_y \tag{5.2}$$

$$z(t_k) = y(t_k) + Gv(t_k), \quad k = 1, 2, \ldots, N \tag{5.3}$$

where x is the $(n_x \times 1)$ state vector, u is the $(n_u \times 1)$ control input vector, and y is the $(n_y \times 1)$ observation (or also called model output) vector. The measurement vector z is sampled at N discrete time points. Matrices A, B, C, and D contain the unknown system parameters. Matrices F and G represent the process and measurement noise distribution matrices, respectively. Based on the discussion of Sec. 3.5, we once again consider the bias terms b_x and b_y for the state and observation equations. Recall that these bias parameters account for nonzero initial conditions and biases in the measurements of the control inputs and output variables automatically. Such a treatment of unknown initial conditions and measurement biases was found to be more efficient than treating them separately. Thus, in a most general case, it is required to estimate the parameter vector Θ consisting of the elements of the matrices A, B, C, D; the bias parameters b_x and b_y; and elements of the noise distribution matrices F and G.

There are two important assumptions that we make for the process and the measurement noise. First, it is assumed that they affect the dynamic system linearly, which allows additive noise representation in Eqs. (5.1) and (5.3). Second, the process noise $w(t)$ is assumed to be characterized by a zero-mean Gaussian noise with an identity spectral density, and the measurement noise $v(t_k)$ is assumed to be characterized by a sequence of Gaussian random variables with zero mean and identity covariance. It is further assumed that the process noise and the measurement noise $w(t)$ and $v(t_k)$ are uncorrelated and mutually independent. The noise distribution matrices F and G are, in general, unknown.

Following the maximum likelihood principle presented in Sec. 4.2, the cost function of Eq. (4.13) in terms of the residuals reduces in the present case to

$$J(\Theta, R) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - \tilde{y}(t_k)]^T\,R^{-1}[z(t_k) - \tilde{y}(t_k)] + \frac{N}{2}\ell n(\det(R)) + \frac{Nm}{2}\ell n(2\pi) \tag{5.4}$$

where $\tilde{y}$ is the system output based on the predicted states incorporating the Kalman filter and R is the steady-state covariance matrix of the residuals (also called innovations). Parameter estimates for Θ and R are obtained by minimization of the likelihood function, Eq. (5.4). To evaluate the cost function, the first task is to compute $\tilde{y}$, which depends upon the estimated states. Although we have made the assumption of process noise being additive, it still complicates the estimation procedure considerably because the system is no longer deterministic. As a consequence, it is not possible to simply integrate the state variables

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(as it was done in the case of the output-error method presented in Chapter 4). Rather, it is now necessary to incorporate some suitable state estimator. For linear systems, the Kalman filter provides an optimal state estimator [14, 15]; see Appendix F for the derivation of the Kalman filter algorithm.

The Kalman filter (state estimator) consists of two steps, and for the linear model postulated in Eqs. (5.1) and (5.2), it is given by

Prediction step:

$$\tilde{x}(t_{k+1}) = \Phi \hat{x}(t_k) + \Psi B \overline{u}(t_k) + \Psi b_x \tag{5.5}$$

$$\tilde{y}(t_k) = C \tilde{x}(t_k) + D u(t_k) + b_y \tag{5.6}$$

Correction step:

$$\hat{x}(t_k) = \tilde{x}(t_k) + K [z(t_k) - \tilde{y}(t_k)] \tag{5.7}$$

where $\tilde{x}$ and $\hat{x}$ denote the predicted and corrected state vectors respectively, $\overline{u}$ the average of the control inputs at the two discrete time points, $\tilde{y}$ the predicted output variables, $[z(t_k) - \tilde{y}(t_k)]$ the residuals (innovations), $\Phi = e^{A\Delta t}$ the state transition matrix, Ψ the integral of the state transition matrix (see Sec. 3.8.2), K the Kalman filter gain matrix, and Δt the sampling interval. This filter is used only for the purpose of obtaining the true state variables from the noisy measurements, that is, for the state estimation only. In Eq. (5.5), we have omitted the initial conditions because, as already pointed out earlier, they are zero. We will discuss later in this chapter the exact sequence of computing Eqs. (5.5–5.7).

In many applications, particularly when the system under investigation is time-invariant and the deviations from the nominal trajectory are small, it is often adequate to use a steady-state filter for state estimation. This simplification results in a significant reduction of the computational burden. The Kalman gain matrix K is a function of the covariance matrix of the residuals R, of the covariance matrix of the state-prediction error P, and of the observation matrix C. It is given by

$$K = P\, C^T R^{-1} \tag{5.8}$$

The covariance matrix of the innovations (residuals) R is related to the measurement noise covariance matrix GG^T through the covariance matrix of the state-propagation error P and the observation matrix C.

$$R = GG^T + C\, P\, C^T \tag{5.9}$$

The covariance matrix of the state-prediction error P is obtained by solving the Riccati equation, which we will discuss in Sec. 5.4.1. The state-prediction error covariance matrix P accounts for the process noise and uncertainties in the initial conditions on the states and thereby improves the state estimates and in turn the parameter estimates.

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5.3 PROCESS NOISE FORMULATIONS

We have now all of the details to discuss the implications of the filter-error method and possible various techniques of parameter estimation in the presence of process and measurement noise. There are three different approaches that have evolved, based on the way the noise distribution matrices F and G or instead the related matrices R and K are estimated [6, 16].

5.3.1 NATURAL FORMULATION

In any optimization problem, the very first approach that comes *naturally* to our mind is to treat the parameters as elements of the unknown parameter vector and then apply the best possible optimization method to minimize the cost function with respect to the unknown parameter vector [6, 7]. The natural formulation results from this basic idea in which the unknown covariances of both process and measurement noise are explicitly treated as unknowns to be estimated along with the other system parameters appearing in the system matrices A, B, C, and D. The complete vector of unknown parameters in this case is given by

$$\begin{aligned}\Theta = [&(A_{ij},\, i = 1,\, n_x;\, j = 1,\, n_x)^T \quad (B_{ij},\, i = 1,\, n_x;\, j = 1,\, n_u)^T \\ &(C_{ij},\, i = 1,\, n_y;\, j = 1,\, n_x)^T \quad (D_{ij},\, i = 1,\, n_y;\, j = 1,\, n_u)^T \\ &(F_{ij},\, i = 1,\, n_x;\, j = 1,\, n_x)^T \quad (G_{ij},\, i = 1,\, n_y;\, j = 1,\, n_y)^T]^T\end{aligned} \tag{5.10}$$

where i and j are the general indices, and each term on the right-hand side within the brackets, that is, $(\cdots)$, a column vector of appropriate size. Although Eq. (5.5) is in terms of the state transition matrix Φ, we prefer to deal with the system matrix A using the relationship $\Phi = e^{A\Delta t}$. In general, the bias terms b_x and b_y appearing in Eqs. (5.5) and (5.6) will have to be included in Eq. (5.10), but they have been dropped from the general discussion in this section without loss of generality because they are treated similarly to other parameters and hence do not affect these investigations. Estimation of parameters appearing in Eq. (5.10) is a brute-force approach to which we can apply any one of the optimization methods. We have already discussed in some depth different methods in Secs. 4.6 through 4.14 and found that the best choice is either one of the Gauss–Newton methods or the Levenberg–Marquardt method.

The two properties of the natural formulation that need some consideration are 1) convergence and 2) computational burden. The maximum likelihood cost function defined in Eq. (5.4) is derived in terms of the covariance matrix of the residuals. The noise distribution matrices appear only indirectly through the relation given in Eq. (5.9). Recalling our discussion in Sec. 4.5 on the reasons for choosing the relaxation strategy, we will face the same difficulties here too, if not more so, because of the complex functional relationship between the various system parameters and F and G. This leads to serious convergence problems. The choice of the initial values for the elements of G is also

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critical; very low values affect the convergence whereas too large values can lead to physically unrealistic values and in a worst case to divergence [6]. The second factor that also does not favor this approach is the increased size of the unknown parameters and the need for the corresponding gradients, resulting in a more computational burden. Although this was also a serious consideration in the 1960s and 1970s, with the availability of powerful computers and workstations, it is just of minor nuisance today. Based on these considerations, although the natural formulation might be theoretically a possible approach, it is not a practical one.

5.3.2 INNOVATION FORMULATION

This approach is very similar to the one followed in the case of the output-error method [2, 6, 9]. It uses the relaxation strategy, in which the covariance matrix of the residuals is obtained by a general closed-form solution that was derived in Sec. 4.5.2. This derivation is valid in the present case as well because it is based on the residuals, and not on the way the residuals are computed, that is, whether they stem from linear or nonlinear system, or from a system with process noise or with both measurement and process noise. In the present case, the maximum likelihood estimate of R, obtained by setting $\partial J/\partial R = 0$, is given by

$$R = \frac{1}{N}\sum_{k=1}^{N}[z(t_k) - \tilde{y}(t_k)]\,[z(t_k) - \tilde{y}(t_k)]^T \tag{5.11}$$

This explicit solution for R overcomes the convergence problems encountered in the natural formulation and also reduces the number of unknown parameters to be estimated in the second step applying the Gauss–Newton or another method.

In the second step, the innovation formulation tries to further simplify the parameter optimization problem by considering just the three equations, namely, Eqs. (5.5–5.7), and estimating along with the system parameters the elements of gain matrix K directly, instead of obtaining the same through Eq. (5.8). The parameter vector in this approach is given by

$$\begin{aligned}\Theta = [&(A_{ij},\, i = 1, n_x; j = 1, n_x)^T \quad (B_{ij},\, i = 1, n_x; j = 1, n_u)^T \\ &(C_{ij},\, i = 1, n_y; j = 1, n_x)^T \quad (D_{ij},\, i = 1, n_y; j = 1, n_u)^T \\ &(K_{ij},\, i = 1, n_x; j = 1, n_y)^T]^T\end{aligned} \tag{5.12}$$

The basic idea behind this approach has been to eliminate the most complex part of the algorithm required otherwise to estimate the process noise distribution matrix F, which appears indirectly in K given by Eq. (5.8) through P. The covariance matrix of the state prediction error P is obtained by solving the Riccati equation. Furthermore, the optimization algorithm needs gradients of P as well.

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(We will talk about these details in the next section.) These two steps are the most complex part of the algorithm.

The preceding choice of estimation K eliminates the most complex part of the process noise method, namely, computing P and its gradients. Although from a computational point of view, the approach might appear appealing, it has two drawbacks. First, for better identifiability and accuracy of the estimates, we usually have more observation variables than state variables, which means that the size of K (that is, $n_x \times n_y$) is larger than that of F (that is, $n_x \times n_x$). Thus, we might have to estimate a larger number of gain matrix parameters to account for the process noise. It can lead to identifiability problems. Moreover, the elements of K have no direct physical meaning, making interpretation and judgment about the estimates difficult. Secondly, the gain matrix K as well as the system parameters and the covariance matrix of residuals R are estimated independent of each other in two separate steps given by Eqs. (5.12) and (5.11), respectively. This can lead to incompatible R and K estimates. They might not satisfy the necessary condition of $KC \leq 1$. (Discussion of this requirement is deferred to the next section.) Constrained nonlinear optimization is necessary to account for this inequality. For larger dimensioned K this becomes very tedious.

5.3.3 COMBINED NATURAL WITH INNOVATION FORMULATION

The combined formulation proposed by Maine and Iliff [6, 17] takes advantage of the best features of both the natural formulation (that is, estimation of the process noise matrix using the Gauss–Newton method) and of the innovation formulation (that is, explicit estimation of the covariance matrix of residuals). Instead of the gain matrix K as in the innovation formulation, here the process noise distribution matrix F is treated as unknown along with the other system parameters. The complete parameter vector is then given by

$$\begin{aligned}\Theta = [&(A_{ij},\, i = 1,\, n_x;\, j = 1,\, n_x)^T \quad (B_{ij},\, i = 1,\, n_x;\, j = 1,\, n_u)^T \\ &(C_{ij},\, i = 1,\, n_y;\, j = 1,\, n_x)^T \quad (D_{ij},\, i = 1,\, n_y;\, j = 1,\, n_u)^T \\ &(F_{ij},\, i = 1,\, n_x;\, j = 1,\, n_x)^T]^T\end{aligned} \tag{5.13}$$

We apply the two-step relaxation strategy, estimating explicitly the covariance matrix of the residuals given by Eq. (5.11), and then estimate the preceding parameter vector of Eq. (5.13) applying the Gauss–Newton method. Estimation of R does not pose any difficulty. In the second step, we have a much lesser number of parameters to be estimated than in the innovation formulation (F is smaller dimensioned than K). The main advantage is that the elements of F have physical meaning. In this case too, the inequality constraint of $KC \leq 1$ will have to be met, requiring constrained optimization, but of a reduced dimension. The small penalty that we pay here is that we have to solve for the Riccati equation to obtain the covariance matrix of the state prediction error P.

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However, this formulation has been found to be more practical with regard to convergence, parameter estimates, and computational burden.

5.4 FILTER-ERROR ALGORITHM

Based on the comparative discussion of the three possible formulations, we prefer and restrict our attention hereafter to the combined formulation. In such an algorithm, it is required to minimize the cost function defined in Eq. (5.4) subject to the system equations formulated in Eqs. (5.1–5.3). The important steps of such an algorithm are as follows [16]:

1. Specify some suitable starting values for the system parameters Θ, that is, Eq. (5.13) consisting of elements of the matrices A, B, C, D, and also of the process noise distribution matrix F.
2. Specify suitable starting values for R, or generate the estimate of R. (We will talk about this in Sec. 5.6.)
3. Do the state estimation applying the Kalman filter [requires computation of gain matrix K, state transition matrix Φ, its integral Ψ, and of course the predicted observation variables $\tilde{y}(t_k)$, that is, Eqs. (5.5–5.8)].
4. Compute the covariance matrix of the residuals, given by Eq. (5.11).
5. Apply the Gauss–Newton method to update Θ, which requires computation of the necessary gradients and checking for the inequality constraint on KC.
6. Iterate on steps 3 to 5 until convergence.

As already stated, the Kalman filter provides the optimal state estimator for linear systems, the computational details of which are given by Eqs. (5.5–5.9). The procedure to compute the state transition matrix Φ and its integral Ψ is given by Eqs. (3.38) and (3.39), Sec. 3.8.2. It now remains to study procedures to compute the covariance matrix of the state prediction error P and to compute the response gradients and those of P required in the Gauss–Newton algorithm.

5.4.1 SOLUTION TO RICCATI EQUATION

The covariance matrix of the state prediction error P is obtained by solving a Riccati equation. We use here a steady-state form of the Riccati equation. Furthermore, the continuous-time Riccati equation, which can be solved efficiently, is preferred over the discrete-time equation [18]. The first-order approximation of this equation is obtained as

$$AP + PA^T - \frac{1}{\Delta t} PC^T R^{-1} CP + FF^T = 0 \tag{5.14}$$

Equation (5.14) is solved using the well-known Potter's method [19], based on the eigenvector decomposition. We present here just the essential details of the

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method to understand the necessary computational steps to solve Eq. (5.14) for P. We start by constructing the Hamiltonian matrix H defined as

$$H = \left[\begin{array}{c|c} A & -FF^T \\ \hline -\dfrac{1}{\Delta t} C^T R^{-1} C & -A^T \end{array}\right] \tag{5.15}$$

H is a real, a general nonsymmetric matrix of the dimensions $2n_x \times 2n_x$, where n_x is the number of state variables. Now, compute the eigenvalues and eigenvectors of H, which can be complex quantities.

Let the eigenvalues and the matrix of eigenvectors of H be denoted by Λ and E, respectively. The eigenvector matrix E is now partitioned into two equal parts such that the left half contains the eigenvectors corresponding to unstable eigenvalues, that is, corresponding to eigenvalues with positive real part. The controllability by the process noise and observability ensure that exactly half the number of eigenvalues will have positive real parts and that the eigenvectors can be determined uniquely. The rearranged matrix of eigenvectors is further decomposed into $(n_x \times n_x)$ size matrices as

$$E = \left[\begin{array}{c|c} E_{11} & E_{12} \\ \hline E_{21} & E_{22} \end{array}\right] \tag{5.16}$$

The solution to the steady-state Riccati equation is then given by

$$P = -E_{11}\, E_{21}^{-1} \tag{5.17}$$

It can be proved that the solution for P given by Eq. (5.17), which is a $(n_x \times n_x)$ matrix, is positive definite because as already said the eigenvectors are uniquely determined [19]. The covariance matrix of the state prediction error P is a steady-state matrix and as such does not depend on time.

Potter's method assumes the partitioned eigenvector matrices to be nonsingular. If the eigenvalues are nearly equal, the method might show poor numerical stability because, although the partitioned eigenvector matrices are still nonsingular, they remain ill-conditioned. In our specific case, because the gradients of P are obtained through numerical difference approximations (to be addressed in Sec. 5.5) using small perturbations in the system matrices, it can lead to ill-conditioning. Small perturbations in the state matrix elements can lead to drastic changes in the partitioned eigenvector matrix, which, in turn, causes poor numerical stability. In such cases gradients might not be reliable or accurate. An improved method, which is more stable and not overly sensitive to small changes, has been demonstrated by Holley and Wei [20]. It alleviates the problems caused by an ill-conditioned partitioned eigenvector matrices. However, in several cases that we have analyzed the classical Potter's method as elaborated in Eqs. (5.14–5.17) proved quite adequate.

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5.4.2 PARAMETER UPDATE

Having computed the predicted observation variables applying the Kalman filter, the computation of the covariance matrix of the residuals is straightforward; it is just a simple summation over the N data points given by Eq. (5.11). In the next step we apply the Gauss–Newton method to update the parameter vector Θ of Eq. (5.13), which yields

$$\Theta_{i+1} = \Theta_i + \Delta\Theta \quad \text{and} \quad \mathcal{F}\,\Delta\Theta = -\mathcal{G} \tag{5.18}$$

where i is the iteration index and $\mathcal{F}$ and $\mathcal{G}$ are given by

$$\mathcal{F} = \sum_{k=1}^{N} \left[\frac{\partial \tilde{y}(t_k)}{\partial \Theta}\right]^T R^{-1} \left[\frac{\partial \tilde{y}(t_k)}{\partial \Theta}\right] \tag{5.19}$$

$$\mathcal{G} = -\sum_{k=1}^{N} \left[\frac{\partial \tilde{y}(t_k)}{\partial \Theta}\right]^T R^{-1}\,[z(t_k) - \tilde{y}(t_k)] \tag{5.20}$$

It is exactly the same equation that we had derived for the output-error method in Eqs. (4.29) and (4.30), except that y is replaced through $\tilde{y}$. Equation (5.19) is an approximation to the second gradient of the cost function $\partial^2 J/\partial\Theta^2$; we have already discussed in Sec. 4.6 the reasons for and suitability of this simplification. As in the case of the output-error method, the Gauss–Newton method can show local intermediate divergence. In such cases we apply the heuristic procedure of parameter-halving that we had discussed in Sec. 4.10.1.

From Eq. (5.11), we already know how to compute R. We also know the procedure, given by Eqs. (5.5) to (5.8) and (5.14) to (5.17), to compute the predicted responses. To compute the parameter update $\Delta\Theta$ applying Eqs. (5.18) to (5.20), we now need a procedure to compute the sensitivity matrix, that is, the response gradients, defined as

$$\left(\frac{\partial \tilde{y}}{\partial \Theta}\right)_{ij} = \frac{\partial \tilde{y}_i}{\partial \Theta_j} \tag{5.21}$$

In a classical way the sensitivity matrix $\partial\tilde{y}/\partial\Theta$ is obtained by solving the sensitivity equations, which are derived by partial differentiation of the system equations with respect to each element of the unknown system parameters. Partial differentiation of Eq. (5.5) leads to

$$\begin{aligned}\frac{\partial \tilde{x}(t_{k+1})}{\partial \Theta} = \Phi\,\frac{\partial \hat{x}(t_k)}{\partial \Theta} + \frac{\partial \Phi}{\partial \Theta}\,\hat{x}(t_k) + \Psi\,\frac{\partial B}{\partial \Theta}\,\overline{u}(t_k) \\ + \frac{\partial \Psi}{\partial \Theta}\,B\,\overline{u}(t_k) + \Psi\frac{\partial\, b_x}{\partial \Theta} + \frac{\partial \Psi}{\partial \Theta}\,b_x\end{aligned} \tag{5.22}$$

Equation (5.22) gives the exact gradient of Eq. (5.5) and requires $\partial\Phi/\partial\Theta$ and $\partial\Psi/\partial\Theta$, the gradients of the state transition matrix Φ and of its integral Ψ.

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The computational burden to obtain $\partial\Phi/\partial\Theta$ and $\partial\Psi/\partial\Theta$ is significant. Hence, we approximate the gradient computation in Eq. (5.22) through [6]

$$\frac{\partial\tilde{x}(t_{k+1})}{\partial\Theta} \approx \Phi\,\frac{\partial\hat{x}(t_k)}{\partial\Theta} + \Psi\frac{\partial A}{\partial\Theta}\overline{x}(t_k) + \Psi\frac{\partial B}{\partial\Theta}\overline{u}(t_k) + \Psi\frac{\partial b_x}{\partial\Theta}, \quad \frac{\partial\tilde{x}(1)}{\partial\Theta} = 0 \tag{5.23}$$

where, as already pointed out in Sec. 3.8.1, $\overline{u}$ is the average of the inputs at t_{k+1} and t_k, and likewise $\overline{x} = 1/2[\tilde{x}(t_{k+1}) + \hat{x}(t_k)]$. Equation (5.23) is accurate to the order of $(\Delta t)^2$ and found adequate because flight data are usually sampled at a high rate, that is, with small sampling time. This simplification leads to sensitivity equations that have the same state transition matrix Φ and the same integral of the state transition matrix Ψ, both of which are anyway evaluated during the computation of the system responses from Eqs. (5.5–5.7).

Partial differentiation of Eqs. (5.6) and (5.7) yields

$$\frac{\partial\tilde{y}(t_k)}{\partial\Theta} = C\frac{\partial\tilde{x}(t_k)}{\partial\Theta} + \frac{\partial C}{\partial\Theta}\tilde{x}(t_k) + \frac{\partial D}{\partial\Theta}u(t_k) + \frac{\partial b_y}{\partial\Theta} \tag{5.24}$$

$$\frac{\partial\hat{x}(t_k)}{\partial\Theta} = \frac{\partial\tilde{x}(t_k)}{\partial\Theta} - K\frac{\partial\tilde{y}(t_k)}{\partial\Theta} + \frac{\partial K}{\partial\Theta}[z(t_k) - \tilde{y}(t_k)] \tag{5.25}$$

Equations (5.23–5.25) form the set of sensitivity equations. All of the terms appearing in Eqs. (5.23–5.25) have been defined, except for $\partial K/\partial\Theta$, the gradient of the Kalman gain matrix.

Partial differentiation of Eq. (5.8) with respect to Θ yields

$$\frac{\partial K}{\partial\Theta} = \frac{\partial P}{\partial\Theta}C^T R^{-1} + P\,\frac{\partial C^T}{\partial\Theta}R^{-1} \tag{5.26}$$

Note that $\partial K/\partial\Theta$ and $\partial P/\partial\Theta$ and some other terms in the following are three-dimensional arrays. During a particular step of the Gauss–Newton method, the covariance matrix R is fixed, and hence only partial differentiations of P and C appear on the right-hand side of Eq. (5.26). Computation of $\partial K/\partial\Theta$ using Eq. (5.26) requires the gradient of P, the covariance matrix of the state prediction error. It is obtained by partial differentiation of the Riccati equation, Eq. (5.14):

$$\begin{aligned} &A\frac{\partial P}{\partial\Theta} + \frac{\partial A}{\partial\Theta}P + P\frac{\partial A^T}{\partial\Theta} + \frac{\partial P}{\partial\Theta}A^T - \frac{1}{\Delta t}\frac{\partial P}{\partial\Theta}C^T R^{-1}CP \\ &- \frac{1}{\Delta t}P\frac{\partial C^T}{\partial\Theta}R^{-1}CP - \frac{1}{\Delta t}PC^T R^{-1}\frac{\partial C}{\partial\Theta}P \\ &- \frac{1}{\Delta t}PC^T R^{-1}C\frac{\partial P}{\partial\Theta} + F\frac{\partial F^T}{\partial\Theta} + \frac{\partial F}{\partial\Theta}F^T = 0 \end{aligned} \tag{5.27}$$

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Combining together the (first and eighth), (fourth and fifth), (second, seventh, and tenth) and (third, sixth, and ninth) terms on the left-hand side of Eq. (5.27), and after simple manipulations using the matrix algebra results $(A+B)^T = A^T + B^T$ and $(AB)^T = B^T A^T$, as well as P and R being symmetrical, $P^T = P$, $R^T = R$, and $(R^{-1})^T = R^{-1}$, it can be shown that Eq. (5.27) can be rewritten in the form of

$$\overline{A}\frac{\partial P}{\partial \Theta} + \frac{\partial P}{\partial \Theta}\overline{A}^T = \overline{C} + \overline{C}^T \tag{5.28}$$

where

$$\overline{A} = A - \frac{1}{\Delta t} PC^T R^{-1} C = A - \frac{1}{\Delta t} KC \tag{5.29}$$

and

$$\overline{C} = -\frac{\partial A}{\partial \Theta} P + \frac{1}{\Delta t} PC^T R^{-1} \frac{\partial C}{\partial \Theta} P - \frac{\partial F}{\partial \Theta} F^T \tag{5.30}$$

For each element of Θ, we obtain one equation in the preceding form of Eq. (5.28). For the n_q unknown parameters, this leads to a set of Lyapunov equations of the form $AX + XA^T = B$. There are several general procedures to solve this set of equations. However, because the matrix $\overline{A}$ remains the same for the complete set, they are solved more efficiently by using the transformation

$$\overline{A}' = T^{-1}\,\overline{A}\,T \tag{5.31}$$

and

$$(\overline{C} + \overline{C}^T)' = T^{-1}(\overline{C} + \overline{C}^T)\,T^{*-1} \tag{5.32}$$

where T is the matrix of eigenvectors of $\overline{A}$. This similarity transformation makes $\overline{A}'$ diagonal and reduces the Lyapunov equation (5.28) to

$$\overline{A}'\left(\frac{\partial P}{\partial \Theta}\right)' + \left(\frac{\partial P}{\partial \Theta}\right)'\overline{A}'^* = (\overline{C} + \overline{C}^T)' \tag{5.33}$$

Because $\overline{A}'$ is diagonal, Eq. (5.33) can be solved for $(\partial P/\partial \Theta)'$ easily. In a general case, the solution to $AX + XA^T = B$ for a diagonal A is given by $X_{ij} = B_{ij}/(A_{ii} + A_{jj})$.

The gradient of the covariance matrix P is then obtained through back transformation:

$$\frac{\partial P}{\partial \Theta} = T\left(\frac{\partial P}{\partial \Theta}\right)' T^* \tag{5.34}$$

As in the case of the Hamiltonian matrix, Eq. (5.15), the matrix $\overline{A}$ is a general real matrix, and as such the eigenvectors can be complex quantities.

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Equations (5.18–5.20) for the parameter update using the Gauss–Newton method yield the unconstrained minimum of the cost function, in which all of the parameters are treated independent of each other. In the mixed formulation discussed here, direct estimation of R instead of explicit F and G matrices eliminates the convergence difficulties. However, because R is a function of F and G, it is necessary to ensure that the measurement noise covariance matrix GG^T as will be obtained indirectly from Eq. (5.9) is positive semi-definite (to be physical meaningful). To meet this condition, all of the eigenvalues of KC must be less than or equal to one [6]. However, KC tends to be diagonal dominant, and hence the eigenvalue constraint is approximated by constraining the diagonal elements of KC to be less than one. Thus, it is now required to verify whether the preceding unconstrained solution satisfies the constraints of the diagonal elements of KC to be less than unity. These nonlinear inequality constraints lead to a nonlinear programming problem. The quadratic programming method, which requires a linear approximation of the constraints, is used to solve this complex optimization problem [21]. The linear approximation to the constraints is given by

$$(KC)_{ii} + \frac{\partial (KC)_{ii}}{\partial \Theta} \Delta\Theta \leq 1 \tag{5.35}$$

where the gradient of KC is given by

$$\frac{\partial (KC)}{\partial \Theta} = \frac{\partial K}{\partial \Theta} C + K \frac{\partial C}{\partial \Theta} \tag{5.36}$$

Let ℓ be the number of constraints violated by the unconstrained solution given by Eq. (5.18). In this case an auxiliary vector S and a matrix M are formed such that

$$S_i = 1 - (KC)_{ii} \tag{5.37}$$

and

$$M(i, j) = \frac{\partial (KC)_{ii}}{\partial \Theta}, \quad i = 1, \ell; \quad j = 1, n_q \tag{5.38}$$

where n_q is the total number of unknown parameters.

The constrained solution is then given by

$$\Delta\Theta = \Delta\Theta - \left(\frac{\partial^2 J}{\partial \Theta^2}\right)^{-1} M^T \left\{ M \left(\frac{\partial^2 J}{\partial \Theta^2}\right)^{-1} M^T \right\}^{-1} S \tag{5.39}$$

To solve a general nonlinear programming problem using quadratic programming, it is usually required to iterate on Eqs. (5.35–5.39) until all of the inequality

constraints are simultaneously satisfied. In the present case, however, such iterations are not necessary because the final solution converges in the iterative unconstrained optimization loop.

The maximum likelihood estimate of R is obtained by equating the gradient of the cost function, Eq. (5.4), with respect to R to zero. Because innovations are functions of the Kalman gain matrix K, the exact equation for R is complex and computationally tedious. An asymptotic approximation to R is given by

$$R = \frac{1}{N} \sum_{k=1}^{N} [z(t_k) - \tilde{y}(t_k)][z(t_k) - \tilde{y}(t_k)]^T \tag{5.40}$$

which is the same as Eq. (5.11). Computation of R using Eq. (5.40) corresponds to the innovation formulation. In the case of maximum likelihood estimation accounting for measurement noise only, the gain matrix K is zero. The residuals are independent of R, and Eq. (5.40) itself provide the maximum likelihood estimate of the noise covariance.

The two steps to compute R explicitly and $\Delta\Theta$ applying the Gauss–Newton method, Eqs. (5.40) and (5.18), are carried out independently. They do not account for the influence of each on the estimates of the other. This can yield strongly correlated matrices F and R, which affects the convergence. To account for this correlation, [6] suggests the following heuristic procedure to compensate the estimates of the F matrix, whenever R is revised.

$$F_{il}^{new} = F_{il}^{old} \left(\frac{\sum_{j=1}^{n_y} C_{ij}^2 \, r_j^{old} \sqrt{r_j^{old}/r_j^{new}}}{\sum_{j=1}^{n_y} C_{ij}^2 r_j^{old}} \right) \tag{5.41}$$

where r_j is the jth diagonal element of R^{-1}, C_{ij} the (i,j)th element of C, and the superscripts "old" and "new" denote the previous and revised estimates respectively. Equation (5.41) implies that the elements $(l = 1, n_x)$ of the ith row of F are multiplied by the factor within the bracket on the right-hand side. The correction given by Eq. (5.41) is applied only to elements of F, which are being estimated.

In practice, it could happen that the correction of F using the new values of R^{-1} according to Eq. (5.41) can lead to slight local divergence. The reasons are twofold: 1) the estimates of F and R are correlated and 2) the heuristic procedure of Eq. (5.41) is a first-order approximation. In such a case of local divergence resulting from correction of F, we once again apply to the corrected F values the parameter-halving procedure that we had discussed in Sec. 4.10.1. Because the local divergence here, if any, is due to correlation between R and F, very few halving steps are generally enough; we allow just two halving steps. In the eventuality that two halving steps of F corrections are not sufficient to yield a smaller cost function value, then it is recommended to proceed to the next

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full step of Gauss–Newton without performing the F correction during the particular iteration.

From the preceding development, it is now obvious that the formulation adapted makes use of the innovation formulation as a first step to estimate the noise covariance of the residuals. In the second step, instead of the gain matrix K, the state noise distribution matrix F is estimated along with the other system parameters applying the Gauss–Newton method. This is similar to the procedure of the natural formulation. The main difficulties in the estimation of the gain matrix encountered in the innovation formulation are thus circumvented. Although F is a $(n_q \times n_q)$ matrix, it is a common practice to treat the same as a diagonal matrix. This simplification helps not only to reduce the additional computational burden, but also helps to avoid any identification problem.

Although the whole procedure is fairly complex, the detailed step-by-step treatment in this section provides us an insight into the computational complexity of the algorithm based on an analytical approach. Efficient matrix operations provided capabilities to handle conveniently different models postulated in the form of Eqs. (5.1–5.2). It is probably a little more accurate compared to the numerical approach that we will follow in the next section for the nonlinear system, but then it is limited to linear systems only.

If it is required to account for measurement noise only, then in such a case of $F = 0$, the Kalman filter simply integrates the state equations. The predicted and corrected states $\tilde{x}$ and $\hat{x}$ are identical. The gain matrix K is zero. (There is a subtle difference in such cases, but we will talk more about it in Sec. 5.9.) The corresponding equations, namely, from Eqs. (5.14–5.17) and from (5.26–5.39), are redundant. Thus, neglecting the process noise simplifies significantly the estimation algorithm. Such an algorithm accounting for measurement noise only, referred to as an output-error method that was discussed in the preceding chapter, is a special case of the filter-error method.

5.5 FILTER-ERROR METHOD FOR NONLINEAR SYSTEMS

Having discussed the filter-error method for linear systems, we now turn our attention in this section to nonlinear systems. In a general case, the mathematical model in state space is given by the stochastic equations:

$$\dot{x}(t) = f[x(t), u(t), \beta] + Fw(t), \quad x(t_0) = x_0 \tag{5.42}$$

$$y(t) = g[x(t), u(t), \beta] \tag{5.43}$$

$$z(t_k) = y(t_k) + v(t_k) \tag{5.44}$$

where f and g are the n_x and n_y dimensional general nonlinear real valued vector functions. They are assumed to have sufficient differentiability to be able to invoke Taylor-series expansion. The rest of the variables have been defined in Sec. 5.2. As in the case of linear systems, in the present case too, the noise is assumed to be

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additive, and the noise distribution matrices F and G are considered to be time invariant as well as independent of each other.

The basic difference compared to the linear systems is the way we now compute the states and the gradients. Difficulties of extending the filter-error algorithm to nonlinear systems are twofold:

1. To efficiently implement the algorithms those provide flexibility to handle conveniently different model structures without software modifications
2. To derive a suitable filter for nonlinear state estimation

From Sec. 4.8, we recall that the algebraic differentiation of nonlinear model postulates might be difficult, and more importantly any changes in the model structure entail rederivation of the sensitivity equations and subsequent programming changes. The numerical approach of finite difference approximation, which was discussed in the case of the output-error method and also found efficient, can be extended to the filter-error method to compute the response gradients. Because optimal filters for nonlinear systems are practically unrealizable except for some simple trivial cases, an extended Kalman filter (EKF) based on a first-order approximation of the state and measurement equations can be used for nonlinear filtering. A short treatment on EKF as a state estimator is provided in Appendix F.

In many cases, for example, when system models are time invariant, contain weak to moderate nonlinearities, and the deviations from the nominal trajectory are not large, it might be adequate to use a steady-state filter for state estimation. This simplification results in a significant reduction of the computational burden. However, if the system response is dominated by nonlinearities or when the deviations from the nominal trajectory are large, it might be necessary to incorporate a time-varying filter. Both of these possibilities are discussed here.

5.5.1 STEADY-STATE FILTER

The mathematical details of the maximum likelihood estimation using a steady-state filter as a state estimator are similar to the one we already discussed in the foregoing section. The cost function given by Eq. (5.4) is to be minimized subject to the system equations formulated in Eqs. (5.42–5.44). As in the case of linear systems, we follow the combined formulation of Sec. 5.3.3 and apply the two-step procedure to estimate R using Eq. (5.11) and update the parameters applying the Gauss–Newton method as in Eqs. (5.18–5.20). The important steps of such an algorithm can be summarized as follows [10–12]:

1. Specify some suitable starting values for the parameter vector Θ consisting of parameters appearing in the system functions f and g, initial conditions x_0, and elements of the process noise distribution matrix F.
2. Specify suitable starting values for R, or generate the estimate of R. (We will talk about this in Sec. 5.6.)

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3. Do the state estimation applying extended Kalman filter [requires computation of gain matrix K, numerical integration of the state equations, and of course computation of the predicted observation variables $\tilde{y}(t_k)$].
4. Compute the covariance matrix of the residuals, Eq. (5.11).
5. Apply the Gauss–Newton method, Eqs. (5.18–5.20), to update Θ, which requires computation of the necessary gradients and checking for the inequality constraint on KC.
6. Iterate on steps 3 to 5 until convergence.

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As already pointed out, because optimal filters for nonlinear systems are practically unrealizable, an extended Kalman filter is used for nonlinear state estimation. To retain the system nonlinearities in the state estimation as far as possible, a mixed version incorporates a prediction step with the nonlinear model and a correction step based on a first-order approximation of the state and measurement equations. Such a two-step nonlinear constant-gain filter can be represented as follows:

Prediction step:

$$\tilde{x}(t_{k+1}) = \hat{x}(t_k) + \int_{t_k}^{t_{k+1}} f[x(t), \overline{u}(t_k), \beta]\mathrm{d}t, \quad \hat{x}(t_0) = x_0 \tag{5.45}$$

$$\tilde{y}(t_k) = g[\tilde{x}(t_k), u(t_k), \beta] \tag{5.46}$$

Correction step:

$$\hat{x}(t_k) = \tilde{x}(t_k) + K\,[z(t_k) - \tilde{y}(t_k)] \tag{5.47}$$

where $\overline{u}$ could be the average of the control inputs at the two discrete time points or, as discussed in Sec. 3.8.1, appropriately interpolated values between $u(t_k)$ and $u(t_{k+1})$ for the intermediate steps of the numerical integration formula.

The steady-state gain matrix K is a function of the steady-state R, the steady-state P, and C of the linearized system. It is given by

$$K = P\,C^T\,R^{-1} \tag{5.48}$$

where

$$C = \left[\frac{\partial g[x(t), u(t), \beta]}{\partial x}\right]_{t=t_0} \tag{5.49}$$

A steady-state form of the Riccati equation is used to obtain P. We adopt here exactly the same procedure consisting of Eqs. (5.14–5.17) that has already been described in Sec. 5.4.1, except that we use linearized system matrices A and C.

For a small perturbation δx_j in each of the n_q state variables, the elements of system matrices A and C are approximated using the central-difference formulas as

$$A_{ij} \approx \left.\frac{f_i[x + \delta x_j\, e^j, u, \beta] - f_i[x - \delta x_j\, e^j, u, \beta]}{2\delta x_j}\right|_{x=x_0} \tag{5.50}$$

$$C_{ij} \approx \left.\frac{g_i[x + \delta x_j e^j, u, \beta] - g_i[x - \delta x_j e^j, u, \beta]}{2\delta x_j}\right|_{x=x_0} \tag{5.51}$$

where e^j is the column vector with one in the jth row and zeros elsewhere. In the case of nonlinear systems, x_0 are updated iteratively in the parameter-estimation loop. Linearization of the system equations at each iteration about this convenient point x_0 can be used to compute the constant-gain matrix. We would like to highlight once again that the actual postulated nonlinear system equations, functions f and g, are used to extrapolate the state estimates $\tilde{x}$ by numerical integration and for computation of $\tilde{y}$, the predicted system responses. However, the state variable correction in Eq. (5.47), which depends on K, is based on a first-order approximation.

In contrast to the filter-error method for linear systems, in the present case all of the required response gradients are approximated using the finite difference approximation and not by solving the analytically derived sensitivity equations. For a small perturbation $\delta\Theta_j$ in each of the n_q unknown variables of the parameter vector Θ, the perturbed response variable y_i^p for each of the unperturbed variables y_i is computed. The corresponding sensitivity coefficient is then approximated by

$$\left[\frac{\partial\tilde{y}(t_k)}{\partial\Theta}\right]_{ij} = \frac{\tilde{y}_i^p(t_k) - \tilde{y}_i(t_k)}{\delta\Theta_j} \quad i = 1, \ldots, n_y;\;\; j = 1, \ldots, n_q \tag{5.52}$$

The perturbed response variables $\tilde{y}^p$ are obtained from a set of perturbed system equations, similar to Eqs. (5.42–5.43), obtained by replacing Θ through $\Theta + \delta\Theta_j e^j$. For each element of Θ, the state and observation variables can be computed using a two-step nonlinear filter:

Prediction step:

$$\tilde{x}_j^p(t_{k+1}) = \hat{x}_j^p(t_k) + \int_{t_k}^{t_{k+1}} f[x_j^p(t), \overline{u}(t_k), \beta + \delta\beta_j e^j]\mathrm{d}t \tag{5.53}$$

$$\tilde{y}_j^p(t_k) = g[\tilde{x}_j^p(t_k), u(t_k), \beta + \delta\beta_j e^j] \tag{5.54}$$

Correction step:

$$\hat{x}_j^p(t_k) = \tilde{x}_j^p(t_k) + K_j^p[z(t_k) - \tilde{y}_j^p(t_k)] \tag{5.55}$$

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where the superscript p denotes the perturbed variables and the subscript j the index for the parameter being varied.

Computation of the predicted perturbed-state variables $\tilde{x}^p$ from Eq. (5.53) by numerical integration and of perturbed output variables $\tilde{y}^p$ from Eq. (5.54) is straightforward. However, computation of the corrected perturbed-state variables in Eq. (5.55) requires the perturbed gain matrix K_j^p. It is given by

$$K_j^p = P_j^p \, C_j^{pT} R^{-1} \tag{5.56}$$

where the covariance matrix of the perturbed state-prediction error P_j^p for the $\Theta + \delta\Theta_j e^j$ perturbed parameter vector is obtained by solving the Riccati equation (5.14) with perturbed system matrices A_j^p and C_j^p computed for the corresponding perturbation. These linearized perturbed matrices are once again approximated by central-difference formulas given in Eqs. (5.50) and (5.51) evaluated at $\Theta + \delta\Theta_j e^j$.

The extension of finite difference approximation to the filter algorithm thus involves not only the numerical integration of the perturbed state equations but also additionally the computation of the perturbed gain matrices for each element of the unknown parameter vector Θ. The perturbed output variables $\tilde{y}^p$ computed from Eq. (5.54) using perturbed states $\tilde{x}^p$ and perturbed gain matrix K^p will automatically account for the respective gradients. Thus, in comparison to the previous formulation for linear systems, solution to a set of Lyapunov equations for the gradients of P and, from that the computation of gradients of K, has been replaced in the current approach through the solution to the perturbed Riccati equations. The computational burden in the two approaches is of similar order of magnitude, except that the finite difference approximations lead to a flexible algorithm, besides being applicable to general nonlinear models; the linear model, being a simplified case, requires no special treatment.

In Sec. 5.4, it has been shown that to ensure physically meaningful values for indirectly obtained G, the diagonal elements of the matrix KC must be constrained to be less than unity. It was also shown that minimization of the cost function, Eq. (5.4), subject to these nonlinear inequality constraints leads to a nonlinear programming problem, which is solved by a quadratic programming method. In the present case, a similar approach has to be adopted with the further simplification that the elements of KC are constrained to be less than unity, where the observation matrix C is obtained by a first-order system approximation. Because the algorithmic details, covered in Eqs. (5.35–5.39), remain the same, they are not repeated here. It is also required to follow the procedure of Eq. (5.41) to correct the estimates of F whenever R is updated. If we encounter local divergence during the F correction, we also follow the same philosophy of parameter-halving as elaborated in Sec. 5.4.

To highlight the differences between the two approaches for linear and nonlinear systems respectively, the computational steps in the two cases are summarized in Table 5.1.

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TABLE 5.1 SUMMARY OF FILTER-ERROR-METHOD COMPUTATIONS FOR LINEAR AND NONLINEAR SYSTEMS

$\dot{x}(t) = Ax(t) + Bu(t)$	$\dot{x}(t) = f[x(t), u(t), \beta]$
Integrate state equations by state transition matrix and its integral: $\tilde{x}(t_{k+1}) = \Phi\hat{x}(t_k) + \Psi B\bar{u}(t_k)$	Numerical integration of state equations, typically Runge–Kutta 2nd or 4th order
Compute observation variables: $\tilde{y} = C\tilde{x}(t_k) + Du(t_k) + b_y$	Compute observation variables: $\tilde{y} = g[\tilde{x}(t_k), u(t_k), \beta] + b_y$
Compute residuals: $[z(t_k) - \tilde{y}(t_k)]$ Compute maximum likelihood estimate of R, Eq. (5.11)	
Solve steady-state Riccati equation, Eq. (5.14), for P Compute gain matrix: $K = PC^TR^{-1}$	
Compute corrected states: $\hat{x}(t_k) = \tilde{x}(t_k) + K[z(t_k) - \tilde{y}(t_k)]$	
Compute gradients: $\frac{\partial\tilde{x}(t_{k+1})}{\partial\Theta} \approx \Phi\frac{\partial\hat{x}(t_k)}{\partial\Theta} + \Psi\frac{\partial A}{\partial\Theta}\bar{x}(t_k) + \Psi\frac{\partial B}{\partial\Theta}\bar{u}(t_k) + \Psi\frac{\partial b_x}{\partial\Theta}$ $\frac{\partial\tilde{y}(t_k)}{\partial\Theta} = C\frac{\partial\tilde{x}(t_k)}{\partial\Theta} + \frac{\partial C}{\partial\Theta}\tilde{x}(t_k) + \frac{\partial D}{\partial\Theta}u(t_k) + \frac{\partial b_y}{\partial\Theta}$ $\frac{\partial\hat{x}(t_k)}{\partial\Theta} = \frac{\partial\tilde{x}(t_k)}{\partial\Theta} - K\frac{\partial\tilde{y}(t_k)}{\partial\Theta} + \frac{\partial K}{\partial\Theta}[z(t_k) - \tilde{y}(t_k)]$ The gradients of the gain matrix K, $\frac{\partial K}{\partial\Theta} = \frac{\partial P}{\partial\Theta}C^TR^{-1} + P\left(\frac{\partial C}{\partial\Theta}\right)^T R^{-1}$ This requires $\partial P/\partial\Theta$, which are obtained by solving a set of ***Lyapunov equations*** [see Eqs. (5.27–5.34)]	Numerical approximation of response gradients from perturbed system equations: $\dot{\hat{x}}^p(t) = f[\hat{x}^p(t), u(t), \beta + \delta\beta]$ $\tilde{y}^p = g[\tilde{x}^p(t_k), u(t_k), \beta + \delta\beta]$ $\hat{x}^p(k) = \tilde{x}^p(k) + K^p[z(k) - \tilde{y}^p(k)]$ $\left(\frac{\partial\tilde{y}}{\partial\Theta}\right)_{ij} \approx \frac{\tilde{y}_i(\Theta + \delta\Theta_j e^j) - \tilde{y}_i(\Theta)}{\delta\Theta_j}$ for each parameter perturbation, that is, extended Kalman filter for perturbed state equations: – compute perturbed $\tilde{x}^p$, $\tilde{y}^p$ and $\hat{x}^p$. This requires K^p, which is obtained from P^p. Note: Because gradients of $\tilde{x}$ and $\hat{x}$ are not explicitly used (as in the case of linear systems), the gradients of K are not required. This implies that gradients of P, and hence the Lyapunov equations, are not explicitly required. These effects are indirectly taken care of through perturbed gain matrices K^p and calls for solving ***perturbed Riccati equations*** for each parameter.
Compute parameter update $\Delta\Theta$ by Gauss–Newton method; Eqs. (5.18–5.20).	
Check for the constraints $KC < 1$; update $\Delta\Theta$ if necessary; Eqs. (5.35–5.39).	
Compensate F estimates for new R, Eq. (5.41).	
Iterate until convergence.	

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5.5.2 TIME-VARYING FILTER

In a few cases, for example, when system parameters are time varying or the response is dominated by nonlinearities or when the deviations from the nominal trajectory are large, it might become necessary to incorporate a time-varying filter for state estimation. The formulation runs parallel to the one we just discussed for the steady-state filter. It can be shown that in the present case the maximum likelihood cost function is given by [10, 12, 13]

$$J(\Theta, R) = \frac{1}{2}\sum_{k=1}^{N}[z(t_k) - \tilde{y}(t_k)]^T\, R^{-1}(t_k)[z(t_k) - \tilde{y}(t_k)] + \frac{1}{2}\sum_{k=1}^{N}\ell n(\det(R(t_k))) \tag{5.57}$$

Starting from suitably specified initial values of Θ, the new updated estimates are obtained applying the Gauss–Newton algorithm:

$$\Theta_{i+1} = \Theta_i + \Delta\Theta \quad \text{and} \quad \mathcal{F}\Delta\Theta = -\mathcal{G} \tag{5.58}$$

where i is the iteration index and $\mathcal{F}$ and $\mathcal{G}$ are given by

$$\mathcal{F} = \sum_{k=1}^{N}\left[\frac{\partial\tilde{y}(t_k)}{\partial\Theta}\right]^T R^{-1}(t_k)\left[\frac{\partial\tilde{y}(t_k)}{\partial\Theta}\right] \tag{5.59}$$

$$\mathcal{G} = -\sum_{k=1}^{N}\left[\frac{\partial\tilde{y}(t_k)}{\partial\Theta}\right]^T R^{-1}(t_k)\,[z(t_k) - \tilde{y}(t_k)] \tag{5.60}$$

The iterative update equation (5.58) is exactly the same as Eq. (5.18). Equations (5.59) and (5.60) for the information matrix $\mathcal{F}$ and gradient $\mathcal{G}$ are very similar to Eqs. (5.19) and (5.20), except that we now use the time-varying residual covariance matrix $R(t_k)$ at each discrete time point, instead of the steady-state version R.

As in the case of the steady-state filter, we now discuss techniques to propagate the states and compute the output variables. For nonlinear state estimation, we once again use the extended Kalman filter (EKF), based on the nonlinear state equations for prediction and the correction using a first-order approximation of the state and measurement equations. For the system defined in Eqs. (5.42–5.44), the EKF consisting of an extrapolation and an update step can be summarized as follows:

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Extrapolation:

$$\tilde{x}(t_{k+1}) = \hat{x}(t_k) + \int_{t_k}^{t_{k+1}} f[x(t), \overline{u}(t_k), \beta]\mathrm{d}t \tag{5.61}$$

$$\tilde{y}(t_k) = g[\tilde{x}(t_k), u(t_k), \beta] \tag{5.62}$$

$$\widetilde{P}(t_{k+1}) \approx \Phi\hat{P}(t_k)\Phi^T + \Delta t\, FF^T \tag{5.63}$$

where $\Phi = e^{A\Delta t}$ is the transition matrix with sampling time $\Delta t = t_k - t_{k-1}$ and A is given by

$$A(t_{k+1}) = \left.\frac{\partial f[x(t), u(t), \beta]}{\partial x}\right|_{x=\hat{x}(t_k)} \tag{5.64}$$

Update:

$$K(t_k) = \widetilde{P}(t_k)C^T(t_k)[C(t_k)\widetilde{P}(t_k)C^T(t_k) + GG^T]^{-1} \tag{5.65}$$

$$\hat{x}(t_k) = \tilde{x}(t_k) + K(t_k)[z(t_k) - \tilde{y}(t_k)] \tag{5.66}$$

$$\begin{aligned}\hat{P}(t_k) &= [I - K(t_k)C(t_k)]\,\widetilde{P}(t_k) \\ &= [I - K(t_k)C(t_k)]\,\widetilde{P}(t_k)\,[I - K(t_k)C(t_k)]^T \\ &\quad + K(t_k)\,GG^TK^T(t_k)\end{aligned} \tag{5.67}$$

where C is given by

$$C(t_k) = \left.\frac{\partial g[x(t), u(t), \beta]}{\partial x}\right|_{x=\tilde{x}(t_k)} \tag{5.68}$$

Equation (5.63) for propagation of the covariance of the state prediction error is an approximation obtained by assuming that Δt is small. In Eq. (5.67), two equivalent forms are provided to compute the updated covariance matrix of the state error $\hat{P}$. For the reasons elaborated in Appendix F, we use the recommended longer "Joseph" form, which is better conditioned for numerical computations and helps to ensure that P remains positive definite.

The time-varying filter further requires the initial state covariance matrix P_0 to be specified. In the case of nonlinear systems, x_0 are estimated along with the other unknown parameters. In such a case P_0 is zero.

The response gradients $\partial y/\partial\Theta$ required in Eqs. (5.59) and (5.69), as well as the gradients of the system functions, Eqs. (5.64) and (5.68), can be evaluated using either the analytical differentiation or finite difference approximation. For

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the reasons that we have already discussed in this chapter, they are approximated using finite differences [10, 12].

For a small perturbation δx_j in each of the n_q state variables, the gradients in Eqs. (5.64) and (5.68) are approximated using the central difference formulas as follows:

$$A_{ij}(t_{k+1}) = \frac{1}{2\delta x_j(t_k)} \left[f_i[x(t_k) + \delta x_j e^j, u(t_k), \beta] - f_i[x(t_k) - \delta x_j e^j, u(t_k), \beta] \right]_{x=\hat{x}(t_k)} \tag{5.69}$$

$$C_{ij}(t_k) = \frac{1}{2\delta x_j(t_k)} \left[g_i[x(t_k) + \delta x_j e^j, u(t_k), \beta] - g_i[x(t_k) - \delta x_j e^j, u(t_k), \beta] \right]_{x=\hat{x}(t_k)} \tag{5.70}$$

Likewise, for a small perturbation $\delta\Theta_j$ in each component of Θ, the perturbed response variables $\tilde{y}_j^p$ are computed. The corresponding sensitivity coefficient is then approximated by

$$\left[\frac{\partial \tilde{y}(t_k)}{\partial \Theta} \right]_{ij} \approx \frac{\tilde{y}_j^p(t_k) - \tilde{y}_i(t_k)}{\delta\Theta_j} \tag{5.71}$$

which is the same as Eq. (5.62). The perturbed responses $\tilde{y}^p(t_k)$ are obtained from the perturbed system equations. These equations have a form similar to those of Eqs. (5.61–5.68), except that each time Θ is replaced through perturbed parameter vector $(\Theta + \delta\Theta_j e^j)$. The reader can easily write these equations. It is easy to infer from Eq. (5.71) that we have to propagate the perturbed state variables $x^p(t_k)$ and the perturbed error-covariance matrix $P^p(t_k)$ for each element of Θ.

All of the quantities required in Eq. (5.58–5.60) to compute iteratively the parameter improvement $\Delta\Theta$ are now defined. The residuals $[z(t_k) - \tilde{y}(t_k)]$ required in Eq. (5.60) are readily available as a part of Eq. (5.66) in the filter implementation. The covariance matrix of residuals $R(t_k)$ can be obtained without any additional computation by making use of the relation:

$$R(t_k) = C(t_k)P(t_k)C^T(t_k) + GG^T \tag{5.72}$$

where the right-hand side is already evaluated in Eq. (5.65). In this section hitherto, it has been assumed that the state noise and measurement noise distribution matrices F and G are known and that the filter algorithm of Eqs. (5.61–5.70) is formulated accordingly in terms of these F and G matrices and not in terms of R, the covariance matrix of the residuals. It becomes possible to include the elements of the F matrix in the unknown parameter vector Θ; but we do not estimate the elements of G as unknown parameters Θ applying the Gauss–Newton step for the reasons of convergence that we have discussed several times now.

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For the filter-error method using a steady-state filter, the filter implementation was done in terms of R (as already discussed in the foregoing section). This was found to be convenient because the cost function in Eq. (5.4) is defined in terms of R, and also the steady-state R can be estimated using a closed-form solution. It is also possible to implement the steady-state filter in terms of G. From the explicitly estimated steady-state R, the measurement noise covariance matrix GG^T can be derived with only minor additional computations, but was not necessary, because, as just mentioned, the filter is implemented in terms of R.

For a time-varying filter, the maximum likelihood estimate of $R(t_k)$ can be obtained by setting the partial derivative of Eq. (5.57) with respect to $R(t_k)$ to zero. This yields

$$R(t_k) = [z(t_k) - \tilde{y}(t_k)]\ [z(t_k) - \tilde{y}(t_k)]^T \tag{5.73}$$

The time-varying filter implementation, however, requires knowledge of the G matrix. Although Eq. (5.73) provides the maximum likelihood estimate of $R(t_k)$ at each time point, it appears that a simple procedure to obtain updated G estimates from the updated $R(t_k)$, similar to the one in the steady-state case of Sec. 5.5.1, is not feasible for a time-varying filter.

In view of this restriction, we do not consider estimation of GG^T using a time-varying filter. This limitation is a drawback of the time-varying filter method. In this sense, this approach does not lead to a complete, integrated solution yielding maximum likelihood estimates of all relevant quantities affecting the system. Nevertheless, reasonable information about the G matrix can be obtained from the laboratory calibration of the various measurement sensors or applying Fourier smoothing techniques [22]. Although this is a feasible approach, we do not go into the details because such approaches have been used in the past only on sample test cases and not yet proved as applicable on a routine basis. Moreover, any inaccuracies in these estimates of G will affect the estimates of the system parameter Θ. It is from this aspect that we tend to prefer the well-established filter-error method using the steady-state filter.

5.6 INITIAL NOISE COVARIANCE MATRIX

In the case of the output-error method accounting for measurement noise only, there is no state update equation, that is, in other words the Kalman gain matrix K is zero. The predicted and the corrected states are one and the same, which are obtained by integration of the system equations. The residuals are not functions of R. This allows for a given Θ (either specified arbitrarily or obtained through some startup procedure) to compute the covariance matrix R using the Eq. (4.15). This is then followed by the Gauss–Newton step to compute the new Θ. Thus, the two-step procedure in this case amounts to the following:

1. Choose suitable initial values for Θ.
2. Estimate R using Eq. (4.15).

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3. Update parameter vector Θ applying the Gauss–Newton method, Eqs. (4.29) and (4.30).
4. Iterate on step 2 until convergence.

In the case of the filter-error method accounting for both process and measurement noise, a slightly different approach is necessary. To compute the states and the residuals, it is first required to obtain the state noise prediction error matrix P and the Kalman gain matrix K. This, in turn, requires the matrix R to be specified; see Eq. (5.8). The matrix cannot be assumed to be a null matrix because it will then lead to numerical difficulties in the solution of the Riccati equation for P. However, once an initial guess for R is provided, the two-step procedure can be implemented as shown:

1. Choose suitable initial values for Θ.
2. Generate or specify initial values for R.
3. Update parameter Θ by the Gauss–Newton method, Eqs. (5.18–5.20).
4. Estimate R using Eq. (5.11); compensate the estimates of F, Eq. (5.41).
5. Iterate on step 3 until convergence

How to specify the starting values for R might not be obvious in each case. Such information can come from laboratory calibration, from Fourier smoothing, or from prior estimation results. Just a rough order of magnitude is generally good enough. It also becomes possible to automatically generate this information by incorporating an option to compute the matrix R in the first pass, if R is not specified [16]. In such a case, we initially treat the complete noise as measurement noise, that is, basically running the algorithm in the output-error mode. This can be easily done for a given Θ, assuming F and in turn K to be zero. In the subsequent iterations, F gets estimated and R automatically adjusted in the maximum likelihood sense through the details provided. This easy and pragmatic approach is found to work very well in several of the aircraft applications and is hence recommended to be followed for the filter-error method based on the steady-state filter. It is also recommended to carry out the a few initial iterations in which R is kept constant.

5.7 EXTENSION OF FILTER-ERROR METHOD TO MULTIPLE EXPERIMENTS

We have briefly discussed in Chapter 3 the necessity of performing separate flight maneuvers (experiments) to excite different modes of aircraft motion, and in Chapter 4 we have shown that the output-error method can be readily extended to analyze multiple maneuvers simultaneously, yielding one set of parameters. In the case of the filter-error method, however, the analysis of multiple experiments simultaneously requires some considerations.

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Recall from Sec. 3.3 that the option of multiple experiments treats the initial conditions and biases separately for each of the maneuvers. In such a case, dropping the process noise matrix F, we had represented in Eq. (3.12) the single set of unknown parameters Θ as

$$\Theta = [\beta^T \ \alpha_1^T \ \alpha_2^T \ldots \alpha_{nE}^T \ \gamma_1^T \ \gamma_2^T \ldots \gamma_{nE}^T \ \delta_1^T \ \delta_2^T \ldots \delta_{nE}^T]^T \tag{5.74}$$

where n_E is the number of experiments to be analyzed simultaneously. Now, when we include the elements of the process noise distribution matrix in the unknown parameters Θ, Eq. (5.74) leads to

$$\Theta = [\beta^T \ \lambda^T \ \alpha_1^T \ \alpha_2^T \ldots \alpha_{nE}^T \ \gamma_1^T \ \gamma_2^T \ldots \gamma_{nE}^T \ \delta_1^T \ \delta_2^T \ldots \delta_{nE}^T]^T \tag{5.75}$$

where λ denotes the elements of the process noise distribution matrix F. The reader may note that Eq. (5.75) is based on the assumption that the process noise matrix remains the same for all of the experiments, that is, we have just one set of λ, like the system parameters Θ. The steady-state Riccati equation

$$AP + PA^T - \frac{1}{\Delta t} P C^T R^{-1} C P + F F^T = 0 \tag{5.76}$$

which is the same as Eq. (5.14), can be solved once to yield a single steady-state P, and subsequently from Eq. (5.8) or (5.48) a single gain matrix K, common to all of the time segments, can be obtained. In such a case, we can apply the filter-error algorithm that we have discussed in the foregoing sections without any difficulty. The unconstrained optimization and the constrained optimization to ensure KC to be less than one do not pose any difficulties.

The appropriateness of this assumption of the same process noise for all of the experiments is, however, questionable. If the multiple experiments are carried out under similar atmospheric conditions with low magnitude turbulence, this assumption, although not strictly correct, might still be considered acceptable because the estimates are somewhat insensitive to the process noise distribution matrix. In many cases this might be the case because we usually carry out several maneuvers in a sequence during a single flight, or different flights might have been performed when atmospheric conditions were stable. In the other case, that is, when turbulence level is medium to high and different from experiment to experiment, then multiple experiment analysis based on the parameter vector of Eq. (5.75) poses some practical and some fundamental difficulties.

For the sake of argument, consider that we wish to estimate matrix F separately for each experiment, and then the unknown parameter can be represented as

$$\Theta = [\beta^T \ \lambda_1^T \ \lambda_2^T \ldots \lambda_{nE}^T \ \alpha_1^T \ \alpha_2^T \ldots \alpha_{nE}^T \ \gamma_1^T \ \gamma_2^T \ldots \gamma_{nE}^T \ \delta_1^T \ \delta_2^T \ldots \delta_{nE}^T]^T \tag{5.77}$$

Treating elements of F separately for each experiment leads to an increased number of parameters, assuming F to be diagonal, $n_x \cdot n_E$ instead of n_x for the single F common to all experiments. But this is just an additional nuisance, requiring increased memory and computational time. It would be possible to compute the Kalman gain matrices for each time segment appropriately, compute the

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system responses from Eq. (5.46), and compute the covariance matrix of the innovations from Eq. (5.11). The first step of the Gauss–Newton method, Eqs. (5.18–5.20) including the perturbed gain matrices, Eq. (5.56), could be computed. After the unconstrained optimization, the inequality constraints on *KC* have to be checked, and, if violated, a constrained optimization according to Eqs. (5.35–5.39) performed.

The gain matrix *K*, which is a function of *F*, will be different for each time segment being analyzed. This difference is further aggravated in the case of nonlinear systems because we will have to compute the observation matrix *C*, linearized at the corresponding initial conditions [α in Eq. (5.77)], which will also be different from experiment to experiment. Thus, *KC* will be different due to different *F* as well as *C*. The constrained optimization given in Eqs. (5.35–5.39) does not cater to such a case. Constraining the solution of *KC* for any one particular time segment may not guarantee that constraints are satisfied for all of the n_E experiments. This is a fundamental difficulty, and how to account for variable multiple constraints is not yet resolved. Because of these practical and theoretical difficulties, the only pragmatic approach would be to deal with a single process noise distribution matrix by judiciously selecting the experiments to be analyzed simultaneously. In several cases this should be adequate, because, as suggested in Sec. 3.3, we analyze multiple experiments carried out specifically for system identification purposes, which are from the same flight or from flights under similar atmospheric conditions.

5.8 EXPLICIT MODELING OF GUST SPECTRUM

The filter-error method as presented in the foregoing sections is a generic approach in which the atmospheric turbulence $w(t)$ was treated as a white noise process, and we identified the process noise distribution matrix *F*, affecting the system state variables, along with the other unknown system parameters. It is sometimes argued that the characterization of atmospheric turbulence as a white noise is a poor representation and that the generic approach does not capitalize on the possible a priori knowledge about the model structures for the turbulence. Therefore, first we briefly indicate such an alternative approach based on the "state vector augmentation" and then evaluate the pros and cons of both of them.

For illustration purposes, the approach of state-vector augmentation is at best demonstrated by considering a simplified model pertaining to the longitudinal motion valid for small perturbations about the trim.

$$\begin{bmatrix} \dot{u} \\ \dot{\alpha} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_\alpha & 0 & -g \\ Z_u/U_0 & Z_\alpha/U_0 & 1 & 0 \\ 0 & M_\alpha & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ \alpha \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_{\delta e} \\ Z_{\delta e}/U_0 \\ M_{\delta e} \\ 0 \end{bmatrix} \delta_e \tag{5.78}$$

where the state vector is given by $x = [u\ \alpha\ q\ \theta]^T$; $u = \delta_e$ is the elevator input; U_0 the velocity component of the trim speed in the x direction; and $X_{()}$, $Z_{()}$,

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and $M_{()}$ denote the dimensional derivatives pertaining to the longitudinal force, vertical force, and pitching moment coefficients, respectively. The observation variables are given by $[a_x, a_z, u, \alpha, q, \theta]$.

Now, in order to account for the turbulence in the longitudinal and vertical directions, we model these stochastic disturbances using a Dryden spectrum assuming stationary and homogenous gusts [23, 24]. To incorporate such an explicit model in the parameter estimation procedure, we have to arrive at a model form compatible with the state-space model of our system, for example, Eq. (5.78). It can be shown that the Dryden spectrum can be described through a first-order Gauss–Markov process of the form:

$$\dot{x} = -a\,x + b\,\xi \tag{5.79}$$

where x is an arbitrary variable, representing either longitudinal or vertical gust component and $\xi(t) \sim \mathcal{N}(0, q_x)$ is the white noise (Gaussian distributed with zero mean and variance q_x). For a stationary Gauss–Markov process, the power spectral density is given by [7]

$$S_x(\omega) = \frac{b^2 q_x}{a^2} \frac{1}{1 + (\omega/a)^2} \tag{5.80}$$

Equation (5.80) yields a typical spectrum for $a = \omega_E U_0/L$, $b = U_0\sqrt{2\omega_E/\pi L}$, where L is the characteristic length in meters, ω_E the corner frequency in rad/s, and q_x is the mean square intensity of the gust. For longitudinal and vertical gusts, typically, ω_E is 1 and 2.4 rad/s, respectively [7]. Thus, they are characterized through the mean square intensities and scale (characteristic) lengths $(\overline{u}_g^2, L_u)$ and $(\overline{w}_g^2, L_w)$. Combining Eqs. (5.79) and (5.80), and introducing them in Eq. (5.78) leads to an extended model representation of the form

$$\begin{bmatrix} \dot{u} \\ \dot{\alpha} \\ \dot{q} \\ \dot{\theta} \\ \dot{u}_g \\ \dot{w}_g \end{bmatrix} = \begin{bmatrix} X_u & X_\alpha & 0 & -g & -X_u & -X_\alpha/U_0 \\ Z_u/U_0 & Z_\alpha/U_0 & 1 & 0 & -Z_u/U_0 & -Z_\alpha/U_0^2 \\ 0 & M_\alpha & M_q & 0 & 0 & -M_\alpha/U_0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \dfrac{-U_0}{L_u} & 0 \\ 0 & 0 & 0 & 0 & 0 & \dfrac{-\omega_E U_0}{L_w} \end{bmatrix} \begin{bmatrix} u \\ \alpha \\ q \\ \theta \\ u_g \\ w_g \end{bmatrix} + \begin{bmatrix} X_{\delta e} \\ Z_{\delta e}/U_0 \\ M_{\delta e} \\ 0 \\ 0 \\ 0 \end{bmatrix} \delta_e + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ U_0\sqrt{2/\pi L_u} & 0 \\ 0 & U_0\sqrt{2\omega_E/\pi L_w} \end{bmatrix} \begin{bmatrix} \xi_{ug} \\ \xi_{wg} \end{bmatrix} \tag{5.81}$$

Thus, we have now two additional state variables u_g and w_g representing the longitudinal and vertical gust velocities. The unknown parameters are the intensities and scale lengths of the gusts ($\overline{u}_g^2$, L_u) and ($\overline{w}_g^2$, L_w). The inputs ξ_{ug} and ξ_{wg} are the white noise processes, which are generated using the random number generator. Thus, through state variable extension, it becomes possible to include longitudinal and vertical gusts based on the knowledge about the form of the gust spectrum as described here or using other forms of representation [25].

Parameter estimation based on such an extended model is by no means simpler than that using the generic approach. We still have to apply the complex filter-error method incorporating a suitable state estimator. Moreover, if we assume that not only the longitudinal motion is affected by the gusts, but also the lateral-directional motion as well, extensions to coupled motion would require further additional states. Even when we restrict our attention to the longitudinal motion, it has been found that the estimation of the scale lengths and gust intensities posed convergence problems and provided inconsistent estimates compared to the expected values based on physical understanding of the atmosphere. Therefore, in some cases attempts to estimate statistical properties of turbulence were considered futile [8].

To summarize, we can say that based on a Dryden spectrum, it is possible to incorporate explicit gust models in parameter estimation. This approach of state-vector augmentation is, however, not flexible enough because it is coupled to the form of the model chosen to represent the system. The system representation should be pertinent to the physical phenomenon being investigated and can vary from time to time and case to case. Moreover, it leads to a larger size state vector. The number of additional parameters required in the state-augmentation and the generic approach remains typically of the same order. However, the generic approach has the advantages that no additional states are required to account for the turbulence in the parameter estimation and is hence computationally faster. It proved to be flexible having quite good convergence properties. Besides these advantages, because our major interest is to identify aircraft parameters, the form of the turbulence model is of secondary importance, and as such it is of little importance which approach we follow. Based on our past experience that clearly showed the generic approach to be a better one, we recommend that the generic approach based on Sec. 5.5.1 for more reliable and accurate estimation of aerodynamic characteristics from flight data in turbulence be applied.

5.9 ON THE EQUIVALENCE OF OUTPUT-ERROR AND FILTER-ERROR METHODS

Having studied the output-error and the filter-error methods, let us go more deeply into the exact differences between the two and try to understand precisely when the two methods are mathematically equivalent. We discuss the case when there is no process noise ($F = 0$). Besides the assumption of control inputs u being

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known exactly, if we further assume that the initial conditions are also known precisely, then there would be no (state) estimation problem [26]. In the case of linear systems, the state prediction given by Eq. (5.5), which is repeated here for convenience

$$\tilde{x}(t_{k+1}) = \Phi\hat{x}(t_k) + \Psi B\overline{u}(t_k) + \Psi b_x, \quad x(t_0) = 0 \tag{5.82}$$

would provide the exact solution.

We come across loosely made statements that the output-error and filter-error methods are the same when there is no process noise. Mathematically, however, this is true only in the exercises that deal with simulated data because in such a case we know the initial conditions of the states exactly, and we would have switched off the process noise while generating the data to be analyzed. However, in any exercise dealing with flight measured data the assumptions of initial conditions x_0 being known exactly and the process noise F being zero are never met in practice. For the sake of argument, if we presume that the process noise might be neglected, even then we never know the true values of the initial conditions for the data being analyzed because of the measurement errors and measurement noise. The two methods treat statistically the initial conditions differently.

In the sequel, when we consider the filter-error method with process noise distribution matrix $F = 0$, it is not strictly equivalent to the output-error method because we still account for the uncertainties in the initial conditions. Because of this reason, we will get some small differences in the numerical estimates of the parameters when we set $F = 0$ and apply the filter-error method, compared to the pure output-error method (as discussed in the Chapter 4, which completely neglects the statistics of w and x_0). The differences are usually very small and within an acceptable order of the numerical accuracy. In fact, it is from this subtle aspect, it can be argued, that the estimates using the filter-error algorithm with $F = 0$ are more likely to be reliable and better, at least from the statistical view point, because the uncertainties in the state are accounted for.

5.10 FEM SOFTWARE

As in the case of the output-error method, we provide in this book a preliminary version of the software for the filter-error method using a steady-state filter that runs under Matlab®. It caters to estimating the system parameters, process noise distribution matrix, and measurement noise distribution matrix from a single maneuver. We provide here software packages for both types of system model implementations, namely, for linear models in terms of system matrices and for general nonlinear models in terms of state and observation equations. A detailed README is found under the main directory /FVSysID2/chapter05/. The software packages provided help to understand more closely the various algorithmic steps that we have so far studied in this chapter. Furthermore, it allows us generating the results that we will discuss here.

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Figure 5.2 shows a flowchart of the computational procedure for the filter-error method catering to linear systems. It is based on the details provided in Secs. 5.2 and 5.4. The function names are provided at the bottom of each block to trace the programming details. The main program is called "ml_fem_lin2." The model definition is in terms of system matrices A, B, C, D, F, and bias vectors b_x and b_y. It would suffice here to mention that "ml_fem_lin2" is provided to gain deeper understanding of the complex computational procedure that was followed in the past [16, 17]; for practical purposes, we recommend the use of the software "ml_fem2" that is capable of handling both linear and nonlinear systems. It is for this reason that we defer the detailed description of the software limited to linear systems only to /FVSysID2/chapter05/fem_linear/README_Linear.

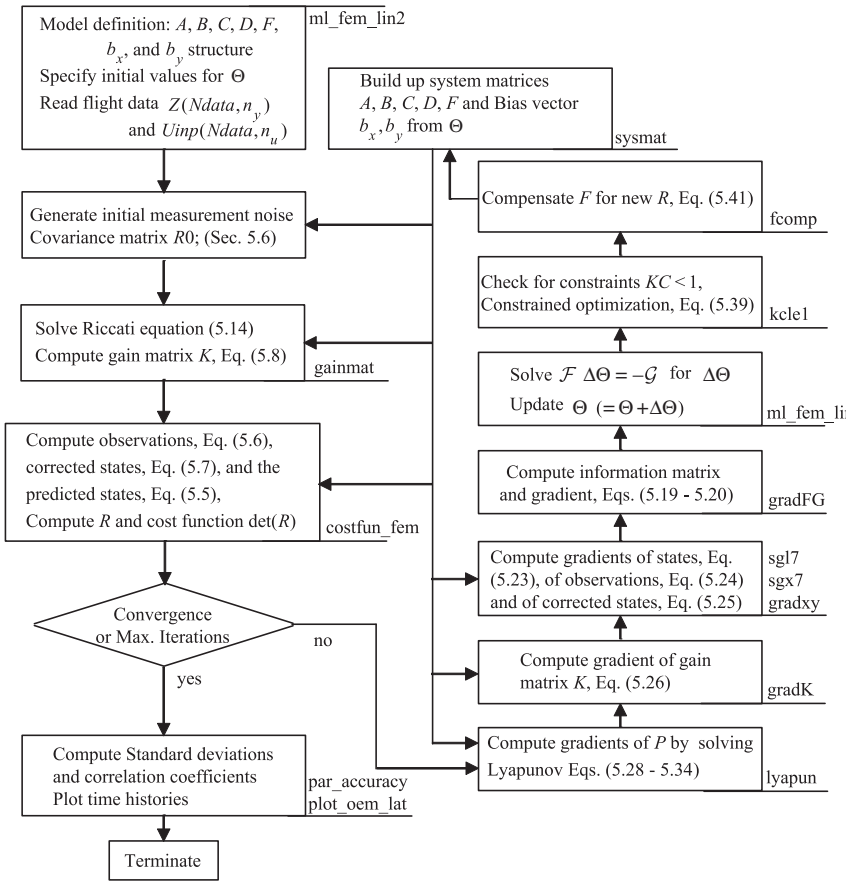

Fig. 5.2 Details of implemented filter-error method for linear systems.

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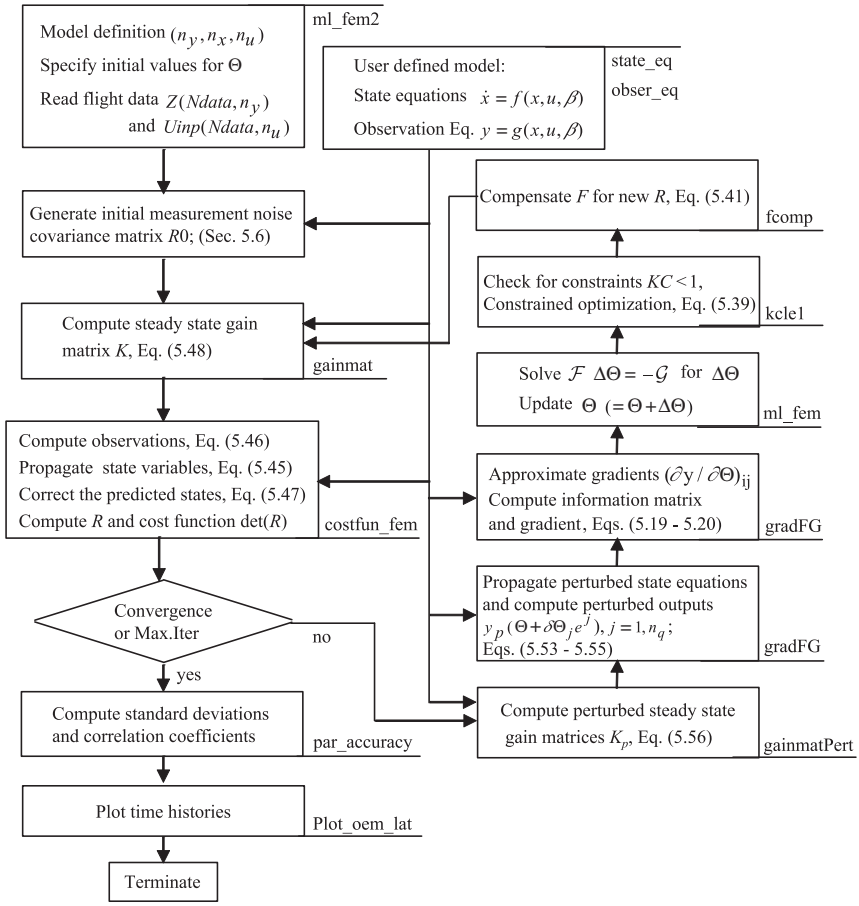

Fig. 5.3 Details of filter-error method for general nonlinear systems.

Figure 5.3 shows a flowchart of the computational procedure for the filter-error method applicable to general nonlinear systems. It is based on the details provided in Sec. 5.5.1. In the case of nonlinear systems, we estimate the system parameters and noise covariances, but prespecify the initial conditions and keep them fixed. Extension of the software to estimate x_0 for nonlinear systems is, of course, possible, but it is left to the readers [27]. While applying this software to linear systems, bias parameters of state and observation equations can be included and estimated. The software provides an option to selectively keep parameters included in the postulated model fixed and estimate the remaining. A comparison of Figs. 5.2 and 5.3 reflects the differences that were highlighted in Table 5.1. It turns out that implementation of the software "ml_fem2," based

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on Sec. 5.5.1, is somewhat more straightforward than that of the "ml_fem_lin2," particularly for computation of various gradients.

From Fig. 5.3 it can be seen that we code the postulated model in functions "state_eq" and "obser_eq." The gain matrix K is computed in "gainmat." The function "costfun_fem" propagates the states and computes the observations and the cost function. The perturbed gain matrices are computed in "gainmatPert"; the function "gradFG" propagates the perturbed state equations and computes the perturbed system responses, the gradient $\mathcal{G}$, and the information matrix $\mathcal{F}$; and the parameter updates $\Delta\Theta$ are computed in the main program "ml_fem2" applying the Gauss–Newton method. We prefer to use Cholesky factorization to solve for $\Delta\Theta$. Then, we check for the constraints in "kcle1" and perform constrained optimization, if necessary. The F estimates are compensated for new R in "fcomp," but only after iterFupR iterations (recommended value 3). The convergence checking is performed in "ml_fem2," so also the step-size control including halving of the parameter steps to overcome intermediate divergence. At the end, we compute the standard deviations and correlation coefficients in "par_accuracy" and finally make plots of the time histories and of the estimates. The source code (Matlab® m-files) for the filter-error method is provided in the directory /FVSysID2/chapter05/.

The starting point is the main program called "ml_fem2." It provides the user interface to define the model and read the flight data. Different system models to be analyzed are denoted through the integer flag "test_case":

test_case	index for the test case

which is to be uniquely associated with the state-space model and with the user-provided interface in the form of a function defining the model parameters, flight data, and other relevant details. Different state-space models, to be coded by the user, can be invoked by defining the following flags and strings in a function mDefCase** called in "ml_fem2," where ** denotes the test case index:

state_eq	function name (to code the right-hand side of the state equations)
obser_eq	function name (to code the right-hand side of the observation equation)
Nx	number of state variables (n_x)
Ny	number of output variables (n_y)
Nu	number of input variables (n_u)
Nparsys	number of system parameters
Nparam	total number of unknown parameters (n_q)

The variable names used in the program "ml_fem2" and in the called functions are shown on the left side, followed by the description and the notation used to denote these variables in the text. As in the case of OEM software elaborated in Sec. 4.19, default values are defined in the main program "ml_fem2" for

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the maximum number of iterations (niter_max), the convergence criterion (tolR), and the step size for the parameter perturbations (par_step). The default value is also provided for "iterFupR," the iteration count from which the F-estimate will be corrected using the latest R-estimate according to Eq. (5.41). They can be suitably adapted, if necessary, in a specific case. It is obvious that the model specification in terms of (Nx, Ny, Nu, Nparam) must match with those coded in the preceding two user functions for the postulated model.

As discussed in Sec. 5.6, an option has been provided to generate the residual covariance matrix. If required, the same can be specified by the user by setting the integer flag iSD and defining the standard deviations of the output errors. It is also required to specify the starting values for the parameters and the integer flags to indicate whether the particular parameter is free (to be estimated) or fixed (to be kept fixed).

iSD	index for the initial residual covariance matrix =0: Default (Recommended); =1: User specified standard deviations of output errors
param	starting values for unknown parameters ($\boldsymbol{\Theta}$)
parFlag	flags for free and fixed parameters =1: free parameters (to be estimated) =0: fixed parameters (not to be estimated)
NparID	total number of unknown parameters, that is, nonzero elements of parFlag (n_q)
x0	initial conditions on state variables (kept fixed)
SDyError	initial standard deviations of output errors, if iSD = 1

The flight data to be analyzed are also to be loaded in the function for the model definition and require specification or assignment of the following information:

Ndata	number of data points
dt	sampling time
t	time vector [=0 to (Ndata-1)*dt]
Z(Ndata,Ny)	flight data for measured outputs (N, n_y)
Uinp(Ndata,Nu)	flight data for measured control inputs (N, n_u)

While loading the flight data in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu), it is obvious that the dimensions of the variables must match with those used in the postulated model and programmed in the state and observation functions state_eq and obser_eq, respectively.

5.11 EXAMPLES

We now turn our attention to the application of the filter-error method to test cases. To enable critical evaluation of the filter-error method and to compare

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the estimates with the nominal parameter values, we first consider simulated aircraft responses and a linear model. The second example pertains to estimation of the lift, drag, and pitching-moment coefficients from a nonlinear model postulate, and the third example analyzes flight data gathered from several flights under different atmospheric conditions.

5.11.1 APPLICATION OF FILTER-ERROR METHOD TO DATA WITH PROCESS NOISE

In Sec. 4.20.2, we had analyzed data with a moderate to high level of turbulence generated through simulation [10–12, 16]. The same test case, pertaining to the aircraft lateral-directional motion, is now analyzed here applying the filter-error method.

The model pertaining to the lateral-directional motion is postulated as follows:

State equations:

$$\begin{aligned} \dot{p} &= L_p\, p + L_r\, r + L_{\delta a}\, \delta_a + L_{\delta r}\, \delta_r + L_v\, v + b_{x\dot{p}} \\ \dot{r} &= N_p\, p + N_{r}\, r + N_{\delta a}\, \delta_a + N_{\delta r}\, \delta_r + N_v\, v + b_{x\dot{r}} \end{aligned} \tag{5.83}$$

Observation equations:

$$\begin{aligned} \dot{p}_m &= L_p\, p + L_r\, r + L_{\delta a}\, \delta_a + L_{\delta r}\, \delta_r + L_v\, v + b_{y\dot{p}} \\ \dot{r}_m &= N_p\, p + N_{r}\, r + N_{\delta a}\, \delta_a + N_{\delta r}\, \delta_r + N_v\, v + b_{y\dot{r}} \\ a_{ym} &= Y_p\, p + Y_{r}\, r + Y_{\delta a}\, \delta_a + Y_{\delta r}\, \delta_r + Y_v\, v + b_{yay} \\ p_m &= p + b_{yp} \\ r_m &= r + b_{yr} \end{aligned} \tag{5.84}$$

The unknown parameter vector Θ for the preceding model consists of the dimensional derivatives and the bias parameters. It is given by

$$\begin{aligned} \Theta^T = [&L_p\ L_r\ L_{\delta a}\ L_{\delta r}\ L_v\ N_p\ N_r\ N_{\delta a}\ N_{\delta r}\ N_v \\ &Y_p\ Y_r\ Y_{\delta a}\ Y_{\delta r}\ Y_v\ b_{x\dot{p}}\ b_{x\dot{r}}\ b_{y\dot{p}}\ b_{y\dot{r}}\ b_{yay}\ b_{yp}\ b_{yr}\ f_{pp}\ f_{rr}] \end{aligned} \tag{5.85}$$

The reader might notice that we have, for the convenience of notation, not shown explicitly the process noise distribution matrix in Eq. (5.83), but it is implicitly assumed that it is considered here, as evident from the unknown parameter vector of Eq. (5.85), which shows, compared to Eq. (4.92), two additional parameters f_{pp} and f_{rr} corresponding to the diagonal process noise distribution matrix.

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Thus, the postulated model is defined as follows:

	No. of Variables		Function Name
States	2	p, r	"xdot_TC02_dhc2_lat"
Outputs	5	$\dot{p}$, $\dot{r}$, a_y, p, r	"obs_TC02_dhc2_lat"
Inputs	3	δ_a, δ_r, v	

The right-hand sides of the state and observation equations, Eqs. (5.83) and (5.84), are programmed in the functions "xdot_TC02_dhc2_lat" and "obs_TC02_dhc2_lat." respectively. Thus, the model definition for this example provided by the function "/FVSysID2/chapter05/mDefCase02.m," which is called from the main program "ml_fem2," is as follows:

```
test_case = 2;                          % Index for the test case
state_eq  = "xdot_TC02_dhc2_lat";       % Function for state equations
obser_eq  = "obs_TC02_dhc2_lat";        % Function for observation equations
Nx        = 2;                          % Number of states
Ny        = 5;                          % Number of observation variables
Nu        = 3;                          % Number of input (control)
                                          variables
NparSys   = 15;                         % Number of system parameters
Nparam    = NparSys + Nx + Ny + Nx;     % Total number of parameters
dt        = 0.05;                       % Sampling time
```

The data to be analyzed, that is, the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu) are loaded from the file /FVSysID2/flt_data/y13aus_da1.asc, which is also provided. In short, we have the same model definition as in the case of the output-error method, Sec. 4.20.2, except that we have two additional parameters for the process noise distribution matrix. Caveat, the model definition file "mDefCase02.m" for the filter-error method is very similar to that for the output-error method, but not exactly the same. We use the default option (iSD = 0) to generate automatically the initial residual covariance matrix. Some suitable starting values have been specified for the Nparam parameters and the corresponding integer flags parFlags set to one because we estimate all of them.

The results of parameter estimation applying both the software packages, ml_fem_lin2 and ml_fem2, are summarized in Table 5.2. Nominal values of the derivatives are also provided in the same table and also those estimated in Sec. 4.20.2 applying the output-error method. The filter-error methods are found to converge smoothly within 10 iterations and yield estimates that are close to the nominal values. No numerical or convergence problems were encountered. On the other hand, the output-error method required 42 iterations, had severe convergence problems at each iteration, and yielded estimates that are not anywhere near the nominal values. This is evident from the comparison of the fourth and fifth columns of Table 5.2. The two programs "ml_fem_lin2" and "ml_fem2"

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TABLE 5.2 ESTIMATES APPLYING FILTER-ERROR AND OUTPUT-ERROR METHODS, TEST_CASE = 2

Derivative	Nominal Value	Estimates (Standard Deviation in %) Obtained by Applying		
		Filter-Error Method		Output-Error Method
		ml_fem_lin2	ml_fem2	ml_oem2
L_p	−5.820	−5.716 (6.6)	−5.722 (6.6)	−2.3294 (30.2)
L_r	1.782	1.720 (9.1)	1.721 (9.1)	0.6284 (35.6)
$L_{\delta a}$	−16.434	−14.946 (11.1)	−14.974 (11.1)	−0.0413 (1136)
$L_{\delta r}$	0.434	0.210 (206)	0.210 (206)	−1.1342 (11.9)
L_v	−0.097	−0.0883 (12.8)	−0.0884 (12.9)	0.0031 (37.6)
N_p	−0.665	−0.620 (9.4)	−0.620 (9.4)	−7.8203 (29.0)
N_r	−0.712	−0.723 (3.4)	−0.723 (3.4)	2.0246 (34.6)
$N_{\delta a}$	−0.428	−0.420 (59.9)	−0.424 (59.4)	0.4457 (147.8)
$N_{\delta r}$	−2.824	−2.831 (2.4)	−2.831 (2.4)	−2.6080 (6.7)
N_v	0.0084	0.0092 (19.0)	0.0092 (19.0)	0.0027 (109.3)
Y_p	−0.278	−0.297 (27.7)	−0.297 (27.7)	1.8696 (46.2)
Y_r	1.410	1.416 (2.5)	1.416 (2.5)	0.9587 (26.3)
$Y_{\delta a}$	−0.447	−0.528 (71.0)	−0.528 (71.0)	0.4415 (120.9)
$Y_{\delta r}$	2.657	2.694 (3.7)	2.694 (3.7)	3.1632 (4.74)
Y_v	−0.180	−0.180 (1.5)	−0.180 (1.5)	−0.1664 (3.0)
f_{pp}	0.200	0.167 (4.2)	0.167 (4.2)	–
f_{rr}	0.200	0.125 (3.7)	0.125 (3.7)	–
$\|R\|$		2.145×10^{-12}	2.1451×10^{-12}	2.3558×10^{-9}
Iterations		7	7	42

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yield the same results, as evident from the third and the fourth columns. Figure 5.4 shows the time history plots of the control inputs and the comparison of the measured data with the model estimated responses. Compared to Fig. 4.9, the agreement is now significantly improved. Thus, for data with an appreciable level of turbulence, the filter-error method is far superior.

A critical reader might notice that in Table 5.2 the estimates provided by the filter-error method agree much better with the nominal values for some of the derivatives than those for a few others. This is attributed to the fact that we have analyzed just a single maneuver, and the estimates of the secondary

derivatives are somewhat less accurate than the primary derivatives. In the present case the estimates of the stability derivatives, that is, the derivatives associated with the state variables, namely, L_p, L_r, N_p, and N_r, are significantly improved by the filter-error method. This directly confirms the statement that we had made in Sec. 5.2, saying the process noise and the covariance matrix of the state prediction error P improve the model. Also, the smooth progress of the optimizations with a significantly less number of iterations confirms that the filter-error method is less sensitive to stochastic disturbances and leads to a more nearly linear optimization problem, which has fewer local minima and a faster convergence rate.

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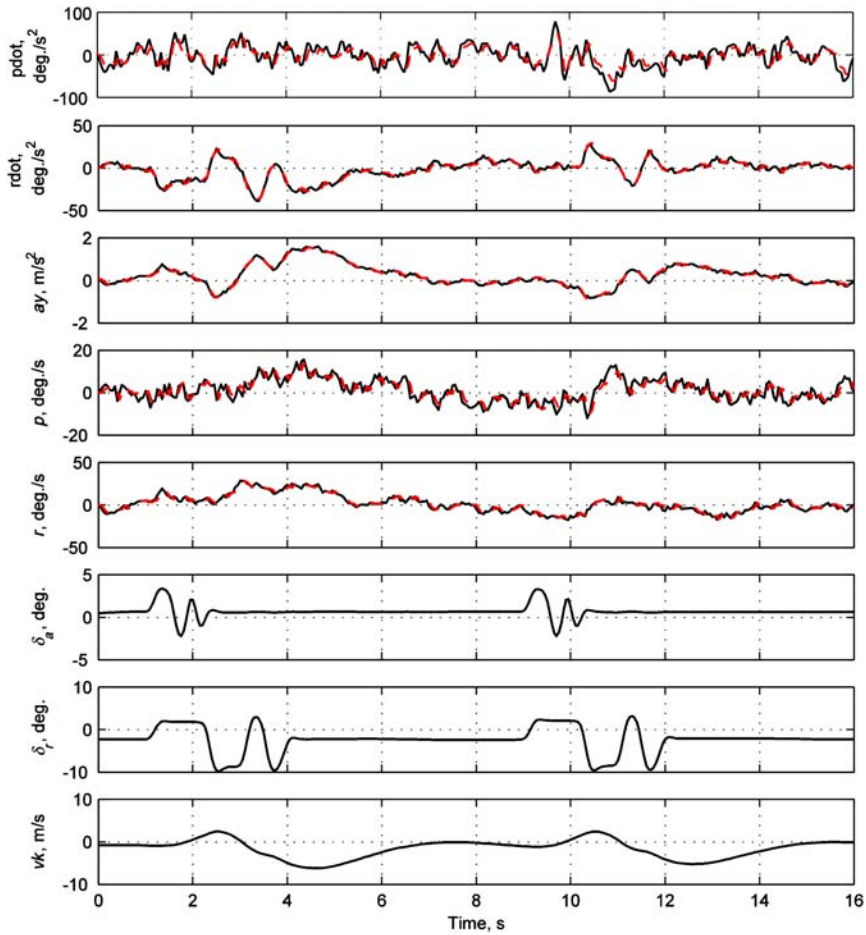

Fig. 5.4 Filter-error method applied to data with moderate to high level of turbulence – test case 2: ——, flight measured; -----, model predicted.

5.11.2 ESTIMATION OF LONGITUDINAL DERIVATIVES

The second example pertains to determination of the lift, drag, and pitching-moment coefficients of the research aircraft HFB-320. Flight tests were carried out to excite the longitudinal motion through a multistep elevator input resulting in short-period motion and a pulse input leading to phugoid motion [28]. The following model has been postulated to estimate the nondimensional derivatives:

State equations:

$$\begin{aligned}
\dot{V} &= -\frac{\bar{q}S}{m}\, C_D + g\, \sin(\alpha - \theta) + \frac{F_e}{m}\cos(\alpha + \sigma_T) \\
\dot{\alpha} &= -\frac{\bar{q}S}{mV}\, C_L + q + \frac{g}{V}\cos(\alpha - \theta)\, - \frac{F_e}{mV}\sin(\alpha + \sigma_T) \\
\dot{\theta} &= q \\
\dot{q} &= \frac{\bar{q}S\bar{c}}{I_y}\, C_m + \frac{F_e}{I_y}\,(\ell_{tx}\sin\sigma_T + \ell_{tz}\cos\sigma_T)
\end{aligned} \tag{5.86}$$

where the lift, drag, and pitching-moment coefficients are modeled as

$$\begin{aligned}
C_D &= C_{D0} + C_{DV}\,\frac{V}{V_0} + C_{D\alpha}\,\alpha \\
C_L &= C_{L0} + C_{LV}\,\frac{V}{V_0} + C_{L\alpha}\,\alpha \\
C_m &= C_{m0} + C_{mV}\,\frac{V}{V_0} + C_{m\alpha}\,\alpha + C_{mq}\,\frac{q\bar{c}}{2V_o} + C_{m\delta e}\,\delta_e
\end{aligned} \tag{5.87}$$

Observation equations:

$$\begin{aligned}
V_m &= V \\
\alpha_m &= \alpha \\
\theta_m &= \theta \\
q_m &= q \\
\dot{q}_m &= \frac{\bar{q}\,S\bar{c}}{I_y}\, C_m + \frac{F_e}{I_y}(\ell_{tx}\sin\sigma_T + \ell_{tz}\cos\sigma_T) \\
a_{xm} &= \frac{\bar{q}S}{m}\, C_X + \frac{F_e}{m}\cos\sigma_T \\
a_{zm} &= \frac{\bar{q}S}{m}\, C_Z - \frac{F_e}{m}\sin\sigma_T
\end{aligned} \tag{5.88}$$

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where the longitudinal and vertical force coefficients C_X and C_Z are given by

$$\begin{aligned} C_X &= C_L \sin\alpha - C_D \cos\alpha \\ C_Z &= -C_L \cos\alpha - C_D \sin\alpha \end{aligned} \tag{5.89}$$

where V is the true airspeed, α the angle of attack, θ the pitch attitude, q the pitch rate, δ_e the elevator deflection, F_e the thrust, σ_T the inclination angle of the engines, l_{tx}, l_{tz} its location from the center of gravity, $\overline{q}(=1/2\ \rho V^2)$ the dynamic pressure, m the mass, S the wing area, $\bar{c}$ the wing chord, I_y the moment of inertia, and ρ the density of air. The subscript m on the right-hand side of Eq. (5.88) denotes the measured quantities.

The preceding system equations in terms of variables in the stability axes (V, α) contain not only the common trigonometric and multiplicative nonlinearities, but, in addition, the variable dynamic pressure $\overline{q}(=1/2\ \rho V^2)$, which multiplies all of the aerodynamic derivatives and introduces an additional nonlinearity. Furthermore, inversion of the state variable V leads to further nonlinearities in the state equation for α.

The unknown parameter vector Θ consisting of the nondimensional derivatives is given by

$$\Theta = [C_{D0}\ C_{DV}\ C_{D\alpha}\ C_{L0}\ C_{LV}\ C_{L\alpha}\ C_{m0}\ C_{mV}\ C_{m\alpha}\ C_{mq}\ C_{m\delta e}]^T \tag{5.90}$$

From a set of flight maneuvers carried out with small inputs, a particular record that appears to contain atmospheric turbulence is analyzed here. A record length of 60 s with a sampling time of 0.1 s is used.

Thus, the postulated model for the program "ml_fem2" is defined as follows:

	No. of Variables	Function Name
States	4: V, α, θ, q	"xdot_TC04_hfb_lon"
Outputs	7: V, α, θ, q, $\dot{q}$, a_x, a_z	"obs_TC04_hfb_lon"
Inputs	2: δ_e, F_e	

Because the variations in the altitude during the maneuver were small, the density of air ρ is assumed to be constant for the duration of the maneuver. We use a value of $\rho = 0.792\ \text{kg/m}^3$, corresponding to a nominal altitude of roughly 4200 m. The values for various geometrical quantities (mass, moments of inertia, reference area, chord, etc.) for the present case are found in the preceding two functions. The right-hand sides of the state equation, Eq. (5.86), are programmed in the function "xdot_TC04_hfb_lon" and those of the observation equation, Eqs. (5.87–5.89), in "obs_TC04_hfb_lon." Thus, the model definition for this example provided by the function "/FVSysID2/chapter05/mDefCase 04.m," called from the main program "ml_fem2," is as follows:

```
test_case = 4;                    % Integer flag for the test case
state_eq  = "xdot_TC04_hfb_lon";  % Function for state equations
obser_eq = "obs_TC04_hfb_lon";    % for observation equations
```

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```
Nx     = 4;                  % Number of states
Ny     = 7;                  % Number of observation variables
Nu     = 2;                  % Number of input (control) variables
NparSys = 11;                % Number of system parameters
Nparam = NparSys + Nx;       % Total number of parameters
dt     = 0.1;                % Sampling time
```

Once again, we use the default value for iSD = 0, specify suitable parameter values, and set the integer flags parFlag to one for all parameters.

To this case we once again apply both the output-error method and the filter-error method. Several different initial starting values for the parameters were tried out. It turns out that the output-error method (/FVSySID2/chapter04/ml_oem2 with test_case = 4) either did not converge or had convergence problems when the starting values were far from the optimum. However, starting from parameters somewhat closer to the optimum, the output-error method converged, yielding the plots shown in Fig. 5.5. The match is reasonable, showing clear indications of turbulence seen from the plots for α and a_z, particularly for the first few seconds, and for V over the complete record.

On the other hand, the filter-error method (/FVSySID2/chapter05/ml_fem2 with test_case = 4) worked quite well for several sets of starting values. Starting from the same set of values as used while applying the output-error method, the filter-error method yields the response match shown in Fig. 5.6. The agreement between the flight measured and model estimated responses is found to be very good and clearly brings out the improvement obtained by accounting for the process noise.

We now analyze the spectra of residuals from the two methods, shown in Fig. 5.7 for the four state variables. From Fig. 5.7a, it is found that they are not flat for the output-error method. The residuals show more energy content at lower frequencies. On the other hand, Fig. 5.7b shows for all of the four residuals flat spectra, thus indicating that the filter-error method is better suited to account for the atmospheric disturbances and unavoidable modeling errors.

5.11.3 ESTIMATION FROM FLIGHT DATA IN SEEMINGLY STEADY ATMOSPHERE

For flight data gathered in turbulence, the filter-error methods are necessary, because the output-error method is known to yield biased estimates in the presence of atmospheric turbulence [6, 13]. Even in the case of flight maneuvers in smooth air, the filter-error method could lead to better estimation results because some of the unavoidable modeling errors are then treated as process noise characterized by low-frequency contents rather than as measurement noise. Moreover, although it is generally argued that the flight tests for aircraft parameter estimation should be carried out in calm air, in any practical exercise one has no control over the prevailing atmospheric conditions or due to very

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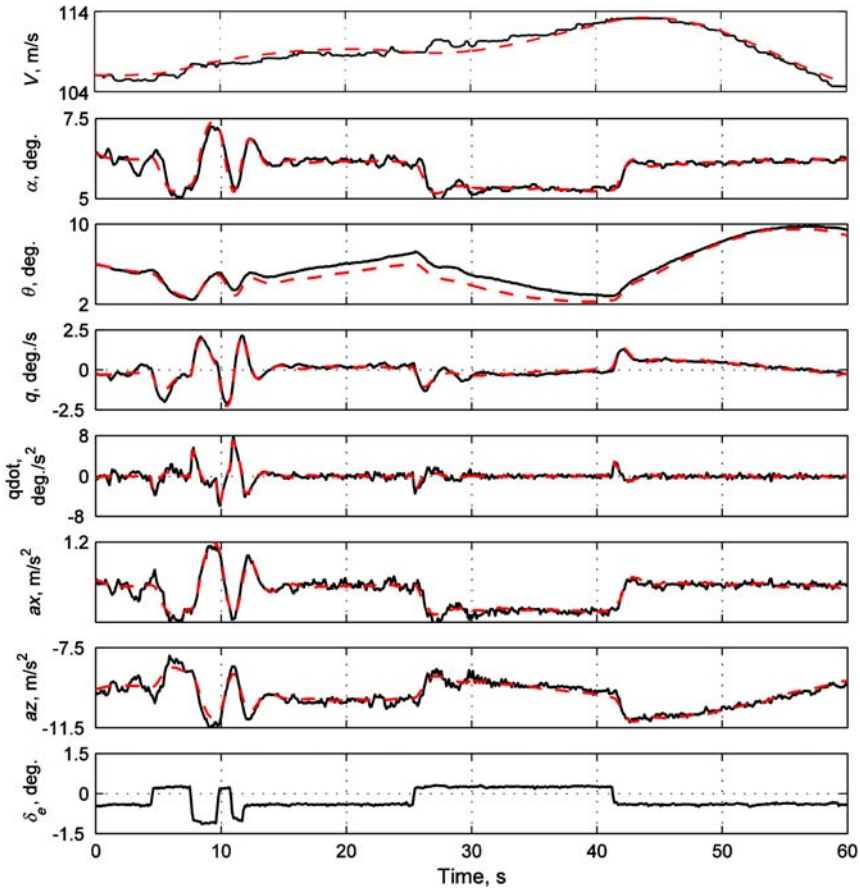

Fig. 5.5 Estimation of nondimensional derivatives by output-error method: ——, flight measured; -----, model predicted.

tight time schedules and due to cost factors involved in a time-bound project very little choice of waiting for steady atmospheric conditions [29].

As a typical example, the estimates of the weathercock stability derivative $C_{n\beta}$, obtained by applying the output-error and the filter-error method to the same set of flight data, are provided in Fig. 5.8a [30]. The C-160 data analyzed here were gathered from eight flights carried out during a span of less than two weeks, seven of them being in a seemingly steady atmosphere whereas one encountered a moderate amount of turbulence. It is clearly visible that the estimates provided by the output-error method, particularly those for the flight 223 during which moderate turbulence was encountered, differ much from those of other flights at the same nominal flight conditions. Moreover, a fair amount of

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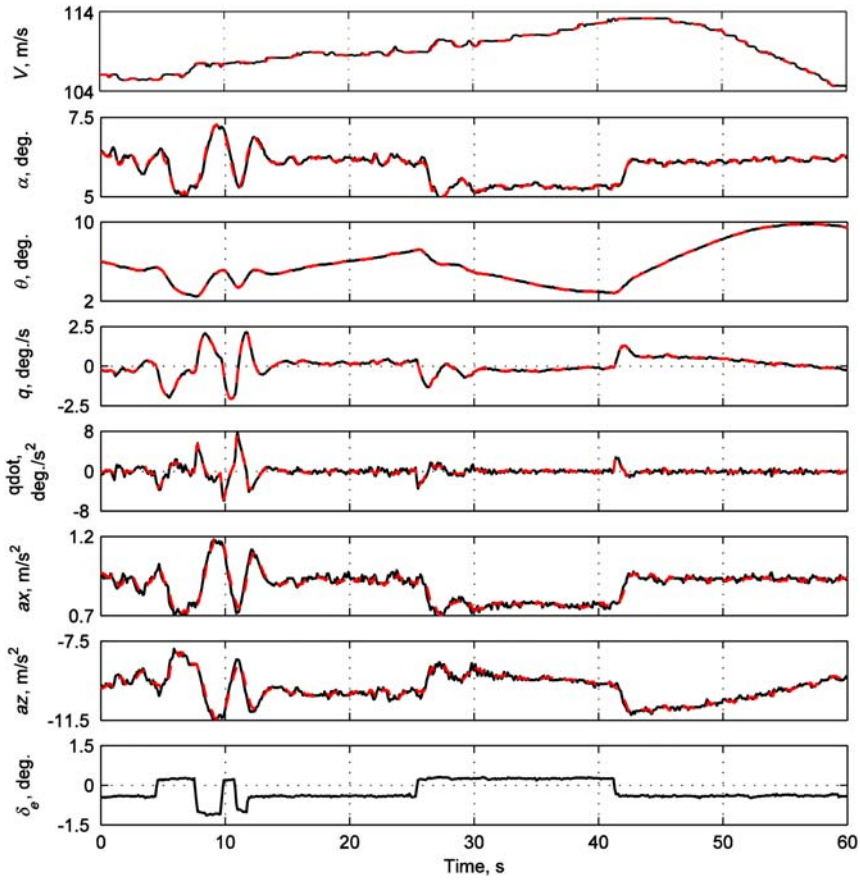

Fig. 5.6 Estimation of nondimensional derivatives by filter-error method: ——, flight measured; -----, model predicted.

scatter is observed in the estimates from other flights in a seemingly steady atmosphere, making a final conclusion regarding the nature of the nonlinearity or fairing of data difficult. On the other hand, the filter-error method yields clearly grouped estimates with much less scatter, and the estimates from the flight 223 match well with the other estimates. The nonlinear dependency of the weathercock stability on the angle of attack is now to be observed much better.

Figure 5.8b shows the estimates for the rudder effectiveness derivative $C_{n\delta r}$. The estimates of the control derivatives applying the output-error method even for flight in moderate turbulence were found to be of similar magnitude as those from other flights for corresponding flight conditions, subject to a small scatter with the scatter being somewhat smaller for the filter-error method.

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However, in an overall sense, the two methods provided comparable estimates of the control derivatives.

The improvements obtained in the estimates of the stability derivatives applying the filter-error method were significant; see Fig. 5.8a, whereas they are marginal in the case of control derivatives, as seen from Fig. 5.8b. The better performance of the filter-error method compared to the output-error method in estimating the stability derivatives is attributed to the fact that these derivatives depend on the aircraft motion variables, that is, on the estimated states. The estimated states are affected by stochastic inputs such as atmospheric turbulence. On the other hand, the comparable performance of the two methods in the case of control derivatives is attributed to the fact that the control derivatives are identified from the deterministic inputs applied to the aircraft. The aircraft response to these control inputs is adequate enough to enable better identification, from flights both in seemingly steady air and with moderate turbulence. Somewhat lower scatter among the estimates from the filter-error method could be partly due to the improved stability derivatives and partly due to accounting for the measurement noise in the control inputs itself; see Fig. 5.1. The noise level in the control inputs is low; otherwise, the output-error method would not have yielded acceptable results in most of the flight conditions analyzed [30].

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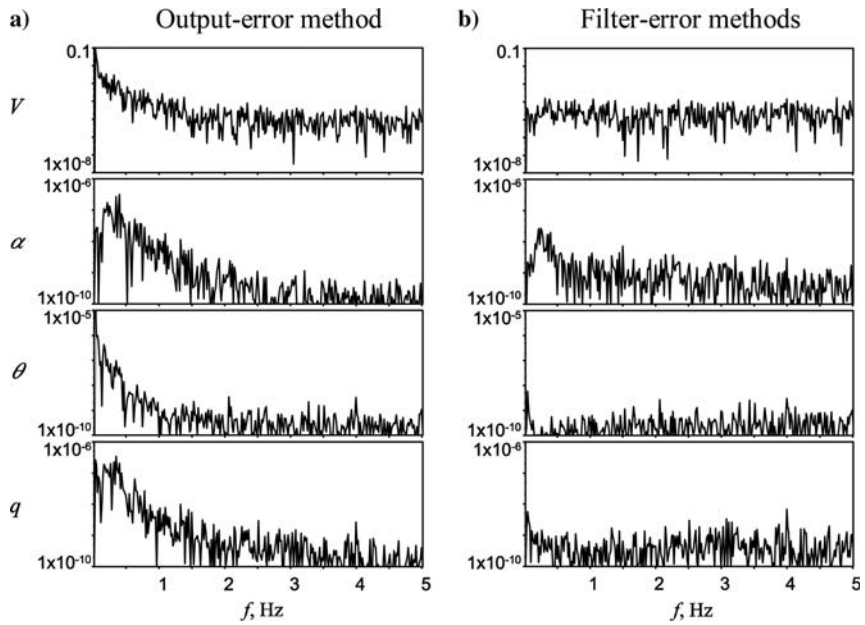

Fig. 5.7 Power spectral densities of residuals.

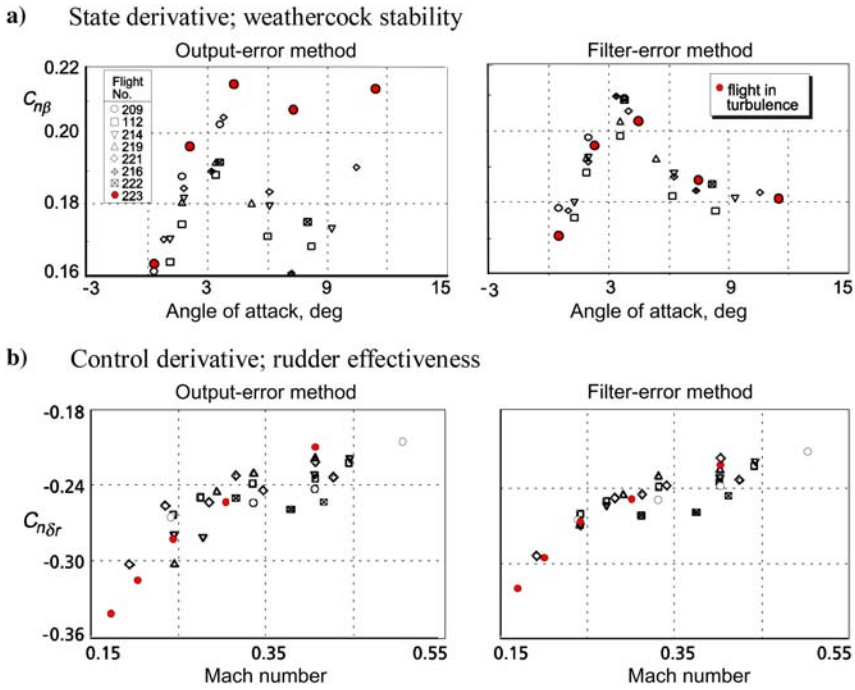

Fig. 5.8 Output-error and filter-error method applied to flight data in seemingly steady atmospheric conditions.

5.12 CONCLUDING REMARKS

In this chapter we discussed the most complex stochastic approach to aircraft parameter estimation that provides capabilities to analyze flight data gathered under turbulent atmospheric conditions. Various process noise formulations were discussed, and it was argued that the combined natural with innovation formulation is the best choice. Through a linear system representation, the various steps of the filter-error method were elaborated. Subsequently, we discussed the extensions of the filter-error method to general nonlinear systems. It has been demonstrated that a numerical approximation of gradients combined with a mixed state estimator using a nonlinear system model for state prediction and a correction based on a first-order approximation is a viable approach. We highlighted the exact differences between the algorithms for linear and nonlinear systems. As in the case of the output-error method, we adopt here the relaxation strategy leading to a two-step algorithm to optimize the maximum likelihood cost function. The system parameters and process noise distribution matrix were updated by the Gauss–Newton algorithm. For state estimation both steady-state

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and time-varying filters were presented. In the majority of the cases, we prefer to use the steady-state filter implementation. Finally, it is shown that the generic approach to estimate the noise distribution matrix is preferable over the other approach based on state augmentation using an explicit gust spectrum model.

Basic software for the filter-error method using a steady-state (constant gain) estimator implemented under Matlab® has been provided to follow the various computational steps of the algorithm. It has been applied to three typical aircraft examples that were presented to demonstrate the performance of the filter-error method. Samples of data with process noise have been supplied. The comparison of the estimates obtained applying the two methods indicates that the filter-error method yields significantly improved results compared to the output-error method for flight data with atmospheric turbulence. Even in the case of flight data in seemingly smooth air, the filter-error method provided estimates of the stability derivatives with a much smaller scatter than that from the output-error method. The estimates of the control derivatives provided by the two methods agreed fairly well.

The examples provide an answer to the question often raised regarding the practical utility of the filter-error method. It can be pragmatically concluded that these methods can yield better estimates, are no more limited to linear systems, and are indispensable for many applications. These advantages outweigh the disadvantage of higher computational overheads. The limitations of these methods are mainly twofold:

1. They allow estimation of a single set of the process noise distribution matrix, which implies they are mostly applicable to a single flight maneuver or to multiple maneuvers carried out under similar atmospheric conditions. As elaborated in this chapter, the difficulties in extending the filter-error method to multiple experiments treating process noise distribution matrix separately for each maneuver are not yet resolved.
2. The filter-error method always yields a good match for the responses, which is a necessary condition for parameter estimation, but not a sufficient one for parameters and model structure to be correct. The good response match tends to mask the discrepancies that can result from unaccounted aerodynamic or other, for example, aeroelastic effects, and as a consequence the model improvements may not become very obvious. In the worst case, we may obtain a good response match even when an important parameter has been inadvertently omitted.

Thus, some care is necessary while applying the advanced filter-error method. To this end, some practical tips have been provided. It is for the reasons just mentioned that the filter-error method is used in practice for special cases only; the majority of applications, including those covered later in a separate chapter on selected advanced examples, are based on the use of output-error or least-squares methods.

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REFERENCES

[1] Balakrishnan, A. V., "Stochastic System Identification Techniques," *Stochastic Optimization and Control*, edited by H. F. Karreman, Wiley, London, 1968.

[2] Mehra, R. K., "Maximum Likelihood Identification of Aircraft Parameters," *Proceedings of the 11th Joint Automatic Control Conference*, Atlanta, GA, 1970, pp. 442–444.

[3] Mehra, R. K., "Identification of Stochastic Linear Dynamic Systems Using Kalman Filter Representation," *AIAA Journal*, Vol. 9, No. 1, 1971, pp. 28–31.

[4] Tyler, J. S., Powell, J. D., and Mehra, R. K., "The Use of Smoothing and Other Advanced Techniques for VTOL Aircraft Parameter Identification," Naval Air Systems Command Contract Systems Control, Inc., Final Rept., N0019-69-C0534, Palo Alto, CA, June 1970.

[5] Iliff, K. W., "Identification and Stochastic Control with Applications to Flight Control in Turbulence," Ph. D. Dissertation, Univ. of California, Los Angeles, CA, May 1973.

[6] Maine, R. E., and Iliff, K. W., "Formulation and Implementation of a Practical Algorithm for Parameter Estimation with Process and Measurement Noise," *SIAM Journal of Applied Mathematics*, Vol. 41, No. 3, 1981, pp. 558–579.

[7] Schulz, G., "Maximum-Likelihood-Identifizierung Mittels Kalman-Filterung – Kleinste Quadrate Schätzung. Ein Vergleich bei der Bestimmung von Stabilitätsderivativa unter Berücksichtigung von Böenstörungen," DLR-FB 75-54, Germany, Aug. 1975 (in German).

[8] Foster, G. W., "The Identification of Aircraft Stability and Control Parameters in Turbulence," Royal Aircraft Establishment, RAE TR-83025, March 1983.

[9] Yazawa, K., "Identification of Aircraft Stability and Control Derivatives in the Presence of Turbulence," AIAA Paper 77-1134, Aug. 1977.

[10] Jategaonkar, R. V., and Plaetschke, E., "Estimation of Aircraft Parameters Using Filter-Error Methods and Extended Kalman Filter," DFVLR-FB 88-15, Germany, March 1988.

[11] Jategaonkar, R. V., and Plaetschke, E., "Identification of Moderately Nonlinear Flight Mechanics Systems with Additive Process and Measurement Noise," *Journal of Guidance, Control, and Dynamics*, Vol. 13, No. 2, 1990, pp. 277–285.

[12] Jategaonkar, R. V., and Plaetschke, E., "Algorithms for Aircraft Parameter Estimation Accounting for Process and Measurement Noise," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 360–372.

[13] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[14] Gelb, A., *Applied Optimal Control*, The MIT Press, Cambridge, MA, 1974.

[15] Grewal, M. S., and Andrews, A. P., *Kalman Filtering Theory and Practice*, Prentice–Hall, Upper Saddle River, NJ, 1993.

[16] Jategaonkar, R. V., and Plaetschke, E., "Maximum Likelihood Estimation of Parameters in Linear Systems with Process and Measurement Noise," DFVLR-FB 87-20, Germany, June 1987.

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[17] Maine, R. E., and Iliff, K. W., "User's Manual for MMLE3, a General FORTRAN Program for Maximum Likelihood Parameter Estimation," NASA TP-1563, Nov. 1980.

[18] Vaughan, D. R., "A Nonrecursive Algebraic Solution for the Discrete Riccati Equation," *IEEE Transactions on Automatic Control*, Vol. AC-15, Oct. 1970, pp. 597–599.

[19] Potter, J. E., "Matrix Quadratic Solutions," *SIAM Journal of Applied Mathematics*, Vol. 14, 1966, pp. 496–501.

[20] Holley, W. E., and Wei, S. Y., "Improved Method for Solving the Algebraic Riccati Equation," *Journal of Guidance and Control*, Vol. 3, No. 2, 1980, pp. 190–192.

[21] Bazaraa, M. S., and Shetty, C. M., *Nonlinear Programming: Theory and Algorithm*, Wiley, New York, 1979.

[22] Morelli, E. A., "Estimating Noise Characteristics from Flight Data Using Optimal Fourier Smoothing," *Journal of Aircraft*, Vol. 32, No. 4, 1995, pp. 689–695.

[23] Dryden, H. L., *A Review of the Statistical Theory of Turbulence*, Interscience Publishers, New York, 1961.

[24] Houbolt, J. C., "Atmospheric Turbulence," *AIAA Journal*, Vol. 11, No. 4, 1973, pp. 421–437.

[25] Iliff, K. W., "Identification and Stochastic Control of an Aircraft Flying in Turbulence," *Journal of Guidance and Control*, Vol. 1, No. 2, 1978, pp. 101–108.

[26] Sorenson, H. W., "Kalman Filtering Techniques," *Advances in Control Systems*, Vol. 3, edited by C. T. Leondes, Academic Press, New York, 1966, pp. 219–239.

[27] Jategaonkar, R. V., "A Modular and Integrated Software Tool for Parameter Estimation and Simulation of Dynamic Systems – User's Manual, Version 1.0," DLR-IB 111-2001/29, Germany, July 2001.

[28] Mackie, D. B., "A Comparison of Parameter Estimation Results from Flight Test Data Using Linear and Nonlinear Maximum Likelihood Methods," DFVLR-FB 84-06, Germany, Dec. 1983.

[29] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[30] Jategaonkar, R. V., "A Comparison of Output-Error and Filter-Error Methods from Aircraft Parameter Estimation Results," *Proceedings of the NAL-DLR Symposium on "System Identification*," DLR-Mitt. 93-14, Dec. 1993, pp. 63–87.

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CHAPTER 6

Equation-Error Methods

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6.1 INTRODUCTION

Least-squares estimation, originally stimulated by astronomical studies, is one of the oldest problems in estimation theory with numerous engineering applications, including flight vehicles. It was invented and applied during the period of 1795 to 1806 independently by A.-M. Legendre and C.-F. Gauss to describe the planetary motion [1, 2]. Historians have, however, acknowledged Gauss as the inventor, who applied his method to a meager set of three well-chosen observations to obtain a preliminary estimate and used further observations made over a period of 41 days to improve the prediction of the orbital position of the asteroid Ceres discovered by the astronomer Piazzi in January 1801 and thereby helped precisely relocate Ceres after a period of several months in December 1801 [3]. Application of his approach to several other planetary bodies located by astronomers proved the generality of the technique and marks the beginning of estimation theory.

Least-squares (LS) techniques, also called regression analysis, belong to a class of methods called the equation-error methods because they minimize a cost function defined directly in terms of input–output equation. The cost function is not based on probability theory as was the case with the maximum-likelihood method discussed in Chapter 4. The linear least-squares technique is characterized by its mathematical simplicity, in the sense that the estimates are obtained applying matrix algebra operations in a one-shot computational procedure. The regression techniques can be applied to nonlinear models as well because the basic principle is to minimize the sum of squares of the errors between the measurements and model response. As for the cost function, it is immaterial whether the errors result from a linear or nonlinear model. Only the minimization procedure is model type dependent. Nonlinear least-squares problems can be solved only iteratively.

As the least-squares method characterizes the cause–effect relation solely in terms of independent and dependent variables, its performance and applicability is strongly influenced by the data quality, assumptions made regarding the

noise, and availability of the necessary variables. The classical ordinary least-squares (OLS) method assumes the independent variables to be error and noise free and dependent variables corrupted by uniformly distributed noise. An extension called weighted least-squares (WLS) accounts for nonconstant scatter in the residuals. The major limitation of these methods is that they yield asymptotically biased and inconsistent estimates in the presence of measurement errors and noise in the independent variables [4–7]. Nevertheless, this method has found several applications to aircraft parameter estimation, providing acceptable results compared to more complex methods, mainly for two reasons. First, high-quality sensors and instrumentation systems minimize these errors. Second, prior to applying the regression method, more reliable signals can be generated through a data preprocessing step [7–13]. An extension called total least-squares (TLS) is also possible to account for noise in the independent variables.

As covered in Chapters 3–5, for several applications dealing with dynamical systems including flight vehicles, the models are postulated in state space. The least-squares method can be applied to such models provided all of the states and their time derivatives are measured. The use of measured states and state derivatives eliminates the state estimation problem, which had to be solved by integration in the case of the output-error method and by a suitable state estimator incorporating a Kalman filter or an extended Kalman filter in the case of the filter-error methods. However, the requirement that all states and state derivatives be measured directly, as well as the assumption of only the state derivatives being corrupted by noise can, in general, be restrictive in a practical application. On the other hand, because the LS method does not rely on the temporal relation defining explicit sequential ordering in time of states/variables between the data points, it facilitates analysis of large-amplitude maneuvers applying linear regression to several smaller portions through a process called "data partitioning." Furthermore, the least-squares method is directly amenable to a step-by-step procedure of selecting the best independent variables in the regression equation.

In this chapter we derive the least-squares estimator from basic principles and study its properties under theoretical assumptions and based on practical considerations. Applicability to state-space models is presented, which allows direct comparison with the approaches covered in the preceding chapters. The weighted least-squares method, which accounts for variability in the errors in the dependent variables, and the total least squares, which helps to eliminate the effects of noise in the independent variables, are presented. Nonlinear and multi-output regression approaches have been elaborated, and their similarity with the output-error method has been brought out. A related procedure of instrumental variable is illustrated briefly. Stepwise regression has been elaborated, looking pragmatically at its relevance in the presence of a priori knowledge about the model structure. A simple example of estimating aerodynamic characteristics from flight data has been presented to bring out the necessary steps to

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compute force and moment coefficients from flight measured data. Application of a one-shot procedure as well as an iterative procedure considering linear and nonlinear equations has been demonstrated using the software and flight data provided.

6.2 LEAST-SQUARES METHOD

In general, the least-squares estimation problem can be stated as follows. We have been given N discrete data samples $[y_i(1); y_i(2); \ldots, y_i(N)]$ of an ith dependent variable, and $[x(1); x(2); \ldots, x(N)]$ of independent variables, where $x = (x_1\ x_2 \ldots x_{nq})^T$ is a $n_q \times 1$ vector. We assume that the measured dependent variable $y_i(k)$ at each discrete time point k depends linearly on the independent variables:

$$y_i(k) = \theta_1 x_1(k) + \theta_2 x_2(k) + \cdots + \theta_{nq} x_{nq}(k) + \varepsilon(k); \quad k = 1, 2, \ldots, N \qquad (6.1)$$

where $\theta = (\theta_1\ \theta_2 \ldots \theta_{nq})^T$ denotes the vector of unknown parameters and ε is the equation error representing modeling discrepancies and/or noise in the dependent variable y. It is not possible to measure the parameters θ directly; hence, it is required to estimate the system parameters θ. In our specific case of flight vehicle system identification, θ s are the unknown stability and control derivatives to be estimated from the measured (flight data derived) aerodynamic force and moment coefficients denoted as y. The independent variables x comprise aircraft motion variables like angular rates, flow angles, and control surface deflections. The θ are assumed constant over all data samples N, and x is assumed to be measured exactly and error free. A block schematic of the least-squares approach is shown in Fig. 6.1. Although not included explicitly in Eq. (6.1), the procedure allows estimation of bias terms (e.g., aerodynamic zero parameters) by simply adding a vector of all ones as an independent variable.

The just-formulated problem deals with a dependent variable, which is a function of more than one independent variables, and as such represents a problem called "multiple linear regression." In our specific case of flight vehicle system identification, as a rule, this is the case. The single-dimensional case with just one independent variable is of theoretical interest only, and hence we do not elaborate on it here. Accordingly, whenever we say least-squares or regression analysis, it is automatically implied that we are talking about multiple linear regression analysis. In the literature on estimation theory, different terminologies are used to characterize the least-squares problem. The dependent variable y, as it will be called in this book, is also termed as "response variable" or "observation." The independent variables x are alternatively termed as "regressors," "explanatory variables," or "predictor variables." The deterministic part of Eq. (6.1), defining the relationship between the dependent and independent variables, is commonly called "regression equation"; sometimes it is also called "smoothing function." The

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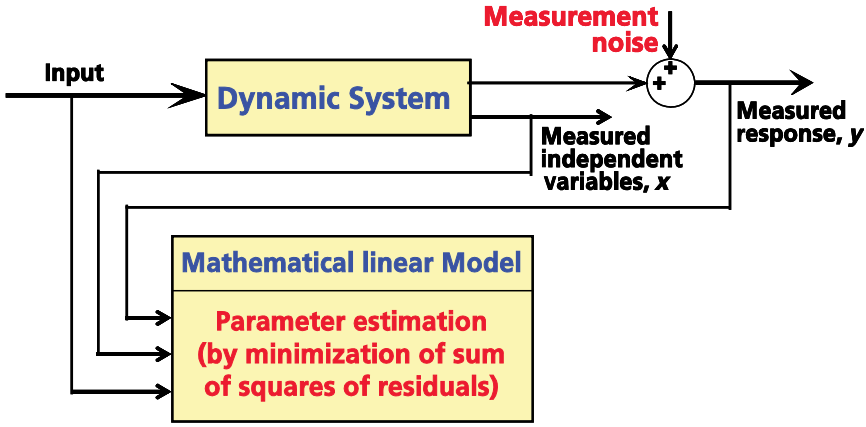

Fig. 6.1 Block schematic of least-squares method.

OLS method is also sometimes called, particularly in neural-network literature, as least-mean-squares estimator (LMS).

In Eq. (6.1), we have denoted the independent variable as y_i, that is, y with subscript i, implying that we might have more than one dependent variable. For the moment, however, to develop the estimation procedure and to study their properties, we restrict our attention to a single dependent variable because each of them is usually modeled separately. For such a one-dimensional case, we drop for the convenience of notation the subscript i from the preceding equation.

In Eq. (6.1) the y and x are provided (as measured data) and hence known; it contains n_q unknowns, namely, $(\theta_i, i = 1, n_q)$, which can be solved in an ideal case from exactly n_q pairs of dependent and independent variables, provided $\varepsilon = 0$, that is, in the absence of noise and error. This is the minimum number of data samples that is required for the determination of the unknown parameters. However, because of the presence of errors in the measurements of dependent variables, that is, $\varepsilon \neq 0$, we cannot find exact values of θ. We rather try to find the best approximations based on some suitable criterion, which requires more than n_q samples of data to estimate the parameters and leads to an overdetermined problem, loosely speaking, to average out the noise effects on the parameter estimates. Intuitively, the larger the number of such redundant data, the better will be the averaging-out process, yielding more reliable estimates.

Equation (6.1) can be rewritten in matrix notation as

$$y(k) = x^T(k)\theta + \varepsilon(k) \tag{6.2}$$

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Thus, for each of the N discrete time points at which the measurements are available, Eq. (6.2) can be written as

$$\begin{aligned} y(1) &= x^T(1)\theta + \varepsilon(1) \\ y(2) &= x^T(2)\theta + \varepsilon(2) \\ &\cdots \\ &\cdots \\ y(N) &= x^T(N)\theta + \varepsilon(N) \end{aligned} \tag{6.3}$$

Equation (6.3) can be rewritten in matrix notation as

$$Y = X\theta + \varepsilon \tag{6.4}$$

where $Y = [y(1)\ y(2)\ \ldots\ y(N)]^T$ and $\varepsilon = [\varepsilon(1)\ \varepsilon(2)\ \ldots\ \varepsilon(N)]^T$ are $N \times 1$ size vectors and X is the $(N \times n_q)$ matrix of independent variables:

$$X = \begin{bmatrix} x_1(1) & x_2(1) & \ldots & x_{nq}(1) \\ x_1(2) & x_2(2) & \ldots & x_{nq}(2) \\ & & & \\ x_1(N) & x_2(N) & \ldots & x_{nq}(N) \end{bmatrix} \tag{6.5}$$

The errors (also called residuals) ε are assumed to be uncorrelated with the independent variables. Furthermore, the scatter in the residuals is assumed to be constant across the values of independent variables, that is, the standard deviation of the random errors in y is constant. The residuals are assumed to be a white noise process with zero mean and variance σ^2, that is, $E\{\varepsilon\} = 0$ and $E\{\varepsilon\varepsilon^T\} = \sigma^2 I$, where I is the identity matrix.

From Eq. (6.2), the equation error at the discrete time point k is given by

$$\varepsilon(k) = y(k) - x^T(k)\theta \tag{6.6}$$

The errors at N discrete time points can be written from Eq. (6.6) or from Eq. (6.4) in vector form as

$$\begin{aligned} \varepsilon &= [\varepsilon(1)\ \varepsilon(2)\ \ldots\ \varepsilon(N)]^T \\ &= Y - X\theta \end{aligned} \tag{6.7}$$

The least-squares estimates of the unknown parameters θ are obtained by minimizing the sum of the squares of the residuals, or alternatively the weighted sum of the squares of the residuals about which we will talk about in the next section. In both the cases the basic principle is, however, not based on the probability theory as was the case with the maximum likelihood method discussed

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in Chapter 4. Accordingly, the cost function for the least-squares estimation is defined as

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{N} \varepsilon^2(k) = \frac{1}{2}\varepsilon^T\varepsilon$$

$$= \frac{1}{2}[Y - X\theta]^T[Y - X\theta] = \frac{1}{2}[Y^T - \theta^T X^T][Y - X\theta] \tag{6.8}$$

The gradient of the cost function with respect to the parameters is given by

$$\frac{\partial J(\theta)}{\partial \theta} = -Y^T X + \theta^T(X^T X) \tag{6.9}$$

Because the error $\varepsilon(k)$ is a linear function of the parameters θ [see Eq. (6.6)], the minimum of the cost function $J(\theta)$ defined in Eq. (6.8) is obtained by setting the gradient of $J(\theta)$ with respect to θ to zero. Accordingly, equating Eq. (6.9) to zero, taking the transpose, and denoting $\hat{\theta}$ the estimates of the true parameter values θ, we arrive at the so-called *normal equation*:

$$(X^T X)^T \hat{\theta} = X^T Y \tag{6.10}$$

Assuming $(X^T X)$ invertible, a unique solution to Eq. (6.10) is given by

$$\hat{\theta} = (X^T X)^{-1} X^T Y \tag{6.11}$$

which are the least-squares estimates of θ. In case $(X^T X)$ happens to be singular, then there will be multiple solutions. For parameter estimation purposes from specially conducted flight maneuvers, the data matrix is, in general, nonsingular.

The matrix $X^T X$ is called information matrix because it is a measure of the information content in the data being analyzed. Thus, based on the postulated model of Eq. (6.1), the estimates of the system parameters are obtained through matrix algebra operations given by Eq. (6.11) with the data matrix X of the independent variables and Y the vector of dependent variables. We do not go into the various techniques of matrix-inversion, except to say that a procedure based on Cholesky factorization or singular-value decomposition is numerically more efficient. Under the Matlab® environment, the job is even simpler because standard procedures are readily available. In particular, the use of the backslash operator is recommended, mainly because of better error detection properties.

6.2.1 PROPERTIES OF LEAST-SQUARES ESTIMATES

As in the case of the maximum likelihood method, accuracy of the least-squares estimates can be studied in terms of bias, covariance of the parameter

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estimation error, and consistency. For the assumptions made, namely, 1) residuals are uncorrelated with the independent variables and represented by a white noise process with $E\{\varepsilon\} = 0$ and $E\{\varepsilon\varepsilon^T\} = \sigma^2 I$; and 2) the independent variables are error free, the least-squares estimates are unbiased, efficient, and consistent.

The bias in the least-squares estimates is given by $E\{\hat{\theta}\}$, the expected values of the estimates $\hat{\theta}$. Substituting Eq. (6.11) for $\hat{\theta}$, we get

$$E\{\hat{\theta}\} = E\{(X^TX)^{-1}X^TY\} \tag{6.12}$$

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Now substitution of Eq. (6.4) for Y in terms of the true value θ, and the use of linearity of expectations (namely, expected value of a sum or difference is the sum or difference of the expected values) leads to

$$\begin{aligned} E\{\hat{\theta}\} &= E\{(X^TX)^{-1}X^T(X\theta + \varepsilon)\} \\ &= E\{(X^TX)^{-1}X^TX\theta\} + E\{(X^TX)^{-1}X^T\varepsilon\} \\ &= E\{\theta\} + E\{(X^TX)^{-1}X^T\varepsilon\} \end{aligned} \tag{6.13}$$

But, by definition we know that $E\{\theta\}$ is nothing but its true value θ. Because X is assumed to be noise free, the matrix of independent variables is known exactly. Equation (6.13) then simplifies to

$$E\{\hat{\theta}\} = \theta + (X^TX)^{-1}X^TE\{\varepsilon\} \tag{6.14}$$

Now, making use of the assumptions made regarding the noise statistics, we know that $E\{\varepsilon\}$ is zero. Accordingly, the last term on the right-hand side of Eq. (6.14) vanishes, yielding

$$E\{\hat{\theta}\} = \theta \tag{6.15}$$

Equation (6.15) implies that the least-squares estimation yields unbiased estimates, subject to the assumptions made.

To verify whether the least-squares procedure is an efficient (that is, minimum variance) estimator or not, we need to derive an expression for the parameter error covariance matrix P, that is, $E\{(\hat{\theta} - \theta)(\hat{\theta} - \theta)^T\}$, which can be written by expanding the terms within the brackets as

$$\begin{aligned} P = E\{(\hat{\theta} - \theta)(\hat{\theta} - \theta)^T\} &= E\{\hat{\theta}\hat{\theta}^T - \theta\hat{\theta}^T - \hat{\theta}\theta^T + \theta\theta^T\} \\ &= E\{\hat{\theta}\hat{\theta}^T\} - \theta E\{\hat{\theta}^T\} - E\{\hat{\theta}\}\theta^T + \theta\theta^T \end{aligned} \tag{6.16}$$

Now making use of the property that the least-squares estimates are unbiased, substituting $E\{\hat{\theta}\} = \theta$ from Eq. (6.15) in the preceding equation leads to

$$P = E\{\hat{\theta}\hat{\theta}^T\} - \theta\theta^T \tag{6.17}$$

Substituting the least-squares solution of Eq. (6.11) in Eq. (6.17), making use of the matrix algebra result $(AB)^T = B^T A^T$ successively, and knowing that the data matrix X is known exactly from the measurements lead to

$$\begin{aligned} P &= E\{[(X^TX)^{-1}X^TY][(X^TX)^{-1}X^TY]^T\} - \theta\theta^T \\ &= E\{(X^TX)^{-1}X^TYY^TX(X^TX)^{-1}\} - \theta\theta^T \\ &= (X^TX)^{-1}X^TE\{YY^T\}X(X^TX)^{-1} - \theta\theta^T \end{aligned} \tag{6.18}$$

As in the case of the derivation for unbiased estimates, we now substitute the regression equation (6.4) for Y, which results in

$$\begin{aligned} P &= (X^TX)^{-1}X^TE\{(X\theta+\varepsilon)(X\theta+\varepsilon)^T\}X(X^TX)^{-1} - \theta\theta^T \\ &= (X^TX)^{-1}X^TE\{X\theta\theta^TX^T + \varepsilon\theta^TX^T + X\theta\varepsilon^T + \varepsilon\varepsilon^T\}X(X^TX)^{-1} - \theta\theta^T \\ &= (X^TX)^{-1}X^T[X\theta\theta^TX^T + E\{\varepsilon\}\theta^TX^T + X\theta E\{\varepsilon^T\} + E\{\varepsilon\varepsilon^T\}]X(X^TX)^{-1} - \theta\theta^T \end{aligned} \tag{6.19}$$

Because the errors ε are assumed to have zero mean value, $E\{\varepsilon\} = 0$, and constant covariance, $E\{\varepsilon\varepsilon^T\} = \sigma^2 I$, this assumption leads to

$$\begin{aligned} P &= (X^TX)^{-1}X^T[X\theta\theta^TX^T + \sigma^2 I]X(X^TX)^{-1} - \theta\theta^T \\ &= (X^TX)^{-1}X^TX\theta\theta^TX^TX(X^TX)^{-1} + \sigma^2(X^TX)^{-1}X^TX(X^TX)^{-1} - \theta\theta^T \\ &= \theta\theta^T + \sigma^2(X^TX)^{-1} - \theta\theta^T \\ &= \sigma^2(X^TX)^{-1} \end{aligned} \tag{6.20}$$

Thus, the inverse of the information matrix is the covariance matrix, representing the expected error in the parameter estimates. Usually, σ^2 is unknown. Its estimate is given by

$$\hat{\sigma}^2 = \frac{1}{N - n_q}\sum_{k=1}^{N}[\,y(k) - X(k)\,\hat{\theta}\,]^2 \tag{6.21}$$

To investigate the asymptotic consistency of the estimates, we have to check the behavior of P for a large number of data samples. For this purpose we

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rewrite the covariance matrix given by Eq. (6.20) as

$$P = \frac{1}{N}\sigma^2\left(\frac{1}{N}X^TX\right)^{-1} \tag{6.22}$$

Now taking the limit as $N \to \infty$, it is obvious that the first term on the right-hand side, namely, $1/N$ vanishes asymptotically, yielding

$$\begin{aligned}\lim_{N\to\infty} P &= \lim_{N\to\infty}\left\{\frac{1}{N}\sigma^2\left(\frac{1}{N}X^TX\right)^{-1}\right\} \\ &= 0\end{aligned} \tag{6.23}$$

where it is assumed that limit of $(X^TX)^{-1}$ exists. Equation (6.23) implies that the parameter error covariance matrix P asymptotically converges to zero, which, in turn, implies that there is no error in the estimates $\hat{\theta}$, and hence $\hat{\theta} = \theta$, the true values. Thus, the least-squares estimator is asymptotically consistent.

6.2.2 PRACTICAL CONSIDERATIONS

The least-squares procedure developed in Sec. 6.2 and the properties derived in Sec. 6.2.1 are based on the assumption that the measurements of the independent variables x are exact and noise free, and accordingly the variables x and matrix X hitherto represent the true values. In reality, this is never the case; every recorded variable contains unavoidable measurement noise as well as systematic errors like bias or scale factor; recall our discussion on systematic errors in Chapter 3. Let us denote the measured independent variables as x_m, the matrix of x_m as X_m (note that the subscript m is used in this section to denote explicitly the measurements and not the mth variable), the systematic errors as Δx, and the noise in the independent variable as μ, which we also assume to be additive and independent of ε. The true value of the independent variables is then given by

$$X = X_m - (\Delta X + \mu) \tag{6.24}$$

where ΔX is the matrix of Δx, which are assumed constant over the N data samples. Substituting Eq. (6.24) into Eq. (6.4) leads to

$$Y = (X_m - \Delta X - \mu)\theta + \varepsilon \tag{6.25}$$

Premultiplying Eq. (6.25) with $(X_m^TX_m)^{-1}X_m^T$ yields

$$(X_m^TX_m)^{-1}X_m^TY = (X_m^TX_m)^{-1}X_m^T[(X_m - \Delta X - \mu)\theta + \varepsilon] \tag{6.26}$$

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But for the change of notation, the left-hand side of Eq. (6.26) is nothing but $\hat{\theta}$, the estimate of unknown parameters. Expanding the terms on the right-hand side of Eq. (6.26) gives

$$\begin{aligned}\hat{\theta} &= (X_m^T X_m)^{-1} X_m^T X_m \theta - (X_m^T X_m)^{-1} X_m^T (\Delta X + \mu)\theta + (X_m^T X_m)^{-1} X_m^T \varepsilon \\ &= \theta + (X_m^T X_m)^{-1} X_m^T (\varepsilon - \Delta X \theta - \mu\theta)\end{aligned} \tag{6.27}$$

Based on Eq. (6.27), we can now check the properties of the least-squares estimator in the presence of systematic errors and noise in the independent variables. From Eq. (6.27), the expected value of the estimates is obtained as

$$E\{\hat{\theta}\} = \theta + E\{(X_m^T X_m)^{-1} X_m^T (\varepsilon - \Delta X \theta - \mu\theta)\} \tag{6.28}$$

Now applying the linearity of expectation and the multiplicative rule to the expectation of a product (namely, expected value of a product of two independent variables is the product of the expected values), based on the assumptions of noise ε in the dependent variables being zero mean and μ independent of ε and θ, Eq. (6.28) simplifies to

$$E\{\hat{\theta}\} = \theta - E\{(X_m^T X_m)^{-1} X_m^T (\Delta X + \mu)\}\theta \tag{6.29}$$

Following a procedure similar to that used in Eqs. (6.18–6.20), in the present case, it can be shown that the covariance of the parameter estimation error P is given by

$$\begin{aligned}P = {} & E\{(X^T X)^{-1} X^T \varepsilon \varepsilon^T X (X^T X)^{-1}\} \\ & + E\{(X^T X)^{-1} X^T (\Delta X + \mu)\theta\theta^T (\Delta X + \mu)^T X (X^T X)^{-1}\}\end{aligned} \tag{6.30}$$

Comparing Eq. (6.29) with Eq. (6.15), it becomes clear that the estimates are now affected by the combined errors $(\Delta x + \mu)$ and in general yield biased estimates. The bias depends only on the errors in the measurement of the independent variables and not on the noise in the dependent variable. The variance is affected by the noise levels of dependent as well as independent variables.

Let us further analyze Eq. (6.29) in terms of the two components of possible corruptions, namely, μ (noise) and Δx (systematic errors/bias or scale factor). For argument's sake, if we assume that there are no systematic errors, that is, $\Delta x = 0$, then the estimates will be unbiased, provided the noise μ in the independent variables is zero mean; such a case is dealt with in Sec. 6.5. In the other case, that is, in the presence of systematic errors Δx, the estimates will always be biased. It is this undesirable property that needs consideration while applying the least-squares method to flight data and, in turn, necessitates critical checking of recorded

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data to eliminate such errors before the data could be used for regression purposes. Thus, in practice, application of the least-squares procedure calls for a data preprocessing step to eliminate the errors and thereby leads to a two-step procedure for aerodynamic parameter estimation. As already mentioned in Chapter 2 and will be pursued in Chapter 10, it becomes possible at some additional computational costs to check for the kinematic consistency of data applying the flight-path reconstruction technique based either on filtering techniques or the maximum likelihood method and to eliminate such errors from the measured data.

In an analogous manner, the presence of atmospheric turbulence (process noise) can be investigated. In such cases it turns out that the covariance of the parameter estimation error will be affected by process noise as well. As in the case of measurement noise μ in the independent variables, if we assume that the process noise is a white noise process having zero mean, then the estimates will be unbiased, provided systematic errors are eliminated. It is this property that implies that the least-squares estimator accounts for the process noise because the measured variables contain responses due to deterministic inputs as well as those due to nonmeasurable stochastic inputs.

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6.2.3 APPLICABILITY TO STATE-SPACE MODELS

To bring out the applicability of the least-squares principle to models represented in state space, consider the following linear system representation:

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{6.31}$$

Now assuming that the measurements of all of the states x and of their time derivatives $\dot{x}$ are available and denoted as x_m and $\dot{x}_m$ respectively, the equation error can be expressed as

$$\varepsilon_{\mathrm{EE}} = \dot{x}_m - Ax_m(t) - Bu(t) \tag{6.32}$$

where the subscript m is used to denote measured variables; note that we have not used subscript m for the control inputs because the control inputs have to be measured in any case, whether we apply the equation-error method or any other and as such the subscript m is implicit for the u. Furthermore, note that $\dot{x}_m$ is not a differential operation of the state variables, but the measurements. This implies that the least-squares criterion does not involve solution of the system differential equations, that is, there is no state estimation required. Instead, the system equations are used with the measured data.

For N discrete measurements, the cost function to be minimized can be written as

$$J_{\mathrm{LS}}(\theta) = \sum_{k=1}^{N} [\dot{x}_m(k) - Ax_m(k) - Bu(k)]^T [\dot{x}_m(k) - Ax_m(k) - Bu(k)] \tag{6.33}$$

Combining the control vector u with the measured state vector x_m and also the matrices B and A into a single partitioned matrix as

$$\theta = \{B : A\} \tag{6.34}$$

Equation (6.33) can be rewritten as

$$J_{\mathrm{LS}} = \sum_{k=1}^{N} \left[\dot{x}_m(k) - \theta \begin{pmatrix} u \\ x_m \end{pmatrix}\right]^T \left[\dot{x}_m(k) - \theta \begin{pmatrix} u \\ x_m \end{pmatrix}\right] \tag{6.35}$$

Setting the gradient of Eq. (6.35) with respect to θ to zero yields

$$\theta^T = \left\{\sum_{k=1}^{N} \begin{pmatrix} u \\ x_m \end{pmatrix} \begin{pmatrix} u \\ x_m \end{pmatrix}^T\right\}^{-1} \left\{\sum_{k=1}^{N} \begin{pmatrix} u \\ x_m \end{pmatrix} \dot{x}_m^T\right\} \tag{6.36}$$

Equation (6.36) is an alternate form of the least-squares solution of Eq. (6.11). It is now apparent that the least-squares procedure can be applied to the state-space model provided the measurements of all of the states and the derivatives of all of the states with respect to time are available.

6.3 WEIGHTED LEAST-SQUARES METHOD

The least-squares method dealt hitherto yields unbiased estimates when, as pointed out in Sec. 6.2, the standard deviations of the random errors ε in the dependent variables are constant over all of the values of the independent variables; see constant scatter band on the left-hand side of Fig. 6.2 for a single-dimensional case. In some cases the least-squares fit would yield residuals with nonconstant variability, which can be, as shown on the right-hand side of Fig. 6.2, increasing with the amplitude (level) of the independent variable or may not be randomly distributed. In such cases the weighted least-squares method, which is an extension of the ordinary least-squares method, is preferable. In this extension, weights are associated to each data point, reflecting the degree of confidence in (or relative precision of) the individual measurements. Instead of minimizing

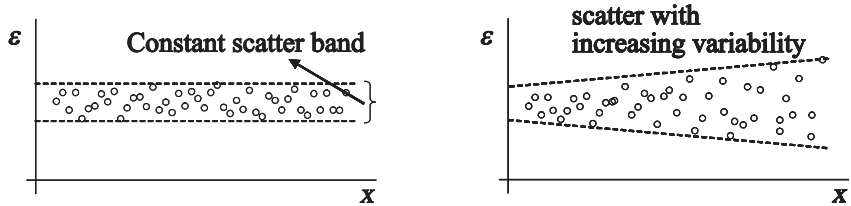

Fig. 6.2 Residuals from simple linear regression.

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the error $\varepsilon^T\varepsilon$ in Eq. (6.8), we now minimize $\varepsilon^T W\varepsilon$. But for this change leading to the weighted sum of squares, we have the same set up as in the preceding case.

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{N} w_k\varepsilon^2(k) = \frac{1}{2}\varepsilon^T W\varepsilon$$
$$= \frac{1}{2}[Y - X\theta]^T W[Y - X\theta] \tag{6.37}$$

where W is the given weighting matrix, assumed symmetric positive definite. The gradient of the cost function with respect to the parameters is given by

$$\frac{\partial J(\theta)}{\partial\theta} = -Y^T WX + \theta^T X^T WX \tag{6.38}$$

Setting the gradient of $J(\theta)$ with respect to θ to zero and taking the transpose, we arrive at the modified *normal equation*:

$$(X^T WX)\hat{\theta} = X^T WY \tag{6.39}$$

where $\hat{\theta}$ denotes the estimates of the true parameters θ. Assuming $(X^T W^T X)$ invertible and after simple rearrangement, the estimates are obtained as

$$\hat{\theta} = (X^T WX)^{-1}X^T WY \tag{6.40}$$

which can be solved in one shot through simple matrix operations.

Another way to arrive at the WLS solution is through the transformation $\widetilde{Y} = C^T Y$, where C is the Cholesky square root, that is, the lower triangular matrix, such that $W = CC^T$. Now, substituting Eq. (6.4) for Y in $\widetilde{Y} = C^T Y$, we obtain [5]

$$\widetilde{Y} = C^T X\theta + C^T\varepsilon \tag{6.41}$$

which can be rewritten in terms of the transformed variable $\widetilde{X} = C^T X$ and transformed error $\tilde{\varepsilon} = C^T\varepsilon$ as

$$\widetilde{Y} = \widetilde{X}\theta + \tilde{\varepsilon} \tag{6.42}$$

Equation (6.42) pertains to the transformed variables, but otherwise is exactly in the same form as of Eq. (6.4). Applying the ordinary least-squares method, that is, Eq. (6.11), studied in Sec. 6.2, the solution to Eq. (6.42) can be readily written down as

$$\hat{\theta} = (\widetilde{X}^T\widetilde{X})^{-1}\widetilde{X}^T\widetilde{Y} \tag{6.43}$$

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Now resubstitution of $\widetilde{X} = C^T X$ and $\widetilde{Y} = C^T Y$ in Eq. (6.43) and further minor simplification leads to the solution

$$\hat{\theta} = (X^T W X)^{-1} X^T W Y \tag{6.44}$$

which is the same as Eq. (6.40).

There is a minor issue that needs to be considered while applying the weighted least-squares technique because such an estimator is derived based on the basic assumption that the weights are known exactly. Based on the practical considerations presented in Sec. 6.2.2, we are aware that such an assumption represents an ideal case, and, in reality, it is hard to meet. Therefore, the practical approach is to estimate the weights applying some suitable procedure, hoping that the performance of the estimator is not significantly affected. The influence will be reduced when such weights are estimated from a large number of data samples. A common procedure is to choose the weights inversely proportional to the variance at each level of the independent variables.

The weighted least-squares method is often called generalized least-squares (GLS) method. We prefer the explicit terminology of WLS because the other term GLS has also been used in other connotations such as total least squares, which we will address in Sec. 6.5.

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6.4 NONLINEAR AND MULTI-OUTPUT REGRESSION

In the regression equation that we had so far considered, the dependent variable y was linearly related to the independent variables x. Linear models are usually valid over a smaller range of excursions around an operating point. Models valid over a wider range might be necessary in some applications and might involve a nonlinear response relationship. There are two approaches to handle such cases depending upon the type of nonlinearity in the postulated model. They are as follows:

1. Model being nonlinear in the independent variables, but linear in parameters; for example, $y = \theta_1 x_1 + \theta_2 x_2^2 + \cdots$
2. Model being nonlinear in the parameter itself; for example, represented as $y = f(x, \theta)$, where f is a general nonlinear function.

In the first case, the problem can be transformed into a linear least-squares problem by precomputing the nonlinear terms and treating them as a pseudo independent variable. This is possible because the independent variables are available. To this problem in terms of new independent variables, the model then being linear in them, we can apply the classical least-squares or weighted-least-squares methods we have already covered in Secs. 6.2 and 6.3, respectively. This approach is found to work fairly well.

The second case of $y = f(x, \theta)$ is more complex and leads to a nonlinear regression problem. Although we have indicated a possibility to handle the first case through classical linear least-squares approach, in its basic form it also really falls under the category of nonlinear regression problems. In such cases, making the same assumptions regarding the noise ε, the following cost function is minimized:

$$J_{\mathrm{LS}} = \sum_{k=1}^{N} [y(k) - f(x(k), \theta)]^2 \tag{6.45}$$

Minimization of the cost function in Eq. (6.45) can be performed only iteratively. Thus, instead of the one-shot solution of Eq. (6.11), we have to apply now some iterative optimization procedure. Before going into the details of such an algorithm, we discuss yet another case, namely, applying the equation-error method to several dependent variables, which can also be more efficiently solved through an iterative optimization algorithm.

We recall from Eq. (6.1) that the dependent variable y is actually the ith variable, but we had dropped the subscript i because each dependent variable was then treated separately. In case we have more than one dependent variable, then for each of them it is necessary to form the vector Y and matrix X and to solve Eq. (6.11) each time. In our specific case of flight vehicle system identification, we would, in general, wish to model the three force and three moment coefficients, which are the dependent variables. They are functions of the independent variables x comprising the aircraft motion variables like angular rates and flow angles as well as control surface deflections.

Because the aerodynamic force and moment coefficients depend on some variables from a set of independent variables, the six equations can be written in matrix form as

$$y(k) = \Theta x(k) + e(k) \quad k = 1, 2, \ldots, N \tag{6.46}$$

where y is now a $n_y \times 1$ vector (e.g., $n_y = 6$ for aerodynamic modeling) and Θ contains the unknown parameters $[\theta_{11} \ldots \theta_{1nq}; \ldots; \theta_{ny1}, \ldots, \theta_{nynq}]^T$. The problem formulated in Eq. (6.46) is called the multivariate multiple regression (MMR), which is typical of engineering applications. For the linear MMR case of Eq. (6.46), the matrix Y of the dependent variables is of the size $N \times n_y$. Assuming the same set of independent variables for each of the dependent variables, the matrix of the independent variables X becomes of the size $N \times n_q$. The $n_q \times n_y$ dimensional matrix of unknown parameters Θ can be estimated applying the one-shot procedure of Eq. (6.11). Although this is a viable procedure, it is seldom practiced because treating the same set of independent variables for all of the y s introduces cross-coupling derivatives Their estimation might not be reliable if adequate information is not available in X and Y. Furthermore, some of the other coefficients might not be physically realistic. The least-squares estimates are commonly obtained by treating each of them separately because

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the one-shot procedure is simple enough and can be easily repeated by appropriately choosing different independent variables for each dependent variable. To such a case and also to the nonlinear MMR, the iterative procedure can be effectively applied as discussed next.

In the case of nonlinear MMR, that is, when the model is nonlinear either in the parameters Θ or in the independent variables x, Eq. (6.46) can be rewritten as

$$y(k) = f(x(k), \Theta) + e(k) \quad k = 1, 2, \ldots, N \tag{6.47}$$

Equation (6.47) is exactly in the form of an observation equation of the state-space model postulated in Chapter 4, Eqs. (4.11–4.12). As such, we can apply the same iterative algorithm that was developed there for the output-error method [14]. The iterative procedure described in Sec. 4.5.2 for the output-error method caters to estimation of the covariance matrix of the residuals, which is used in the parameter-update step as a weighting matrix. Thus, application of the same algorithm to Eq. (6.47) consequently yields weighted least-squares estimates. In other words, as already pointed out in Sec. 6.2.3, algorithmically the only difference between the output-error and weighted least-squares estimation is the way we obtain the information about the states (independent variables); in one case it is by incorporating a suitable state estimator and in the other case by using the measured states directly. The block schematic of the weighted least-squares procedure is not explicitly shown because it is very similar to that of the output-error method (Fig. 4.1) with the exception of using the measured states.

6.5 TOTAL LEAST SQUARES

It is pointed out in Sec. 6.2.2 that the ordinary least-squares technique yields biased estimates in the presence of systematic errors and noise in the independent variables. Although there is no general procedure within the framework of regression techniques to account for systematic errors, the total least-squares method allows accounting for the noise in the independent variables. It is also termed as orthogonal distance regression (ODR) or error-in-variable (EIV) modeling. The differences between OLS and TLS are brought out best by considering the dependent variable as a function of a single independent variable. The OLS amounts to minimizing the difference along the y axis (ordinate) between the model (regression equation) and the measured values of the dependent variables, that is, δy_{OLS} in Fig. 6.3. On the other hand, while considering errors in both, that is, in the dependent variables (along y axis) and the independent variables (along x axis), a clear assignment between them becomes difficult, and hence minimization of the orthogonal distance, which contains errors in both x and y, is more appropriate; see Fig. 6.3.

To account for noise in X, we rewrite the regression equation (6.4) as

$$Y = (X - \mu)\theta + \varepsilon \tag{6.48}$$

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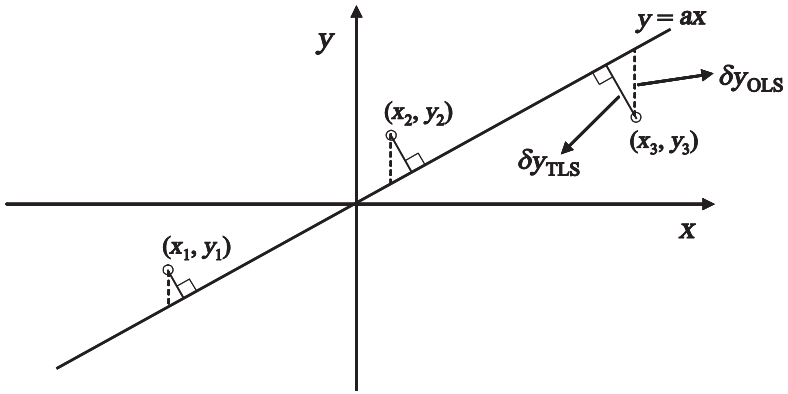

Fig. 6.3 Schematic of OLS and TLS principle.

By multiplying out the terms on the right-hand side, by bringing all of them to one side, and by regrouping, Eq. (6.48) can be rewritten as

$$[[X \quad Y] \quad -[\mu \quad \varepsilon]]\begin{bmatrix}\theta\\-1\end{bmatrix} = [\widetilde{X} \quad -\tilde{\Delta}]\begin{bmatrix}\theta\\-1\end{bmatrix} = 0 \tag{6.49}$$

where $\widetilde{X} = [X \; Y]$ is the compounded data matrix of size $(N \times n_q + 1)$ formed by adjoining the column vector Y to the columns of X on the right, and $\tilde{\Delta} = [\mu \; \varepsilon]$ is the compounded noise vector. In the literature on TLS, the compounded matrix has been denoted as $[X|Y]$; we denote the same as $[X \; Y]$, which is more compatible with the computational implementation. The TLS solution to θ is obtained by solving Eq. (6.49).

Without going into detailed development, we just take a look at the computational steps of obtaining TLS estimates, for which a simple linear solution as in Eq. (6.11) is not possible. The solution proposed by Golub and van Loan based on the singular value decomposition is the most commonly applied approach to solve the TLS problem [15]. Considerable advancements have been carried out by van Huffel and Vanderwalle [16] and others. The compounded matrix $[X \; Y]$ can be written as

$$[X \; Y] = U\Sigma V^T \tag{6.50}$$

where $\Sigma = (diag(\sigma_1, \sigma_2, \ldots, \sigma_{nq+1}))$ is the $(n_q + 1 \times n_q + 1)$ diagonal matrix of singular values such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{nq+1}$; and U and V are the left and right singular matrices. Note that $[X \; Y]$ is of size $(N \times n_q + 1)$, but it suffices to compute only the first $n_q + 1$ columns of U corresponding to the first $n_q + 1$ nonzero singular values.

It can be shown that the smallest of the singular values $(\sigma_1, \sigma_2, \ldots, \sigma_{nq+1})$, which happens to be the $n_q + 1$ th value (that is, the last in the sorted list),

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corresponds to the minimum. The total least-squares solution is then obtained from the column corresponding to the smallest singular value, that is, the last column of V and is given by

$$\theta_{\text{TLS}} = -\upsilon/\lambda \tag{6.51}$$

where λ is the last element and υ is the vector of the first n_q elements of the last column of V. Equation (6.51) corresponds alternatively to [16]

$$\theta_{\text{TLS}} = (X^T X - \sigma_{n+1}^2 I)^{-1} X^T Y \tag{6.52}$$

where σ_{nq+1} is the smallest singular value. Comparing Eq. (6.52) with Eq. (6.11), it becomes evident that the total least-squares method incorporates a correction term to account for the noise in the independent variables. It has already been pointed out in Sec. 6.2.2 that the TLS estimates are unbiased, provided the noise μ is zero mean.

Although a little more complex than the OLS solution, the classical TLS problem, as presented in Eqs. (6.49–6.52), can be solved analytically. Both the OLS and TLS assume $N > n_q$, that is, an overdetermined set of linear equations. TLS caters to noise in the data matrix X as well as in the observation vector Y, minimizing the sum of squares of residuals on all of the (independent and dependent) variables in the equations and not just on the dependent variable.

The analytical solution of TLS can be extended to MMR having n_y observation (dependent) variables; see Sec. 6.4. In such a case, Y will be a matrix of size $N \times n_y$ instead of a column vector, and the compounded matrix is $(N \times n_q + n_y)$ sized instead of $(N \times n_q + 1)$ sized for the case with a single dependent variable. It is assumed that the same set of independent variables X is used. The singular value decomposition of the compounded matrix $[X\ Y]$ is then given by

$$[X \quad Y] = U \Sigma V^T = [U_S \quad U_N] \begin{bmatrix} \Sigma_S & 0 \\ 0 & \Sigma_N \end{bmatrix} \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}^T \tag{6.53}$$

where the matrices are partitioned into what is termed as system subspace (subscript S) and a null subspace (subscript N). As in the preceding case, it suffices to compute only the first $n_q + n_y$ columns of U corresponding to the first $n_q + n_y$ nonzero singular values. It turns out that the TLS solution is given by [15, 16]

$$\theta_{\text{TLS}} = -V_{12} V_{22}^{-1} \quad \text{if } n_q \geq n_y; \quad \text{else} \quad \theta_{\text{TLS}} = (V_{11}^T)^{-1} V_{21} \tag{6.54}$$

For the commonly addressed problem with $n_q \geq n_y$, the partitioned matrices V_{12} and V_{22} required in the solution are of the sizes $(n_q \times n_y)$ and $(n_y \times n_y)$ respectively, corresponding to the singular values in the null subspace. Following the Matlab® notation, they are given by $V_{12} = V(1\!:\!nq,\ nq+1\!:\!nq+ny)$ and $V_{22} = V(nq+1\!:\!nq+ny,\ nq+1\!:\!nq+ny)$. It is obvious that Eq. (6.51) is a special case of Eq. (6.54) for $n_y = 1$.

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TLS is not to be confused with the WLS that was discussed in Sec. 6.3, which assumes error-free independent variables and minimizes the distance along the y coordinate as the weighted sum of squares of the errors to account for variations in the reliability of the dependent variables. Several extensions of TLS are possible, for example, weighted TLS to include a weighting matrix based on the square root of the covariance matrix [17, 18] and generalized TLS; but we do not go into the details of such techniques, which can be found in [16]. The TLS problem can also be solved using iterative techniques [18, 19].

Applications of the TLS to flight vehicle system identification have been very few [17, 20]. Nevertheless, it offers an option to obtain better least-squares estimates compared to the ordinary least-squares procedure in the presence of noise. Because the procedure does not cater to systematic errors in the variables, it once again requires rigorous data preprocessing to eliminate such errors. In this sense, it has the same practical limitations as OLS. On the other hand, the methods discussed in Chapters 4 and 5 are more general and allow simultaneous estimation of system parameters and systematic errors as well as noise statistics.

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6.6 INSTRUMENTAL-VARIABLE METHOD

The ordinary least-squares method developed in Sec. 6.2 assumes the equation errors (residuals) to be a white noise process, that is, random and uncorrelated. If the residuals are correlated, the least-squares procedure yields biased estimates. In such cases, an instrumental-variable (IV) method is applied to account for correlated noise. In this method the so-called "instrument variables" are introduced to cancel out the effect of correlated noise. Figure 6.4 shows a schematic of this method. It is required to choose signals $z = (z_1, \ldots, z_{nq})^T$ such that $z(k)$ are strongly correlated with the independent variables $x(k)$ but are uncorrelated with an equation error $\varepsilon(k)$, that is,

$$E\{z(k)x^T(k)\} \text{ is a positive definite matrix; } E\{z(k)\varepsilon(k)\} = 0 \tag{6.55}$$

Now, multiplying Eq. (6.6) by $z(k)$ and summation and averaging over N data samples yields

$$\frac{1}{N}\sum_{k=1}^{N} z(k)\varepsilon(k) = \frac{1}{N}\sum_{k=1}^{N} z(k)y(k) - \frac{1}{N}\sum_{k=1}^{N} z(k)x^T(k)\theta \tag{6.56}$$

From the assumption of $z(k)$ being uncorrelated with $\varepsilon(k)$, for large N we know

$$\frac{1}{N}\sum_{k=1}^{N} z(k)\varepsilon(k) = 0 \tag{6.57}$$

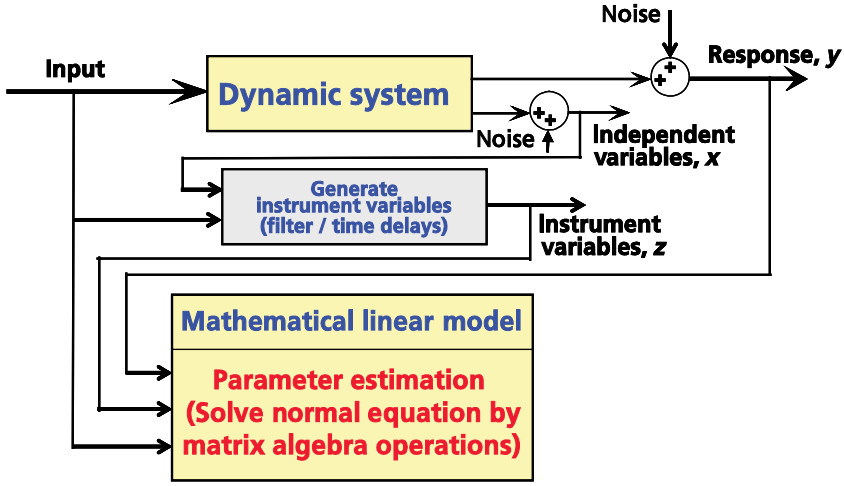

Fig. 6.4 Block schematic of instrumental-variable method.

which is the left-hand side of Eq. (6.56). Combining Eqs. (6.56) and (6.57), that is, equating the right-hand side of Eq. (6.56) to zero, and solving for θ yield the estimates

$$\hat{\theta} = \left[\frac{1}{N}\sum_{k=1}^{N} z(k)x^T(k)\right]^{-1}\left[\frac{1}{N}\sum_{k=1}^{N} z(k)y(k)\right] \tag{6.58}$$

which can be conveniently written in an equivalent matrix form as

$$\hat{\theta} = (Z^T X)^{-1} Z^T Y \tag{6.59}$$

where X and Y are as defined as before by Eqs. (6.4) and (6.5) and the matrix of instrument variables Z:

$$Z = \begin{bmatrix} z_1(1) & z_2(1) & \dots & z_{nq}(1) \\ z_1(2) & z_2(2) & \dots & z_{nq}(2) \\ & & & \\ z_1(N) & z_2(N) & \dots & z_{nq}(N) \end{bmatrix} \tag{6.60}$$

Equation (6.59) for estimates applying the instrumental-variable method is similar in form to that of the least-squares solution of Eq. (6.11). The estimates from Eq. (6.59) can be obtained once again through simple matrix algebra

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operations. It can be shown that these estimates are asymptotically unbiased [21], but we do not go into these details. The major difficulty associated with the instrumental-variable method is, however, in finding an appropriate set of such variables z. Any choice that satisfies the conditions specified in Eq. (6.55) will yield unbiased estimates. Most of the procedures [21, 22] to generate z are based on some form of filter incorporating either a priori values of the unknown parameters θ or estimates obtained through a least-squares procedure. Another possibility is to use time-delayed values of the independent variables as instrument variables. Although the method yields unbiased estimates, it has not found widespread use in flight vehicle system identification, mainly because of the difficulty just discussed.

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6.7 DATA PARTITIONING

Modeling of aerodynamic characteristics over a large range of values of the influencing independent variable, for example, angle of attack, might be of interest in specific cases, for example, fighter aircraft with high angle-of-attack capabilities. Estimation of aerodynamic parameters of linear models and from small-amplitude maneuvers at such flight conditions applying the classical output-error method might require a large number of flight maneuvers at several trim conditions. Moreover, it might not always be possible to trim the aircraft at some of these extreme flight conditions. On the other hand, estimation of aerodynamic characteristics from a single large-amplitude maneuver, such as sweeping the angle-of attack, will require complex nonlinear model postulates, whose form might not be known or anticipated. The estimation can pose several difficulties due to an inadequate model structure. It is also likely that at high angles of attack, stochastic vortex shedding at the aircraft nose might be dominant, necessitating an estimation method capable of accounting for process noise thereby further aggravating the difficulties. In these cases the least-squares technique combined with "data partitioning," which will be discussed in the following, provides a simpler alternative approach.

The linear regression analysis, being based on linear models, might not be directly applicable to large-amplitude maneuvers. In such cases, the data partitioning is adopted to divide a maneuver that covers a large range of some variables into several portions, each of which spans a smaller range of that variable [8, 23, 24]. To each segment the regression analysis can be applied to estimate the unknown derivatives. A schematic explanation of data partitioning is shown in Fig. 6.5. Typically, the angle of attack is chosen as the variable for data partitioning to select those data points from the complete maneuver for which the measured angle of attack is in the specified range, that is, $\alpha_{\min} \leq \alpha \leq \alpha_{\max}$.

For the case depicted in Fig. 6.5, the range from 50 to 55 deg of angle of attack is chosen to demonstrate the procedure. The upper part shows the complete flight maneuver covering the angle of attack from 20 deg to

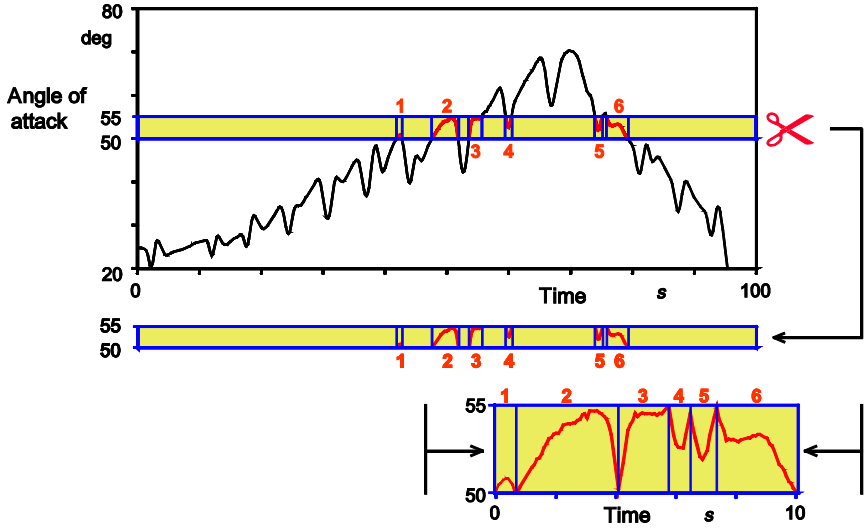

Fig. 6.5 Schematic of data partitioning.

roughly 65 deg. It is observed that there are six portions, marked 1 through 6, that span the chosen range of 50 to 55 deg. These six data segments are cut from the complete maneuver (see middle part of the figure) and concatenated for evaluation purposes as shown in the bottom part of Fig. 6.5. Principally, it would be possible to apply the output-error or filter-error method to these six time slices because these methods provide a multirun evaluation capability. The disadvantage is, however, that the initial conditions will have to be estimated for each time segment. The short record lengths and large number of parameters pose convergence problems and mostly result in correlated estimates. On the other hand, least-squares techniques can be readily applied to such partitioned data sets, estimating the aerodynamic parameters directly without the need of estimating the initial conditions. This is particularly useful when different model structures and different data partitioning will have to be tried out. The procedure usually gives improved parameter estimates and reduced correlations.

Although the procedure of data partitioning has been illustrated based on a single parameter, in the present case the angle of attack, it is possible to partition the data considering multiple parameters simultaneously. Angle of attack and angle of sideslip is one such combination that might be useful in estimating linear models successively to arrive at a model valid for large ranges of these variables from large-amplitude maneuvers.

Basically, data partitioning is a preprocessing step followed by parameter estimation applying the conventional linear regression method. This approach is

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feasible because the regression analysis does not rely on the temporal relation between the data points. The temporal relation defines the explicit sequential ordering in time of states/variables, defining the current states as some function of the previous states. It is fundamental to the state-space model representation, in which the states evolve over time, but not for the least-squares estimation. Processing of N data points in any arbitrary order of sequence will lead to the same least-squares estimates.

6.8 MODEL STRUCTURE DETERMINATION

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Model structure determination is one of the widely addressed topics in the abundant literature on the regression analysis. There are several dedicated publications addressing just this aspect. What we cover in this section is by no means exhaustive. Rather, we take a cursory glance at the general philosophy and methodology as well as at some of the evaluation criteria and various issues involved. Finally, we will take a pragmatic look at this approach as applicable to our interest of flight vehicle system identification, particularly in light of the fact that considerable a priori knowledge about the aerodynamic model structure is available in many of the cases.

The two typical problems encountered in the application of the hitherto discussed regression approach are the following:

1. Some of the independent variables x might be slightly correlated with the dependent variable y.
2. Some of the independent variables might be redundant because they may be correlated with other independent variables (collinearity).

Such independent variables need to be detected to avoid their inclusion in the regression model because the resulting inaccuracies due to their inclusion outweigh the increase in the model fidelity [25]. Collinearity is very often the cause for poor regression results because while considering two variables, it shows strong influence for each when considered separately (as can be indicated by a large t-ratio, which is simply the ratio of estimated coefficient to its standard deviation), but a small t-ratio when both variables are included implying large uncertainty in the estimates. Such estimates might show poor predictive capability when applied to other data sets. Hence, some procedure is necessary to overcome these practical difficulties, in order to include the most significant variables, so that the overall model has as few variables as possible yet has adequate ability to effectively predict the dependent variable y.

Three basic approaches to model structure determination are as follows:

1. Forward selection
2. Backward elimination
3. Stepwise regression

In the forward selection procedure, starting from the simplest of the models, containing just the bias (aerodynamic zero term), we add to the regression equation first an independent variable having highest correlation with the dependent variables. Subsequently, at each step contribution, each of the remaining variables is evaluated and added to the model one at a time provided the corresponding t-ratio is greater than the specified threshold value. When none of the remaining variables have t-ratio greater than a specified threshold, the process is terminated. Other statistical test criteria, for example, partial F-statistics, which we will cover later in this section, have also been used instead of t-ratio to determine the significance of the parameter being entered. The forward selection is a unidirectional procedure, based purely on evaluating the t-ratio. Once a particular variable has been selected, the choice is ultimate in the sense that the selection procedure does not evaluate the degradation that can result due to entering other variables subsequently.

The backward elimination, as the name implies, is the reverse process of forward selection. Here, we start with an oversized model, including all of the potential variables in the model. At each step, the effect of dropping one variable is evaluated. If the corresponding t-ratio is smaller than the specified threshold and is the smallest of the t-ratios of all variables still in the model, the corresponding variable is then removed. If none of these two conditions are satisfied, the model retains the particular variable.

Stepwise regression combines the forward selection and backward elimination. It consists of starting from a simplified model with just one or two primary independent variables, then introducing other independent variables sequentially one at a time, and evaluating the significance of the new addition by suitable statistical criteria. At each step, just like in the backward elimination procedure, all of the parameters included there are reassessed. It could happen that, due to correlation among the variables, some of the variables introduced earlier in the model development process drop out later. Thus, the stepwise regression procedure enables determination of an appropriate ("best") subset of independent variables from the potential variables that have to be included in the regression equation (6.1) to adequately model the observations. In other words, it helps to arrive at the parsimonious representation. In general, the stepwise regression can lead to a subset of variables that is different from the one obtained by forward or backward elimination procedure. Because stepwise regression is more commonly used for aerodynamic model structure determination, we study this procedure in more detail in the next subsection.

In the case of multivariate multiple regression, that is, when y is a $n_y \times 1$ vector, there are two possible approaches, namely, 1) we treat the n_y dependent variables one at a time and perform the model selection procedure n_y times separately, and 2) perform the multivariate model selection procedure to arrive at a common set of predictors giving the best combination over all of the n_y dependent variables. In the case of aerodynamic modeling, we know from the physics of the

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phenomenon that, in general, we come across the problem of the first type because the six aerodynamic coefficients are functions of different sets of independent variables. As such, the simpler approach leading to n_y different subsets of predictors is more appropriate to our specific case.

6.8.1 STEPWISE REGRESSION

The stepwise regression procedure of variable selection can be summarized as follows:

1. To start, define a set of potential independent variables, compute the correlations between each of the independent variables x_i and the output y, and choose the variable with the highest correlation as the first entry.
2. Add another variable from the remaining set with highest partial correlation coefficient.
3. Check partial F values of all up to then included variables; eliminate those variables having F values below the prespecified threshold.
4. Jump back to step 2 to repeat the procedure until none of the remaining variables lead to model improvement.

Mathematically, the preceding four steps lead to the following computational procedure:

Step 1: Compute the correlation coefficients between y (dependent variable) and each independent variable x_i from the set of n_p potential variables:

$$\rho_i = r_{yx_i} = \frac{\sum_{k=1}^{N} (y(k) - \overline{y})(x_i(k) - \overline{x}_i)}{\sqrt{\sum_{k=1}^{N} (y(k) - \overline{y})^2 \sum_{k=1}^{N} (x_i(k) - \overline{x}_i)^2}} \tag{6.61}$$

where the subscript i corresponds to the ith independent variable, and $\overline{y} = 1/N[\sum_{k=1}^{N} y(k)]$ and $\overline{x}_i = 1/N[\sum_{k=1}^{N} x_i(k)]$ denote the respective mean values.

Check whether the correlation coefficients ρ_i are significant or not (that is, greater than a preselected threshold). If not, none of the potential variables can enter the model, yielding the mean as the best fit. Otherwise, select the independent variable, say, jth variable, with the highest correlation coefficient as the first entry and postulate the probable regression model as

$$y = \theta_0 + \theta_j x_j + \varepsilon \tag{6.62}$$

and proceed with the model of Eq. (6.62)

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Step 2: Compute the partial correlations $r_{yx_i^* x_j}$, $i \neq j$, for each of the remaining independent variables x_i, $(i = 1, \ldots, j-1, j+1, \ldots, n_p)$. The procedure to compute the partial correlations with one or more variables held fixed is described subsequently.

Choose an independent variable with the largest partial correlation, say, the kth independent variable. Then fit the appended model:

$$y = \theta_0 + \theta_j x_j + \theta_k x_k + \varepsilon \tag{6.63}$$

Step 3: Now, compute the partial F values F_j and F_k corresponding to the jth and kth variable. Comparing the smallest of F_j and F_k with the preselected threshold F_{Th} leads to two possibilities:

1. $\min(F_j, F_k) < F_{Th}$, suggesting that the contribution due to the corresponding variable is not significant. Accordingly, delete either the jth or kth independent variable corresponding to the smaller of them. Jump back to step 2, and continue with the remaining variables.
2. $\min(F_j, F_k) > F_{Th}$, suggesting that the kth variable has a significant contribution to y and hence can be retained in the extended model of Eq. (6.63). Jump back to step 2, and proceed to examine the partial correlations of y on the remaining variables, given j and k are in the model.

At each step, these two substeps can lead to either increasing or decreasing the variables included in the model. The partial correlations will have to be computed keeping multiple variables already entered in the model held fixed. The procedure automatically stops when the variables once entered in the model cannot be removed and when addition of any new variables does not lead to improvement in the model fit.

6.8.2 STATISTICAL TEST CRITERIA

The correlation coefficient defined in Eq. (6.61) lies in the range $[-1 \;\; 1]$; it is an indicator of the degree of influence of the particular independent variable on the dependent variables (observation). The higher the correlation coefficient, the larger is the dependence of y on that particular independent variable. The correlation coefficient defined in Eq. (6.61) is, however, not to be confused with the correlation coefficient between the parameters that we had discussed in connection with the properties of parameter estimates in Sec. 4.14.

Partial correlation determines, in general, the correlation between any two of the variables, if the other variables are held constant. In our case, it is an indicator of the degree of influence of the particular independent variable, say, x_i, on the dependent variables y, when all of the x except x_i are held fixed. In other words, it predicts the influence of the variables, which are held

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fixed on the other two variables, that is, on y and x_i. If partial correlation approaches zero, it implies that there is no significant link between the two variables being tested because the control variables (that is, variables held fixed) are the primary or intervening variables. Hence, the partial correlation can be used as a measure of relative importance of each independent variable in the regression equation.

Given any three variables y, x_i, and x_j, computation of the partial correlation keeping just one variable fixed is fairly straightforward. In this case the partial correlation can be computed from the simple correlation coefficient:

$$r_{yx_i * x_j} = \frac{r_{yx_i} - r_{yx_j} r_{x_i x_j}}{\sqrt{(1 - r_{yx_j}^2)(1 - r_{x_i x_j}^2)}} \tag{6.64}$$

where the correlation coefficients r_{yx_i}, r_{yx_j}, and $r_{x_i x_j}$ are computed from the expression given in Eq. (6.61) replacing y and x_i with appropriate combinations of y, x_i, and x_j. However, when there are multiple variables entered in the model, computation of the partial correlation for the rest, keeping those already in the model fixed, is somewhat tedious. Basically, it is based on calculating 1) the residuals after fitting y using entered variables (representing the part that has not been so far explained) and 2) the residuals after fitting the candidate variable x_i using the same set of entered variables (representing the part of x_i that cannot be predicted by the entered variables). The correlation between these two residuals gives the desired partial correlation. To describe the process mathematically, for the general model postulated in Eq. (6.1), the partial correlation of x_j on y, denoted by $r_{yx_j *}(x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_{nq})$, is computed as follows. First, we fit the model by dropping the jth independent variable:

$$y_{(-j)} = \theta_1 x_1(k) + \cdots + \theta_{j-1} x_{j-1}(k) + \theta_{j+1} x_{j+1}(k) + \cdots + \theta_{nq} x_{nq}(k) + \varepsilon(k)$$
$$k = 1, 2, \ldots, N \tag{6.65}$$

where $y_{(-j)}$ denotes the independent variable y, but without the contribution due to the jth independent variable, as evident from the term $\theta_j x_j(k)$ missing on the right-hand side of Eq. (6.65). Note specifically the notation $y_{(-j)}$; it is not the jth element of y; we are dealing here with a single dependent variable.

Now fit a similar model to the dropped variable, that is, jth, independent variable using the same set of independent variables as in Eq. (6.65):

$$x_j = \theta_1 x_1(k) + \cdots + \theta_{j-1} x_{j-1}(k) + \theta_{j+1} x_{j+1}(k) + \cdots + \theta_{nq} x_{nq}(k) + \varepsilon(k)$$
$$k = 1, 2, \ldots, N \tag{6.66}$$

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Denoting the fit errors in the two cases resulting from the model fit of Eqs. (6.65) and (6.64) as $e^{y(-j)}$ and e^{x_j} respectively, the partial correlation is then given by

$$r_{yx_j*}(x_1,\ldots,x_{j-1},\,x_{j+1},\ldots,x_{nq}) = \frac{\sum_{k=1}^{N}(e^{y(-j)}(k)-\bar{e}^{y(-j)})(e^{x_j}(k)-\bar{e}^{x_j})}{\sqrt{\sum_{k=1}^{N}(e^{y(-j)}(k)-\bar{e}^{y(-j)})^2\sum_{k=1}^{N}(e^{x_j}(k)-\bar{e}^{x_j})^2}} \tag{6.67}$$

where $\bar{e}^{y(-j)} = 1/N\sum_{k=1}^{N} e^{y(-j)}(k)$ and $\bar{e}^{x_j} = 1/N\sum_{k=1}^{N} e^{x_j}(k)$ are the respective mean values of the residuals.

The F value used in the stepwise regression procedure is a measure of the significance of the independent variables and relates the goodness of fit (that is, the value of the cost function) to the fit error, weighted by the degrees of freedom in the data (that is, the number of independent samples from which the variances is calculated). It can be computed from the partial correlations as follows:

$$F = \frac{r_{yx_i{}^*x_j}}{(1-r_{yx_i{}^*x_j})}\frac{N-n_q}{n_q-1} \tag{6.68}$$

If more than one independent variable is held fixed, then the partial correlation $r_{yx_i{}^*}(x_1,\ldots,x_{j-1},\,x_{j+1},\ldots,x_{nq})$ is to be used to compute the corresponding F values.

Some other useful criteria are the coefficient of determination (R-Square), the adjusted coefficient of determination, Mallow's Cm statistics, and Akaike's Information Criterion. We just give a brief statement on the first two of these additional tests. The coefficient of determination (R-Square) is given by

$$R^2 = \frac{\sum_{k=1}^{N}(\hat{y}(k)-\bar{y})^2}{\sum_{k=1}^{N}(y(k)-\bar{y})^2} = \frac{\hat{\theta}^T X^T Y - N\bar{y}^2}{Y^T Y - N\bar{y}^2} \tag{6.69}$$

where $\hat{y}$ represents the predicted output (dependent variable) using the estimated model parameters $\hat{\theta}$ and y the measured data samples and as already defined $\bar{y}$ the mean value. As the number of estimated parameters (elements of $\hat{\theta}$) will vary from step to step, R-square statistics gives the proportion of the variation in the dependent variable y that is explained by the regression model and has a value between 0 and 1. The larger the value of R-Square, the better is the model fit. However, it tends to be overestimated when the number of samples N is not large compared to n_q, the number of independent variables. Furthermore, for the same number of data samples, R-Square always increases with increasing the number of independent variables, even when some of them might not have

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sufficient influence to justify their inclusion. This might not lead to a parsimonious model representation.

The adjusted coefficient of determination $AdjR^2$ is a modified version that corrects Eq. (6.69) for the number of parameters being estimated and does not necessarily increase when we add extra variables:

$$AdjR^2 = 1 - (1 - R^2)\frac{N - 1}{N - n_q - 1} \tag{6.70}$$

6.8.3 PRACTICAL ASPECTS

Although we have covered in some depth the stepwise regression procedure as a possible means to model structure determination, the actual procedure is plagued with some of the common problems that we encounter in system identification. It might look quite easy and promising at a first glance, but for complex problems with a large number of independent variables and complex nonlinear relations, it requires some skill and judgment to arrive at the final model. Among the difficulties associated with the model structure determination using stepwise regression, the four most important ones are as follows:

1. In the presence of data collinearity, the variable selection process is inefficient and inconsistent.
2. The R-square statistics are usually overestimated.
3. The threshold values used to judge the statistical significance are situational, that is, they may vary from case to case.
4. If two data sets at almost similar operating conditions differ in variability, the selected variable might not turn out to be a unique set.

The choice of the potential independent variables, including all possible cross combinations, has to be prespecified. An inadvertent omission of a particular primary influencing variable might not be detected, leading to a good match of the data fitted through combinations of other variables. Most likely, such models will not show good predictive capabilities for data not used in fitting the model. The specification of a partial F-statistics threshold to accept a particular variable will also affect the final model.

Furthermore, the best choice of variables is usually guided by physical understanding and good judgment of prime variables of the process. From this aspect we quote Judd and McClelland's observation on the stepwise regression procedure [26]:

> It is our experience and strong belief that better models and a better understanding of one's data result from focused data analysis, guided by substantive theory.

Although the observation pertains to a completely different phenomenon being analyzed, it brings out some concern that one should keep in mind while applying

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such procedures. For flight vehicles, the a priori information about the typical structure of aerodynamic forces and moments is usually available, particularly when we try to estimate linear models valid for small excursions around the trim point. In such cases the advantages of stepwise regression are minimal. Furthermore, the commonly used set of potential independent variables, like α, β, p, q, r, δ_e, δ_a, δ_r and their various combinations such as $\beta\alpha$, $\alpha\delta_e$, might not be quite adequate when the model is nonlinear within the range of variable excursions. We will come across such a case in Sec. 12.8.1 showing $C_{n\beta}$ to be nonlinearly dependent on the angle of attack. The efficacy of straightforward model structure determination, without additional considerations such as data partitioning, is not quite evident in such cases. In addition, the least-squares method and the stepwise regression approach have certain limitations, which have been brought out earlier.

Based on some of these practical considerations, we generally prefer to use a more general approach to modeling based on the following: 1) incorporation of the basic a priori knowledge, which we mostly possess, about the modes of aircraft motion, aerodynamic effects, and associated model structure; and 2) estimation of parameters applying more powerful methods such as the output-error method described in Chapter 4. In special cases like modeling at high angles of attack and unstable aircraft, however, stepwise regression combined with the option of data partitioning can be useful. In such cases, techniques to detect collinearity and procedures to account for them in the least-squares estimation are necessary. Such procedures will be covered in Chapter 9 dealing specifically with unstable aircraft identification.

6.9 EXAMPLES

Because we are mainly concerned in this book with aircraft parameter estimation, we demonstrate the least-squares method to estimate the stability and control derivatives from flight data. To bring out the various aspects studied in this chapter, it suffices to analyze a single operating point based on simple model. A more complex application is presented in Chapter 12.

6.9.1 FLIGHT-DERIVED AERODYNAMIC FORCE AND MOMENT COEFFICIENTS

Although the main advantage of the regression analysis is its simplicity, a fair amount of computation during a preprocessing step is required in the case of aircraft parameter estimation because the aerodynamic forces and moments (that is, the dependent variables) are not directly measured, but have to be obtained from the measurements of the related variables, namely, linear accelerations and angular rates. Actually, one needs angular accelerations, but in practice in most of the cases angular rates are measured and not the angular accelerations. It is also possible to treat the measured linear and angular accelerations as dependent variables, in which case the coefficients pertaining to the aerodynamic forces

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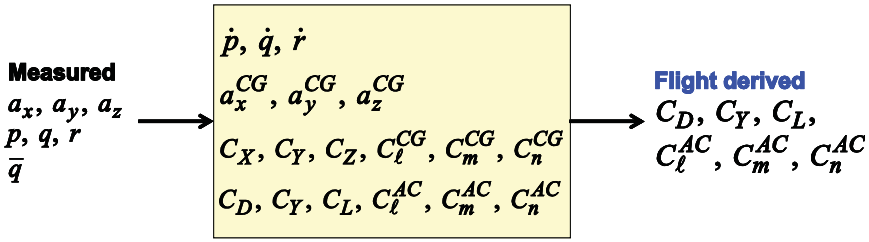

Fig. 6.6 Schematic of data preprocessing to compute aerodynamic force and moment coefficients from flight measured data.

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and moments will have to be obtained through proper transformations after the estimation. Generally, the first approach of precomputing the force and moment coefficients is preferred. A block schematic of the computational procedure presented is depicted in Fig. 6.6.

Considering the flight vehicle as a point mass, we know that the resultant aerodynamic forces and moments act at the center of gravity (c.g.). Accordingly, to apply the regression method, one needs to know the linear acceleration at the c.g. Because of the restrictions of the mechanical mountings, it is not always possible to mount the sensors exactly at the c.g.; they are usually located at some convenient locations, away from the c.g. Moreover, the c.g. location depends on the loading conditions and may vary from flight to flight. The actual c.g. location is, however, computed from the flight loading manifest, and thus the actual distances from the c.g. to the sensor assembly can be determined accurately. If the measurement of actual fuel quantity (or alternatively of fuel consumption rate) is available, the effects of variations in the total fuel quantity during the flight on the c.g. location can also be accounted for. Depending upon the specific application, it might also be necessary to account for fuel slosh effects. Because the accelerometers mounted away from the c.g. measure local accelerations that contain the components due to the angular rates and accelerations, it is necessary to obtain the linear accelerations at c.g.

The body-fixed accelerations ($a_x^{CG}, a_y^{CG}, a_z^{CG}$) at the c.g. are computed from the translational accelerations ($a_{xm}^{AS}, a_{ym}^{AS}, a_{zm}^{AS}$) measured at the accelerometer sensor position (AS) through the following transformation:

$$
\begin{aligned}
a_x^{CG} &= a_{xm}^{AS} + (q^2 + r^2)x_{\mathrm{ASCG}} - (pq - \dot{r})y_{\mathrm{ASCG}} - (pr + \dot{q})z_{\mathrm{ASCG}} \\
a_y^{CG} &= a_{ym}^{AS} - (pq + \dot{r})x_{\mathrm{ASCG}} + (p^2 + r^2)y_{\mathrm{ASCG}} - (qr - \dot{p})z_{\mathrm{ASCG}} \\
a_z^{CG} &= a_{zm}^{AS} - (pr - \dot{q})x_{\mathrm{ASCG}} - (qr + \dot{p})y_{\mathrm{ASCG}} + (p^2 + q^2)z_{\mathrm{ASCG}}
\end{aligned}
\tag{6.71}
$$

where (p, q, r) are the measured angular rates, $(\dot{p}, \dot{q}, \dot{r})$ the angular accelerations obtained by numerical differentiation of the measured angular rates, and

$(x_{\text{ASCG}}, y_{\text{ASCG}}, z_{\text{ASCG}})$ the position of the AS with respect to the c.g. along the three body-fixed coordinate axes, respectively.

Having obtained the linear accelerations at the c.g., the three body axes aerodynamic force coefficients are given by

$$\begin{aligned} C_X &= (ma_x^{CG} - F_{\text{eng}} \cos \sigma_{\text{eng}})/(\overline{q}S) \\ C_Y &= (ma_y^{CG})/(\overline{q}S) \\ C_Z &= (ma_z^{CG} + F_{\text{eng}} \sin \sigma_{\text{eng}})/(\overline{q}S) \end{aligned} \tag{6.72}$$

where m is the aircraft mass, F_{eng} the total thrust, σ_{eng} the inclination angle of the engines, $\overline{q}$ the measured dynamic pressure, and S the reference area.

The lift, drag, and side-force coefficients are derived from the body-axes force coefficients C_X, C_Y, C_Z as follows:

$$\begin{aligned} C_L &= C_X \sin \alpha - C_Z \cos \alpha \\ C_D &= -C_X \cos \alpha - C_Z \sin \alpha \\ C_Y &= C_Y \end{aligned} \tag{6.73}$$

Computation of the three moment coefficients is a little more involved and requires the angular accelerations $(\dot{p}, \dot{q}, \dot{r})$. As already pointed out, because angular accelerations are not directly measured, they have to be obtained by numerical differentiation of the measured angular rates (p, q, r). Numerical differentiation can introduce some additional noise due to discretization errors and noise in the measured rates. Several techniques of numerical differentiation are readily available, as discussed in Sec. 2.5.2, hence not elaborated here further.

The body-axes rolling, pitching, and yawing coefficients C_ℓ, C_m, C_n referred to c.g. are given by

$$\begin{aligned} C_\ell^{CG} &= (I_X \dot{p} - I_{XZ}\dot{r} - I_{XZ}pq - (I_Y - I_Z)qr - \Delta F_{\text{eng}} \sin \sigma_{\text{eng}} y_{\text{ENCG}})/(\overline{q}Sb) \\ C_m^{CG} &= (I_Y \dot{q} + I_{XZ}(p^2 - r^2) - (I_Z - I_X)pr \\ &\quad - F_{\text{eng}} \cos \sigma_{\text{eng}} z_{\text{ENCG}} - F_{\text{eng}} \sin \sigma_{\text{eng}} x_{\text{ENCG}})/(\overline{q}Sl) \\ C_n^{CG} &= (I_Z \dot{r} - I_{XZ}\dot{p} + I_{XZ}qr - (I_X - I_Y)pq - \Delta F_{\text{eng}} \cos \sigma_{\text{eng}} y_{\text{ENCG}})/(\overline{q}Sb) \end{aligned} \tag{6.74}$$

where I_X, I_Y, I_Z are the moments of inertia about the x, y, z axes, respectively; I_{XZ} the cross moment of inertia; ΔF_{eng} the differential thrust (left minus right engine thrust); y_{ENCG} the lever arm (distance of the right engine from the centerline along y axis); $(x_{\text{ENCG}}, z_{\text{ENCG}})$ the position of the engines with respect to the c.g. along the x and z axes; b the reference length for the lateral-directional coefficient; and l the reference length for the longitudinal coefficient. All other quantities have been defined earlier. The reference lengths for the lateral-directional and longitudinal coefficients are usually the wing span (or half the wing span) and the wing mean aerodynamic chord, respectively. In a few specific cases the

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mean aerodynamic chord is used for both of the modes. What is actually used as the reference length is not really important, but has to be kept in mind when comparing derivatives and when using the identified aerodynamic model. The aerodynamic model contains not only the equations for the coefficients, but also the application rules (containing the numerical values of the reference areas and lengths) to calculate the forces and moments. This is one of the common sources of error that is often overlooked, when results from different sources are compared or combined together.

The moment coefficients referred to the aerodynamic center (AC), also some time called moment reference point, are now computed from Eqs. (6.74) and (6.72). They are given by

$$\begin{aligned} C_\ell^{AC} &= C_\ell^{CG} - C_Z \frac{y_{\text{ACCG}}}{b} + C_Y \frac{z_{\text{ACCG}}}{b} \\ C_m^{AC} &= C_m^{CG} - C_X \frac{z_{\text{ACCG}}}{l} + C_Z \frac{x_{\text{ACCG}}}{l} \\ C_n^{AC} &= C_n^{CG} - C_Y \frac{x_{\text{ACCG}}}{b} + C_X \frac{y_{\text{ACCG}}}{b} \end{aligned} \tag{6.75}$$

where (C_X, C_Y, C_Z) are the force coefficients given by Eq. (6.72), and $(x_{\text{ACCG}}, y_{\text{ACCG}}, z_{\text{ACCG}})$ is the position of the AC with respect to the c.g.

It is obvious that the preceding data preprocessing step involves several important aspects, like computation of exact c.g. location, accounting for sensor offsets, the engine thrust, etc. An astute reader will realize that these steps are mostly in the reverse order of those which are required while identifying the model applying the output-error method. Thus, we do not really avoid the computations, except that they are performed at different stages of the algorithm being applied. In the output-error method the aerodynamic model is embedded in the equations of motion and constitutes a part of the postulated state-space model and the computations carried out using the integrated states. Depending upon the numerical integration method, they will be repeated at each data point a number of times. This contributes significantly to the computational burden of the output-error method. On the other hand, while using the regression technique, there are no state equations and no integrations; the forces and moments are computed a priori from the measurements of the corresponding variables. This data preprocessing step needs to be done only once. Combined with the use of measured states in the estimation, it leads to a significant reduction in the computational burden.

Although not considered explicitly in Eqs. (6.71–6.75), it is straightforward to see that any errors in the measurements of the linear accelerations, angular rates, and dynamic pressure will directly affect the computed force and moment coefficients. These flight-derived coefficients C_L, C_D, C_Y, C_ℓ, C_m, C_n are the dependent variables to be modeled to estimate the stability and control derivatives. Any errors in these flight-derived coefficients will directly affect the estimates of the unknown aerodynamic parameters. Although the LS estimation accounts for

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noise in the dependent variables, we have to eliminate the systematic errors in them, such as bias, scale factor, or time delays. Elimination of such errors in these dependent variables as well as those in the independent variables (comprising of angular rates, flow angles and control surface deflections) is critical to the next step of parameter estimation.

6.9.2 ESTIMATION OF AERODYNAMIC PARAMETERS OF A LINEAR MODEL

For demonstration purposes we consider here analysis of flight data recorded with the research aircraft ATTAS. Three flight maneuvers performed at a nominal altitude of 16,000 ft and at 200 kt are considered. The first maneuver is a multistep elevator input exciting the short-period motion, the second is an aileron input resulting in the bank-to-bank motion, and the third is a rudder doublet input exciting the Dutch roll motion. The three maneuvers are 25, 30, and 30 s long, respectively. The same set of three maneuvers is analyzed in Sec. 10.5 to check the kinematic consistency of the recorded flight data. The results of data consistency checking generated in that section (test_case = 22) showed the following biases in the measurements of linear accelerations and angular rates:

$$\Delta a_x = 0.0067\,\mathrm{m/s^2}, \quad \Delta a_y = -0.0015\,\mathrm{m/s^2}, \quad \Delta a_z = 0.0839\,\mathrm{m/s^2}$$
$$\Delta p = 0.0010\,\mathrm{rad/s}, \quad \Delta q = 0.0034\,\mathrm{rad/s}, \quad \Delta r = 0.0010\,\mathrm{rad/s}$$

The measurements of these variables are corrected for the corresponding biases before they are used in the data preprocessing step described in Sec. 6.9.1 and for parameter estimation in the following. The engine thrusts F_{eL} and F_{eR} due to the left and right engines were computed from the measured engine parameters and stored in the flight data files along with the other measured data.

The models for aerodynamic force and moment coefficients are now postulated as follows:

$$\begin{aligned} C_D &= C_{D0} + C_{D\alpha}\alpha \\ C_L &= C_{L0} + C_{L\alpha}\alpha \\ C_m &= C_{m0} + C_{m\alpha}\alpha + C_{mq}q^* + C_{m\delta e}\delta_e \\ C_Y &= C_{Y0} + C_{Y\beta}\beta \\ C_\ell &= C_{\ell 0} + C_{\ell\beta}\beta + C_{\ell p}p^* + C_{\ell r}r^* + C_{\ell\delta a}\delta_a \\ C_n &= C_{n0} + C_{n\beta}\beta + C_{np}p^* + C_{nr}r^* + C_{n\delta r}\delta_r \end{aligned} \tag{6.76}$$

where δ_e is the elevator deflection, δ_a the aileron deflection, δ_r the rudder deflection, α the angle of attack, β the angle of sideslip, (p^*, q^*, r^*) the normalized angular rates, and $(C_{D()}, C_{L()}, C_{m()}, C_{Y()}, C_{\ell()}, C_{n()})$ the unknown aerodynamic parameters. The normalized angular rates are given by

$$p^* = pb/V; \quad q^* = ql/V; \quad r^* = rb/V \tag{6.77}$$

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where V is the true airspeed; other quantities have already been defined. We identify the unknown aerodynamic derivatives applying the LS method in both the forms, namely, 1) iterative solution elaborated in Sec. 6.4, using the OEM software developed in Chapter 4; and 2) classical one-shot OLS and TLS solutions using matrix algebra operations, discussed in Secs. 6.2 and 6.5, respectively.

To apply the iterative OEM program "/FVSysID2/chapter04/ml_oem2.m," we first carry out the data preprocessing step described in Sec. 6.9.1 and then provide the details for parameter estimation in the model definition function "mDefCase23.m" and set the integer flag test_case = 23.

The input and output variables for the data preprocessing step to compute the aerodynamic force and moment coefficients are given by

Input variables:

$$\begin{aligned} \text{ZAccel} &\quad a_x, a_y, a_z, p, q, r \\ \text{Uinp} &\quad \delta_e, \delta_a, \delta_r, p, q, r, \alpha, \beta, V, \rho, F_{\text{eL}}, F_{\text{eR}}, \dot{\alpha}, Ma \end{aligned}$$

Output variables:

$$\text{Z} \quad C_D, C_L, C_m, C_Y, C_\ell, C_n$$

where ρ is the density of air, F_{eL} the left engine thrust, and F_{eR} the right engine thrust. The dynamic pressure required in the precomputation of the force and moment coefficients is given by $0.5\rho V^2$. The angular accelerations $\dot{p}$, $\dot{q}$, $\dot{r}$ required to compute the moment coefficients C_m, C_ℓ, C_n are obtained by numerical differentiation of measured angular rates p, q, r. The input array Uinp contains variables that are required during both the steps of preprocessing and parameter estimation. Not all signals from Uinp are required in the present case, but for the sake of similarity with the other cases addressed in other chapters it is made oversized.

The data to be preprocessed and the input variables, that is, the arrays ZAccel(Ndata,Ny) and Uinp(Ndata,Nu), are loaded from three data files:

```
load ..\flt_data\fAttasElv1
load ..\flt_data\fAttasAil1
load ..\flt_data\fAttas Rud1
```

It is, of course, necessary to provide the input and output variables in proper units. Because we are analyzing three maneuvers simultaneously, we additionally define the following:

```
Nzi =3;                              % Number of time segments
izhf=[Nts1; Nts1 + Nts2; Nts1        % Cumulative index
      + Nts2 + Nts3];
```

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where Nts1, Nts2, and Nts3 are the number of data points in the three individual time segments and izhf is the cumulative index at which the maneuvers end when they are concatenated.

Having defined the necessary details, the aerodynamic force and moment coefficients are computed according to Eqs. (6.71–6.75) by calling the data preprocessing function "umr_reg_attas.m":

$$[Z, Uinp] = umr_reg_attas(ZAccel, Uinp, Nzi, izhf, dt, test_case)$$

Using the flight-derived aerodynamic coefficients now available from the foregoing preprocessing step in array Z and the input in the array Uinp, we now formulate the model for parameter using the OEM method as follows:

	No. of Variables		Function Name
States	0	–	–
Outputs	6	C_D, C_L, C_m,C_Y, C_ℓ, C_n	"obs_TC23_attas_regLonLat"
Inputs	14	δ_e, δ_a, δ_r, p, q, r, α, β, V, ρ, F_{eL}, F_{eR}, $\dot{\alpha}$, Ma	

As there are no state equations, we do not actually need a function to code the right-hand sides of the state equations, but because we propose to use here the same algorithm that was developed for the output-error method, to avoid problems associated with the number of arguments being passed, we either supply a dummy function, called "xdot_attas_reg," in which there are no computations, or alternatively we define the function name for state equation as an empty string, as will be done here. The right-hand sides of the regression equation, namely, Eq. (6.76), are coded in "obs_attas_regLonLat.m." Thus, the model definition for this example provided by the function "/FVSysID2/chapter04/mDefCase23.m," which is called from the main program "/FVSysID2/chapter04/ml_oem2," is as follows:

```
Test_case = 23;                          % Test case number
state_eq  = [ ];                         % Empty string for state equations
obser_eq                                 % Function for observation equations
= "obs_TC23_attas_regLonLat";
Nx        = 0;                           % Number of states
Ny        = 6;                           % Number of observation variables
Nu        = 14;                          % Number of input variables
Nparam    = 20;                          % Total number of parameters
dt        = 0.04;                        % Sampling time
```

We specify zero starting values for all of the 20 parameters and run the program "/FVSysID2/chapter04/ml_oem2.m." The determinant of the covariance matrix of the residuals (cost function) corresponding to these starting parameter values is 2.2856×10^{-20}. The iterative procedure is terminated after

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two iterations with the cost function value of 5.7088×10^{-33}. Actually, the model being linear in parameters and in the independent variables, the minimum is reached after the first iteration. But the iterative procedure needs another iteration to check whether convergence has been achieved or not.

The control surface deflections (δ_e, δ_a, δ_r), angular rates, (p, q, r) and flow angles (α, β), which are inputs to the model (independent variables), are shown in Fig. 6.7. The variation in the flow angles about the trim values is of the order of ± 3.3 and ± 4 deg, respectively. A comparison of the flight-derived (that is, measured) force and moment coefficients with those estimated by the model postulated in Eq. (6.76) is shown in Fig. 6.8; it is observed that the linear model captures the major cause–effect relations nicely. Small correlation was observed between the parameter pairs (C_{D0}, $C_{D\alpha}$) and (C_{L0}, $C_{L\alpha}$), which is attributed to the fact that we have analyzed just one time record pertaining to the longitudinal motion. Moreover, the drag derivatives are better estimated from longer

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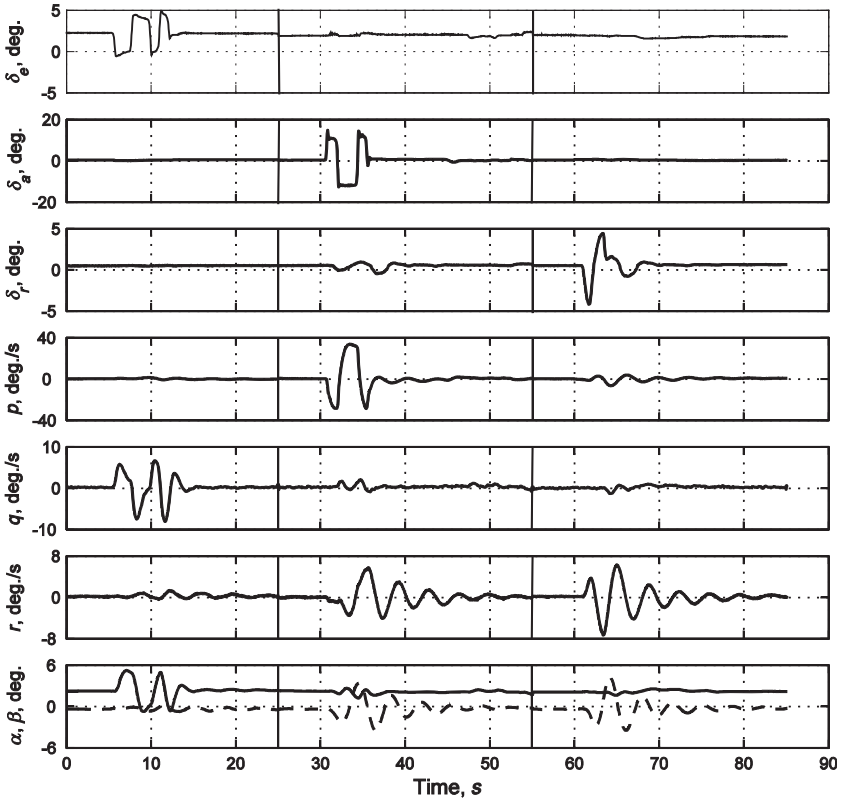

Fig. 6.7 Time histories of independent variables.

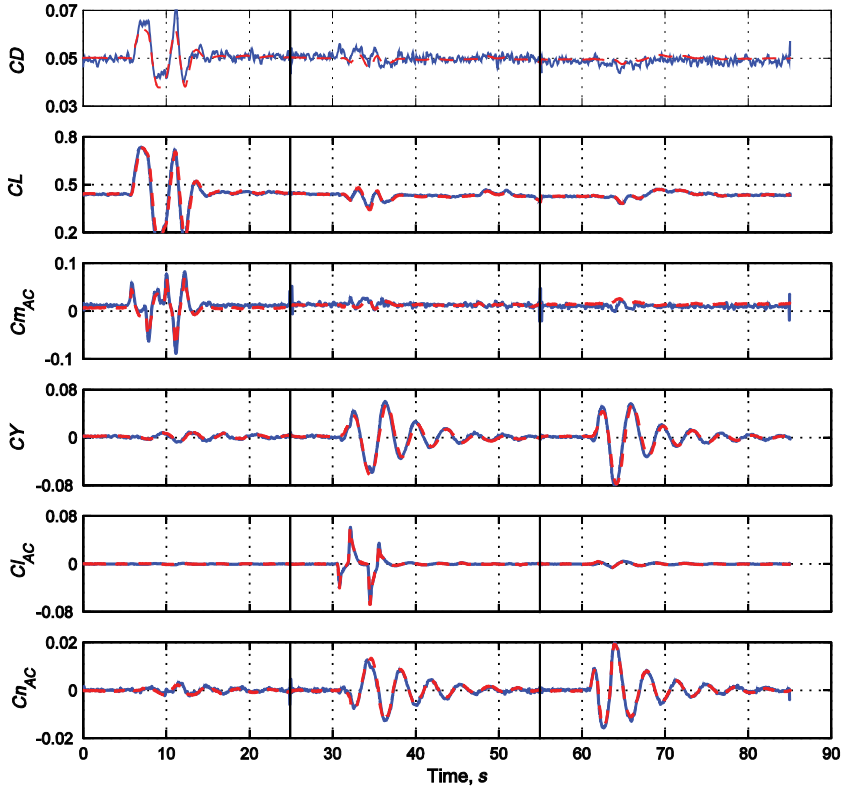

Fig. 6.8 Time histories of measured and estimated outputs: ——, flights measured; -----, model predicted.

duration maneuvers, including phugoid motion. Nevertheless, the example is quite adequate to demonstrate the procedure of applying iterative regression method to flight data, to show the use of the same software that was developed in Chapter 4, and to bring out the two-step approach consisting of data preprocessing and followed by LS parameter estimation.

To this linear regression model, we also apply the classical one-shot least-squares method. The data preprocessing step is the same, but the estimates are now obtained from Eq. (6.11) using matrix algebra operations, that is, by solving the normal equation. The one-shot procedure is applied six times considering the dependent variables one at a time; it requires building the output vector Y and data matrix X appropriately each time, constituted of different sets of variables. As summarized in Table 6.1, the data matrix is of different size each time, as it depends upon the number of independent variables appearing in each regression equation being estimated; see Eq. (6.76).

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As already pointed out in Sec. 6.2, the procedure allows estimation of aerodynamic zero terms by simply associating a vector of all ones as an additional independent variable. This is denoted by "1" in Table 6.1, and it is sometimes called "bias vector."

We run this case by calling the program "/FVSysID2/chapter06/attas_regLS," which is a very rudimentary implementation of the OLS method. There are many elegant, sophisticated least-squares programs available, providing more flexibility. The basic principle behind them, however, remains the same. The simple version is quite sufficient to trace the procedure elaborated in Sec. 6.2 and to compare the results with the iterative version. To run this program, we load the three time segments, define Nzi and izhf, assign the arrays ZAccel (Ndata,Ny) and Uinp(Ndata,Nu), and then call the data preprocessing function "umr_reg_attas" exactly in the same way as before. This is followed by building the output vector Y and data matrix X for each coefficient according to Table 6.1. The parameter estimates are obtained by solving Eq. (6.11) through simple matrix operations. The estimates of the parameters and their absolute and relative standard deviations are printed out. The time history plots of computed and estimated aerodynamic coefficients are generated, but not shown here.

As expected, the one-shot and the iterative procedures yield the same results. This is evident from the second and the third column of Table 6.2, providing estimates of a few selected parameters. The results for the other (pitching and yawing moment) coefficients can be readily verified by running the two cases. The iterative procedure is however more flexible as it can be applied to nonlinear model postulates.

To the same problem, we also apply the TLS method by calling the program "/FVSysID2/chapter06/attas_regTLS." This basic version provided herewith helps to trace the computational steps elaborated in Sec. 6.5, Eqs. (6.50–6.51) and to compare the results generated with OLS. As in preceding case, the estimates of the parameters and their absolute and relative standard deviations are printed out, and the time history plots of computed and estimated aerodynamic

TABLE 6.1 SUMMARY OF VARIABLES FOR ONE-SHOT LEAST-SQUARES ESTIMATION

Coefficient of:	Dependent Variable Y	Independent Variables X	Parameters θ
Drag	C_D	1, α	C_{D0}, $C_{D\alpha}$
Lift	C_L	1, α	C_{L0}, $C_{L\alpha}$
Pitching moment	C_m	1, α, q^*, δ_e	C_{m0}, $C_{m\alpha}$, C_{mq}, $C_{m\delta e}$
Side force	C_Y	1, β	C_{Y0}, $C_{Y\beta}$
Rolling moment	C_ℓ	1, β, p^*, r^*, δ_a	$C_{\ell 0}$, $C_{\ell\beta}$, $C_{\ell p}$, $C_{\ell r}$, $C_{\ell\delta a}$
Yawing moment	C_n	1, β, p^*, r^*, δ_r	C_{n0}, $C_{n\beta}$, C_{np}, C_{nr}, $C_{n\delta r}$

TABLE 6.2 LEAST-SQUARES ESTIMATES FOR FEW SELECTED PARAMETERS

Parameter	One-shot LS; Linear Model: "attas_regLS"	Iterative LS; Linear Model: "ml_oem2" Test_case = 23	Iterative LS; Nonlinear Model: "ml_oem2" Test_case = 24	One-shot TLS; Linear Model: "attas_regTLS"
C_{D0}	0.03985	0.03985	0.03887	0.03962
$C_{D\alpha}$	0.26040	0.26040	NA*	0.26638
e	NA	NA	0.77813	NA
C_{L0}	0.23496	0.23496	0.23492	0.23389
$C_{L\alpha}$	5.22568	5.22568	5.22666	5.25327
C_{Y0}	−0.00491	−0.00491	−0.00491	−0.00499
$C_{Y\beta}$	−1.00403	−1.00403	−1.00403	−1.01777
$C_{\ell 0}$	0.000184	0.000184	0.000184	0.000191
$C_{\ell\beta}$	−0.10268	−0.10268	−0.10268	−0.10244
$C_{\ell p}$	−0.75976	−0.75976	−0.75976	−0.76643
$C_{\ell r}$	0.23767	0.23767	0.23767	0.25191
$C_{\ell\delta a}$	−0.19247	−0.19247	−0.19247	−0.19365

*NA: Not applicable

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coefficients are generated, but not shown here in the text. Comparing the second and the last column of Table 6.2, it is observed that the TLS estimates differ only slightly from the OLS estimates; in general, for the data analyzed, they are fairly comparable. This is attributed to the fact that the noise in the measurements of the dependent and independent variables is quite low and that the systematic errors were eliminated in the data preprocessing step. Under such conditions, the two methods tend to be equivalent. The low noise level is attributed to the high-quality sensors and instrumentation used to gather the flight data. The data analyzed here is a typical sample of flight recordings and representative of noise levels generally present in the data used for parameter estimation. In the presence of low-cost sensors which can introduce a higher level of noise, the estimates from the two methods might differ more.

The performance of the OLS and TLS methods depends critically on the adequacy of the postulated model. This aspect was not very apparent from the modeling of the aerodynamic coefficients C_D, C_L, C_Y, C_l, and C_n, for which, as just discussed, the estimates and the match from the OLS and TLS methods were comparable. In these cases the cross-coupling effects were not very strong, and hence the postulated models in Eqs. (6.76) were reasonable for the qualitative analysis performed here. However, the TLS method applied to the pitching moment

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coefficient C_m showed some very discernible discrepancies in the match for C_m compared to that from the OLS method. The estimates were also not comparable. This can be verified from the first plot obtained by running "/FVSysID/chapter06/attas_regTLS." This anomaly is attributed to the fact that three time segments with elevator, aileron, and rudder inputs are analyzed simultaneously whereas the model for C_m is based on longitudinal variables only, that is, α, q, δ_e as independent variables; see Eq. (6.76). It does not account for the cross-coupling effects from to the lateral-directional motion variables β, p, r, δ_r into the pitching motion, which happen to be significant in the present case. The variations in these lateral-directional variables are large in the second and third maneuvers. Because they are missing in the model, they are treated as noise. These discrepancies in the postulated model lead to wrong TLS estimates of pitching-moment derivatives from three maneuvers. To analyze this case correctly, it would require model augmentation (that is, introduction of appropriate independent variables). Because our aim is restricted to demonstration of TLS principle, such model improvements are not pursued here. The preceding discussion can be verified in an alternative way by analyzing separately the first maneuver with an elevator input based on the simplified model of Eq. (6.76). This has been carried out in the same program, where we apply both OLS and TLS to this single maneuver, and plot the results in the second figure. It turns out that the OLS and TLS estimates and the match for time history of C_m are now comparable. This aspect of model adequacy needs to be ascertained while applying the OLS or TLS methods to data from multiple maneuvers.

The foregoing OLS and TLS estimates were obtained by treating each dependent variable separately and by suitably choosing different independent variables each time (see Table 6.1). As discussed in Secs. 6.4 and 6.5, the multivariate OLS and multivariate TLS procedures are mathematically feasible, but not practiced because the same set of independent variables is usually not suitable for all of the dependent variables. Nevertheless, to demonstrate these techniques, two script files, namely, 1) "attasLonLat_mV_LS.m" and 2) "attasLonLat_mV_TLS.m" (both found in the directory /FVSysID2/chapter06/) are provided. These functions are tailored to the combined longitudinal and lateral-directional motion with (C_D, C_L, C_m, C_Y, C_ℓ, C_n) as outputs, that is, $n_y = 6$, and (1, α, q, δ_e, β,p, r, δ_a, δ_r) as inputs, that is, $n_q = 9$. The unknown parameter Θ is a ($n_q \times n_y$) matrix. Additional functions (attasLon_mV_LS.m, attasLat_mV_LS.m, attasLon_mV_TLS.m, and attasLat_mV_TLS.m) for the decoupled longitudinal and lateral-directional motion are also provided for reference purposes. For the same set of independent variables, the multivariate or univariate OLS provide the same results. This can be verified by running any one of the preceding multivariate cases. However, the same is not true in the case of TLS. To illustrate the point, let us consider the longitudinal motion. The independent variables are (1, α, q, δ_e), and the three dependent variables are (C_D, C_L, C_m). The multivariate TLS is based on the composed data matrix [X C_D C_L C_m], where X is the data matrix of the just-defined independent variables. Because of the

presence of noise in the independent variables X, the TLS problem based on $[X\ C_D\ C_L\ C_m]$ is not the same as solving $[X\ C_D]$, $[X\ C_L]$, $[X\ C_m]$ separately. For the reasons pointed out in Sec. 6.4, we prefer to treat the dependent variables separately.

6.9.3 ESTIMATION OF AERODYNAMIC PARAMETERS FROM A NONLINEAR MODEL

To demonstrate the nonlinear regression method, we consider the following improved modeling of the drag coefficient:

$$C_D = C_{D0} + \frac{C_L^2}{e\pi AR} \tag{6.78}$$

where e is the Oswald factor and AR the aspect ratio. The Oswald factor characterizes the increase in drag over the ideal condition caused by nonelliptical lift distribution and interference [27]. It is typically less than one. The regression model is nonlinear in the parameters and in the independent variables as well. Equations for the rest of the coefficients, namely, for C_L, C_Y, C_ℓ, C_m, C_n, are exactly the same as those in Eq. (6.76). The model definition required for the program "ml_oem2" is almost the same as that for the preceding case presented in Sec. 6.9.2, except for the change of test_case number and the function for observation equation (see the function "/FVSysID2/chapter04/mDefCase24.m"):

```
test_case = 24;                          % Test case number
obser_eq = "obs_TC24_attas_regNL";       % Function for observation
                                           equations
```

Except for the change in the model for the drag coefficient, the function "obs_TC24_attas_regNL.m" is the same as "obs_TC23_attas_regLonLat.m." In the case of the linear least-squares model estimated in the preceding section, all of the 20 parameters were initialized to zero. In the present case, to avoid the division by zero, it is necessary to specify a nonzero starting value for e. Furthermore, because of the nonlinearity in the postulated model, reasonable starting values are required for the lift curve slope $C_{L\alpha}$. By setting the flag for test cases to 24, we run the program "/FVSysID2/chapter04/ml_oem2" to estimate the 20 aerodynamic parameters.

It is found that starting from the initial values of 1 for e, 5 for $C_{L\alpha}$, and 0 for the rest of the 18 parameters, the optimization of the cost function requires six iterations to converge, yielding a final cost function value of 4.2773×10^{-33}. Additionally, specifying initial values of 0.03 for C_{D0} and 0.2 for C_{L0} results in convergence to the same minimum value after four iterations. The results for this model for a few selected derivatives are summarized in Table 6.2. The estimates of side-force and rolling-moment coefficients, as well as those for the pitching and yawing moment, are exactly the same as those provided by the cases analyzed

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in the foregoing section. This should be the case because these four output variables are treated independently of each other. On the other hand, the drag coefficient is now not independent, but a nonlinear function of the lift coefficient, which itself is one of the dependent variables. Although the lift coefficient is modeled linearly as in Eq. (6.76), the coupling between the drag and the lift coefficients in Eq. (6.78) influences the estimates of both of these coefficients. Accordingly, some minor differences are found in the estimates of C_{L0}, $C_{L\alpha}$, and C_{D0}; the estimates of $C_{D\alpha}$ in Eq. (6.76) and the Oswald factor e in Eq. (6.78) cannot be directly compared.

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6.10 CONCLUDING REMARKS

In this chapter we have discussed various equation error approaches to parameter estimation and studied the properties of least-squares estimates under ideal conditions and practical considerations. It is shown that the ordinary linear least-squares estimation is a one-shot procedure based on simple matrix algebra operations and does not require initial parameter values to be specified. The basic principle is applicable to nonlinear regression equations as well. It is also possible to extend the least-squares procedure to multiple outputs and to models represented in state space, provided all of the states and state derivatives are measured. It has been argued that this might be a restriction in some of the practical cases, as all of them might not be directly measurable. Obtaining the state derivatives from the measured state variables by numerical differentiation is likely to introduce noise. Because the least-squares procedure yields biased and inconsistent estimates in the presence of systematic errors and noise in the independent variables, a rigorous data preprocessing is generally necessary to eliminate such undesirable effects. From this aspect, the output error is more versatile, and hence the equation-error method is generally used as a startup procedure to generate good initial parameter values for the output-error method. Three aspects that favor least-squares estimation are as follows: 1) somewhat simpler approach to analyze large-amplitude maneuvers through a series of linear models using data partitioning, 2) somewhat simpler mechanism to judge the appropriateness of individual parameters through stepwise regression, and 3) applicability to unstable aircraft (though not covered in this chapter; will be discussed in Chapter 9). These aspects can outweigh some of the limitations that have been mentioned, as evident from several successful applications of aircraft parameter estimation using the least-squares method. Related topics of weighted and total least squares and stepwise selection of variables have been discussed, bringing out each time the pros and cons from the viewpoint of practical applicability.

The least-squares theory and its practical application have been illustrated on a typical example of estimating stability and control derivatives based on linear as well as nonlinear regression models. Estimation of the stability and control derivatives applying the least-squares method requires a fairly elaborative data

preprocessing step to derive the aerodynamic force and moment coefficients from flight measurements of the linear accelerations and angular rates. These computations are mostly in the reverse order of the computations performed in the state equations postulated for the output-error method. It has been demonstrated that the iterative algorithm developed in Chapter 4 can be applied to the multivariate multiple regression model.

REFERENCES

[1] Gauss, C.-F., *Theoria Motus Corporum Coelestium in Sectionibus Conicis Solem Ambientium*, Perthes and Besser, Hamburg, Germany, 1809 (in Latin); also *Theory of the Motion of the Heavenly Bodies Moving About the Sun in Conic Section*, Dover, New York, 1963.

[2] Sorenson, H. W., "Least-Squares Estimation from Gauss to Kalman," *IEEE Spectrum*, Vol. 7, July 1970, pp. 63–68.

[3] Tennenbaum, J., "Gauss' Determination of the Orbit of Ceres, Parts 1–3," *The American Almanac*, Dec. 22, 1997.

[4] Goodwin, G. C., and Payne, R. L., *Dynamic System Identification: Experiment Design and Data Analysis*, Academic Press, New York, 1977.

[5] Sorenson, H. W., *Parameter Estimation - Principles and Problems*, Marcel Dekker, New York, 1980.

[6] Mendel, J. M., *Lessons in Digital Estimation Theory*, Prentice–Hall, Englewood Cliffs, NJ, 1987.

[7] Klein, V., "Identification Evaluation Methods," AGARD LS-104, Paper No. 2, Nov. 1979.

[8] Greenberg, H., "A Survey of Methods for Determining Stability Parameters of an Airplane from Dynamic Flight Measurements," NACA TN-2340, April 1951.

[9] Gerlach, O. H., "Determination of Performance, Stability and Control Characteristics from Measurements in Non-Steady Manoeuvres," AGARD-CP-17, Sept. 1966, pp. 499–523.

[10] Klein, V., "Estimation of Aircraft Aerodynamic Parameters from Flight Data," *Progress in Aerospace Sciences*, Pergamon, Oxford, England, U.K., Vol. 26, No. 1, 1989, pp. 1–77.

[11] Mulder, J. A., Sridhar, J. K., and Breeman, J. H., "Identification of Dynamic Systems – Applications to Aircraft, Part 2: Nonlinear Analysis and Manoeuvre Design," AGARD AG-300, Vol. 3, Part 2, May 1994.

[12] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD AG-300, Vol. 2, Jan. 1985.

[13] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[14] Plaetschke, E., Jategaonkar, R. V., Rohlf, D., and Weiss, S., "Methoden zur Schätzung der Parameter eines Flugzeugs im Post-Stall-Bereich," *Proceedings of the 4th Aerospace Symposium on 'Safety in Air Traffic'*, Technical Univ. of Braunschweig, Germany, 13–15 Sept. 1994, pp. 297–306 (in German).

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[15] Golub, O. H., and Van Loan, C. F., "An Analysis of the Total Least Squares Problem," *SIAM Journal of Numerical Analysis*, Vol. 17, No. 6, 1980, pp. 883–893.

[16] Van Huffel, S., and Vanderwalle, J., *The Total Least Squares Problem: Computational Aspects and Analysis*, SIAM, Philadelphia, 1991.

[17] Laban, M., and Masui, K., "Total Least Squares Estimation of Aerodynamic Model Parameters from Flight Data," *Journal of Aircraft*, Vol. 30, No. 1, 1993, pp. 150–152.

[18] Markovsky, I., and Van Huffel, S., "A Matlab Toolbox for Weighted Total Least Squares Approximation," Catholic Univ. Leuven, Report No. ESAT-SISTA/TR 2004-220, Leuven-Heverlee, Belgium, Nov. 2004.

[19] Davis, T. G., "Total Least-Squares Spiral Curve Fitting," *Journal of Surveying Engineering,* Vol. 125, No. 4, 1999, pp. 159–176.

[20] Pinkelman, J. K., Batill, S. M., and Kehoe, M. W., "Total Least-Squares Criteria in Parameter Identification for Flight Flutter Testing," *Journal of Aircraft*, Vol. 33, No. 4, 1996, pp. 784–792.

[21] Wong, K. Y., and Polak, E., "Identification of Linear Discrete Time Systems Using the Instrumental Variable Method," *IEEE Transactions on Automatic Control*, Vol. AC-12, No. 6, 1967, pp. 707–718.

[22] Van den Boom, A. J. W., "On the Relation Between Weighted Least-Squares Estimators and Instrumental Variable Estimators," *Proceedings of the IFAC 4th Symposium on Identification and System Parameter Estimation*, Sept. 1976, pp. 457–466.

[23] Batterson, J. G., and Klein, V., "Partitioning of Flight Data for Aerodynamic Modeling of Aircraft at High Angles of Attack," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 334–339.

[24] Weiss, S., Friehmelt, H., Plaetschke, E., and Rohlf, D., "X-31A System Identification Using Single Surface Excitation at High Angles of Attack," *Journal of Aircraft*, Vol. 33, No. 3, 1996, pp. 485–490.

[25] Spark, R. S., Zucchini, W., and Coutsourides, D., "On Variable Selection in Multivariate Regression," *Communications in Statistics – Theory and Methods*, Vol. 14, No. 7, 1985, pp. 1569–1587.

[26] Judd, C. M., and McClelland, G. H., *Data Analysis: A Model-Comparison Approach*, Harcourt Brace Jovanovich, San Diego, CA, 1989.

[27] Oswald, W. B., "General Formulas and Charts for the Calculation of Airplane Performance," NACA Report No. 408, 1932.

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CHAPTER 7

Recursive Parameter Estimation

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7.1 INTRODUCTION

In general, motion of a flight vehicle can be represented through well-defined equations of motion, whereby adequate nonlinear models with time-invariant parameters can be postulated and updated to characterize aerodynamic effects. Accordingly, the major applications of aircraft parameter estimation have been in an off-line mode, applying methods discussed in Chapters 4 to 6, which assume availability of data set over a fixed interval of time. Such methods are also called batch mode or postflight analysis techniques, which implicitly assume system parameters constant over the period of observation. All data points are processed together at a time, yielding parameters representing average system behavior. As opposed to batch processing, recursive estimation methods utilize the data point by point as they become available. Such methods are alternatively called as sequential, adaptive, or real-time algorithms [1, 2]. We use the terminology of recursive methods, which is commonly followed in the control applications.

Historically, it appears that C.-F. Gauss, the inventor of the least-squares method (see Sec. 6.1), had also suggested the recursive least-squares (RLS) approach [3]. The RLS method was, however, not widely pursued until 1950, when R. L. Plackett rediscovered the algorithm [4]. This has been followed by a widespread use of RLS and development of other recursive algorithms. The basic ideas of practical engineering applications of recursive parameter estimation (RPE) originate from control applications in chemical and thermal power generating industrial processes, where online adaptation of model parameters is desired to increase the overall plant efficiency. In such applications, we encounter not only changing operating conditions but also several different types of system disturbances. Moreover, the control algorithms are usually based on a simplified representation of a highly complex process, for which exact mathematical representation might require prohibitive effort or might not even be possible. In the case of flight vehicles, the primary motivation for recursive estimation could be to obtain immediate knowledge of aircraft model, which is essential

for designing adaptive control, for fault detection and reconfiguration of control system following damages. These real-time applications necessarily call for recursive estimation methods. During the 1970s, initial attempts were undertaken to apply such techniques in flight. However, because of the limited computational power of onboard computers, this approach did not find widespread use then [5, 6]. During the recent years, the research has once again been focused on development of real-time parameter estimation techniques to address applications in the area of adaptive control, fault tolerant systems, and experiment design [7–9].

To some extent, such methods can also be useful for onboard verification of recorded data and for iterative experiment design. If done onboard, the RPE results from a nonoptimal maneuver could be used to design the subsequent test maneuver and so on throughout the flight test until acceptable results are achieved. This adaptive experiment design and data analysis could provide a higher level of confidence and ensure that sufficient high-quality data are collected for the modeling task before landing. The flight testing in this way would, therefore, increase the efficiency and effectiveness because the complete identification cycle would be carried out in flight in an automated fashion. However, as discussed in Chapter 2, based on the existing a priori knowledge about the flight vehicles and using some simple guidelines, adequate experiments in most of the cases can be designed. Furthermore, near real-time processing capabilities could also serve the intended purposes in some of these cases. With powerful computers being now available, it becomes possible to gather data and apply the classical off-line algorithms implemented on an onboard computer before proceeding to the next maneuver during the flight. Such techniques were applied for safer envelope expansion through real-time comparison of flight tests data with predicted responses [10] and for onboard estimation of aerodynamic characteristics [11–13].

RPE methods are characterized by the following features:

1. They are approximations of the more elaborative nonrecursive methods that we have already studied in the foregoing chapters.
2. By nature, they cater to systems with time-varying parameters.
3. Computer memory requirements are small because storage of past data is not required, and hence they are suitable for online implementation even on a small onboard computer or a chip.

The second attribute, namely, the ability to track time-varying parameters, helps us indirectly in aerodynamic modeling. Estimates evolve over the time, that is, parameter estimates vary in time, even when they may represent a constant effect, which is, as already argued, usually the case for flight vehicles. This leads to accounting for nonlinearities in system model, at least to some extent, although the postulated model might be linear. Likewise, it also helps to account for the changes in the flight conditions, which implies that the dependencies of some of the aerodynamic derivatives on other motion variables like angle of attack or

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Mach number or configuration changes like flap setting are covered automatically by the same linear model.

Although the last two just-mentioned behaviors are in favor of RPE, there are certain limitations associated with these techniques. For example,

1. The convergence of standard RPE methods, like recursive least squares, is slow and might not be adequate for real-time fault detection or to detect sudden changes in the dynamics.
2. Convergence can be improved by incorporating a forgetting factor to discard older data and thereby relying more on recent data. This, however, leads to increased noise sensitivity. Shorter records are necessary for faster adaptation whereas longer records are necessary to distinguish noise. A compromise between the rate of tracking parameter changes and noise sensitivity is necessary. Wrong choice of forgetting factor can result in estimates oscillating around the true values.
3. Uninterruptedly running recursive estimation procedure can face numerical problems because of lack of or limited information content pertaining to dynamic motion. During parts of the flight phases, like steady level flight, the control and motion variables could be below noise level.
4. Verification of data collinearity is not a part of RPE methods. The nearly correlated states and controls affect estimation of stability and control derivatives.

Determination of instrument errors such as biases, scale factors, and time delays is yet another issue that needs some consideration. We will address the techniques of flight-path reconstruction in Chapter 10; if needed, those methods can be implemented in real time to perform flight-path reconstruction [14] to estimate biases and scale factors and to calibrate flow variables. Estimation of time delays using recursive approach is basically feasible, but is usually not considered in the application pertaining to aircraft flight-path reconstruction or parameter estimation from flight data applying such recursive algorithms. Off-line estimation procedures based on the output-error method, which can also be implemented onboard in near real time, can cope up with such problems more easily. As will be demonstrated in Chapter 10, elaborative off-line procedures are more powerful to address such nonlinear estimation problems. In general, the presence of such errors can adversely affect the estimates. Because of this reason, we restrict ourselves here to recursive estimation of aerodynamic parameters only, assuming that measured data are of adequate quality and free of major instrument errors. Furthermore, because of some of the aforementioned limitations, we prefer more accurate and powerful off-line methods unless there is a need of online estimation, for example, in terms of fault detection and online control or reconfiguration. Accordingly, we further restrict ourselves mainly to models adequate to demonstrate the methodology behind recursive estimation methods.

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In this chapter, we start with recursive least-squares (RLS) method, which is a recursive version of the ordinary least-squares method that was studied in Chapter 6. This is followed by a recursive weighted least-squares method based on a forgetting factor to deemphasize older data. We then look at a so-called sequential batch procedure based once again on weighted least squares, but incorporating a check to judge the information content in the new incoming data before discarding the old one to include the new one. This is followed by yet another method called Fourier transform regression (FTR) based on the least-squares approach, but now implemented in the frequency domain using the discrete Fourier transform. We then turn our attention to a different class of techniques based on filtering approach, which is widely used for state estimation in the presence of process and measurement noise. Augmenting the basic system states with system parameters and applying a nonlinear filter to the augmented states allows recursive estimation of parameters. Two nonlinear filter algorithms are discussed. The first one is the commonly applied extended Kalman filter (EKF) based on linearization of system dynamics, and the second one is a more recently introduced nonlinear filter, called unscented Kalman filter (UKF) having better performance than that of the EKF. Two different implementations of UKF are presented. This is followed by a state estimator without requiring the knowledge of the noise covariances, called extended forgetting factor recursive least squares (EFRLS) [15], which is an extension of the RLS to linear dynamic systems.

Finally, we test these recursive algorithms in an off-line environment on flight data pertaining to the longitudinal and lateral-directional modes of research aircraft VFW-614 ATTAS and HFB-320. The RLS results are compared with LS method discussed in Chapter 6. For state-space models, the results from the standard iterative output-error method (OEM) discussed in Chapter 4 or the filter-error method (FEM) elaborated in Chapter 5 form the basis for evaluating the performance of these recursive methods. Their performance in terms of parameter convergence, robustness in the presence of noise, and the computational effort is evaluated [16]. As in the case of earlier chapters, the flight data and the software are appended to this book.

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7.2 LEAST-SQUARES BASED RECURSIVE METHODS

There are many variants of recursive algorithms based on the least-squares principle. First, we consider the ordinary least-squares method to show how an off-line algorithm can be modified to a recursive computational form and to bring out the fact that such methods are approximations of the more elaborative nonrecursive methods. In general, following a similar approach any off-line method can be converted into an equivalent recursive form. For example, the output-error method based on the maximum likelihood principle, discussed in Chapter 4, can be translated into an equivalent recursive maximum likelihood (RML) method [14]. Because of certain limitations of RLS, which we will discuss in our treatment

on the subject, more commonly applied algorithms are based on the use of a forgetting factor to deemphasize the older data. Such methods have a better adaptation (convergence) rate and allow estimating parameters in time-varying systems.

7.2.1 RECURSIVE LEAST SQUARES

From Sec. 6.2, we recall that given a set of k samples of independent variables $x = (x_1\ x_2 \ldots x_{nq})^T$ and of dependent variable y, the least-squares estimates of θ are obtained as

$$\begin{aligned}\hat{\theta}(k) &= (X_k^T X_k)^{-1} X_k^T Y_k \\ &= P(k) X_k^T Y_k \end{aligned} \tag{7.1}$$

where $X_k = [x(1)\ x(2)\ \ldots\ x(k)]^T$, $Y_k = [y(1)\ y(2)\ \ldots\ y(k)]^T$, and the inverse of the information matrix $(X_k^T X_k)^{-1} = P(k)$. We note the change from N for the total number of data points used in Sec. 6.2 for off-line algorithm to k in this chapter because we wish to derive an algorithm to process data sample by sample as they become available.

Now, if we assume that one more sample point denoted by $(x(k+1), y(k+1))$ becomes available, then similar to Eq. (7.1), the off-line least-squares estimates from the complete set of 1 to $(k+1)$ data points will be given by

$$\hat{\theta}(k+1) = P(k+1) X_{k+1}^T Y_{k+1} \tag{7.2}$$

where $X_{k+1} = [X_k;\ x^T(k+1)]$ and $Y_{k+1} = [Y_K;\ y(k+1)]$. Equation (7.2) can be rewritten as

$$\hat{\theta}(k+1) = P(k+1)[X_k^T Y_k + x(k+1) y(k+1)] \tag{7.3}$$

Using $X_{k+1} = [X_k;\ x^T(k+1)]$ and through simple matrix multiplications, we obtain $P(k+1)$ appearing in Eq. (7.3) in terms of $P(k)$:

$$P(k+1) = [X_{k+1}^T X_{k+1}]^{-1} = [P^{-1}(k) + x(k+1) x^T(k+1)]^{-1} \tag{7.4}$$

Now, applying the so-called matrix inversion lemma, namely, for matrices A, B, C, and D of compatible dimensions,

$$[A + BCD]^{-1} = A^{-1} - A^{-1} B [D A^{-1} B + C^{-1}]^{-1} D A^{-1} \tag{7.5}$$

and substituting $A = P^{-1}(k)$, $B = x(k+1)$, $C = I$, and $D = x^T(k+1)$, Eq. (7.4) leads to

$$P(k+1) = P(k) - P(k) x(k+1) [x^T(k+1) P(k) x(k+1) + I]^{-1} x^T(k+1) P(k) \tag{7.6}$$

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which can be expressed in terms of the gain matrix K as

$$P(k+1) = P(k) - K(k+1)x^T(k+1)P(k) \tag{7.7}$$

where the gain matrix K is defined as

$$K(k+1) = P(k)x(k+1)[x^T(k+1)P(k)x(k+1) + I]^{-1} \tag{7.8}$$

Now, substituting Eq. (7.7) in Eq. (7.3), we derive the recursive formula for updating the parameters θ:

$$\begin{aligned}\hat{\theta}(k+1) &= P(k+1)[X_k^T Y_k + x(k+1)y(k+1)] \\ &= [P(k) - K(k+1)x^T(k+1)P(k)][X_k^T Y_k + x(k+1)y(k+1)] \\ &= P(k)X_k^T Y_k + P(k)x(k+1)y(k+1) - K(k+1)x^T(k+1)P(k)X_k^T Y_k \\ &\quad - K(k+1)x^T(k+1)P(k)x(k+1)y(k+1)\end{aligned} \tag{7.9}$$

But, from Eq. (7.1) we know that $P(k)X_k^T Y_k = \hat{\theta}(k)$. Accordingly, Eq. (7.9) can be further simplified as

$$\begin{aligned}\hat{\theta}(k+1) = \hat{\theta}(k) &+ P(k)x(k+1)y(k+1) - K(k+1)x^T(k+1)\hat{\theta}(k) \\ &- K(k+1)x^T(k+1)P(k)x(k+1)y(k+1)\end{aligned} \tag{7.10}$$

Postmultiplying Eq. (7.8) with $[x^T(k+1)P(k)x(k+1)+1]$ and substituting the result for $P(k)\ x(k+1)$ in the second term of Eq. (7.10) and minor sorting of the terms lead to

$$\begin{aligned}\hat{\theta}(k+1) &= \hat{\theta}(k) - K(k+1)x^T(k+1)\hat{\theta}(k) + P(k)\,x(k+1)\,y(k+1) \\ &\quad + \{K(k+1)x^T(k+1)P(k)\ x(k+1) + K(k+1) \\ &\quad -K(k+1)x^T(k+1)P(k)\ x(k+1)\}y(k+1) \\ &= \hat{\theta}(k) - K(k+1)x^T(k+1)\hat{\theta}(k) + K(k+1)y(k+1) \\ &= \hat{\theta}(k) + K(k+1)[y(k+1) - x^T(k+1)\hat{\theta}(k)]\end{aligned} \tag{7.11}$$

Thus, we have now all of the information necessary to implement the recursive least-squares method. To facilitate easy referencing in the text, the RLS algorithm consisting of Eqs. (7.11), (7.8), and (7.7) is summarized here:

$$\hat{\theta}(k+1) = \hat{\theta}(k) + K(k+1)[y(k+1) - x^T(k+1)\hat{\theta}(k)] \tag{7.12}$$

$$K(k+1) = \frac{P(k)\,x(k+1)}{1 + x^T(k+1)P(k)\,x(k+1)} \tag{7.13}$$

$$P(k+1) = P(k) - K(k+1)x^T(k+1)P(k) \tag{7.14}$$

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where we have reformulated Eq. (7.8) into (7.13) because the last term in the square bracket on the right-hand side of Eq. (7.8) is a scalar and amounts to simple division and not a matrix inversion. Thus, we note that the matrix inversion required in the off-line algorithm has been replaced through scalar division at each step in the recursive version. Thus, the RLS algorithm consists of simple matrix multiplications and additions whereas the one-shot off-line LS procedure involves matrix inversion. To start the algorithm, initial values need to be specified for parameters θ and matrix P. In the absence of any a priori knowledge, we can start with the zero initial values for the parameters; this is similar to the one-shot LS method. The initial P can be assumed to be diagonal matrix with large values, indicating large variances.

It may be recalled from Sec. 6.2.1 that the estimate of the expected variance in the least-squares estimates of n_q parameters from N data points is given by

$$\hat{\sigma}^2 = \frac{1}{N - n_q} \sum_{k=1}^{N} [y(k) - X(k)\hat{\theta}]^2 \tag{7.15}$$

In the present case, as the estimates evolve over the time with number of data points processed k increasing continuously, the estimate of σ tends to be highly inaccurate initially because the nonconverged estimates will be used to estimate covariance matrix. This does not affect the estimates themselves, which are given by Eq. (7.12), but affects only the computation of σ in (7.15) for k data points. This is one of the limitations of the recursive least-squares approach. To obtain an estimate of the accuracy of the estimates, we usually apply Eq. (7.15) once the estimates are converged or when the recursive procedure is terminated.

7.2.2 RECURSIVE WEIGHTED LEAST SQUARES

The RLS algorithm just described is the simplest of the various recursive procedures. However, in practice the convergence of RLS is generally poor. To allow RLS to adapt rapidly with time, the past information must be quickly discarded. This is often accomplished using a forgetting factor λ, which introduces an exponentially decaying weights on the past measurements [1, 2]. Weighting functions for different values of λ are computed using the function "/FVSysID2/chapter07/weightFun_Lamda.m" and plotted in Fig. 7.1. It is observed that smaller values of λ tend to neglect more and more data points in the farther past. The cost function minimized in this case is given by

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{k} \lambda^{k-i} \varepsilon^2(i) \tag{7.16}$$

Compared to Eq. (6.37), the exponential weighting structure is necessary to arrive at a recursive version. Without going into the exact derivation, which is very

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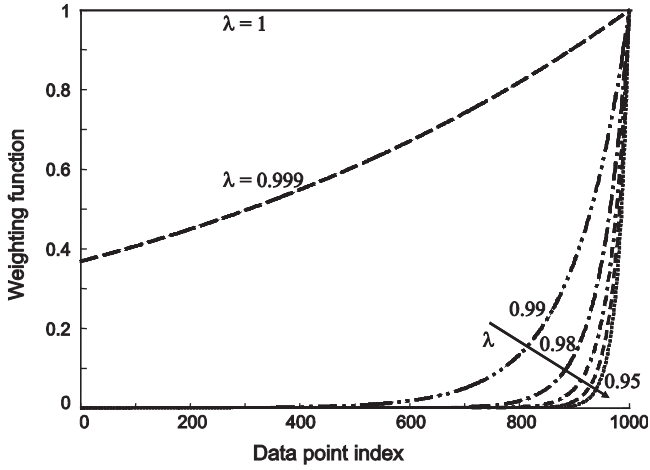

Fig. 7.1 Weighting functions for different forgetting factors.

similar to that demonstrated in Sec. 7.2.1 for RLS, we simply state here the final algorithmic steps [1].

The RLS with a forgetting factor λ consists of the following computational steps:

$$\hat{\theta}(k+1) = \hat{\theta}(k) + K(k+1)[y(k+1) - x^T(k+1)\hat{\theta}(k)] \tag{7.17}$$

$$K(k+1) = \frac{P(k)x(k+1)}{\lambda + x^T(k+1)P(k)x(k+1)} \tag{7.18}$$

$$P(k+1) = \frac{1}{\lambda}\{P(k) - K(k+1)x^T(k+1)P(k)\} \tag{7.19}$$

We can easily deduce that the standard RLS is a special case of weighted least squares with forgetting factor $\lambda = 1$. Further discussion on the effect of λ and guidelines to choose the same is deferred to Sec. 7.3.3.

7.2.3 LOCALLY WEIGHTED REGRESSION METHOD

Yet another weighted least-squares approach is to minimize Eq. (6.37) yielding estimates given by Eq. (6.40) by considering step-by-step sets of data, that is, N, of limited size. Because online estimation problems may impose constraints of storing large size data matrices, the procedure can be implemented for a fixed number of data points and repeated each time after collecting these data points. It leads to a sequential batch approach. It is also possible to define a weighting matrix in an exponential form to place more importance on data points closer

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to the current point than on the points in the past, which is the same principle of forgetting factor. Such a diagonal weighting matrix is given by [9]

$$W_{ii} = e^{-(d_i/2\kappa^2)} \tag{7.20}$$

where d_i is the norm of the difference between the current point and ith one and κ is the time-varying "Gaussian window width," which can be varied to change to effect data points in the past. The smaller the κ, the smaller is the set of data points weighted in the cost function, which implies that more emphasis will be given to current data points and vice versa. For the weighting matrix defined in Eq. (7.20), through proper substitution of W according to Eq. (7.20) in Eq. (6.40), it can be easily verified that the so-called locally weighted regression (LWR) estimates are given by [9, 17, 18]

$$\hat{\theta} = (X^T W^2 X)^{-1} X^T W^2 Y \tag{7.21}$$

The simplest approach to implement the least-squares estimation in a batch-sequential mode, with suitably defined weighting matrix, is to simply collect data for a predefined length and obtain the estimates by solving for example Eq. (7.21). Each time the data matrix is rebuilt completely anew without checking for the information content of the previous data matrix. This is a very inefficient approach because we may blindly discard useful information, for example, that corresponding to dynamic maneuver, which is really essential for estimation. A slightly improved strategy, compared to complete rebuild of the data matrix, would be to delete sequentially the oldest data from the data matrix WX. In such a case, we might once gain delete rows whose information content is large. Therefore, a much better strategy would be to check the rows of the data matrix for their relative importance and selectively remove that which contributes less to be replaced through the new data point. The deleted row could be located anywhere in the data being processed. This is possible because, as already pointed in Sec. 6.7, the least-squares estimation does not depend on the temporal relation between the data points.

One criterion to judge the quality of the incoming data will be to evaluate the rows of data matrix WX and delete that which contributes least (and even lesser than the new data) to the trace of the inverse of the data information matrix, namely, $(X^T W^2 X)^{-1}$, which is the first term on the left-hand side of Eq. (7.21). Let us denote WX as follows [9]:

$$WX = \begin{bmatrix} Z \\ H \end{bmatrix} \tag{7.22}$$

where H is the row to be deleted. Then, we have

$$(X^T W^2 X)^{-1} = \left(\begin{bmatrix} Z^T & H^T \end{bmatrix} \begin{bmatrix} Z \\ H \end{bmatrix} \right)^{-1} = (Z^T Z + H^T H)^{-1} \tag{7.23}$$

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Now, denoting $P = Z^T Z + H^T H$ and $R = Z^T Z$ implies $P = R + H^T H$ or, in other words, $R = P - H^T H$. Now applying the matrix inversion lemma of Eq. (7.5), we get

$$R^{-1} = [P - H^T H]^{-1} = P^{-1} - P^{-1} H^T (HP^{-1} H^T - I)^{-1} H\ P^{-1} \tag{7.24}$$

Taking trace (that is, sum of the diagonal elements, which is also sum of the eigenvalues) of Eq. (7.24) leads to

$$tr(R^{-1}) = tr(P^{-1}) - tr[P^{-1} H^T (HP^{-1} H^T - I)^{-1} H\ P^{-1}] \tag{7.25}$$

Because $HP^{-1}H^T$ is a scalar, we can write Eq. (7.25) as

$$tr[P^{-1} H^T (HP^{-1} H^T - I)^{-1} H\ P^{-1}] = \frac{tr[P^{-1} H^T H\ P^{-1}]}{HP^{-1} H^T - I} \tag{7.26}$$

Now, using the matrix algebra rule, namely, if A and B are matrices such that AB is a square matrix, then $tr(AB) = tr(BA)$; we can rewrite Eq. (7.26) as

$$tr[P^{-1} H^T (HP^{-1} H^T - I)^{-1} HP^{-1}] = \frac{H\ P^{-1}\ P^{-1} H^T}{HP^{-1} H^T - I} = \frac{H(P^2)^{-1} H^T}{HP^{-1} H^T - I} \tag{7.27}$$

Note that trace is not required on the right-hand side of Eq. (7.27) because, as already mentioned, $HP^{-1}H^T$ is a scalar. Now, denoting the singular value decomposition of a real, symmetric, positive-definite matrix $P = USV^T$, it follows that $P^2 = P^T P = VS^2 V^T$ and $P^{-2} = VS^{-2} V^T$, which reduces Eq. (7.27) to

$$tr[P^{-1} H^T (HP^{-1} H^T - I)^{-1}\ H\ P^{-1}] = \frac{HVS^{-2} V^T H^T}{HVS^{-1} V^T H^T - I} \tag{7.28}$$

The problem now reduces to finding the row H among all of the rows of WX such that the trace in Eq. (7.28) is maximized. Denoting $C = HV$ and making use of the fact that S is diagonal, we can rewrite the right-hand side of Eq. (7.28) as follows:

$$tr[P^{-1} H^T (HP^{-1} H^T - I)^{-1}\ H\ P^{-1}] = \frac{\sum_{j=1}^{n_q} (c_{ij}^2 / s_{jj}^2)}{\sum_{j=1}^{n_q} (c_{ij}^2 / s_{jj}) - 1} \equiv F(i) \tag{7.29}$$

Thus, it now remains to find the index of the row having maximum value, that is, max $F(i)$, which is straightforward and needs no further explanation. Although the procedure has been elaborated considering H to be a row vector, it can be extended to a more general case of multiple rows.

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The preceding elaborated procedure is computationally little involved because we need to compute the SVD twice, once of matrix P for retention and deletion scheme based on Eq. (7.29) and once to compute the estimates using Eq. (7.21). The number of data points used in Eq. (7.21) for sequential batch update and subsequent test for refreshing the same will influence the computational time; 50 samples at every 10 time steps have been used in practice [9]. The preceding procedure is appealing because it leads to inclusion of the new data only when it leads to improved information compared to that which is already available in data set of fixed length size; in turn, it results in reduction of the variances of the estimated parameters. The approach also automatically retains the best portions of data, which are processed in the past, and, loosely speaking, helps to filter out the noise. In other words, the converged estimates, obtained from a dynamic maneuver performed for system identification purposes, do not tend to diverge in spite of the long phases of the maneuver in which there is not much dynamic motion. Nevertheless, if recursive estimation runs over a considerable length of time, it might be become necessary to reinitialize the data matrix or introduce a forgetting factor to deemphasize older data. Otherwise, the data matrix might contain information that is too far in the past, and this can affect the ability to track the changes in the system behavior.

7.2.4 FOURIER TRANSFORM REGRESSION

In the earlier chapters and preceding sections of this chapter, we have been dealing with methods, both off-line and recursive, in the time domain. In this section we slightly deviate from our set goal to address a particular method in the frequency domain, not only because it is based on the equation error approach we had studied in Chapter 6, but also because it appears to be quite promising and advantageous compared to other recursive algorithms [8, 16, 19, 20]. The basis for this Fourier transform regression (FTR) method is the discrete Fourier transform of the time domain data into the frequency domain. This is carried out using sampled time-domain data and a simple Euler approximation of the finite Fourier integral. In the following theoretical development, we draw on the work of [8].

The Fourier transform of an arbitrary signal $x(t)$ is given by

$$\tilde{x}(\omega) = \int_0^T x(t) e^{-j\omega t} \mathrm{d}t \tag{7.30}$$

where $\sim$ denotes the discrete Fourier transform of the corresponding variable. The simple Euler approximation to the preceding Fourier transforms yields

$$\tilde{x}(\omega) \approx \Delta t \sum_{k=0}^{N-1} x_k e^{-j\omega k \Delta t} \tag{7.31}$$

where k indicates the discrete time index, Δt the sampling interval, and N the total number of data points. Denoting the summation on the right-hand side of Eq. (7.31) as the discrete Fourier transform $\widetilde{X}(\omega)$, that is,

$$\widetilde{X}(\omega) = \sum_{k=0}^{N-1} x_k \, e^{-j\omega k \Delta t} \tag{7.32}$$

we can rewrite Eq. (7.31) as follows:

$$\tilde{x}(\omega) \approx \widetilde{X}(\omega) \, \Delta t \tag{7.33}$$

Equation (7.32) is a first-order Euler approximation of the finite Fourier transform of Eq. (7.30), which is valid for a small sampling time Δt. If this is not true, then some corrections will be necessary to reduce the errors [21]. From Sec. 3.8, we recall here our discussion on the Euler integration method and its limitations.

Now, applying the Fourier transform to the linear model of Eqs. (3.24) and (3.25) and neglecting the process noise $w(t)$, we obtain an equivalent system represented in the frequency domain as

$$j\omega\tilde{x}(\omega) = A\tilde{x}(\omega) + B\tilde{u}(\omega) \tag{7.34}$$

$$\tilde{y}(\omega) = \tilde{x}(\omega) \tag{7.35}$$

where we have considered only the states as the observation variables. In contrast to Eqs. (3.27) and (3.28), we have omitted here the initial conditions and measurement biases for the reasons that we will discuss shortly.

Now, assuming that the measurements of the state, output, and input variables are available, we can make use of the equation error formulation discussed in Chapter 6 to estimate parameters appearing in the matrices A and B appearing in Eq. (7.34). Accordingly, similar to Eq. (6.33) in the time domain, we can write the least-squares cost function in the frequency domain for the kth state equation as [22]

$$J_k = \frac{1}{2} \sum_{n=1}^{m} |\, j\omega_n \tilde{x}_k(n) - A_k \tilde{x}(n) - B_k \tilde{u}(n)|^2 \tag{7.36}$$

where m is the number of frequencies [instead of N the data points in Eq. (6.33)], A_k and B_k the kth rows of matrices A and B respectively, and $\tilde{x}_k(n)$ the kth element of vector $\tilde{x}$ for frequency ω_n. Symbols $\tilde{x}(n)$ and $\tilde{u}(n)$ denote the Fourier transform of the state and control vectors for frequency ω_n. Similar cost expressions can be written down for other output equations. The number of frequencies m over which the cost function is evaluated depends upon the range of interest, which is usually selected as evenly spaced between $\omega_{\min}$ and $\omega_{\max}$. For a rigid-body aerodynamic model, it is usually sufficient to consider frequencies from 0.01 to 1.5 Hz. Furthermore, because we are mainly interested in the stability and control

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derivatives, we can neglect the zero frequency, which corresponds to the trim and measurement biases. Thus, excluding the zero frequency in the cost function eliminates the need to estimate initial conditions and measurement biases. This is one of the advantages of the frequency-domain approach.

In matrix notation of Eq. (6.3), the standard regression problem with complex data can be formulated as

$$\tilde{Y} = \tilde{X}\Theta + \tilde{\varepsilon} \tag{7.37}$$

where $\tilde{\varepsilon}$ represents the complex equation error in the frequency domain, Θ the vector of unknown model parameters appearing in matrices A_k and B_k, and the matrix of independent variables and vector of dependent variable are given by

$$\tilde{X} = \begin{bmatrix} \tilde{x}^T(1) & \tilde{u}^T(1) \\ \tilde{x}^T(2) & \tilde{u}^T(2) \\ & \vdots \\ \tilde{x}^T(m) & \tilde{u}^T(m) \end{bmatrix} \quad \text{and} \quad \tilde{Y} = \begin{bmatrix} j\omega_1\tilde{x}_k(1) \\ j\omega_2\tilde{x}_k(2) \\ \vdots \\ j\omega_m\tilde{x}_k(m) \end{bmatrix} \tag{7.38}$$

Similar to Eq. (6.8) in the time domain approach, in the present case we can write the least-squares cost function defined in Eq. (7.36) in terms of $\tilde{X}$ and $\tilde{Y}$ as

$$J = \frac{1}{2}(\tilde{Y} - \tilde{X}\Theta)^*(\tilde{Y} - \tilde{X}\Theta) \tag{7.39}$$

where the superscript * denotes the complex conjugate transpose. Following the similar procedure as in Sec. 6.2, minimization of the cost function of Eq. (7.39) yields [22]

$$\hat{\Theta} = [\mathrm{Re}(\tilde{X}^*\,\tilde{X})]^{-1}\,\mathrm{Re}(\tilde{X}^*\,\tilde{Y}) \tag{7.40}$$

The statistical properties of the least-squares estimates given by Eq. (7.40) are similar to those discussed in Sec. 6.2.1. The standard deviations of the estimates are given by the square root of the diagonal elements of the parameter error covariance matrix $P = E\{(\hat{\Theta} - \Theta)(\hat{\Theta} - \Theta)^T\}$, which is given by

$$P = \mathrm{cov}(\hat{\Theta}) = \sigma^2\,[\mathrm{Re}(\tilde{X}^*\,\tilde{X})]^{-1} \tag{7.41}$$

where the equation error covariance σ^2 can be estimated from the residuals

$$\sigma^2 = \frac{1}{(m - n_q)}[(\tilde{Y} - \tilde{X}\hat{\Theta})^*(\tilde{Y} - \tilde{X}\hat{\Theta})] \tag{7.42}$$

and n_q is the number of unknown parameters Θ.

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Equation (7.40) provides a one-shot solution to the least-squares problem for data transformed in the frequency domain. Because we are interested in a recursive version of the algorithm, we need to arrive at a procedure to update point by point the discrete Fourier transform defined in Eq. (7.32). We know that at any discrete time point k, the discrete Fourier transform $\widetilde{X}_k(\omega)$ is given by

$$\widetilde{X}_k(\omega) = \sum_{i=0}^{k-1} x_i\, e^{-j\omega i\Delta t} \tag{7.43}$$

We specifically note that $\widetilde{X}(\omega)$ has been appended on the left-hand side with subscript k to denote the variables corresponding to those up to the kth point. Likewise, we also note that the upper index of summation N in Eq. (7.32) has been replaced through k for the same reason, and as a consequence the running index is changed to i. Now, if we assume that one more sample point becomes available, then similar to Eq. (7.43), the discrete Fourier transform $\widetilde{X}_{k+1}(\omega)$ at the $(k+1)$ data points will be given by

$$\widetilde{X}_{k+1}(\omega) = \sum_{i=0}^{k} x_i\; e^{-j\omega i\Delta t} \tag{7.44}$$

which can be rewritten by taking out the last point from the summation and then through substitution of Eq. (7.43) as

$$\begin{aligned}\widetilde{X}_{k+1}(\omega) &= \sum_{i=0}^{k-1} x_i\, e^{-j\omega i\Delta t} + x_{k+1}\, e^{-j\omega(k+1)\Delta t} \\ &= \widetilde{X}_k(\omega) + x_{k+1} e^{-j\omega(k+1)\Delta t}\end{aligned} \tag{7.45}$$

Let us take a look at the computational aspects of implementing Eq. (7.45), which are of particular interest to us. For this purpose, we further expand on the exponential term appearing in the right-hand side of Eq. (7.45) as

$$e^{-j\omega(k+1)\Delta t} = e^{-j\omega k\Delta t} e^{-j\omega\Delta t} \tag{7.46}$$

For a given sampling time Δt and frequency ω, the second term on the right-hand side of Eq. (7.46) is a constant, which can be computed once outside the recursive loop and stored. Thus, it is fairly straightforward to note the update of discrete Fourier transform at any time point needs just one addition in Eq. (7.45) and two multiplications: one in Eq. (7.46) and one in Eq. (7.45). Thus, the computational requirements are marginal for computing the Fourier transforms, but this is just one part of the overall algorithm.

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Equation (7.45) provides a simple recursive formula for updating the discrete Fourier transform at any time point. Once we have obtained the updated discrete Fourier transform $\widetilde{X}_{k+1}(\omega)$, the rest of the procedure is the same as that presented in Eqs. (7.38–7.40), which is implemented at each time point. The preceding FTR approach, although recursive, is slightly different than that of RLS presented in Sec. 7.2.1. Both of the methods do not require storage of time histories in the past. However, computationally the time-domain approach of RLS given by Eqs. (7.12–7.14) involves a simple division by a scalar and matrix multiplications whereas the frequency-domain approach of FTR needs an inverse of a matrix or equivalently solving linear algebraic equation by a suitable procedure such as Cholesky factorization or singular value decomposition. This is the part of the algorithm that requires the most computational effort. To speed up the computations, the FTR update can be done at some predefined interval by skipping points, instead of updating the estimates at each discrete time point. The number of points that can be skipped without adversely affecting the results depends upon the sampling frequency and the frequencies of interests included in the Fourier transform. At each update point, the information up to the preceding sample, including that of the skipped points, is automatically accounted for by the fixed term appearing in Eq. (7.46), that is, as already mentioned earlier, computed ahead of the recursive loop.

Besides simplicity and fairly simple computations, the FTR method is associated with a few other advantages: 1) no tuning parameters are required to be specified and 2) by omitting the zero frequency, the trim conditions and measurement biases are eliminated. Like any other least-squares approach, no starting values of parameters are necessary to start the algorithm. Furthermore, the information from the preceding maneuvers or flights can be used in the current recursive estimation without any further overheads by continuing the recursive calculation of the Fourier transform, thereby improving accuracy of the estimates. The approach is, however, based on the assumption that the measurements of the state variables are available and are error free; we had discussed both of these aspects in connection with the least-squares technique in Chapter 6. The method is limited to linear models only. It is also plagued by some of the common problems encountered in the recursive least-squares procedure, namely, deemphasizing of data in the distant past though the forgetting factor is necessary to enable tracking of rapid changes in the dynamics, and also that some dynamic excitation is necessary resulting in motion having amplitudes larger than the noise level. This is particularly critical in the fault detection applications. With regard to the last aspect, the FTR is found to perform better compared to other methods presented in this section because by restricting the range of frequencies to be evaluated within a limited band of $\omega_{\min}$ to $\omega_{\max}$, the algorithm filters out automatically the noise outside the range of interest.

The last aspect of FTR algorithm that needs some consideration pertains to the treatment of trim values, which basically correspond to the zero frequency. We

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had argued earlier that by omitting the zero frequency in the evaluation we can reduce the number of parameters to be estimated because the initial conditions and bias terms can be neglected. The system equations (7.34) and (7.35) then truly involve perturbed quantities. We recall here our discussion in Sec. 3.5.2 in terms of the lumped bias parameters in which the nonzero initial conditions were accounted for indirectly through lumped biases. In the present case, this is not possible, and, as a consequence to conform to perturbed equations, it is necessary to remove the trim values from the measurements of the motion variables as well as controls before using them in the algorithm. Failure to do so leads to erratic behavior of the FTR algorithm, resulting from disproportionately large component at zero frequency. There are different procedures that are possible to remove the trim values; for example, we can simply subtract the value of the first data point or reduce noise effects from the average of the first few data points, from the measurements before starting the recursion, or by using a simple high-pass filter. The other repercussion of neglecting the zero frequency is that the FTR procedure allows estimation of the dynamic derivatives only and that we cannot estimate the zero aerodynamic terms. Both of these limitation are, in general, of minor consequence, particularly for the end goal of real-time estimation.

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7.3 FILTERING METHODS

Filtering techniques belong to a class of stochastic methods and are primarily used for state estimation in the presence of noise. They use the measured outputs y and inputs u, together with the postulated system model and assumed process and measurement noise statistics, to compute the states and their error covariances. Fundamental to all such methods is a two-step procedure consisting of prediction, also called time update, followed by correction, also called measurement update. The procedure developed by Kalman provides an optimal linear state estimator, which operates point by point and hence is ideally suited for recursive implementation [23]. Derivation of the Kalman filter is provided in Appendix F. Here, we consider two extensions of the Kalman filter, which allow state estimation in nonlinear systems. There are many different implementations possible for the Kalman filter and of the extensions leading to numerically more efficient realizations. We restrict ourselves to standard techniques that provide a basic understanding of these methods.

Estimation of parameters applying the filtering approach is an indirect procedure, which needs transformation of the parameter estimation problem into a state estimation problem. We do this by artificially defining the unknown parameters as additional state variables. Considering the system parameter vector Θ as the output of an auxiliary dynamic system

$$\dot{\Theta} = 0 \tag{7.47}$$

and by defining an augmented state vector

$$x_a = [x^T \quad \Theta^T]^T \tag{7.48}$$

we can represent the extended system as

$$\dot{x}_a(t) = \begin{bmatrix} f[x_a(t),\ u(t)] \\ 0 \end{bmatrix} + \left[\begin{array}{c|c} F & 0 \\ \hline 0 & 0 \end{array}\right] \begin{bmatrix} w(t) \\ 0 \end{bmatrix}$$

$$= f_a[x_a(t),\ u(t)] + F_a\, w_a(t) \tag{7.49}$$

$$y(t) = g_a[x_a(t),\ u(t)] \tag{7.50}$$

$$z(k) = y(k) + G\, v(k) \tag{7.51}$$

where the subscript a denotes the augmented variables, f_a and g_a are system functions of the state and observation equations of the augmented systems, and F_a is the corresponding process noise matrix. As in the preceding cases covered in Sec. 5.2, the process noise w and measurement noise v are assumed to be zero mean, uncorrelated, and mutually independent; F and G are the corresponding noise distribution matrices. We also note here that the extended model will be nonlinear in the extended states x_a, involving products of the states x and parameters Θ, even though the basic system model might have been linear in x. Thus, irrespective of the type of model, parameter estimation using the filtering approach necessarily leads to a nonlinear filtering problem.

7.3.1 EXTENDED KALMAN FILTER

The extended Kalman filter (EKF) gives an approximate solution to the nonlinear state estimation problem, also called as nonlinear filtering problem. It also provides a solution to the combined state and parameter estimation problem [24] and has been used in the past to estimate aerodynamic derivatives [25–30]. It can also be used for the specific case of unstable aircraft identification [31–33], which we will address in Chapter 9. The EKF involves linearization of the postulated nonlinear model about some suitable point and the use of the Kalman filter. Appendix F provides the basic procedural details of the EKF. In general, the nonlinearities in the model affect the EKF performance. Therefore, to retain the system nonlinearities in the state estimation as far as possible, we use a mixed version incorporating state prediction with a nonlinear model and covariance propagation based on a first-order approximation of the system dynamics. As already pointed out in Chapter 5, such a mixed nonlinear filter is found to work satisfactorily in practical cases, including those of flight-path reconstruction and aerodynamic model identification.

We can apply the EKF computational procedure elaborated in Appendix F to the extended system defined in Eqs. (7.49–7.51) to obtain the estimates of the

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augmented states x_a, which contain the unknown system parameters. The two steps of the EKF are given by the following:

Prediction (time update) step:

$$\tilde{x}_a(k+1) = \hat{x}_a(k) + \int_{t_k}^{t_{k+1}} f[\hat{x}_a(t), \bar{u}(k)]\,\mathrm{d}t \tag{7.52}$$

$$\tilde{P}_a(k+1) \approx \Phi_a(k+1)\hat{P}_a(k)\Phi_a^T(k+1) + \Delta t F_a F_a^T \tag{7.53}$$

with initial conditions

$$\hat{x}_a(1) = x_{a0}, \quad \hat{P}_a(1) = P_{a0} \tag{7.54}$$

and

$$\Phi_a(k+1) = e^{A_a(k)\Delta t} \approx \mathrm{I} + A_a(k)\,\Delta t + A_a^2(k)\,\frac{\Delta t^2}{2!} + \cdots \tag{7.55}$$

$$A_a(k) = \left.\frac{\partial f_a[x_a(t), u(t)]}{\partial x_a}\right|_{x_a=\hat{x}_a(k)} = \left[\begin{array}{c|c} \partial f/\partial x & \partial f/\partial \Theta \\ \hline 0 & 0 \end{array}\right]_{x_a=\hat{x}_a(k)} \tag{7.56}$$

Correction (measurement update) step:

$$\tilde{y}(k) = g_a[\tilde{x}_a(k), u(k)] \tag{7.57}$$

$$K_a(k) = \tilde{P}_a(k)C_a^T[C_a\tilde{P}_a(k)C_a^T + R]^{-1} \tag{7.58}$$

$$\hat{x}_a(k) = \tilde{x}_a(k) + K_a(k)[z(k) - \tilde{y}(k)] \tag{7.59}$$

$$\hat{P}_a(k) = [I - K_a(k)C_a]\tilde{P}_a(k)[I - K_a(k)C_a]^T + K_a(k)RK_a^T(k) \tag{7.60}$$

where

$$C_a(k) = \left.\frac{\partial g_a[x_a(t), u(t)]}{\partial x_a}\right|_{x_a=\tilde{x}_a(k)} = \left[\begin{array}{c|c} \dfrac{\partial g}{\partial x} & \dfrac{\partial g}{\partial \Theta} \end{array}\right]_{x_a=\tilde{x}_a(k)} \tag{7.61}$$

where $\bar{u}$ denotes the average or interpolated values of the inputs between the time points k and $k+1$, $\sim$ the predicted variables, $\hat{}$ the updated variables, K_a the Kalman gain matrix, and $R = GG^T$ the measurement noise covariance matrix.

Depending upon the number of unknown parameters, the matrices A_a and C_a can become much larger than those for the original system. From Eq. (7.56) we observe that many elements of the augmented matrix A_a are zero; this fact can be used while programming the computation of state transition matrix Φ_a

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defined in Eq. (7.55). Because these computations are performed at every data point, considerable savings in the computational time can be achieved, particularly when the number of unknown parameters far exceeds the number of system states. For propagation of the states across the two discrete time points according to Eq. (7.52), we use one of the Runge–Kutta methods discussed in Sec. 3.8.1.

The first-order approximation of the covariance propagation in Eq. (7.53) is valid for small sampling time, which is usually the case we encounter while dealing with flight data analysis. The initial conditions x_{a0} are required to be specified. It is also required to specify the initial covariance matrix $P_a(0)$ for the augmented state vector. The initial values for the covariances corresponding to the auxiliary state variables, that is, corresponding to the unknown parameters, should reflect the confidence in the starting values of the parameters Θ. In the absence of any a priori knowledge, it is common to assume fairly high values for the initial covariance matrix $P_a(0)$, indicating lesser confidence. Such a choice also aids in better tracking of values that can deviate much from the initial guess values. The covariance matrix P_a provides the information about the accuracy of the augmented states, which include the unknown parameters. Standard deviations of the parameter estimates are readily obtained as square root of the corresponding diagonal elements of P_a.

A priori specification of the process and measurement noise covariance matrices $F_a F_a^T (=Q)$ and $GG^T (=R)$ is necessary to use the EKF as a parameter estimator. The problem of estimating the unknown noise statistics for a known system dynamics is often referred to in the literature as "adaptive filtering." Several approaches have been adopted in practice, one of them being developed by Myers and Tapley [34], which is more intuitive and more commonly used. We do not address any of these techniques here. For good EKF performance proper values or tuning of these matrices might be necessary.

The measurement noise covariance matrix R can usually be obtained in a relatively straightforward manner directly from the measurements using the Fourier decomposition approach [35] or from the characteristics of the instrumentation used. The process noise covariance matrix Q is, however, more difficult to determine. In practice, Q is usually treated as a diagonal matrix. Its first n_x elements for the states should reflect the process noise (turbulence) level whereas the following n_q values corresponding to the parameters depend upon the nature of the parameters. For the case considered in Eqs. (7.47–7.51), the parameters are treated as constant over the complete period of observation, which is similar to the approach of off-line analysis. In such a case, the corresponding elements of Q are set to zero; see Eq. (7.49) with a matrix containing F and zeros. This procedure is followed for the test cases addressed in Sec. 7.5. However, if the system parameters change significantly during the period of observation, for example, as a result of change of configuration, due to malfunction or failure, or for any other reason, the covariances for the parameters should then be large enough to allow tracking of variations in system parameters. Many attempts are being

made in recent times to estimate the noise covariances adaptively but at the cost of increased computational burden, thus making it undesirable for real-time application.

7.3.2 UNSCENTED KALMAN FILTER

As elaborated in Appendix F, the EKF is a nonoptimal solution based on a first-order approximation of the nonlinear system dynamics. The EKF performance is sensitive to nonlinearities in the system model and to the process and measurement noise covariances, which are required to be specified a priori. Strong system nonlinearities, or in other words the higher-order terms neglected in the propagation of states and error covariances, and wrong values of noise statistics can result in biased estimates and in a worst-case lead to divergence [36]. The mixed filter implementation followed in Sec. 7.3.1 using nonlinear equations for propagation of the system states and a linearized model for propagation of error covariances mitigates the problem somewhat; however, the fundamental problem of not accounting for nonlinear transformation the random variables undergo remains, which basically affects the accuracy. Various techniques, such as iterated Kalman filter or second- and higher-order filters, are possible and applied in specific cases to reduce the estimation errors, but they are much more involved. More recently, yet another approach, called the sigma point filters (SPF), has evolved, which retains the standard Kalman filter form, involves no local iterations, and has a better performance [37–39]. We discuss here just one of them, called the unscented Kalman filter (UKF).

The UKF is based on propagating a finite set of points, called sigma points, through the nonlinear dynamics, and by approximating the distribution (mean and covariance) through a weighted sum and outer (cross) product of the propagated points. In contrast to the first-order approximation used in the EKF for covariance propagation, in the UKF nonlinear dynamics are used without approximations. Figure 7.2 (adapted from [39]) depicts schematically the two approaches for a two-dimensional case, showing on the left the true mean and covariance propagation using Monte Carlo sampling, in the center the linear propagation in the EKF, and on the right the nonlinear propagation as well as mean and covariance as weighted sums in the UKF. The flaw in the EKF of calculating mean and covariance of random variables undergoing a nonlinear transformation is thus eliminated in the UKF. This difference leads to a better performance of the UKF, which is shown to be equivalent to the second-order EKF, but without calculating the Jacobian or Hessian, that is, first- and second-order approximations of the system dynamics. This characteristic, namely, propagation, does not require computations of first- or higher-order approximations of system functions and, together with the resulting improved performance, makes it better suited for real-time applications and for nondifferentiable functions. Needless to say, Kalman filter, EKF, and UKF are all equivalent for linear models.

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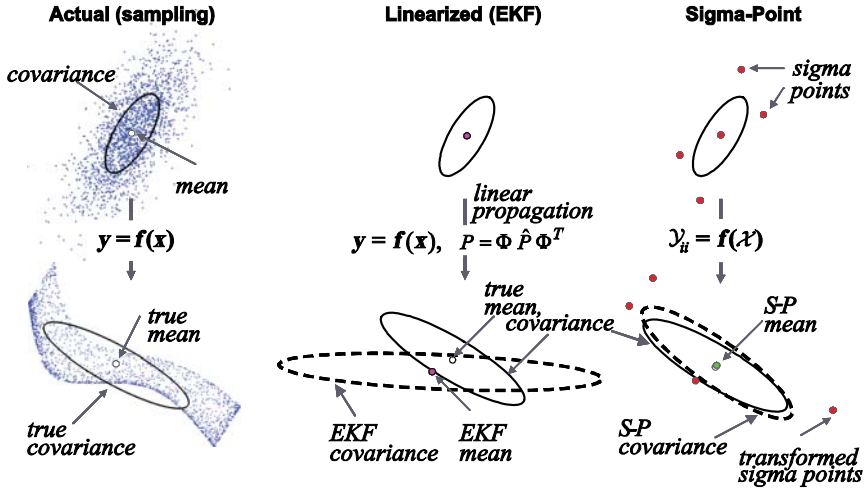

Fig. 7.2 Schematic of linear and nonlinear propagation (Source: [39]).

To implement the UKF, it is necessary to define $(2n_a + 1)$ sigma points, where n_a is the total number of states to be estimated, which include the basic system state, the unknown system parameters, as well as the process and measurement noise disturbances. We note that each point consists of a vector. One of the sigma vectors is the expected value of the augmented state vector, and the remaining $2n_a$ points are obtained from the columns of the matrix square root $\pm(\gamma P)_i$ for $i = 1, 2, \ldots, n_a$, where P is the covariance matrix of the augmented state vector x_a, and $\gamma = \sqrt{n_a + \lambda}$ and $\lambda = \alpha^2(n_a + \kappa) - n_a$ are scaling parameters. The constant α determines the spread of the sigma points around the estimated x_a; it is set equal to a small positive value less than one, typically in the range from 0.001 to 1, and $\kappa = 0$ or $3 - n_a$ for state or parameter estimation [37, 38]. The weights required in the computation of mean and covariance are defined as

$$\begin{aligned} W_0^m &= \lambda/(n_a + \lambda) \\ W_0^c &= \lambda/(n_a + \lambda) + (1 - \alpha^2 + \beta) \\ W_i^m &= W_i^c = 1/\{2(n_a + \lambda)\}, \quad i = 1, 2, \ldots, 2n_a \end{aligned} \tag{7.62}$$

where the subscript 0 corresponds to the estimated states and $i = 1, 2, \ldots, 2n_a$ the other sigma points; and the superscripts m and c indicate weights for the computation of mean and covariance, respectively. The constant β is used to incorporate prior knowledge of the distribution of x in the computation of weights for covariances W_0^c; the optimum value is $\beta = 2$ for Gaussian distribution. The scale

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parameters are, in general, problem dependent; some guidelines to choose them are provided in [40].

To illustrate the UKF procedure, let us first consider a discrete-time representation widely used in the applications of the UKF. Accordingly, let the discrete-time nonlinear state-space model be given by

$$x_{k+1} = f_d(x_k, \Theta_k, u_k, w_k) \tag{7.63}$$

$$y_k = g_d(x_k, \Theta_k, u_k, v_k) \tag{7.64}$$

where x_k is the $(n_x \times 1)$ state vector, Θ_k the $(n_q \times 1)$ vector of unknown parameters, u_k the $(n_u \times 1)$ vector of exogenous inputs, y_k the $(n_y \times 1)$ model output vector, and f_d and g_d the corresponding state and output functions. In a general case, the process noise w_k and the measurement noise v_k are assumed to enter the model nonlinearly as implied by the Eqs. (7.63) and (7.64). They are $(n_w \times 1)$ and $(n_v \times 1)$ size vectors, respectively. The simpler case of additive noise will be treated at the end of this section. The measurement vector is denoted by z_k. The augmented state vector of the size $(n_a \times 1)$ is given by

$$x_k^a = [x_k^T \quad \Theta_k^T \quad w_k^T \quad v_k^T]^T \tag{7.65}$$

where the superscript a denotes the augmented state vector, $n_a = n_{xp} + n_w + n_v$ and $n_{xp} = n_x + n_q$. Usually, we consider cases where $n_w = n_{xp}$ and $n_v = n_y$. The noise processes w_k and v_k are assumed to be zero mean and with covariance matrices Q and R, respectively. The starting values of the augmented state vector and its covariance are given by

$$\begin{aligned}\hat{x}_0^a &= E\{x_0^a\} = E\{[\hat{x}_0^T \quad \Theta^T \quad w_0^T \quad v_0^T]\} \\ &= [\hat{x}_0^T \quad \Theta_0^T \quad 0 \quad 0]\end{aligned} \tag{7.66}$$

$$P_0^a = E\left\{(x_0^a - \hat{x}_0^a)(x_0^a - \hat{x}_0^a)^T\right\} = \begin{bmatrix} P_{x0} & 0 & 0 & 0 \\ 0 & P_{\Theta 0} & 0 & 0 \\ 0 & 0 & Q & 0 \\ 0 & 0 & 0 & R \end{bmatrix} \tag{7.67}$$

Following the guidelines already discussed, we choose the constants α, β and the scaling parameters κ, γ, and λ and compute the set of weights W_i^m and W_i^c for $i = 0, 1, \ldots, 2n_a$.

Without going into mathematical derivation, the standard UKF algorithm can be summarized as follows [39]:

1. Set discrete time point $k = 1$, and build up the augmented state vector $\hat{x}_k^a$ and the corresponding covariance matrix $\hat{P}_k^a$ for the initial values according to Eqs. (7.66) and (7.67), respectively.

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2. Calculate the $(2n_a + 1)$ sigma points:

$$\hat{\chi}_k^a = \left[\hat{x}_k^a \quad \hat{x}_k^a - \gamma\sqrt{\hat{P}_k^a} \quad \hat{x}_k^a + \gamma\sqrt{\hat{P}_k^a}\right] \tag{7.68}$$

3. Prediction (time update):

$$\tilde{\chi}_{k+1}^x = f_d[\hat{\chi}_k^x, u_k, \hat{\chi}_k^w] \tag{7.69}$$

$$\tilde{x}_{k+1} = \sum_{i=0}^{2n_a} W_i^m \, \tilde{\chi}_{i,k+1}^x \tag{7.70}$$

$$\tilde{P}_{k+1} = \sum_{i=0}^{2n_a} W_i^c \, [\tilde{\chi}_{i,k+1}^x - \tilde{x}_{k+1}] \, [\tilde{\chi}_{i,k+1}^x - \tilde{x}_{k+1}]^T \tag{7.71}$$

4. Measurement update:

$$\mathcal{Y}_{k+1} = g_d[\tilde{\chi}_{k+1}^x, \; u_{k+1}, \; \tilde{\chi}_{k+1}^v] \tag{7.72}$$

$$\tilde{y}_{k+1} = \sum_{i=0}^{2n_a} W_i^m \, \mathcal{Y}_{i,k+1} \tag{7.73}$$

$$P_{\tilde{y}\tilde{y}_{k+1}} = \sum_{i=0}^{2n_a} W_i^c \, [\mathcal{Y}_{i,k+1} - \tilde{y}_{k+1}] \, [\mathcal{Y}_{i,k+1} - \tilde{y}_{k+1}]^T \tag{7.74}$$

$$P_{\tilde{x}\tilde{y}_{k+1}} = \sum_{i=0}^{2n_a} W_i^c \, [\tilde{\chi}_{i,k+1}^x - \tilde{x}_{k+1}] \, [\mathcal{Y}_{i,k+1} - \tilde{y}_{k+1}]^T \tag{7.75}$$

$$K_{k+1} = P_{\tilde{x}\tilde{y}_{k+1}} \, P_{\tilde{y}\tilde{y}_{k+1}}^{-1} \tag{7.76}$$

$$\hat{x}_{k+1} = \tilde{x}_{k+1} + K_{k+1} \, (z_{k+1} - \tilde{y}_{k+1}) \tag{7.77}$$

$$\hat{P}_{k+1} = \tilde{P}_{k+1} - K_{k+1} \, P_{\tilde{y}\tilde{y}_{k+1}} \, K_{k+1}^T \tag{7.78}$$

5. Increment k, and jump back to step 2 to continue.

In Eqs. (7.68–7.78), $\chi^a = [(\chi^x)^T \; (\chi^w)^T \; (\chi^v)^T]$ denotes the sigma points, χ^x corresponds to the system states and unknown parameters, $\sim$ (tilde) the predicted values, and $\hat{}$ (hat) the corrected values.

The foregoing UKF development is valid for a general case of noise disturbances entering the system nonlinearly and for discrete-time models; see Eqs. (7.63) and (7.64). Now, getting back to our specific interest, let us focus on

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the mixed continuous-discrete-time state-space models with additive noise disturbances.

$$\dot{x}(t) = f(x(t), u(t), \Theta) + w(t) \tag{7.79}$$

$$y(t) = g(x(t), u(t), \Theta) \tag{7.80}$$

$$z(t_k) = y(t_k) + v(t_k) \tag{7.81}$$

For such a model the UKF steps are very similar to those of Eqs. (7.68–7.78), but for the two modifications:

1. Propagation of sigma points through numerical integration
2. Simplification resulting from additive noise disturbances.

As in the case of the EKF, to propagate the sigma points based on the continuous-time nonlinear equation, we use one of the numerical integration methods instead of Eq. (7.69):

$$\widetilde{\chi}^{\,a}_{k+1} = \hat{\chi}^{a}_{k}(k) + \int_{t_k}^{t_{k+1}} f[\chi^{a}(t), \bar{u}(k)]\, \mathrm{d}t \tag{7.82}$$

For additive zero-mean noise disturbances, it turns out that augmentation of the state vector through noise vectors is not necessary. In such a case, the augmented state vector is given by $x^a_k = [x^T_k \;\; \Theta^T_k]^T$, exactly as considered in the case of EKF, the initial covariance matrix $P^{\,a}_0$ in Eq. (7.67) corresponds to that for the states and system parameters only, and the total number of states to be estimated is $n_a = n_{xp}$. The respective noise covariances Q and R get simply added on the right sides of Eqs. (7.71) and (7.74), yielding

$$\tilde{P}_{k+1} = \sum_{i=0}^{2n_a} W^c_i [\widetilde{\chi}^{\,x}_{i,k+1} - \tilde{x}_{k+1}]\, [\widetilde{\chi}^{\,x}_{i,k+1} - \tilde{x}_{k+1}]^T + Q \tag{7.83}$$

$$P_{\tilde{y}\tilde{y}_{k+1}} = \sum_{i=0}^{2n_a} W^c_i\, [\mathcal{Y}_{i,k+1} - \tilde{y}_{k+1}]\, [\mathcal{Y}_{i,k+1} - \tilde{y}_{k+1}]^T + R \tag{7.84}$$

The reduced size of the augmented state vector (and of sigma points) leads to a reduction in the computational burden.

In the case of UKF, some care is necessary while specifying the initial covariance matrix $P^{\,a}_0$ because the sigma points are generated from the specified initial augmented states $x^{\,a}_0$ and $P^{\,a}_0$; see Eq. (7.68). Each sigma point comprises system states and parameters. Completely inappropriate values may result in large deviations in the computed responses compared to the measured outputs for the first very few data points. Usually, this is not critical because the updated covariances are used to generate sigma points at each data point subsequently. In general, it

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does not affect the parameter convergence over a large number of data points. This phenomenon is specific to UKF only and does not arise in the case of EKF where usually large values are specified for the initial covariances.

To recapitulate, Eq. (7.69) or Eq. (7.82) propagates the sigma points using the nonlinear dynamics, the mean is computed as a weighted sum given by Eq. (7.70), and the error covariance is computed by Eqs. (7.71) or (7.83). The outputs for the sigma points using the nonlinear model are provided by Eq. (7.72), and the system output is the weighted sum given by Eq. (7.73). This is followed by computing the covariance and cross-correlation in Eqs. (7.74) or (7.84) and (7.75) respectively, which are required to compute the Kalman gain [Eq. (7.76)]. Once the gain is calculated, the predicted states and their covariances can be updated according to Eqs. (7.77) and (7.78). By comparing the UKF steps just given with those of the EKF in the preceding section, we notice that the major changes pertain to computing the mean and covariances of states and outputs whereas the measurement update is basically similar. It is also apparent that the first- or second-order approximations of the system dynamics are not required.

Thus, we have two versions of the unscented Kalman filter:

1. General case of noise disturbance entering the system nonlinearly, represented by the augmented vector $x_k^a = [x_k^T \;\; \Theta_k^T \;\; w_k^T \;\; v_k^T]^T$ to which we apply Eqs. (7.66–7.68), (7.82), and (7.70–7.78).
2. Special case of additive noise, represented by the state vector $x_k^a = [x_k^T \;\; \Theta_k^T]^T$ in which the simplified equations, (7.83) and (7.84), are used to compute covariances.

For an arbitrary choice of the scaling parameter κ, the two versions based on augmentation and nonaugmentation of states through noise vectors can result in some difference; however, for the specific choice of $\kappa = 3 - n_a$, they are theoretically equivalent [41]. The assumption of additive process and measurement noise corresponds to the cases we had hitherto considered in the other chapters.

Even though the sigma point approach is more accurate and helps to mitigate the nonlinearity effects, as in the case of the EKF, proper specification of the noise covariances is necessary. The augmentation of the state vector through noise vectors in Eq. (7.65) and of the corresponding covariances in Eq. (7.67) is necessary in a general case when noise affects the system nonlinearly, but it does not imply tuning of noise covariances Q and R. This is evident from the additive noise case, which clearly shows that the covariances just get added and are kept fixed; see Eqs. (7.83) and (7.84). For such a case of additive noise where the covariances appear explicitly in the covariance prediction equations, tuning techniques similar to those for the EKF or other Kalman filters can be applied. Adaptive tuning is also possible in the other general case of UKF in which the process noise covariance is implicit in the prediction equation, Eq. (7.71) [42]. However, as already mentioned in Sec. 7.3.1, we do not address any of these techniques here.

7.3.3 STATE ESTIMATOR WITHOUT KNOWLEDGE OF NOISE COVARIANCES

In general, inaccurate noise statistics and cross-correlated noise can lead to degraded performance of the Kalman filter algorithms. To circumvent these difficulties, an alternative approach is sometimes attempted based on an extension of the RLS with forgetting factor incorporating dynamic system matrix. Such an algorithm, called EFRLS (extended forgetting factor recursive least squares) does not require knowledge of the noise covariances [15, 43]. Other algorithms that do not require prior knowledge of measurement errors are also available [43], but we restrict ourselves here to the EFRLS only. In its basic form, the EFRLS is developed for the linear dynamic system defined as

$$x(k+1) = \Phi(k)\, x(k) + w(k) \tag{7.85}$$

$$y(k) = C(k)\, x(k) \tag{7.86}$$

$$z(k) = y(k) + v(k) \tag{7.87}$$

where k is the discrete time index, Φ the system matrix, w and v the zero-mean random variable. Without going into exact derivation of the algorithm that is found in [15], the recursive estimate of the states of the preceding discrete system are given by

$$x(k+1) = \Phi(k)\, x(k) + \Phi(k)\, L(k+1)\, [z(k+1) - C(k+1)\Phi(k)x(k)] \tag{7.88}$$

$$L(k+1) = P(k)\, \Phi^T(k)\, C^T(k+1) \times [\lambda I + C(k+1)\, \Phi(k)\, P(k)\, \Phi^T(k)\, C^T(k+1)]^{-1} \tag{7.89}$$

$$P(k+1) = \frac{1}{\lambda}\, \Phi(k)\, [I - L(k+1)\, C(k+1)\, \Phi(k)]\, P(k)\, \Phi^T(k) \tag{7.90}$$

where λ is the forgetting factor (generally between 0.98 and 0.995), which is an extension of the forgetting factor used in the standard least-squares method. Like in the recursive weighted least-squares procedure, in the present case too the performance of EFRLS is determined by the choice of the forgetting factor The investigations show that the EFRLS is less sensitive to the choice of the forgetting factor.

The EFRLS computational steps are somewhat similar to the EKF discussed in Sec. 7.3.1. However, there is a subtle difference between the two procedures. EKF incorporates the noise processes explicitly and has theoretically two tuning parameters corresponding to process and measurement noise matrices, compared to the single adaptation parameter λ of EFRLS. In the case of EKF, the performance is mainly influenced by the ratio between the process noise and measurement noise covariances. By lowering or increasing the measurement noise covariance, we can put increased or reduced faith in the measurements respectively; in the first case, it would lead to larger variations in the state estimates from sample to sample whereas a larger value would lead to smoother estimates, which also

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would mean rapid fluctuations are not tracked. The reverse is true of the process noise influence.

By deemphasizing the older data, the forgetting factor leads to a faster adaptation of time-varying parameters. The smaller the forgetting factor, the better is the adaptation to current data. The faster convergence is, however, associated with increased sensitivity to the noise. In fact, if λ is much less than one, the estimates that might have converged to the true value tend to diverge or oscillate as a result of small disturbances or noise acting as inputs. Possible different approaches to overcome the high sensitivity to noise are through time-variant forgetting factor or Kalman filter [44].

The choice of λ is based on the following considerations [45]. If the process noise variance is expected to be large, λ should be small because the past data are not giving more information on the current augmented states. If the process noise variance is relatively smaller than the measurement noise variance, λ should be of a larger value. This implies that more data should be used to average out the effect of the noise on measurements. Hence, λ should be very close to one but less than one. We can also look at the forgetting factor in terms of the so-called memory index. The memory index (MI) of the RLS filter can be defined as $MI = 1/(1-\lambda)$. Thus, if $\lambda = 1$, then MI is infinity, implying that the filter has infinite memory. This means that the entire data are given equal weighting, which is the case with the standard recursive least-squares method. It is also possible to arrive at a reasonable λ based on the condition number (ratio of the magnitudes of the largest to smallest eigenvalue) of the data matrix. A large condition number indicates an ill-conditioned matrix, and accordingly less weight should be given the past data. Readers might recall that we had used in Sec. 4.10.3 a similar concept of condition number to detect the ill-conditioning of the information matrix and possibly neglecting the search directions to avoid intermediate convergence of the Gauss–Newton or other optimization methods.

Although the EFRLS algorithm is similar to the Kalman filter, there is a subtle difference in the two cases. Let us take a look at 1) the applicability or extensions to more general nonlinear models and 2) the matrix P appearing in the algorithms and computation of the standard deviations.

As already discussed in Sec. 7.3.1, the Kalman filter algorithm can be readily extended to any general nonlinear state-space model. Extension of the EFRLS algorithm to a continuous-time state-space model of the form $\dot{x}(t) = Ax(t) + w(k)$ is fairly straightforward. The first term on the right-hand side of Eq. (7.88), namely, $\Phi(k)\ x(k)$, gets simply replaced through states obtained by numerical integration, and Φ appearing in Eqs. (7.88–7.90) through the state transition matrix $\Phi = e^{A\Delta t}$. However, extension of the EFRLS to a more general case with deterministic exogenous inputs u represented by

$$\dot{x}(t) = A(t)\ x(t) + B(t)\ u(t) + w(t) \tag{7.91}$$

$$y(k) = C(k)\ x(k) + D(k)\ u(k) + v(k) \tag{7.92}$$

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is more involved. For linear systems, the principle of superposition allows us to decompose the preceding system into two, first one stochastic and the other deterministic. Denoting $x(k) = x_s(k) + x_d(k)$ and $y(k) = y_s(k) + y_d(k)$, it can be easily verified that the system given by Eqs. (7.85–7.87) can be decomposed into 1) $x_s(k+1) = \Phi(k)\, x_s(k) + w(k)$; $y_s(k) = C(k)\, x_s(k) + v(k)$ and 2) $x_d(k+1) = \Phi(k)\, x_d(k) + \Psi B u(k)$; $y_d(k) = C(k)\, x_d(k) + D(k)u(k)$, where Ψ is the integral of Φ. We know that $w(t)$ and $w(k)$ are not the same, but because we are dealing here with an estimator without knowledge of noise covariances, the difference is immaterial. To the first system in terms of the state and observation variables (x_s, y_s), we can apply the EFRLS algorithm given by Eqs. (7.88–7.90), and the second system in terms of (x_d, y_d) being deterministic amounts to simple simulation. The preceding approach, although feasible for linear systems, is not very flexible. Furthermore, it cannot be readily extended to nonlinear model postulates. Thus, the EFRLS algorithm in its present form appears to have a limited scope of applicability.

In the case of Kalman or extended Kalman filter, the matrix P represents the covariance of the state prediction error, and hence the standard deviation of the estimates can be readily computed as square root of the diagonal elements as the propagation progresses in time. On the other hand, the matrix P appearing in Eqs. (7.90) is a data matrix and not the covariance matrix of the state prediction error. Therefore, the standard deviations cannot be obtained directly for EFRLS as for the case of the Kalman filter. As in the case of RLS, the standard deviations can be computed based on Eq. (7.15) using the complete set of data after completing the EFRLS estimation.

7.4 ALGORITHMIC IMPLEMENTATION AND SOFTWARE

As in the case of Chapters 4 to 6, we provide here simple implementations of the software for recursive parameter estimation based on the RLS, FTR, EKF, UKF (in two versions), and EFRLS methods discussed in the foregoing sections; these programs enable closer understanding of the algorithmic steps that we have so far studied in this chapter. Furthermore, it allows us to generate the results that we will discuss here. The source codes (Matlab® m-files) for recursive parameter estimation are provided in the directory /FVSysID2/chapter07/.

In the case of the EKF and both versions of the UKF, we have implemented here the algorithms in their so-called standard versions. Numerically, more efficient implementations based on UD factorization are possible. Particularly for UKF, the square-root implementation has a further advantage because the square root of the covariance matrix is a part of the algorithm, which is also needed in the computation of the sigma points [46, 47]. Thus, a considerable reduction in the computational overhead is possible. Computational speed can also be enhanced by vectorizing the matrix operations at some of the steps involved in the weighted sum for mean and covariances by treating the weights

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as a matrix having the same weights in all of the columns. All such extensions are left to the reader. We deal here with standard versions and treating weights as a column vector because it allows tracing back the algorithmic steps given in the text and also because it leads to a more readable code.

The program "attas_regLS_RLS" is a simple implementation of the RLS algorithm described in Sec. 7.2.2 for a test case that we had analyzed in Sec. 6.9.2. For recursive estimation based on state-space models, the main program "mainRPE" is the starting point. It provides an option to invoke different recursive algorithms, to define the test case to be analyzed, and to call the function for model definition:

test_case	integer flag for the test case	
method	flag to choose the RPE method	
	EKF:	Extended Kalman Filter
	UKF:	Unscented Kalman Filter (simplified case of additive process and measurement noise, i.e., $x_k^a = [x_k^T \;\; \Theta_k^T]^T$)
	UKFaug:	Augmented unscented Kalman filter (general case of state augmentation through process and measurement noise, that is, $x_k^a = [x_k^T \;\; \Theta_k^T \;\; w_k^T \;\; v_k^T]^T$)
	EFRLS:	State estimator without knowledge of noise covariances
	FTR:	Fourier transform regression
	ALL:	Sequential application of EKF, UKF, UKFaug, EFRLS, and FTR

Depending upon the selected method, a few constants might be necessary. For example, in the case of the EFRLS, it is required to specify the forgetting factor "lambda"; for FTR, it is required to specify the update rate for FFT and the range of frequencies to be accounted for, and in the case of UKF or UKFaug it is required to specify the scaling parameters α, β, κ. It is possible to apply any one of the five algorithms to the specified problem by specifying the flag "method" appropriately. Otherwise, setting it to ALL will apply each of them in a sequence; this option will be used for the examples presented in the next section because it allows generating comparative results more readily. The choice of the method will depend upon the type of the system model.

Different system models to be analyzed are denoted through the integer flag "test_case," which is to be uniquely associated with the state-space model and with a user-provided interface in the form of a function defining the model parameters, flight data, and other relevant details. All but the FTR method need functions for coding the defined state-space model. The model definition comprises the following information:

state_eq	Function name: to code the right-hand side of the state equations
obser_eq	Function name: to code the right-hand side of the observation equations

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Nx	Number of state variables (n_x)
Ny	Number of output variables (n_y)
Nu	Number of input variables (n_u)
NparSys	Number of system parameters
Nparam	Total number of system and bias parameters

The preceding list shows on the left side the variable names used in the program mainRPE and in other functions called therein, followed by the description and the notation used to denote these variables in the text. In the case of the FTR, which is based on a least-squares approach, there are no state equations. Accordingly, the preceding two functions are not necessary.

The functions to code the state and observation equations used in the UKFaug are extended versions of the preceding two functions. They are denoted as "state_eq_xyNoise" and "obser_eq_xyNoise," respectively. In these functions, the noise vectors obtained from the sigma points appear explicitly in the state and observation equations. The major part remains the same as in the equations "state_eq" and "obser_eq"; only the noise needs to be incorporated appropriately.

It is also required to specify the starting values for the parameters, integer flags to indicate whether the particular parameter is free or fixed, initial conditions, the perturbations for numerical approximations, and covariances of measurement noise, augmented states, and state propagation error:

param	starting values for unknown parameters (Θ_0)
parFlag	flags to define free and fixed parameters = 1: free parameters (to be estimated) = 0: fixed parameters (not to be estimated)
NparID	total number of unknown parameters, that is, nonzero elements of parFlag (n_q)
x0	initial conditions on state variables x
delxa	perturbation in augmented states for numerical approximations of system matrices
rr	diagonal elements of measurement noise covariance matrix
qq	diagonal elements of process noise covariance matrix
p0	initial state propagation error covariance matrix

The number of augmented states Nxp is given by Nx + NparID, and the augmented states by xah = $[x_0^T\ \Theta_0^T]^T$. It is obvious that the qq and p0 are to be specified for the Nxp augmented states.

The flight data to be analyzed are also to be loaded in the model definition function and require specification or assignment of the following information:

Ndata	Number of data points
dt	Sampling time
Z(Ndata,Ny)	Flight data for measured outputs (N, n_y)
Uinp(Ndata,Nu)	Flight data for measured control inputs (N, n_u)

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While loading the flight data in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu), it is obvious that the units of the variables must match with those used in the postulated model and programmed in the state and observation functions "state_eq" and "obser_eq," respectively. The model specification in terms of (Nx, Ny, Nu, NparSys, Nxp) must also match with that coded in the preceding two user functions for the postulated model.

At the end of the state with parameter estimation, depending upon the method and the model, plots are generated of the time histories, convergence of system states, or of parameters as appropriate. Additionally, comparison is made of the parameter estimates provided by the different estimation methods.

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7.5 EXAMPLES

We consider here four examples to demonstrate some of the recursive algorithms detailed in the preceding sections. The first example pertains to a model in an equation-error form (that is, without state equations) whereas the other three pertain to state-space models. The second and the third examples are based on linear models in terms of dimensional derivatives. The fourth example considers a nonlinear model pertaining to lift, drag, and pitching moment. Based on the last three examples, the various RPE algorithms are evaluated considering the following issues: 1) computational effort required, 2) parameter convergence to reference values and the time taken, 3) robustness in the presence of noise, and 4) computed variance of the estimation error. The estimates obtained applying these algorithms are compared against those obtained from other off-line nonrecursive methods discussed in earlier chapters.

The data analyzed in the first three examples were gathered from flight tests carried out with the research aircraft ATTAS [48]. Several repeat runs were performed and analyzed; we restrict ourselves to just one typical data set in each case. In the fourth case we analyze data that were gathered with test aircraft HFB-320. The flight data analyzed in these cases are supplied to the readers as sample flight data for verification and to try out other options and possibly to verify software that they can modify or develop themselves. These data files are found under the directory /FVSysID2/flt_data/.

7.5.1 ESTIMATION OF AERODYNAMIC PARAMETERS OF A LINEAR MODEL

To demonstrate the RLS method, we turn our attention to the example considered in Sec. 6.9.2 in which the LS method was applied to flight data from three concatenated maneuvers performed at 16,000 ft and at 200 kt with the test aircraft ATTAS. The first maneuver is a multistep elevator input exciting the short-period motion, the second is an aileron input resulting in a bank-to-bank motion, and the third is a rudder doublet input exciting the Dutch roll motion. The three maneuvers are 25, 30, and 30 s long, respectively. The LS estimates for the aerodynamic

model postulated in Eq. (6.76), obtained by running the program "/FVSysID2/chapter06/attas_regLS," were provided in Table 6.2.

Now, we run the same case by calling the program "/FVSysID2/chapter07/attas_regLS_RLS," which includes the same data preprocessing step as described in Sec. 6.9.2 to compute the aerodynamic force and moment coefficients and also the procedures to apply the LS method. These two steps remain unchanged. For each aerodynamic coefficient, the RLS algorithm of Eqs. (7.17–7.19) is appended in the new program "attas_regLS_RLS." The models for the aerodynamic force and moment coefficients consist of

$$\begin{aligned} C_D &= C_{D0} + C_{D\alpha}\,\alpha \\ C_L &= C_{L0} + C_{L\alpha}\,\alpha \\ C_m &= C_{m0} + C_{m\alpha}\,\alpha + C_{mq}\,q^* + C_{m\delta e}\,\delta_e \\ C_Y &= C_{Y0} + C_{Y\beta}\,\beta \\ C_\ell &= C_{\ell 0} + C_{\ell\beta}\,\beta + C_{\ell p}\,p^* + C_{\ell r}\,r^* + C_{\ell\delta a}\,\delta_a \\ C_n &= C_{n0} + C_{n\beta}\,\beta + C_{np}\,p^* + C_{nr}\,r^* + C_{n\delta r}\,\delta_r \end{aligned} \tag{7.93}$$

As in the case of the least-squares method, the aerodynamic derivatives appearing in Eq. (7.93) are estimated by considering the drag, lift, pitching-moment, side-force, rolling-moment, and yawing-moment coefficients one at a time. The forgetting factor is chosen to be $\lambda = 1$. As a typical case, the estimates of a few selected parameters are plotted in Fig. 7.3, which shows the RLS estimates as time histories in continuous lines. For comparison purposes, the one-shot ordinary least-squares (OLS) estimates are shown in dotted lines as constant values for the complete duration of data analyzed.

As expected, it is observed that the RLS estimates at the end of the maneuver match with the OLS estimates. In most the cases it can be observed that the convergence is good. For example, as seen in the figure, the estimates of drag and lift parameters converge during the first 25 s, in which the short-period motion is excited. Although not shown in the figure, by running the program "attas_regLS_RLS," it can be verified that the parameters pertaining to the rolling motion show variations during the first 25 s, but they get adjusted once the appropriate information becomes available during the following aileron input maneuver. The parameters of the yawing motion are also estimated during this maneuver because of the coupling between the rolling and yawing motion. Some further adjustments occur once the information from the rudder input is evaluated. The convergence of the parameters of the pitching-moment coefficient, particularly the pitch damping derivative C_{mq}, was rather slow. The preceding discussion, the effect of the forgetting factor, and plots of the other aerodynamic parameters appearing in other force and moment coefficients can be verified by running the test case as just described.

For the reasons already elaborated in Sec. 7.2.1, the standard deviations are not plotted in Fig. 7.3 as the estimates evolve in time. Instead, they are computed after

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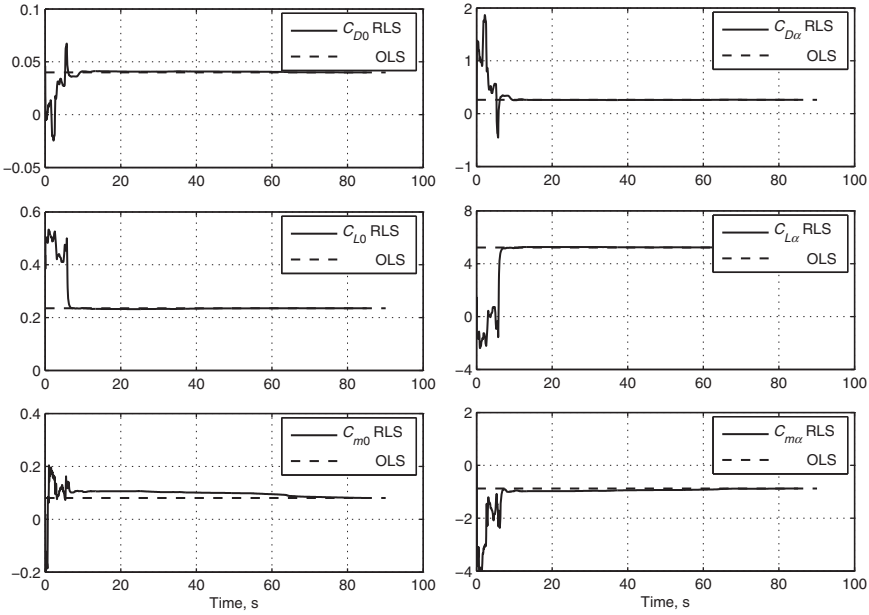

Fig. 7.3 Estimates of lift, drag, and pitching-moment parameters applying RLS method.

the estimation based on the complete length of data using Eq. (6.21) as done for the least-squares method. The standard deviations in the two cases of LS and RLS were practically the same.

7.5.2 LONGITUDINAL MOTION

A second-order linear model pertaining to the aircraft short-period motion excited through an elevator input is given by the following:

State equations:

$$\begin{aligned} \dot{\alpha} &= Z_0 + Z_\alpha\ \alpha + (1 + Z_q)\ q + Z_{\delta e}\ \delta_e \\ \dot{q} &= M_0 + M_\alpha\ \alpha + M_q\ q + M_{\delta e}\ \delta_e \end{aligned} \tag{7.94}$$

Observation equations:

$$\begin{aligned} \alpha_m &= \alpha \\ q_m &= q \end{aligned} \tag{7.95}$$

where the subscript m in Eq. (7.95) refers to the measured values. The angle of attack α and pitch rate q are the two state variables, the measured α and q are the two output variables, and elevator deflection δ_e is the control input. The

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dimensional parameters pertaining to vertical force $Z_{()}$ and pitching motion $M_{()}$ are to be estimated. The unknown parameter vector Θ, consisting of the dimensional derivatives and aerodynamic zero terms, is given by

$$\Theta = [Z_0 \;\; Z_\alpha \;\; Z_q \;\; Z_{\delta e} \;\; M_0 \;\; M_\alpha \;\; M_q \;\; M_{\delta e}]^T \tag{7.96}$$

Thus, the postulated model is defined as follows:

	No. of Variables		Function Name
States	2	α, q	"xdot_TC11_lon_sp"
Outputs	2	α, q	"obs_TC11_lon_sp"
Inputs	1	δ_e	

The right-hand sides of the state equation (7.94) are programmed in the function "xdot_TC11_lon_sp" and those of the observation equation (7.95) in "obs_TC11_lon_sp." The preceding postulated model is designated as the eleventh test case. Thus, the model definition for this example, provided by the function "mDefCase11.m" called from the main program "mainRPE," is as follows:

```
test_case = 11;                    % Index for the test case
state_eq  = "xdot_TC11_lon_sp";    % Function for state equations
obser_eq  = "obs_TC11_lon_sp";     % Function for observation equations
Ny        = 2;                     % Number of observation variables
Nx        = 2;                     % Number of states
Nu        = 1;                     % Number of input (control) variables
NparSys   = 8;                     % Number of system parameters
Nxp       = Nx + NParSys;          % Total number of states
                                     (system states + parameters)
dt        = 0.04;                  % Sampling time
```

Deviating from the preceding model definition, the state and observation equations used in the UKFaug are coded in "xdot_TC11_lon_sp_xyNoise" and "obs_TC11_lon_sp_xyNoise," respectively (see elaboration in Sec. 7.4).

To increase the length of the data segment being processed recursively, two sets of short period data were cascaded. The data are loaded from the following two files:

```
load ..\flt_data\fAttasElv1
load ..\flt_data\fAttasElv2
```

and stored in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu). The storage of data is not essential for recursive algorithms because states and parameters are updated point by point. However, for comparison purposes the various RPE algorithms are

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run in an off-line environment, that is, after the data have been recorded. The stored data allow us to make plots of the time histories as well.

First, we generate the reference values applying the output-error method, that is, "/FVSysID2/chapter04/ml_oem2," for which we use the same flag test_case = 11 and the same functions for the state and observation equations, but the following additional information concerning the individual time segments is necessary:

```
Nzi = 2;                    % Number of time segments
izhf = [Nts1; Nts1 + Nts2]; % Cumulative index
```

where Nts1 and Nts2 are the number of data points in the two individual time segments and izhf is the cumulative index at which the maneuvers end when they are concatenated. The estimates obtained applying "ml_oem2" are summarized in Table 7.1 along with those from the other methods. The "ml_oem2" estimates in Fig. 7.4 are plotted as a constant value over the complete

TABLE 7.1 PARAMETER ESTIMATES FROM FLIGHT DATA

Parameter	Reference Values (OEM)	Estimates				
		EFRLS	FTR	EKF	UKF	UKFaug
Z_0	−0.009 (7.25)*	−0.0075	–	−0.0075 (15.2)	−0.0077 (14.6)	−0.0090 (2.78)
Z_α	−0.483 (1.36)	−0.500	−0.477 (3.74)	−0.496 (2.28)	−0.496 (2.24)	−0.484 (0.50)
Z_q	0.104 (5.28)	0.109	0.105† (1.29)	0.101 (9.66)	0.103 (9.31)	0.108 (1.88)
Z_{δ_e}	0.676 (1.76)	0.669	0.665 (5.47)	0.647 (3.58)	0.653 (3.48)	0.680 (0.67)
M_0	0.475 (0.3)	0.465	–	0.467 (0.28)	0.468 (0.26)	0.474 (0.10)
M_α	−4.927 (0.33)	−4.846	−4.929 (1.52)	−4.873 (0.26)	−4.873 (0.25)	−4.913 (0.11)
M_q	−2.006 (0.57)	−1.972	−2.031 (2.96)	−1.969 (0.57)	−1.969 (0.54)	−2.014 (0.20)
M_{δ_e}	−7.208 (0.35)	−7.129	−7.254 (2.11)	−7.085 (0.38)	−7.081 (0.36)	−7.222 (0.12)

*Values in parentheses denote relative standard deviation in %.
†Obtained from FTR estimate $(1 + Z_q)$.

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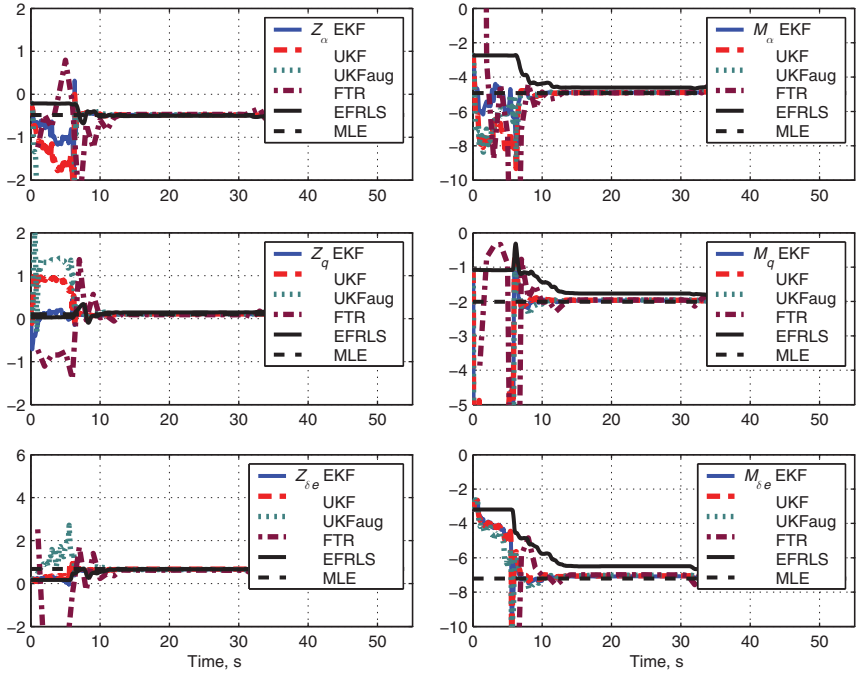

Fig. 7.4 Recursive estimates of longitudinal derivatives.

duration of data in the figures, which show convergence of the estimates from other methods.

Now, we set the flag "method=ALL" in "/FVSysID2/chapter07/mainRPE" to apply the five recursive methods, namely, EKF, UKF, UKFaug, EFRLS, and FTR. This yields the results summarized in Table 7.1, which gives average values over the last few data points, typically 10 for the EKF, UKF, UKFaug, and EFRLS, and 5 for the FTR. Representative values have been specified for the measurement noise covariance matrix (see function "mDefCase11.m"), and the process noise covariance matrix is also specified as diagonal with values of 1×10^{-7} for states and zero for the extended states, that is, parameters, according to Eq. (7.49). In each case the time history match between the flight-measured data and model output was found to be satisfactory; these plots as well as those of the parameter estimates from individual RPE methods are also not included in the text here, but can be easily generated and verified running this case. The estimates of the six derivatives are plotted graphically in Fig. 7.4.

Let us now compare the numerical values and accuracies of the estimates from the various methods. The estimates of all of the parameters obtained applying various methods appear plausible. They are within reasonable ranges

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and are comparable to the reference values provided by the output-error method. The standard deviations for some of the estimates by EKF, UKF, and FTR are somewhat higher than those for the off-line method or those from UKFaug. In an overall sense, the UKFaug estimates compare best with the OEM results.

As already discussed in Sec. 7.2.4, in the case of FTR, the aerodynamic bias parameters Z_0 and M_0 are not estimated. Moreover, trim values were also subtracted from the measured data. (This is done in the function "ftr_mod" implementing the FTR.) The large fluctuations in the estimates prior to initiation of the flight maneuver (at about $t = 6$ s) are due to lack of information in the data content up to then. However, as can be observed from the parameter time history plots, they tend to converge to the reference values as more and more information becomes available in the data.

The robustness of the EKF, UKF, UKFaug, and EFRLS algorithms has been tested by initiating the recursion with arbitrary starting values, whereby the algorithms converged without any numerical difficulties. The parameters are well estimated once the input is applied, here from 6 to 12 s, which is followed by the next excitation from 31 to 37 s. During the period between the two maneuvers, which corresponds to flight without any specific control input, the converged estimates show no divergence. The additional excitation leads to hardly any changes in the EKF, UKF, UKFaug, and EFRLS estimates whereas the FTR estimates get marginally adjusted during the maneuver itself.

7.5.3 LATERAL-DIRECTIONAL MOTION

A second-order linear model pertaining to the lateral-directional motion excited through aileron and rudder inputs is given by the state equations

$$\begin{aligned} \dot{p} &= L_p p + L_r r + L_{\delta a}\delta_a + L_{\delta r}\delta_r + L_\beta \beta + L_0 \\ \dot{r} &= N_p p + N_r r + N_{\delta a}\delta_a + N_{\delta r}\delta_r + N_\beta \beta + N_0 \end{aligned} \tag{7.97}$$

and by the observation equations

$$\begin{aligned} p_m &= p \\ r_m &= r \end{aligned} \tag{7.98}$$

where p is the roll rate, r the yaw rate, β the sideslip angle, δ_a the aileron deflection, and δ_r the rudder deflection. In this simplified model, we treat the angle of sideslip as an input variable, and hence it does not appear as a state equation. $L_{()}$ and $N_{()}$ are the dimensional parameters pertaining to the rolling and yawing motion, which are to be estimated. The unknown parameter vector Θ consisting of the dimensional parameters is given by

$$\Theta = [L_p \;\; L_r \;\; L_{\delta a} \;\; L_{\delta r} \;\; L_\beta \;\; L_0 \;\; N_p \;\; N_r \;\; N_{\delta a} \;\; N_{\delta r} \;\; N_\beta \;\; N_0]^T \tag{7.99}$$

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Thus, the postulated model is defined as follows:

	No. of Variables		Function Name
States	2	p, r	"xdot_TC03_lat_pr"
Outputs	2	p, r	"obs_TC03_lat_pr"
Inputs	3	$\delta_a, \delta_r, \beta$	

The right-hand sides of the state equation (7.97) are programmed in the function "xdot_TC03_lat_pr" and those of the observation equation (7.98) in "obs_TC03_lat_pr." The preceding postulated model is designated as the third test case. Thus, the model definition provided by the function "mDefCase03.m," which is called from the main program "mainRPE" is as follows:

```
test_case = 3;                      % Flag for the test case
state_eq  = "xdot_TC03_lat_pr";     % Function for state equations
obser_eq = "obs_TC03_lat_pr";       % Function for observation equations
Ny        = 2;                      % Number of observation variables
Nx        = 2;                      % Number of system states
Nu        = 3;                      % Number of input (control) variables
NparSys  = 12;                      % Number of system parameters
Nxp       = Nx + NParSys;           % Total no. of states (system
                                      states + parameters)
dt        = 0.04;                   % Sampling time
```

Deviating from the preceding model definition, the state and observation equations used in the UKFaug are coded in "xdot_TC03_lat_pr_xyNoise" and "obs_TC03_lat_pr_xyNoise," respectively (see elaboration in Sec. 7.4).

Parameters pertaining to the lateral-directional motion are estimated from a bank-to-bank maneuver and a Dutch roll motion. As in the case of the preceding example, we analyze two sets of cascaded flight data, each with an aileron input followed by a rudder input. The data are loaded from the following two files:

```
load ..\flt_data\fAttasAilRud1
load ..\flt_data\fAttasAilRud2
```

and stored in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu). Because the general procedure of data analysis is similar to that followed for the preceding example, we just give here the bare minimum details because the general remarks are valid in this case too. The reference values are generated applying the output-error method, that is, "/FVSysID2/chapter04/ml_oem2" by setting flag test_case = 3. The information about the two time segments (Nzi and izhf) is defined as in the case of the preceding example.

Table 7.2 summarizes the estimates obtained applying the output-error method "/FVSysID2/chapter04/ml_oem2," along with those from the recursive

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TABLE 7.2 ESTIMATES OF PARAMETERS PERTAINING TO LATERAL-DIRECTIONAL MOTION

Parameter	Reference Values (OEM)	Estimates				
		EFRLS	FTR	EKF	UKF	UKFaug
L_p	−2.067 (0.21)*	−2.094	−2.046 (1.70)	−2.077 (0.16)	−2.072 (0.15)	−2.077 (0.12)
L_r	1.029 (0.84)	1.051	1.041 (6.96)	1.021 (0.64)	1.019 (0.62)	1.042 (0.48)
$L_{\delta a}$	−6.306 (0.17)	−6.391	−6.367 (1.23)	−6.349 (0.13)	−6.334 (0.12)	−6.324 (0.10)
$L_{\delta r}$	1.145 (1.84)	1.238	1.277 (16.59)	1.195 (1.43)	1.194 (1.38)	1.065 (1.17)
L_β	−3.844 (0.38)	−3.909	−3.786 (3.28)	−3.802 (0.29)	−3.800 (0.28)	−3.835 (0.22)
L_0	0.010 (3.45)	0.0085	–	0.0103 (2.76)	0.0104 (2.65)	0.0093 (2.16)
N_p	−0.184 (1.34)	−0.197	−0.162 (6.97)	−0.171 (1.16)	−0.170 (1.14)	−0.176 (0.26)
N_r	−0.486 (0.96)	−0.458	−0.433 (5.44)	−0.425 (0.97)	−0.425 (0.95)	−0.460 (0.20)
$N_{\delta a}$	−0.391 (1.64)	−0.408	−0.352 (7.21)	−0.380 (1.25)	−0.378 (1.23)	−0.383 (0.31)
$N_{\delta r}$	−1.761 (0.63)	−1.773	−1.747 (3.94)	−1.716 (0.61)	−1.719 (0.60)	−1.764 (0.13)
N_β	3.031 (0.30)	3.029	3.074 (1.31)	3.029 (0.23)	3.031 (0.22)	3.025 (0.06)
N_0	0.0346 (0.45)	0.035	–	0.034 (0.55)	0.034 (0.54)	0.034 (0.09)

*Values in parentheses denote relative standard deviation in %.

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methods. The relative standard deviations of the estimates are provided in brackets. The OEM estimates are the reference values for evaluating the RPE algorithms. They are plotted as a constant value over the complete duration of data in Fig. 7.5.

Having generated the reference values, we now apply the RPE algorithms "/FVSysID2/chapter07/mainRPE" by setting the flags "method=ALL" and "test_case=3." The results are provided in Table 7.2 giving average values over the last few data points. In the case of FTR, the aerodynamic bias parameters L_0 and N_0 are not estimated, and trim values were subtracted from the measured

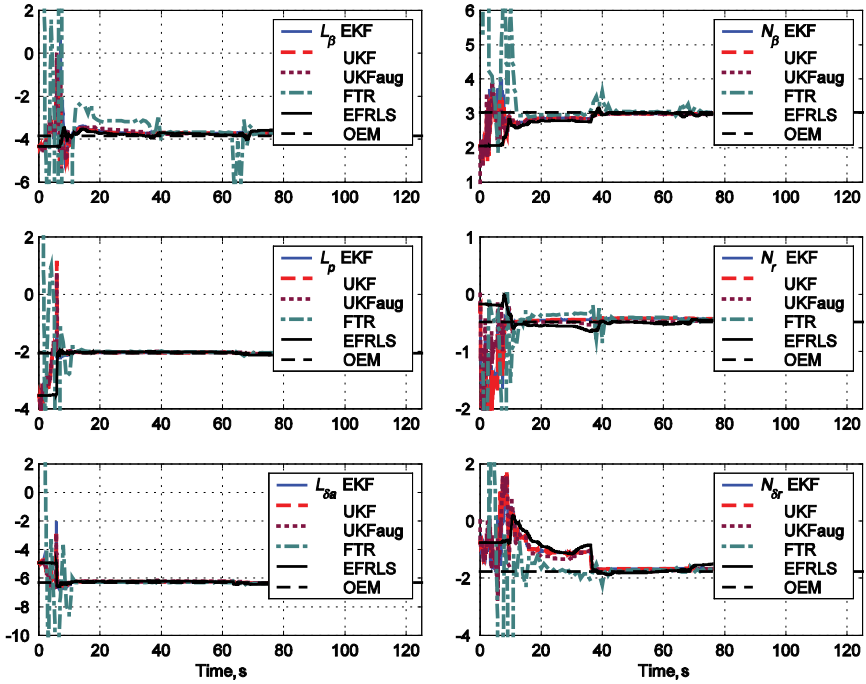

Fig. 7.5 Recursive estimates of lateral-directional derivatives.

data. (See function "ftr_mod" implementing the FTR.) As in the case of the preceding example, representative values were specified for the measurement noise covariance matrix (see function "mDefCase03.m"), and the process noise covariance matrix is also specified as diagonal with values of 1×10^{-7} for states and zero for the extended states, that is, parameters, according to Eq. (7.49).

The same characteristic behavior of the five RPE algorithms discussed in Sec. 7.5.3 repeats in the present case. Figure 7.5 gives the plots of some selected derivatives of the lateral-directional motion. From Table 7.2 and Fig. 7.5, we note that the estimates from the various RPE algorithms compare reasonably well with the reference values from OEM. The performance of all of the RPE algorithms was satisfactory and encountered no numerical difficulties.

7.5.4 ESTIMATION OF NONDIMENSIONAL LONGITUDINAL DERIVATIVES

The fourth example pertains to determination of the lift, drag, and pitching-moment coefficients of the research aircraft HFB-320 considered in Sec. 5.11.2. As already pointed out there, the system equations (5.86) in terms of variables in the stability axes (V, α) contain not only the common trigonometric and multiplicative nonlinearities but also the variable dynamic pressure $\bar{q}$ ($=1/2\,\rho V^2$),

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which multiplies all of the aerodynamic derivatives and introduces an additional nonlinearity. The observations in Eq. (5.88) are coupled through C_X and C_Z. The maneuver analyzed is the longitudinal motion excited through a multistep elevator input resulting in short-period motion and a pulse input leading to phugoid motion. The state vector is $x = [V\ \alpha\ \theta\ q]^T$, and the unknown parameters appearing in Eq. (5.87) are $\Theta = [C_{D0}\ C_{DV}\ C_{D\alpha}\ C_{L0}\ C_{LV}\ C_{L\alpha} C_{m0}\ C_{mV}\ C_{m\alpha}\ C_{mq}\ C_{m\delta e}]^T$.

Because it is now attempted to estimate the augmented state vector $x^a = [x^T\ \Theta^T]^T$ subject to the model postulated in Eqs. (5.86–5.89), we require recursive methods that cater to nonlinear state-space models. Accordingly, the choice of methods reduces to EKF, UKF, and UKFaug because the RLS procedure is based on a different model structure, FTR is applicable to linear models only, and the EFRLS algorithm presented in Sec. 7.3.3 does not directly cater to nonlinear state and observation equations. For the EKF, UKF, and UKFaug methods, the postulated model is defined as follows:

	No. of Variables		Function Name
States	4	$V,\ \alpha,\ \theta,\ q$	"xdot_TC04_hfb_lon"
Outputs	7	$V,\ \alpha,\ \theta,\ q,\ \dot{q},\ a_x,\ a_z$	"obs_TC04_hfb_lon"
Inputs	2	$\delta_e,\ F_e$	

Thus, the model definition for this example provided by the function "/FVSysID2/chapter07/mDefCase04.m," called from the main program "mainRPE," is as follows:

```
test_case = 4;                    % Integer flag for the test case
state_eq  = "xdot_TC04_hfb_lon";  % Function for state equations
obser_eq  = "obs_TC04_hfb_lon";   % Function for observation equations
Nx        = 4;                    % Number of states
Ny        = 7;                    % Number of observation variables
Nu        = 2;                    % Number of input (control)
                                    variables
NparSys   =11;                    % Number of system parameters
Nparam    = NparSys + Nx;         % Total number of parameters
dt        = 0.1;                  % Sampling time
```

Deviating from the preceding model definition, the state and observation equations used in the UKFaug are coded in "xdot_ TC04_hfb_lon_xyNoise" and "obs_ TC04_hfb_lon_xyNoise," respectively (see elaboration in Sec. 7.4).

In Sec. 5.11.2, it has been demonstrated that the data being analyzed contain atmospheric turbulence and that the filter-error method "ml_fem" provided better results than the output-error method. The estimates obtained running the program "/FVSysID2/chapter05/ml_fem2" for test_case = 4 are provided in Table 7.3. To enable comparison of the recursive estimates with those from the off-line filter-error method, we specify the process and measurement noise

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TABLE 7.3 ESTIMATES OF NONDIMENSIONAL LONGITUDINAL DERIVATIVES

Parameter	Filter-error Method (FEM)	RPE Methods		
		EKF	UKF	UKFaug
C_{D0}	0.123 (2.45)*	0.124 (2.50)	0.123 (2.64)	0.124 (2.55)
C_{DV}	−0.0645 (3.95)	−0.0652 (4.01)	−0.0642 (4.27)	−0.0653 (4.09)
$C_{D\alpha}$	0.320 (2.26)	0.319 (2.33)	0.320 (2.40)	0.316 (2.37)
C_{L0}	−0.0929 (21.1)	−0.0853 (23.5)	−0.087 (22.9)	−0.099 (20.0)
C_{LV}	0.149 (11.1)	0.144 (11.7)	0.147 (11.4)	0.157 (10.6)
$C_{L\alpha}$	4.328 (1.08)	4.303 (1.14)	4.289 (1.14)	4.303 (1.13)
C_{m0}	0.112 (3.27)	0.112 (4.28)	0.112 (4.29)	0.115 (3.40)
C_{mV}	0.0039 (82.1)	0.0046 (90.5)	0.0045 (92.2)	0.0022 (153)
$C_{m\alpha}$	−0.968 (1.12)	−0.971 (1.54)	−0.970 (1.54)	−0.983 (1.24)
C_{mq}	−34.710 (2.27)	−34.937 (2.85)	−35.363 (2.82)	−35.098 (2.27)
$C_{m\delta e}$	−1.529 (1.27)	−1.533 (1.65)	−1.539 (1.65)	−1.552 (1.31)

*Values in parentheses denote relative standard deviation in %.

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covariances obtained by running the filter-error method "ml_fem." This has been carried out in "/FVSysID2/chapter07/mDefCase04.m," whereby it can be recalled that the filter-error method is based on the mixed continuous-discrete system, and accordingly the factor of $1/\Delta t$ has been taken care of while specifying the process noise covariance.

We observe that the recursive methods EKF, UKF, and UKFaug converge well without encountering any numerical difficulties. Figure 7.6 shows convergence plots of a few selected parameters. The error bounds are not shown in this figure, but can be verified easily by running this test case. The numerical values of all of the derivatives, provided in Table 7.3, appear plausible and compare well with those from the filter-error method. Looking at the numerical estimates,

at their convergence plots and at the time histories of the model outputs, it appears that the performance of these methods is comparable.

By running the test case, it can be verified that the time history plots from EKF show a good match between the measured and model estimated responses. The match for the UKF and UKFaug is also good, except for the large deviations in some of the variables for the first data point only. For the reasons discussed in Sec. 7.3.2, these deviations are attributed to the large initial covariance matrix P_0^a. We note that the state variables α, θ, q in the programmed model are in terms of radians and radians/s, whereas the corresponding initial covariances are specified as 10. The subsequent parameter convergence is, however, not affected. More realistic initial covariance values in the case of UKF eliminate these discrepancies in the match and yield very similar final estimates. The same large initial covariance values were used in all three cases to bring out this specific aspect and also to allow comparison of results starting from the same conditions.

In this section we have not presented the results provided by the EFRLS method for the reasons already elaborated, namely, the method does not directly cater to nonlinear state and observation equations. By setting method = EFRLS,

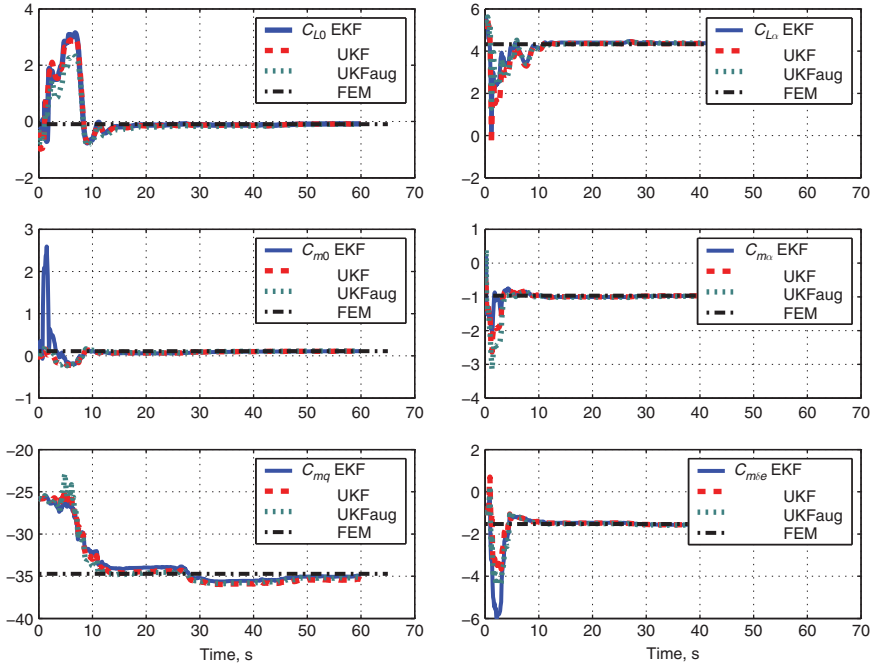

Fig. 7.6 Recursive estimates from a nonlinear state-space model.

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by changing the "if statement" in the main program "mainRPE.m" appropriately and running the same, it can be verified that the convergence is far from satisfactory.

7.6 COMPARATIVE EVALUATION OF RECURSIVE ALGORITHMS

Based on the four examples presented here, we can now try to arrive at a consolidated performance evaluation and some general conclusions. In general, the various recursive algorithms work satisfactorily in practice on real flight data, whereby certain care had to be exercised in each case.

The performance of the RLS or that of the EFRLS is affected by the forgetting factor used to deemphasize the older data. The forgetting factor leads to a faster adaptation of variations in the system, but a nonoptimal choice can lead to a poor performance or divergence. The EFRLS performed reasonably well when applied to the second and the third examples (see Tables 7.1 and 7.2). In these cases the observation model included only the states, and the state-space model is linear. In the fourth example, the performance of EFRLS was very poor. This is attributed to the fact that the state-space model is nonlinear and specifically to the observation equations, which include not only states but also other nonlinear terms and contributions from exogenous inputs. The EFRLS as implemented here does not cater to this general case. As already elaborated, the RLS and EFRLS methods do not directly yield the accuracies of the estimates as the states are propagated; the standard deviations are usually obtained once the recursive estimation is completed.

In the FTR method there are no tuning parameters, but trim values have to be removed before computing the discrete Fourier transforms, and zero aerodynamic terms cannot be estimated. Removal of trim values from the measured data is a trivial numerical computation, and inability to estimate zero aerodynamic terms is not a serious limitation. The FTR method is found to be consistent and outperforms the other methods. With lesser computational requirements and with no necessity of tuning parameters, FTR appears to be more appealing for online applications than the EFRLS, EKF, UKF, and UKFaug. It is, however, applicable to linear model postulates only.

The EKF, UKF, and UKFaug, by their basic formulation, automatically cater to linear or nonlinear state-space models. Furthermore, additional observation variables in any form, for example, the angular accelerations $\dot{p}$ and $\dot{r}$ or other output variables such as flow angles, can be easily included in the estimation, which will lead to better convergence and accuracy of the estimates.

It is noticed that the UKF performance was comparatively insensitive to the choice of the scaling factors β and κ. They help to improve the higher-order contributions, which are negligible for the models considered here. The two UKF algorithms are more involved than the EKF procedure. Let us now compare the computational time required by the three filtering algorithms. For the second

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example discussed in Sec. 7.5.2, the EKF, UKF, and UKFaug required on an average 4, 8.8, and 11.8 s, respectively; for the third example presented in Sec. 7.5.3 the required time was roughly 14.2, 32.4, and 39 s whereas for the more complex fourth model they needed 3.4, 7.6, and 14.3 s. The computational time depends upon the size of the model and on the complexity of the state equations. The preceding comparative time requirements indicate that the computational load of the more advanced UKF in its simplified version is at least twice that of the conventional EKF whereas that of the UKFaug catering to a more general case of noise affects the system nonlinearly three to four times.

The three nonlinear state estimators worked also when the initial values were fairly far off from the final estimates. Only in a very few cases when the initial errors are very large and the nonlinearities are very strong, the UKF tended to converge whereas the EKF showed convergence problems. Interestingly, the performance of the EKF and that of the UKF in terms of the convergence and parameter accuracies was very similar for the test cases presented here as well as for several others. From Tables 7.1 and 7.2, it is observed that the standard deviations of the estimates were consistently lower for the UKF than for the EKF case, but only marginally. For these two cases, they conform to the concept of improving accuracy through the sigma point propagation. But the same cannot be said for the last example presented in Table 7.3, as there is no clear trend that can be observed. In such a more realistic practical case with moderate nonlinearities and nonnegligible process and measurement noise, it is not very obvious whether any one of these methods performs better than the others. However, we recall that the process and measurement noise covariances were optimally prespecified for the filtering methods whereas filter-error method estimates noise covariances along with the system parameters. In the absence of accurate knowledge about these noise characteristics, the performance of EKF, UKF, and UKFaug can be affected. In general, it was observed that the performance of these three state estimators was not overly sensitive to noise covariances, and reasonable values usually are sufficient. As already pointed out earlier in this chapter, tuning of noise covariances is not simple.

Thus, the UKF, although theoretically quite interesting because it eliminates the flaws in EKF resulting from the linearization used in the propagation, did not show any significant advantages in terms of the convergence or achieved accuracy over the EKF. Based on our investigations, it appears that the EKF will most likely serve the purpose of estimating aerodynamic parameters from flight data. Such a conclusion has also been reached in yet another application as well [49].

7.7 CONCLUDING REMARKS

In this chapter we have discussed several basic issues related to the recursive parameter estimation, which processes the data point by point and are suitable for online applications. Both pros and cons of the basic methodology have been presented in general; in addition, critical numerical aspects are elaborated for a few

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specific algorithms. The techniques addressed here belong to the two broad categories, namely, least-squares and nonlinear filtering approach. A comparative study has been carried out on flight data analyzed in an off-line environment to estimate dimensional derivatives pertaining to the short-period and lateral-directional motion, as well as the nondimensional derivatives pertaining to lift, drag, and pitching-moment coefficients. The results from the standard off-line, iterative methods form the basis for evaluating the performance of these recursive methods. The various RPE algorithms detailed in this chapter, namely, 1) extended Kalman filter (EKF), and 2) unscented Kalman filter (UKF and UKFaug), 3) extended forgetting factor recursive least squares (EFRLS), and 4) Fourier transform regression (FTR), were evaluated for convergence, accuracy, and numerical effort. A consolidated performance evaluation together with general conclusions has been presented, which helps to arrive at a suitable choice of a method in a given case.

Balancing between the advantages, disadvantages, and limitations of recursive parameter estimation, in general, we tend to favor the off-line estimation methods discussed in Chapters 4 through 6, which are much more extensive, accurate, and robust. They are able to cover a more complex and broad spectrum of problems. The near real-time processing performed onboard using off-line algorithms on stretches of data may be an alternative. Some of these ideas were elaborated in this chapter. In some specific applications, however, recursive algorithms will be necessary, for example, when certain action is to be taken immediately in real time affecting the aircraft behavior in flight, for example, fault detection followed by adaptive control and control reconfiguration. This choice between the off-line nonrecursive and online recursive approach in a specific case underlines the fundamental philosophy of system identification, namely, we should first define the goal and the most appropriate methods should be chosen accordingly.

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REFERENCES

[1] Ljung, L., and Söderström, T., *Theory and Practice of Recursive Identification*, MIT Press, Cambridge, MA, 1983.

[2] Haykin, S., *Adaptive Filter Theory*, Prentice–Hall, Englewood Cliffs, NJ, 1991.

[3] Young, P., *Recursive Estimation and Time-Series Analysis*, Springer-Verlag, Berlin, 1984.

[4] Plackett, R. L., "Some Theorems in Least Squares," *Biometrika*, Vol. 37, No. 1/2, 1950, pp. 149–157.

[5] Sidar, M., "A Recursive on-Line Estimation Method with Application to Aircraft Dynamics Parameter Identification," *Israel Journal of Technology*, Vol. 14, 1976, pp. 56–65.

[6] "Mini-Issue on NASA's Advanced Control Law Program for the F-8 DFBW Aircraft," *IEEE Transactions on Automatic Control*, Vol. AC-22, No. 5, 1977, pp. 752–804.

[7] Ward, D. G., Monaco, J. F., and Bodson, M., "Development and Flight Testing of a Parameter Identification Algorithm for Reconfigurable Control," *Journal of Guidance, Control, and Dynamics*, Vol. 21, No. 6, 1998, pp. 948–956.

[8] Morelli, E. A., "Real-Time Parameter Estimation in the Frequency Domain," *Journal of Guidance, Control, and Dynamics*, Vol. 23, No. 5, 2000, pp. 812–818.

[9] Song, Y., Campa, G., Napolitano, M., Seanor, B., and Perhinschi, M. G., "On-Line Parameter Estimation Techniques Comparison Within a Fault Tolerant Flight Control System," *Journal of Guidance, Control, and Dynamics*, Vol. 25, No. 3, 2002, pp. 528–537.

[10] Bauer, J. E., Crawford, D. B., Gera, J., and Andrisani, D., "Real-Time Comparison of X-29A Flight Data and simulation Data," *Journal of Aircraft*, Vol. 26, No. 2, 1989, pp. 117–123.

[11] Quanwei, J., and Qiongkang, C., "Dynamic Model for Real-Time Estimation of Aerodynamic Characteristics," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 315–321.

[12] Keller, B. S., "On-Line Physical Parameter Identification and Adaptive Control of Launch Vehicle," Ph.D. Dissertation, Stanford Univ., Stanford, CA, March 1993.

[13] Laban, M., "On-Line Aircraft Aerodynamic Model Identification," Ph.D. Dissertation, Delft Univ. of Technology, Delft, The Netherlands, May 1994.

[14] Chu, Q. P., Mulder, J. A., and van Woerkom, P. T. L. M., "Modified Recursive Maximum Likelihood Adaptive Filter for Nonlinear Aircraft Flight-Path Reconstruction," *Journal of Guidance, Control, and Dynamics*, Vol. 19, No. 6, 1996, pp. 1285–1295.

[15] Zhu, Y., "Efficient Recursive State Estimator for Dynamic Systems Without Knowledge of Noise Covariances," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 35, No.1, 1999, pp. 102–114.

[16] Basappa, K., and Jategaonkar, R. V., "Evaluation of Recursive Methods for Aircraft Parameter Estimation," AIAA Paper 2004-5063, Aug. 2004.

[17] Neter, J., Kutner, M. H., Nachtsheim, C. J., and Wasserman, W., *Applied Linear Regression Models*, 3rd ed., Richard, D. Irwin, Inc., Burr Ridge, IL, 1996.

[18] Atkeson, C. G., Moore, A. W., and Schaal, S., "Locally Weighted Learning for Control," *Artificial Intelligence Review*, Vol. 11, 1997, pp. 75–113.

[19] Seanor, B., Song, Y., Napolitano, M. R., and Campa, G., "Comparison of On-Line and Off-Line Parameter Estimation Techniques Using the NASA F/A-18 HARV Flight Data," AIAA Paper 2001-4261, Aug. 2001.

[20] Moes, T. R., Smith, M. S., and Morelli, E. A., "Flight Investigation of Prescribed Independent Surface Excitations for Real-Time Parameter Identification," AIAA Paper 2003-5702, Aug. 2003.

[21] Morelli, E. A., "High Accuracy Evaluation of the Finite Fourier Transform Using Sampled Data," NASA TM 110340, June 1997.

[22] Klein, V., "Aircraft Parameter Estimation in the Frequency Domain," AIAA Paper 78-1344, Aug. 1978.

[23] Kalman, R. E., "A New Approach to Linear Filtering and Prediction Problems," *Transactions of the ASME – Journal of Basic Engineering*, Vol. 82, Series D, March 1960, pp. 35–45.

[24] Kashyap, R. L., "A New Method of Recursive Estimation in Discrete Linear Systems," *IEEE Transactions on Automatic Control*, Vol., AC-15, No. 1, 1970, pp. 18–24.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[25] Chen, R. T. N., Eulrich, B. J., and Lebacqz, J. V., "Development of Advanced Techniques for the Identification of V/STOL Aircraft Stability and Control Parameters," Cornell Aeronautical Lab., Inc., Rept. No. BM-2820-F-1, 1971.

[26] Eulrich, B. J., and Rynaski, E. G., "Identification of Nonlinear Aerodynamic Stability and Control Parameters at High Angles of Attack," AGARD CP-172, May 1975, Paper 2.

[27] Klein, V., and Schiess, J. R., "Compatibility Check of Measured Aircraft Responses Using Kinematic Equations and Extended Kalman Filter," NASA TN D-8514, Aug. 1977.

[28] Speyer, J. L., and Crues, E. Z., "On-Line Aircraft State and Stability Derivative Estimation Using the Modified-Gain Extended Kalman Filter," *Journal of Guidance and Control*, Vol. 10, No. 3, 1987, pp. 262–268.

[29] Jategaonkar, R. V., and Plaetschke, E., "Estimation of Aircraft Parameters Using Filter Error Methods and Extended Kalman Filter," DFVLR-FB 88-15, Germany, March 1988.

[30] Jategaonkar, R. V., and Plaetschke, E., "Algorithms for Aircraft Parameter Estimation Accounting for Process and Measurement Noise," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 360–372.

[31] Kokolios, A., "Use of a Kalman Filter for the Determination of Aircraft Aerodynamic Characteristics from Flight Test Data," AIAA Paper 94-0010, Jan. 1994.

[32] Plaetschke, E., Jategaonkar, R. V., Rohlf, D., and Weiss, S., "Methoden zur Schätzung der Parameter eines Flugzeugs im Post-Stall-Bereich," *Proceedings of the Symposium on 'Sicherheit im Luftverkehr,'* Technical Univ. of Braunschweig, Germany, Sept. 1994, pp. 297–306 (in German).

[33] Garcfa-Velo, J., and Walker, B. K., "Aerodynamic Parameter Estimation for High Performance Aircraft Using Extended Kalman Filtering," *Journal of Guidance, Control, and Dynamics*, Vol. 20, No. 6, 1997, pp. 1257–1259.

[34] Myers, K. A., and Tapley, B. T., "Adaptive Sequential Estimation with Unknown Noise Statistics," *IEEE Transactions on Automatic Control*, Vol. 21, No. 4, 1976, pp. 520–523.

[35] Morelli, E. A., "Estimating Noise Characteristics from Flight Test Data Using Optimal Fourier Smoothing," *Journal of Aircraft*, Vol. 32, No. 4, 1995, pp. 689–695.

[36] Ljung, L., "Asymptotic Behavior of the Extended Kalman Filter as a Parameter Estimator for Linear Systems," *IEEE Transactions on Automatic Control*, Vol. AC-24, No. 1, 1979, pp. 36–50.

[37] Julier, S., Uhlmann, J., and Durrant-Whyte, H. F., "A New Method for the Nonlinear Transformation of Means and Covariances in Filters and Estimators," *IEEE Transactions on Automatic Control*, Vol. 45, No. 3, 2000, pp. 477–482.

[38] Wan, E. A., and van der Merwe, R., "The Unscented Kalman Filter for Nonlinear Estimation," *Proceedings of the IEEE Symposium 2000 on Adaptive Systems for Signal Processing, Communication and Control AS-SPCC*, Lake Louise, Alberta, Canada, Oct. 2000, pp. 153–158.

[39] van der Merwe, R., Wan, E. A., and Julier, S., "Sigma-Point Kalman Filters for Nonlinear Estimation and Sensor-Fusion – Applications to Integrated Navigation," AIAA Paper 2004-5120, Aug. 2004.

[40] Julier, S., and Uhlmann, J. K., "Unscented Filtering and Nonlinear Estimation," *Proceedings of the IEEE*, Vol. 92, March 2004, pp. 401–422.

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[41] Wu, Y., Hu, D., Wu, M., and Hu, X., "Unscented Kalman Filtering for Additive Noise Case: Augmented Versus Nonaugmented," *IEEE Signal Processing Letters*, Vol. 12, No. 5, 2005, pp. 357–360.

[42] Lee, D.-J., and Alfriend, K. T., "Adaptive Sigma Point Filtering for State and Parameter Estimation," AIAA Paper 2004-5101, Aug. 2004.

[43] Bousson, K., and Paglione, P., "On-Line Aircraft Stability Derivative Estimation," *Proceedings of the 2001 Aerospace Congress*, Society for Automotive Engineers, Paper 2001-01-2982, Sept. 2001.

[44] Pouliezos, A. D., and Stavrakakis, G. S., *Real Time Fault Monitoring of Industrial Processes*, Kluwer Academic, Dordrecht, The Netherlands, 1994.

[45] Escobet, T., and Travé-Massuyès, L., "Parameter Estimation Methods for Fault Detection and Isolation," Bridge Workshop, Via Lattea, Italy, March 2001, pp. 1–11; also Lab. for Analysis and Architecture of Systems, LAAS Rept. No. 01471, Toulouse, France.

[46] van der Merwe, R., Wan, E. A., and Julier, S., "The Square-Root Unscented Kalman Filter for State and Parameter Estimation," *Proceedings of the International Conference on Acoustics, Speech, and Signal Processing*, Salt Lake City, Utah, May 2001, pp. 3461–3464.

[47] Brunke, S., and Campbell, M. E., "Square Root Sigma Point Filtering for Real-Time, Nonlinear Estimation," *Journal of Guidance and Control and Dynamic*, Vol. 27, No. 2, 2004, pp. 314–317.

[48] Jategaonkar, R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DLR FB 90-40, Germany, Sept. 1990.

[49] Wendel, J., Maier, A., Metzger, J., and Trommer, G. F., "Comparison of Extended and Sigma-Point Kalman Filters for Tightly Coupled GPS/INS Integration," AIAA Paper 2005-6055, Aug. 2005.

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CHAPTER 8

Artificial Neural Networks

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8.1 INTRODUCTION

In the preceding chapters we focused on parameter estimation methods suitable for phenomenological models based on the physical insight. Developing such models, which we usually prefer because it leads to better understanding of the underlying physics, can be highly demanding. This is particularly the case when the phenomenon being investigated is highly nonlinear, for example, in our case, aerodynamic effects at separated flow conditions. Because we wish to estimate and validate such models from measured input–output data, the artificial neural networks (ANN) provide an alternative approach to model building [1–7]. The ANNs are neuroscience-inspired computational tools extensively used for pattern recognition. They provide a general framework for nonlinear functional mapping of the input–output subspace and as such are part of the system identification methodology that deals implicitly with the measured input–output data. The ANNs, or simply termed NNs, are alternatively called computational neural networks to clearly differentiate them from the biological neural networks. They are also referred to as "universal approximators" or "connectionist models."

Historically, the concept of neural networks can be traced to McCulloch and Pitts, a neuroanatomist and a mathematician, who showed in 1943 that principally any arithmetic or logical function can be computed using a network consisting of simple neurons [8]. The neurons were binary switches, which were triggered in the absence of any inhibitory input when the sum of excitatory inputs exceeded a threshold. This led to realization of the Boolean functions such as "AND," "OR," or "XOR," the basic building blocks of a network. The first learning rule was subsequently formulated in 1949 by Hebb [9]. Pursuing these findings, the first perceptron, a single neuron associated with an effectiveness (weight) that could be adjusted, was developed in 1958 by Rosenblatt, a psychologist, who is widely regarded as the founder of the neural networks [10]. He also developed a learning procedure based on an error correction in the Hebbean sense and applied perceptrons for pattern recognition. The scope of applications of such

perceptrons was limited mainly because the learning was restricted to linearly separable patterns. Introduction of hidden layers increased the efficiency, but posed the problems of deriving a suitable learning rule that required a procedure to compute the desired output, which was known only at the output layer and not for the hidden layer. The major advancement was made in 1974 by Werbos, who developed the technique, called backpropagation, that allowed adjustment of weights in a hidden layer [11, 12]. After more than a decade, the procedure was re-invented in 1986 by Rumelhart et al. [13]. It turns out that the backpropagation is the same as the Kelly–Bryson gradient algorithm used in the discrete-time optimal control problems [14]. The backpropagation is an iterative procedure that allows determination of weights of an arbitrarily complex neural network.

The unique feature of ANNs is its highly interconnected structure spread over multiple layers, each with a large number of simple processing elements. A typical ANN consists of an input layer, one or more hidden layers, and an output layer. The number of nodes (that is, neurons or processing elements) in the input and output layers are automatically fixed through the numbers of input–output variables, that is, the data subspace to be matched, whereas the number of nodes in the hidden layers vary from case to case. The processing elements in the hidden layer are invariably continuous, nonlinear functions; these sigmoidal (S-shape) functions basically provide the global approximation capability to ANNs. The elements in the output layers can be linear or nonlinear. The ANNs are trained by exposing them to measured input–output data (also called patterns). Training, also called learning, is to adjust the free parameters, namely, the weights associated with the various nodal connections so that the error between the measured output and network output to the same inputs will be minimized and thus amounts a classical optimization problem. The knowledge acquired through learning gets stored in the weights thus determined. The ability of ANNs to generalize the system behavior from limited data makes them an ideal tool for characterizing complex nonlinear processes [15–17].

Several types of neural networks result from different ways of interconnecting the neurons and the way the nodes function in hidden layers. The three networks most commonly used in system identification are as follows:

1. Feedforward neural network (FFNN)
2. Recurrent neural network (RNN)
3. Radial basis function (RBF) neural network

As the name implies, FFNNs process the information in only one direction through the various layers. They consist of input, output, and usually one or two hidden layers. They are the simplest of the NNs. On the other hand, RNNs are characterized by a bidirectional flow, that is, having one or more neurons that feed data back into the network and thereby modify the inputs to the neurons. Among this class, RNN developed by Hopfield [18] is more commonly used in practice [19–23]. Because of the feedback structure, they are amenable to

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state-space model representations. A RBF is a multivariable functional interpolation that depends upon the distance of input signal vector from the center, which is determined with reference to the input distribution [24, 25]. The basic structure of RBFs, which contain only one hidden layer and a linear output layer, is similar to that of single-layer perceptron, but with a difference that the hidden-layer inputs are not weighted; instead, the hidden nodes perform the radial basis functions. In general, RBFs are powerful, efficient, and less prone to local minima whereas the performance tends to be dominated by the spread of the input–output subspace.

Yet another type of network that finds application in modeling is the so-called local model network (LMN) developed by Johansen and Foss [26]. The basic idea behind the approach is to partition the multidimensional input space into several subspaces, to which we fit linear models. They are then pieced together in a form of a simple network. A weighted sum of these local models gives the overall network output [26–29]. Use of a Gaussian weighting function ensures continuity over the partitions. Two issues involved in the use of LMN are 1) the structure characterized by a number of input space partitions, their sizes, and their location; and 2) the estimation of parameters in each of these models. An adaptive partitioning procedure allows efficient breakdown of the input space into a least number of parts, having as large a range of each variable as possible without sacrificing the accuracy of linear models. The estimation of parameters can be carried out applying any standard algorithm such as equation-error or output-error method. The approach has some advantages over the classical FFNN, namely, that physical meaning can be assigned to estimated parameters and that a priori information can be made use of. Applicability of the approach to model hinge-moment coefficients from flight data [28] and also to model nonlinearities in the lift coefficients due to stall hysteresis has been demonstrated [29], but otherwise LMNs have not yet found widespread use in aerodynamic modeling from flight data.

In the case of flight vehicles, NNs are used for a variety of different applications, such as 1) structural damage detection and identification [30], 2) fault diagnostics (detection and isolation) leading to compensation of control surface failures [31–33], 3) modeling of aerodynamic characteristics from flight data [34–39], 4) generalized reference models for six-degrees-of freedom motion simulation using global aerodynamic model including unsteady aerodynamics and dynamic stall [40–42] and for detection of unanticipated effects such as icing [43], and 5) autopilot controllers and advanced control laws for applications such as carefree maneuvering [44, 45]. Other applications include, for example, calibration of flush air data sensors [46], multisensor data fusion, mission planning and generation of mission-plan in real time [47], and modeling of unstable aircraft or aeroservoelastic effects. Although FFNNs have been used for the preceding and other purposes, engineers well versed with the so-called conventional approach always had in the past some hesitations in applying these techniques. The reservations on the overall suitability and efficacy of the NN approach stem partly from the background and outlook of the analysts, partly from the

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empirical character and black-box model structure without explicit formulation of dynamic system equations and of aerodynamic phenomenon, and partly from the uncertainties associated with extrapolation capabilities of model trained on a limited set of data. The last issue, which is still an open issue, is critical for databases used in certification and for flight safety aspects to guarantee sufficient reliability. Furthermore, incremental updates to existing NN or tuning of submodels is difficult. Nevertheless, in several applications fairly good results can be easily obtained through simple NNs, which can serve the intended purpose of mapping the overall cause–effect relationship.

It is beyond the scope of this book to provide a detailed account of various types of NN and their capabilities. We restrict ourselves only to the FFNNs, with a goal to acquire the basic understanding of these techniques for the specific purpose of aerodynamic modeling. The treatment provided here should serve as a starting point to this rapidly evolving approach, which is complementary to the conventional techniques. In this chapter we start with basics of the neural processing, concentrate on forward propagation and network training using backpropagation. Two modifications to improve the convergence are presented. This is followed by summarizing a study that suggests possible choices for the network tuning parameters such as input–output data scaling, number of nodes, learning rate, and slope of sigmoidal activation function. A brief description of a simple software implementation is provided. Finally, we address three examples of increasing complexity to demonstrate FFNN modeling capabilities. The first example considers a fairly simple case of modeling lateral-directional motion; the second example deals with modeling lift, drag, and pitching-moment coefficients from data with atmospheric turbulence. The same two examples considered in the preceding chapters provide a comparative background. The third example pertains to significantly more complex nonlinear phenomenon of aircraft undergoing quasi-steady stall.

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8.2 BASICS OF NEURAL-NETWORK PROCESSING

The basic building block of a neural network comprises a computational element called *neuron*, shown in Fig. 8.1. Each neuron receives inputs from multiple sources, denoted as u, a $(n_u \times 1)$ vector. Usually, an additive bias representing nonmeasurable input is appended to the external inputs, which is denoted by b and has some constant value. These inputs are multiplied by effectiveness factors called weights (W_i, $i = 1, \ldots, n_u$) and added up to yield the weighted sum, which passes through an activation function f to provide an output. Thus, a neuron represents a multi-input, single-output subsystem. Obviously, the input–output characteristic of a neuron depends on the weights W and the activation function f. Different types of activation function have been used in NN applications, such as linear, linear with a limiter, switching type, hyperbolic tangent, and logistic function [2]. For the reasons that we will discuss subsequently, the most common choice is the one shown in Fig. 8.1.

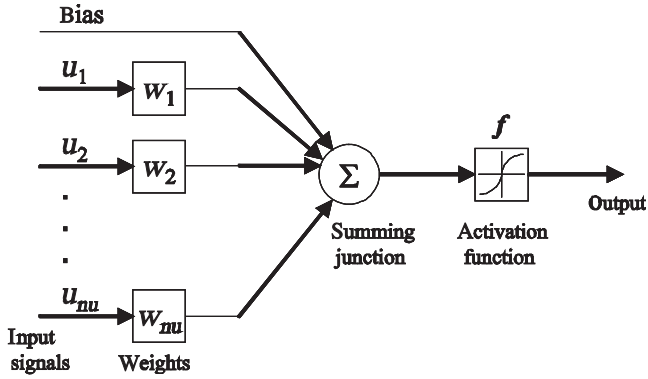

Fig. 8.1 Schematic of a single neuron.

A feedforward neural network comprises a large number of such neurons that are interconnected and arranged in multiple layers. Each node in a layer is connected to each node in the next layer. Figure 8.2 shows a schematic of FFNN with one hidden layer. The input–output space to be mapped consists of n_u inputs and n_y outputs. Accordingly, the number of nodes in the input and output layers is fixed automatically. The internal behavior of the network is

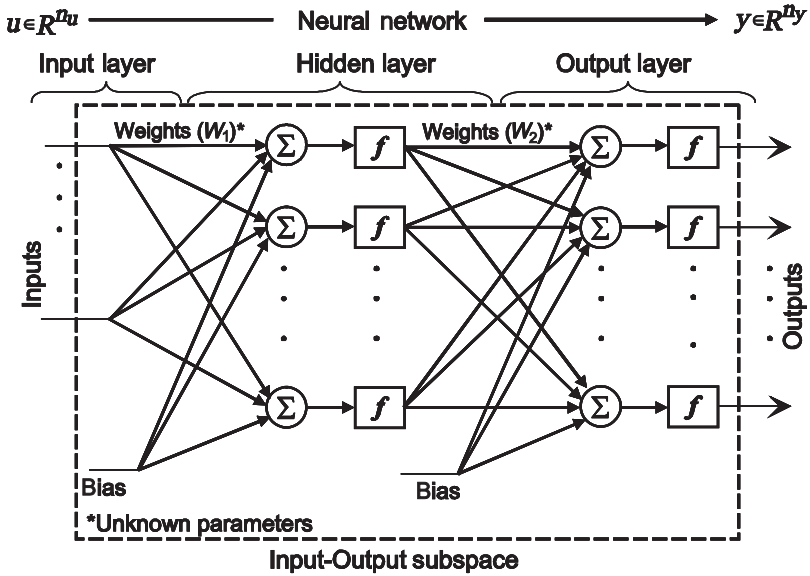

Fig. 8.2 Schematic of a feedforward neural network with one hidden layer.

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characterized by the inaccessible (hidden) layer. To be able to cater to any general nonlinear problem, it is mandatory that the activation function of the hidden layer must be nonlinear whereas that in the output layer can be linear or nonlinear. If all neurons in a NN, including those in the hidden layer, had linear activation function, the scope of modeling gets restricted only to linearly separable subspaces. In the case of NN with more than one hidden layer, the basic procedure remains the same as depicted in Fig. 8.2, wherein the output of the first hidden layer feeds the second hidden layer, the output of which then passes through the next hidden layer or through the output layer.

The neural-network performance, that is, its ability to accurately duplicate the data used in training and having adequate prediction capability, depends on the size of the networks, that is, on the number of hidden layers and on the number of neurons in each hidden layer. The larger the size, the larger is the computational effort. NN networks having more than two hidden layers are extremely rare in practice. In general, it has been shown that a FFNN with a single hidden layer and any continuous sigmoidal activation function can arbitrarily well approximate any given input–output subspace [48, 49]. The so-called sigmoidal function $f(u)$ is smooth around zero and has values between [0, 1] for inputs in the range $[-\infty, \infty]$. It is also easily differentiable, a property that is very useful in the training algorithm, which we will address in the next section. The practical investigations reported in the past have also shown that a single hidden layer FFNN with adequate number of neurons is quite sufficient for aerodynamic modeling from flight data [37]. Accordingly, we consider in this book FFNNs with only one hidden layer.

The choice of a number of neurons in the hidden layer is little trickier because the optimum number of neurons required in a specific case depends on several factors, such as number of input and outputs, the amount of noise in the data to be matched, and the complexity of the input–output subspace. Surprisingly, it also depends on the training algorithm. In the NN literature several rules of thumb have been suggested for n_h, the number of neurons in the hidden layer, such as 1) somewhere between n_u and n_y, 2) two-thirds of $(n_u + n_y)$, or 3) less than $2n_u$. The efficacy of these rules is doubtful because they do not account for the aforementioned issues of noise and complexity [7]. Too few neurons can lead to underfitting whereas too many can result in overfitting the complex subspace, in both cases affecting the generalization, that is, overall performance and predictive capability. The best approach to determine the optimum number of n_h, which we have mostly followed in the past, appears to be the one based on numerical investigations of trying out many networks with a different number of hidden neurons and choosing the one yielding minimum estimated error.

Application of FFNN to a given data consists of two parts, namely, training and prediction:

1. During the first part called *training*, the network is exposed to a given data set. The free parameters of the network, that is, the weights, are determined by

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applying some suitable algorithm. The weights are continuously adjusted until some convergence criterion is met. This learning phase, which essentially leads to characterizing the input–output characteristics, corresponds to the modeling part of the system identification.

2. During the second part called *prediction*, the same set of data or possibly other data not used in training is passed through the preceding network, keeping the weights fixed, and checking the prediction capability. In other words, this step corresponds to the model validation.

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8.3 TRAINING ALGORITHMS

Having elaborated on the basics of the NN, we now turn our attention to procedures of determining the weights. Backpropagation (BP) is the most commonly used method for this purpose; the essential idea behind this approach is to view the error as a function of network weights or parameters and to perform gradient descent in the parameter space to search for a minimum error between the desired and estimated value. Strictly speaking, backpropagation in itself is not an optimization procedure; it provides a means to compute the gradients, but we start with the output layer and apply the chain rule to obtain gradients at the previous (hidden) layers. In other words, we propagate the error backwards and hence the name of the algorithm. The actual minimization is based on the steepest descent method. The training algorithm comprises two steps:

1. Forward pass, which basically corresponds to simulation, that is, keeping the weights fixed so that the input signal is propagated through the network, which is necessary to compute the error between measured outputs and network computed responses.
2. Backward pass, which propagates the error backwards though various layers, computes the gradients, and updates the weights.

There are two common types of BP learning algorithms: 1) batch or sweep through and 2) sequential or recursive. The batch BP updates the network weights after presentation of the complete training data set. Hence, a training iteration incorporates one sweep through of all of the training patterns. In the case of recursive BP method, also called *pattern learning*, the network weights are updated sequentially as the training data set is presented point by point. We address here only the recursive BP, which is more convenient and efficient compared to the batch BP [2, 13]. A single recursive pass through the given data set is not sufficient to minimize the error; hence, the process of recursively processing the data is repeated a number of times, thus leading to a recursive-iterative process.

8.3.1 FORWARD PROPAGATION

As already pointed point, forward pass refers to the computational procedure of applying the measured input data to the network and computing the outputs. To begin with, it is required to specify starting values for the weights. They are usually initialized randomly. Let us denote the weights of a FFNN with a single hidden layer shown in Fig. 8.2 as follows [37, 50]:

W_1 weight matrix between input and hidden layer ($n_h \times n_u$)

W_{1b} bias weight vector between input and hidden layer ($n_h \times 1$)

W_2 weight matrix between hidden and output layer ($n_y \times n_h$)

W_{2b} bias weight vector between hidden and output layer ($n_y \times 1$)

where n_u is number of nodes in the input layer, n_h is the number of nodes in the hidden layer, and n_y is the number of nodes in the output layer.

Denoting the given input vector as u_0, propagation from input layer to hidden layer yields

$$y_1 = W_1 u_0 + W_{1b} \tag{8.1}$$

$$u_1 = f(y_1) \tag{8.2}$$

where y_1 is the vector of intermediate variables (output of the summing junctions), u_1 is the vector of node outputs at the hidden layer, and f is the vector of nonlinear sigmoidal node activation functions defining the node characteristics. Hyperbolic tangent function is chosen here.

$$f_i(y_1) = \tanh\left(\frac{\gamma_1}{2}\, y_1(i)\right) = \frac{1 - e^{-\gamma_1 y_1(i)}}{1 + e^{-\gamma_1 y_1(i)}}, \qquad i = 1, 2, \ldots, n_h \tag{8.3}$$

where γ_1 is the slope (gain) factor of the hidden layer activation function. All of the nodes at any particular layer have, in general, the same slope factor.

The node outputs u_1 at the hidden layer form the inputs to the next layer. Propagation of u_1 from the hidden layer to the output layer yields

$$y_2 = W_2 u_1 + W_{2b} \tag{8.4}$$

$$u_2 = f(y_2) \tag{8.5}$$

where y_2 is the vector of intermediate variables (output of the summing junctions), u_2 is the vector of node outputs at the output layer, and f is the vector of sigmoidal node activation functions. Once again, the hyperbolic tangent function is chosen for the node activation function:

$$f_i(y_2) = \tanh\left(\frac{\gamma_2}{2}\, y_2(i)\right) = \frac{1 - e^{-\gamma_2 y_2(i)}}{1 + e^{-\gamma_2 y_2(i)}}, \qquad i = 1, 2, \ldots, n_y \tag{8.6}$$

where γ_2 is the slope (gain) factor of the output layer activation function.

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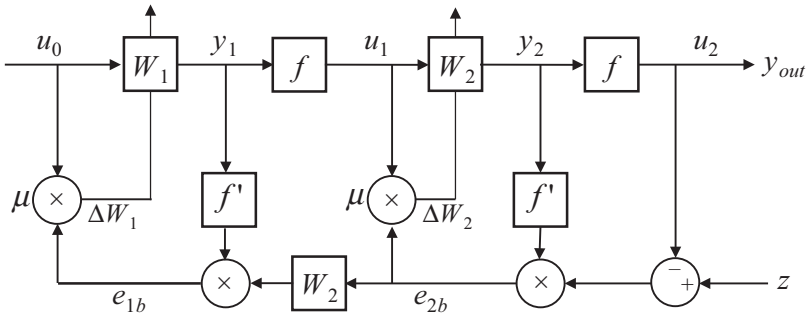

Fig. 8.3 Schematic of forward pass and backpropagation computations.

Thus, propagation of the inputs u_0 through the input, hidden, and output layers using known (fixed) weights yields the network estimated outputs u_2. The preceding computational steps of the forward pass and those of the backpropagation discussed next are shown schematically in Fig. 8.3, where the bias is not shown explicitly. The goal of optimization boils down to finding optimal values for the weights such that the outputs u_2 match with the measured system responses z.

8.3.2 STANDARD BACKPROPAGATION ALGORITHM

The backpropagation learning algorithm is based on optimizing a suitably defined cost function. At each point, the local output-error cost function, which is the sum of the squared errors, is given by

$$E(k) = \frac{1}{2}[z(k) - u_2(k)]^T[z(k) - u_2(k)] = \frac{1}{2}e^T(k)e(k) \tag{8.7}$$

where k is the discrete data index, z the measured response vector, u_2 the network estimated response vector, and $e = (z - u_2)$ the error.

Minimization of Eq. (8.7) applying the steepest descent method yields

$$W_2(k+1) = W_2(k) + \mu\left(-\frac{\partial E(k)}{\partial W_2}\right) \tag{8.8}$$

where μ is the learning rate parameter and $\partial E(k)/\partial W_2$ is the local gradient of the error cost function with respect to W_2. The learning rate is equivalent to a step size and determines the speed of convergence. A judicious choice of μ, greater than zero, is necessary to ensure reasonable convergence rate.

Substitution of Eqs. (8.4) and (8.5) in Eq. (8.7), and the partial differentiation with respect to the elements of matrix W_2, yields the gradient of the cost function.

$$\frac{\partial E(k)}{\partial W_2} = -f'[y_2(k)][z(k) - u_2(k)]u_1^T(k) \tag{8.9}$$

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where $f'[y_2(k)]$ is the derivative of the output node activation function, that is, of Eq. (8.6). Now, substituting Eqs. (8.9) in Eq. (8.8) and by defining e_{2b} as

$$e_{2b}(k) = f'[y_2(k)][z(k) - u_2(k)] \tag{8.10}$$

the weight-update rule for the output layer is obtained as

$$W_2(k+1) = W_2(k) + \mu\, e_{2b}(k)\, u_1^T(k) \tag{8.11}$$

Similarly, substituting Eqs. (8.1) and (8.2) in Eq. (8.7) and partial differentiation with respect to W_1 yields

$$\frac{\partial E(k)}{\partial W_1} = -f'[y_1(k)]\; W_2^T\; e_{2b}(k)\; u_0^T(k) \tag{8.12}$$

where $f'[y_1(k)]$ is the derivative of the hidden layer activation function, that is, of Eq. (8.3). Now, once again following a similar procedure and by defining e_{1b} as

$$e_{1b}(k) = f'[y_1(k)]\; W_2^T\; e_{2b}(k) \tag{8.13}$$

the weight-update rule for the hidden layer is obtained as

$$W_1(k+1) = W_1(k) + \mu e_{1b}(k)\, u_0^T(k) \tag{8.14}$$

For the hyperbolic tangent function, chosen as the hidden and output layer node activation functions respectively in Eqs. (8.3) and (8.6), the derivatives $f'[y_1(k)]$ and $f'[y_2(k)]$ are given by

$$f'(y_i) = \frac{\gamma_l}{2}\left[1 - \tanh^2\left(\frac{\gamma_l y_i}{2}\right)\right] = \frac{2\,\gamma_l\, e^{-\gamma_l y_i}}{(1 + e^{-\gamma_l y_i})^2} \tag{8.15}$$

where l (=1 or 2) is the index for the input-to-hidden and hidden-to-output layers.

To summarize, the complete algorithm comprises the following:

1. Computation of the hidden and output node activation functions and outputs, that is, Eqs. (8.1–8.2) and (8.4–8.5)
2. Computation of the derivatives of the node activation functions, that is, of Eq. (8.15) for hidden and output layers
3. Computation of the error functions, that is, Eqs. (8.10) and (8.13)
4. Computation of the new weights from Eqs. (8.11) and (8.14)

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These steps are recursively repeated for $k = 1, \ldots, N$, where N is number of data points. At the end of the recursive loop, the mean square error (MSE) is computed:

$$\sigma^2 = \frac{1}{N\, n_y} \sum_{k=1}^{N} \sum_{j=1}^{n_y} [z_j(k) - u_j(k)]^2 \tag{8.16}$$

Several such training iterations are carried out until the MSE satisfies some convergence criterion, for example, the relative change from iteration to iteration less than a specified value or until the maximum number of iterations is reached. Usually, a large number of iterations, ranging from few hundreds to thousands, are required to achieve good training and predictive performance.

The convergence rate of the steepest descent optimization is slow, affected by poor initial weights, and prone to finding a local minimum. It turns out that the standard BP algorithm discussed in this section might not yield training error to a level near the globally optimum value [7, 51]. Increasing the number of neurons in the hidden layer usually helps to find a good local optimum. In such a case it is necessary to ascertain the predictive capability of the oversized network. Other approaches are to modify the optimization method. There are several possible techniques, including the more advanced Levenberg–Marquardt algorithm. We restrict ourselves here to the two modifications that are commonly used in the NN applications.

8.3.3 BACKPROPAGATION ALGORITHM WITH MOMENTUM TERM

The performance of the standard BP algorithm described in the foregoing section is determined by the step size (learning rate) μ. A very small value implying very small steps leads to extremely slow convergence. On the other hand, larger value results in faster learning, but can lead to rapid changes in the direction of descent from discrete point to point resulting in parasitic oscillations, particularly when the cost function has sharp peaks and valleys. To overcome these difficulties, the update equations (8.11) and (8.14) are modified as follows:

$$W_2(k+1) = W_2(k) + \mu\, e_{2b}(k)\, u_1^T(k) + \Omega\, [W_2(k) - W_2(k-1)] \tag{8.17}$$

$$W_1(k+1) = W_1(k) + \mu\, e_{1b}(k)\, u_0^T(k) + \Omega\, [W_1(k) - W_1(k-1)] \tag{8.18}$$

where Ω is called momentum factor. As implied by the difference $[(W_l(k) - W_l(k-1)]$, the additional term in the update equations makes the optimization move along the temporal average direction used in the past iteration. Overrelaxation of the gradient update, that is, of Eqs. (8.11) and (8.14), through the last terms on right-hand side of Eqs. (8.17) and (8.18), respectively called the *momentum term*, helps to damp out parasitic oscillations in the weight

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update. As Ω is chosen greater than zero and less than one, approximately it amounts to increasing the learning rate from μ to $\mu/(1-\Omega)$ without magnifying the parasitic oscillations [2]. For this reason, the last term in the preceding equations is also called an *acceleration term*.

8.3.4 MODIFIED BACKPROPAGATION ALGORITHM

Yet another modified BP algorithm results from minimization of the mean-squared error with respect to the summing junction outputs (that is, inputs to the nonlinearities). This is in contrast to the standard BP procedure that minimizes the mean squared error with respect to the weights. We draw here heavily on the development presented in [52] to summarize the principle behind the algorithm and give only the computational steps necessary for implementation. Briefly, the error signals generated through backpropagation are used to estimate the inputs to the nonlinearities. These estimated inputs to the nonlinearities and the inputs to the respective nodes are used to produce updated weights through a set of linear equations, which are solved using a Kalman filter at each layer. The overall training procedure is similar to that of the standard BP algorithm discussed in Sec. 8.3.2, except for the weight update formulas, which require computation of the Kalman gains at each layer. Hence, we provide here only those steps that differ from the ones given in Sec. 8.3.2.

The update equations for the output layer and hidden layer are given by

$$W_2(k+1) = W_2(k) + [d(k) - y_2(k)]\, K_2^T(k) \tag{8.19}$$

$$W_1(k+1) = W_1(k) + \mu\, e_{1b}(k)\, K_1^T(k) \tag{8.20}$$

where K_2 and K_1 are the Kalman gain vectors of the size $(n_h+1, 1)$ and $(n_u+1, 1)$ associated with the hidden layer and output layer, respectively, and d is desired summation output given by

$$d(k) = \frac{1}{\gamma}\, \ell n\left(\frac{1+z(k)}{1-z(k)}\right) \tag{8.21}$$

The Kalman gains for the two layers are given by

$$K_1(k) = \frac{D_1(k)\, u_0(k)}{\lambda_1 + u_0^T(k)\, D_1(k)\, u_0(k)} \tag{8.22}$$

$$K_2(k) = \frac{D_2(k)\, u_1(k)}{\lambda_2 + u_1^T(k)\, D_2(k)\, u_1(k)} \tag{8.23}$$

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and the matrices D, representing the inverse of the correlation matrices of the training data, are given by

$$D_1(k+1) = \frac{D_1(k) - K_1(k)\, u_0^T(k)\, D_1(k)}{\lambda_1} \tag{8.24}$$

$$D_2(k+1) = \frac{D_2(k) - K_2(k)\, u_1^T(k)\, D_2(k)}{\lambda_2} \tag{8.25}$$

where λ_1 and λ_2 denote the forgetting factors. These factors allow the new information to dominate. If any one of the measured reposes $z(k)$ happens to be exactly unity, then the term $[1 - z(k)]$ in the denominator of Eq. (8.21) leads to numerical difficulties. This can be avoided by detecting such cases and simply adding a very small fraction. The other option is to use a linear activation function in the output layer [50].

The experimental investigations pertaining to neural-network aerodynamic modeling from flight data confirm the improved convergence rate compared to standard BP or that with momentum term. Although the computational overhead at each node is higher than that for the other two algorithms, the modified algorithm required lesser total overhead. It is also less sensitive to the choice of the network parameters, such as slope of the activation function or initial weights, and yields training error to a level near the globally optimum value.

Interestingly, in the context of a different estimation technique we have already come across this property of improved convergence through incorporation of Kalman gain. We recall from Sec. 5.11.1 that compared to the standard output-error method, the more advanced filter-error method incorporating Kalman or extended Kalman filter provided significantly improved convergence. In general, such approaches are less sensitive to stochastic disturbances and lead to a more nearly linear optimization problem, which has fewer local minima and a faster convergence rate.

8.4 OPTIMAL TUNING PARAMETERS

The standard BP or BP with a momentum term are characterized by a slow convergence and high sensitivity to parameters like learning rate and initial weights. There are several interrelated parameters that influence the FFNN performance: 1) input and output data scaling, 2) initial network weights, 3) number of hidden nodes, 4) learning rate, 5) momentum parameter, and 6) slope (gain) factors of the sigmoidal function. Although some heuristic rules of thumb are found in the literature, the choice of these parameters is usually based on the experimental investigations. Based on a study carried out on a typical flight data pertaining to lateral-directional motion, a FFNN with six inputs (sideslip angle β, roll rate p, yaw rate r, aileron deflection δ_a, spoiler deflection δ_{SP}, and rudder deflection δ_r) and three outputs (side-force coefficient C_Y, rolling-moment

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coefficient C_ℓ, and yawing-moment coefficient C_n), the following general guidelines are suggested [37]:

Number of hidden layers	1	(1)
Number of hidden nodes	6	(5 to 8)
Data scaling range	−0.5 to 0.5	(−0.5 to 0.5)
Nonlinear function slopes	0.85/0.6	(0.6 to 1.0)
Learning rate parameter	0.125	(0.1 to 0.3)
Momentum parameter	0.5	(0.3 to 0.5)
Initial random weights	0.3	(0.0 to 1.0)

Typical values are provided in the second column. In general, it is observed that it is adequate to choose these parameters within a certain band listed in the brackets.

For the reasons elaborated in Sec. 8.2 and supported by the preceding investigations based on networks with one and two hidden layers, FFNN with a single hidden layer is considered adequate for aerodynamic modeling from flight data. In most of the cases, less than eight neurons in the hidden layer were sufficient to have good predictive capability. In some cases, where the input–output relationship is highly nonlinear, more nodes might be necessary.

In general, it is known that minimization of an error function is affected by numerical values of the input–output data, particularly if the values differ much in magnitude. In these cases data scaling is recommended while applying the optimization methods. This is found to be particularly the case while training FFNN. Here, scaling refers to arranging the values between the chosen lower and upper limits such that all of the variables have similar order of magnitudes. It is our experience that data scaling improves significantly the convergence of FFNN training to a reasonable number of iterations. The scaling range of −0.5 to 0.5 is found to be optimal for both input and output variables.

Because the weights appearing in a FFNN have no physical significance, we cannot make use of any a priori information that might be available from classical approaches. The initial weights are normally set to small random numbers to avoid saturation in the neurons. It was observed that the algorithm does not work properly if the initial weights are either zero or poorly chosen nonzero values. Random weights in the range of zero to one led to best FFNN convergence whereas the random weights of magnitude more than one led to poor convergence in some cases. Larger weights on hidden-to-output layer than those for the input-to-hidden usually help to improve the overall training process. However, no significant differences were observed when the same factor, as suggested in the preceding list, was used.

Larger learning rate μ leads to a faster convergence, but the resulting NN might not have sufficient prediction capabilities. In addition, inappropriate

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selection of other influencing parameters in combination with a large value of learning rate can lead to oscillations. By analogy to human brain, it can be said that the network is unable to retain the complete information if it is fed in at a very fast rate. From this aspect, we tend to prefer smaller values for μ, which calls for a large number of training iterations. Learning rates in the range of 0.1 to 0.3 are found to be optimal.

The effect of momentum term, which is introduced to improve the convergence rate and damp out the larger oscillations in the network weights, is not quite obvious. If the other parameters like learning rate and initial weights are chosen properly, the influence of this parameter on the network convergence is negligible. If at all, then Ω in the range from 0.3 to 0.5 is suggested. However, in some cases, momentum parameter above 0.8 value led to complete divergence.

With regard to the slopes of the nonlinear activation functions of the hidden layer nodes (γ_1) and of the output layer nodes (γ_2), it was observed that very small slope factors close to zero have an adverse effect. It was also found that the performance was not satisfactory for the slope factor values of more than one. In general, the slope factor γ_1 of the hidden layer will have more influence on network convergence than the slope factor γ_2 of output layer nonlinear function. This observation conforms to the basic FFNN property, namely, the hidden layer provides primarily the characterization of the internal behavior of the network and global functional approximation. The type of activation function, linear or nonlinear, for the output layer plays only a secondary role. In our study we have retained nonlinear activation function, mainly because it leads to a more general representation. It has also been observed that different slope factors for hidden and output layer functions help to improve the convergence. Accordingly, we suggest using slopes of 0.85 and 0.6 for the hidden and output layers, respectively.

Finally, we once again recall from Sec. 8.3.4 that the modified BP algorithm using Kalman gains tends to be more robust to the choice of initial weights and learning rates. The momentum term is not involved, but instead we need to specify the forgetting factors λ_1 and λ_2. They should be very close to one. In typical cases, which we will address subsequently in this chapter, forgetting factor of 0.999 is used. Smaller values provided quite erratic results. The choice of a number of neurons and the effect of data scaling remains the same as already discussed. The learning rate μ may differ in the present case from that suggested earlier for the other algorithms.

8.5 EXTRACTION OF STABILITY AND CONTROL DERIVATIVES FROM TRAINED FFNN

The inability to provide physical interpretations to the weights of a trained FFNN is one of the limitations of the neural processing. In a limited sense, one can

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attempt to overcome this limitation by extracting the linearized derivatives from a trained network. This becomes possible by using the basic concepts of numerical perturbations and the definitions of the stability and control derivatives appearing in the aerodynamic force and moment coefficients. These derivatives can be thought of as the change in the aerodynamic force and moment coefficients due to a small variation in one of the motion or control variable while the rest of the variables are held constant.

The starting point to apply this approach, called Delta method [39, 53] is the availability of a trained FFNN with appropriate inputs and outputs. As considered in the preceding section, such a network for the lateral-directional motion can consist of six inputs (β, p, r, δ_a, δ_{RSP}, δ_r) and three outputs (C_Y, C_ℓ, C_n). Now, let us consider an extraction of dihedral effect, the derivative $C_{\ell\beta}$. For this purpose, we perform FFNN predictions, which consist only of the forward pass keeping the weights fixed, with two sets of perturbed β input variable, keeping the other inputs as used in the training unchanged. Let the perturbation be denoted by a constant value $\Delta\beta$. The same perturbation is used for all of the N data points. Denoting the FFNN outputs corresponding to perturbed inputs $\beta + \Delta\beta$ and $\beta - \Delta\beta$ as C_ℓ^+ and C_ℓ^- respectively, the derivative $C_{\ell\beta}$ can be approximated as

$$C_{\ell\beta} = \frac{C_\ell^+ - C_\ell^-}{2\,\Delta\beta} \tag{8.26}$$

For the reasons that we have already elaborated in Chapter 3, central differencing is preferred over the one-sided forward or backward difference approximation.

Application of Eq. (8.26) to complete data yields N values or, in other words, time histories, for the derivative to be extracted. These extracted values are plotted as histograms, which usually show a near normal distribution, from which the mean representing the aerodynamic derivative can be determined. Equivalently, a simple averaging process can also be used. Following the preceding procedure, any other derivative can be extracted from the trained FFNN.

Although the Delta method may be a viable approach, let us take a look at the practical implications. First, if we are interested in extracting linear derivatives from a given set of input–output data, the classical approach of least squares or total least squares covered in Chapter 6 is more suitable and simpler, as well as quite sufficient. The regression approach is more direct and would yield more reliable estimates than those through the roundabout way of using FFNN. Linear regression is, in fact, equivalent to a simplest form of FFNN without any hidden layer and having linear activation function. The least-squares estimates are also obtained without any a priori knowledge about the starting values. Second, we are well aware that the validity and range of applicability of linear derivatives is restricted; in some cases the linearized derivative might not even make any sense. The main advantage and strength of FFNN modeling is in its

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ability to capture highly nonlinear complex phenomenon in a global sense for which we might not be in a position to postulate models with physical understanding. In such cases only the advantages of FFNN will be fully exploited.

8.6 FFNN SOFTWARE

There are many commercial as well as free software packages available, covering different types of NNs and training algorithms [7]. Instead of using one of those, we develop simple software here based on the two backpropagation training algorithms discussed earlier in this chapter. This simple version provided here is quite sufficient to trace the procedures elaborated in Sec. 8.2 and to generate the results that we will discuss here. It also helps to provide a somewhat uniform flight data interface. The source codes (Matlab® m-files) for feedforward neural network with backpropagation are provided in the directory /FVSysID2/chapter08/.

The main program "mainFFNN" is the starting point. It provides an option to invoke test case to be analyzed and an interface to model definition function specifying the input–output data space to be modeled. Specification or assignment of the following information is necessary:

test_case	Integer flag for the test case
Zout (Ndata,Ny)	Flight data for measured outputs (N, n_y)
Xin (Ndata,Nu)	Flight data for measured control inputs (N, n_u)
dt	Sampling time

Thus, Xin and Zout define the subspace to be matched. The following related information is automatically generated from the sizes of the preceding arrays.

Nu	Number of input variables (n_u)
Ny	Number of output variables (n_y)
Ndata	Number of data points used in the training phase (N)
NdataPred	Number of data points to be used for prediction

The preceding list shows on the left side the variable names used in the program "mainFFNN" and in other functions labeled within the program followed by the description and the notation used to denote these variables in the text. If prediction is performed on the same data set used in the training, then usually NdataPred is the same as Ndata. If validation is to be performed on a completely different data set, it will be necessary to run only the forward pass, for which minor modifications will be necessary, which are left to the reader. The sampling time dt is not used in the algorithm, but used for generating time history plots. Although there is no restriction on the units of the various input and output variables, we usually load the flight data in Zout(Ndata,Ny) and Xin(Ndata,Nu) having consistent units.

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The neural-network related parameters (γ_1, γ_2, μ, Ω) are set to the recommended default values in the main program "mainFFNN.m":

γ_1	Slope of activation function from input to hidden layer
γ_2	Slope of activation function from hidden to output layer
μ	Learning rate parameter
Ω	Momentum parameter
iScale	Integer flag to scale the data
SCmin	Lower limit for data scaling
SCmax	Upper limit for data scaling

These tuning parameters might have to be adapted in a specific case to obtain optimal convergence during training and to result in a good predictive capability.

The network size (number of nodes in the hidden layer), the maximum number of iterations of the recursive-iterative training, and the training algorithm are specified interactively by keying in the following information:

NnHL1	Number of nodes in the hidden layer
itMax	Maximum number of iteration
trainALG	Training algorithm: =0: Backpropagation with momentum term =1: Modified backpropagation using Kalman gains

At the end of the neural-network training and prediction cycles, depending upon the test case, plots of the time histories of measured and estimated responses are generated, showing the predictive capability of the trained network.

8.7 EXAMPLES

To demonstrate the ability of FFNNs to model any input–output subspace, we consider here three examples of different complexities. The first one pertains to a fairly simple and classical example of modeling lateral-directional motion. In the second example, we consider the case of estimating lift, drag, and pitching-moment coefficients from flight data with atmospheric turbulence. The last example addresses a highly nonlinear aerodynamic phenomenon of quasi-steady stall.

8.7.1 MODELING OF LATERAL-DIRECTIONAL AERODYNAMICS

As a first test case, we once again turn our attention to the example considered in Secs. 6.9.2 and 7.5.1 in which the LS and RLS methods were applied to flight data from three concatenated maneuvers performed at 16,000 ft and at 200 kt nominal speed with the test aircraft ATTAS. Recall that the first maneuver is a multistep

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elevator input exciting the short-period motion, the second is an aileron input resulting in the bank-to-bank motion, and the third is a rudder doublet input exciting the Dutch roll motion. The three maneuvers are 25, 30, and 30 s long, respectively. The control inputs applied during these three flight maneuvers are found in Fig. 6.7, and a comparison of the flight-derived (that is, measured) force and moment coefficients with those estimated by the least-squares method using the model postulated in Eq. (6.76) is shown in Fig. 6.8. To these data, we apply here the neural-network technique to model the lateral-directional aerodynamics.

Now, we run this case by calling the program "/FVSysID2/chapter08/mainFFNN," which includes the same data preprocessing step as described in Sec. 6.9.2 to compute the aerodynamic force and moment coefficients This step of deriving aerodynamic force and moments coefficients C_D, C_L, C_m, C_Y, C_ℓ, C_n from measured accelerations and angular rates remains unchanged. The details are not repeated here, but they can be easily traced in the data definition function "mDefCase23.m," which calls the data preprocessing function "umr_reg_attas." We designate this case as test_case = 23.

Having derived the aerodynamic force and moment coefficients from relevant measured data, we now define the input–output subspace to be modeled using FFNN as follows:

Number of dependent variables:	3
Dependent variables:	C_Y, C_ℓ, C_n
Number of independent variables:	5
Independent variables:	β, p, r, δ_a, δ_r

The corresponding data are stored in the arrays Zout(N, n_y) and Xin(N, n_u), respectively. Accordingly, the number of nodes in the input layer are $n_u = 5$, and those in the output layer are $n_y = 3$. The length of the data (Ndata) for training is 2128, and the same length is used for prediction purpose. The recommended default values ($\gamma_1 = 0.85$, $\gamma_2 = 0.6$, $\mu = 0.125$, $\Omega = 0.5$) are used for the neural-network parameters. The number of nodes in the hidden layer $NnHL1$ (=8) and the maximum number of iterations $itMax$ (=2000) are specified interactively by keying in the values. Both training algorithms were investigated.

The choice of $NnHL1 = 8$ and $itMax = 2000$ provided typical results for predictive capability shown in Fig. 8.4. The two plots from the bottom show the aileron and ruder deflections (δ_a, δ_r), and the three plots from the top match for the side-force, rolling-moment, and yawing-moment coefficients. Because initial weights are initiated randomly, it will not be possible to reproduce exactly the same results, but several repeated trials yielded similar model quality for the match between the measured data and predicted responses. The residual error after the training phase is not the same as that obtained from the prediction step on the same data used in the training. This is because

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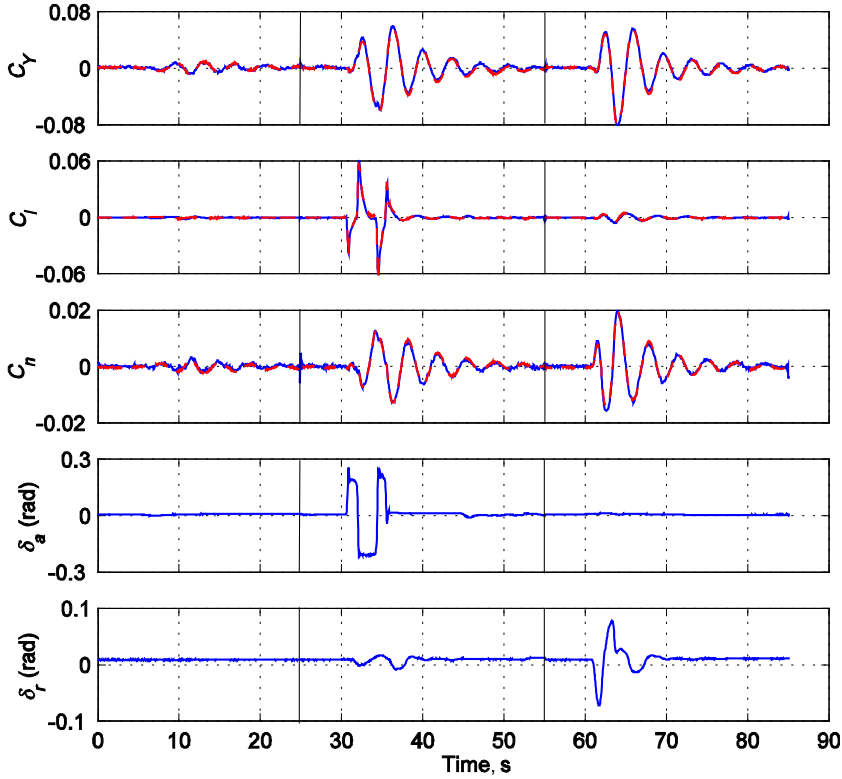

Fig. 8.4 Predictive capability of network trained on lateral-directional data: ———, flight measured; - - - - -, FFNN predicted.

the weights are altered recursively for each point during training whereas during the prediction step the estimated weights are kept fixed. It is for this reason that the prediction step needs to be performed to verify the performance over the complete data of N points.

In this particular training run, starting from the cost function value of 3×10^{-7}, the backpropagation algorithm reduced the error significantly to roughly 5×10^{-13} in less than 50 iterations, reaching a cost function value of 4.7×10^{-13} after 500 iterations and 3.5×10^{-13} after 2000 iterations. The application of the modified backpropagation algorithm using Kalman gain showed a very fast decrease in the first few iterations, but the stringent tolerances for terminations were not reached even after few hundred iterations. This is attributed to the fact that data are corrupted with noise, and after capturing the main system characteristics, the network tries to account for random noise, leading to small oscillations in the cost function. To avoid such a phenomenon, other termination

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criteria should be introduced. We also make a note of the fact that the initial cost function during the training phase at the starting iteration, even when using the same starting weights, is different for the two methods because the backpropagation that adjusts the weights recursively over N data points is done differently. The cost function minimum from the modified algorithm was observed to be slightly lower than that from the standard backpropagation with momentum term. This supports the observation made earlier that the network size and ability to capture the input–output relation also depends on the training algorithm.

8.7.2 MODELING OF LONGITUDINAL DATA WITH PROCESS NOISE

The second example pertains to modeling of lift, drag, and pitching-moment coefficients of the research aircraft HFB-320 considered in Secs. 5.11.2 and 7.5.4. It is now attempted to model these force and moment coefficients using a feedforward neural network. Because they are not directly measured, as in the preceding cases, a two-step approach is necessary.

We designate this case at Test_case = 4. The data preprocessing and definition of input–output subspace is performed in the function "mDefCase04.m," which is called from the main program /FVSysID2/chapter08/mainFFNN. The input and output variables of the data preprocessing step are given by the following:

Input variables:

ZAccel $a_x, a_y, a_z, \dot{p}, \dot{q}, \dot{r}$
Uinp $\delta_e, \delta_a, \delta_r, p, q, r, V, \alpha, \beta, \overline{q}, F_{eL}, F_{eR}$

Output variables:

Z $C_L, C_D, C_Y, C_\ell, C_m, C_n$

where all other variables have been defined in the examples discussed in the earlier chapters. Although measured signal for the dynamic pressure is available, it is computed $\overline{q} = \rho V^2/2$ using measured true airspeed signal assuming a constant value for the density of air ρ.

The flight data were gathered during a longitudinal motion excited through a multistep elevator input resulting in short-period motion and a pulse input leading to phugoid motion. The recorded data are loaded from the data files:

```
load ..\flt_data\hfb320_1_10.asc
```

and stored in the arrays Z(Ndata,Ny) and Uinp(Ndata,Nu). Because we are analyzing one maneuver and because the angular accelerations ($\dot{p}, \dot{q}, \dot{r}$) were available

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from the data-processing step, the following information pertaining to number of time segments being analyzed is not mandatory; however, for the sake of uniformity with the other examples, we specify it as follows:

```
Nzi = 1;        % Number of time segments
izhf = [Nts1];  % Cumulative index
 dt = 0.1       % Sampling time
```

where Nts1 is the number of data points and izhf is the cumulative index at which the maneuvers end when more time segments are concatenated. In the present case the total number of data points is $N = 601$. The preceding information is needed in the function "ndiff_Filter08.m" to numerically differentiate measured angular rate using described in Sec. 2.5 for concatenated multiple maneuvers. Not all signals from the input array Uinp are required in the present case, but for the sake of similarity with other similar cases it is made oversized.

Having defined the necessary details, the aerodynamic force and moment coefficients are computed according to Eqs. (6.71–6.74) in "mDefCase04.m" by calling the data-preprocessing function "umr_reg_hfb":

```
[Z, Uinp] = umr_reg_hfb(ZAccel, Uinp, Nzi, izhf, dt, test_case)
```

The computational steps in "ume_reg_hfb.m" are very similar to those described in Sec. 6.9.1 and carried out in the function "umr_reg_attas.m" for the examples discussed in Sec. 6.9.2.

Using the flight-derived aerodynamic coefficients now available from the foregoing preprocessing step in array Z and the input in the array Uinp, we now formulate the input–output subspace to be modeled by a FFNN as follows:

Number of dependent variables:	3
Dependent variables:	C_D, C_L, C_m
Number of independent variables:	4
Independent variables:	δ_e, α, q, V

The data to be modeled are stored in the arrays Zout(N, n_y), and Xin(N, n_u), respectively. Accordingly, the number of nodes in the input layer are $n_u = 4$, and those in the output layer are $n_y = 3$.

We use the complete length of the time segment (Ndata = 601) for training as well as for prediction purposes. We specify interactively the number of neurons *NnHL*1=6 and the maximum number of iterations *itMax*. Starting from the default values for the network parameters (γ_1, γ_2, μ, Ω), through trial and error more suitable values for these parameters yielding good approximation to input–output subspace as well as having good prediction capabilities were determined. They turn out to be ($\gamma_1 = 0.5$, $\gamma_2 = 0.5$, $\mu = 0.125$, $\Omega = 0.5$). Thus, smaller gains were found to be better in the present case.

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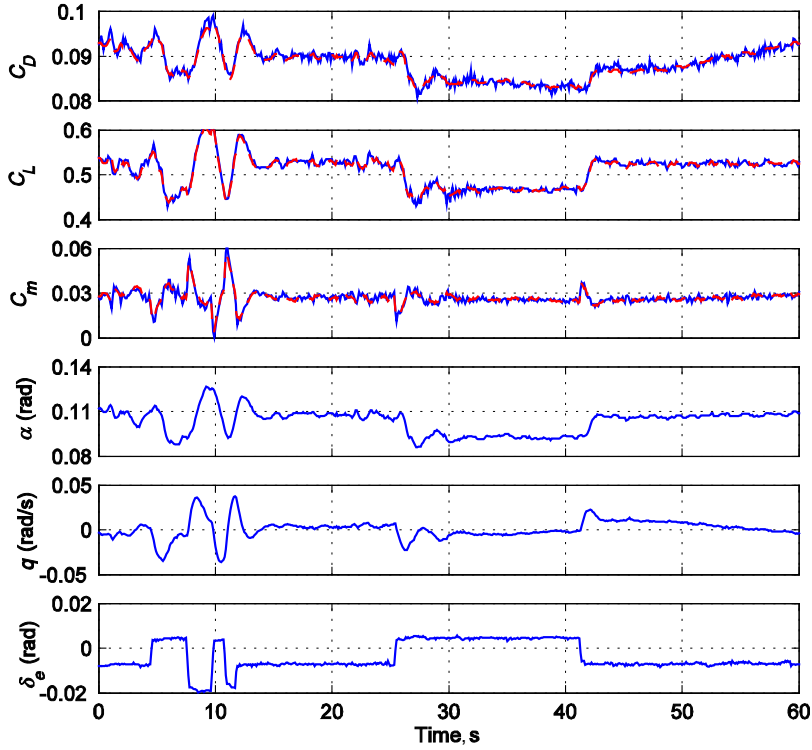

Fig. 8.5 Predictive capability of network trained on longitudinal data with atmospheric turbulence: ——, flight measured; -----, FFNN predicted.

A neural network with $NnHL1 = 6$ and $itMax = 2000$ provided typical results for predictive capability shown in Fig. 8.5. The three plots from the bottom show the input variables δ_e, q, α. (The fourth input variable V is not shown in the figure, but can be seen in Fig. 5.5.) The three plots from the top show the match for the drag, lift, and pitching-moment coefficients, respectively. The predictive capability of the trained FFNN for data with atmospheric turbulence is found to be good, which is comparable to that obtained filter-error method (see Fig. 5.6, which shows match for motion variables). Although plot is provided with $itMax = 2000$, even fewer iterations provided acceptable modeling capabilities. Both training algorithms, that is, BP with momentum term and modified BP, were tried and led to similar conclusions as pointed out for the preceding example.

8.7.3 QUASI-STEADY STALL MODELING

The last example pertains to modeling of a complex nonlinear aerodynamic phenomenon. Flight data have been gathered with the test aircraft VFW-614

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ATTAS at an altitude of 16,000 ft and for clean configuration undergoing quasi-steady stall. In this section we apply the FFNN model without any reference to physics behind the stall hysteresis. We will reconsider the same example later in Chapter 12 and apply the classical parameter estimation method to demonstrate that through advanced modeling concepts it becomes possible to extract models based on the physics of the unsteady aerodynamics.

As in the preceding two examples, the first step consists of data preprocessing to compute the aerodynamic force and moment coefficients that we wish to model. These precomputations are performed in the function "umr_reg_attas.m." The exact details can be easily traced, which are very similar to the procedure described in Sec. 6.9.2 and also used in Sec. 8.7.1. We designate this case as test_case = 27, and accordingly these steps, including definition of input–output subspace, are performed in the function "mDefCase27.m" called from the main program /FVSysID2/chapter08/mainFFNN. The recorded data from two stall maneuvers (data files \FVSysID2\flt_data\fAttas_qst01.asc and fAttas_qst02.asc) are analyzed to obtain the input–output model representation through FFNN. The number of time segments is set to Nzi = 2 and the cumulative index izhf set accordingly to [Nts1; Nts1 + Nts2], where Nts1 and Nts2 are the number of data points in the two individual time segments. We restrict ourselves to longitudinal motion only.

Having derived the aerodynamic force and moment coefficients from the relevant measured data, we now define the input–output subspace to be modeled using FFNN as follows:

Number of dependent variables:	3
Dependent variables:	C_D, C_L, C_m
Number of independent variables:	5
Independent variables:	α, q, δ_e, Ma, $\dot{\alpha}$

The data to be modeled are stored in the arrays Zout(N, n_y) and Xin(N, n_u), respectively. Accordingly, the number of nodes in the input layer are $n_u = 5$, and those in the output layer are $n_y = 3$.

We use the complete length of the data (Ndata = 5052) for the training as well as for prediction purposes. After setting the integer flag test_case = 27, we run the program "mainFFNN.m." The number of neurons $NnHL1$ and the maximum number of iterations $itMax$ are specified interactively. Through repeated trials, neural-network parameters yielding good match between measured data and FFNN outputs during training, and having adequate model predictive capability, were determined. In general, it was observed that a somewhat larger number of neurons in the hidden layer was necessary, and the gains of the activation functions were smaller.

It was also observed that finding a suitable FFNN architecture with adequate predictive capability for the lift and drag coefficients C_L and C_D was simpler and possible with network tuning parameter close to default values. The modeling

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of pitching-moment coefficient C_m, having adequate predictive capability, was more difficult to achieve. Because the goal here is limited to demonstrating the FFNN applicability to nonlinear systems and not the exact numerical results, we have not performed rigorous optimization of network tuning parameters and run the training until convergence. For a typical case with ($\gamma_1 = 0.1$, $\gamma_2 = 0.1$, $\mu = 0.3$, $\Omega = 0.5$) and $NnHL1 = 12$, a FFNN trained over 2000 iterations yields prediction plots shown in Fig. 8.6 giving the time histories. The cross plot of the lift coefficient vs angle of attack is shown in Fig. 8.7. The ability of the trained FFNN to reproduce the stall hysteresis is evident from these figures.

Both of the training algorithms were investigated. As in the preceding two examples, but for the initial faster convergence of the modified backpropagation algorithm, the two algorithms required a large number of iterations. The rate of

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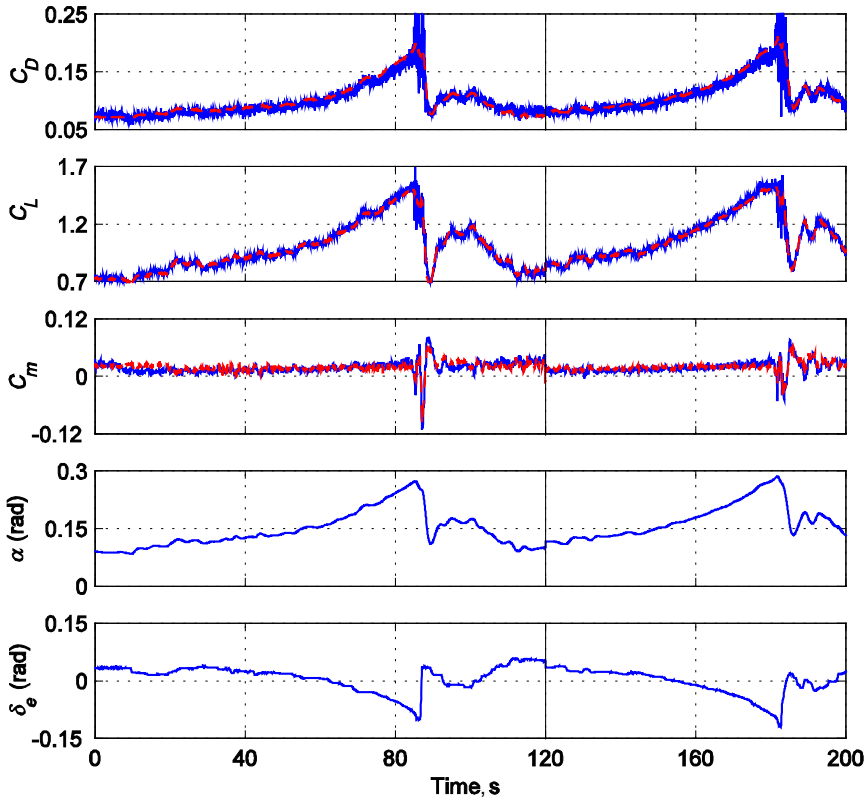

Fig. 8.6 Modeling of quasi-steady stall time histories: ———, flight derived; - - -, model estimated.

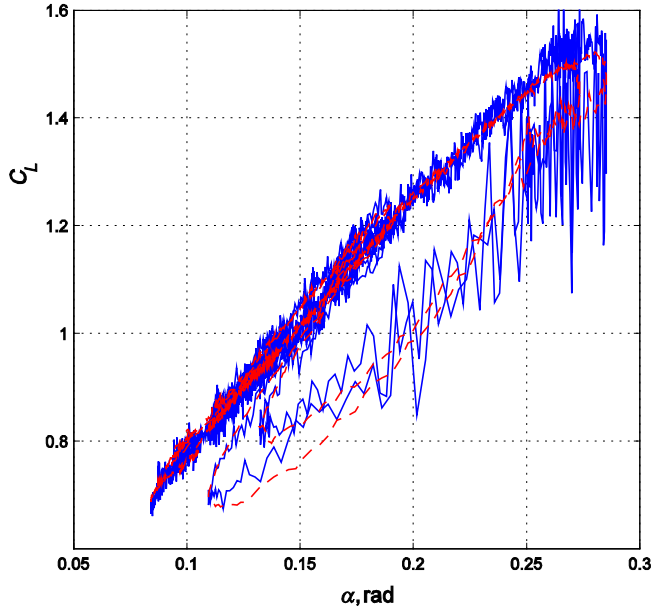

Fig. 8.7 Stall hysteresis: ———, flight derived; - - - - -, estimated.

change of angle of attack is used as one of the input variables. Although $\dot{\alpha}$ is not measured directly, it is generated through numerical differentiation in the data preprocessing step. The other alternative would to use time signals from the past, that is, $\alpha(k-1)$, $\alpha(k-2)$ etc. during the kth step [41].

In the preceding analysis, we developed a single NN with 12 nodes in the hidden layer providing functional approximation to a multi-input, multi-output ($n_u = 5$, $n_y = 3$) subspace. As mentioned earlier in this section, tuning of the pitching-moment coefficient was considerably more difficult than that of the lift and drag coefficients. To simplify the overall training task, we can use the concepts of modular neural networks (MNN) [54]. Instead of matching the multiple outputs through a single network, multiple modules are used, each consisting of a suitable neural network characterizing just one aerodynamic coefficient. Thus, the problem is broken down to submodels, each representing a multi-input, single-output subspace. This approach provides more flexibility, and it might turn out that smaller-size neural networks are adequate for each module. Usually, the training task is also simpler. It is also possible to further expand on the MNN concept to incorporate the a priori information based on the physical understanding in the network leading to structured neural networks (SNN). In such networks, instead of modeling the total coefficient as an output of a network, each derivative appearing in the aerodynamic coefficient is modeled as

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a NN. Of course, the training of such SNNs is more involved than that of the simpler FFNNs we considered here [55].

We once again highlight the fact that in none of the three examples discussed here, we have postulated any models in the classical form as required by the other estimation methods. As demonstrated here, the FFNN approach is solely based on the input–output matching using a black-box approach. It is also evident that the approach to arrive at an appropriate NN architecture, also called topology, is more or less a trial-and-error procedure. We attempt to avoid under- or overfitting the input–output subspace, which is necessary to ensure good prediction on data not used in the training and also for extrapolation on data outside the range of training data set. This generalization property depends, in general, on the network topology in terms of number of inputs and number of neurons. In other words, the degrees of freedom can be approximated as $n_h(n_u + 2)$. For good generalization, the size of the training set should be much greater than the degrees of freedom [5, 42]. Two different techniques, called *regularization* and *early stopping*, are used to improve this important property. Regularization approach minimizes a cost function comprising an error term and sum of squares of weights and thereby penalizes large weights that usually lead to poor extrapolation. The early stopping, as the name implies, is an empirical approach and aims at capturing the main input–output characteristics. In general, it is not necessary that the larger the number of training iterations the better is the trained FFNN because the network starts fitting the noise as well. Recall our observations made in Sec. 8.7 regarding oscillations in the weights as the optimum is approached. For better generalization of FFNN, the early stopping tries to avoid fitting the noise. We do not cover these and other techniques here; they can be found in any standard literature dealing explicitly with neural networks [5, 7].

8.8 CONCLUDING REMARKS

In this chapter we looked at the basic ideas behind the ANN processing and briefly touched on some of the commonly applied networks. We concentrated on just one of them, namely, the FFNN, which processes the data unidirectionally. The FFNNs are trained using a gradient-based algorithm using backpropagation. We derived the standard version of the same, which optimizes an error cost function with respect to the weights. This is followed by a short description of two extensions. The first one is based on adding a momentum term to damp out the parasitic oscillations resulting from rapid changes in the direction of descent. The other modification is based on optimizing an error cost function with respect to inputs to the activation function instead of weights, leading to incorporation of Kalman gain. These extensions are intended to speed up the convergence. A recursive-iterative approach has been used to train the FFNNs. The convergence of the modified backpropagation algorithm with Kalman gain is faster during the initial iterations and leads to a lower minimum during training

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phase. Usually, a large number of iterations are necessary to obtain trained network having adequate prediction capabilities. Near the optimum the optimization might show a tendency to oscillate, particularly when the data are noisy. Without going into exact details, possible approaches to overcome such difficulties were indicated.

In general, noise in the measurements, complexity of the data subspace, and network parameters play a significant role in NN tuning, affecting the overall convergence and performance. Based on a study pertaining to the lateral-directional motion, some typical values have been suggested for the tuning parameters, which are rough guidelines. Mostly, these parameters have to be adjusted in each specific case, as done for the three examples discussed. Several trial-and-error attempts might be necessary to arrive at an optimal combination of network tuning parameters. A more formal and systematic procedure is necessary to simplify this task.

The flexibility of FFNNs to provide global functional approximation capability is at the cost of higher overheads during the training. For the same size of the input–output data, training of FFNNs requires significantly larger computational time compared to the classical method based on the state-space model. Several hundreds to thousands of iterations are not uncommon. On the other hand, once having obtained such a NN, then the network responses can be computed more easily and with much less computational overhead. Hence, for cases in which we are not particularly interested in physics behind the phenomenon, but only in duplicating the overall input–output characteristics, the black-box model can be an alternative. Because of the universal approximation feature, the potentials of FFNN are enormous, particularly for highly complex nonlinear systems. Such networks have found certain acceptance in control applications and also in flight-simulation development. Their suitability for aerodynamic modeling from flight data has been demonstrated, but not yet evolved to an acceptance level enjoyed by the other estimation methods we have covered in the other chapters. The generalization (extrapolation) property, which depends on the NN architecture, remains an important issue. To improve upon this aspect, proper excitation of system is necessary, covering a wide range of static and dynamic data used in training. Generation of such a data rich in information contents usually needs an understanding of the physics behind the process.

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REFERENCES

[1] Zurada, J. M., *Introduction to Artificial Neural Systems*, West Publishing Co., New York, 1992.

[2] Cichocki, A., and Unbehauen, R., *Neural Networks for Optimization and Signal Processing*, Wiley, New York, 1993.

[3] Masters, T., *Practical Neural Network Recipes in C++*, Academic Press, San Diego, CA, 1993.

[4] Haykin, S., *Neural Networks – a Comprehensive Foundation*, Macmillan, New York, 1994.

[5] Hassoun, M. H., *Fundamental of Artificial Neural Networks*, MIT Press, Cambridge, MA, 1995.

[6] Reed, R. D., and Marks, R. J., II, *Neural Smithing: Supervised Learning in Feedforward Artificial Neural Networks*, MIT Press, Cambridge, MA, 1999.

[7] Sarle, W. S. (ed.), "Neural Network FAQ, Part 1 to 7," Periodic posting to the Usenet newsgroup comp.ai.neural-nets, 1997, ftp://ftp.sas.com/pub/neural/FAQ.html.

[8] McCulloch, W. S., and Pitts, W., "A Logical Calculus of the Ideas Immanent in Nervous Activity," *Bulletin of Mathematical Biophysics*, Vol. 5, 1943, pp. 115–133; reprinted McCulloch, W. S., *Embodiments of Mind*, MIT Press, Cambridge, MA, 1965, pp. 19–39.

[9] Hebb, D. O., *The Organization of Behavior: A Neuropsychological Theory*, Wiley, New York, 1949.

[10] Rosenblatt, F., "The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain," *Psychological Review*, Vol. 65, No. 6, 1958, pp. 386–408.

[11] Werbos, P. J., "Beyond Regression: New Tools for Prediction and Analysis in the Behavioral Sciences," Ph.D. Dissertation, Applied Mathematics, Harvard Univ., Cambridge, MA, 1974.

[12] Werbos, P. J., *The Roots of Backpropagation: From Ordered Derivatives to Neural Networks and Political Forecasting*, Wiley, New York, 1994.

[13] Rumelhart, D. E., Hinton, G. E., and Williams, R. J., "Learning Internal Representations by Error Propagation," *Parallel Distributed Processing: Explorations in the Microstructure Cognition, Volume 1: Foundations*, edited by D. E. Rumelhart, and J. L. McClelland, MIT Press, Cambridge, MA, 1986, pp. 318–362.

[14] Dreyfus, S. E., "Artificial Neural Networks, Back Propagation, and the Kelly-Bryson Gradient Procedure," *Journal of Guidance, Control, and Dynamics*, Vol. 13, No. 5, 1990, pp. 926–928.

[15] Narendra, K. S., and Parthasarathy, K., "Identification and Control of Dynamical Systems Using Neural Networks," *IEEE Transactions on Neural Networks*, Vol. 1, No. 1, 1990, pp. 4–27.

[16] Chu, S. R., Rahamat, S., and Tenorio, M., "Neural Networks for System Identification," *IEEE Control Systems Magazine*, April 1990, pp. 31–35.

[17] Sjöberg, J., Hjalmarsson, H., and Ljung, L., "Neural Networks in System Identification," *Proceedings of the IFAC Symposium on System Identification and Parameter Estimation*, Vol. 2, 1994, pp. 49–72.

[18] Hopfield, J. J., "Neural Networks and Physical Systems with Emergent Collective Computational Abilities," *Proceedings of the National Academy of Science*, Vol. 79, 1982, pp. 2554–2558.

[19] Raol, J. R., "Parameter Estimation of State Space Models by Recurrent Neural Networks," *IEE Proceedings on Control Theory Applications*, Vol. 142, No. 2, 1995, pp. 385–388.

[20] Raol, J. R., and Jategaonkar, R. V., "Aircraft Parameter Estimation Using Recurrent Neural Networks – A Critical Appraisal," AIAA Paper 95-3504, Aug. 1995.

[21] Shen, J., and Balakrishnan, S. N., "A Class of Modified Hopfield Networks for Aircraft Identification and Control," AIAA Paper 1996-3428, Aug. 1996.

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[22] Faller, W. E., Smith, W. E., and Huang, T. T., "Applied Dynamic System Modeling: Six Degree-of-Freedom Simulation of Forced Unsteady Maneuvers Using Recursive Neural Networks," AIAA Paper 1997-336, Jan. 1997.

[23] Hu, Z., and Balakrishnan, S. N., "Parameter Estimation in Nonlinear Systems Using Hopfield Neural Networks," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 41–53.

[24] Powell, M. J. D., "Radial Basis Functions for Multivariable Interpolations: A Review," *Algorithms for Approximations*, edited by J. C. Mason, and M. G. Cox, Oxford Univ. Press, Oxford England, U.K., 1987, pp. 143–167.

[25] Broomhead, D. S., and Lowe, D., "Multivariable Functional Interpolation and Adaptive Networks," *Complex Systems*, Vol. 2, 1988, pp. 321–355.

[26] Johansen, T. A., and Foss, B. A., "A NARMAX Model Representation for Adaptive Control Based on Local Models," *Modeling, Identification and Control*, Vol. 13, No. 1, 1992, pp. 25–39.

[27] Murray-Smith, R., "A Local Model Network Approach to Nonlinear Modeling," Ph.D. Dissertation, Univ. of Strathclyde, Glasgow, Scotland, U.K., 1994.

[28] Weiss, S., and Thielecke, F., "Aerodynamic Model Identification Using Local Model Networks," AIAA Paper 2000-4098, Aug. 2000.

[29] Giesemann, P., "Identifizierung Nichtlinearer Statischer und Dynamischer System mit Lokalmodell-Netzen," DLR FB 20001-32, Germany, Jan. 2002 (in German).

[30] Tsou, P., and Shen, M.-H. H., "Structural Damage Detection and Identification Using Neural Networks," *AIAA Journal*, Vol. 32, No. 1, 1994, pp. 176–183.

[31] Rauch, H. E., Kline-Schoder, R. J., Adams, J. C., and Youssef, H. M., "Fault Detection, Isolation, and Reconfiguration for Aircraft Using Neural Networks," AIAA Paper 1993-3870, Aug. 1993.

[32] Napolitano, M. R., Neppach, C., Casdorph, V., and Naylor, S., "Neural-Network-Based Scheme for Sensor Failure Detection, Identification, and Accommodation," *Journal of Guidance, Control, and Dynamics*, Vol. 18, No. 6, 1995, pp. 1280–1286.

[33] De Weerdt, E., Chu, Q., and Mulder, J., "Neural Network Aerodynamic Model Identification for Aerospace Reconfiguration," AIAA Paper 2005-6448, Aug. 2005.

[34] Hess, R. A., "On the Use of Back Propagation with Feed-Forward Neural Networks for the Aerodynamic Estimation Problem," AIAA Paper 93-3638, Aug. 1993.

[35] Youssef, H. M., and Juang, J.-C., "Estimation of Aerodynamic Coefficients Using Neural Networks," AIAA Paper 93-3639, Aug. 1993.

[36] Linse, D. J., and Stengel, R. F., "Identification of Aerodynamic Coefficients Using Computational Neural Networks," *Journal of Guidance, Control, and Dynamics*, Vol. 16, No. 6, 1993, pp. 1018–1025.

[37] Basappa, and Jategaonkar, R. V., "Aspects of Feed Forward Neural Network Modeling and Its Applications to Lateral-Directional Flight Data," DLR IB 111-95/30, Germany, Sept. 1995.

[38] Amin, S. M., Gerhart, V., and Rodin, E. Y., "System Identification via Artificial Neural Networks: Applications to on-Line Aircraft Parameter Estimation," AIAA Paper 97-5612, Oct. 1997.

[39] Ghosh, A. K., Raisinghani, S. C., and Khubchandani, S., "Estimation of Aircraft Lateral-Directional Parameters Using Neural Networks," *Journal of Aircraft*, Vol. 35, No. 6, 1998, pp. 876–881.

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[40] Faller, W. E., and Schreck, S. J., "Neural Networks: Applications and Opportunities in Aeronautics," *Progress in Aerospace Sciences*, Vol. 32, 1996, pp. 433–456.

[41] Rokhsaz, K., and Steck, J. E., "Use of Neural Networks in Control of High-Alpha Maneuvers," *Journal of Guidance, Control, and Dynamics*, Vol. 16, No. 5, 1993, pp. 934–939.

[42] Scharl, J., and Mavris, D., "Building Parametric and Probabilistic Dynamic Vehicle Models Using Neural Networks," AIAA Paper 2001-4373, Aug. 2001.

[43] Johnson, M. D., and Rokhsaz, K., "Using Artificial Neural Networks and Self-Organizing Maps for Detection of Airframe Icing," *Journal of Aircraft*, Vol. 38, No. 2, 2001, pp. 224–230.

[44] Napolitano, M. R., and Kincheloe, M., "On-Line Learning Neural-Network Controllers for Autopilot Systems," *Journal of Guidance, Control, and Dynamics*, Vol. 33, No. 6, 1995, pp. 1008–1015.

[45] Yavrucuk, I., Prasad, J. V. R., and Calise, A., "Adaptive Limit Detection and Avoidance for Carefree Maneuvering," AIAA Paper 2001-4003, Aug. 2001.

[46] Rohloff, T. J., Whitmore, S. A., and Catton, I., "Fault Tolerant Neural Network Algorithm for Flush Air Data Sensors," *Journal of Aircraft*, Vol. 36, No. 3, 1999, pp. 541–549.

[47] "Knowledge-Based Guidance and Control Functions," AGARD, AR-325, Jan. 1995.

[48] Cybenko, G., "Approximation by Superposition of a Sigmoidal Function," *Mathematics of Control, Signals, and Systems*, Vol. 2, 1989, pp. 303–314.

[49] Hornik, K., Stinchcombe, M., and White, H., "Universal Approximation of an Unknown Mapping and Its Derivatives Using Multilayer Feedforward Networks," *Neural Networks*, Vol. 3, 1990, pp. 551–560.

[50] Raol, J. R., and Jategaonkar, R. V., "Artificial Neural Networks for Aerodynamic Modeling," DLR IB 111-94/41, Germany, Oct. 1994.

[51] Lawrence, S., Giles, C. L., and Tsoi, A. C., "What Size Neural Network Gives Optimal Generalization? Convergence Properties of Backpropagation," Inst. for Advanced Computer Studies, Univ. of Maryland, Technical Repts. UMIACS-TR-96-22 and CS-TR-3617, College Park, MD, 1996.

[52] Scalero, R. S., and Tepedelenlioglu, N., "A Fast New Algorithm for Training Feedforward Neural Networks," *IEEE Transactions on Signal Processing*, Vol. 40, No. 1, Jan. 1992, pp. 202–210.

[53] Raisinghani, S. C., Ghosh, A. K., and Kalra, P. K., "Two New Techniques for Aircraft Parameter Estimation Using Neural Networks," *Aeronautical Journal*, Vol. 192, No. 1011, 1998, pp. 25–29.

[54] Jordan, M. I., "Modular and Hierarchical Learning Systems," *The Handbook of Brain Theory and Neural Networks*, edited by M. A. Arbib, MIT Press, Cambridge, MA, 1995.

[55] Keeman, V., "System Identification Using Modular Neural Network with Improved Learning," *Proceedings of the International Workshop on Neural Networks for Identification, Control, Robotics, and Signal/Image Processing* (NICROSP '96), IEEE Computer Society, 1996, p. 40–48.

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CHAPTER 9

Unstable Aircraft Identification

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9.1 INTRODUCTION

The demands of high-performance characteristics, for example, high maneuverability of modern flight vehicles, have led to aerodynamically unstable aircraft configurations. Advancements in modern control theory, microelectronics, and computer techniques make such basically unstable aircraft flyable. As depicted in Fig. 9.1, there are two possible approaches to unstable aircraft identification, namely, closed-loop and open-loop identification [1]. Both the approaches pose certain problems; however, the nature of the difficulties encountered is characteristically different in the two cases. The classical closed-loop identification approach to model the command to response behavior does not pose any numerical problems because the overall system is stable and any standard parameter estimation method like, for example, output error can be applied without encountering any serious difficulty. The main limitation of this approach is that it yields an "equivalent" model, giving overall system representation.

Although highly unstable aircraft can be flown only with the aid of a flight controller, that is, in closed loop, determination of aerodynamic characteristics of the basic unstable aircraft, that is, of the open-loop plant, is of primary interest in several instances: 1) validation of wind-tunnel and theoretical estimates and, if necessary, update of these predictions with those from flight data are particularly essential for new conceptual designs; 2) for flight envelope expansions; and 3) to support flight evaluation of the control systems and provide more accurate models for design modifications of the stability and control augmentation system. Open-loop parameter estimation in the time domain using the classical output-error method poses numerical difficulties because numerical integration of inherently unstable and highly sensitive system can lead to numerical divergence in the simulation and optimization [2, 3]. Such difficulties are not encountered in the analysis of marginally unstable phugoid or spiral modes. In the case of highly unstable systems, special techniques and modifications are necessary to prevent the growth of errors introduced by poor initial guess values, roundoff or discretization errors, and propagated by inherent instabilities of the system equations. Such

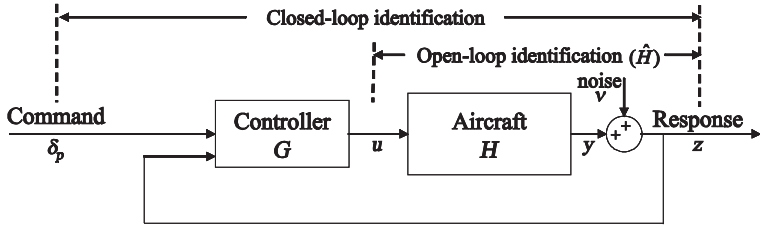

Fig. 9.1 Closed-loop and open-loop identification of unstable aircraft.

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approaches are based either on limiting the integration interval or making more efficient use of the observed data.

Assuming that we know the controller exactly (for that matter even inaccurately), identification of open-loop unstable aircraft via closed-loop identification, although feasible, is rather impractical. To obtain the open-loop parameters of the basic aircraft from such an attempt, it would require incorporating the models for the controller and actuator dynamics in the estimation procedure. As already pointed out, the overall system being stable, any standard parameter estimation method would work. With the current state of the art, even the increased model size should not be a serious problem. The primary difficulty is to obtain the exact description of the complex control laws containing discrete nonlinearities and of the actuator performance and controller gains that might be flight dependent (gain scheduling). Moreover, this approach might result in open-loop parameter estimates with low accuracy. The primary difficulties related to identifiability of the parameters are threefold:

1. The controller reacts to the motion and suppresses the oscillatory and transient motion. That is what the controller is designed for; it does its job nicely. It is, however, detrimental to parameter estimation because it reduces drastically the information content required for estimating the parameters [4].
2. Feedback leads to correlated inputs and motion variables.
3. Because the measured variables get fed back, the measurement noise in these variables acts as input thereby introducing a component of stochastic input to the system.

Apart from the modeling of aerodynamics at high angles of attack, various challenges encountered in the unstable or highly augmented aircraft identification can be broadly classified into two categories:

1. Choice of a suitable parameter estimation method
2. Improvement of parameter identifiability

Besides the least-squares techniques, extended Kalman filter, and filter-error methods, several variants based on the output-error approach are possible such as [4] a) s-plane transformation [5], b) regression startup procedure [6, 7], c) output-error method with artificial stabilization [2, 3], d) equation decoupling [8], and e) multiple-shooting based on solving a two-point boundary-value problem [7, 9]. The problems due to the correlated inputs and correlated motion variables can be mitigated by directly deflecting the control surfaces, a technique often called in the literature as separate surface excitation [4, 10]. The separate surface excitation might, however, be a complex procedure requiring hardware modifications and often flight certification. Otherwise, the alternative approach would be to attempt parameter estimation based on detection of data collinearity and mixed estimation. In such cases, however, it might be possible to obtain unbiased estimates of only a subset of parameters or combined derivatives. Moreover, the basic problems of insufficient excitation still persist.

In this chapter, we deal with the various issues related to identification of unstable or highly augmented aircraft and describe several variants of parameter estimation methods applicable to unstable aircraft and techniques to overcome the difficulties encountered in the process. They employ different techniques with different justifying arguments and with varying degrees of complexity to overcome the aforementioned numerical difficulties. These time-domain methods are applicable to both linear and nonlinear systems. In some cases the linearized form of the system equations is used, but only for the sake of clarity. We cover in some depth the issues related to parameter identifiability and suggest possible solutions. Various estimation techniques are applied to the simulated unstable aircraft data, which is provided along with this book for verification. In the majority of the cases, we use the algorithms and software that were already discussed and provided in the book. A more complex example is presented to bring out a practical application.

9.2 BASICS OF UNSTABLE AIRCRAFT IDENTIFICATION

The two possible approaches to unstable aircraft identification, namely, the closed-loop and open-loop identification, are shown in Fig. 9.1. The open-loop plant is denoted as H, the controller as G, pilot commands as δ_p, and the input to the aircraft (that is, output of the controller) as u. The process noise (atmospheric turbulence) has not been explicitly shown in the figure. The measured system response y is assumed to be corrupted by noise v. It is obvious that open-loop identification implies determination of the best approximation $\hat{H}$ of the plant model with y as the output and u as the input, that is, input/output relation y/u. The plant input u usually corresponds to the measured control surface deflections and output y to the aircraft motion variables such as angular rates, attitude angles, flow angles, and accelerations. On the other hand, closed-loop identification implies characterization of the overall system as a lumped model, that is, from

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pilot input δ_p to the measured output y. This lumped model combines the open-loop plant and controller without explicit representation of each of them. Such a lumped model characterizing the overall response is commonly termed as equivalent model.

As elaborated in Chapters 2 and 4, adequate excitation of the system and exogenous inputs (i.e., inputs generated independent of the system output) are necessary for reliable and accurate parameter estimation. In the present case of unstable aircraft, first we try to gain more insight into these basic requirements because feedback can dominate the system performance. In the following discussion, we draw heavily on the several related aspects presented in [1].

Considering the linear system representation, we can represent the basic aircraft dynamics as

$$\dot{x} = Ax + Bu + w \tag{9.1}$$

where x is the state vector, u the input (control surface deflections), and w the process noise. Assuming a proportional feedback, the control input u comprises a linear combination of state feedback and pilot command

$$u = Kx + \delta_p \tag{9.2}$$

where K is a feedback matrix. Now, rewriting Eq. (9.2) as $Kx + \delta_p - u = 0$, multiplying it with an arbitrary matrix Λ, adding the product to Eq. (9.1), and simple rearranging yield

$$\dot{x} = (A + \Lambda K)x + (B - \Lambda)u + (w + \Lambda\delta_p) \tag{9.3}$$

When open-loop identification is attempted from the measurements of x and u, the estimation method treats the last term in Eq. (9.3), that is, the sum $(w + \Lambda\delta_p)$, as noise. Provided the pilot input δ_p is large, the elements of Λ become very small and might, therefore, be neglected in all terms of Eq. (9.3). In such a case, it is then approximately the same as Eq. (9.1), and the feedback, consequently, has only little influence on the estimation results. In other words, this implies that the basic aircraft parameters (matrices A and B) can be estimated.

On the other hand, if the pilot input δ_p during the observation period is small or restricted to very short duration, the term $\Lambda\delta_p$ will be negligible compared to the system noise w even for rather large values of Λ, which cannot then be neglected. In such a case, Eq. (9.3) still has the structure similar to that of Eq. (9.1), but the estimation procedure would yield the combined matrices $(A + \Lambda K)$ and $(B - \Lambda)$, with any arbitrary Λ, instead of desired A and B.

These difficulties are caused by the fact that for small δ_p the input u (control surface deflections) is linearly dependent on the states x; see Eq. (9.2). It can be shown that in the worst case with $\delta_p = 0$ it becomes possible to estimate only the feedback gain matrix and not the system matrices [11]. In practice, where usually both pilot inputs and process noise are present, the estimation of open-loop system from closed-loop responses results in biased estimates. The ratio of

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the aircraft response due to pilot inputs and that due to noise from various sources (for example, feedback noise, atmospheric turbulence, or that generated by canards) plays a very important role and is crucial to extracting the aerodynamic derivatives accurately and reliably. Based on numerical studies, a minimum ratio of three is necessary for this purpose [11, 12].

Similar difficulties can arise when aircraft responses are correlated or become linear combinations. This correlation can be expressed in the following form:

$$x = Kx, \quad \text{with } K \neq I \tag{9.4}$$

Multiplying Eq. (9.4) with an arbitrary matrix Λ and adding the product to Eq. (9.1) yield

$$\dot{x} = [A + \Lambda(K - I)]\, x + Bu + w \tag{9.5}$$

It is obvious that an arbitrary matrix Λ is involved in the system matrix, the first term in the square brackets on the right-hand side of Eq. (9.5). Following the arguments as in the preceding case, here again it becomes difficult to determine the basic system matrix A in the presence of correlated system responses. The preceding procedure can also be applied to investigate the influence of other types of controllers, like those with differential or integrating feedback. Having obtained the basic understanding of the issues involved, we do not go into their details, except to state that the problem becomes a little more complex. How to deal with such identifiability problems is discussed in the following sections.

As already pointed out in Sec. 9.1, besides the identifiability issues just discussed, the choice of estimation method is also important for unstable aircraft. In practice, measurement noise is always present, and the feedback of such noise-corrupted measured outputs acts as a stochastic input to the system. This stochastic input introduced via feedback acts in addition to the true process noise, which might be caused by atmospheric turbulence or flow disturbances resulting from flight at high angles of attack. It leads to a much more complex situation. The simplest approach to identification of unstable aircraft is to use the regression technique or the maximum likelihood method implemented in the frequency domain. Several variants of the classical output-error method in the time domain as well as the extended Kalman filter are possible and, of course, the more advanced approach of the filter-error method.

9.3 LEAST-SQUARES METHOD

The least-squares method discussed in Chapter 6 can be readily applied to unstable aircraft because it characterizes the input–output behavior solely from the measured data and does not involve integration of the state equations. As such, the question of numerical divergence of the open-loop plant integration

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resulting from the system instability does not arise. We may recall from Sec. 6.2 that the LS estimates of the regression equation

$$Y = X\theta + \varepsilon \tag{9.6}$$

are given by the noniterative solution

$$\hat{\theta} = (X^T X)^{-1} X^T Y \tag{9.7}$$

where $Y = [y(1)\ y(2) \ldots y(N)]^T$ is the $N \times 1$ vector of dependent variable y, $\varepsilon = [\varepsilon(1)\ \varepsilon(2) \ldots \varepsilon(N)]^T$ vector of residuals, X the $(N \times n_q)$ matrix of n_q independent variables, and N the number of data points. Besides the simple noniterative solution provided by Eq. (9.7), the least-squares method has another advantage in the present case, namely, it accounts for the process noise (that is, system excitation through stochastic input) that can result, for example, from stochastic vortex shedding at the aircraft nose. As already pointed out in Sec. 6.2.2, we once again specifically draw the attention to the fact that the LS estimation accounts for the process noise because the measured variables contain responses due to deterministic inputs as well as those due to nonmeasurable stochastic inputs. Process noise is different from measurement noise. The measurement noise in the dependent variables is accounted for, but not in the independent variables. This is possible in the total least-squares estimation.

In Sec. 6.8, we addressed some of the problems encountered in the least-squares estimation, indicating that data collinearity (that is, correlation between two independent variables) leads to poor regression results [13]. In the case of unstable aircraft, data collinearity is highly pronounced due to the feedback. The integrated flight system causes high correlations between the aircraft states and controls. Hence, special care needs to be exercised while selecting the proper independent variables in the regression equation.

9.3.1 DETECTION OF DATA COLLINEARITY

Collinearity in the independent variables is mostly detected using eigensystem analysis or singular value decomposition. The correlation matrix of the independent variables or the coefficient of determination (see Sec. 6.8.2) is also sometimes used for this purpose. Theil's measure of multicollinearity is based on the coefficient of determination (that is, R-Square) from multiple partial regressions [14]. It is given by

$$R^2 - \sum_{j=1}^{n_q} (R^2 - R^2_{-j}) \tag{9.8}$$

where R^2 is the coefficient of determination given by Eq. (6.62) for the full model with n_q independent variables and R^2_{-j} that for the same model excluding the jth independent variable. The difference $(R^2 - R^2_{-j})$ is a measure of the contribution of the jth independent variable. Values of Theil's measure close

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to zero are indicators of absence of collinearity. In this connection, we might recollect some of the limitations of the R-Square criterion discussed in Sec. 6.8.2, which are likely to indirectly affect the preceding measure of multicollinearity. The correlation coefficients defined in Eq. (6.54) are also an indicator of the individual contributions. However, the correlation matrix of the independent variables is rarely used because the computed correlation coefficients are not reliable in the presence of multiple coexisting near dependencies, and the prediction of data collinearity from such a correlation matrix might not be accurate and reliable.

Eigensystem analysis based on the condition number is a more efficient and reliable approach to assess data collinearity. Here, we can recall from Sec. 4.10.3 the concept of dominant directions based on eigenvalues to avoid the ill-conditioning of the information matrix used in the Gauss–Newton optimization method. Eigenvalues smaller than ε (a small positive number), indicating so-called bad directions, were neglected to improve the convergence properties. Data collinearity is usually the most common cause of such ill-conditioning. In the present case, we apply the same philosophy to detect the ill-conditioning of the data matrix X^TX, which is the information matrix for the LS problem. The $(n_q \times n_q)$ data matrix X^TX being symmetric, it can be written as follows:

$$X^TX = T\Lambda T^{-1} \tag{9.9}$$

where Λ is the diagonal matrix of eigenvalues and T is the matrix of eigenvectors. In the present case, very small or close-to-zero eigenvalues indicate high correlation among the independent variables.

Alternatively, we can use the condition number that is a well-known measure of a matrix. It is defined as the ratio of the largest to the smallest (nonzero) eigenvalue, of the given matrix. In the present case the ratio C_i of the largest eigenvalues to the eigenvalues pertaining to each independent variable

$$C_i = \frac{|\lambda_{\max}|}{|\lambda_i|} \tag{9.10}$$

which represents a partial condition number, is used for the same purpose. Equation (9.10) yields n_q such individual condition numbers, which can be used to judge coexistence of multiple dependencies. Condition numbers close to one are the most desirable, and those very much greater than one are indicators of an ill-conditioned matrix, that is, data collinearity in the present case. We hesitate here to specify a fixed number or to suggest a rule of thumb to decide on the ratio as being large or not because condition number is relative and a function of the problem size. A wide range of values ranging from 100 to 100,000 or higher have been suggested for the ratio. Thus, some care and engineering judgment is necessary in using this criterion to avoid discarding a perfectly valid independent variable from the regression model.

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It is also possible to use the singular values of the matrix X of the independent variables, instead of the eigenvalues of X^TX. Following the notation used in Sec. 6.5, we can write

$$X = U\Sigma V^T \tag{9.11}$$

where $\Sigma = [diag(\sigma_1, \sigma_2, \ldots, \sigma_{nq})]$ is the $(n_q \times n_q)$ diagonal matrix of singular values such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{nq}$; and U and V are the $(N \times n_q)$ and $(n_q \times n_q)$ size left and right singular matrices, respectively. The condition index defined as the ratio of the largest singular values (σ_1) to the individual (nonzero) singular values is given by

$$C_i = \frac{\sigma_1}{\sigma_i} \tag{9.12}$$

Once again, a condition index close to one suggests a well-conditioned matrix or the absence of data collinearity, and large values suggest otherwise. As in the preceding case of condition numbers based on eigenvalues, here too the choice of a proper decision value is relative and not unique. Although there is no hard and fast rule to judge the condition index, ratios larger than 50 mostly indicate strong data collinearity.

It is apparent from the issues discussed in this subsection that different criteria are possible to investigate the data collinearity, but in an absolute sense none is conclusive. Proper care is necessary in the use of these and other similar test criteria to prevent elimination of a possible independent variable from the regression model.

9.3.2 ESTIMATION IN THE PRESENCE OF DATA COLLINEARITY

The general approach to deal with the data collinearity is to diagnose the cause and then react accordingly. The two main reasons leading to data collinearity are the following:

1. Chosen maneuvers (data) do not contain independent excitation of the independent variables.
2. Variables can be linearly dependent, for example, α, Θ, γ.

The possible approaches to overcome these difficulties are as follows:

1. Analyze multiple maneuvers simultaneously or other data sets.
2. Reformulate the model to reduce the size, that is, less number of parameters (less number of independent variables), probably leading to combined derivatives.
3. Use a priori information about parameter values, if available.
4. Redesign the experiment with better system excitation.

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The first choice is a trivial one because if additional data were available, they should be used anyway to improve the estimates. Additional data collected from similar flight conditions using similar inputs are not likely to improve the estimates because the cause of difficulties being data collinearity will continue to persist. The second choice based on estimating a reduced number of derivatives will be possible, but it will lead to combined derivatives; one may not get estimates of individual derivatives unless some of them are kept fixed at predefined values. Yet another option that incorporates a priori information is commonly called *mixed estimation*, which will be addressed mainly in this section. The last option of redesigning the experiment with better system excitation is probably the best choice to overcome the data collinearity problem. It is no doubt more involved and will be treated separately in Sec. 9.14.

The basic principle behind mixed estimation is very similar to that of Bayesian estimation introduced in Sec. 4.16, where the probabilistic formulation is used to incorporate the a priori values together with the associated error covariance matrix in the maximum likelihood function. Because the least-squares estimation is not based on probabilistic theory (see Sec. 6.2), we resort to an indirect way of constrained optimization to include prior information in the least-squares cost function. Referring to Eq. (9.6), if we assume that the a priori values are available for n_{ap} number of parameters ($n_{ap} < n_q$), this prior knowledge is formulated as

$$\theta^* = A\theta + \varsigma \tag{9.13}$$

where A is a $(n_{ap} \times n_q)$ matrix of known constants based on the a priori information available, ς is a vector of random variables with $E\{\varsigma\} = 0$ and $E\{\varsigma\varsigma^T\} = \sigma^2 W$, and W is a known weighting matrix. Combining Eqs. (9.6) and (9.13) leads to the mixed regression model:

$$\begin{bmatrix} Y \\ \theta^* \end{bmatrix} = \begin{bmatrix} X \\ A \end{bmatrix} \theta + \begin{bmatrix} \varepsilon \\ \varsigma \end{bmatrix} \tag{9.14}$$

Assuming σ^2 to be known [can be estimated using Eq. (6.21)] and applying the ordinary least-squares procedure to Eq. (9.14), the estimate of θ is obtained as

$$\hat{\theta}_{\mathrm{ME}} = \left(X^T X + A^T W^{-1} A\right)^{-1} \left(X^T Y + A^T W^{-1} \theta^*\right) \tag{9.15}$$

where subscript ME is introduced to denote the mixed estimates of parameters θ. A thorough treatment on mixed estimation including properties of such estimates is found in [15]. We do not go any further into these details because of the practical limitations of applying this method discussed next.

Addition of the a priori information helps to reduce the variance of the estimates and thereby improve the reliability of the estimates. Thus, mixed estimation

is an indirect way to handle the problems encountered due to data collinearity. Although mixed estimation appears to be a viable approach, we can recollect our pragmatic discussion in Sec. 4.16 on the limitations of such an approach, namely, specifying the weighting matrix and a priori values correctly. In the absence of precise knowledge about θ^*, σ, and W, Eq. (9.15) yields biased estimates [13].

9.4 TOTAL LEAST-SQUARES METHOD

As studied in Sec. 6.5, the total least-squares (TLS) method is an extension of the ordinary least-squares method and accounts for measurement noise in both the dependent variables as well as in the independent variables (regressors) [16, 17]. The TLS solution is also obtained by an analytical noniterative procedure and does not involve any integration of the state equations and hence can be readily applied to an unstable aircraft. As formulated in Eq. (6.48), the regression equation modified to account for the noise in X is given by

$$Y = (X_m - \mu)\theta + \varepsilon \tag{9.16}$$

which is solved by rank reduction using the singular value decomposition of the compounded matrix $[X\ Y]$:

$$[X\ Y] = U\Sigma V^T \tag{9.17}$$

where $\Sigma = [diag(\sigma_1, \sigma_2, \ldots, \sigma_{nq+1})]$ is the $(n_q + 1 \times n_q + 1)$ diagonal matrix of singular values such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{nq+1}$; and U and V are the left and right singular matrices. The total least-squares solution is then obtained from the column corresponding to the smallest singular value, that is, the last column of V and is given by

$$\hat{\theta}_{\text{TLS}} = -\upsilon/\lambda \tag{9.18}$$

where λ is the last component and υ is the vector of the first n_q elements of the last column of V.

Because TLS accounts for measurement noise in both the dependent variables as well as in the independent variables, theoretically, it is likely to yield better results compared to the LS method. However, in practice, the level of measurement noise introduced by the high-quality sensors is very low. Furthermore, stochastic vortex shedding at the aircraft nose might be dominant, which acts as process noise and not as measurement noise. But, we know that the measured variables automatically account for the response due to such stochastic inputs. Thus, from both these aspects, in practice the improvements obtained from TLS compared to LS might turn out to be only marginal. As regards identifiability of parameters, the problems due to presence of data collinearity persist, whether we use OLS or TLS.

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9.5 COMBINED OUTPUT-ERROR AND LEAST-SQUARES APPROACH

In Sec. 4.15 we introduced the regression startup procedure to generate better initial values for the output-error method [6, 7]. The approach was based on the general idea of using measured states in combination with parameters that were to be estimated, so that starting values were not required to be specified for these parameters. For the other constant terms or fixed parameters appearing in the system model, the integrated states were used without affecting the performance. The use of measured states makes the algorithm behave like the least-squares method, and because the LS method does not diverge, inclusion of such state measurements in the output-error method tends to prevent the numerical difficulties encountered otherwise. The same method or its variant as below can be applied to unstable aircraft. Although not guaranteed in all cases, the approach seems to work in practice.

Assuming that we know the primary state variable resulting in divergence of a particular mode, for example, vertical speed component w associated with longitudinal static instability, and an independent measurement of that state variable is available, the system equations postulated in Eqs. (4.75–4.76) can be reformulated as

$$\dot{x}(t) = A_S\, x(t) + [B(\beta); \quad A_U(\beta)] \begin{bmatrix} u(t) \\ x_m(t) \end{bmatrix} \tag{9.19}$$

$$y(t) = C_S\, x(t) + [D(\beta); \quad C_U(\beta)] \begin{bmatrix} u(t) \\ x_m(t) \end{bmatrix} \tag{9.20}$$

The state matrices A and C are resolved into two matrices each, namely, $(A_S; A_U)$ and $(C_S; C_U)$. The matrices with the subscripts S and U contain the parameter corresponding to stable and unstable states, respectively. This is in contrast to the regression startup procedure of Sec. 4.15 in which the system matrices were partitioned based on fixed and free (to be estimated) coefficients. Usually, the parameters leading to system instability are fewer than the total number of parameters to be estimated. Thus, the partitioning based on stable and unstable states uses measured states to a lesser extent. In other words, it will tend to behave more like the output-error method. Accordingly, good initial starting values will be required for such an approach. The regression startup presented in Sec. 4.15 is more robust from this viewpoint.

9.6 EQUATION DECOUPLING METHOD

The basic idea behind this method is as follows: with the aid of state measurements, reformulate the state model in such a way that each differential equation can be integrated independently [7, 8]. Continuing with the linear system representation of Eq. (4.73), partition the system matrix A into two submatrices

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denoted by A_D and A_{OD}, where A_D is diagonal containing the diagonal elements of A and the matrix A_{OD} the off-diagonal elements. Now, augmenting the control input vector u with the measured states x_m, the state equations can be rewritten as

$$\dot{x}(t) = A_D(\beta)x(t) + [B(\beta); \quad A_{\mathrm{OD}}(\beta)]\begin{bmatrix} u(t) \\ x_m(t) \end{bmatrix} \tag{9.21}$$

Because the reformulated system is decoupled, as implied by the diagonal system matrix A_D, it might change the original unstable system into a decoupled stable system. Although, strictly speaking, such a reformulation does not guarantee a stable decoupled system, in general, it is found that it does lead to stable systems in practical cases. It is possible to consider various degrees of decoupling, depending upon the resolution of matrix A.

To illustrate the concept on a typical example, we consider a system with four state variables and two control inputs. We denote the corresponding system as [18]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \tag{9.22}$$

For notational simplicity, we assume that we have measurements of just four state variables. Thus, the observation equation is given by

$$[y_1 \quad y_2 \quad y_3 \quad y_4]^T = [x_1 \quad x_2 \quad x_3 \quad x_4]^T \tag{9.23}$$

and the corresponding measurement vector by

$$[z_1 \quad z_2 \quad z_3 \quad z_4]^T = [y_1 + \nu_1 \quad y_2 + \nu_2 \quad y_3 + \nu_3 \quad y_4 + \nu_4]^T \tag{9.24}$$

where ν is the measurement noise.

Following the principle of equation decoupling introduced in Eq. (9.21), Eq. (9.22) can now be rewritten as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & 0 \\ 0 & 0 & 0 & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} 0 & a_{12} & a_{13} & a_{14} \\ a_{21} & 0 & a_{23} & a_{24} \\ a_{31} & a_{32} & 0 & a_{34} \\ a_{41} & a_{42} & a_{43} & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} \tag{9.25}$$

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Because the measurements z are corrupted by noise [see Eq. (9.24)], from Eq. (9.25) it is obvious that the off-diagonal matrix A_{OD} introduces process noise. For this reason, application of the output-error method accounting for measurement noise only (described in Chapter 4) might pose some difficulties; at least it will affect the accuracy of the estimates unless the measurement noise is so low that it can be neglected. Furthermore, in a few cases, it will also involve a consideration based on the model definition and measurements available; for example, when the postulated model contains the angle of attack or of sideslip as a state variable. These vane measurements are usually less accurate and noisy.

To overcome these uncertainties resulting from process noise, one can consider applying the filter-error method that we studied in Chapter 5. This is more likely to yield accurate estimates [18]. However, even under such a case we face a theoretical formulation difficulty. From Eqs. (9.24) and (9.25), one can infer that the process noise introduced is cross correlated. The reader might recall from Chapters 4 and 5 the assumption of independent process and measurement noise, which is not strictly met here. Nevertheless, because noise statistics are of secondary importance and also because the filter-error method is relatively insensitive to the actual distribution of process and measurement noise, the filter-error method applied to equation decoupling is likely to perform well. The necessity of applying the filter-error method, however, raises another interesting aspect that is seldom discussed. As soon as we have to apply the filter-error method, then there is no necessity of equation decoupling to integrate the states because the filter-error methods, as will be discussed in Sec. 9.8, are numerically stable due to inherent feedback. Similarly, equation decoupling implemented in the frequency domain is superfluous because the output-error method implemented in the frequency domain, as will be discussed in Sec. 9.13, is by nature stable and does not warrant any further measures to overcome numerical difficulties encountered in the time domain implementation [19].

The equation decoupling method, although illustrated on a linear system, is equally valid and applicable to nonlinear systems. The equation decoupling method, as it becomes evident, is a variation of the regression method.

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9.7 EIGENVALUE TRANSFORMATION METHOD

One of the earliest approaches to identification of unstable aircraft is based on an extension of a controller design philosophy applied to unstable systems. It is based on finding a suitable s-plane transformation to transform the basic open-loop unstable plant into a stable plant. Eigensystem analysis is one way to investigate the system stability; we know that the eigenvalues and eigenvectors of the system state matrix completely characterize the system behavior; the eigenvalues determine the nature of the transient response whereas the eigenvectors

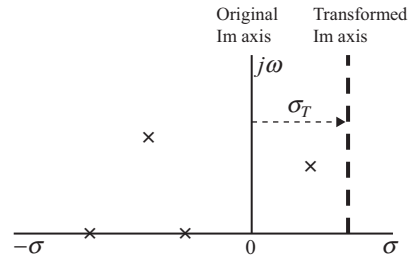

Fig. 9.2 Eigenvalue transformation approach.

determine the amplitude of that response. For an unstable plant, one or more eigenvalues have positive real parts and lie in the right half of the σ-jω plane. Shifting the eigenvalue corresponding to the largest positive real part to the left half of the s-plane and scaling the complete system proportionately would lead to a transformed system that is open-loop stable [5]. This principle of eigenvalue transformation is shown in Fig. 9.2.

Selecting a real positive σ_T larger than the largest real part of the most unstable eigenvalue, the state variables x, the observation variables y, and the control variables u are transformed through an exponential function:

$$\tilde{x}(t) = e^{-\sigma_T t}x(t), \quad \tilde{y}(t) = e^{-\sigma_T t}y(t), \quad \tilde{u}(t) = e^{-\sigma_T t}u(t) \tag{9.26}$$

where $\tilde{x}$, $\tilde{y}$, $\tilde{u}$ are the transformed state, observation, and input variables, respectively. Because we are dealing with measured data containing noise, it is also imperative to transform the measurement equation and for the same reason the process noise affecting the state variables. Accordingly,

$$\tilde{z}(t) = e^{-\sigma_T t}z(t), \quad \tilde{w}(t) = e^{-\sigma_T t}w(t), \quad \tilde{v}(t) = e^{-\sigma_T t}v(t) \tag{9.27}$$

Substituting Eqs. (9.26) and (9.27) in the system model postulate of Eqs. (3.24) and (3.25) and through simple derivation, it can be shown that the transformed system can be represented as

$$\dot{\tilde{x}}(t) = (A - \sigma_T I)\tilde{x}(t) + B\tilde{u}(t) + F\tilde{w}(t) \tag{9.28}$$

$$\tilde{y}(t) = C\tilde{x}(t) + D\tilde{u}(t) \tag{9.29}$$

$$\tilde{z}(t_k) = \tilde{y}(t_k) + G\tilde{v}(t_k) \tag{9.30}$$

The preceding transformation modifies the eigenvalues, but the relative location in the s-plane is not altered. It is obvious that the system matrix $\tilde{A} = (A - \sigma_T I)$ will have all stable eigenvalues, that is, having negative real parts because we have chosen σ_T larger than the eigenvalue with largest positive real part.

Thus, the transformed system being stable, any standard estimation algorithm, for example, the output-error method discussed in Chapter 4 can be applied without encountering numerical difficulties. For parameter estimation purposes, the given set of flight data, that is, time histories at discrete time points, of the output variables z and of the input variables u are multiplied by $e^{-\sigma_T t}$. To these transformed time histories, we fit the model given by Eqs. (9.28–9.30). Because we are interested in applying the output-error method, we neglect the process noise $w(t)$. Estimation accounting for process noise requires the filter-error

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method, which is, as will be discussed in Sec. 9.8, by its formulation numerically stable, and will not require the technique of eigenvalue transformation. It is necessary only when we would like to apply the output-error method.

In such a case, the estimates of the parameters are obtained by applying the Gauss–Newton method:

$$\sum_{k=1}^{N}\left[\frac{\partial\tilde{y}}{\partial\Theta}\right]^{T}\tilde{R}^{-1}\left[\frac{\partial\tilde{y}}{\partial\Theta}\right]\Delta\Theta=\sum_{k=1}^{N}\left[\frac{\partial\tilde{y}}{\partial\Theta}\right]^{T}\tilde{R}^{-1}\left[\tilde{z}(k)-\tilde{y}(k)\right] \tag{9.31}$$

where $\Delta\Theta$ contains the elements of the matrices $\tilde{A}$ $(=A-\sigma_T I)$, B, C, D, and possible biases. The covariance matrix of the noise $\tilde{v}$ is obtained as

$$\tilde{R}=\frac{1}{N}\sum_{k=1}^{N}\left[\tilde{z}(t_k)-\tilde{y}(t_k)\right]\left[\tilde{z}(t_k)-\tilde{y}(t_k)\right]^{T} \tag{9.32}$$

which is related to the measurement covariance matrix R through $\tilde{R}=e^{-2\sigma_T t}R$.

From Eqs. (9.28) and (9.29), we note that only the diagonal elements of state matrix A are modified through σ_T. This needs to be properly taken care of while coding the state and the observation equations. If the state equations are programmed as $\dot{\tilde{x}}=\tilde{A}\tilde{x}+B\tilde{u}$, then some care needs to be taken while coding the observation equations because the matrix C in Eq. (9.29) contains the original parameters whereas the parameters in $\tilde{A}$ appearing in Eq. (9.28), which will be estimated, are for the transformed system matrix. Therefore, if any diagonal elements appearing in the state matrix also appear in the matrix C, then they should be backtransformed before using them in the computation of the observation variables. Such a case arises when we consider accelerations as the observation variables. Failure to do so will lead to a mismatch in those particular parameters in the state and observation equations. After the estimation, for further use the parameters of the original state matrix can be obtained by backtransformation $A=\widetilde{A}+\sigma_T I$. Alternatively, we can program the state equations exactly as in Eq. (9.28), that is, as $\dot{\tilde{x}}=(A-\sigma_T I)\tilde{x}+B\tilde{u}$, in which case the data will be processed in terms of the transformed system, but the numerical estimates will correspond directly to those of the original system. We will follow the second approach, which is more direct, in the example presented in Sec. 9.16.1.

Although the approach looks promising, it has one limitation, namely, the data transformation is based on an exponentially decaying function $e^{-\sigma_T t}$, which approaches zero as t becomes larger. This limits the time length of data that can be analyzed using this method [5], not so much because of the noise properties, but mainly because the transformed variables $\tilde{z}$ and $\tilde{u}$ approach zero and thereby tend to numerically loose information content. The larger the instability, the larger will be σ_T, and the smaller will be the record length that can be appropriately analyzed using this method.

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9.8 FILTER-ERROR METHOD

The filter-error approach to parameter estimation introduced in Chapter 5 is the most general one, which basically caters to the need of analyzing flight data gathered in turbulent atmosphere [20, 21]. Two versions considering steady state and time-varying filters were introduced, and their applicability to general nonlinear systems brought out. Because the exact algorithm has already been presented in full details, we recapture here just the essentials of the state estimator using a steady-state filter discussed in Sec. 5.5.1. The state estimator consists of a prediction and a correction step given by

$$\tilde{x}(t_0) = x_0 \tag{9.33}$$

$$\tilde{y}(t_k) = g[\tilde{x}(t_k), u(t_k), \beta] \tag{9.34}$$

$$\hat{x}(t_k) = \tilde{x}(t_k) + K[z(t_k) - \tilde{y}(t_k)] \tag{9.35}$$

$$\tilde{x}(t_{k+1}) = \hat{x}(t_k) + \int_{t_k}^{t_{k+1}} f[x(t), \overline{u}(t), \beta]\mathrm{d}t \tag{9.36}$$

where $\tilde{x}$ is the predicted state vector, $\hat{x}$ the updated state vector, and K the filter–gain matrix. The zero shifts Δu and Δy have been not shown in the preceding equation but, as discussed in the earlier chapters, will have to be accounted for during the estimation.

As evident from Eq. (9.35), the state estimators incorporate measurements of the observation variables $z(t_k)$. A feedback proportional to the fit error updates the predicted state variables. This feedback stabilizes numerically the filter algorithm and also helps to improve the convergence properties of the estimation algorithm. Because of this stabilizing property of the filter-error algorithm, it can be readily applied to unstable systems without encountering numerical divergence problems. The filter feedback matrix K need not bear any relation to the control system feedbacks that actually help fly the unstable aircraft.

Thus, because of its basic formulation, the filter-error method is very attractive for unstable systems for two reasons: 1) it accounts for the stochastic input (process noise) resulting, for example, from stochastic vortex shedding at the aircraft nose; and 2) without any additional modifications it helps to overcome the divergence problems that are likely to be encountered in the numerical integration. Because the filter-error method allows estimation of process noise as well as measurement noise and simultaneously the systematic errors in the measurements as well, it provides a complete solution to the estimation problem. In practical applications, however, the approach has certain limitations discussed in Secs. 5.12 and 5.7, namely, 1) good response match tends to mask the modeling discrepancies and 2) usually the method is applicable to single maneuver or to maneuvers carried out under similar atmospheric conditions. In the present case the first limitation

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will, as in general a case, affect the model development process. The second restriction is not a serious drawback because the major problem we are addressing is to avoid numerical divergence due to system instability during analysis or maneuvers carried out under calm air or similar weather conditions.

9.9 EXTENDED AND UNSCENTED KALMAN FILTERS

For the very same reasons that we have just discussed in the case of the filter-error method, namely, incorporation of a feedback proportional to the residual error, the extended or unscented Kalman filter can also be applied to open-loop identification of unstable aircraft. The difference between the filter-error approach and the nonlinear filtering approach is that the former is a direct approach that minimizes iteratively the maximum likelihood cost function applying the Gauss–Newton method whereas EKF/UKF is an indirect approach to parameter estimation treating the unknown parameters as additional states; see Sec. 7.3.

Any suitable implementation of the EKF and UKF would serve the purpose here, for example, as given in Eqs. (7.52–7.61). Of course, the problems of specifying the measurement noise and process noise distribution matrices still persist in this case too. Improper values of such noise distribution matrices might influence the estimates adversely. Nevertheless, it is a viable approach that does not face numerical problems.

9.10 OUTPUT-ERROR METHOD

The output-error method is undoubtedly the most commonly used parameter estimation method for stable systems. Its application to unstable systems encounters many practical difficulties, not due to the theoretical formulation but mainly due to the numerical integration of unstable system in open loop, which is sensitive to the errors in the initial conditions and system parameters. In our case, because the aerodynamic parameters are unknown, added to the unknown or inaccurately known initial conditions, it can lead to a diverging solution, more often than not exceeding the numerical range of the computer. This behavior is caused by characteristic attributes of the single-shooting approach, that is, of the initial-value problems, incorporated in the output-error method.

Although extremely laborious in practice, one can attempt to apply the output-error method to unstable systems by analyzing a given maneuver in multiple steps. Starting from a very small initial portion, estimating the parameters, and then successively using the improved estimates and extending the duration of the maneuver being analyzed, in some cases it does become possible to apply the output-error method to sensitive cases such as unstable systems. In some rare cases such an approach has been applied [22]. Besides the approach being laborious, it does not guarantee convergence in every case, and it is also difficult to apply to a multirun evaluation. A more convenient approach for routine applications is desirable.

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9.11 OUTPUT-ERROR METHOD WITH ARTIFICIAL STABILIZATION

To overcome the practical difficulties just discussed in the preceding section, by analogy to the filter-error methods possessing stabilizing properties, the output-error method is extended to provide artificial stabilization [2, 3]. Although such an artificial stabilization is arbitrary and does not have the sound theoretical basis as the filter-error method, the approach is seen as a practical solution to extend the commonly accepted output-error method for parameter estimation to unstable aircraft.

Such an algorithm is parallel to the filter-error algorithm of Eq. (9.35), except that the updated state vector is now computed according to

$$\hat{x}(t_k) = x(t_k) + S[z(t_k) - y(t_k)] \tag{9.37}$$

where S is an arbitrarily selected stabilization matrix of size $(n_x \times n_y)$, assumed to be independent of the system parameters and initial conditions. This overcomes the need for computation of the filter gain matrix, which is the computationally complex and time consuming part of the filter-error method.

Two limiting values of S are the null and the identity matrix, provided the units of states x, measurements z, and model output y variables are compatible with each other. For $S = 0$, the algorithm corresponds to the output-error method. In the other case of $S = I$, it reduces to the equation-error method, provided only states are used as the observation variables. This is obvious from Eq. (9.37), in which substitution of $y = x$ in the right-hand side leads to $\hat{x} = z$, the measured states. In a more general case, when the observation vector y contains not only the states but other variables as well, in the limiting case the matrix S will not be an identity matrix, but will have to be specified appropriately.

Introduction of the stabilization matrix S affects the parameter estimates. This influence, however, will be small provided the elements of S are selected to be small. Its influence will in any case be minimized when the output error is small, that is, when modeling errors are small. Furthermore, the choice of the type of stabilization, that is, which fit error to be fed back to which state variable, is also neither defined nor obvious. This may call for some engineering judgment. Usually, fit error in the states as outputs is fed back to the states being integrated thereby increasing the static stability or the system damping. The choice of variables to be fed back might also depend on the degree of confidence in the particular measurement. The measurements of rates are more reliable than those of the flow angles. Stabilization by feeding back other variables is possible, but not so direct. Some additional care will be required in specifying the values for elements of S, if the states, measurements, and model outputs differ in their units.

In addition to stabilizing the system solution, stabilization is also necessary of the solutions to the perturbed state equations used in numerically approximating the sensitivity coefficients required in the optimization (see Sec. 4.8) [3, 7].

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By partially differentiating Eq. (9.37), it can be shown that the same stabilization matrix is to be used to stabilize the perturbed state solutions.

9.12 MULTIPLE SHOOTING METHOD

The commonly applied technique to solve estimation problems is fitting the unknown parameters in an iterative procedure incorporating a solution of an initial-value problem (single shooting). In this approach, optimization of a scalar function does not make efficient use of the primary information available to the estimation procedure, which is defined by the measured trajectories. In contrast, the multiple shooting approach accounts more efficiently for the provided information.

The basic concept of the multiple shooting approach is to subdivide the integration interval $[t_0, t_f]$ by a suitable choice of a grid:

$$g^m : \{t_0 = \tau_1 < \tau_2 < \cdots < \tau_m = t_f\} \tag{9.38}$$

into $(m - 1)$ intervals; see Fig. 9.3. The corresponding initial-value problems

$$\dot{x}(t) = f(t, x, \Theta), \quad x(\tau_j) = \sigma_j, \quad t \in [\tau_j, \tau_{j+1}] \tag{9.39}$$

with the additional variables $(\sigma_1, \ldots, \sigma_m)$ as estimates of the states $x(\tau_j)$ can be solved independently with classical integration methods, such as the Runge–Kutta algorithm. To ensure continuity of the final solution over the entire time interval, additional constraints

$$h_j(\sigma_j, \sigma_{j+1}, \Theta) = x(\tau_{j+1} | \sigma_j, \Theta) - \sigma_{j+1} = 0; \quad j = 1, \ldots, (m-1) \tag{9.40}$$

have to be satisfied. This technique transforms the original identification problem into a constrained least-squares problem:

$$\min_{\tilde{\Theta}} J(\tilde{\Theta}) \quad \text{subject to } h_j(\sigma_j, \sigma_{j+1}, \Theta) = 0 \tag{9.41}$$

where $\tilde{\Theta} = [\sigma_1^T \ \sigma_2^T \ldots \sigma_m^T \ \Theta^T]^T$ is the extended unknown parameter vector and the equality constraints h_j are defined in Eq. (9.40).

A direct solution to this problem is provided by constrained optimization. A more efficient approach is, however,

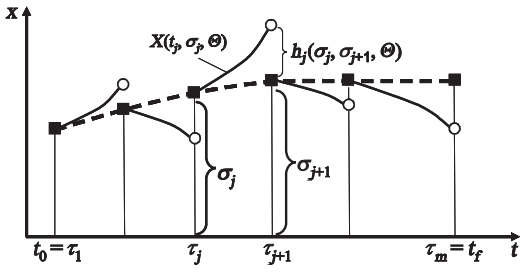

Fig. 9.3 Basic principle of the multiple shooting method.

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to recognize the special structure and to apply the Gauss–Newton method in combination with a condensation algorithm. The advantage of the multiple shooting method is that by defining additional grid points (σ_j), the computed solution remains close to the observed data. Exponential growth of the parasitic components in the solution of unstable differential equations can be restricted. The choice of grid points is one critical aspect in the application of this method influencing both estimates and the convergence. The grid points can be equidistant and can be either specified a priori or can also be determined automatically by specifying a tolerance limit for the error growth.

The foregoing approach of subdividing the integration interval into several segments is not the same as that of evaluating simultaneously multiple time records. In the multiple run evaluation applying the output-error method, the individual time segments are treated independently whereas the multiple shooting approach enforces continuity of the solution at the grid points. Although multiple shooting is a viable approach, for the reasons to be elaborated in Sec. 9.15, we have restricted the description here to indicate the approach without going into specific methodical details. A detailed description of the formulation is found in [9, 23, 24].

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9.13 OUTPUT-ERROR METHOD IN FREQUENCY DOMAIN

Because the main difficulties of the output-error method when applied to unstable systems originate from the diverging solution resulting from the integration of the state equations, an alternative approach would be to implement the method in the frequency domain. The transformation of the system equations to the frequency domain leads to a set of algebraic equations. Thus, there is no integration of the states involved in the frequency domain, which automatically eliminates the numerical divergence problems resulting from the unstable eigenvalues. Assuming periodic signals, that is, $x(0) = x(T)$, and applying the Fourier transformation, the system equations (3.24) and (3.25) get transformed into [4, 25, 26]

$$j\omega x(\omega) = A(\beta)\, x(\omega) + B(\beta)\, u(\omega) \tag{9.42}$$

$$y(\omega) = C(\beta)\, x(\omega) + D(\beta)\, u(\omega) \tag{9.43}$$

$$z(\omega_l) = y(\omega_l) v(\omega_l), \quad l = 1, \ldots, N \tag{9.44}$$

where $x(\omega)$, $u(\omega)$, $y(\omega)$ are the Fourier transformed state, input, and observation variables; $\omega_l = 2\pi\, l/T$ is the lth discrete frequency; and $T = N\Delta t$. The cost function to be minimized is then given by

$$J_{\mathrm{FR}} = \sum_{l=1}^{M} [z(\omega_l) - y(\omega_l)]^* S_{vv}^{-1} [z(\omega_l) - y(\omega_l)] + \log|S_{vv}| \tag{9.45}$$

where M is the number of frequencies to be evaluated, S_{vv} is the spectral density matrix of the measurement noise, and the subscript FR is appended to the cost function to distinguish between the frequency domain and time domain. Minimization of Eq. (9.45) by the Gauss–Newton method yields the maximum likelihood estimates of the parameters β, fitting the Fourier coefficients of the measured data and the model response. Note that the estimated parameters β are still the same stability and control derivatives as in the time-domain method and have the same physical definition. They do not represent or correspond in any form to transfer function models, which are commonly used for model representation in the frequency domain. The distinctive features of such an approach are as follows:

1. Optimization amounts to fitting the frequency spectra, involving algebraic equations; hence, instabilities do not affect the estimation.
2. Without affecting the estimation results, the zero frequency can be neglected in the evaluation, which is advantageous in eliminating the need to estimate a large number of initial conditions and bias terms, and thereby help to reduce the problems of correlation between the bias parameters and aerodynamic derivatives. For multirun evaluations, bias parameters and initial conditions often far exceed the number of aerodynamic derivatives.
3. As recorded flight data usually do not meet the requirements of being periodic, extensions to Eqs. (9.42–9.45) are necessary to include corrections terms that allow use of nonperiodic data [26, 27].

Although the estimated parameters are directly the stability and control derivatives, the computed responses are in the frequency domain. For comparison of time histories, they have to be backtransformed into the time domain, which is not critical because it does not involve numerical integration. Although the output-error method implemented in the frequency domain has some advantages, the major limitation is that it is applicable to only linear systems. For unstable aircraft, however, it provides a useful alternative approach.

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9.14 SEPARATE SURFACE EXCITATION

As pointed out in Sec. 9.3.2, better system excitation is probably the best choice to reduce data collinearity. Better excitation of unstable aircraft for the specific purpose of parameter estimation is possible through separate surface excitation (SSE), also sometimes termed as single surface excitation [10]. The basic idea behind this approach is to feed separate inputs, bypassing the flight control system, to directly deflect the individual control surfaces. In most of the cases, it is expected that with minor modifications or through judicious use of existing hardware it would be possible to inject such controlled inputs. For example, in the case of the X-31A aircraft, the flutter test box was used advantageously for this

purpose by integrating a dedicated signal generating card; see Fig. 9.4. Although theoretically it would be possible to inject extra inputs simultaneously to multiple control surfaces, it is preferable to excite them one at a time for the very reason of reducing collinearity.

Reduction of data collinearity through SSE is shown in Fig. 9.5. This figure shows time history plots of 1) pitch and PID command, 2) angle of attack, 3) canard deflection, and 4) thrust-vector (TV) deflection and symmetric trailing-edge flaps for three independent flight maneuvers, demarcated by vertical lines. The first maneuver pertains to a pitch doublet commanded by the pilot. It is seen that the TV deflection and symmetric trailing-edge flaps are completely correlated with each other and also with the canard deflection. The second and the third maneuvers show SSE inputs, first to the canard and then to the symmetric trailing-edge flaps (that is, elevator). From the second maneuver it is observed that the canard deflection is now different compared to the TV and symmetric trailing-edge flaps. During the third maneuver, a separate input to trailing-edge flaps forces elevator deflection uncorrelated with the TV deflection. Thus, combining these two SSE maneuvers provides adequate information to estimate the respective control surface effectiveness.

It is obvious that during such SSE input maneuvers the feedback is active, which will react in due time to damp out the dynamic motion resulting from SSE and also result in correlated inputs and motion variables. However, initial aircraft response to the directly fed inputs is characteristically different compared to that if the same inputs were fed through the flight control system. This initial response and decoupled control surface deflections allow accurate estimation of the individual control surface effectiveness. Furthermore, improved estimation of the control surface effectiveness combined with reduced correlation

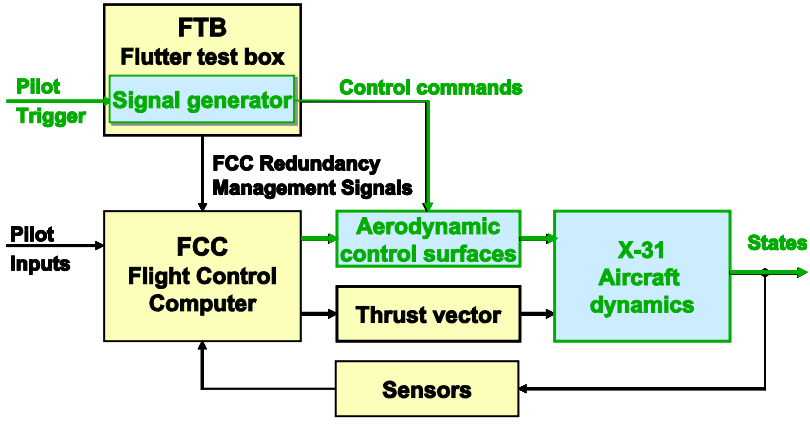

Fig. 9.4 Realization of separate surface excitation.

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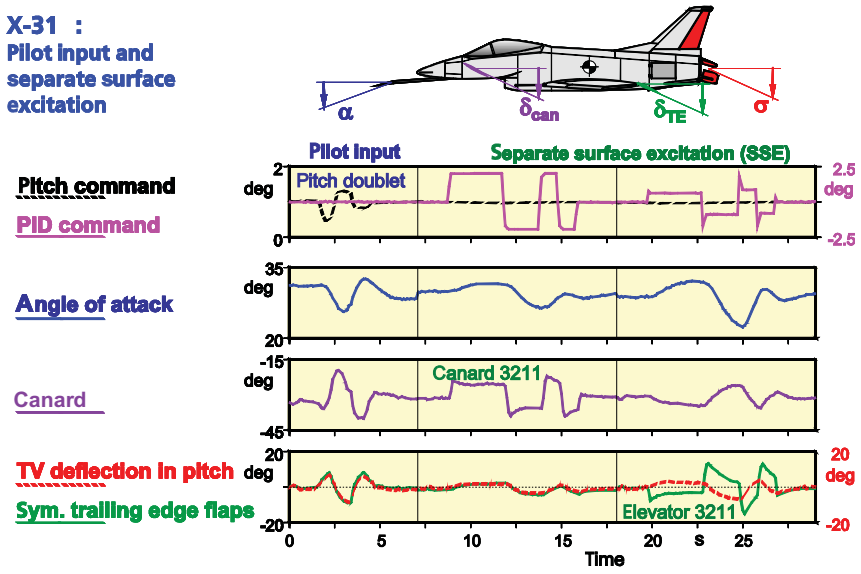

Fig. 9.5 Reduction of data collinearity through separate surface excitation.

between the motion variables and control surface deflections leads to improved estimation of the other derivatives as well. Figure 9.6 shows the estimate $C_{m\delta can}$, pitching moment due to canard, obtained from pilot input maneuvers, and those from SSE input maneuvers [4, 10]. The vertical bars denote the standard deviations. It is clearly seen that the pilot input maneuvers yield estimates with large standard deviations and the scatter in the estimates is large too. On the other hand, SSE input maneuvers provide more accurate and reliable estimates as implied by the low standard deviations (vertical bars) and significantly reduced scatter.

The brief discussion of SSE provided in this section is typical of the issues pertaining to excitation of highly augmented unstable aircraft. For a detailed treatment of various practical issues in connection with the X-31A aircraft, the reader is referred to [10, 28, 29].

9.15 PROGRAMMING CONSIDERATIONS

To compare the complexities of modifying existing estimation software or writing new software to provide capabilities of analyzing unstable systems, it is reasonable to assume that software for the output-error algorithm is available, including the one provided in this book. Application of regression startup or of the equation

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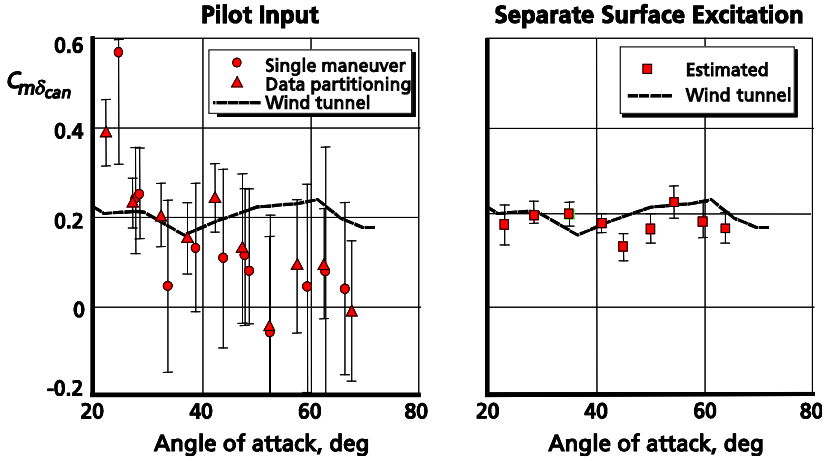

Fig. 9.6 Improved estimation through separate surface excitation.

decoupling method does not warrant any estimation program modifications. Only the postulated model needs to be properly coded. This will be highlighted on a test example presented in the next section. To include artificial stabilization in the output-error method requires some software modifications. These changes are comparatively minor and can be incorporated in the existing software without much developmental time. The basic output-error software described in Sec. 4.19 and provided with the book has been extended to provide such an option through a flag "iArtifStab." For the default value of iArtifStab = 0, the program runs in the mode of classical output-error method. Artificial stabilization is invoked by setting this flag to any value greater than zero. In such cases it is also necessary to specify the stabilization matrix S(Nx, Ny) appearing in Eq. (9.37). The direct and simplest approach is to feed back errors in the observed states to the respective states.

On the other hand, the filter-error algorithm might not be universally available. New development of such software and its validation is time consuming and costly. However, a few estimation software packages do exist [20, 21], and experience gained can be used to minimize the developmental costs. A basic version of the software for the filter-error method has been provided here. The multiple shooting method for aircraft parameter estimation is a relatively new approach. The programming complexity of the multiple shooting method with condensation algorithm is of the same order as that of the filter-error method. Although some investigations help to establish the feasibility of the approach [7, 9], we do not go into any further details of such an approach or software using this technique because other options are more direct and well established.

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9.16 EXAMPLES

The various algorithms discussed in the foregoing sections are applied to two test cases. The first one pertains to simulated unstable aircraft responses to critically evaluate the performance of the estimation methods under a controlled environment. As in the case of Sec. 5.11.1 dealing with analysis of data in turbulence, we once again resort to simulated data, mainly because 1) it provides data generated under known external initial conditions and noise characteristics and 2) no modeling errors are involved in the postulated model, both of which affect the estimates. This is all the more true in the case of unstable systems. The second example pertains to estimation of lateral-directional derivatives of X-31A, which is a highly control augmented fighter with enhanced maneuverability.

9.16.1 SIMULATED UNSTABLE AIRCRAFT RESPONSE DATA

A simplified model pertaining to the short-period motion of the test aircraft de Havilland DHC–2 "BEAVER" is considered here [30]. The nominal values of the aerodynamic derivatives used to generate the responses were obtained by parameter estimation from flight data at a typical flight condition. The static stability parameter M_w is, however, so adjusted as to result in an unstable system with time-to-double of 1 s. To provide realistic control inputs, the pilot input actually applied in a particular flight is used [30]. A total of 12.5 s of data with a sampling time of 0.05 s has been generated [3]. The state and observation equations are given by the following:

State equations:

$$\begin{aligned} \dot{w} &= Z_w w + (U_0 + Z_q) q + Z_{\delta e} \delta_e \\ \dot{q} &= M_w w + M_q q + M_{\delta e} \delta_e \end{aligned} \tag{9.46}$$

Observation equations:

$$\begin{aligned} a_z &= Z_w w + Z_q q + Z_{\delta e} \delta_e \\ w &= w \\ q &= q \end{aligned} \tag{9.47}$$

where w denotes the vertical velocity, U_0 the stationary forward speed, q the pitch rate, a_z the vertical acceleration, and δ_e the elevator deflection. Numerical simulation of such an unstable aircraft is feasible only through incorporation of a suitable controller. A feedback proportional to the vertical velocity is used:

$$\delta_e = \delta_p + Kw \tag{9.48}$$

where δ_p denotes the pilot input. The feedback K is so chosen as to result in the original degree of stability for the closed-loop system [3].

Although the data are generated through a closed-loop simulation, it is attempted to identify the system in open loop, that is, by treating δ_e, the elevator

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deflection, as the control input. The controller is not considered in the estimation procedure. The nominal (true) values of the parameters are provided in the second column of Table 9.1. The eigenvalues of the system matrix

$$A = \begin{bmatrix} Z_w & U_0 + Z_q \\ M_w & M_q \end{bmatrix} \tag{9.49}$$

for the nominal speed U_0 of 44.57 m/s are also provided in the same table. The two modes of the simulated system have the eigenvalues (0.6934, −5.8250), one of them being positive corresponding to the unstable mode, which results in a time to double of roughly 1 s given by

$$T_2 = \frac{\ell n\, 2}{\sigma_T} \tag{9.50}$$

To these data, we now apply the various estimation methods discussed in Secs. 9.3–9.11, namely, 1) least squares (LS), 2) total least squares (TLS), 3) combined least squares and output error (LS/OEM), 4) equation decoupling (EqDecpl), 5) eigenvalue transformation (EigTrans), 6) filter-error method (FEM), 7) extended Kalman filter (EKF), 8) unscented Kalman filter (UKF), 9) classical output-error method (OEM), and 10) artificially stabilized output-error method (SOEM). The results are provided in the upper part of Table 9.1, giving the numerical estimates of the derivatives appearing in Eqs. (9.46–9.47) and the corresponding eigenvalues. The lower part of the table provides details required to run the respective software programs. These programs are started from different folders as follows:

LS	/FVSysID2/chapter09/uAC_regLS	
TLS	/FVSysID2/chapter09/uAC_regTLS	
LS/OEM	/FVSysID2/chapter04/ml_oem2	test_case = 6
EqDecpl	/FVSysID2/chapter04/ml_oem2	test_case = 7
EigTransf	/FVSysID2/chapter04/ml_oem2	test_case = 10
FEM	/FVSysID2/chapter05/ml_fem2	test_case = 8
EKF	/FVSysID2/chapter07/mainRPE	test_case = 8
UKF	/FVSysID2/chapter07/mainRPE	test_case = 8
OEM	/FVSysID2/chapter04/ml_oem2	test_case = 8
SOEM	/FVSysID2/chapter04/ml_oem2	test_case = 9

The data file containing the time histories is found under the directory /FVSysID2/flt_data/unStabAC_sim.asc.

The programs uAC_regLS and uAC_regTLS are specially tailored for the specific case being analyzed applying the least-squares and total least-squares techniques. These programs are fairly straightforward to follow and do not

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TABLE 9.1 SUMMARY OF ESTIMATES AND MODEL DEFINITIONS FOR VARIOUS ESTIMATION METHODS: SIMULATED UNSTABLE SHORT-PERIOD MOTION

	Nominal Value	LS	TLS	LS/OEM	EqDecpl	EigTransf	FEM	EKF	UKF	OEM	SOEM
Z_w	−1.4249	−1.4249	−1.4249	−1.4249	−1.4277 (0.01)†	−1.4294 (0.01)	−1.4278 (0.01)	−1.4252 (0.0)	−1.4252 (0.0)	−1.3086 (2.48)	−1.4268 (0.01)
Z_q	−1.4768	−1.4768	−1.4768	−1.47680*	−1.47680*	−1.4768*	−1.4768*	−1.4768*	−1.4768*	−1.4768*	−1.4768*
$Z_{\delta e}$	−6.2632	−6.2632	−6.2632	−6.2632	−6.1681 (0.01)	−6.12825 (0.01)	−6.16658 (0.01)	−6.2622 (0.01)	−6.2653 (0.01)	−9.9477 (8.72)	−6.1711 (0.02)
M_w	0.2163	0.2148	0.2148	0.21634	0.21616 (0.01)	0.21776 (0.01)	0.21717 (0.01)	0.21716 (0.04)	0.21713 (0.04)	0.05216 (5.85)	0.21702 (0.01)
M_q	−3.7067	−3.6905	−3.6907	−3.71215	−3.7086 (0.0)	−3.72645 (0.01)	−3.71967 (0.0)	−3.7373 (0.03)	−3.7343 (0.03)	−1.9277 (4.04)	−3.7201 (0.01)
$M_{\delta e}$	−12.784	−12.7056	−12.7059	−12.7687	−12.7733 (0.01)	−12.8262 (0.01)	−12.8152 (0.01)	−12.8247 (0.02)	−12.8122 (0.02)	−7.7214 (1.87)	−12.8148 (0.01)
Eigen-values	0.6934 −5.8250	0.6887 −5.8042	0.6887 −5.8043	0.6919 −5.8290	0.6900 −5.8262	0.6936 −5.8495	0.6930 −5.8405	0.6913 −5.8538	0.6898 −5.8493	−0.0873 −3.1490	0.6925 −5.8395
Program name		uAC_regLS	uAC_regTLS	ml_oem2	ml_oem2	ml_oem2	ml_fem	mainRPE	mainRPE	ml_oem2	ml_oem2
Test case		–	–	6	7	10	8	8	8	8	9
Function name for state and observation equations		–	–	xdot_TC06_uAC_ RegSt obs_TC06_ uAC_ RegSt	xdot_TC07_ uAC_EqDecoup obs_TC08_ uAC	xdot_TC10_uAC_ EigT obs_ TC08_ uAC	xdot_TC08_ uAC obs_TC 08_uAC	xdot_TC08_ uAC obs_TC 08_uAC	xdot_TC08_ uAC obs_ TC 08_uAC	xdot_TC08_ uAC obs_ TC 08_uAC	xdot_TC08 _ uAC obs_ TC 08_uAC
Nx; Ny; Nu		0; 1; 3	0; 1; 3	2; 3; 3	2; 3; 3	2; 3; 1	2; 3; 1	2; 3; 1	2; 3; 1	2; 3; 1	2; 3; 1
NparSys		3; 3	3; 3	6	6	6	6	6	6	6	6

*Parameter held fixed. †Relative standard deviation in %.

need any further explanation. The least-squares and total least-squares method do not need starting values to be specified. From Table 9.1 it is observed that the LS and TLS methods yield numerically the same estimates. This is not really surprising because in the absence of noise in the independent variables TLS is equivalent to LS. These estimates agree very well with the nominal parameter values.

To run the other programs, it is necessary to specify test_case the integer flag for test case number, Nx the number of states, Ny the number of observation variables, Nu the number of input (control) variables, and NparSys the number of system parameters. The total number of parameters to be estimated is given by Nparam = NparSys, and in the case of EKF or UKF applying the program "mainRPE," it is Nxp = Nx + Nparam. In each case the model definition is provided by the user-defined function "mDefCase**.m" where, as given earlier,** is set to 6 to 10 as the case may be. For example, to apply the equation decoupling method described in Sec. 9.6, we specify the various details as follows:

```
test_case = 7                                  % Flag for test case
state_eq  = "xdot_TC07_uAC_EqDecoup";          % Function for state equations
obser_eq  = "obs_TC08_uAC";                    % Function for observation equations
Nx        = 2;                                 % Number of states
Ny        = 3;                                 % Number of observation variables
Nu        = 3;                                 % Number of input variables
NparSys   = 6;                                 % Number of system parameters
Nparam    = NparSys;                           % Total number of parameters
dt        = 0.05;                              % Sampling time
param                                          % Starting values for parameters
parFlag                                        % Flags for free/fixed parameters
```

For the other methods a similar procedure is to be followed, each time specifying the details as provided in Table 9.1. In all of these cases, the same initial guess values for the parameters were used, which were about 50% off the nominal values. In most cases due to high correlation, it was necessary to keep the parameter Z_q fixed; hence, the integer flag "parFlag" has to be defined appropriately.

From Table 9.1, it is obvious that the programs FEM, EKF, UKF, OEM, and SOEM use the same functions, namely, xdot_TC08_uAC, for the state equations, and obs_TC08_uAC for the observation equations. Thus, once the postulated model is coded in these two functions, different techniques can be readily applied. The function xdot_TC07_uAC_EqDecoup called *the equation decoupling method* is a modified version of xdot_TC08_uAC using integrated state values for the diagonal terms only and measured states for the off-diagonal terms; see Eq. (9.21). Accordingly, in this function the derivatives Z_q and M_w are multiplied by q_m and w_m, that is, by the respective measured quantities, which are provided as

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additional input variables. But for these two differences, there are no other changes. The function xdot_TC10_uAC_EigT called *the eigenvalue transformation method* includes, as elaborated in Sec. 9.7, the transformation σ_T in the diagonal elements of state matrix whereas the observation function obs_TC08_uAC in terms of original system parameters is used. In the combined LS/OEM method using regression startup procedure, we make more use of the measured variables by multiplying all of the free derivatives with the corresponding measured states and not just the off-diagonal terms as in the case of the equation decoupling method. Accordingly, we use the function xdot_TC06_uAC_RegSt. In a similar way, except for the regression startup, all other methods call the same function, namely, obs_TC08_uAC, to compute the observation variables a_z, w, and q according to Eq. (9.47).

Figure 9.7 shows for two cases, namely, the OEM and SOEM, a comparison of data to be matched (continuous lines) and model output (dashed lines) for the specified starting parameter values. A diverging solution resulting from the integration in the classical OEM is observed in Fig. 9.7a (obtained by running ml_oem2, for test_case = 8 and niter_max = 0). On the other hand, the artificial feedback prevents such a divergence; see Fig. 9.7b (obtained by running ml_oem2, for test_case = 9 and niter_max = 0). The filter-error method provided a similar match as that in Fig. 9.7b. In the case of the other methods too, starting from the specified initial parameter values the tendency to diverge was not observed. Next, by setting niter_max to 10, various cases are run to estimate the parameters. The various methods, except for the classical output error, provided a very good match between measured data and model responses. Because

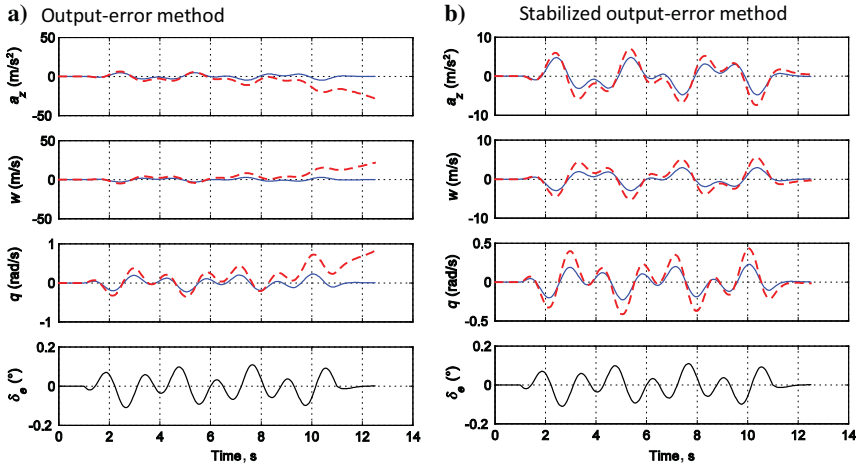

Fig. 9.7 Comparison of system (———) and estimated (--------) responses for starting parameter values.

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there were no observable differences in the match, a single typical plot is provided in Fig. 9.8.

In the case of the eigenvalue transformation method, we preprocess the output and input variables prior to estimation by multiplying them with $\exp(-0.7^{*}t)$, which corresponds to the transformation of Eq. (9.26) with $\sigma_T = 0.7$, which is slightly higher than the largest positive eigenvalue. This is done in the model definition function "mDefCase10.m." We note that the vertical acceleration a_z is one of the observation variables. To properly account for this general case, as elaborated in Sec. 9.7, we adopt the alternative way of programming the state equations by incorporating σ_T in the diagonal elements in the state equations. This has been carried out in the function "xdot_TC10_uAC_eigT.m" for the parameters Z_w and M_q. In such a case no further backtransformation is necessary, and, accordingly, these estimates are provided in Table 9.1. From Fig. 9.9, it can be observed that the exponential function tends to reduce the information content as time increases. The eigenvalues of the transformed system matrix, that is, of $(\tilde{A} = A - \sigma_T I)$, were found to be $(-0.0064; -6.5495)$ implying a stable transformed system.

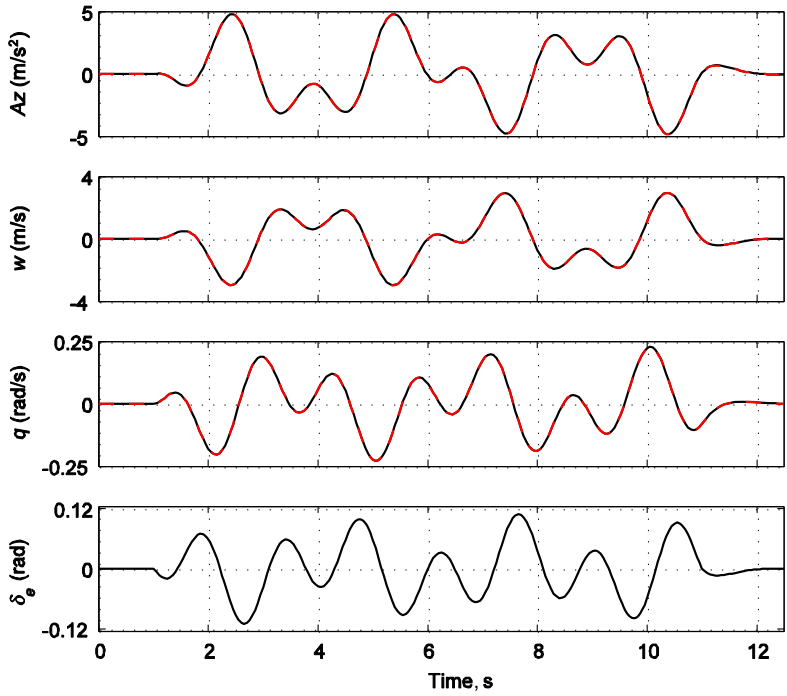

Fig. 9.8 Comparison of system (——) and estimated (------) responses applying a typical estimation method for unstable systems (simulated data).

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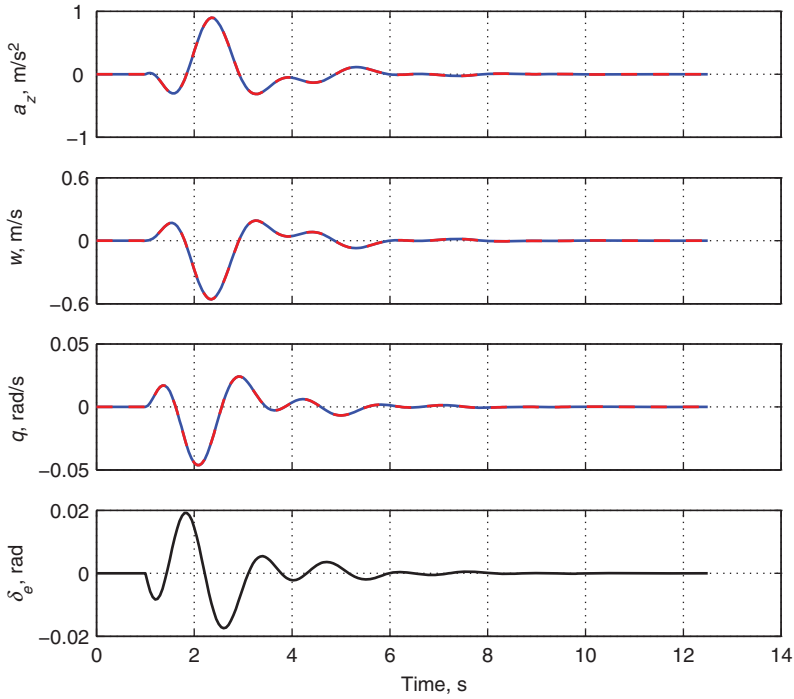

Fig. 9.9 Comparison of system (——) and estimated (-------) responses applying the equation decoupling method.

In the case of the extended Kalman filter (invoked by calling the program "/FVSysID2/chapter07/mainRPE" with test_case = 8), the measurement noise covariance matrix is specified in "mDefCase08.m" to be diagonal with values of 5×10^{-8}, and the process noise covariance matrix is also specified as diagonal with values of 1×10^{-9} for states and zero for the extended states, that is, parameters, according to Eq. (7.49). In the present case, specification of these noise matrices was no problem because we are analyzing simulated data generated without any process or measurement noise; very small nonzero values were used to avoid numerical inconsistencies. In general, in the presence of measurement and process noise, inaccurate specification of their covariances will affect the estimates. The convergence plot of the estimates is provided in Fig. 9.10; it is observed that convergence is achieved within the first few seconds. An inset shows an enlarged plot of the last few seconds for just two parameters. Table 9.1 gives values that are the averages of the last 10 data points, obtained from an arbitrarily specified initial covariance matrix P. Different starting values for initial P were used; they provided very comparable estimates.

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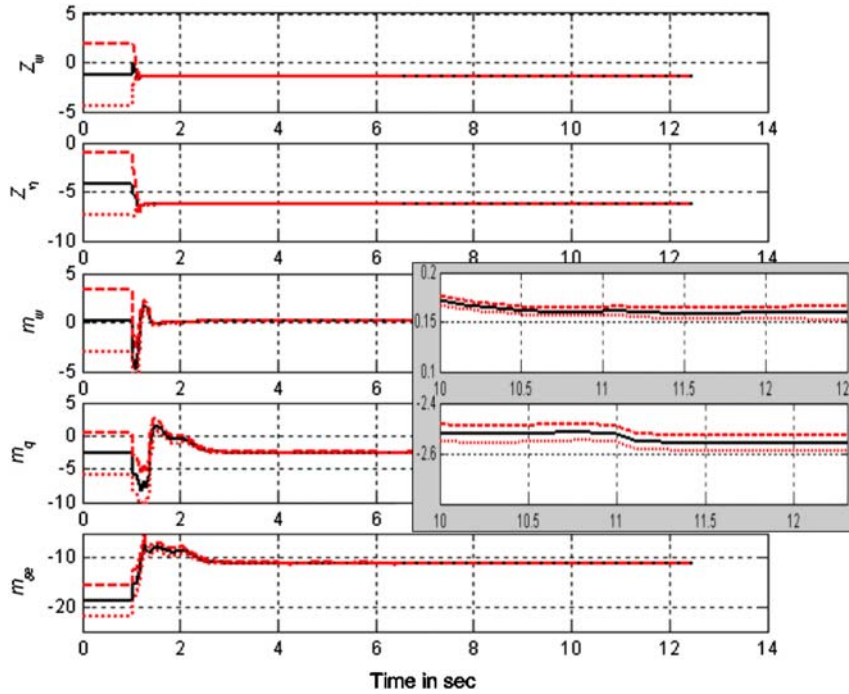

Fig. 9.10 Estimates of parameters applying EKF: ——, estimate; ------, estimate ± standard deviation.

Using the same values for the noise covariances as just defined for the EKF, the UKF is applied by setting the flag method = UKF. This is done in mainRPE. From Table 9.1, it is observed that the estimates from UKF and EKF are comparable.

The filter-error method is applied by calling the program "/FVSysID2/chapter05/ml_fem2" with test_case = 8. In the present case, elements of the process noise distribution matrix F cannot be estimated because the simulated data are generated without process and measurement noise. Attempts to estimate F elements lead to numerical problems. They are kept fixed at very small values of 10^{-6}, practically zero. The results are tabulated in the eighth column of Table 9.1. As expected, no numerical problems were encountered. The estimates match fairly well with the expected values.

The standard output-error method, described in Sec. 4.6, is applied by calling the program "/FVSysID2/chapter04/ml_oem2" and setting test_case = 8. For several sets of starting values, some even closer to the nominal values, the algorithm faced numerical divergence problems in the very first iteration, making further estimation impossible. For a few sets of starting values, the algorithm did converge, but to wrong values. One such case provided the estimated

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parameters shown in the eleventh column. It is observed that the numerical values of all of the parameters are completely wrong. The algorithm somehow manipulates the parameter values leading to a convergence and matches the response variables (see Fig. 9.11), but to completely wrong parameter values. The eigenvalues corresponding to the final estimates, as observed from Table 9.1, are -0.0873 and -3.1490, both of which are not correct. They suggest a stable system whereas the system being analyzed is unstable. The standard deviations of the estimates provided in the same column range from roughly 2 to 9% and at first glance might not appear to be too large. However, the data being analyzed are simulated data with no process and measurement noise; they are very high. All other methods yielded estimates with extremely low standard deviations of less than 0.02%. Standard deviation is a measure of the accuracy of the estimates.

On the other hand, the artificially stabilized output-error method described in Sec. 9.11 is applied by calling the same program, namely, "ml_oem2," with the flag test_case = 9 and by setting the flag iArtiStab and defining the artificial stabilization matrix as follows:

```
iArtifStab = 1;                % Artificial stabilization
StabMat = [0 0.05 0; 0 0 0];   % Stabilization matrix
```

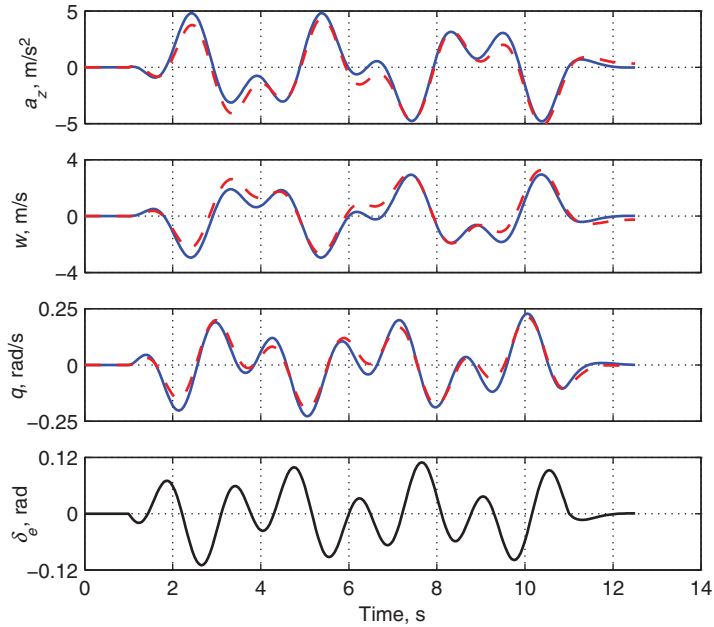

Fig. 9.11 Comparison of system (——) and estimated (------) responses applying an output-error method.

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The preceding values of stabmat correspond to the stabilization matrix S of Eq. (9.37) given by

$$S = \begin{bmatrix} 0 & 0.05 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{9.51}$$

for the two states (w, q) and three output variables (a_z, w, q) according to Eqs. (9.46) and (9.47). This implies that we stabilize the integrated state variable w by feeding back the error in the observation variable w. A very low stabilization factor is sufficient to overcome the numerical problems. The same choice of stabilization matrix was found to be adequate for several different sets of starting parameter values. The response match was as shown in Fig. 9.8.

The eigenvalues for the models identified applying different algorithms are computed using the program "/FVSysID2/chapter09/eig_uAC_ID.m" and the results provided as the eighth row of Table 9.1. It is clearly observed that all of the methods, except for the classical OEM, yield eigenvalues close to the expected values of the unstable simulated system. The foregoing discussion and results presented in this section corroborate the finer aspects that were brought out while describing the algorithms in Secs. 9.3–9.11. It is apparent that these methods use the measured states to varying degrees. Through minor modifications of the function for the state equations, it can be easily verified that the feedback of state variable w is quite sufficient for this example. This is not really surprising because we know that the instability was introduced in the simulation through the derivative M_w.

9.16.2 IDENTIFICATION OF X-31A LATERAL-DIRECTIONAL MOTION

X-31A is a highly control augmented aircraft with enhanced maneuverability. Poststall maneuvering is enabled applying advanced technologies such as high-angle-of-attack aerodynamics and flight control system integrated thrust vectoring. The challenges encountered in the X-31A system identification are typical of unstable aircraft, namely, 1) inherently unstable open-loop plant; 2) highly correlated control surface deflections, thrust vectors vanes and states due to integrated flight control laws; and 3) insufficient and nonoptimized excitation. The first and the second aspects have already been dealt with in the various foregoing sections bringing out the utility and applicability of several estimation algorithms to inherently unstable systems. The improvements through the separate surface excitation were highlighted in Sec. 9.14. In this section we further address the last aspect of estimating the parameters from pilot input maneuvers. A bank-to-bank roll around the velocity vector at 54-deg angle of attack, executed by the pilot through a manual roll command input, is considered here [28].

The dynamic model describing the lateral-directional motion is formulated as follows without the usual simplifications to account for larger sideslip angles and

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possible cross-coupling effects. The state equations are given by

$$\begin{aligned}
\dot{\beta} &= \frac{g}{V}\left\{-[(n_x - \sin\theta)\cos\alpha + (n_z + \cos\phi\cos\theta)\sin\alpha]\sin\beta \right. \\
&\qquad \left. + \left[\frac{1}{mg}(\bar{q}SC_Y + Y_{\text{FIneng}}) + \sin\phi\cos\theta\right]\cos\beta\right\} + p\sin\alpha - r\cos\alpha \\
I_x\dot{p} - I_{xz}\dot{r} &= \bar{q}SbC_\ell^{CG} + L_{\text{FIneng}} + I_{xz}\,p\,q + (I_y - I_z)\,q\,r \\
-I_{xz}\dot{p} + I_z\dot{r} &= \bar{q}SbC_n^{CG} + N_{\text{FIneng}} - I_{xz}\,q\,r + (I_x - I_y)\,p\,q \\
\dot{\phi} &= p + (q\sin\phi + r\cos\phi)\,\tan\theta \\
\dot{\psi} &= (q\sin\phi + r\cos\phi)/\cos\theta
\end{aligned} \tag{9.52}$$

Because most of the variables appearing in Eq. (9.52) are according to the standard notation, we do not explain each of them explicitly here. The forces and moments due to engine—Y_{FIneng}, L_{FIneng}, N_{FIneng}—were computed using the appropriate measured variables prior to parameter estimation and treated as additional input variables. They include components due to engine inlet and position only, not those due to the thrust vector deflections. The variables pertaining to the longitudinal motion—$V, \alpha, q, \theta, n_x, n_z, \bar{q}$—are treated as additional pseudo control inputs. As discussed in Sec. 3.6, the approach of pseudo control inputs helps to account for the effects of other motion variables that are not treated as state variables.

The coefficients of rolling and yawing moments in Eq. (9.52) are referred to the center of gravity, as implied by the superscript c.g. They are given by

$$\begin{aligned}
C_\ell^{CG} &= C_\ell^{AC} - C_Y\frac{z_{\text{ACCG}}}{b} \\
C_n^{CG} &= C_n^{AC} + C_Y\frac{x_{\text{ACCG}}}{b}
\end{aligned} \tag{9.53}$$

where (x_{ACCG}, z_{ACCG}) denote the position of the aerodynamic center (a.c.) with respect to the c.g. We consider here the aerodynamic model referred to a.c. to facilitate comparison of the estimated moment derivatives with the wind-tunnel derived aerodynamic data set (ADS) and also of estimates obtained from different tests performed with different c.g. locations. Considering linear model postulates, the side-force, rolling-moment, and yawing-moment coefficients are modeled as

$$\begin{aligned}
C_Y &= C_{Y0} + C_{Y\beta}\,\beta + C_{Yp}\frac{pb}{2V} + C_{Yr}\frac{rb}{2V} + C_{Y\delta a}\,\delta_a + C_{Y\delta r}\delta_r + C_{Y\kappa}\frac{\kappa F}{\bar{q}S} \\
C_\ell^{AC} &= C_{\ell 0} + C_{\ell\beta}\,\beta + C_{\ell p}\frac{pb}{2V} + C_{\ell r}\frac{rb}{2V} + C_{\ell\delta a}\,\delta_a + C_{\ell\delta r}\,\delta_r \\
C_n^{AC} &= C_{n0} + C_{n\beta}\,\beta + C_{np}\frac{pb}{2V} + C_{nr}\frac{rb}{2V} + C_{n\delta a}\,\delta_a + C_{n\delta r}\,\delta_r + C_{n\kappa}\frac{\kappa F}{\bar{q}S}
\end{aligned} \tag{9.54}$$

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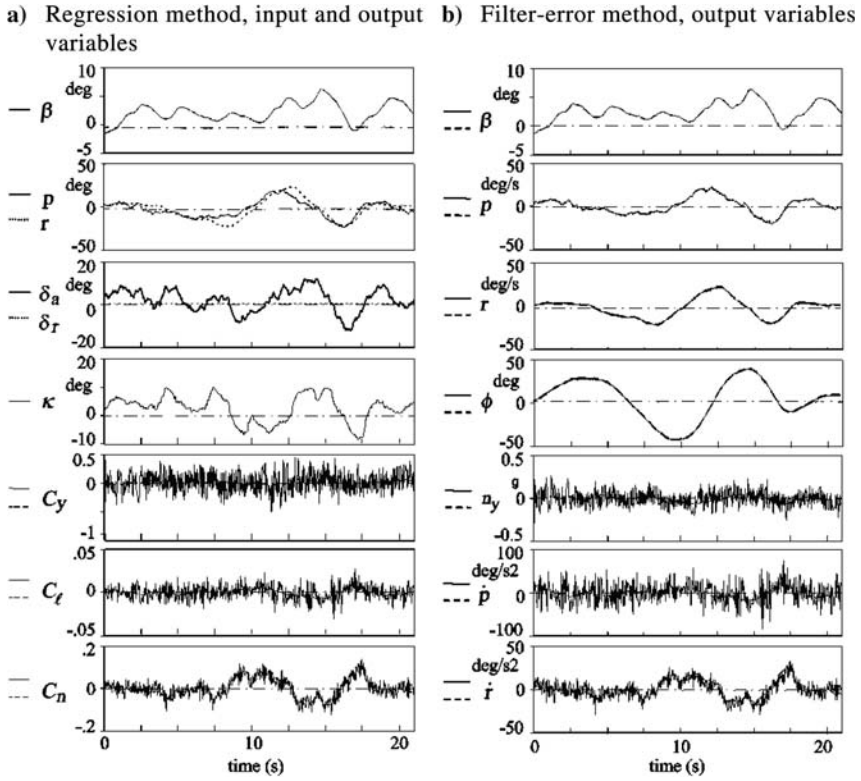

Fig. 9.12 Identification of a bank-to-bank maneuver at 54-deg angle of attack: ——, ... flight measured data; ------, model output.

It is now our goal to estimate the various parameters $C_{Y()}$, $C_{\ell()}$, $C_{n()}$ appearing in Eq. (9.54) by matching the observation variables n_y, β, p, r, ϕ, and ψ. Additionally, the angular accelerations, $\dot{p}$ and $\dot{r}$ (though not measured directly, but generated by numerical differentiation of the rates), are also matched as observation variables because inclusion of such angular accelerations is found to speed up the convergence significantly (recall our arguments in Sec. 2.5 on this aspect). Besides the already pointed out pseudo inputs, the primary input variables are the aileron deflection (asymmetric trailing-edge flap deflection) δ_a, the rudder deflection δ_r, and the thrust deflection angle in yaw κ.

Attempts to estimate the various parameters appearing in Eq. (9.54) were not successful; estimates were either highly correlated or were not reliable as implied by large standard deviations. This was rather expected because as evident from upper four plots of Fig. 9.12a, it is clearly observed that aileron deflection δ_a

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and thrust deflection angle in yaw κ are highly correlated. Likewise, the roll rate p is correlated with yaw rate r. Furthermore, the rudder deflection δ_r is zero because the rudder is faded out by the flight control laws for angles of attack above 40 deg. At such flight conditions the lateral control is possible only through the thrust vector deflection. Based on these inferences, we now reduce the model of Eq. (9.54) to analyze the specific flight maneuver as follows: 1) remove the derivatives due to rudder without effecting the estimation because rudder deflection is anyway zero; 2) keep thrust-vector effectiveness in yaw $C_{n\kappa}$ fixed at values identified from other flight maneuvers to overcome problems due to correlated δ_a and κ, thereby enable estimation of $C_{n\delta a}$; and 3) introduce the following combined derivatives because correlated roll and yaw do not allow independent estimation of these derivatives:

$$\begin{aligned} C^*_{Yp} &= C_{Yp} + C_{yr}\, r/p \\ C^*_{\ell p} &= C_{\ell p} + C_{\ell r}\, r/p \\ C^*_{nr} &= C_{nr} + C_{np}\, p/r \end{aligned} \tag{9.55}$$

The combined derivatives defined in Eq. (9.55) are, in general, valid only for cases when the roll rate p and yaw rate r are correlated.

Thus, a judicious combination of mixed estimation concept and of combined derivatives leads to the following aerodynamic model:

$$\begin{aligned} C_Y &= C_{Y0} + C_{Y\beta}\,\beta + C^*_{Yp}\,\frac{pb}{2V} + C_{Y\delta a}\,\delta_a + C_{Y\kappa}\,\frac{\kappa F}{\bar{q}S} \\ C^{AC}_{\ell} &= C_{\ell 0} + C_{\ell\beta}\,\beta + C^*_{\ell p}\,\frac{pb}{2V} + C_{\ell\delta a}\,\delta_a \\ C^{AC}_{n} &= C_{n0} + C_{n\beta}\,\beta + C^*_{nr}\,\frac{rb}{2V} + C_{n\delta a}\,\delta_a + C_{n\kappa}\,\frac{\kappa F}{\bar{q}S} \end{aligned} \tag{9.56}$$

Detailed investigations clearly brought out the limitations of the artificially stabilized output-error method. Moreover, as observed from the time history of roll rate p in Fig. 9.12a, the roll rate is corrupted by turbulence. The roll rate p has a considerable influence on the sideslip angle β. Although for conventional aircraft the derivative C_{Yp} is not important and often neglected, in the present case it is necessary to include the same. As a consequence, due to the presence of noise in p, the sideslip angle modeling was also not accurate. This has been confirmed in a separate step by treating the roll rate as an input variable instead of state, which provided good match for the sideslip angle [28]. These considerations indicate the necessity to account for process noise in the estimation. Accordingly, to this maneuver we apply three parameter estimation methods, namely: 1) least squares, 2) filter error, and 3) extended Kalman filter.

Filter-error and EKF methods can be applied directly to the model postulated in Eqs. (9.52–9.56). However, application of the LS method requires computation

of aerodynamic forces and moments; the general procedure to compute them from the measured linear accelerations and angular rates has been described in Sec. 6.9.1. Figure 9.12 shows the complete bank-to-bank roll evaluated applying a) a least-squares method and b) a filter-error method [28, 29]. Depending upon the method being applied, different variables are plotted in Figs. 9.12a and 9.12b. In the first case, the aerodynamic coefficients C_Y, C_ℓ, and C_n are matched as seen from the three plots from bottom in Fig. 9.12a (shown by continuous and dashed lines). The four plots from the top show measured β, p, r, δ_r, δ_r, and κ (continuous and dotted lines), which are input (independent) variables in the least-squares method. On the other hand, the filter-error method matches the output variables n_y, $\dot{p}$, $\dot{r}$, β, p, r, and ϕ. The flight-measured model and estimated time histories for these variables are shown in Fig. 9.12b by continuous and dashed lines. In the case of the filter-error method, the fit between flight-test data and model output is perfect for all variables. The match in the lateral, roll, and yaw accelerations is of the same quality as the match in the corresponding aerodynamic coefficients obtained from regression.

Figure 9.13 shows the time history plots of the derivatives estimated applying the EKF from the same bank-to-bank maneuver at 54-deg angle of attack [29]. It is clearly observed that the lateral force derivatives converge poorly. This was to be expected because, as already pointed, out the flight control laws suppress the lateral force, and as such the data contain practically no information that is necessary to enable estimation of these derivatives.

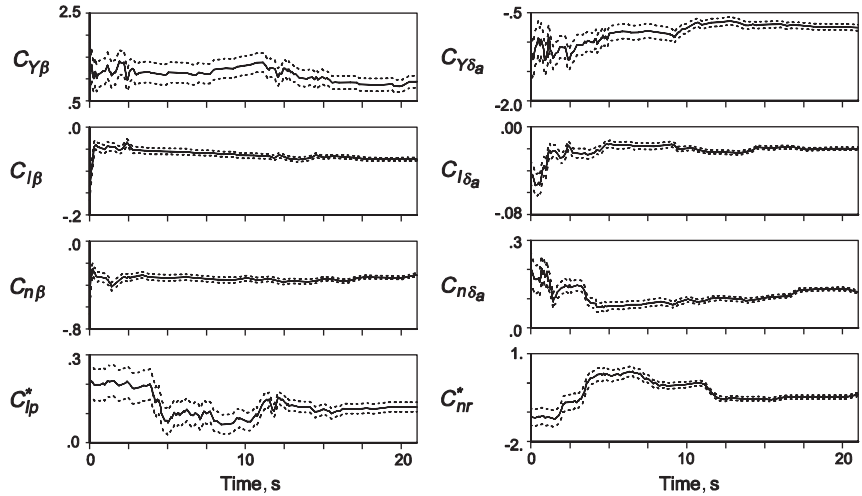

Fig. 9.13 Identification of a bank-to-bank maneuver at 54-deg angle of attack applying EKF: ——, estimate; -------, estimate ± standard deviation.

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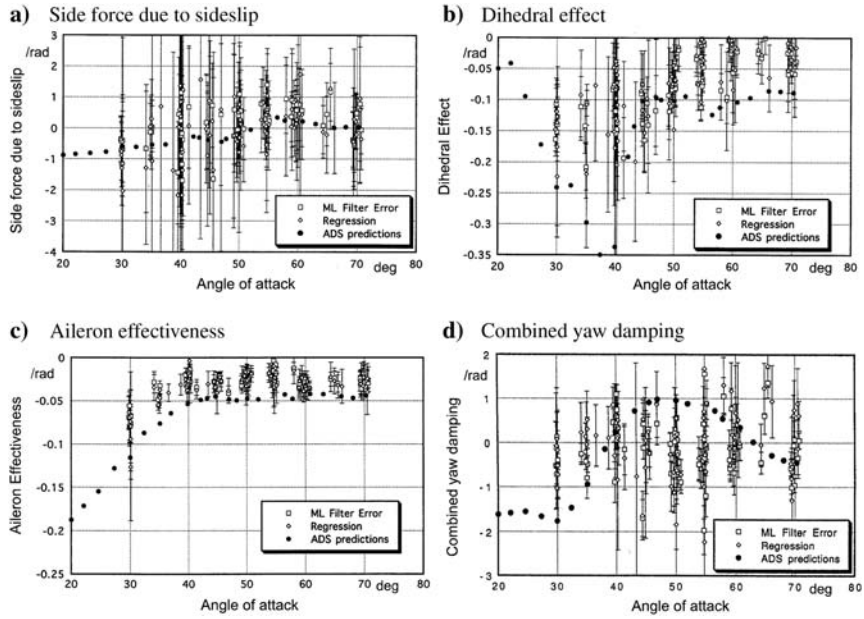

Fig. 9.14 Estimates of aerodynamic derivatives.

Having discussed the problems encountered in the estimation of aerodynamic parameters from pilot input maneuvers at high angles of attack, we now turn our attention to study the performance of the least-squares and filter-error method over a wide range of 30- to 70-deg angle of attack in the poststall regime. Identification results are presented in Fig. 9.14. For comparison, also the ADS predictions are plotted. The sideforce due to sideslip $C_{Y\beta}$ in Fig. 9.14a shows large scatter and large uncertainty levels indicated by the error bars. These estimates are most unreliable. As already pointed out, this is to be expected as there is nearly no excitation in the lateral acceleration. The estimates of dihedral effect $C_{\ell\beta}$ in Fig. 9.14b and of aileron effectiveness $C_{\ell\delta a}$ in Fig. 9.14c show some deviations compared to the ADS wind-tunnel predictions. The aileron effectiveness was the best identifiable parameter. The estimates indicate that the decline in aileron effectiveness starts at about 30-deg angle of attack, which is earlier than predicted at about 40 deg. The combined yaw damping C^*_{nr} in Fig. 9.14d also has larger scatter due to insufficient excitation of the aircraft eigenmotion resulting from pilot input because, as pointed out in Sec. 9.1, the controller suppresses the oscillatory motion.

The low quality of the estimation results from pilot input maneuver is attributed to the fact that the aircraft motion is insufficiently excited. Effective excitation of dynamic motion suitable for parameter estimation at high angles of attack from pilot inputs is prevented by the X-31A flight control laws and would be achieved only through separate surface excitation. This has already been substantiated

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through typical results in Sec. 9.14. It is observed from Fig. 9.14 that no significant differences can be noticed between the results obtained from regression and filter-error method, apart from the fact that the uncertainty levels from regression are somewhat larger. This indicates that the simplest and the most sophisticated algorithm produce approximately the same results. This similarity in the two results suggests that the process noise must be dominant compared to the measurement noise. The process noise results from atmospheric turbulence and in the present case from stochastic shedding of vortices whereas measurement noise results from the sensors. High-quality sensors have fairly low noise levels. It may be recalled that the least-squares method accounts for the process noise, but not for the measurement noise in the independent variables whereas the filter-error method accounts for both the process and measurement noise. Some more results pertaining to the X-31A aerodynamic model validation and update will be discussed in Sec. 12.9. The parameter estimation techniques described in this chapter have also been applied to other unstable and high-angle-of-attack aircraft, for example, the X-29A and F-18 High Angle of Attack Research Vehicle [2, 31–34].

9.17 CONCLUDING REMARKS

In this chapter we have clearly demonstrated that the difficulties encountered in the unstable aircraft parameter estimation in open loop are related to practical issues of applying system identification and not to the issues of estimation methodology being applicable or not. This is evident from the fact that we use mostly the same estimation algorithms that we already covered in the previous chapters or some suitable variants of them based on the principle of using measured data within the estimation algorithm itself. Several applicable techniques have been elucidated, bringing out advantages and limitations each time. From the sample example analyzing simulated unstable aircraft responses, it is found that all of them serve the intended purpose.

While analyzing flight data, a diligent choice of an algorithm from those presented in this chapter would aid the data analysis. It has been clearly brought out that an algorithm accounting for process noise is generally preferable. They are 1) the least-squares method, 2) the filter-error method, 3) the extended Kalman filter method, and 4) the unscented Kalman filter method. Because tuning of the noise matrices in EKF/UKF might be laborious, and a wrong choice might adversely affect the estimates, we prefer the filter-error method or the least-squares method. The simple least-squares technique provided estimates comparable to those from the complex filter-error method. As a general trend, the least-squares approach, combined with data partitioning if necessary, appears to be adequate, provided the data analyzed is verified for consistency and corrected for systematic errors.

The problems resulting from correlated states and inputs are more difficult to handle. The best choice is separate surface excitation, that is, introducing inputs

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directly after the controller. The substantial improvements in the accuracy of the estimates obtained from SSE have been clearly demonstrated. But, it might be difficult to implement SSE in practice. In the absence of SSE, accurate and reliable estimation of all of the unknown derivatives might not be possible. The alternative would be to apply techniques based on detecting data collinearity and subsequent least-squares estimation in terms of combined derivatives or mixed estimation incorporating a priori information. The difficulties encountered in the parameter estimation of pilot input maneuvers have been brought out by analyzing flight maneuvers at high angles of attack in the poststall regime.

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REFERENCES

[1] Koehler, R., and Wilhelm, K., "Closed Loop Aspects of Aircraft Identification," AGARD, LS-104, Paper 10, Nov. 1979.

[2] Maine, R. E., and Murray, J. E., "Application of Parameter Estimation to Highly Unstable Aircraft," *Journal of Guidance, Control, and Dynamics*, Vol. 11, No. 3, 1988, pp. 213–219.

[3] Plaetschke, E., "Identifizierung Instabiler Flugmechanischer Systeme mit einem Ausgangsfehlerverfahren," *ZFW*, Vol. 12, No. 4, 1988, pp. 233–240 (in German).

[4] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[5] Lebacqz, J. V., and Govindaraj, K. S., "Implicit Model Following and Parameter Identification of Unstable Aircraft," *Journal of Guidance and Control*, Vol. 3, No. 2, 1980, pp. 119–123.

[6] Jategaonkar, R. V., "Startup Algorithm for Maximum Likelihood Estimation of Parameters in Nonlinear Systems," DFVLR-IB 111-86/67, Germany, Dec. 1986.

[7] Jategaonkar, R. V., and Thielecke, F., "Evaluation of Parameter Estimation Methods for Unstable Aircraft," *Journal of Aircraft*, Vol. 31, No. 3, 1994, pp. 510–519.

[8] Preissler, H., and Schäufele, H., "Equation Decoupling – A New Approach to the Aerodynamic Identification of Unstable Aircraft," *Journal of Aircraft*, Vol. 28, No. 2, 1991, pp. 146–150.

[9] Thielecke, F., "Identifizierung Instabiler Flugmechanischer Systeme mit Hilfe der Mehrzielmethode," DLR-IB 111-91/27, Germany, Oct. 1991 (in German).

[10] Weiss, S., Friehmelt, H., Plaetschke, E., and Rohlf, D., "X-31A System Identification Using Single-Surface Excitation at High Angles of Attack," *Journal of Aircraft*, Vol. 33, No. 3, 1996, pp. 485–490.

[11] Kaletka, J., and Fu, K.-H., "Frequency-Domain Identification of Unstable Systems Using X-31A Aircraft Flight Test Data," AIAA Paper 93-3635, Aug. 1993.

[12] Tischler, M. B., "Frequency Response Identification of XV-15 Tilt-Rotor Aircraft Dynamics," NASA TM-89428, May 1987.

[13] Klein, V., "Estimation of Aircraft Aerodynamic Parameters from Flight Data," *Progress in Aerospace Sciences*, Pergamon, Oxford, U.K., Vol. 26, No. 1, 1989, pp. 1–77.

[14] Theil, H., *Principles of Econometrics*, Wiley, Hoboken, NJ, 1971.

[15] Toutenburg, H., *Prior Information in Linear Models*, Wiley, New York, 1982.

[16] Golub, G. H., and Van Loan, C. F., "An Analysis of the Total Least Squares Problem," *SIAM Journal of Numerical Analysis*, Vol. 17, No. 6, 1980, pp. 883–893.

[17] Laban, M., and Masui, K., "Total Least Squares Estimation of Aerodynamic Model Parameters from Flight Data," *Journal of Aircraft*, Vol. 30, No. 1, 1993, pp. 150–152.

[18] Naihong, L., Yaohua, W., and Pingyuan, C., "Comment on 'Equation Decoupling – A New Approach to the Aerodynamic Identification of Unstable Aircraft,'" *Journal of Aircraft*, Vol. 30, No. 3, 1993, p. 431.

[19] Singh, J., and Raisinghani, S. C., "An Equation Decoupling Technique for Identification of Unstable Aircraft with Unsteady Aerodynamic Modelling," *Aeronautical Journal*, Vol. 97, No. 969, Nov. 1993, pp. 321–327.

[20] Maine, R. E., and Iliff, K. W., "Formulation and Implementation of a Practical Algorithm for Parameter Estimation with Process and Measurement Noise," *SIAM Journal of Applied Mathematics*, Vol. 41, No. 3, 1981, pp. 558–579.

[21] Jategaonkar, R. V., and Plaetschke, E., "Algorithms for Aircraft Parameter Estimation Accounting for Process and Measurement Noise," *Journal of Aircraft*, Vol. 26, No. 4, 1989, pp. 360–372.

[22] Balakrishna, S., and Niranjana, T., "Wind Tunnel Dynamic Flying Study of the Pitching Moment Derivatives of the Standard Dynamic Model in Active Control," AIAA Paper 87-2626, Aug. 1987.

[23] Stoer, J., and Bulirsch, R., *Numerische Mathematik 2*, 3rd ed., Springer-Verlag, Berlin, 1990.

[24] Bock, H. G., "Numerical Solution of Nonlinear Multipoint Boundary Value Problems with Applications to Optimal Control," *ZAMM*, Vol. 58, 1978, pp. 407–409.

[25] Klein, V., "Aircraft Parameter Estimation in the Frequency Domain," AIAA Paper 78-1344, Aug. 1978.

[26] Fu, K.-H., and Marchand, M., "Helicopter System Identification in the Frequency Domain," *9th European Rotorcraft Forum*, Paper 96, Stresa, Italy, Sept. 1983.

[27] Marchand, M., and Fu, K.-H., "Frequency Domain Parameter Estimation of Aeronautical Systems Without and with Time Delays," *Proceedings of the 7th IFAC Symposium on Identification and System Parameter Estimation*, York, England, U.K., 1985, pp. 669–674.

[28] Plaetschke, E., Weiss, S., and Rohlf, D., "Identification at High Angles of Attack Applied to X-31A Flight Test Data," DLR-Mitt. 93-14, Germany, Dec. 1993, pp. 181–198.

[29] Plaetschke, E., Jategaonkar, R. V., Rohlf, D., and Weiss, S., "Methoden zur Schätzung der Parameter eines Flugzeugs im Post-Stall-Bereich," *Proceedings of the 4th Aerospace Symposium on Safety in Air Traffic*, SFB-212 Technical Univ. of Braunschweig, Germany, Sept. 1994, pp. 297–306 (in German).

[30] Plaetschke, E., Mulder, J. A., and Breeman, J. H., "Results of Beaver Aircraft Parameter Identification," DFVLR-FB 83-10, Germany, March 1983.

[31] Klein, V., Cobleigh, B. R., and Noderer, K. D., "Lateral Aerodynamic Parameters of the X29 Aircraft Estimated from Flight Data at Moderate to High Angles of Attack," NASA TM 104155, Dec. 1991.

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[32] Iliff, K. W., "X-29A Lateral-Directional Stability and Control Derivatives Extracted from High-Angle-of-Attack Flight Data," NASA TP-3664, Dec. 1996.

[33] Iliff, K. W., and Wang, K. C., "Flight-Determined Subsonic Longitudinal Stability and Control Derivatives of the F-18 High Angle of Attack Research Vehicle (HARV) with Thrust Vectoring," NASA/TP-97-206539, Dec. 1997.

[34] Iliff, K. W., and Wang, K. C., "Flight-Determined, Subsonic, Lateral-Directional Stability and Control Derivatives of the Thrust-Vectoring F-18 High Angle of Attack Research Vehicle (HARV), and Comparison to the Basic F-18 and Predicted Derivatives," NASA/TP-1999-206573, Jan. 1999.

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CHAPTER 10

Data Compatibility Check

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10.1 INTRODUCTION

In general, parameter estimation methods assume availability of measured data containing adequate information about the cause–effect relationship with a minimum amount of corruption caused by systematic errors like scale factor, zero shift biases, and time lags. The process to generate experimental data with adequate information contents has been dealt with in Chapter 2 on data gathering. In this chapter, we deal with the means to check and improve the quality of the recorded data. The presence of noise is treated differently by the various methods. As discussed in Chapter 6, the least-squares estimates are sensitive to such systematic errors and noise in the independent variables. The output-error method of Chapter 4 and filter-error method of Chapter 5 principally allow estimation of noise statistics and incorporation of corrections for systematic errors as unknown parameters, but it can lead to correlations among the initial conditions or other aerodynamic derivatives, which affects the convergence and accuracy of the estimates. Thus, a data check, independent of system parameter estimation, is usually desirable. Furthermore, in the context of aerodynamic characterization of an aircraft, a large number of variables is usually measured and recorded during flight-test programs. Before using the raw flight-test data, it is often necessary and time saving to verify whether the recorded data are compatible or not. The basis for verifying the compatibility of measured channels is the use of kinematic relationships. Such a general procedure, which, in addition, allows for estimation of systematic instrument errors [1, 2], is commonly called *compatibility checking*.

The aim of a data compatibility check is to ensure that the measurements used for subsequent aerodynamic model identification are consistent and error free. For example, the measured angle of attack and angle of sideslip must match with those reconstructed from the measured linear accelerations and angular rates. Such a verification is possible in the case of flight data because the well-defined kinematic equations of aircraft motion provide a convenient means to bootstrap

the information through a numerical procedure. The primary goals of this data preprocessing step, also termed as flight-path reconstruction (FPR), are twofold:

1. Determine systematic instrument errors such as scale factors, zero shifts, and time delays in the measurements of flight variables.
2. Generate accurate aircraft states, which basically implies state estimation. The states to be estimated may comprise, for example, the velocity components, or other motion variables.

The FPR is also a necessary first step of the two-step approach using the least-squares technique to estimate aerodynamic derivatives [3] and of the similar approach referred to as estimation-before-modeling [4–6].

In general, there are two approaches to flight-path reconstruction: 1) a rigorous one in the stochastic framework based on the extended Kalman filter and 2) a simpler one in the framework of deterministic systems based on the output-error method, which we found adequate for many practical cases and suitable for more routine use. In the context of methods belonging to the general class of output error, state estimation is a part of the estimation procedure, and hence, during the data preprocessing step that we are currently addressing, it is usually sufficient to focus on the first goal only. Commonly, it aims at calibrating flow variables from flight data because accurate knowledge about the flow angles, that is, angle of attack and angle of sideslip, is necessary for parameter estimation and for other applications such as in-flight simulation incorporating model following control, gain scheduling, and direct flight control augmentation. These variables, conventionally measured using vanes or a five-hole probe, are prone to calibration, bias, and alignment errors and, as a rule, are of lower quality. Laboratory calibration is mostly inadequate, making necessary the determination of calibration characteristics, which can also be configuration dependent.

In this chapter we adapt the kinematic equations of aircraft motion to include systematic errors, postulate sensor models, and address both the aforementioned techniques of state and parameter estimation. We then deal with the estimation-before-modeling approach. This is followed by discussions on calibrating a five-hole airflow probe and on determination of the position error in static-pressure measurements from dynamic and tower flyby maneuvers. Finally, a more recent approach to calibrate the same variables from a special flight maneuver called wind-box maneuver applying Kalman filter technique and incorporating GPS information is presented. We also demonstrate the procedure of kinematic consistency checking and calibrating flow angles applying the output-error method developed in Chapter 4 and describe the details of model definition for the software and flight data provided.

10.2 KINEMATIC EQUATIONS

The kinematic model for the data compatibility check is derived from the equations of translational motion of an aircraft [7, 8]. For the purpose of

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simplification, it is assumed that the aircraft is a rigid body. The translational equations of motion in the body-fixed axes system with origin at the aircraft center of gravity are given by

$$\begin{aligned} m(\dot{u} + qw - rv) + mg\sin\theta &= F_X \\ m(\dot{v} + ru - pw) - mg\cos\theta\sin\phi &= F_Y \\ m(\dot{w} + pv - qu) - mg\cos\theta\cos\phi &= F_Z \end{aligned} \tag{10.1}$$

where F_X, F_Y, F_Z are the external (aerodynamic and thrust) forces. Rewriting the preceding equation in state variable form, suitable for parameter estimation purposes, yields

$$\begin{aligned} \dot{u} &= -qw + rv - g\sin\theta + a_x \\ \dot{v} &= -ru + pw + g\cos\theta\sin\phi + a_y \\ \dot{w} &= -pv + qu + g\cos\theta\cos\phi + a_z \end{aligned} \tag{10.2}$$

The terms a_x, a_y, and a_z in Eq. (10.2) are the same as F_X/m, F_Y/m, and F_Z/m, respectively. They represent the accelerations along the three axes. The angles ϕ and θ denote the roll and pitch angles, respectively.

The additional kinematical equations relating the Euler angles ϕ, θ, ψ to the body-fixed rotational rates p, q, r are given by

$$\begin{aligned} \dot{\phi} &= p + q\ \sin\phi\tan\theta + r\cos\phi\tan\theta \\ \dot{\theta} &= q\cos\phi - r\sin\phi \\ \dot{\psi} &= q\sin\phi\ \sec\theta + r\cos\phi\ \sec\theta \end{aligned} \tag{10.3}$$

The last set of kinematic relations for the position is given by

$$\begin{aligned} \dot{x}_g &= u\cos\psi\ \cos\theta + v(\cos\psi\ \sin\theta\ \sin\phi - \sin\psi\ \cos\phi) \\ &\quad + w(\cos\psi\ \sin\theta\ \cos\phi + \sin\psi\ \sin\phi) \\ \dot{y}_g &= u\sin\psi\cos\theta + v(\sin\psi\ \sin\theta\ \sin\phi + \cos\psi\ \cos\phi) \\ &\quad + w(\sin\psi\ \sin\theta\ \cos\phi - \cos\psi\ \sin\phi) \\ \dot{h} &= u\sin\theta - v\cos\theta\ \sin\phi - w\cos\theta\ \cos\phi \end{aligned} \tag{10.4}$$

Equations (10.2–10.4) represent the complete set of kinematical relationships used for FPR. These nonlinear equations are exact in terms of motion variables (u, v, w), (ϕ, θ, ψ), (x_g, y_g, h), (p, q, r), and (a_x, a_y, a_z) because the aerodynamic force and moment coefficients in terms of derivatives do not explicitly appear in these equations. Instead, we use equivalently the linear accelerations and angular rates. The first three triples are the state variables, whose values will be obtained from Eqs. (10.2–10.4). Assuming that the measurements of the last two triples are available, and hence known, we have all of the necessary

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information required for state estimation. Initial conditions are either estimated or determined from the measured data as discussed in Sec. 3.7. Thus, through the use of measured linear accelerations (a_x, a_y, a_z) and angular rates (p, q, r), it becomes possible to estimate the state variables (u, v, w, ϕ, θ, ψ, x_g, y_g, h) using the just-defined kinematic equations [9].

Once the states of the aircraft motion are known, it is fairly straightforward to derive other variables, such as angle of attack α, angle of sideslip β, true airspeed V, and dynamic pressure $\overline{q}$.

$$\begin{aligned} V &= \sqrt{u^2 + v^2 + w^2} \\ \alpha &= \tan^{-1}(w/u) \\ \beta &= \sin^{-1}(v/V) \\ \overline{q} &= \frac{1}{2}\rho V^2 \end{aligned} \tag{10.5}$$

This approach was initially used in dynamic flight testing to eliminate the need for direct measurement of angle of attack [1]. The density of air ρ can be assumed constant, if the variations in altitude during the test maneuver are small and can be obtained from standard atmosphere tables. Otherwise, it can be computed from the actual measurement of the static pressure p_s, using the universal gas law:

$$\rho = \frac{p_s}{RT_s} \tag{10.6}$$

where R is the gas constant and T_s the static temperature. The procedure to calibrate static pressure measurements will be discussed in Sec. 10.7.

In many cases the measurements of flow angles α and β are available. However, as already mentioned in the preceding section, these variables, measured generally using vanes or a five-hole probe, are prone to calibration, bias and alignment errors, and are inherently of lower quality. Because effects of flight configuration under operating conditions cannot be tested realistically in a laboratory, calibration from flight data is necessary to eliminate the errors. This is possible by appending a sensor model to the variables appearing in Eq. (10.5). Considering a simple sensor model in terms of scale factor, bias, and time delay as elaborated in Secs. 3.3 and 3.4, any general measured variable, say y, can be expressed as

$$y_m(t) = K_y\, y(t - \tau) + \Delta y \tag{10.7}$$

where K_y is the calibration factor, Δy the unknown instrument bias, and τ the time delay in the recorded signal. The subscript m on the left-hand side refers to the measured quantity. Ideally, the scale factor should be unity and the bias and time delay negligible. The instrument errors are usually assumed to be constant over the period of observation, that is, over the length of the time segment being analyzed. Thus, by considering the variables V, α, β, and $\overline{q}$ as the output variables and incorporating the sensor models similar to Eq. (10.7) in Eq. (10.5), we have a state-space model suitable for parameter estimation. Apart from the variables V, α, β, $\overline{q}$, the variables ϕ, θ, ψ, and h are also additionally

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considered as output variables. They are also necessary to enable estimation of the corresponding initial conditions, which might be unknown and, hence, will have to be estimated as well. A sensor model of Eq. (10.7) can be applied to output variables as well as input variables of the state-space model.

We also notice that state equations for geodetical positions x_g and y_g are decoupled, that is, these variables do not appear in the other equations in Eqs. (10.2–10.4). Hence, they can be omitted without affecting the results for the classical flight-path reconstruction purposes pertaining to variables of motion with respect to atmosphere only. Similarly, the $\dot{\psi}$ equation is also not explicitly coupled into the other equations. For the same reason as before, it would be possible to neglect this variable too, but it is usually retained as it aids estimation of bias in the yaw rate r. The state vector thus reduces to $[u\ v\ w\ \phi\ \theta\ \psi\ h]^T$.

There is one subtle point that is rarely brought out in connection with the preceding model. As postulated in Eqs. (10.2) and (10.5), it is a mix of inertial and aerodynamic coordinate frames; integration of the inertial accelerations yields incremental changes in an inertial coordinate frame, but they are used to compute the true airspeed in aerodynamic frame. This is valid only for constant wind, which would be accounted for in the initial condition. Under such an assumption, the equations are compatible for the intended purpose of determining instrument errors. An alternative would be to use ground speed. Likewise, in deriving the kinematical relations, apart from the assumption of rigid body, additional assumptions of flat and stationary Earth were implicitly incorporated. It is possible to include the effect of flexible aircraft and of a spherical rotating Earth. For an extended use of flight-path reconstruction, for example, verification of the onboard navigation, however, the full set of kinematic equations will be necessary, including a spherical rotating Earth. However, such extensions are not considered here because we are primarily addressing the problems dealing with information necessary for identification of aerodynamic model parameters.

The different motion variables in the kinematic equations (10.2–10.5) are with respect to the center of gravity. In Sec. 6.9.1, we have already covered the practical aspects of linear accelerometers mounted at off-c.g. locations, and transformations to compute accelerations referred to the c.g. The choice of c.g. as the reference point in Eqs. (10.2–10.5) was just for the simplicity sake; any other arbitrary fixed point can be used as well, but then accounting for the geometrical locations of the various sensors from this chosen reference point, instead of from the c.g. We stick to the c.g. as the reference point because it is easier to comprehend. Recalling from Sec. 6.9.1, the linear accelerations (a_x^{CG}, a_y^{CG}, a_z^{CG}) at the c.g. are computed from the accelerations (a_{xm}^{AS}, a_{ym}^{AS}, a_{zm}^{AS}) measured by the acceleration sensor (AS) at a point away from the c.g. through the following relation:

$$
\begin{aligned}
a_x^{CG} &= a_{xm}^{AS} + (q^2 + r^2)\, x_{\mathrm{ASCG}} - (pq - \dot{r})\, y_{\mathrm{ASCG}} - (pr + \dot{q})\, z_{\mathrm{ASCG}} - \Delta a_x \\
a_y^{CG} &= a_{ym}^{AS} - (pq + \dot{r})\, x_{\mathrm{ASCG}} + (p^2 + r^2)\, y_{\mathrm{ASCG}} - (qr - \dot{p})\, z_{\mathrm{ASCG}} - \Delta a_y \\
a_z^{CG} &= a_{zm}^{AS} - (pr - \dot{q})\, x_{\mathrm{ASCG}} - (qr + \dot{p})\, y_{\mathrm{ASCG}} + (p^2 + q^2)\, z_{\mathrm{ASCG}} - \Delta a_z
\end{aligned}
\tag{10.8}
$$

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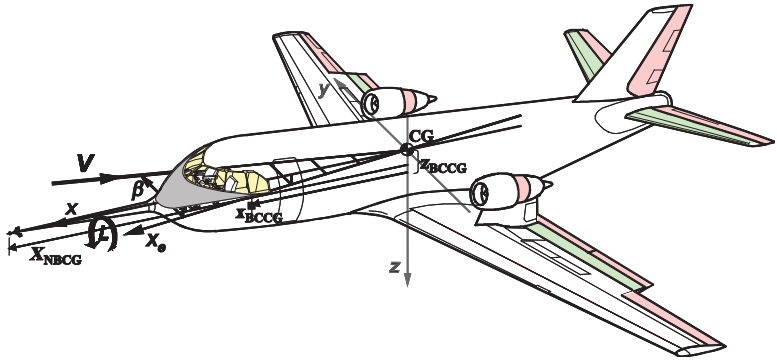

Fig. 10.1 Locations of off-c.g. sensors.

where the m in the subscript stands for the measured quantities and (x_{ASCG}, y_{ASCG}, z_{ASCG}) denote the position of the accelerometer with respect to the c.g. in the body-fixed coordinates; see Fig. 10.1. Exact location of the c.g. during the flight is determined from the instantaneous fuel quantity and weight and balance sheet. The biases in the measurements of (a_{xm}^{AS}, a_{ym}^{AS}, a_{zm}^{AS}) are denoted by (Δa_x, Δa_y, Δa_z), and the angular rates (p, q, r) appearing in Eq. (10.2) are given by ($p_m - \Delta p$, $q_m - \Delta q$, $r_m - \Delta r$) obtained from the measured rates (p_m, q_m, r_m) corrected for the biases (Δp, Δq, Δr). The variables ($\dot{p}$, $\dot{q}$, $\dot{r}$) required in Eq. (10.8) are obtained by numerical differentiation of the measured angular rates.

Besides the preceding computations to account for the off-c.g. location, another type of correction that might have to be performed results from the accelerometer misalignment. If the accelerometers are not aligned exactly along the body-fixed coordinates, it leads to cross coupling between the axes. An error in the orientation causes acceleration along the axis of sensitivity to be measured along the other orthogonal axes as well. This is schematically depicted in Fig. 10.2 for the alignment error of ε_θ in the pitch axis. It is obvious that the correct accelerations along the body-fixed coordinates will be given by

$$\begin{aligned} a_x^B &= a_{xm}^{AS} \cos\varepsilon_\theta + a_{zm}^{AS} \sin\varepsilon_\theta \\ a_z^B &= -a_{xm}^{AS} \sin\varepsilon_\theta + a_{zm}^{AS} \cos\varepsilon_\theta \end{aligned} \tag{10.9}$$

where the superscript B denote the accelerations along the body-fixed coordinates. Denoting the misalignments with respect to the three body-fixed X, Y, and Z axes as ε_ϕ, ε_θ, and ε_ψ respectively, and for small angles in radians approximating the sine and cosine function through zero and one, accelerations along the body-fixed

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coordinates can be obtained from the measurements provided by accelerometer triple as

$$\begin{bmatrix} a_x^B \\ a_y^B \\ a_z^B \end{bmatrix} = \begin{bmatrix} 1 & -\varepsilon_\psi & \varepsilon_\theta \\ \varepsilon_\phi \varepsilon_\theta + \varepsilon_\psi & 1 - \varepsilon_\phi \varepsilon_\theta \varepsilon_\psi & -\varepsilon_\phi \\ \varepsilon_\phi \varepsilon_\psi - \varepsilon_\theta & \varepsilon_\theta \varepsilon_\psi + \varepsilon_\phi & 1 \end{bmatrix} \begin{bmatrix} a_{xm}^{AS} \\ a_{ym}^{AS} \\ a_{zm}^{AS} \end{bmatrix} \tag{10.10}$$

The first term on the right-hand side of Eq. (10.10) is the coordinate transformation matrix for misalignments defined with respect to the body-fixed coordinates; see Fig. 10.2. It is usually possible to mount the triaxis accelerometer very precisely; the misalignments are extremely small. Furthermore, in many cases these corrections components are smaller than the measurement noise and vibrations. For these reasons, in practice, we usually do not consider this correction for accelerations. The preceding principle of correcting for alignment errors can be applied to any sensor. Like accelerometers, the alignments errors in rate gyro are also very small and negligible. If at all, exact alignment of an external long noseboom might pose some difficulties. If such an orientation error is known, it is best to correct it using the preceding equation. It would not be possible to estimate such alignment errors applying estimation methods, as they are correlated with the constant bias errors included in Eq. (10.7).

Strictly speaking, we should also include the scale factors for linear accelerations and angular rates as well, but they are omitted in Eq. (10.8) for the reasons of identifiability; they tend to get highly correlated with biases. Furthermore, scale factor errors in the inertial measurements from high-precision linear accelerometers and rate gyros are very small and their laboratory calibration more precise. For the same reason the bias errors are also very small, but we include them in the model to avoid drifts in the integrated variables.

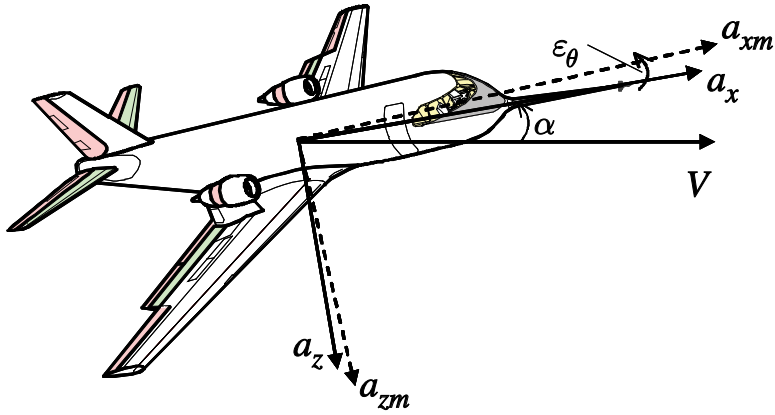

Fig. 10.2 Alignment error in pitch axis.

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Thus, flight-path reconstruction amounts to state estimation using kinematic relations, Eqs. (10.2–10.5), subject to systematic errors as in Eq. (10.7), and accounting for sensor locations as represented in Eq. (10.8). The complete set of nonlinear system equations is obtained by including, as just discussed, the sensor models in Eqs. (10.2–10.4), which leads to the following state equations [9, 10]:

$$
\begin{aligned}
\dot{u} &= -(q_m - \Delta q)w + (r_m - \Delta r)v - g\sin\theta + a_x^{CG}, && u(t_0) = u_0 \\
\dot{v} &= -(r_m - \Delta r)u + (p_m - \Delta p)w + g\cos\theta\sin\phi + a_y^{CG}, && v(t_0) = v_0 \\
\dot{w} &= -(p_m - \Delta p)v + (q_m - \Delta q)u + g\cos\theta\cos\phi + a_z{}^{CG}, && w(t_0) = w_0 \\
\dot{\phi} &= (p_m - \Delta p) + (q_m - \Delta q)\sin\phi\tan\theta + (r_m - \Delta r)\cos\phi\tan\theta, && \phi(t_0) = \phi_0 \\
\dot{\theta} &= (q_m - \Delta q)\cos\phi - (r_m - \Delta r)\sin\phi, && \theta(t_0) = \theta_0 \\
\dot{\psi} &= (q_m - \Delta q)\sin\phi\sec\theta + (r_m - \Delta r)\cos\phi\sec\theta, && \psi(t_0) = \psi_0 \\
\dot{h} &= u\sin\theta - v\cos\theta\sin\phi - w\cos\theta\cos\phi, && h(t_0) = h_0
\end{aligned}
\tag{10.11}
$$

where $(a_x^{CG}, a_y^{CG}, a_z^{CG})$ are defined in Eq. (10.8). The measurement equations are given by

$$
\begin{aligned}
V_m &= \sqrt{u^2 + v^2 + w^2} \\
\alpha_{\mathrm{NB},m} &= K_\alpha \tan^{-1}\left(\frac{w_{\mathrm{NB}}}{u_{\mathrm{NB}}}\right)_\tau + \Delta\alpha_{\mathrm{NB}} \\
\beta_{\mathrm{NB},m} &= K_\beta \sin^{-1}\left(\frac{v_{\mathrm{NB}}}{\sqrt{u_{\mathrm{NB}}^2 + v_{\mathrm{NB}}^2 + w_{\mathrm{NB}}^2}}\right)_\tau + \Delta\beta_{\mathrm{NB}} \\
\phi_m &= \phi \\
\theta_m &= \theta \\
\psi_m &= \psi \\
h_m &= h
\end{aligned}
\tag{10.12}
$$

The velocity components along the three body-fixed axes at an off-c.g. location, say at the noseboom (subscript NB), denoted by $(u_{\mathrm{NB}}, v_{\mathrm{NB}}, w_{\mathrm{NB}})$, are computed from (u, v, w) as follows:

$$
\begin{aligned}
u_{\mathrm{NB}} &= u - (r_m - \Delta r)\, y_{\mathrm{NBCG}} + (q_m - \Delta q)\, z_{\mathrm{NBCG}} \\
v_{\mathrm{NB}} &= v - (p_m - \Delta p)\, z_{\mathrm{NBCG}} + (r_m - \Delta r)\, x_{\mathrm{NBCG}} \\
w_{\mathrm{NB}} &= w - (q_m - \Delta q)\, x_{\mathrm{NBCG}} + (p_m - \Delta p)\, y_{\mathrm{NBCG}}
\end{aligned}
\tag{10.13}
$$

where $(x_{\mathrm{NBCG}}, y_{\mathrm{NBCG}}, z_{\mathrm{NBCG}})$ denote the offset distances from the center of gravity to the flow angle sensor mounted on the noseboom.

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The quantities appearing on the right-hand side of Eq. (10.12) are either the state variables (u, v, w, ϕ, θ, ψ, h) obtained by integration of the state equations (10.11) or from derived quantities (u_{NB}, v_{NB}, w_{NB}) using Eq. (10.13). The parameters K_α and K_β denote the scale factors for the angle of attack and angle of sideslip, and the biases in these variables are denoted as $\Delta\alpha_{NB}$ and $\Delta\beta_{NB}$, respectively. If the noseboom is not sufficiently long, these scale factors might be configuration dependent. We can also add bias corrections to other observation variables, but usually it is not possible because initial conditions are estimated.

In Eq. (10.12), the subscript τ implies that the computed angles of attack and sideslip, $\alpha_{NB,C} = \tan^{-1}(w_{NB}/u_{NB})$ and $\beta_{NB,C} = \sin^{-1}(v_{NB}/V_{NB})$, are to be time shifted through delays τ_α and τ_β, respectively. Using the notation of Eq. (3.14), the exact representation, for example, for angle of attack would be $\alpha_{NB,m}(t) = \alpha_{NB,C}(t - \tau_\alpha) + \Delta\alpha_{NB}$; but for a notational simplicity we have dropped the time t in the preceding set of equations. Usually, time delays are found in the measurements of attitude angles ϕ, θ, ψ and also in the flow variables V, α, β, $\bar{q}$. The angular rates can also contain time delays although not shown explicitly. For the estimation of delays, a time reference is necessary. This is provided by assuming that the linear accelerations have no time delays. This is a reasonable assumption because they are measured accurately and generally recorded directly. Thus, time delays are usually estimated relative to the time frame of the acceleration measurements.

Independent estimation of scale factor and bias parameters using Eq. (10.12) from a limited set of flight maneuvers needs some consideration because of the differences in order of magnitude between the trim (nominal) value and bias error. Small variations in scale factor (which is usually around one) multiplied by the large trim value [for example, true airspeed or angle of attack appearing on the right-hand sides of Eq. (10.12)] can compensate for small zero shifts without affecting the match significantly. This leads to a correlation between the two parameters. One possible approach to avoid such a correlation is to evaluate simultaneously several maneuvers at different trim conditions with good dynamic variations. As an example, Fig. 10.3 shows a combination of maneuvers that provided very good information to estimate the scale factors and biases independently; here we observe dynamic maneuvers at different trim points and quasi-steady maneuvers. The other alternative would be to reformulate the observation equations to include trim values explicitly and associate the scale factors with the variations about the nominal values. For example, the observation equation for angle of attack can be written as

$$\alpha_m = K_\alpha(\alpha - \alpha_0) + \alpha_0 + \Delta\alpha \tag{10.14}$$

where we have now introduced α_0 representing the trim value. As just elaborated, the angle of attack α will have to be computed appropriately from integrated states w and u, accounting for position of the sensor. The nominal value of the angle of attack α_0 is predefined for each time section being analyzed and held fixed. It can

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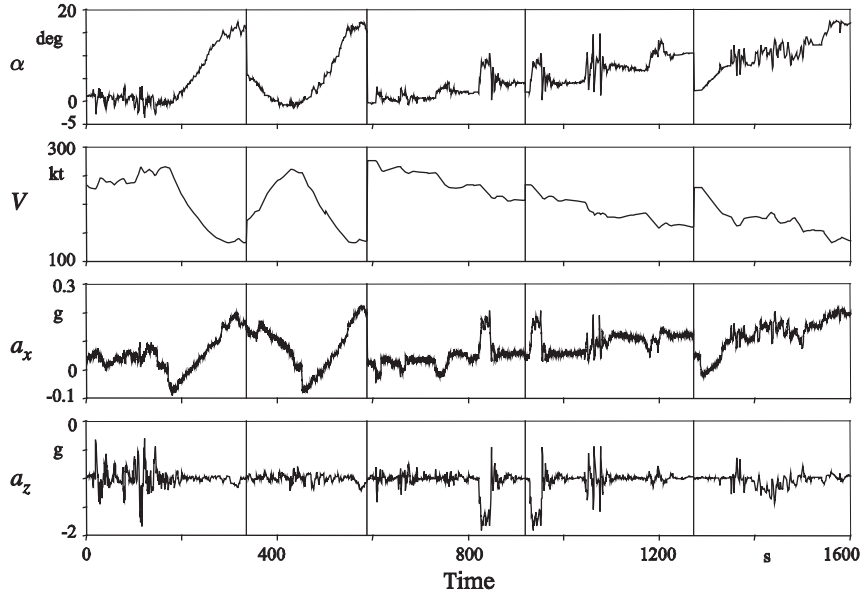

Fig. 10.3 Maneuver combination to avoid scale factor and bias correlation.

be obtained from the measured data (see Sec. 3.7 on treatment of initial conditions). A better approach would be to obtain it as $\alpha_0 = \tan^{-1}(w_0/u_0)$ from the initial conditions of states, which will be estimated along with the other instrument errors. However, the estimates of the scale factors in the two cases, namely, Eqs. (10.12) and (10.14), will be different. A similar formulation can be applied to other variables. The same approach can be adopted in the estimation of aerodynamic derivatives to overcome the correlation among the various derivatives, particularly that with the aerodynamic zero term [11].

Equation (10.12) is the minimum set of observations, provided we perform checking based on the six-degrees-of-freedom equations. In case further variables are to be checked, for example, calibrated airspeed, then one can add these as further observation variables, which might need some additional quantities like static air temperature, static pressure, and density of air.

10.3 FLIGHT-PATH RECONSTRUCTION TECHNIQUES

Equations (10.11) and (10.12) are in the general form of a state-space model defined in Chapter 3. Hence, it is evident that we are dealing with a problem that has been addressed a number of times in the preceding chapters. To recapitulate, the problem is to obtain the estimates of the instrument errors Θ and

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in some cases also of the state vector x subject to the postulated model of Eqs. (10.11–10.13). Depending upon the assumptions we make regarding the noise in the input variables (a_x, a_y, a_z, p, q, r, $\dot{p}$, $\dot{q}$, $\dot{r}$), we need either the extended Kalman filter or the output-error method to solve the preceding problem.

10.3.1 DETERMINISTIC APPROACH

In many practical applications, particularly when high-precision low-noise accelerometers and rate gyros are used to measure linear accelerations and angular rates, noise in these measurements is very small. Neglecting the noise in these variables, which are input variables in the model for flight-path reconstruction, makes the model for kinematic checking deterministic. It is apparent from the parameter estimation methods discussed in Chapters 4 and 5 that deterministic systems are much easier to handle. The state estimation problem reduces to simple numerical integration. Under such an assumption, the model for estimation of instrument errors is given by

$$\begin{aligned} \dot{x}(t) &= f[x(t), u(t), \Theta], \quad x(t_0) = x_0 \\ y(t) &= g[x(t), \Theta] \\ z(t_k) &= y(t_k) + v(t_k) \end{aligned} \tag{10.15}$$

For such a system model, the state, input, and observation vectors for data compatibility are given by

$$\begin{aligned} x &= [u \;\; v \;\; w \;\; \phi \;\; \theta \;\; \psi \;\; h]^T \\ u &= [a_x \;\; a_y \;\; a_z \;\; p \;\; q \;\; r]^T \\ y &= [V \;\; \alpha \;\; \beta \;\; \phi \;\; \theta \;\; \psi \;\; h]^T \end{aligned} \tag{10.16}$$

The unknown parameters are

$$\Theta = [\Delta a_x \;\; \Delta a_y \;\; \Delta a_z \;\; \Delta p \;\; \Delta q \;\; \Delta r \;\; K_\alpha \;\; \Delta\alpha \;\; K_\beta \;\; \Delta\beta]^T \tag{10.17}$$

In addition, it might be required to estimate the initial conditions $x_0 = [u_0 \;\; v_0 \;\; w_0 \;\; \phi_0 \;\; \theta_0 \;\; \psi_0 \;\; h_0]^T$.

To this state-space model, we can readily apply the output-error method for nonlinear systems discussed and developed in Chapter 4. The iterative procedure allows estimation of the parameter vector of Eq. (10.17) and also of the covariance matrix of the measurement noise R. Thus, in this approach the problem of data compatibility check is solved as a parameter estimation problem considering a parameter vector consisting of unknown initial conditions and the instrument errors [9, 12, 13]. Because the output-error method has been elaborately presented and already applied to a few examples, we do not go into any further discussion of the algorithm.

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10.3.2 STOCHASTIC APPROACH

A rigorous approach to the flight-path reconstruction is based on the EKF (extended Kalman filter) that accounts for measurement noises in both input and output variables [14–20]. Introduction of the noise in the input variables makes the system represented in Eq. (10.15) stochastic. In such a case, the model of Eq. (10.15) gets modified to

$$\begin{aligned} \dot{x}(t) &= f[x(t), u_m(t) - w(t), \Theta], \quad x(t_0) = x_0 \\ y(t) &= g[x(t), \Theta] \\ z(t_k) &= y(t_k) + v(t_k) \end{aligned} \tag{10.18}$$

where u_m is the measured input vector, $w(t)$ represents the noise in the measured inputs (a_x, a_y, a_z, p, q, r), and $v(t_k)$ that in the observation variables. The noise processes are characterized by Gaussian white noise with zero mean and covariance matrices Q and R respectively, and the two noise processes are uncorrelated, that is,

$$\begin{aligned} E\{w(t_k)\} &= 0, \quad E\{w(t_k)\, w(t_\ell)^T\} = Q\delta_{k\ell} \\ E\{v(t_k)\} &= 0, \quad E\{v(t_k)\, v(t_\ell)^T\} = R\delta_{k\ell} \\ E\{w(t_k)v^T(t_\ell)\} &= 0 \end{aligned} \tag{10.19}$$

Before going into the details of an estimation procedure, let us try to correlate few variables appearing in Eq. (10.18) to those that we have used so far. Hitherto, the measurements of input variables were assumed to be noise free; hence, u_m was simply denoted as u in Eq. (10.15). Now, in the present case, we try to account for the noise in the inputs, hence the term $u_m(t) - w(t)$ instead of $u(t)$ only. Furthermore, the noise $w(t)$ in this context is not the same as the process noise (atmospheric turbulence) considered in Chapter 5 although we have denoted both by the same variable w. In the present case, similar to the measurement noise covariance matrix R of the outputs, we use covariance matrix Q to characterize the input noise. On the other hand, the atmospheric turbulence was characterized by process noise distribution matrix F, which was assumed unknown and to be identified.

There are different approaches to solve the preceding general problem of state and parameter estimation. In the simplest form, we prespecify the noise statistics appropriately and apply the extended Kalman filter to the augmented state vector consisting of state variables of the kinematic equations and the various instrument errors, that is, scale factors and bias parameters. A more advanced procedure is based on a variational solution of a nonlinear, fixed-interval

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smoothing problem, minimizing a quadratic cost function in terms of P_0, R, and Q, which allows estimation of the initial conditions x_0 and the noise covariance matrix Q as well [21]. Here, we restrict ourselves to the simpler form based on prespecifying the noise covariances.

As already discussed in Chapter 7 and Appendix F, by defining additional auxiliary differential equations $\dot{\Theta} = 0$ for the unknown parameters, which are assumed constant over the period of observation, and augmenting Θ to x, leads to the augmented state vector:

$$\begin{aligned} x_a &= [x^T \;\; \Theta^T]^T \\ &= [u \;\; v \;\; w \;\; \phi \;\; \theta \;\; \psi \;\; h \\ &\qquad \Delta a_x \;\; \Delta a_y \;\; \Delta a_z \;\; \Delta p \;\; \Delta q \;\; \Delta r \;\; K_\alpha \;\; \Delta\alpha \;\; K_\beta \;\; \Delta\beta]^T \end{aligned} \tag{10.20}$$

The extended Kalman filter for this nonlinear model consists of the following [22]:

Prediction step:

$$\begin{aligned} \tilde{x}_a(k+1) &= \hat{x}_a(k) + \int_{t_k}^{t_{k+1}} f_a[\hat{x}_a(t), \overline{u}_m(k)]\mathrm{d}t \\ \tilde{P}(k+1) &= \Phi(k+1)\hat{P}(k)\Phi^T(k+1) \\ &\qquad + \Psi(k+1)\, B(k)\, Q(k)\, B^T(k)\, \Psi^T(k+1) \end{aligned} \tag{10.21}$$

with initial conditions $\hat{x}_a(0) = x_{a0}$ and $\hat{P}(0) = P_0$.

Correction step:

$$\begin{aligned} K(k) &= \tilde{P}(k)\, C^T(k) \left\{C(k)\tilde{P}(k)C^T(k) + R(k)\right\}^{-1} \\ \hat{x}_a(k) &= \tilde{x}_a(k) + K(k)\{z(k) - g_a[\tilde{x}_a(k)]\} \\ \hat{P}(k) &= \{I - K(k)C(k)\}\tilde{P}(k) \\ &= \{I - K(k)C(k)\}\tilde{P}(k)\{I - K(k)C(k)\}^T + K(k)R(k)K^T(k) \\ &\qquad\qquad \text{for } k = 1, 2, \ldots, N \end{aligned} \tag{10.22}$$

where $\overline{u}$ is the average or interpolated input (refer to Sec. 3.8.1); the notation of $\sim$(tilde) and $\hat{}$ (hat) followed in Chapter 5 has been used in Eqs. (10.21) and (10.22) to denote the predicted and updated states, respectively. In the other commonly used notation, they are denoted as $\hat{x}(k|k-1)$ $[=\tilde{x}(k)]$ and $\hat{x}(k\,|\,k)$ $[=\hat{x}(k)]$. Likewise, the error covariances as $\hat{P}(k\,|\,k-1)$ $[=\tilde{P}(k)]$, $\hat{P}(k\,|\,k)$ $[=\hat{P}(k)]$ and $\Phi(k\,|\,k-1)$ $[=\Phi(k)]$ and $\Psi(k\,|\,k-1)$ $[=\Psi(k)]$. The state transition matrix Φ

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and its integral Ψ at each discrete point are given by Eqs. (3.38–3.39), which are repeated here for convenience:

$$\begin{aligned}\Phi(k+1) &= I + A(k)\Delta t + A^2(k)\,\frac{\Delta t^2}{2!} + \cdots \\ \Psi(k+1) &= I\,\Delta t + A(k)\frac{\Delta t^2}{2!} + A^2(k)\frac{\Delta t^3}{3!} + \cdots\end{aligned} \tag{10.23}$$

The Jacobian matrices appearing in Eqs. (10.21–10.23) are defined as

$$\begin{aligned}A(k) &= \left(\frac{\partial f_a}{\partial x_a}\right)_{\hat{x}_a(k-1),\,u(k-1)}, \quad B(k) = \left(\frac{\partial f_a}{\partial u}\right)_{\hat{x}_a(k-1),\,u(k-1)} \\ C(k) &= \left(\frac{\partial g_a}{\partial x_a}\right)_{\tilde{x}_a(k)}\end{aligned} \tag{10.24}$$

These matrices are approximated using the central difference formula discussed in Sec. 3.5.3. The equation for $\tilde{P}$ is slightly different compared to Eq. (7.53). This is because of the differences we have already discussed at the beginning of this section, namely, between input noise and atmospheric turbulence.

The initial values for $\hat{x}_a(t_0)$ and P_0 need to be specified to complete the algorithm. The initial values for the states are derived by averaging over the first few data points. Determination of P_0 can be obtained from the stationary part of the time segment, preceding the maneuver being filtered. An alternative would be to specify it arbitrarily; usually, a larger value of the initial state error covariance is necessary to allow tracking of the parameters, whose initial values might differ considerably from the final estimates.

The state estimation applying the extended Kalman filter is often followed by a smoothing technique applied to reduce the inherent bias in the EKF estimates [23]. From the three types of smoothing techniques, namely, fixed-interval, fixed-point, and fixed-lag smoothing [22], the Rauch–Tung–Striebel smoother [24, 25] belonging to the fixed-interval type has been mostly used in the aircraft state estimation, mainly because we perform state estimation for a fixed length of recorded time corresponding to specific dynamic maneuvers. The smoothing procedure is based on using the state estimates from all N data points, not just up to the kth point as done in the Kalman filter. It leads to a two-pass procedure, the first pass being the regular extended Kalman filter that we have just discussed, called here *forward Kalman filter*, and the second pass being also a Kalman filter, but running backwards from $N-1$ to 1. Figure 10.4 depicts the schematic of the combined filtering-smoothing technique. The smoothed estimate is a weighted linear combination of the forward and backward Kalman filter estimates.

Without going into the detailed derivation, we specify here just the final equations for the smoothed state estimates and the error covariance matrix

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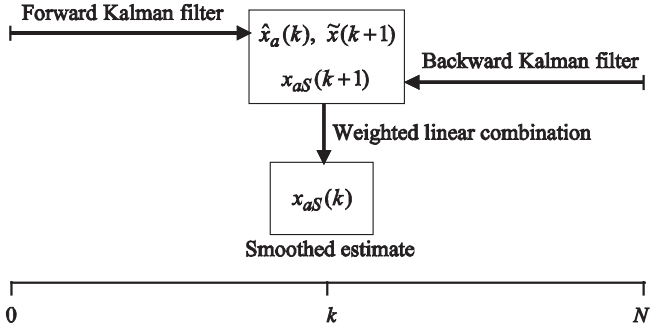

Fig. 10.4 Schematic of filtering and smoothing procedures.

propagation, which can be found in [24], or any standard book on filtering, for example, [22] or [25]:

$$
\begin{aligned}
K_S(k) &= \hat{P}(k)\,\Phi^T(k+1)\tilde{P}^{-1}(k+1)\\
x_{aS}(k) &= \hat{x}_a(k) + K_S(k)\{x_{aS}(k+1) - \tilde{x}_a(k+1)\}\\
P_S(k) &= \hat{P}(k) + K_S(k)\big\{P_S(k+1) - \tilde{P}(k+1)\big\}K_S^T(k) \qquad (10.25)\\
&= \{I - K_S(k)\Phi(k+1)\}\hat{P}(k)\,\{I - K_S(k)\,\Phi(k+1)\}^T\\
&\quad + K_S(k)\big\{\tilde{P}(k+1) + \Psi(k+1)\,B(k)Q(k)\,B^T(k)\Psi^T(k+1)\big\}K_S^T(k)\\
&\qquad\qquad \text{for } k = N-1,\ N-2,\ \ldots
\end{aligned}
$$

where the subscript S is added to denote variables pertaining to the smoother and thereby distinguish between the filter (forward pass) and smoother (backward pass). Accordingly, $x_{aS}(k)$, $K_S(k)$, $P_S(k)$ denote the smoothed states, smoother gain matrix, and the corresponding error covariance matrix. As in the case of the forward filter, these variables are also denoted in the literature as $x_a(k\,|\,N)$ and $P(k\,|\,N)$, where $|N$ implies the reverse filter starting from N-1th point.

It is apparent from the discrete time point index k running from $N-1$ to 1 in Eq. (10.25) that the smoother works backwards in time and requires variables $\hat{x}_a(k)$, $\tilde{x}_a(k+1)$, $\hat{P}(k)$, $\tilde{P}(k+1)$, $\Phi(k+1)$, and $\Psi(k+1)$ from the forward filter. As such, their time histories will have to be stored for the complete N time points during the first pass of forward filtering. As in the case of the forward filter (EKF), we have two forms for the covariance matrix of the state prediction error $P_S(k)$ in Eq. (10.25), the second one being numerically more stable for the reasons that we have already discussed in Chapter 7 and Appendix F.

Solutions based on the extended Kalman filter (EKF) techniques were proposed in the literature for this combined problem of estimation of states and of instrument errors [14–20]. In the simplified approach presented here, the noise statistics

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is generally prespecified, which might not be quite accurate, and thereby might affect the estimates. As already pointed out in Sec. 5.5.2, reasonable information about the R and Q matrices can be obtained from the laboratory calibration of the various measurement sensors. Alternatively, these noise characteristics can also be obtained applying Fourier smoothing techniques [26]. In the context of flight-path reconstruction, it is once again pointed out that the input noise is not be confused with the atmospheric turbulence. It implies noise in the measurements of the linear accelerations and angular rates, which are in this case the input variables. If the EKF-based approach is to be applied to consistency checking of flight data gathered in turbulent atmospheric condition, it requires not only an extended Kalman filter, but also simultaneous modeling of atmospheric turbulence. Prespecification of noise statistics in such cases is poor and can vary from case to case. Alternatively, as discussed in Chapter 5, the filter-error method accounting for both process and measurement noise can be used.

The deterministic approach based on the output-error method, presented in Sec. 10.3.1, was suggested as an alternative to the EKF, which had exhibited convergence problems [12, 13]. Comparison of EKF and ML methods for flight-path reconstruction indicates that both the techniques yield similar results [13, 18, 19]. Furthermore, it is reported in [18] that for higher noise levels the ML method appears to perform better compared to the EKF. From several aircraft cases investigated, it has been our experience that the maximum likelihood output-error method accounting for noise in the output variables only is quite adequate for this purpose. It can be used more routinely than the EKF approach, which is more rigorous but is more involved, has some limitations as well, and may perform poorly in the absence of accurate a priori knowledge about noise statistics.

As discussed in this section, the kinematic consistency checking provides an a posteriori means to verify recorded data and to estimate instrument errors pertaining to the aircraft motion variables. It is an off-line procedure that can be applied during the data analysis after flight testing. Besides these motion variables, aerodynamic model identification also needs the measurements of control surface deflections. The measurements of these deflections and other control inputs also should be verified and corrected for instrument and recording errors. Unfortunately, there is no way to check the accuracy of these measurements because there are no kinematic relations between them and other variables. This is a fundamental problem, for which there is no easy solution. Hence, calibration of these independent inputs needs great care, and considerable effort is spent on laboratory and in situ calibration of these measurements and on reverification at regular intervals.

10.4 ESTIMATION-BEFORE-MODELING APPROACH

The estimation-before-modeling (EBM) methodology is once again a two-step procedure to model aircraft aerodynamics; the first step is that of generating

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smoothed time histories of the vehicle states, and the second is identifying the aerodynamic model parameters, usually applying the linear regression technique discussed in Chapter 6. Compared to the conventional two-step approach based on the flight-path reconstruction discussed in Sec. 10.3 followed by parameter estimation applying regression technique, the EBM differs primarily in the first step concerned with the state estimation only. In Sec. 10.3, we treated the linear accelerations and angular rates as inputs to the kinematic equations, estimating constant scale factors and bias errors in them and generating smoothed time histories for the states. The linear accelerations and angular rates (input variables) were not estimated and smoothed. In contrast, the EBM methodology, although conceptually very similar, generates smoothed estimated of the states and estimates the instrument errors, and additionally the optimal, smoothed time histories of the aerodynamic forces and moments. We pursue here the discussion of the first step only because the second step of identification of aerodynamic model parameters applying the regression method covered in Chapter 6 is the same in both the cases.

The development of the mathematical model for the state estimation in the present case is very similar to that discussed in Sec. 10.3, the only difference being to include additional differential equations for the angular rates p, q, and r. Once again considering the aircraft as a rigid body, the equation of motion in the body-fixed frame referred to the c.g. can be formulated as [4–6]

$$\begin{aligned}
\dot{u} &= -qw + rv - g\sin\theta + X \\
\dot{v} &= -ru + pw + g\cos\theta\sin\phi + Y \\
\dot{w} &= qu - pv + g\cos\theta\cos\phi + Z \\
\dot{p} &= pqC_{11} + qrC_{12} + qC_{13} + L + NC_{14} \\
\dot{q} &= prC_{21} + (r^2 - p^2)C_{22} - rC_{23} + M \\
\dot{r} &= pqC_{31} + qrC_{32} + qC_{33} + LC_{34} + N \\
\dot{\phi} &= p + q\sin\phi\tan\theta + r\cos\phi\tan\theta \\
\dot{\theta} &= q\cos\phi - r\sin\phi \\
\dot{\psi} &= q\sin\phi\sec\theta + r\cos\phi\sec\theta \\
\dot{h} &= u\sin\theta - v\cos\theta\sin\phi - w\cos\theta\cos\phi
\end{aligned} \tag{10.26}$$

where (X, Y, Z) and (L, M, N) are the normalized (that is, divided by mass or appropriate moment of inertia) external aerodynamic forces and moments acting on the aircraft. The constant coefficients C_{ij} appearing in Eq. (10.26) are given by

$$
\begin{aligned}
C_{11} &= \frac{I_{xz}(I_z + I_x - I_y)}{I_x I_z - I_{xz}^2} & C_{21} &= \frac{I_z - I_x}{I_y} \\
C_{12} &= \frac{I_z(I_y - I_z) - I_{xz}^2}{I_x I_z - I_{xz}^2} & C_{22} &= \frac{I_{xz}}{I_y} \\
C_{13} &= 0 & C_{23} &= 0 \\
C_{14} &= \frac{I_{xz}}{I_x} & & \\
C_{31} &= \frac{I_x(I_x - I_y) + I_{xz}^2}{I_x I_z - I_{xz}^2} & C_{32} &= \frac{I_{xz}(I_y - I_z - I_x)}{I_x I_z - I_{xz}^2} \\
C_{33} &= 0 & C_{34} &= \frac{I_{xz}}{I_z}
\end{aligned}
\tag{10.27}
$$

The terms C_{13}, C_{23}, and C_{33} are zero for the case of neglecting the angular momentum due to engines. If the angular momentum due to rotating parts is considerable and cannot be neglected, then these three terms will have to be appropriately considered [4].

Equation (10.26) can be reformulated in the general state variable form:

$$\dot{x}(t) = f[x(t), \chi(t)] + w(t) \tag{10.28}$$

where $w(t)$ is, as in the preceding case, the Gaussian noise vector with zero mean and covariance matrix Q. The state vector x and the vector χ consisting of external forces and moments are given by

$$
\begin{aligned}
x &= [u \;\; v \;\; w \;\; p \;\; q \;\; r \;\; \phi \;\; \theta \;\; \psi \;\; h]^T \\
\chi &= [X \;\; Y \;\; Z \;\; L \;\; M \;\; N]^T
\end{aligned}
\tag{10.29}
$$

In the preceding approach discussed in Sec. 10.2, we had used for χ equivalently the measured accelerations and angular accelerations as inputs. Instead of treating them as inputs, in the present case we treat them as pseudostate variables by modeling them stochastically. This is done for each component of χ by a third-order Gauss–Markov process, which can be represented as [6]

$$\dot{\chi}_i = L_i \chi_i + \xi_i(t); \quad i = 1, 2, \ldots, 6 \tag{10.30}$$

where

$$L_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \tag{10.31}$$

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and ξ is the white zero-mean Gaussian noise, generated using the random number generator. Equations (10.30–10.31) imply that for each component of χ we have three first-order differential equations. For example, the pitching moment M will consist of ($\dot{M} = M_1$; $\dot{M}_1 = M_2$; $\dot{M}_2 = 0$), where (M, M_1, M_2) are the three pseudostates. Without going into the statistical details of Gauss–Markov process, we simply note here that it is used to model the external forces and moments. Equation (10.31) yields a quadratic interpolation polynomial as a function of time; the recursive algorithm updates the polynomial coefficients at each sample. Thus, a total of 18 additional states are introduced, corresponding to three for each of the six forces and moments.

Finally, as in the preceding case, assuming the instrument errors, represented by the combined vector Θ, to be constant over the period of observation, they are modeled by the auxiliary dynamic system:

$$\dot{\Theta} = 0 \tag{10.32}$$

Combining Eqs. (10.28), (10.30), and (10.32) leads to the augmented system:

$$\begin{bmatrix} \dot{x} \\ \dot{\chi} \\ \dot{\Theta} \end{bmatrix} = \begin{bmatrix} f[x(t), \chi(t)] \\ L\chi(t) \\ 0 \end{bmatrix} + \begin{bmatrix} w \\ \xi \\ 0 \end{bmatrix} \tag{10.33}$$

consisting of 10 basic states of aircraft motion variables, 18 pseudostates for the six aerodynamic forces and moments (each modeled as a third-order Gauss–Markov process), and auxiliary states whose number will depend upon the number of instrument errors to be estimated. The matrix L is block-diagonal consisting of six blocks, each of the form of Eq. (10.31).

To complete the model postulate for state and parameter estimation, we define the observation equations

$$\begin{aligned} y(t) &= g[x(t)] \\ z(t_k) &= y(t_k) + \nu(t_k) \end{aligned} \tag{10.34}$$

where the observation vector y consists of the following variables:

$$y = [V \;\; \alpha \;\; \beta \;\; \phi \;\; \theta \;\; \psi \;\; h \;\; p \;\; q \;\; r \;\; a_x \;\; a_y \;\; a_z]^T \tag{10.35}$$

The observation variables appearing in Eq. (10.35) are standard variables; their exact forms have already been defined earlier in the chapter; they can also be found in [6] and [8]. We also specifically note that because (a_x, a_y, a_z, p, q, r) are now treated as output variables biases in their measurements appear in the observation equations, not in the state equation formulated in Eq. (10.26). This is in contrast to Eq. (10.11) of the foregoing approach elaborated in Sec. 10.2. The vector of unknown parameters corresponding to the instrument errors is once again given by

$$\Theta = [\Delta a_x \;\; \Delta a_y \;\; \Delta a_z \;\; \Delta p \;\; \Delta q \;\; \Delta r \;\; K_\alpha \;\; \Delta\alpha \;\; K_\beta \;\; \Delta\beta]^T \tag{10.36}$$

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Additional corrections might have to be introduced, but the choice of corrections given in Eq. (10.36) has proved adequate in most of the cases analyzed. Introducing additional corrections in the observation equation (10.35) and appending the parameter vector of Eq. (10.36) and the auxiliary state equation of (10.32) is a trivial exercise. As common to any parameter estimation problem, care needs to be exercised while adding new parameters to be identified because all of them might not be identifiable; they might be correlated, or there might not be adequate information in the data set being analyzed.

Now, we can apply the extended Kalman filter and smoother, Eqs. (10.21–10.25), discussed in Sec. 10.3.2 to the system defined by Eqs. (10.33–10.36). This provides the optimal smoothed estimates of the basic states x, of the normalized aerodynamic forces and moments (X, Y, Z, L, M, N) that are elements of χ, and of the instrument errors Θ.

Augmentation of the basic aircraft states through 18 additional pseudostates to model the aerodynamic forces and moments makes the model computationally intensive and time consuming compared to the approach presented in the preceding section. Considerable saving in the computational time is possible by making optimum use of the block-diagonal structure of the L matrix appearing in Eq. (10.33) and also of the null matrix corresponding to the auxiliary state equations $\dot{\Theta} = 0$; see the schematic of the system matrix of Eq. (10.33) shown in Fig. 10.5. The nonzero elements appear in blocks shown in gray with

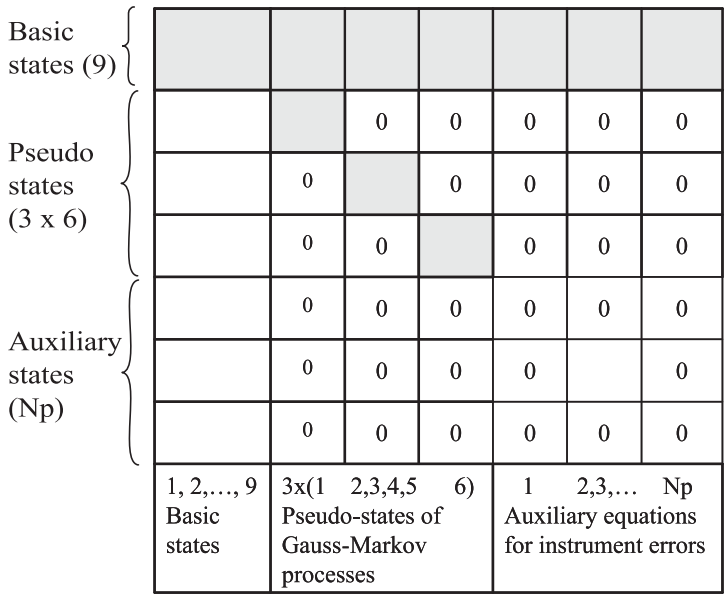

Fig. 10.5 Block structure of system matrix for state estimation using EBM.

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bold borders. The special matrix structure shown in the figure can be used advantageously to reduce the number of numerical computations while computing the state transition matrix Φ, its integral, and the Jacobian matrices defined in Eq. (10.24).

EBM methodology provides smoothed time histories of the external normalized forces and moments, which can be analyzed without any further processing in the second step to estimate aerodynamic derivatives. Thus, numerical differentiation of the angular rates, which is otherwise necessary for the two-step approach based on the procedure of Sec. 10.3, is eliminated. As elaborated in Sec. 6.9.1, computation of aerodynamic forces and moments required angular accelerations, which are not directly measured, but were generated through numerical differentiation. Such procedures might be inaccurate in the presence of noise. But, the advantages of EBM from this aspect might be just marginal because, generally, the noise in the angular rates is low and good digital symmetrical filters are available to smooth the data before numerical differentiation. Furthermore, approximation of the forces and moments using a third-order process is a compromise between total model order and anticipated time rate of change in forces and moments [5].

10.5 EXAMPLE

To demonstrate the procedure of data compatibility checking, we consider here a simple case of checking data recorded during typical flight tests carried out with the research aircraft ATTAS [10]. Three dynamic maneuvers are analyzed, namely, an elevator input maneuver exciting the short-period motion, an aileron input maneuver resulting in bank-to-bank motion, and a rudder input maneuver leading to Dutch roll motion. The first maneuver is 25 s long, and the other two are of 30 s duration each. These maneuvers were executed starting from the same nominal trim condition and performed manually by the pilot. The flight maneuvers were carried out under seemingly steady atmospheric conditions when the turbulence was considered negligible. The data were recorded with a sampling time of 40 ms. To these three maneuvers, which were used already in Chapter 6, we apply the deterministic approach presented in Sec. 10.3.1, using the output-error method and the corresponding estimation program "ml_oem2" elaborated in Chapter 4.

The state, input, and observation variables for the kinematic model consisting of state and observation equations (10.11–10.12) and the corresponding functions to code the right-hand sides are defined as

	No. of variables		Function name
States	7	$u, v, w, \phi, \theta, \psi, h$	"xdot_TC22_fpr"
Outputs	7	$V, \alpha_{NB}, \beta_{NB}, \phi, \theta, \psi, h$	"obs_TC22_fpr"
Inputs	9	$a_x, a_y, a_z, p, q, r, \dot{p}, \dot{q}, \dot{r}$	

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To start with, we assume that the angle of attack and angle of sideslip have been correctly calibrated and require no corrections. Accordingly, we introduce bias corrections for the input variables only. For such a case, the unknown parameter vector Θ consists of

$$\Theta = [\Delta a_x \;\; \Delta a_y \;\; \Delta a_z \;\; \Delta p \;\; \Delta q \;\; \Delta r]^T$$

Designating this test case as 22, the model definition provided by the function /FVSysID2/chapter04/mDefCase22.m, which is called from the main program "/FVSysID2/chapter04/ml_oem2," is as follows:

```
test_case = 22;                 % Test case number
state_eq  = "xdot_TC22_fpr";    % Function for state equations
obser_eq  = "obs_TC22_fpr";     % Function for observation equations
Nx        = 7;                  % Number of states
Nout      = 7;                  % Number of observation variables
Ninp      = 9;                  % Number of input (control) variables
NparSys   = 6;                  % Number of system parameters
Nparam    = NparSys;            % Total number of parameters
dt        = 0.04;               % Sampling time
LineSrch  = 0;                  % Flag for line search option
```

The starting values for the unknown parameters, that is, biases are zeros and the integer flags parFlag are set to one, indicating that all parameters are to be estimated. The flag for line search option LineSrch is set to zero, indicating that the heuristic approach is to be applied to correct the progress of optimization if intermediate divergence is observed while applying Gauss–Newton method.

The data to be analyzed, the arrays Z(Ndata,Nout) and Uinp(Ndata,Ninp), are loaded from three data files:

```
load ..\flt_data\fAttasElv1
load ..\flt_data\fAttasAil1
load ..\flt_data\fAttasRud1
```

Because we are analyzing three separate maneuvers simultaneously, we additionally define

```
Nzi = 3;                        % Number of time segments
izhf = [Nts1; Nts1 + Nts2;      % Cumulative index
        Nts1 + Nts2 + Nts3];
```

where Nts1, Nts2, and Nts3 are the number of data points in the three individual time segments and izhf is the cumulative index at which the maneuvers end when they are concatenated.

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As in the test cases of the preceding chapters, it is necessary to provide the input and output variables in proper units. This has been taken care of in the model definition function "mDefCase22.m," immediately after loading the data files and while assigning the appropriate channels to the arrays Z and Uinp. The angular accelerations are generated by numerical differentiation of the measured angular rates using a differentiator filter "ndiff_Filter08.m" discussed in Sec. 2.5.2, called from the function "ndiff_pqr.m." The initial conditions (u_0, v_0, w_0) for the three time segments are set to (129.4; –0.9; 5.05), (129.4; −0.9; 5.15), and (128.7; −0.9; 4.75) m/s, and those for the other states, namely, (ϕ, θ, ψ, h) to the measured values of the respective first data point. They are assigned for the three time segments as x0(:,1), x0(:,2), and x0(:,3), respectively. The flags for all of the initial conditions for all of the three time segments are set to zero, implying all of the initial conditions are kept fixed at their starting values. The exact procedure of defining initial conditions x_0 and setting the corresponding flags parFlagX0 can be easily traced in "mDefCase22.m."

The biases in the input variables, that is, Δa_x, Δa_y, Δa_z, Δp, Δq, Δr, are defined as the first six elements of the parameter vector param, followed by the scale factor and bias for the angle of attack as seventh and eighth elements and then those for the angle of sideslip as ninth and tenth elements in the model definition function "mDefCase22.m." The starting values for the biases in the input variables are set to zero, and the corresponding flags parFlag are set to ones, implying that all of them are free and are to be estimated. To start with, the scale factors and biases for flow angles are specified as one and zero, respectively. The corresponding parFlags(7:10) are set to zero, implying they are kept fixed at their starting values specified. The upper and lower bounds for the free parameters are set to default values $-\infty$ and ∞, respectively, implying the choice of unconstrained optimization. Setting the counter for the maximum number of iterations niter_max to zero, a simulation is performed by running the program "ml_oem2" from the directory/FVSysID2/chapter04/. The time histories of the input variables, in the present case those of linear accelerations and angular rates, are shown in Fig. 10.6. Time histories for the output variables V, α_{NB}, β_{NB}, ϕ θ, ψ, h are plotted in Fig. 10.7, showing flight measured data in continuous lines, and those simulated using the kinematic equations for the specified initial conditions and without any corrections of the instrument errors in dashed lines. It is apparent that the specified initial conditions are reasonably accurate, but the simulation shows drifts in all of the variables that are attributed to biases in the input variables and small errors in the starting values of the initial conditions. Here, we recall our suggestion made in Sec. 10.2 that biases in the input variables need to always be considered to avoid drifts in the integrated variables, even when such biases may be very small.

Now, setting the counter niter_max to a higher value, say, 10, we restart the estimation program. It is observed that the estimation procedure starting with the initial cost function det(R) value of 1.5138×10^{-12} converges to the minimum of 1.8791×10^{-26} within four iterations. The bias parameters are

estimated to be

$$\Delta a_x = 0.0092 \text{ m/s}^2, \quad \Delta a_y = 0.00024 \text{ m/s}^2, \quad \Delta a_z = 0.0960 \text{ m/s}^2$$
$$\Delta p = 0.0011 \text{ rad/s}, \quad \Delta q = 0.0033 \text{ rad/s}, \quad \Delta r = 0.0010 \text{ rad/s}$$

None of the estimates were correlated (>0.9), and their standard deviations were small, suggesting reliable values. The large relative standard deviation associated with Δa_y is because the bias in this measurement is estimated to be very small in magnitude. The resulting match shown in Fig. 10.8 between the flight measured and reconstructed responses for the variables V, α_{NB}, β_{NB}, ϕ, θ, ψ, h is seen to be acceptable. A careful look at the match for the angle of sideslip β in Fig. 10.8 and an enlarged plot of the angle of sideslip (not shown here) shows some drift, which is attributed mostly to a small mismatch in the initial values that were only roughly specified and held fixed.

We now extend the scope of analysis to estimate the initial conditions on the state variables $x = [u \ v \ w \ \phi \ \theta \ \psi \ h]^T$ for all of the three time segments. For this purpose, we start from the same starting values of the initial conditions, and in the

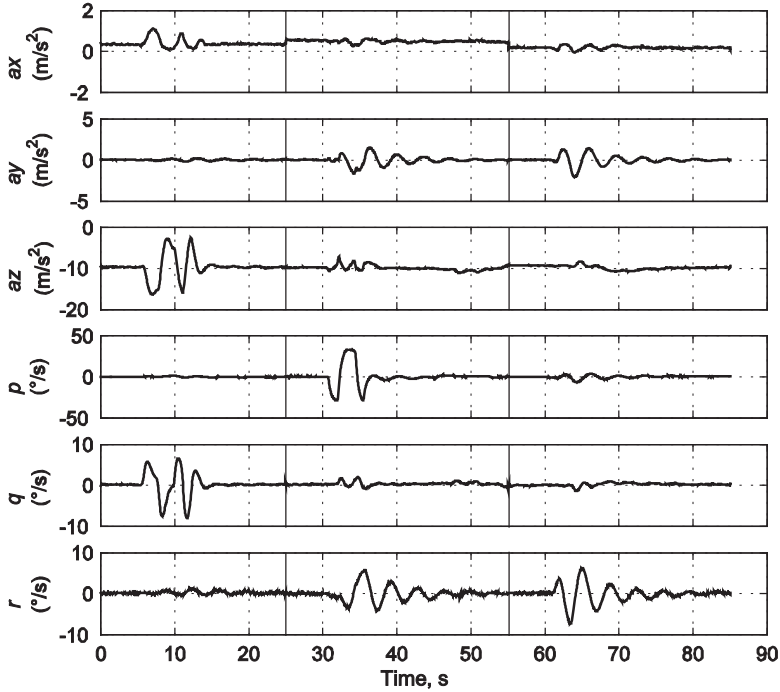

Fig. 10.6 Time histories of input variables for data compatibility check.

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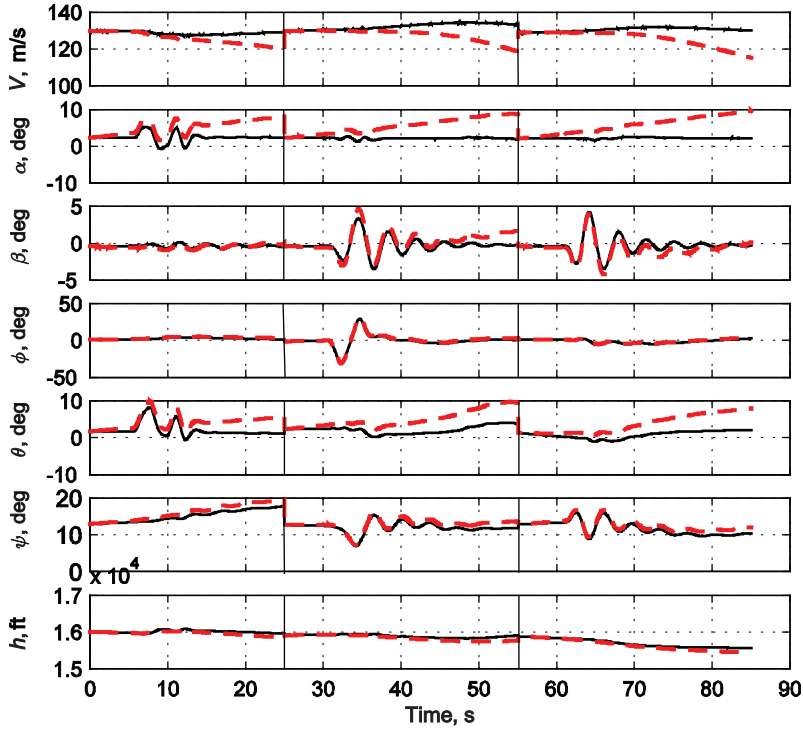

Fig. 10.7 Measured and simulated responses without instrument corrections: ——, flight measured; - - - - -, reconstructed.

model definition function "mDefCase22.m" set appropriately the flags for initial conditions parFlagX0(i,j), i = 1, n_x; j = 1, n_E, presently for all of the three time segments to ones and rerun the case. Any arbitrary combination of keeping the flags fixed or free is possible. It is now observed that the estimation procedure starting with the same initial cost function det(R) value of 1.5138×10^{-12} converges to the minimum of 1.2309×10^{-28} within five iterations. The bias parameters are estimated to be

$$\Delta a_x = 0.0067 \text{ m/s}^2, \quad \Delta a_y = -0.0015 \text{ m/s}^2, \quad \Delta a_z = 0.0839 \text{ m/s}^2$$

$$\Delta p = 0.0010 \text{ rad/s}, \quad \Delta q = 0.0034 \text{ rad/s}, \quad \Delta r = 0.0010 \text{ rad/s}$$

Once again, none of the estimates were correlated (>0.9), and their standard deviations were small. The estimates of the biases are mostly comparable to the preceding case, except for the bias Δa_y, which gets adjusted due to better estimation of initial conditions now. The improvement (reduction) in the cost

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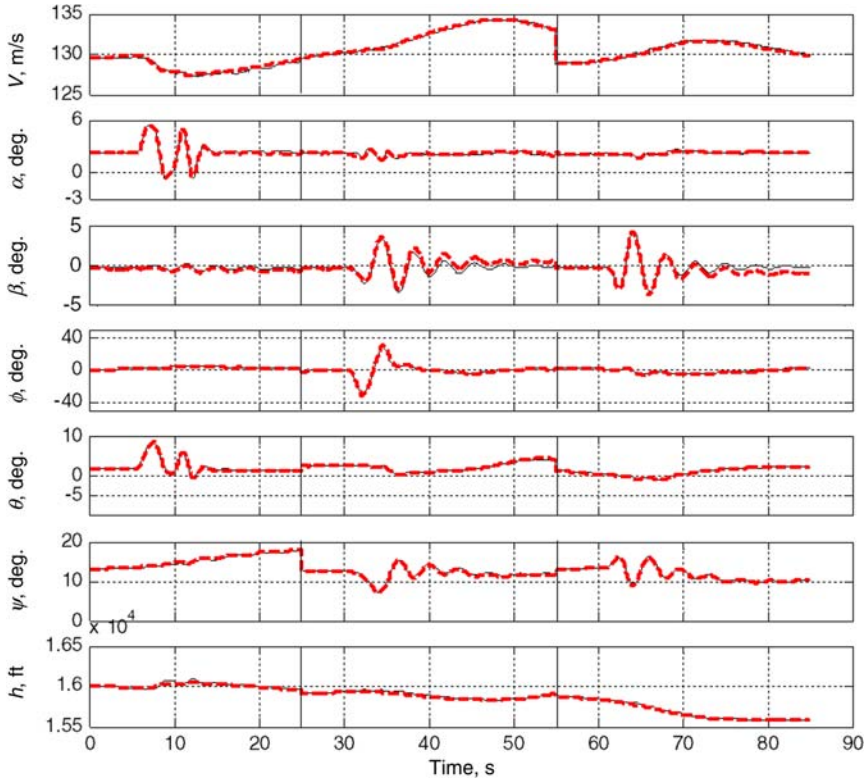

Fig. 10.8 Measured and simulated responses with bias corrections: ——, flight measured; - - - - -, reconstructed.

function from 1.8791×10^{-26} to 1.2309×10^{-28} is significant. The response match for this case is found in Fig. 10.9 from which it is observed that small drifts noticed in Fig. 10.8 in the response match for the angle of sideslip are now eliminated.

Continuing with the same example, we now estimate the scale factors and biases in the measurements of the angle of attack and angle of sideslip. For this purpose, we set the corresponding flags, elements parFlag(7:10), to unity and rerun the case. It observed that the scale factors come out to be 0.9864 and 0.9758 respectively, and the biases 0.00061 and -0.0055 rad, respectively. The cost function was 1.0275×10^{-28}, which is not much different from the preceding case where the scale factors and biases were fixed at one and zero. In the example presented, the reconstructed flow angles matched the recorded data without any scale factor or bias corrections fairly well; this is attributed to the fact that the onboard recording system was calibrated and updated, incorporating

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such corrections that were already determined from preliminary flight data analysis. Otherwise, usually scale factors and bias corrections are necessary for measurements of flow angles. Several repeated checks at different flight conditions and from different maneuvers provided very similar estimates for these bias parameters. Thus, a minimum set of parameters to be estimated in the case of flight-path reconstruction, as discussed in this chapter, consists of initial conditions on the state variables u_0, v_0, w_0, ϕ_0, θ_0, ψ_0, h_0 and biases in the inputs variables Δa_x, Δa_y, Δa_z, Δp, Δq, Δr. All other parameters pertaining to the sensors such as scale factors and biases are to be introduced and estimated based on the error analysis of the residuals.

As discussed in Chapters 3 and 4, the system parameters are usually common to all of the time segments analyzed simultaneously. Accordingly, they are treated as elements of the parameter vector param. Such system parameters are aerodynamic derivatives and in the present case the scale factors and

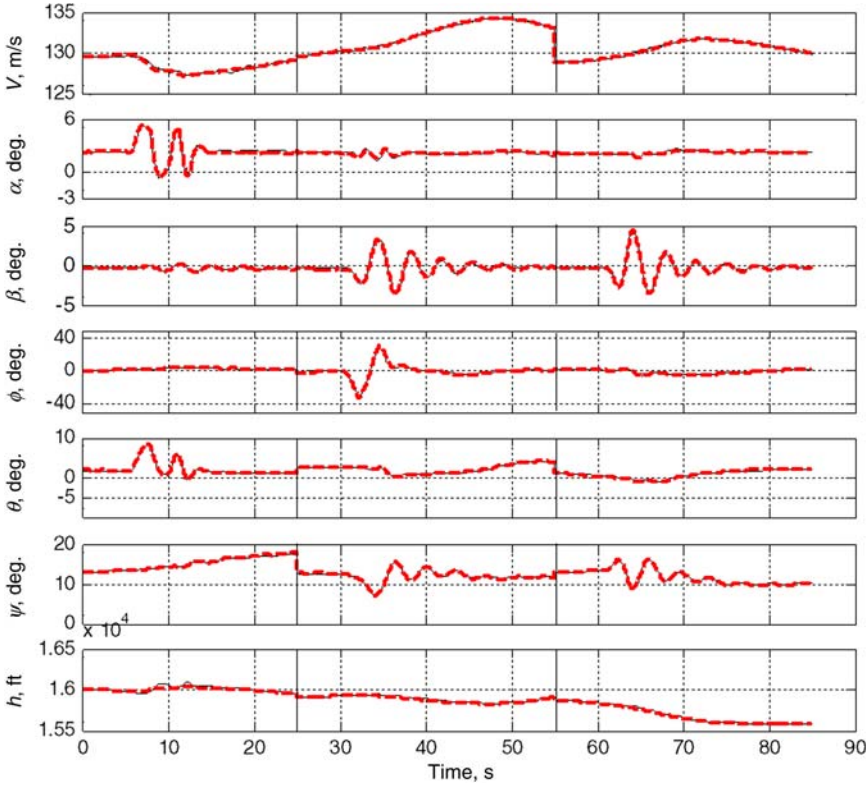

Fig. 10.9 Measured and simulated responses with estimated bias corrections and initial conditions: ———, flight measured; - - - - -, reconstructed.

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biases, assuming that same set of sensors and data recording are used to gather the data. Even in such cases, the biases in the measurements can vary due to various effects such as temperature changes, vibrations, etc. In such specific cases it might be desirable to treat biases in the measurements separately for each time segment. The extended software provided with the book, as discussed in Chapter 4, allows such an option to estimate specified parameters separately for each time segment. These parameters are clubbed into an array bXpar. They can be either kept free and estimated or kept fixed at specified values by defining appropriately the corresponding flags as parFlagBX. Such an example is provided for reference purpose as test_case=32 in the software folder /FVSysID2/Chapter04/. In the test_case=32, the bias parameters are estimated separately for each of the three time segments analyzed; it is an extension of the example discussed in this section so far, namely, data compatibility check (test_case=22) in which the biases in the input variables Δa_x, Δa_y, Δa_z, Δp, Δq, Δr were estimated common to all time segments and hence treated as elements of the parameter vector param. These details can be easily traced from the model definition function mDefCase32.m and the corresponding changes from xdot_TC32_fpr.m and obs_TC32_fpr.m. The results obtained in this case are not much different from those already presented here; hence, we do not go into any further discussion or detailed description of this option. The choice of treating bias parameters common to all time segments or separately for each maneuver depends upon the application. If the same sensors and data gathering system are used for all of the data being analyzed, usually it is adequate to estimate them common to all of the time segments. On the other hand, it is necessary to estimate the initial conditions separately for each time segment because they change from maneuver to maneuver. Because the state variables are propagated starting from the initial conditions, they affect the model responses significantly, as was evident from the current example. For the same reason, generally, good starting values are necessary for the initial conditions.

We once again continue with the same example of flight-path reconstruction (test_case=22), but now expand the observation model to demonstrate estimation of time delay for multiple variables. Recall that the procedure to account for time delay has already been demonstrated on a simpler model in Sec. 4.20.1.3; here we consider estimation of time delays in more than one variable, namely, angle of attack, angle of sideslip, and true heading. We could have easily carried out the changes in the observation model for the test_case=22, but for clarity we call the present case as test_case=33. Because we are considering time delays in the output variables, we need to adapt only the function for observation variables; the function for state equation in this case remains the same. The corresponding model definition function mDefCase33.m needs to be slightly appended with additional parameters for the time delays in the three variables. But for this minor change, the model definition function mDefCase33.m is the same as mDefCase22.m.

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Following the procedure provided in Sec. 3.4, we introduce time delay in angle of attack as follows:

$$[alNBTz, xWS_al, tNewX_al, iNewX_al] = timeDelay(alNBTz, tDelay_al, xWS_al, tNewX_al, iNewX_al, nTdMx)$$

where *alNBTz* is the computed angle of attack to be time shifted by a parameter *tDelay_al* being estimated; *xWS_al, tNewX_al, iNewX_al* are the auxiliary arrays associated with time delay for the particular variable being time shifted. A similar procedure is repeated for the other two variables being considered in this example, namely, for the angle of sideslip and true heading. We need to keep in mind that the auxiliary arrays have to be defined uniquely for each variable and declared as global in the calling function, here in obs_TC33_fpr.m. We assume that the maximum time delay is 40 times the sampling time and accordingly define *nTDMX*=40. If this is not sufficient, the computations are terminated in the function timeDelay with a warning message for the analyst. The exact details of calling timeDelay and defining auxiliary variables can be traced in the function obs_TC33_fpr.m. The three additional parameters are now *tDelay_al*, *tDelay_be*, and *tDelay_psi*; they are included in the parameter vector param in mDefCase33.m and obs_TC33_fpr.m as eleventh to thirteenth parameters. The initial values for the three time delays param(11:13) are set to zero in mDefCase33.m, and the corresponding flags parFlag(11:13) are set to one.

Having postulated the model with time delays as discussed here, we run /FVSysID2/chapter4/ml_oem2 by setting the test case flag to 33. It is observed that the convergence is achieved in six iterations. The cost function turns out to 7.0652×10^{-29}, which is once again somewhat lower than the cost function of 1.2309×10^{-28} for a similar case presented earlier but without time delay. The estimates of the biases in the accelerations and angular rates are very much comparable to the estimates given by other cases discussed so far. The estimates of the scale factors in angle of attack and angle of sideslip are also comparable to the preceding test case. The time delay in the true heading is estimated to be 0.03373 s having a standard deviation of 3.3%. The time delays in the angle of attack and angle of sideslip are estimated to be −0.01175 s and −0.00516 s, respectively; the associated standard deviations are 17.2 and 19.8%.

There are several interesting aspects that we would like to discuss here from the specific case chosen for demonstration purpose to bring them out. First, the time delays are extremely small for the angle of attack and angle of sideslip. They can be neglected without affecting the results. Second, the estimated values turn out to be negative; this is physically not possible for casual system. But, estimation of time delays needs some flexibility, and the function allows numerically negative time delays up to twice the sampling time. It is left to the analyst to note such a peculiar behavior and eliminate the particular time delay. In several other applications, we have observed that the measured flow angles

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have time delays, typically of the order of 100 to 200 ms. We would once like to recall that the time delays are estimated relative to the accelerations, which are inputs to the flight-path reconstruction.

We also observe that the computational time is now comparatively larger; this increase in the computational burden is not only due to the three additional system parameters, but mostly due to the computational intensive part of the interpolation involved in the delay matrix. In several other examples investigated, it has been observed that the progress of the optimization in which time delays are estimated is sensitive to starting values of the parameters and initial conditions; optimization starting from initial cost function value far from the minimum is likely to diverge. In other words, optimization for a model without time delays has a much wider range of convergence and hence is less sensitive to starting values. If we wish to estimate the time delays, we usually get good starting values of other parameters by estimating them for zero time delay first and then extending the estimation to include them in the complete set of parameters. As pointed out in Sec. 3.4, it is also possible to account for the time delays through an approximation by a first-order lag given by Eq. (3.18).

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10.6 CALIBRATION OF FIVE-HOLE FLOW ANGLE PROBE

Having studied different approaches to determine instrument errors in Secs. 10.2 and 10.3 and demonstrated the procedure on a simplified example in Sec. 10.5, we now turn our attention to more detailed practical issues related to calibrating the flow angle sensors. This is one of the prime applications of the data compatibility check in general and, as pointed out in Sec. 10.1, particularly for identification of aerodynamic model parameters using the output-error method. Different types of sensors, such as vanes, five-hole probes, or multihole systems in the form of a flush air data sensor, are used to measure the flow angles. The procedures to calibrate a vane or a five-hole probe are quite similar except for the observation model. We address here the case of calibrating a five-hole probe from flight-test data applying the data compatibility check [27, 28].

A five-hole probe, Fig. 10.10, provides measurements of total pressure p_{tot}, differential pressures for angle of attack $p_{\alpha 1} - p_{\alpha 2}$ $(= p_{d\alpha})$, and for angle of sideslip $p_{\beta 1} - p_{\beta 2}$ $(= p_{d\beta})$. Static pressure p_s is also measured very close to the tip of the noseboom. The angles of attack and sideslip are then given by [29, 30]

$$\alpha = \frac{p_{\alpha 1} - p_{\alpha 2}}{K\ p_{\text{dyn}}}; \quad \beta = \frac{p_{\beta 1} - p_{\beta 2}}{K\ p_{\text{dyn}}} \tag{10.37}$$

where p_{dyn} $(= p_{\text{tot}} - p_s)$ is the dynamic pressure and $K = f(Ma)$ is the sensitivity coefficient specified by the manufacturer, that is, to be flight verified. Other corrections have to be determined, if necessary. In the present case we verify the raw data

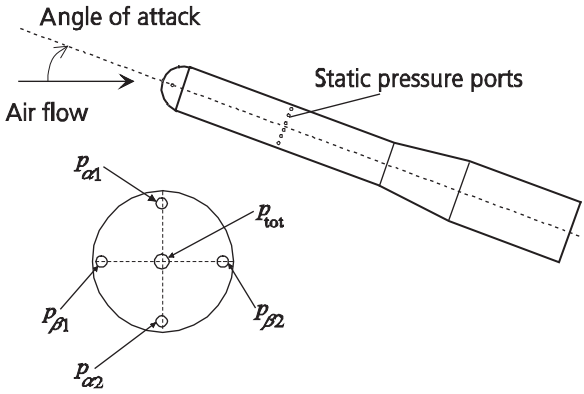

Fig. 10.10 Five-hole probe–port configuration.

for the recorded $p_{d\alpha}$ and $p_{d\beta}$ directly without any onboard corrections or preprocessing. This is possible by reformulating Eq. (10.37) as follows:

$$\begin{aligned} p_{d\alpha m} &= K_\alpha\, p_{\mathrm{dyn}}\alpha_{\mathrm{NB}} + \Delta p_{d\alpha} \\ p_{d\beta m} &= K_\beta\ p_{\mathrm{dyn}}\beta_{\mathrm{NB}} + \Delta p_{d\beta} \\ p_{\mathrm{dyn}m} &= \frac{1}{2}\rho\sqrt{u_{\mathrm{NB}}^2 + v_{\mathrm{NB}}^2 + w_{\mathrm{NB}}^2} \end{aligned} \tag{10.38}$$

where, as before, m denotes the measured quantities, K_α and K_β are the unknown sensitivity coefficients per degree, and Δ are the bias corrections in the respective measurements. The flow variables at the noseboom, namely, the angle of attack α_{NB}, angle of sideslip β_{NB}, and the dynamic pressure p_{dyn}, are obtained from the local velocity components at the noseboom, computed from the reconstructed states by accounting for the sensor location away from the center of gravity. The reconstructed states are the states provided by the kinematic equations, in the present case by integration because we once again apply the deterministic approach, because the flight tests were carried out in steady atmospheric conditions, where the turbulence level is negligible.

To account for possible time delays resulting from the recording system, the quantities on the right-hand side of Eq. (10.38) are appropriately modified. For example, in the case of angle of attack, the exact representation will be

$$\begin{aligned} p_{d\alpha,C} &= K_\alpha\, p_{\mathrm{dyn}}\alpha_{\mathrm{NB}} \\ p_{d\alpha m}(t) &= p_{d\alpha,C}(t - \tau_{pd\alpha}) + \Delta p_{d\alpha} \end{aligned} \tag{10.39}$$

where $\tau_{pd\alpha}$ is the time delays in the difference pressure $p_{d\alpha}$. The time delays $\tau_{pd\beta}$ and $\tau_{p\mathrm{dyn}}$ were also introduced for the difference pressure $p_{d\beta}$ and dynamic pressure through similar equations. The time delays are in general not known, and, hence, have to be estimated along with the unknown scale factors and biases. Thus, the kinematic model consists of state equations (10.11) and observation equations (10.12) in which V, α, β are replaced through $p_{d\alpha}$, $p_{d\beta}$, p_{dyn} as modeled in Eq. (10.38). Such a model is applied to calibrate five-hole probes mounted on a noseboom on two aircraft: 1) VFW-614 ATTAS, Fig. 10.11a;

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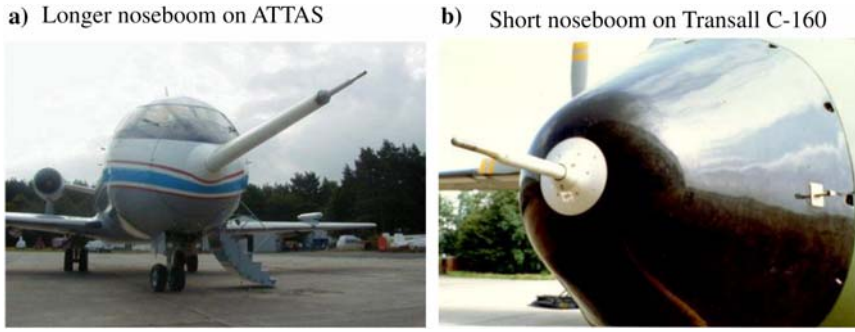

Fig. 10.11 Noseboom mounted five-hole probe for air data.

and 2) Transall C-160, Fig. 10.11b. The results of parameter estimation for the two aircraft are provided in Figs 10.12 and 10.13, respectively [27, 28].

For VFW-614 ATTAS, Fig. 10.12 clearly shows that the flight estimates of sensitivity coefficients match reasonably well with the manufacturer's data [29, 30]. To summarize, the estimated scale factors for the angle of attack and angle of sideslip are 0.0819 for Mach variation up to 0.55 compared to 0.079 specified by the manufacturer. For Mach 0.6, the scale factor for angle of attack is estimated to be 0.0781, and no Mach dependency could be estimated for angle of sideslip. The minor differences compared to the manufacturer's data are attributed to position error, possible errors in measured dynamic pressure, and noseboom length limited to roughly 1.35 times the fuselage diameter, which is probably not sufficient to reach the freestream flow. An empirical rule suggests a boom length of 2.5 to 3 times the fuselage diameter. The sensitivity coefficients and biases were found to be independent of the flap setting and other configuration changes.

The measurement of difference pressure for angle of attack showed a bias of 131 Pa and for the angle of sideslip 200 Pa, resulting mostly from misalignment. As demonstrated in [28], the time delay in both of the measurements was of the order of 130 ms, which was verified in a separate laboratory calibration to result from recording equipment. It is necessary to point out that the scale factors, biases, and time delays are system installation specific. Any generalization to other systems or configurations requires careful consideration.

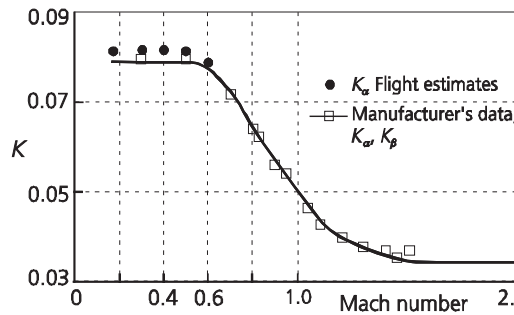

Fig. 10.12 Comparison of flight estimated scale factor for ATTAS with manufacturer's data.

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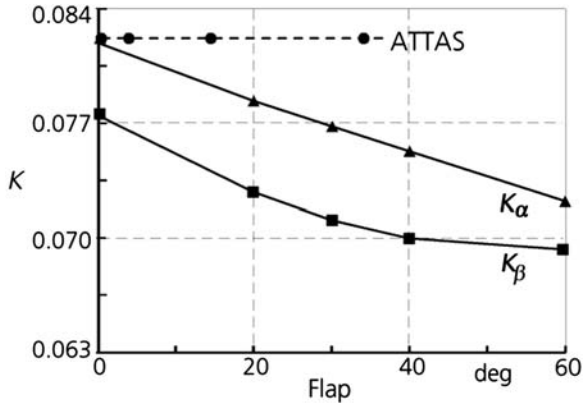

Fig. 10.13 Flight estimated calibration factors of Transall C-160.

In the other case, the same five-hole probe was mounted on a shorter nose-boom on a Transall C-160 aircraft, Fig. 10.11b. The estimated parameters (sensitivity coefficients) are shown in Fig. 10.13, which clearly shows that the scale factors are configuration dependent. This is attributed to the fact that the nose-boom length was just about 0.5 m. It is well known that the flowfield around the aircraft is distorted. A large number of maneuvers was necessary to cover the variations with configuration change.

As already pointed out at the beginning of this section, the procedure just described is applied to calibrate flow angle vanes or a five-hole probe. Five-hole probes have faster response and are more robust, reliable, and better suited for operation in adverse weather conditions. It is known that the dynamics of the vane influences the measurements of the flow angles. Moreover, vanes have mechanical difficulties such as friction and grit in the bearings and are prone to malfunctioning in bad weather conditions. Both of these types of sensors mounted on a boom located at a suitable location such as nose, wing tip, or rudder tip are used on a wide variety of research and prototype aircraft. Booms are sensitive to vibrations and alignment errors. Other aircraft have vanes mounted on the fuselage, typically at the pilot station. In general, they provide adequate measurement of flow angles in the low-to-medium range of angle of attack. In specific cases these two options to measure the flow angles might not be suitable or desirable for these reasons: 1) limited measurement range, 2) external fittings influence the flowfield affecting the aerodynamic performance, and 3) harsh thermal environment during supersonic speeds for entry vehicles. Furthermore, mounting a boom might not be possible due to radar equipment in the nosecone or due to interference with radar performance (radar backscatter). For stealth aircraft boom is highly undesirable because the pointed boom increases the visibility. In such cases a flush air data system (FADS) provides a better option to measure the flow variables.

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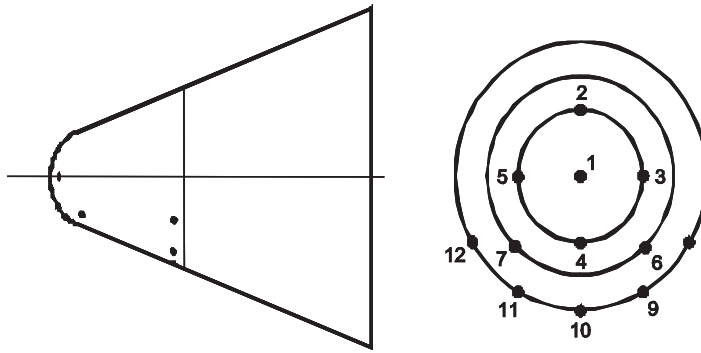

Fig. 10.14 Schematic of 12-hole flush air data system.

The FADS concept is based on a nonprotruding probe, which consists of multiple pressure ports bored on the aircraft nose. They measure the local pressure distributions from which the airflow parameters are determined applying suitable algorithms. Redundancy concepts and failure detection can be incorporated in these algorithms to yield reliable flow variables in the event of malfunctioning of some ports due to blockage or other reasons. These algorithms are derived from extensive wind-tunnel data and subsequently verified from flight data. FADS with multiple number of ports have also been used in the following applications: the SEADS (Shuttle Entry Air Data System) consisting of 20 flush orifices (14 primary and 6 supplementary orifices) [31], or a 25-port HI-FADS suitable for high angle of attack tested on HARV [32], or RT-FADS with 11 orifices on F-18SRA (System Research Aircraft) [33], and FADS with 12 pressure ports (see Fig. 10.14) optimized for high angle-of-attack measurements on VECTOR [34, 35]. Common to all of these systems is the basic layout of the orifices which are located for symmetric nose cap on concentric radial rings and a single nose tip port. The arrangement of the individual ports on the nose cap defined by the cone and clock coordinate angle is optimized for specific aircraft. For the nonsymmetric form of the nose cap, as in the case of the shuttle [31], the orifices are arranged in a cruciform array. A schematic of the pressure port locations of the FADS on VECTOR is shown in Fig. 10.14. The five pressure ports marked 1 through 5 are geometrically the same as those that were found on the five-hole probe we had discussed earlier; see Fig. 10.10. Such a FADS was calibrated applying different algorithms [34].

10.7 CALIBRATION OF STATIC PRESSURE PORTS

Calibration of flow angles measured by a five-hole probe needs the dynamic pressure p_{dyn}, which is given by the difference between the total pressure p_{tot} measured by the central port (see Fig. 10.10), and the static pressure p_s measured

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at a location very close by. Inaccuracies in the measurements of the static pressure at the noseboom directly affect the difference, that is, dynamic pressure, and subsequently the flow angles. Thus, accurate calibration of the static pressure port is a prerequisite for flow angle calibration. Similarly, deriving airspeed and altitude information from total and static pressures also needs accurate measurements of these pressures in freestream air. Besides indirectly influencing the other measured or computed variables, accurate static pressure information is also necessary for vertical spacing, that is, for safety reasons. Generally, the static pressure measured by the pitot static system mounted on an aircraft in flight differs from the freestream pressure [36, 37]. This difference is primarily dependent on the location of the pressure ports, speed, and altitude; secondary effects due to angle of attack and Reynolds number can also exist. Because the static pressure field varies along the length of an aircraft (see Fig. 10.15), it is necessary to choose appropriately the location of the ports mounted on the fuselage. In the case of ports mounted on a noseboom, the length of the boom, which can be limited due to other considerations such as weight, structural vibrations, and stability, is an important criterion.

The commonly adopted procedures to calibrate the static pressure ports are 1) trailing cone and 2) tower flyby [38]. The trailing cone procedure compares the static pressure measured by the aircraft's pitot static system to the freestream static pressure measured behind the aircraft by a static source. A sensor attached to a long cable or tube is necessary to sense the ambient pressure out of the aircraft's wake. The cone helps to stabilize the tube and to keep it taut. Some mechanism is necessary to let out tube and cone in free atmosphere during flight, for example, as done by opening the ramp door in the case of the military transport aircraft Transall C-160; see Fig. 10.16. This might not be possible in a passenger transport aircraft. The overall procedure is somewhat difficult in practice, requires

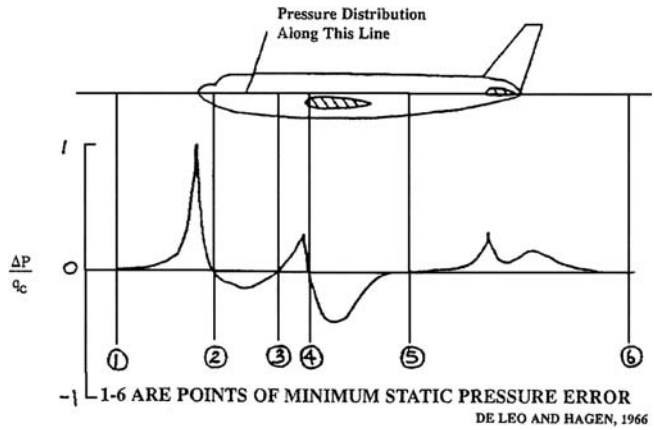

Fig. 10.15 Static pressure field along the length of aircraft (Source: [36]).

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Fig. 10.16 C-160 flight test with trailing cone.

very accurate and steady flight, and is associated with some risk resulting from instabilities, which are difficult to predict. On the other hand, the tower flyby method is a comparatively simpler procedure to accurately calibrate the static pressure measurements. It can be applied to any type of aircraft and is somewhat less risky. The test procedure consists of flying the aircraft past an observation tower and recording the geometrical position of the aircraft, for example, using a camera. The second approach, namely, tower flyby, is adopted in this investigation [28].

10.7.1 TOWER FLYBY AND MEASUREMENTS

The tower flyby tests were performed with ATTAS aircraft at four flap positions covering the speed range from 120 to 240 KIAS (Mach number roughly 0.14 to 0.45). Table 10.1 defines the flight-test matrix of airspeed and flap setting. At each flight condition a tower flyby is performed, keeping as far as possible height above the runway and speed constant. The nominal height above the runway is about 150 ft.

An onboard measurement system installed in the test aircraft ATTAS provides measurements of a large number of signals such as aircraft motion variables, atmospheric conditions, control surface positions, engine parameters, etc. Pertinent to the current aspect of investigations, the onboard measurements of altitude, static pressure, total pressure, and temperature are obtained from a digital air data computer. The altitude AGL (above ground level, that is, runway) is measured using the radio altimeter. Besides the preceding standard onboard measurements, aircraft position above the runway during the flyby is filmed by photo cameras as well as a digital-video camera from the top of the observation tower. The static pressure and temperature were recorded at the observation tower. Furthermore, the QNH, air pressure calculated to MSL (mean sea level), relayed by the control tower is recorded.

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TABLE 10.1 TEST FLIGHT CONDITIONS

KIAS Flaps	120	130	140	145	150	160	170	180	190	200	210	240
14 deg	X		X			X			X			
5 deg		X			X			X			X	
SP (1 deg)			X				X			X		X
IN (-6 deg)				X			X			X		X

10.7.2 DATA ANALYSIS

A schematic of tower flyby maneuver analysis is shown in Fig. 10.17. Knowing the tower height, distance from tower to runway centerline, and to marking lights, it is a simple geometrical procedure of evaluating the photos to obtain the height above the runway. Having determined the AGL, it is straightforward to get the aircraft altitude above sea level from the known runway elevation. Figure 10.18 provides the altitude above sea level as obtained from the geometrically computed height above the runway from the two different cameras.

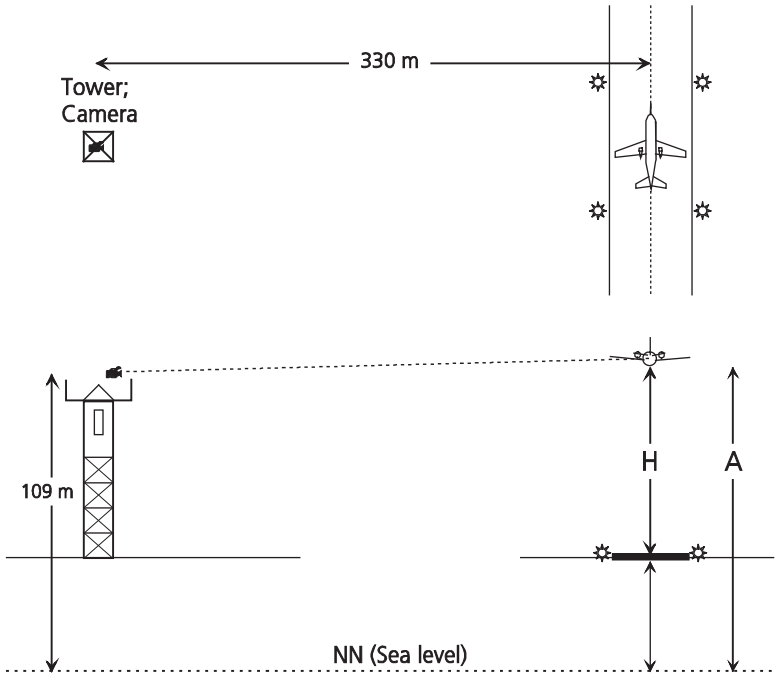

Fig. 10.17 Schematic of tower flyby maneuver analysis.

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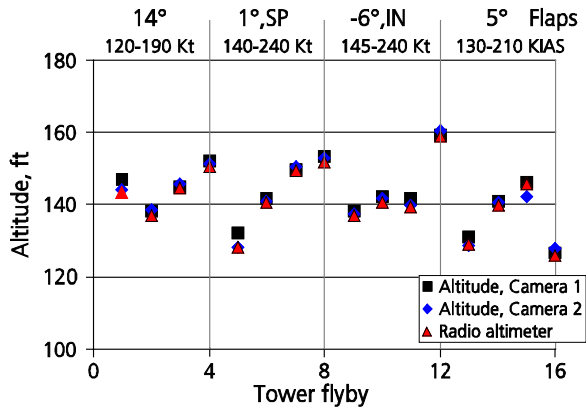

Fig. 10.18 Altitude above runway during tower flyby.

Some small differences are observed in the altitudes obtained from two different cameras. These minor deviations are attributed mostly to the variations in the flight path above the runway, different time points of exposure, and manual evaluation procedure. Nevertheless, these altitudes, obtained from two independent sources, match fairly well with the radio altitude.

From the measured pressure altitude h (in meters) during the tower flyby, the static pressure p_s (in Pascals) is obtained as

$$p_s = p_{0\mathrm{Akt}} \left(1 + \frac{T'}{T_{0\mathrm{Akt}}} h\right)^{5.25588} \tag{10.40}$$

where $p_{0\mathrm{Akt}}$ (in Pascals) and $T_{0\mathrm{Akt}}$ (in Kelvin) are the current QNH and temperature at the time of the particular test; T' is the temperature gradient that is assumed to be –0.0065 K/m according to the standard atmosphere for altitudes up to 11,000 m; and $p_{0\mathrm{Akt}}$ and $T_{0\mathrm{Akt}}$ are noted during each tower flyby.

In the present investigations, it was noted that the variations in $T_{0\mathrm{Akt}}$ during the complete flight test were less than 0.5°C and hence were neglected in the detailed computation of static pressure for each flyby. The average temperature was 18°C. The static pressure at sea level at takeoff was 1013 hPa, and for the most part of the flight maneuvers up to completion of the test program after 1 h 40 min it was 1012 hPa.

Based on the just noted $T_{0\mathrm{Akt}}$ and $p_{0\mathrm{Akt}}$, the static pressure in hPa at the altitude h_{RA} during each flyby is obtained from the relation

$$p_{\mathrm{sRA}} = 1012 \left(1 - 2.23253 \times 10^{-5} \times h_{\mathrm{RA}}\right)^{5.25588} \tag{10.41}$$

These static pressure values p_{sRA} corresponding to the altitude above NN obtained from the radio altimeter measurements shown in Fig. 10.18 are plotted in Fig. 10.19.

Figure 10.19 also gives values of p_{s_m} and p_{sNB_m}, the static pressure measured by the basis system (DADC) and that by the five-hole probe. These mean values are obtained from the time histories for the tower flyby phase during the 16 maneuvers. From Fig. 10.19, it is observed that, in general, the static pressure

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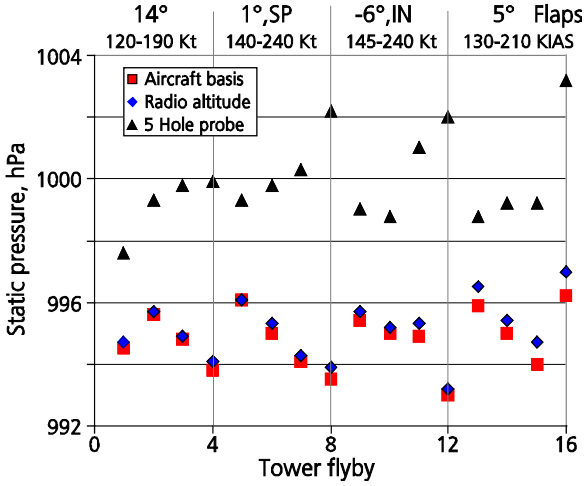

Fig. 10.19 Static pressures during tower flyby.

p_{s_m} from the basis system matches well with p_{sRA}, that derived from the tower flyby tests. On the other hand, the static pressure values measured by the five-hole probe sensor assembly (points marked as triangles in Fig. 10.19) differ from the other two. After it is concluded from Fig. 10.19 that the DADC and altimeter derived values are correct, the position error in the static pressure at the noseboom is the difference between p_{sNB_m} and p_{sRA}. Figure 10.20 shows this position error in static pressure measured at the noseboom as a function of speed. A simple correction factor was identified from this evaluation. As already pointed out in Sec. 10.6, in the case of ATTAS the noseboom length was roughly 1.35 times the fuselage diameter, which is probably not sufficient to reach the freestream flow; see Fig. 10.11a.

10.8 WIND-BOX MANEUVER TECHNIQUE

A special flight-test procedure, called wind-box maneuver, and a Kalman filter algorithm incorporating GPS information was proposed and used at National Research Council (NRC), Canada, to calibrate a flush air data system consisting of five pressure ports [39, 40]. Calibration of such a flush air data system is, in general, configuration and flight condition dependent because the flow at the aircraft nose is strongly distorted. Although in the present

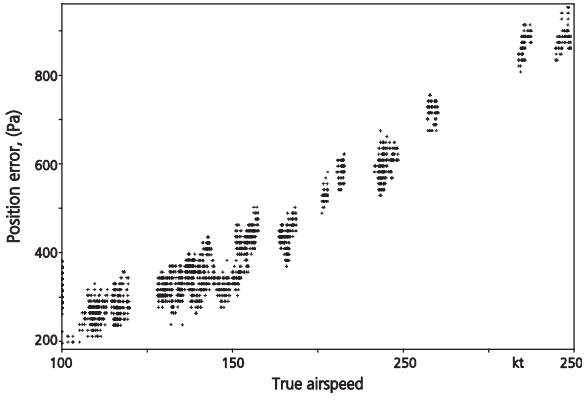

Fig. 10.20 Position error in static pressure measured at noseboom.

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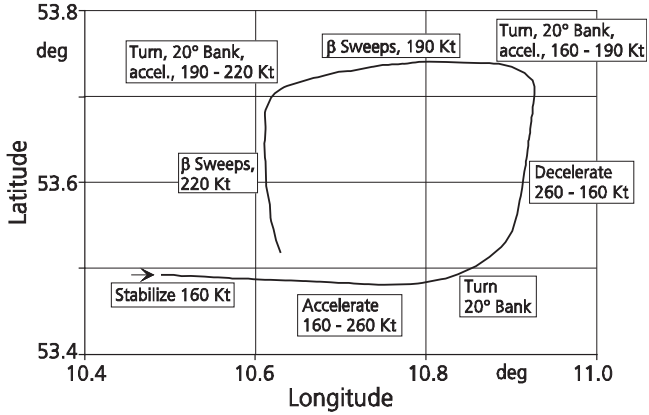

Fig. 10.21 Typical flight track during wind-box maneuver.

investigation the flow angles are not measured by a flush air data system, but rather by a five-hole probe mounted on a noseboom, we compare the standard technique described in Sec. 10.6 and the approach presented in this section.

The wind-box maneuver leads to a typical horizontal flight track shown in Fig. 10.21. It consists of acceleration and deceleration phases leading to angle-of-attack variations, buildup of steady-state sideslips in both directions, and 90-deg turns to traverse the spatial field in both x–y directions. This special type of maneuver helps to eliminate the wind effects. The typical duration of such a maneuver is about 700 s. Flight tests with ATTAS included two wind-box maneuvers, one at 8000 ft and another at 16,000 ft. The following signals are necessary to analyze this maneuver: 1) static pressure p_s; 2) dynamic pressure p_{dyn}; 3) total temperature T_{tot}; 4) difference pressure for angle of attack $p_{d\alpha}$ and for angle of sideslip $p_{d\beta}$; 5) Euler angles ϕ, θ, ψ; 6) angular rates p, q, r; and 7) three GPS inertial speed components.

Two approaches to analyze such a maneuver are possible, one for off-line analysis and one for real-time onboard verification. We consider here the second method, which is based on filter technique incorporating GPS information. As common to all such techniques, it is necessary to postulate suitable models for propagation of states and for observations, in the present case the combined air data and navigation equations. The propagation equations are in terms of the relationships between the errors in the air data navigation and those in the geographic velocity and position. The detailed equations for the Kalman filter algorithm and those for the aircraft velocity components are found in [40]; here we just focus on the part related to air data. The dynamic pressure is modeled as

$$p_{\text{dyn}} = C_{p0} + (1 + C_{p1})\ p_{\text{dyn_}m} \tag{10.42}$$

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where C_{p0} and C_{p1} are the unknown coefficients modeling the position error. The flow angles are modeled as

$$\alpha = C_{A0} + C_{A1}\,\frac{p_{d\alpha}}{p_{\text{dyn}}}$$
$$\beta = C_{B0} + C_{B1}\,\frac{p_{d\beta}}{p_{\text{dyn}}} \tag{10.43}$$

where C_{A0} and C_{B0} account for the misalignment between noseboom and aircraft axes and C_{A1} and C_{B1} are sensitivity factors. As elaborated earlier, $p_{d\alpha}$ and $p_{d\beta}$ are the difference pressures corresponding to angle of attack and angle of sideslip, and p_{dyn} is the dynamic pressure. Although Eq. (10.43) might appear to be formulated differently, it is basically the same as Eq. (10.38). It is obvious that C_{A1} and C_{B1} appearing in Eq. (10.43) are inverses of the sensitivity coefficients K_α and K_β appearing in Eq. (10.38). We sacrifice here the uniformity of notation to facilitate further referencing the corresponding literature.

The Kalman filter algorithm consists of 18 error states (three air navigator position errors, three wind component errors, two calibration errors for angle of attack, two calibration errors for angle of sideslip, two calibration errors for position error correction, three aircraft Euler angle errors, and three GPS position offset errors). All of the error states, other than the three air data navigational position errors, are modeled as first-order Gauss–Markov processes. A schematic of the error feedback Kalman filter is shown in Fig. 10.22.

For a typical wind-box maneuver, Fig. 10.23 shows time histories of the measured and estimated variables. The reconstructed variables match fairly well

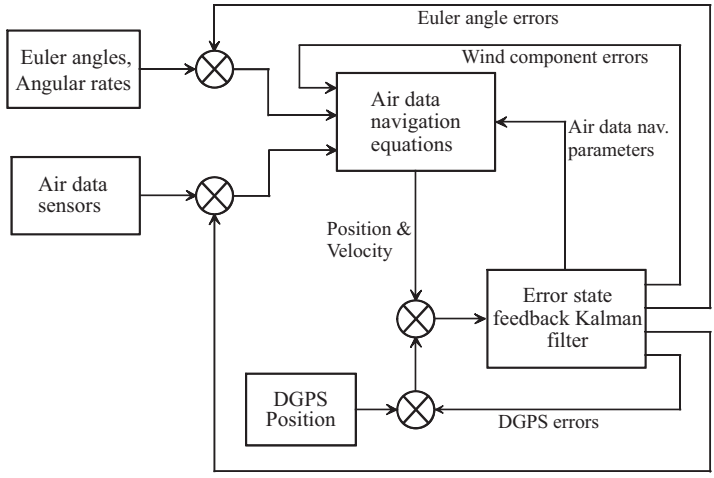

Fig. 10.22 Schematic of error feedback Kalman filter.

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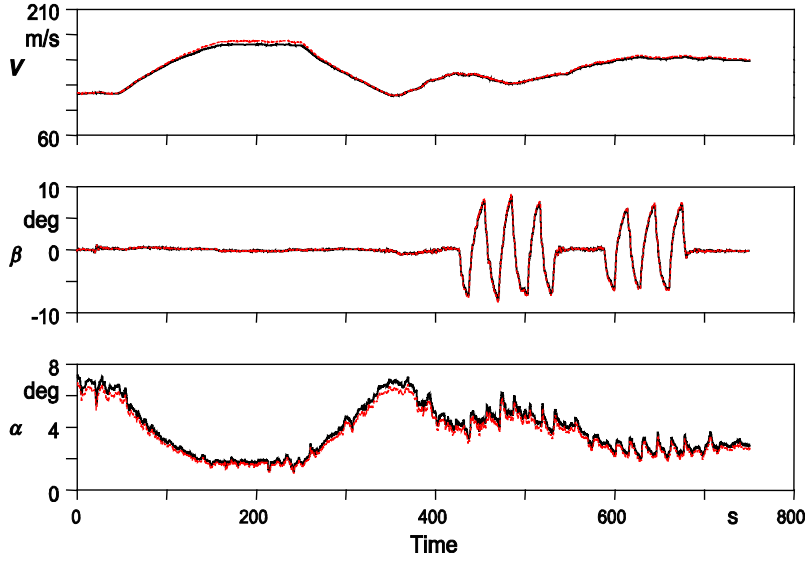

Fig. 10.23 Flight measured and reconstructed time histories for a wind-box maneuver: ———, flight measured; - - - - -, reconstructed.

with the measured data. It is observed that the sensitivity coefficients are estimated with very low standard deviation. The sensitivity coefficients C_{A1} and C_{B1} are estimated as 12.03 and 12.02, respectively.

The results obtained from Eq. (10.43) applying the Kalman filter technique to a wind-box maneuver are compared with those obtained from Eqs. (10.38–10.39) applying the output-error method to dynamic maneuvers. The inverse of the C_{A1} estimate yields 0.0831 ($=1/12.03$), which compares fairly well with the estimate of K_α ($=0.0819$). The same is true for the calibration factors pertaining to angle of sideslip. The minor differences are attributed primarily to the fact that in the Kalman filter technique the time delays in the flow angles were not accounted for. Investigations in the past have clearly demonstrated that omission of time delays can affect estimation results significantly. Furthermore, the noise covariances were prespecified and kept fixed; tuning of noise covariances can influence somewhat the estimates. Nevertheless, the agreement between the two different analysis techniques applied to two different types of flight maneuvers is good and increases the confidence in the results generated from flight data analysis.

The procedure described in this section is a relatively new approach. It can be implemented onboard in real time. As just elaborated, it allows simultaneous determination of flow angles and of position error; see Eq. (10.42). Elimination of position error leads to more reliable information on the dynamic

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pressure and thereby on the true airspeed. The wind-box maneuver provides information necessary for this purpose. In the conventional approach discussed in Sec. 10.2, we had mainly focused on calibration of flow angles from dynamic maneuvers only. As apparent from Fig. 10.23, the duration of the wind-box maneuver is quite long and time consuming; if weather conditions are not stable during the test time, it will become necessary to improve upon the model to include varying wind conditions. Likewise, the Kalman filter performance can be affected by other measurements and might need proper tuning. Although an appealing approach, the maneuvering time and analysis methods appear somewhat cumbersome for repeated check after minor modifications of the complete flight instrumentation system and for application of a routine basis. The conventional approach appears to be more direct and simpler.

10.9 CONCLUDING REMARKS

The well-defined kinematic equations of aircraft motion provide a sound basis for a data compatibility check, which is often called the flight-path reconstruction or aircraft state estimation. The separation of state estimation and estimation of aerodynamic model parameters is commonly termed as two-step method or estimation-before-modeling technique. The subtle differences between the two approaches have been clearly brought out. Strictly speaking, the so-called first step of state estimation is also a combined state and parameter estimation because the procedure includes sensor models and estimates parameters such as calibration factors, biases, and time delays in the recorded signals. The problem might be of smaller size and easier than that of estimation of aerodynamic model parameters; nevertheless, it is still a combined state and estimation problem that we had handled before in Chapters 4 and 5.

A rigorous approach based on the extended Kalman filter and the simplified procedure using the classical output-error method have been presented. The latter is applied to a simple test case first and then to a detailed investigation pertaining to the calibration of a five-hole probe for flow angles from dynamic maneuvers. The importance of flight validation of the sensor laboratory calibrations has been brought out, which might be aircraft configuration dependent. A related aspect of calibrating the static pressure port has been presented. To determine the position error in static pressure, the tower flyby technique was adopted. Finally, an alternative procedure based on a wind-box maneuver incorporating an error-feedback Kalman filter and GPS information has been elaborated, which allows a simultaneous calibration of flow angle, dynamic pressure, and static pressure sensors. A data compatibility check to generate smoothed states and to eliminate instrument errors is an important part of aircraft modeling from flight data because it helps to obtain more reliable and accurate estimates of aerodynamic derivatives.

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REFERENCES

[1] Gerlach, O. H., "Determination of Performance, Stability and Control Characteristics from Measurements in Non-Steady Manoeuvres," AGARD, CP-17, Sept. 1967, pp. 499–523.

[2] Jonkers, H. L., "Application of the Kalman Filter to Flight Path Reconstruction from Flight Test Data Including Instrumental Bias Error Corrections," Ph. D. Dissertation, Delft Univ. of Technology, Delft, The Netherlands, 1976.

[3] Mulder, J. A., Sridhar, J. K., and Breeman, J. H., "Identification of Dynamic Systems, Applications to Aircraft, Part 2: Nonlinear Analysis and Manoeuvre Design," AGARD, AG-300, Vol. 3, Part 2, May 1994.

[4] Stalford, H. L., "High-Alpha Aerodynamic Model Identification of T2C Aircraft Using the EBM Method," *Journal of Aircraft*, Vol. 18, No. 10, 1981, pp. 801–809.

[5] Sri-Jayantha, M., and Stengel, R. F., "Determination of Nonlinear Aerodynamic Coefficients Using the Estimation-Before-Modeling Method," *Journal of Aircraft*, Vol. 25, No. 9, 1988, pp. 796–804.

[6] Hoff, J. C., and Cook, M. V., "Aircraft Parameter Identification Using an Estimation-Before-Modeling Technique," *Aeronautical Journal*, Vol. 100, No. 997, Aug.–Sept. 1996, pp. 259–268.

[7] Etkin, B., *Dynamics of Atmospheric Flight*, Wiley, New York, 1972.

[8] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems – Applications to Aircraft. Part 1: The Output Error Approach," AGARD-AG-300, Vol. 3, Part 1, Dec. 1986.

[9] Jategaonkar, R. V., "Maximum Likelihood Parameter Estimation Method for a Class of Nonlinear Systems and Its Applications to Problems in Flight Mechanics," Ph. D. Dissertation, Indian Inst. of Science, Bangalore, India, Jan. 1986.

[10] Jategaonkar, R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DFVLR-FB 90-40, Germany, July 1990.

[11] Weiss, S., Friehmelt, H., Plaetschke, E., and Rohlf, D., "X-31A System Identification Using Single-Surface Excitation at High Angles of Attack," *Journal of Aircraft*, Vol. 33, No. 3, 1996, pp. 485–490.

[12] Keskar, D. A., and Klein, V., "Determination of Instrumentation Errors from Measured Data Using Maximum Likelihood Method," AIAA Paper 80-1602, Aug. 1980.

[13] Feik, R. A., "A Maximum Likelihood Program for Non-Linear System Identification with Application to Aircraft Flight Data Compatibility Checking," Aeronautical Research Lab., ARL-AERO-Note 411, July 1982.

[14] Mulder, J. A., "Estimation of the Aircraft States in Non-Steady Flight," AGARD, CP-172, Paper No. 19, May 1975.

[15] McBrinn, D. E., and Brassell, B. B., "Aerodynamic Parameter Identification for the A-7 Airplane at High Angles of Attack," *Proceedings of the AIAA 3rd Atmospheric Flight Mechanics Testing Conference*, AIAA, New York, 1976, pp. 108–117.

[16] Klein, V., and Schiess, J. R., "Compatibility Check of Measured Aircraft Responses Using Kinematic Equations and Extended Kalman Filter," NASA TN D-8514, Aug. 1977.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[17] Mulder, J. A., Jonkers, H. L., Horsten, J. J., Breeman, J. H., and Simons, J. L., "Analysis of Aircraft Performance, Stability and Control Measurements," AGARD, LS-104, Paper No. 5, Nov. 1979.

[18] Martin, C. A., and Feik, R. A., "Estimation of Aircraft Dynamic States and Instrument Systematic Errors from Flight Test Measurements," *Proceedings of the 2nd Conference on Control Engineering*, Inst. of Engineers, Australia, 1982, pp. 174–178.

[19] Evans, R. J., Goodwin, G. C., Feik, R. A., Martin, C., and Lozano-Leal, R., "Aircraft Flight Data Compatibility Checking Using Maximum Likelihood and Extended Kalman Filter Estimation," *Proceedings of the 7th IFAC Symposium on Identification and System Parameter Estimation*, Pergamon Press, Oxford, England, U.K., 1985, pp. 487–492.

[20] Plaetschke, E., "Kalman-Filterung," *Lecture Notes: Praktische Einführung in die Systemidentifizierung*, Carl-Cranz-Gesellschaft, Oberpfaffenhofen, Germany, June 1985, Sec. 10 (in German).

[21] Bach, R. E., "State Estimation Applications in Aircraft Flight Data Analysis," NASA RP 1252, March 1991.

[22] Gelb, A., *Applied Optimal Control*, MIT Press, Cambridge, MA, 1974.

[23] Ljung, L., "Asymptotic Behaviour of the EKF as a Parameter Estimator for Linear Systems," *IEEE Transactions on Automatic Control*, Vol. AC-24, No. 1, 1979, pp. 36–50.

[24] Rauch, H. E., Tung, F., and Striebel, C. T., "Maximum Likelihood Estimation of Linear Dynamic Systems," *AIAA Journal*, Vol. 3, No. 8, 1965, pp. 1445–1450.

[25] Haykin, S., "Kalman Filters," *Kalman Filtering and Neural Networks*, edited by Haykin, S., Wiley, New York, 2001, pp. 1–21.

[26] Morelli, E. A., "Estimating Noise Characteristics from Flight Data Using Optimal Fourier Smoothing," *Journal of Aircraft*, Vol. 32, No. 4, 1995, pp. 689–695.

[27] Jategaonkar, R. V., Fischenberg, D., and von Gruenhagen, W., "Aerodynamic Modeling and System Identification from Flight Data – Recent Applications at DLR," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 681–691.

[28] Parameswaran, V., Jategaonkar, R. V., and Press, M., "Calibration of Five-Hole Probe for Flow Angles from Dynamic and Tower Flyby Maneuvers," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 80–86.

[29] "Rosemount Model 858 Flow Angle Sensors," Rosemount, Inc., Bulletin 1014, Burnsville, MN, 1988.

[30] DeLeo, R. V., and Hagen, F. W., "Aerodynamic Performance of Rosemount Model 858AJ Air Data Sensor," Rosemount, Inc., Rept. 8767, Minneapolis, MN, July 1976.

[31] Pruett, C. D., Wolf, H., Heck, M. L., and Siemers, P. M., "Innovative Air Data System for the Space Shuttle Orbiter," *Journal of Spacecraft and Rockets*, Vol. 20, No. 1, 1983, pp. 61–69.

[32] Whitmore, S. A., Moes, T. R., and Larson, T. J., "High Angle-of-Attack Flush Airdata Sensing System," *Journal of Aircraft*, Vol. 29, No. 5, 1992, pp. 915–919.

[33] Whitmore, S. A., Davis, R. J., and Fife, J. M., "In-Flight Demonstration of a Real-Time Flush Airdata Sensing (RT_FADS) System," AIAA Paper 95-3433, Aug. 1995.

[34] Weiss, S., "Comparing Flush Air Data System Algorithms Using VECTOR-FADS Wind Tunnel Data," AIAA Paper 2002-0535, Jan. 2002.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[35] Jost, M., Schwegmann, F., and Köhler, T., "Flush Air Data System – an Advanced Air Data System for the Aerospace Industry," AIAA Paper 2004-5028, Aug. 2004.

[36] DeLeo, R. V., and Hagen, F. W., "Flight Calibration of Aircraft Static Pressure Systems," Federal Aviation Agency, Contract FA 64WA-5025, Rept. No. RD-66-3, Rosemount Engineering Co., Minneapolis, MN, Jan. 1966.

[37] Brown, E. N., "Position Error Calibration of a Pressure Survey Aircraft Using a Trailing Cone," National Center for Atmospheric Research, Technical Note NCAR/TN-313+STR, Boulder, CO, July 1988.

[38] Gallagher, G. L., Higgins, L. B., Khinoo, L. A., and Pierce, P. W., "Fixed Wing Performance," Naval Test Pilot School Flight Test Manual, Rept. USNTPS-FTM-No. 108, Sept. 1992.

[39] Hui, K., Srinivasan, R., and Baillie, S., "Simultaneous Calibration of Aircraft Position Error and Airflow Angles Using Differential GPS," *Canadian Aeronautics and Space Journal*, Vol. 42, No. 4, Dec. 1996, pp. 185–193.

[40] Leach, B. W., and Hui, K., "A DGPS-Based Technique for Real-Time Simultaneous Calibration of Air Data Systems Using Kalman Filtering," *Proceedings of the 13th Symposium on "Navigation,"* Canadian Aeronautics and Space Inst., 2000, pp. 165–174.

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CHAPTER 11

Model Validation

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11.1 INTRODUCTION

Estimates are not the same as the facts. In sequel, model validation is necessary to gain confidence in, or reject, a particular model. This basic fundamental principle applies to all engineering and other processes including economical and political decisions. In our particular case of flight vehicles, as discussed in Chapter 1 and depicted in Figs. 1.2 and 1.5, the parameter estimation and the model validation are integral parts of system identification. From the foregoing chapters, it is apparent to the readers that the parameter estimation methods provide an answer to the question: *"Given the system inputs and responses, what is the model?"* On the other hand, model validation tries to provide an answer to the question [1]: *"How do you know that you got the right answer?"* It is obvious; the answer in this case means the identified model. In this chapter we deal with the issues related to the process of determining the correctness, accuracy, adequacy, and applicability of the identified model. We also attempt to check the validity of the underlying theoretical assumptions that were made in the derivation of the parameter estimation methods applied. An apt definition of model validation is provided by Schlesinger et al. [2]:

> . . . Validation refers to the process of confirming that the conceptual model is applicable or useful by demonstrating an adequate correspondence between the computational results of the model and the actual data (if it exists) or other theoretical data.

The various different aspects of model validation can be broadly classified into three subcategories:

1. Statistical properties of the estimates
2. Residual analysis
3. Model predictive quality

These three techniques provide clues into the effectiveness, or lack thereof, of model parameters. They provide suitable means that are necessary to judge the

adequacy of identified models and their parameters to duplicate the system sufficiently closely.

Careful analysis of each of these aspects is of paramount importance. Some of the criteria used in this methodological process are related to the basic statistical properties whereas others have conceptually evolved keeping in mind the application domain of the identified model. Several criteria, to be used in conjunction with each other, that help to validate the identified model can be broadly classified as follows:

1. Standard deviations of the parameter estimates
2. Correlation coefficients among the estimates
3. Goodness of fit (that is, value of the cost function being minimized, for example, determinant of the covariance matrix of the residuals)
4. Statistical analysis of residuals (bias, variance, covariance, and power spectral density)
5. Model deficiencies in terms of residual control inputs (inverse simulation)
6. Plausibility of estimates
7. Model predictive capability

In most of the identification exercises, it is necessary to predefine the objectives of the model, that is, for what purpose the model is to be used. This dictates in most of the cases the amount of effort spent on arriving at an adequate model for the intended purpose because if the validation process indicates inadequacy of the model, the parameter estimation and model development processes will have to be repeated. Additional experiments might become necessary to improve the estimates or to extend the range of model validity. This iterative procedure of model identification and model validation is to be performed again and again until the validation process proves adequate correspondence of the model.

The techniques that we discuss in this chapter can be applied to validate simple models derived for a particular trim point as well as for global models covering the entire operational flight regime. The basic philosophy in the two cases remains the same.

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11.2 STATISTICAL ACCURACY OF PARAMETER ESTIMATES

Having applied statistical estimation methods, the first and the most natural question that comes to our mind is about the statistical accuracy of these estimates. Common criteria are the standard deviations of the estimates and the correlation coefficients among them. As already pointed out in Sec. 4.3, for the maximum likelihood estimation, the Fisher information matrix, that is, Eq. (4.30), provides

a good approximation to the parameter error covariance matrix [3, 4]. Thus,

$$P \approx \left\{ \sum_{k=1}^{N} \left[\frac{\partial y(t_k)}{\partial \Theta} \right]^T R^{-1} \left[\frac{\partial y(t_k)}{\partial \Theta} \right] \right\}^{-1} \tag{11.1}$$

The estimation error covariance matrix P thus depends upon the model parameters Θ, the data points being analyzed N, and the covariance matrix of the residuals R (that is, on the measurement noise). If a priori information, as discussed in Sec. 4.16, is included in the estimation, it is required to use the system matrix of Eq. (4.86) while computing the parameter error covariance matrix in Eq. (11.1).

The diagonal elements of P, which are the variances of the estimates, are indicators of the accuracy of the estimates. The standard deviations of the estimates, also called Cramér–Rao bounds, can be readily computed from Eq. (11.1) as

$$\sigma_{\Theta i} = \sqrt{p_{ii}} \tag{11.2}$$

and the relative standard deviation as a percentage is then simply given by $100 * \sigma_{\Theta i}/\Theta_i$, where i is the index for the parameter.

In addition to the standard deviations, the correlation coefficients, which are a measure of the statistical dependence between the parameters, can be obtained from the off-diagonal elements of P as

$$\rho_{\Theta i\, \Theta j} = \frac{p_{ij}}{\sqrt{p_{ii} p_{jj}}} \tag{11.3}$$

The estimates are statistically more accurate the smaller the standard deviations and the smaller the correlation coefficients are.

In practice, the estimated error bounds (that is, the standard deviations), obtained from Eq. (11.2), are found to be too optimistic. The cause for this discrepancy has been traced to colored residuals [5–7]. The output-error method lumps together both process and measurement noise and the deterministic modeling error, thus corrupting the noise statistics. In addition, the assumption of Gaussian measurement noise used in the derivation of the estimator might not be true. Some theoretical techniques have been suggested to account for the band-limited (colored) noise in the residuals [5–8]. We do not, however, discuss such options, mainly because these problems are unavoidable. Instead, we use the simpler ad hoc approach of a fudge factor to get a realistic estimate of the errors. It is common practice to use a fudge factor of typically 5 to 10. This is an ad hoc approach, but based on experience. It attempts to bridge the gap between theory (the assumption of independent noise samples) and practice [9]. (White noise is a mathematical assumption, and residual spectra are seldom white.)

Having corrected the error bounds through a fudge factor, one would expect the scatter in the estimates of a particular derivative from the same model and from repeated experiments to agree within a band resulting from their corrected

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standard deviations. Unfortunately, we do come across cases in which this is not the case, that is, the scatter in the estimates being larger than the standard deviations. There is no sound theoretical procedure to overcome this discrepancy, and we have to keep in mind this subtle issue while interpreting these statistical accuracies. This issue is nontrivial because we cannot avoid scatter in the estimates, which could result from a multitude of different reasons like insufficient and improper excitation, inadequate model postulate, measurement errors, atmospheric turbulence, and the wrong choice of the estimation method.

The foregoing brief argumentation is true in the case of correlation coefficients as well. Usually, smaller correlation coefficients among the estimates are simply ignored, but what is meant by "small" is not specified and can vary from application to application. Correlation coefficients greater than 0.9 require some attention, and those greater than 0.95 can be considered to be almost linearly dependent. Unfortunately, this is also not true in all of the cases; in some of the exercises, we have come across estimates that were plausible and in the expected range, but the correlation coefficients were larger than the threshold indicated above or close to one. If parameters are linearly dependent, it implies that we can achieve the same match by compensating the changes due to variations in one parameter through proportional variations in the other. We come across higher correlations mostly between scale factors and measurement biases, when identified from a data with limited information content (for example, few trim conditions or not enough variations in the magnitude). Ideally, such correlations are to be avoided in the estimation. Thus, in a limited sense, subject to the concerns just mentioned, these two statistical criteria help indirectly to improve upon the model.

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11.3 RESIDUAL ANALYSIS

Residual analysis is one of the most commonly used statistical approaches to judge the quality of the fit. The criteria checked are goodness of the fit in terms of cost function value, Theil's inequality coefficient, and whiteness of the residuals.

11.3.1 GOODNESS OF FIT

The value of the cost function is a measure of the model fit. In the case of maximum likelihood method, it is $|R|$ the determinant of the covariance matrix of residuals. Although it is the most direct way to evaluate the model quality, there are some practical difficulties associated with the evaluation of the numerical values. As in the case of the statistical properties of the estimates, requiring that variances of the estimated parameters and that covariances must be small, the cost function should also be small. The cost function value depends upon the number of system outputs and amount of noise in the data analyzed. It also depends upon the units (radians or degrees for angles, rad/s or deg/s for

angular rates, m/s^2 or g for linear accelerations, meter or feet for altitude) of the variables being weighed in the cost function. Because of these reasons, the absolute value alone does not yield any clues to the quality of the model fit. As a consequence, one difficulty that we encounter here is that no consistent workable criteria have been so far put forward to this aspect, which would allow direct comparison of different models. Nevertheless, for a chosen number of system outputs and using consistently the same units, the cost function values can be used in a limited way to judge the improvements that can result from introducing additional aerodynamic parameters.

In the same connection we study yet another peculiar characteristic of the cost function. A very low value of the cost function does not necessarily guarantee that all of the variables are matched equally well. If a better matching of any one output variable is feasible, the optimization procedure might do so even when the match for other variables may deteriorate, provided the overcall cost function value is reduced. In an ideal case, perfect matching of a single output would be sufficient to reach the absolute minimum, namely, zero, and yet the model could be completely inappropriate. Luckily, in practice, we do not come across such a theoretical possibility, mainly because of the presence of measurement noise and modeling errors, both of which are always present. In some cases, a relative scaling of the variables is incorporated to aid matching of various variables equally well. Whether to consider any such scaling in the optimization will depend upon the postulated model and data being analyzed. In all of the cases reported in this book, including the advanced examples of large-size systems or highly nonlinear models, we have used the standard cost function without facing difficulties of the preceding nature.

The mean of the residuals is expected to be zero. Accordingly, the standard deviation σ of the output errors is given by

$$\sigma_i = \sqrt{\frac{1}{N}\sum_{k=1}^{N}[z_i(t_k) - y_i(t_k)]^2}, \quad i = 1, 2, \ldots, n_y \tag{11.4}$$

Because we deal with multi-input, multi-output systems, we have to check the standard deviations for each of the n_y output variables. Furthermore, Eq. (11.4) provides the total standard deviation based on N, that is, for the complete data analyzed. As we have already discussed in Chapter 3, multiple experiment analysis is a common procedure in flight vehicle system identification. In such a case, we have to check the standard deviations for each of the n_E time segments being analyzed. In general, the standard deviation for each time segment must roughly be of the same order of magnitude as that of other experiments or better and also as that given by Eq. (11.4). A significant deviation in any one of them suggests a poor model quality for that particular experiment. Based on this clue, we can attempt to find out the reason for the bad fit, which could be due to inappropriate system excitation in the particular experiment.

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Another possibility is that the model postulate might not be adequate to duplicate the actual system response, for example, a linear model postulate for strongly nonlinear phenomenon. It could also result from an inappropriate choice of the estimation algorithm, for example, output-error method being applied to data with process noise.

According to an old adage, goodness of fit is no criterion; it is a necessary but not a sufficient condition. A perfect fit of one or more experiments is likely, or to say more precisely we hope, to result in good predicting capabilities for other data. But it does not guarantee perfect prediction capability, that is, ability to duplicate system response. We will talk in more detail about prediction capabilities and ways to check them in one of the following sections.

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11.3.2 THEIL'S INEQUALITY COEFFICIENT AND DECOMPOSITION OF FIT ERROR

Yet another form of output statistics for the overall fit is provided by Theil's inequality [10]. Compared to the goodness of fit in terms of the residuals, it provides a little more insight into the correlation between the two time series, in our case the measured aircraft responses z and the model predicted responses y. Theil's inequality coefficient for each of the output variables is defined as

$$U_i = \frac{\sqrt{\frac{1}{N}\sum_{k=1}^{N}[z_i(t_k) - y_i(t_k)]^2}}{\sqrt{\frac{1}{N}\sum_{k=1}^{N}[z_i(t_k)]^2} + \sqrt{\frac{1}{N}\sum_{k=1}^{N}[y_i(t_k)]^2}}, \quad i = 1, 2, \ldots, n_y \tag{11.5}$$

where N is the total number of data points, z the measurement vector, and y the model output vector. Equation (11.5) provides n_y separate Theil's inequality coefficients, U_i, $i = 1, \ldots, n_y$, corresponding to the n_y number of outputs. It is also possible to define a single measure for the overall fit, called *multiple Theils's inequality coefficient*, based on the fit errors in the multiple pairs of time series [11]. However, we prefer to treat the inequality coefficients U separately because it aids in locating outputs that are matched better and those that are not. It is also possible to apply Eq. (11.5) separately to each of the n_E time segments being analyzed simultaneously. It helps to find out maneuvers for which the model shows comparatively larger discrepancies compared to others, which can result from modeling errors or high noise content of that particular maneuver. Theil's inequality coefficient U is a measure of degree of conformance between two time series. In statistical terminology, it represents the ratio of the root-mean-square fit error to the root-mean-square values of the measured and estimated signals summed together.

From Eq. (11.5), it is obvious that U is a normalized index, bounded within the closed interval between zero and unity; this is the main advantage of this procedure over the goodness of fit discussed in the foregoing section. A perfect fit

is implied by $U = 0$ (case of equality), and $U = 1$ corresponds to the case of maximum inequality. Values of U close to one arise when the two time series are significantly different. Thus, Theil's inequality provides insight into the validity of the predicted responses. Although the acceptable value for U depends on the application, in general, as a rule of thumb, a value in the range of 0. 25 to 0.3 indicates a good agreement.

Theil also decomposed the fit error between the two time series, that is, $[z_i - y_i]$, in terms of bias, variance and covariance proportions given by [10, 11]

$$U_i^M = \frac{(\overline{z}_i - \overline{y}_i)^2}{(1/N)\sum_{k=1}^{N}[z_i(t_k) - y_i(t_k)]^2} \tag{11.6}$$

$$U_i^S = \frac{(\sigma_{z_i} - \sigma_{y_i})^2}{(1/N)\sum_{k=1}^{N}[z_i(t_k) - y_i(t_k)]^2} \tag{11.7}$$

$$U_i^C = \frac{2(1-\rho_i)\sigma_{z_i}\sigma_{y_i}}{(1/N)\sum_{k=1}^{N}[z_i(t_k) - y_i(t_k)]^2} \tag{11.8}$$

where, as before, the subscript i ($i = 1, \ldots, n_y$) refers to the ith output variable, $\overline{z}_i$ and $\overline{y}_i$ denote corresponding the mean values, σ and ρ the standard deviations and correlation coefficient respectively of the two time series z and y. They are given by

$$\sigma_{z_i} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}[z_i(k) - \overline{z}_i]^2}, \quad \sigma_{y_i} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}[y_i(k) - \overline{y}_i]^2} \tag{11.9}$$

$$\rho = \frac{1}{\sigma_{z_i}\sigma_{y_i}}\frac{1}{N}\sum_{k=1}^{N}[z_i(k) - \overline{z}_i][y_i(k) - \overline{y}_i] \tag{11.10}$$

Once again, we treat the three proportions defined in Eqs. (11.6–11.8) separately for each output variable.

The bias proportion U_i^M is a measure of the systematic error in the identified model whereas variance proportion U_i^S represents the model's ability to duplicate the variability in the true system. The covariance proportion U_i^C is the measure of nonsystematic error. The preceding decomposition provides an insight into the causes of fit error. It is obvious that the bias and variance proportions should be very small, in an ideal case, zero, whereas the covariance proportion should be close to one. The sum of these three proportions turns out to be 1. For both U_i^M and U_i^S, a large value, typically above 0.1, would be of concern and indicate the need for model update. Theil's inequality and the breakdown fit error have been used regularly for economic forecasting for quite some time [10, 12]; more

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recently it is also being used in flight vehicle system identification [13–15]. In combination with visual inspection of the match between the time histories, these criteria provide slightly better insight into the properties of residuals.

11.3.3 TEST FOR WHITENESS

In an ideal case, the residuals must be white, that is, independent at different time instants. The autocorrelation function and the power spectral density of the residuals are statistical measures to check for whiteness. For a given set of samples, the autocorrelation matrix is estimated as

$$C(\tau) = \frac{1}{N}\sum_{k=\tau}^{N} [v(k) - \bar{v}]\ [v(k-\tau) - \bar{v}]^T \tag{11.11}$$

where v is the fit error (residual), $\bar{v}$ the mean, and $\tau = 1, 2, \ldots$ the lag. If residuals $[v(t_k), k = 1, 2, \ldots, N]$ are white, the autocorrelation coefficients C_i are zero for all τ. However, because we deal in practice with a finite length of data points N, the estimates of mean will not be true, even if the residual is part of a white noise sequence. To overcome these practical limitations of finite N, and also to exclude outliers, a confidence level is introduced in statistical sense. In many applications, it is customary to use a 95% confidence level. In this case, for all τ greater than one, the correlation coefficient should lie in the band $\pm 1.96/\sqrt{N}$ more than 95% of the times. In such a case the population (as N goes to ∞) and sample (finite N) means are not different enough to matter. The choice of 95% confidence is somewhat arbitrary; 90 or 99% intervals have also been used in specific applications, but we do not go into any further details for the reasons that will be discussed next. A more detailed treatment on various aspects of autocorrelation functions and its statistical significance can be found in any standard textbook on time series analysis, for example, [16].

We also know that the white noise has infinite bandwidth and the corresponding power spectrum is flat over the complete range. Accordingly, the power spectral density of the residual gives some rough idea about the noise statistics and of the validity of the theoretical assumptions made. Readers may recall from Sec. 5.11.2 that we used this criterion to compare the filter-error and output-error methods. The test of whiteness is rarely used critically, except to verify whether the process noise option would be more appropriate and the improvements obtained therefrom. However, if the test for whiteness fails, that is, indicates colored noise, we generally do not undertake any steps of model update. Noise statistics is of secondary importance, our primary goal being a model that is sufficiently close to reality and with good predictive capability. Both of these criteria are being verified more closely by other diagnostic tests being discussed in this chapter. The theoretical assumption of noise being Gaussian distributed was anyway made to keep the mathematical procedure tractable. Hence, as already pointed out in Sec. 11.2, we do not

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discuss the techniques that have been suggested to account for band-limited (colored) noise.

11.4 INVERSE SIMULATION

Recalling from Sec. 1.1 the definition of simulation (namely, given the inputs u and the system model f, find the system output y), inverse simulation, as the name implies, is the process of calculating desired controls for the given system model and response. Thus, it is closely similar to the control problem definition (namely, given the system model f and the output y, find the control input u). The subtle difference being that the classical control problem does not require measured system responses whereas inverse simulation explicitly needs measured responses and leads to controls based on residuals.

The principle of inverse simulation is depicted in Fig. 11.1. The measured control inputs are fed to the identified mathematical model. The computed model response is compared to the flight-measured aircraft response; the residuals, which contain the model deficiencies, are fed back to the system through a feedback controller [17]. Usually, a proportional and integral controller is quite adequate for the purpose at hand. The feedback controller ensures that the computed model responses match with the measured system response exactly, whereby the modeling deficiencies now appear in the control inputs.

This approach provides another interpretation of the model deficiencies in terms of inputs instead of outputs. Furthermore, the model does not deviate from the actual flight (trim) condition. For a sufficiently accurate model, the incremental control inputs Δu will be small enough and centered about zero. As a rule of thumb, during aerodynamic model development, deviations in control inputs smaller than half a degree are considered acceptable. Otherwise, the deviation in any particular control variable suggests a modeling error and provides clues on possible model extensions. The modeling task would then reduce to minimizing the controller activity to zero.

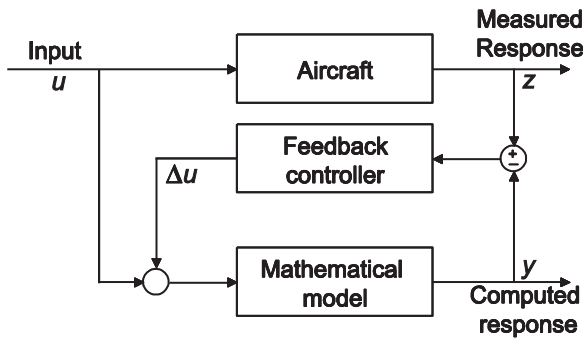

Fig 11.1 Principle of inverse simulation.

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We can extend this approach to detect secondary aerodynamic effects such as those due to landing gear or ground effect. It calls for a flight validated model of the basic aircraft without the landing gear or ground effect [18]. In such a case, the inverse simulation will show discernible controller activity. Thus, inverse simulation provides a suitable tool not only for the classical applications [19, 20], for example, of validating the simulation software, but also to detect the modeling deficits, which can be reduced in the next step through system identification.

11.5 MODEL PLAUSIBILITY

As pointed out in Sec. 11.3.1, because the goodness of fit, that is, the response match, is a necessary but not sufficient condition, it is advisable to check estimates from yet another aspect, namely, the plausibility of the estimates from physics of the dynamical system. An overparameterized model will give a good response match, but not necessarily a good system representation. The most direct way to check the plausibility of the estimated parameters is by comparison with estimates obtained from other sources. Because we are dealing mainly with flight vehicles, aerodynamic databases derived from wind-tunnel measurements on a scaled model are generally available. Hence, a comparison with such wind-tunnel derived databases or with analytical estimates is a widely used approach to check the plausibility of the estimates.

Another commonly applied method is to check and interpret the estimates from physical understanding of the system under investigation. In such cases not only the polarity of the estimates, but also the relative magnitudes of the different derivatives contributing to a particular aerodynamic force or moment coefficient can provide clues to the plausibility. This is particularly true for the secondary derivatives that are difficult to estimate, for example, landing-gear effects that are typically less than 5% of those for the clean configuration. Nevertheless, such effects are very important for critical phases of flight [21].

Yet another example pertains to estimation of $C_{L\delta e}$, the lift-curve slope of the elevator. The lift due to elevator being a very small portion of the total lift and the major portion coming from the wings, the estimates of $C_{L\delta e}$ are often associated with large standard deviations and scatter, both indicating lower confidence in the estimates. From the physics of the problem, that is, by looking at the cause–effect relation, we know that an elevator deflection primarily leads to a pitching moment, resulting from small changes in lift due to elevator, but quite large lever arm (the tail length—the distance from the neutral point of the wing to that of the tail). Because of this reason, the pitching moment due to elevator deflection $C_{m\delta e}$, that is, elevator effectiveness is always well identifiable, which results mainly from matching the pitching motion variables. The two parameters $C_{L\delta e}$ and $C_{m\delta e}$ are physically coupled, and we can use this knowledge to improve the estimates by estimating just one parameter [22], namely, $C_{L\delta e}$, and then

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computing the contribution to the moment due to elevator through the relation:

$$C_{m\delta e} = -C_{L\delta e}\,\frac{r_H}{\bar{c}}\,\frac{S_H}{S} \tag{11.12}$$

where r_H is the tail length, $\bar{c}$ the mean aerodynamic chord, S the wing reference area, and S_H the horizontal tail area.

Yet another procedure to check the plausibility is to look at an equivalent linear system representation, which is helpful to gain insight into the system behavior and also to apply analysis techniques well established for linear systems although the physical phenomenon being modeled might be nonlinear. The S-plane plot of the eigenvalues of the linearized system matrices derived from different identified models helps, for example, to highlight the influence of model extensions such as those due to the effect of rotor dynamics on the rigid-body motion of a helicopter [1]. The most commonly adapted numerical procedure to compute eigenvalues and eigenvectors is based on balancing the system matrix, reduction to upper Hessenberg form by orthogonal transformations and QR decomposition. Standard functions under Matlab® or from the LAPACK library in FORTRAN are available. A typical plot of eigenvalues is shown in Fig. 11.2, showing also the counters of the constant natural frequencies ω_n and constant damping ratios ζ.

The eigenvectors are plotted in Fig. 11.3 as bar charts. We use here the normalized eigenvectors having Euclidean norm equal to one. The vertical bars represent the magnitudes; the phase information is not considered in this form of representation. There are other possibilities to plot the eigenvectors, but the one depicted in Fig. 11.3 is easy to interpret and clearly shows the contribution of different variables to the particular mode. Depending upon the largest

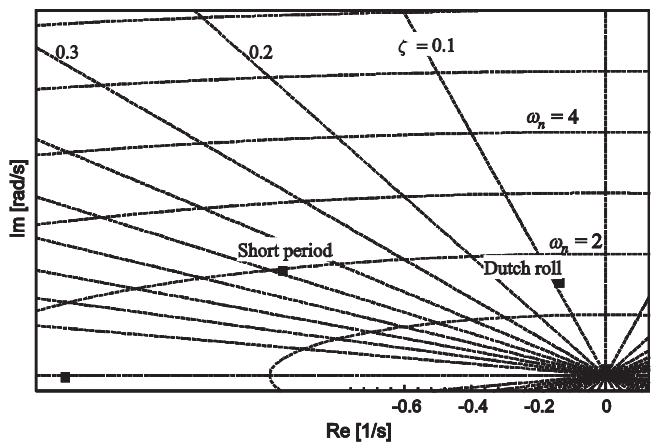

Fig. 11.2 Typical S-plane plot of eigenvalues.

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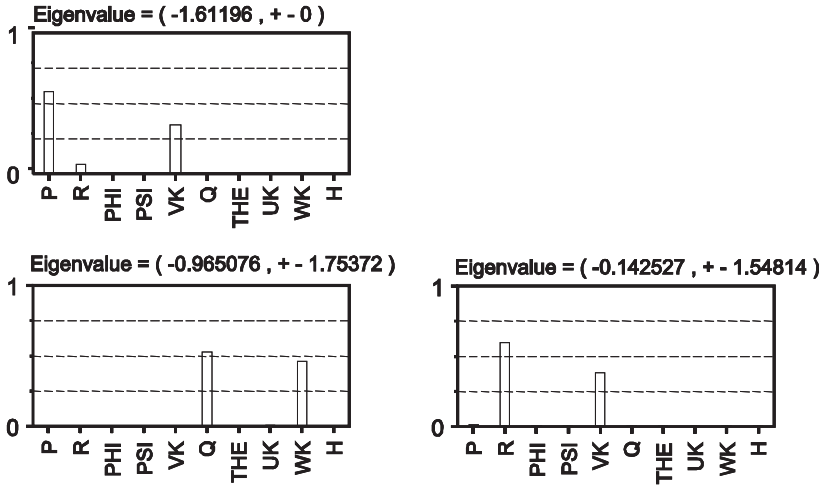

Fig. 11.3 Typical plot of eigenvectors.

contribution or the combination of individual components of the eigenvectors, the eigenvalues (modes of motion) can be determined. For example, the two plots at bottom of Fig. 11.3 with complex eigenvalues pertain to two oscillatory modes: on the left hand, that of the short-period mode with components due to pitch rate q and vertical velocity w, and on the right hand, that of the Dutch roll with dominant components due to yaw rate r and lateral velocity v.

The eigenvalues and eigenvectors of the system state matrix completely characterize the system behavior; the eigenvalues determine the nature of the transient response whereas eigenvectors determine the contributions of each state to that response. The eigenvalues corresponding to the oscillatory modes of the system appear as complex conjugate pairs. The natural frequency ω_n and damping ζ of the oscillatory modes are given by

$$\omega_n = \sqrt{R_e^2 + I_m^2}, \quad \zeta = \frac{-R_e}{\omega_n} \tag{11.13}$$

where R_e and I_m are the real and imaginary parts of the eigenvalues, respectively. Using the preceding formulas, the frequencies and damping of the short period and Dutch roll can be worked out as ($\omega_{SP} \approx 2$ rad/s, $\zeta_{SP} \approx 0.48$) and ($\omega_{DR} \approx 1.55$ rad/s, $\zeta_{DR} \approx 0.092$), respectively. These values can be traced to the two modes marked in Fig. 11.2. The eigensystem analysis is one way to investigate the system stability and to compute the frequency and damping ratio of oscillatory modes required for the proof-of-match cases that we will discuss in the next section.

The other approach is to approximate the oscillatory motion by a simple spring-mass-damper system and to apply one of the parameter estimation

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methods to estimate the damping and period. In such a case, we postulate a second-order system given by

$$\ddot{x} = -\omega_n^2 x - 2\zeta\omega_n \dot{x} + Ku \tag{11.14}$$

which is transformed into two first-order state equations for parameter estimation purposes as

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ K \end{bmatrix} u \tag{11.15}$$

where the two states are x and $\dot{x}$, the position and the velocity, respectively. We can now apply the output-error method we discussed in Chapter 4 to estimate the parameters ω_n and ζ.

11.6 MODEL PREDICTIVE CAPABILITY

The predictive capability of the identified model is determined by comparing the flight measured system responses with those predicted by the model for the same ("identical") control inputs. In flight vehicle applications terminology, this process is often called "proof-of-match" (POM); it is an important part of flight simulator certification and acceptance. In this proof-of-match process, the identified (aerodynamic) model is kept fixed.

Besides the use of identical control inputs, three other issues that play a very important role are the following:

1. Proper choice of the data set to compare against
2. Initial conditions on the state variables (trim)
3. Criteria to check adequate correspondence between model response and measured aircraft outputs

A fundamental principle of empirical sciences suggests, as a general rule, complementary flight data, that is, flight maneuvers not used in the estimation of the aerodynamic database, to check the model predictive capability. Validation on complementary data not used in the estimation is sometimes also termed loosely as "acid test." In most of the model validation exercises, including those for the flight simulators, this approach of separating the data for model development and for demonstration of model fidelity is adopted. However, there is another philosophy, although rarely practiced, that advocates incorporating maximum amount of available flight data in the model development and not to split the flight data in two sets. The logic behind this approach is that the identification process would automatically lead to a model with adequate fidelity, satisfying the validations tests that we will discuss in the following. It can also lead to a fewer iterative steps in the overall model building. We, however, recommend the first approach based on demonstrating the model

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fidelity on complementary data because it provides increased confidence in the model predictive capability.

Conducting the flight tests for validation purposes generally calls for greater precision and caution than those for database development. It is also required to cover the complete operational envelope and to test the relevant modes of aircraft motion. Accordingly, for a full range of proof-of-match to demonstrate adequate model fidelity for simulator applications, the Federal Aviation Administration (FAA) has defined a set of roughly a little more than 100 different cases, covering different modes of aircraft motion and configurations [23, 24]. These tests are usually carried out under calm atmospheric conditions to minimize the aircraft response due to stochastic inputs.

The second issue that plays a very important role is that of setting the proper initial conditions on the state variables to be used in the simulation [25]. Ideally, the simulation is to be started from the same initial conditions as in the flight. The initial conditions have to be suitably adjusted to match the flight conditions being verified. Unfortunately, measurements of the control inputs as well as those of the outputs (states) are subject to measurement errors and noise. Furthermore, even when the maneuvers for proof-of-match can be carried out under seemingly smooth atmospheric conditions, the aircraft is excited by some small amount of nonmeasurable excitation due to turbulence. Likewise, the identified models are not precise. Because of these unavoidable practical difficulties, the proof-of-match turns out to be a nontrivial task. Starting with the initial conditions as measured in flight, in most of the cases it becomes difficult, if not impossible, to meet the tolerances. A pragmatic approach, therefore, is to allow small biases on the selected initial conditions and on the measured control deflections, which are the inputs to the simulation.

To eliminate subjective evaluation of the match between measured system responses and model predicted outputs, FAA has specified guidelines in terms of tolerances for each variable, depending upon the nature of the validation test. As an example, Table 11.1 provides the definition of four tests, giving tolerances, flight conditions to be tested, and comments for each. The reader is referred to [23, 24] for the complete list of validation tests. The corresponding flight measurements with these tolerances define a band within which the model predicted response must lie to meet the necessary fidelity requirements.

The majority of the flight vehicle system identification applications deal with estimation of aerodynamic databases. In such cases, the proof-of-match is performed using the measured control surface deflections as the input to the model. For aircraft with an artificial feel system, this is quite adequate and an accepted procedure. In the case of aircraft with reversible controls (that is, with a mechanical control system), the task is a little more complex. Besides the aerodynamic database, the hinge-moment database is also required. For such cases, to improve the accuracy of each database and also to reduce the computational burden during the database development, the two databases are estimated as

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TABLE 11.1 SAMPLE VALIDATION TESTS (SOURCE: [23])

Tests	Tolerances	Flight Condition	Comments
2c10 Phugoid dynamics	±10% of period ±10% of time to $\frac{1}{2}$ or double amplitude or 0.02 of damping ratio	Cruise	Test should include 3 full cycles (6 overshoots after input completed) or that sufficient to determine time to $\frac{1}{2}$ amplitude, whichever is less.
2c11 Short-period dynamics	±1.5-deg pitch or ±2-deg/s pitch rate ±0.1 g normal acceleration	Cruise	Test in normal and non-normal control state.
2d2 Roll response (rate)	±10% or ± 2-deg/s roll rate	Cruise and approach or landing	Test with normal wheel deflection (about 30%). Airplanes with reversible flight control systems must also plot wheel force (±10% or ±3lb/1,3 daN).
2e1 Normal landing	±3 kt airspeed ±1.5-deg pitch ±1.5-deg angle of attack ±10% altitude or ±10 ft (3 m)	Landing	Test from a minimum of 200 ft (61 m) AGL to nosewheel touchdown. Derotation can be shown as a separate segment from the time of main gear touchdown. Medium, light, and near maximum landing weights must be shown. Airplanes with reversible flight control systems must also plot stick/column force [±10% or 2.2 daN (5 lb)].

so-called stand-alone models, that is, separately, as shown in Fig. 11.4a. On the other hand, it is necessary to demonstrate the end-to-end match based on an integrated model shown in Fig. 11.4b [26, 27].

If the two stand-alone models are identified sufficiently accurately, the integrated model used in the end-to-end match (that is, using six-degrees-of-freedom equations of aircraft motion driven through the dynamic models of the flight controls incorporating the identified databases for rigid-body and hinge moments,

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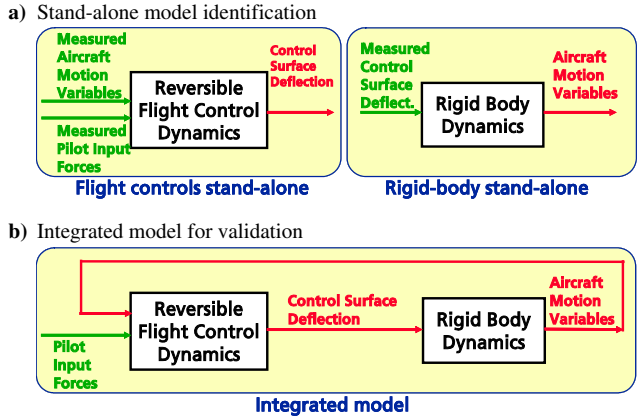

Fig. 11.4 Stand-alone vs integrated models.

respectively) is most likely to yield a match that will be acceptable within the tolerances. Because of highly complex models, some deterioration in the end-to-end match compared to the stand-alone quality is unavoidable although each may have fidelity in far excess of the required tolerances. In such a highly complex and demanding task of end-to-end validation, we are able to appreciate the need of defining the purpose of model identification to minimize the overall effort, an issue that we had discussed in Secs. 1.1 and 11.1. In this task the criteria to judge the adequate correspondence between real process and identified model play an important role, an issue that we have just covered in this section.

The overall POM computational procedure is as follows. In the first attempt, we set the initial conditions of the states to those measured in flight; as already pointed out in Sec. 3.7, the better practice is use of an average of the first few data points to reduce the effects of noise and other measurement errors. Keeping the aerodynamic database fixed, we allow small biases Δu in the measured control inputs. These biases can be adjusted manually or even an estimation algorithm can be applied to determine them. If the resulting match between the measured and simulated responses is within the specified tolerances (Table 11.1), then the model fidelity is established, provided the biases Δu are also within limits. In some cases, small adjustments of the initial conditions x_0 might also become necessary. Depending upon the particular test, pertaining either to the longitudinal or lateral-directional motion, the tolerances are checked stringently for the variables pertaining to the major axis. A list of variables to be matched is provided by certifying authorities [23, 24] (for example, see Table 11.1) or can be defined in the Acceptance Test Guide (ATG). For the other variables, particularly those of the cross axis, a qualitative match, showing correct trends, is usually considered adequate.

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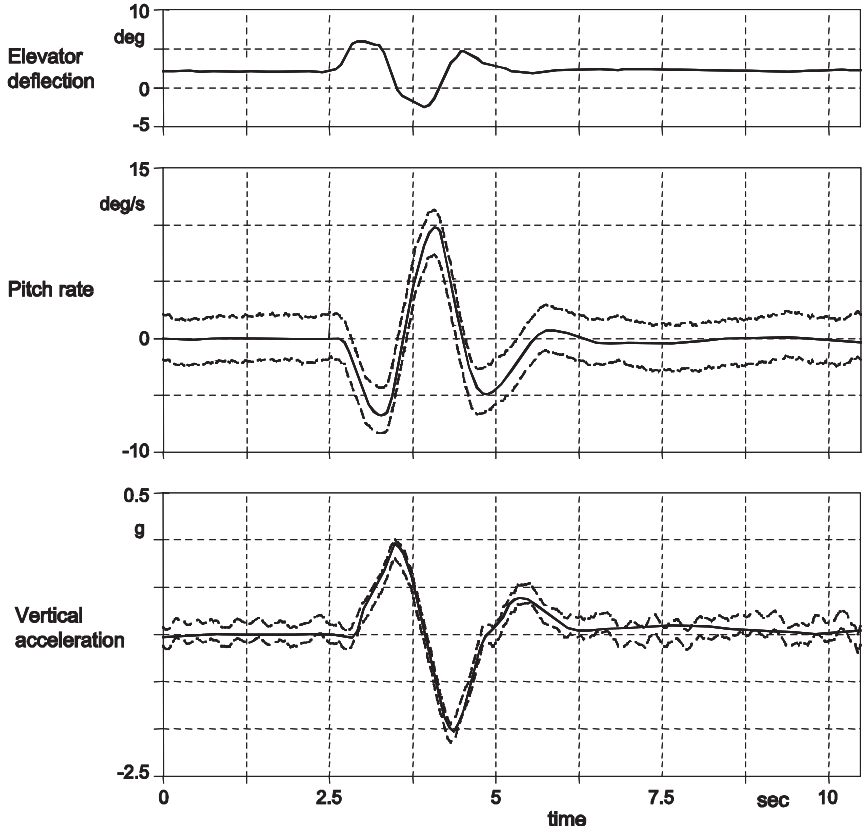

Fig. 11.5 Validation tests 2c11 – short-period dynamics: ------, flight measured ± tolerances; —, model estimated.

The preceding general procedure has been applied to validate the flight-derived databases that we will talk about in Sec. 12.8. Here we demonstrate just the application of preceding basic procedure on a typical example. Figure 11.5 shows a proof-of-match for the validation test 2c11, short-period dynamics. The elevator input and the match for the pitch rate and vertical acceleration are plotted. The model adequacy is quite apparent from the figure. The dashed lines are obtained from the measured data plus/minus the tolerances specified in Table 11.1, and the continuous lines show the model predicted output, which is well within the allowed band for Level-D model fidelity. Roughly 100 to 120 test cases, covering various configurations and maneuvers including special test cases for military aircraft, are usually needed to demonstrate the fidelity of aerodynamic databases for flight simulators.

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11.7 RANGE OF MODEL APPLICABILITY IN FREQUENCY DOMAIN

Although the validation tests are performed in the time domain either through time histories or in terms of period and damping ratios of the oscillatory modes such as phugoid or Dutch roll, it is also possible to extend the verification to the frequency domain, which brings out more clearly the range of applicability of the identified model [28, 29]. This is particularly important for high-authority flight control systems or in cases where aeroservoelastic effects may be dominant.

The procedure is fairly simple. It is illustrated for a single variable, namely, pitch rate, on the example of short-period motion we had considered in Sec. 11.6 and time-domain results shown in Fig. 11.5. In this approach, it is required to compute the frequency responses of the error function q_m/q, that is, of the ratio of measured to computed pitch rate. The Bode plot of magnitude and phase angle provides information on the model fidelity and also on the range of validity in the frequency domain. For a perfect match, we will have 0-dB magnitude and 0-deg phase angle throughout the frequency range, which is possible for hypothetical cases and exercises dealing with simulated data only, but not in any practical application with real data. As in the case of time-domain verification in terms of tolerances, the mismatch boundaries in terms of magnitude and phase are defined. These boundaries are based on the LOES (Low Order Equivalent System) mismatch criteria and provide boundaries of unnoticeable dynamics, that is, any discrepancies within the band are not perceptible to the pilot [30]. Accordingly, for sufficient fidelity, magnitude and phase of the error function should be within these boundaries.

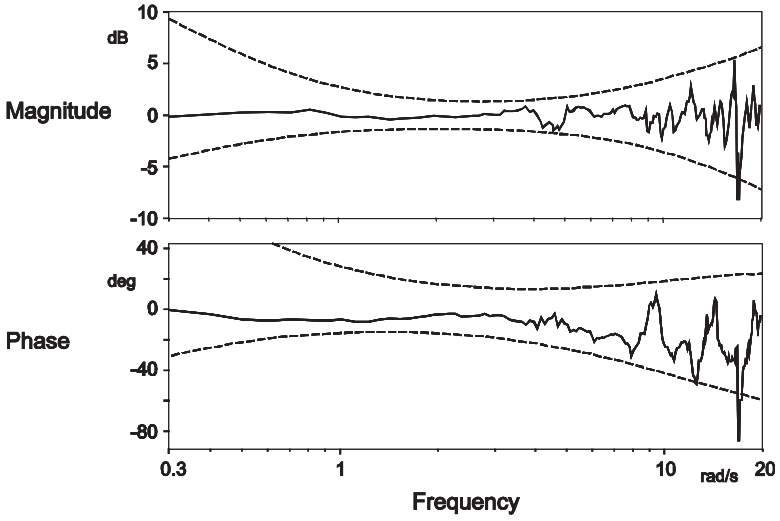

Fig. 11.6 Model validation in frequency domain.

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For the case at hand, namely, short-period motion, the frequency-domain verification is shown in Fig. 11.6, which shows the global model to be sufficiently accurate. The figure shows a more restrictive tolerance band in the range of 1 to 5 rad/s pilot crossover frequency, allows more error outside of it (LOES phase-lead leniency), and are asymmetric at low frequencies (allow more phase-lead error than phase-lag error) to have better fidelity with respect to lag error, which might cause pilot-induced oscillations [31]. The procedure to compute frequency responses is not covered in this book, but can be found in any standard book on frequency-domain techniques. If the recorded data are too noisy, we might have to pay a little more attention while applying these techniques.

Besides looking at the error analysis as just discussed, a comparison of the Bode plot of measured system response and identified model provides useful information on the system dynamics and possibly on model deficiencies. From the linearized system matrices A, B, C, D, the frequency response matrix can be easily computed as given by [32]

$$G(j\omega) = C\,(j\omega I - A)^{-1}\,B + D \tag{11.16}$$

From the measured flight responses, it is possible to approximate frequency responses through classical fast Fourier transform (FFT) techniques. Because this is not our primary goal, we do not cover this subject here anymore. However, as already mentioned, a caution is required while applying techniques to noisy data. Smoothing techniques can falsify the results and interpretations.

11.8 CONCLUDING REMARKS

Besides model estimation, model validation is the other central issue in system identification. In this chapter we discussed some diagnostic tests and approaches to assess the quality of estimated individual parameters and of the model as a whole, both qualitatively and quantitatively. We discussed in some depth procedures to determine the model quality in terms of statistical properties of the estimates and of residuals. Plausibility of the parameter estimates needs to be verified because a good response match is just a necessary condition and not a sufficient one. Such verification is possible through a comparison of flight estimates with wind-tunnel or analytical predictions, or from physical understanding of the dynamic system.

It is pointed out that the demonstration of adequate model predictive capability, particularly for aerodynamic databases for flight simulators, is crucial. The difficulties encountered in the end-to-end match, the treatment of initial conditions on the state variables (trim), biases in the control inputs, and the importance of using complementary data, that is, data not used in the model development, are highlighted. It has been argued that to minimize the effort the application domain of the model be defined in advance and the methodological

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procedures of this chapter applied accordingly to judge the model adequacy. Both the time-domain and the frequency-domain verification have been elucidated using a simple example to demonstrate the basic principle.

Through the discussion of practical issues, it is brought out that there is no single criterion that would give us a clear-cut "yes" or "no" answer about the model adequacy. Nevertheless, some guidelines of the methodological procedure are presented in this chapter. Those can be judiciously applied to gain confidence in, or reject, a particular model.

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REFERENCES

[1] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[2] Schlesinger, S., Buyan, J. R., Callender, E. D., Clarkson, W. K., and Perkins, F. M., "Developing Standard Procedures for Simulation Validation and Verification," *Proceedings of the Summer Computer Simulation Conference*, Vol. 1, Simulation Councils, La Jolla, CA, pp. 927–933.

[3] Plaetschke, E., and Mackie, D. B., "Maximum-Likelihood-Schätzung von Parametern Linearer Systeme aus Flugversuchsdaten – Ein FORTRAN Programm," DFVLR-Mitt. 84-10, Germany, Jan. 1984 (in German).

[4] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985.

[5] Balakrishnan, A. V., and Maine, R. E., "Improvements in Aircraft Estimation Programs," NASA CR-145090, 1975.

[6] Iliff, K. W., and Maine, R. E., "Further Observations on Maximum Likelihood Estimates of Stability and Control Characteristics Obtained from Flight Data," AIAA Paper 77-1133, Aug. 1977.

[7] Maine, R. E., and Iliff, K. W., "Use of Cramér-Rao Bounds on Flight Data with Colored Residuals," *Journal of Guidance and Control*, Vol. 4, No. 2, 1981, pp. 207–213.

[8] Morelli, E. A., and Klein, V., "Accuracy of Aerodynamic Model Parameters Estimated from Flight Test Data," *Journal of Guidance, Control, and Dynamics*, Vol. 20, No. 1, 1997, pp. 74–80.

[9] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems – Applications to Aircraft Part 1: The Output Error Approach," AGARD, AG-300, Vol. 3, Part 1, Dec. 1986.

[10] Theil, H., *Economic Forecasts and Policy*, 4th ed., North-Holland, Amsterdam, 1975.

[11] Kheir, N. A., and Holmes, W. M., "On Validating Simulation Models of Missile Systems," *Simulation*, April 1978, pp. 117–128.

[12] Pindyck, R. S., and Rubinfeld, D. L., *Econometric Models and Economic Forecasts*, 3rd ed., McGraw-Hill, New York, 1991, pp. 336–342.

[13] Murray-Smith, D. J., "Methods for the External Validation of Continuous System Simulation Models: A Review," *Journal of Mathematical and Computer Modelling of Dynamical Systems*, Vol. 4, No. 1, 1998, pp. 5–31.

[14] Jategaonkar, R. V., Fischenberg, D., and von Gruenhagen, W., "Aerodynamic Modeling and System Identification from Flight Data – Recent Applications at DLR," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 681–691.

[15] Paris, A. C., and Bonner, M., "Nonlinear Model Development from Flight Test Data for F/A-18E Super Hornet," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 692–702.

[16] Box, G. E. P., and Jenkins, G. M., *Time Series Analysis – Forecasting and Control*, Holden Day, San Francisco, CA, 1970.

[17] von Grünhagen, W., "Inverse Simulation: A Tool for the Validation of Simulation Programs – First Results," *ZFW*, Vol. 17, No. 3, 1993, pp. 211–219.

[18] Fischenberg, D., "Ground Effect Modeling Using a Hybrid Approach of Inverse Simulation and System Identification," AIAA Paper 99-4324, Aug. 1999.

[19] Hess, R. A., Gao, C., and Wang, S. H., "Generalized Technique for Inverse Simulation Applied to Aircraft Maneuvers," *Journal of Guidance, Control, and Dynamics*, Vol. 14, No. 5, 1991, pp. 920–926.

[20] Rutherford, S., and Thomson, D. G., "Improved Methodology for Inverse Simulation," *Aeronautical Journal*, Vol. 100, No. 993, 1996, pp. 79–86.

[21] Hamel, P. G., and Jategaonkar, R. V., "The Role of System Identification for Flight Vehicle Applications: Revisited," RTO-MP-11, Paper 2, March 1999.

[22] Jategaonkar, R. V., "Determination of Aerodynamic Characteristics from ATTAS Flight Data Gathering for Ground-Based Simulator," DLR-FB 91-15, Germany, May 1991.

[23] "Airplane Simulator Qualification," Federal Aviation Administration, FAA Advisory Circular, AC 120-40C, Draft Version, July 1995.

[24] "Joint Aviation Requirements – Aeroplane Flight Simulators," JAR-STD 1A, Westward Digital, Ltd., Cheltenham, England, U.K., April 1997.

[25] Neville, K. W., "Simulator-to-Flight Validation – Methods, Problems and Dilemmas," *Proceedings of the Symposium on "Data Issues for Flight Simulators – An Ongoing Problem?,"* Royal Aeronautical Society, London, Nov. 1993, pp. 14.1–14.7.

[26] Jategaonkar, R. V., and Mönnich, W., "Identification of DO-328 Aerodynamic Database for a Level D Flight Simulator," AIAA Paper 97-3729, Aug. 1997.

[27] Weiss, S., Gockel, W., Mönnich, W., and Rohlf, D., "Identification of Dornier-328 Reversible Flight Control System," AIAA Paper 98-3729, Aug. 1998.

[28] Hamel, P. G., and Kaletka, J., "Rotorcraft System Identification – An Overview of AGARD FVP Working Group 18," AGARD, CP-552, Paper 18, Oct. 1994.

[29] Tischler, M. B., "System Identification Methods for Aircraft Flight Control Development and Validation," NASA TM 110369, Oct. 1995.

[30] Hoh, R. H., Mitchell, D. G., Ashkenas, I. L., Klein, R. H., Heffley, R. K., and Hodgkinson, J., "Proposed MIL Standard and Handbook – Flying Qualities of Air Vehicles," Air Force Wright Aeronautical Lab., Rept. AFWAL-TR-82-3081, Vol. 2, Dayton, OH, 1982.

[31] Hodgkinson, J., and Mitchell, D., "Handling Qualities," *Flight Control Systems*, edited by R. W. Pratt, AIAA Progress in Astronautics and Aeronautics Series, Vol. 184, AIAA, Reston, VA, 2000, Chap. 4.

[32] Laub, A. J., "Efficient Multivariable Frequency Response Computations," *IEEE Transactions on Automatic Control*, Vol. AC-26, No. 2, 1981, pp. 407, 408.

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CHAPTER 12

Selected Advanced Examples

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12.1 INTRODUCTION

Applications of system identification methodology to flight vehicles have been too numerous to count. This is evident from the plethora of applications reported in the literature, both in terms of sheer number of aircraft configurations being analyzed and also in terms of modeling complexity. A representative summary of such applications is found in the special focus issue on "Flight Vehicle System Identification – Engineering Utility" of the *AIAA Journal of Aircraft* [1]. References [2–6] provide a consolidated account of activities related to aircraft parameter estimation from flight data. Common to most of these advanced applications is the time-domain methodology of system identification, which we have discussed in Chapters 4 through 8. In those chapters, we had also applied them to typical problems to bring out the underlying mathematics, computational details, practical issues, and scope of applicability. Simulated data as well as flight data were analyzed using the software described in the respective chapters. Both the data and software are supplemented to the book. These estimation techniques, either directly or in a suitably modified form, were applied to unstable aircraft in Chapter 9. In general, aerodynamic modeling aspects as well as those pertaining to the data compatibility check using flight-path reconstruction techniques were discussed.

In the present chapter we focus on a few more advanced and comprehensive applications than those reported in the preceding chapters. Specifically, we address the following aspects: 1) modeling of transit time lag effect resulting from angle of attack or direct-lift flap or speed brakes and thrust variations; 2) estimation of aerodynamic effects of secondary order like those due to landing gear; 3) determination of effects due to control surface malfunction, namely, roll-spoiler hardover; 4) modeling and identification of unsteady aerodynamics and quasi-steady stall covering effects in both longitudinal and lateral-directional modes; 5) extraction of ground effects for low-wing and high-wing configurations; 6) generation of high-fidelity aerodynamic databases for Level-D flight simulators of Transall C-160 military transport aircraft, Dornier 328 propeller

aircraft, and VFW-614 ATTAS in-flight simulator; 7) flexible aircraft modeling including rigid-body and structural dynamics; 8) database validation and update of X-31A unstable aircraft; 9) wake-vortex encounter model; 10) verification and update of predicted longitudinal aerodynamics of the reusable launch vehicle demonstrator Phoenix; and 11) rotorcraft modeling including rotor degrees of freedom and wake distortion dynamics. In each case we analyze flight recorded data. We provide the readers the flight data gathered during stall maneuvers with the ATTAS aircraft. In all of these cases, we discuss in detail the modeling issues and results that were obtained from them. The goal of this chapter is to provide an appreciation of a wide variety of aerodynamic modeling and parameter estimation problems.

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12.2 MODELING OF TRANSIT TIME LAG EFFECTS

In the theory of aircraft flight mechanics, it is well known that for a conventional aircraft with wing and horizontal tail, the most significant effect of the wing on the tail is the downward deflection of the flow due to wing circulation, which can be aerodynamically modeled as a downwash angle ε [7–10]. Such effects are present even under the assumption of quasi-steady flow and need to be properly accounted for. As the downwash is caused by circulation, it changes anytime the circulation changes, for example: 1) changes in the wing lift due to changes in the angle of attack; 2) deflection of direct-lift-control flaps, which are a modified part of the landing flaps on the wings; and 3) operation of wing-mounted speed brakes in flight. The modified flow at the wing, however, reaches the empennage after a time interval $\tau = r_H/V$, where V is the airspeed and r_H the tail length. Thus, the effective angle of attack at the tail is changed only after a time delay of τ s. The same transit time effect is also applicable to cases considering vertical wind based on Taylor's hypothesis of treating turbulence as a frozen pattern. Similarly, slipstream transit time affects result from dynamic thrust variations from wing-mounted engines, particularly for propeller aircraft. In this section, we investigate the estimation of the transit time lag effects originating from the aforementioned different cases.

12.2.1 SEPARATION OF PITCH DAMPING DERIVATIVES

Conventionally, as a first-order approximation, the downwash angle is assumed to be proportional to the wing angle of attack and is represented as a constant derivative $\partial\varepsilon/\partial\alpha$. The so-called downwash lag effect can be mathematically expressed as $(\partial\varepsilon/\partial\alpha)\alpha(t-\tau)$. In the context of estimation of aerodynamic derivatives from flight data, two approaches are possible to account for the downwash lag effect [11].

The first approach is based on a linearized aerodynamic model referred to a single point, in which the downwash lag is approximated through derivatives with respect to the translational acceleration in the vertical direction [10]. The derivatives due to the body-fixed translational acceleration in the vertical direction

are equivalent in the stability axis to those with respect to $\dot{\alpha}$, the rate of change of angle of attack. The Taylor-series expansions of the lift force and pitching moment result in an aerodynamic model that is linear in parameters. Typically,

$$C_L = C_{L0} + C_{L\alpha}\,\alpha + C_{L\delta e}\,\delta_e + C_{Lq}\,\frac{q\bar{c}}{V} + C_{L\dot{\alpha}}\,\frac{\dot{\alpha}\bar{c}}{V} \tag{12.1}$$

$$C_m = C_{m0} + C_{m\alpha}\,\alpha + C_{m\delta e}\,\delta_e + C_{mq}\,\frac{q\bar{c}}{V} + C_{m\dot{\alpha}}\,\frac{\dot{\alpha}\bar{c}}{V} \tag{12.2}$$

Focusing on the pitching moment, the derivatives C_{mq} and $C_{m\dot{\alpha}}$ contribute to the pitch damping. In several applications, particularly those related to flight data analysis, $C_{m\dot{\alpha}}$ is usually treated as a part of the total equivalent pitch damping $C_{\mathrm{mqEqv}} = C_{mq} + C_{m\dot{\alpha}}$. However, in a few specific cases the separation of these two components is desirable because, although in some cases closely related, they result from two uncorrelated aerodynamic phenomena. The downwash lag, that is, $C_{m\dot{\alpha}}$, is related to variations in the flow pattern generated by the wing whereas C_{mq} comes dominantly from change in tail angle of attack due to pitching.

Many of the investigations pertaining to aircraft parameter estimation were necessarily restricted to models linear in parameters, mainly because the available estimation programs were capable of handling only the linear models. Although algorithmically it is feasible to estimate the derivatives $C_{L\dot{\alpha}}$ and $C_{m\dot{\alpha}}$, the linearization results in a correlation and affects the independent estimation of these derivatives. In a typical flight maneuver, for example, elevator doublet or multistep input employed in estimating longitudinal derivatives, the pitch rate q is highly correlated to $\dot{\alpha}$ (see Fig. 12.1). This is clear from the $\dot{\alpha}$ equation [10]

$$\dot{\alpha} = -\frac{\bar{q}S}{mV}\,C_L + q + \frac{g}{V}(\cos\theta\,\cos\phi\,\cos\alpha + \sin\theta\,\sin\alpha) - \tan\beta\,(p\,\cos\alpha + r\,\sin\alpha) \tag{12.3}$$

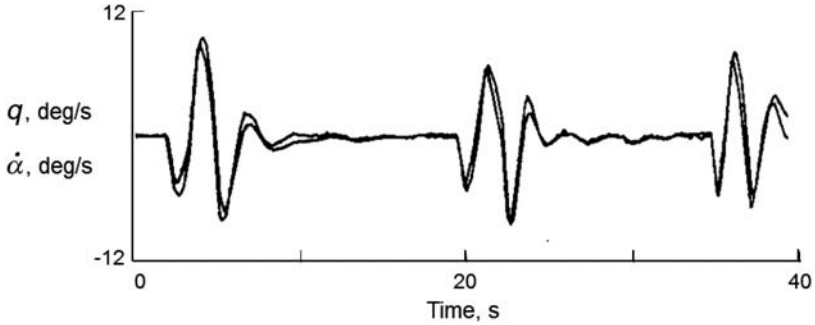

Fig. 12.1 Correlation between pitch rate q and rate of change of angle of attack $\dot{\alpha}$ (Source: [11]).

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In a conventional longitudinal maneuver, variations in the roll angle ϕ are minimal. Because the lift coefficient C_L is a linear function of α, q, and δ_e, the first two terms in Eq. (12.3) yield linearly dependent $\dot{\alpha}$ components. To overcome this linear dependence, special flight maneuvers are necessary. An independent component of $\dot{\alpha}$ can be generated through the gravity term, that is, the third term on the right side of Eq. (12.3). The magnitude of the contribution of this term depends on the excursion in the pitch angle θ or the roll angle ϕ. In addition, simultaneous variation in the airspeed V can be used to augment the contribution. Figure 12.2 clearly indicates that a roll maneuver removes the correlation between the variables q and $\dot{\alpha}$. The variable $\dot{\alpha}$ is not directly measured in flight but is obtained by numerical differentiation only for the qualitative comparison.

For large aircraft, large pitch angle excursions can pose practical difficulties. On the other hand, a roll maneuver is easy to carry out and also provides the necessary separation of q and $\dot{\alpha}$. Reference [10] demonstrates successful estimation of the two pitch damping derivatives from a 180-deg (inverted position) roll maneuver with a highly maneuverable aircraft, indicating limitations of performing such extreme maneuvers with large aircraft. Yet another restriction of the preceding linearized model pertains to implementation. Equation (12.1) can lead to an algebraic loop because computation of C_L appearing in the $\dot{w}$ equation requires $\dot{\alpha}$ (or equivalently $\dot{w}$).

The second approach is based on the multipoint modeling of the aerodynamic effects. For a conventional aircraft with wing and horizontal tail, the lift and drag forces can be assumed to be working at two points, namely, the neutral points of the wing and of the tail. Knowing the forces, the moments can be computed readily because the lever arms are generally known. In a general case, the lift

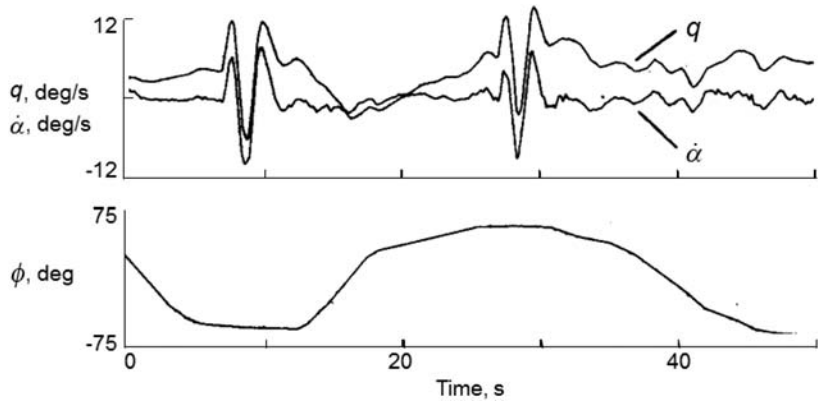

Fig. 12.2 Uncorrelated pitch rate and rate of change of angle of attack from roll maneuver (Source: [11]).

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and pitching moment acting on an aircraft can be represented as [12]

$$C_L = C_{L0} + C_{L\alpha W}\alpha(t) + \frac{S_H}{S} \times \left\{ C_{L\alpha H}\left[\alpha(t) - \frac{\partial\varepsilon}{\partial\alpha}\alpha(t-\tau) + i_H + \alpha_{\text{dyn}}\right] + C_{L\delta e}\delta_e \right\} \tag{12.4}$$

$$C_m = C_{m0} - \frac{r_H^*}{\bar{c}}\frac{S_H}{S} \times \left\{ C_{L\alpha H}\left[\alpha(t) - \frac{\partial\varepsilon}{\partial\alpha}\alpha(t-\tau) + i_H + \alpha_{\text{dyn}}\right] + C_{L\delta e}\delta_e \right\} + C_{mqW}\frac{q\bar{c}}{V} \tag{12.5}$$

where the subscript W refers to the wing/body combination and H to the horizontal tail, S and S_H denote the wing and tail areas, $\bar{c}$ is the mean aerodynamic chord, δ_e the elevator deflection, and i_H the tail incidence angle. The dynamic angle of attack due to pitch rate is given by $\alpha_{\text{dyn}} = \tan^{-1}(qr_H/V)$. The lever arms r_H and r_H^* are, respectively, the distances from the center of gravity and from the neutral point of the wing to the neutral point of the tail. For the sake of clarity, only α is shown to be a function of time although the other variables are also time dependent. If vertical wind is to be accounted for, then the expression within the square brackets on the right-hand sides of these two equations will have to be extended appropriately to include the angle of attack due to gust at wing and at horizontal tail accounting for the transit time effect.

The two-point aerodynamic model necessarily leads to a more complex nonlinear-in-parameters model. The use of such a model to estimate the lift derivatives separately for the wing and tail requires an estimation program capable of handling nonlinear system models as well as a provision to generate delay in a specified variable, in the present case α. Both of these options were available in the estimation program used in these investigations.

Equivalent linear derivatives, if required, say, for comparison purposes, can be obtained from the estimated lift-curve slope of the tailplane $C_{L\alpha H}$ and the downwash parameter $\partial\varepsilon/\partial\alpha$ by simple recomputations. For example,

$$C_{mq} = -\left(C_{L\alpha H}\frac{S_H}{S}\frac{r_H}{\bar{c}}\right)\frac{r_H}{\bar{c}} + C_{mqW} \tag{12.6}$$

$$C_{m\dot{\alpha}} = -\left(C_{L\alpha H}\frac{S_H}{S}\frac{r_H^*}{\bar{c}}\frac{\partial\varepsilon}{\partial\alpha}\right)\frac{r_H}{\bar{c}} \tag{12.7}$$

The test aircraft VFW-614 ATTAS (Advanced Technologies Testing Aircraft System) is a medium-size twin-jet short-haul 44-passenger aircraft (Fig. 12.3), modified and instrumented for the research purposes at DLR. It also serves as an in-flight simulator [13, 14]. The aircraft with a span of 21.5 m, wing area of

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Fig. 12.3 VFW-614 ATTAS.

64 m^2, and full-fuel takeoff weight in the range of 20 tons has maximum roll angle capability up to roughly 70 deg, and the maximum safe load factor is 2.8 g for retracted landing flaps. To investigate estimation of C_{mq} and $C_{m\dot{\alpha}}$, a flight-test program was carried out at an altitude of 5000 m and indicated airspeed of 200 kt. Apart from the conventional maneuvers to excite the longitudinal and lateral-directional motion with rapidly changing 3-2-1-1 multistep inputs, five roll maneuvers were carried out. The first four roll maneuvers were with bank angles of 20, 40, 50, and 60 deg, respectively. In these maneuvers, the variations in airspeed were minimal. In the fifth maneuver, the 60-deg roll is combined with acceleration and deceleration, that is, airspeed variation. Elevator doublets were also applied in the banked position to maximize the information. During the complete maneuver, one must ensure that the load factor is within the allowable safe limits. Although the conventional multistep inputs were generated from the onboard computer and applied automatically, roll maneuvers requiring more coordination and continuous monitoring were carried out manually by the pilot.

The onboard recorded flight data were analyzed off-line using a maximum likelihood estimation program for nonlinear systems. Although a detailed analysis of all of the test maneuvers has been carried out, only typical results sufficient to demonstrate the estimation of downwash effects are presented here. Five roll maneuvers were analyzed separately to find the effect of the roll angle on the separation of C_{mq} and $C_{m\dot{\alpha}}$. In each case, the roll maneuver is combined with conventional multistep input maneuvers, exciting separately the longitudinal and lateral-directional motion.

The results based on the linearized model of Eqs. (12.1–12.2) indicate that it was not possible to estimate separately the derivatives C_{mq} and $C_{m\dot{\alpha}}$ from the roll

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maneuver limited to 20-deg bank angle. The other roll maneuvers with more than 40-deg bank angle provided uncorrelated estimates of these two derivatives, which appeared physically meaningful and also compared well with wind-tunnel predictions. The estimates obtained from the accelerated and decelerated 60-deg roll maneuver are provided in Table 12.1. In Fig. 12.4 showing two maneuvers demarcated by the vertical lines, the agreement between the measured and estimated time responses of typical motion variables is seen to be good. The first segment corresponds to the multistep elevator input and the second to the roll maneuver with superimposed elevator doublet input.

The analysis has been repeated using the two-point aerodynamic model, Eqs. (12.4–12.5). Although this approach does not necessarily require special flight tests, in order to compare the results with the linearized model, the same set of time records including a roll maneuver is analyzed. Because the downwash effect is implicitly modeled, the roll maneuver with only a 20-deg bank angle provided estimates of the downwash parameter $\partial\varepsilon/\partial\alpha$, which agreed well with the estimates from the other roll maneuvers with larger bank angle. From the estimated $C_{L\alpha H}$ and $\partial\varepsilon/\partial\alpha$, the equivalent linear pitch damping derivatives C_{mq} and $C_{m\dot{\alpha}}$ are computed, using Eqs. (12.6–12.7). These values compare well with the estimates from the linearized model and wind-tunnel predictions (see Table 12.1). The match between the flight-measured and estimated responses was essentially the same as in Fig. 12.4. Table 12.1 also provides other pitching-moment derivatives. The elevator effectiveness $C_{m\delta e}$, as estimated by both of the models, is found to be a little higher than the wind-tunnel prediction.

To summarize, the investigations clearly point out that the pitching-moment derivatives estimated from flight data by the two complementary methods and wind-tunnel predictions agreed well [11]. The linearized approach based on the Taylor series requires special flight maneuvers to remove the correlation between the motion variables. The roll maneuvers with more than 40-deg bank angle yielded uncorrelated estimates of the two pitch damping derivatives. The two-point aerodynamic model, based on the forces acting on the wing and tail,

TABLE 12.1 PITCHING-MOMENT DERIVATIVES

Parameter	Wind-tunnel Prediction	Estimates From	
		Linearized Model	Two-point Model[a]
$C_{m\alpha}$	−1.07	−1.18	−1.12
$C_{m\delta e}$	−1.40	−1.66	−1.68
C_{mq}	−8.67	−7.86	−9.03
$C_{m\dot{\alpha}}$	−4.65	−5.23	−4.59

[a]Recomputed estimates.

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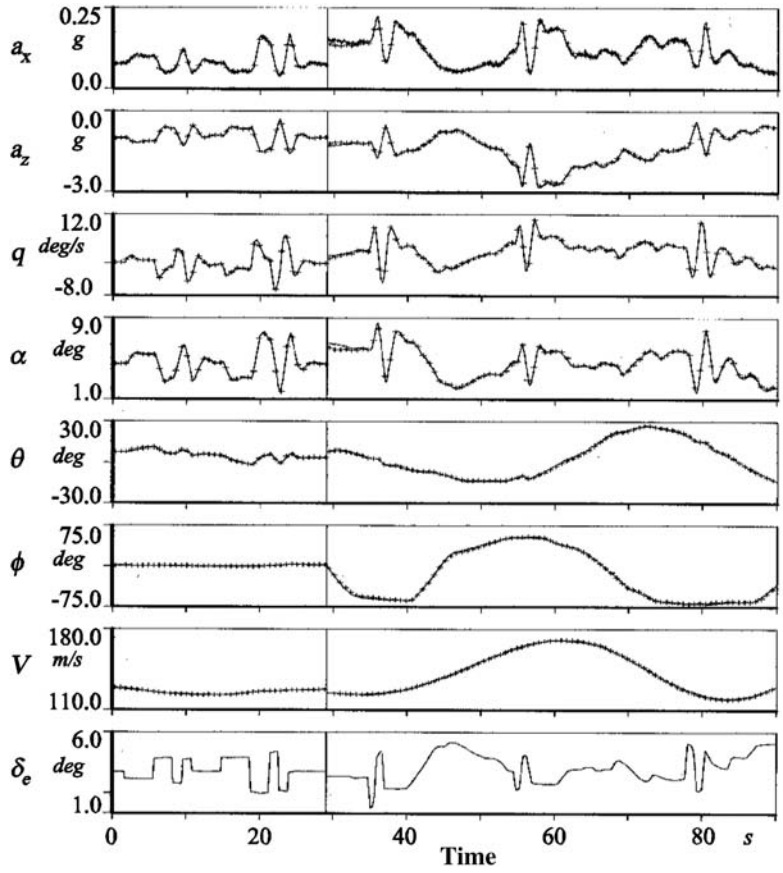

Fig. 12.4 Elevator input and roll maneuvers: ——, flight measured; + + + + +, estimated.

accounts explicitly for the downwash lag effect through a constant derivative and time delay. It requires, however, a more advanced parameter estimation program capable of handling general nonlinear systems, but eliminates the need for special flight maneuvers otherwise necessary to remove correlation between the motion variables. The second approach is more sophisticated for two reasons:

1. It provides a physically more realistic representation of the aerodynamics and accounts for the downwash effect explicitly through the derivatives $\partial\varepsilon/\partial\alpha$ and a time delay τ.
2. It provides an alternative to account for the downwash effects for large aircraft with limited roll angle capabilities, which we had briefly indicated earlier in this section.

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In an analogous treatment of wing/body undergoing lateral perturbations, the resulting forces on the fin lead to a derivative with respect to the rate of change of angle of sideslip $\dot{\beta}$. This derivative, modeling the sidewash effect, contributes to yaw damping. As in the case of longitudinal motion, many applications consider the total equivalent yaw damping $C_{\mathrm{nrEqv}} = C_{\mathrm{nr}} + C_{n\dot{\beta}}$. Estimation of $C_{n\dot{\beta}}$ from flight-test maneuvers is slightly easier than that of $C_{m\dot{\alpha}}$. The classical yawing and rolling maneuvers, which are performed anyway to allow determination of lateral-directional derivatives, contain a small $\dot{\beta}$ component independent of other variables. This is evident from the following equation:

$$\dot{\beta} = \frac{\bar{q}S}{mV} C_Y + p \sin\alpha - r \cos\alpha + \frac{g}{V} \sin\phi \cos\theta \tag{12.8}$$

The flight investigations with ATTAS and the following parameter estimation confirmed that the classical rapid roll and Dutch roll maneuvers did lead to consistent and reliable estimates of the sidewash derivative. A comparison with the wind-tunnel predictions, see far right side plot of Fig. 12.5, clearly shows that the prediction had neglected the contribution due to rate of change of angle of sideslip. This was further confirmed by estimating an equivalent yaw damping derivative C_{nrEqv} from flight maneuvers, yielding a value roughly between -0.33 and -0.28, and not close to the predicted -0.6. On the other hand, separate estimation of the two provided values of -0.568 and -0.281 for C_{nr} and $C_{n\dot{\beta}}$, respectively; the difference between the two roughly matching with the equivalent derivative obtained from flight (note that the r and $\dot{\beta}$ are opposite in phase). The flight estimation of the two components of yaw damping resolved the discrepancies observed in the

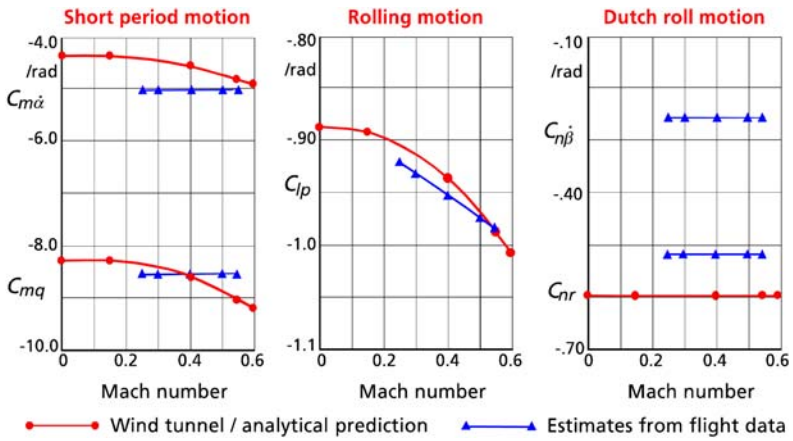

Fig. 12.5 ATTAS damping derivatives.

performance of the model-following control that was optimized on the ground using predicted database and tested in fight.

Figure 12.5 shows a comparison of the other damping derivatives as well. It is observed that the roll damping $C_{\ell p}$ including its dependence on the angle of attack was estimated best, matching very well with the predictions. The estimates of the pitch damping derivatives C_{mq} and $C_{m\dot{\alpha}}$ from two sources were also comparable. The estimates of the dependence of these two derivatives on the Mach number were found to be poor and correlated. Furthermore, their inclusion was also not justified from the quality of the match for all of the output variables. Hence, they are shown as constant values in Fig. 12.5. However, we know from theory that such dependencies do exist in practice, and, hence, some fairing of results between the predictions and flight estimates would be justifiable.

12.2.2 AERODYNAMIC EFFECTS OF DIRECT-LIFT-CONTROL FLAPS

As already pointed out, the VFW-614 ATTAS, Fig. 12.3, is a fly-by-wire aircraft used for in-flight simulation (that is, simulation of other aircraft in flight). One of the major features of ATTAS is that, in addition to the conventional controls, it is equipped with direct-lift-control (DLC) flaps for high-frequency direct-lift modulation [13–15]. They provide an additional independent longitudinal control, which is particularly suitable and necessary for realistic in-flight simulations. In such applications, the feed-forward control laws based on the inversion of the mathematical model of the "host" flight vehicle play an extremely important role. This is because high-bandwidth control systems need more accurate regulation of initial response than of final output error. The fidelity of the in-flight simulator depends to a large extent on the accuracy of the aerodynamic database representing the host aircraft, and as such the demands on the modeling accuracy are very high. Accordingly, system identification of ATTAS has been carried out, including the rigid-body model based on wing and tail separately, modeling DLC, control surface interference, ground effects, and modeling of actuation systems of primary controls [14, 15].

The DLC flaps are derived by modifying a trailing-edge portion (45%) of the landing flaps; see Fig. 12.6. They are divided into six fast moving flaps, three on each wing. Each of the six flaps is driven by an individual electrohydraulic actuation system, which is both rate and force limited for structural safety. The six flaps are designed for a maximum deflection of ± 35 deg and for an actuation rate of 75 deg/s under aerodynamic loads. The DLC flaps can be operated for landing-flap deflections up to 14 deg. Simultaneous variations of the DLC and landing flaps are possible.

For the purpose of aerodynamic modeling, the six DLC flaps are considered in two units, each consisting of three flaps on each wing. The equivalent left and right DLC-flap deflections, denoted by δ^L_{DLC} and δ^R_{DLC}, are computed from the individually measured flap deflections proportional to the surface areas. The lift, drag

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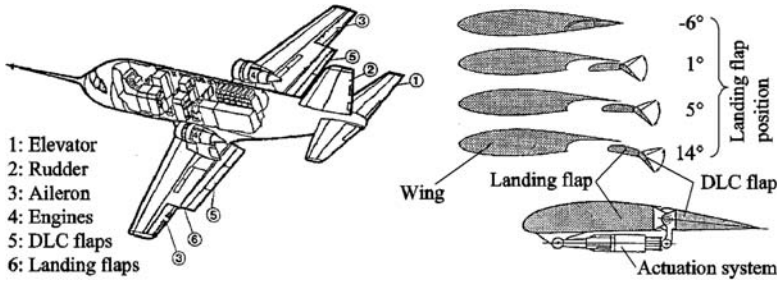

Fig. 12.6 Research aircraft ATTAS and DLC flaps.

and pitching moment generated by the DLC flaps are modeled as [15]

$$\begin{aligned} C_{\mathrm{LDLC}} &= \Delta C^{L}_{\mathrm{LDLC}} + \Delta C^{R}_{\mathrm{LDLC}} \\ C_{\mathrm{DDLC}} &= \Delta C^{L}_{\mathrm{DDLC}} + \Delta C^{R}_{\mathrm{DDLC}} \\ C_{\mathrm{mDLC}} &= \Delta C^{L}_{\mathrm{mDLC}} + \Delta C^{R}_{\mathrm{mDLC}} + \Delta C^{\varepsilon}_{\mathrm{mDLC}} \end{aligned} \tag{12.9}$$

where the contributions due to the left and right DLC flaps are denoted by the superscripts L and R, respectively. In the preceding equation we have considered contributions due to the left and right DLC flaps separately because it was found that the lift and drag differentials resulting from unequal DLC flap deflections on the two sides lead to nonnegligible rolling and yawing motion [14, 15]. We restrict ourselves here to discussion of the longitudinal motion only. The following general model for these contributions is postulated:

$$\begin{aligned} \Delta C_{\mathrm{LDLC}} &= A_1 \delta_{\mathrm{DLC}} + A_2 \delta^2_{\mathrm{DLC}} + A_3 \delta^3_{\mathrm{DLC}} \\ \Delta C_{\mathrm{DDLC}} &= (W_1 + W_{1\alpha}\alpha)\ \delta_{\mathrm{DLC}} + W_2\ \delta^2_{\mathrm{DLC}} \\ \Delta C_{\mathrm{mDLC}} &= M_1 \delta_{\mathrm{DLC}} + M_2 \delta^2_{\mathrm{DLC}} + M_3 \delta^3_{\mathrm{DLC}} \end{aligned} \tag{12.10}$$

where A_i, W_i, and M_i are the unknown derivatives that are to be estimated from flight data. Furthermore, in Eq. (12.9), $\Delta C^{\varepsilon}_{\mathrm{mDLC}}$ denotes the downwash lag effect due to deflection of the DLC flaps. Neglecting this term for the downwash, estimation of the derivatives appearing in Eq. (12.10) yields the pitch rate response due to a multistep 3-2-1-1 input to the DLC flaps shown in Fig. 12.7a. It is observed that the match between the flight measured and estimated response for steady-state level is acceptable, but that for dynamic changes from one steady-state level to another shows larger deviations.

By analogy to the translational acceleration derivatives derived from an approximations of the unsteady aerodynamic effect (in particular, the derivative $C_{m\dot{\alpha}}$ modeling the downwash lag due to the variations in the angle of attack), in the present case the downwash lag due to a DLC-flap deflection can be

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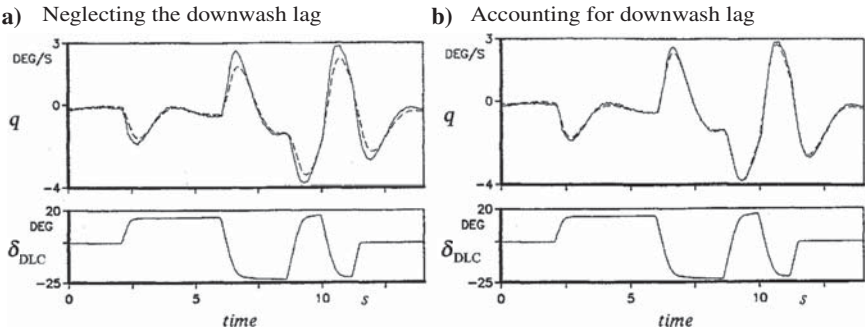

Fig. 12.7 Pitch rate in response to DLC flap input: ——, measured; --------, estimated.

approximated through a first-order linear pitching moment derivative with respect to rate of change of DLC-flap deflection $\dot{\delta}_{\mathrm{DLC}}$. This would call for measurement of $\dot{\delta}_{\mathrm{DLC}}$ or computation of the same from measured δ_{DLC} by numerical differentiation. However, as discussed in Sec. 12.2.1, we adopt here the more elegant approach based on a two-point aerodynamic model incorporating a time-delayed signal in parameter estimation. Accordingly, the downwash lag effect is modeled as

$$\Delta C^{\varepsilon}_{\mathrm{mDLC}} = C_{L\alpha H}\,\frac{S_H}{S}\,\frac{r^*_H}{\bar{c}}\,M_\tau\,\delta_{\mathrm{DLC}}(t-\tau) \tag{12.11}$$

where $C_{L\alpha H}$ is the tail lift-curve slope, M_τ the unknown downwash parameter, and $\delta_{\mathrm{DLC}}(t-\tau)$ the time-delayed signal, where $\tau = r_H/V$ denotes the transit time required for flow modifications generated at the wing to reach the tail; other variables have been defined earlier. Estimation of parameters appearing in Eqs. (12.10–12.11) yields the match for the pitch rate shown in Fig. 12.7b. The improvements in the match in Fig. 12.7b compared to Fig. 12.7a are, without any further explanation, self-illustrative of the downwash lag effect and efficacy of the modeling approach. Thus, the cubic model for the pitching moment due to DLC flaps in Eq. (12.10) needs to be augmented with Eq. (12.11) for the downwash lag effect.

The estimates of the DLC flap effectiveness are shown in Fig. 12.8. It is observed that the estimated lift coefficient C_{LDLC} is smaller than the predicted, the difference being pronounced for larger positive deflections. The influence of flow separation for deflections above 10 deg is to be observed. It is clearly evident that the DLC flaps are somewhat less effective in generating direct lift than designed for and predicted from wind-tunnel tests. The flight-validated static aerodynamic characteristics together with the downwash transient effects provided a high-fidelity model for the in-flight simulation.

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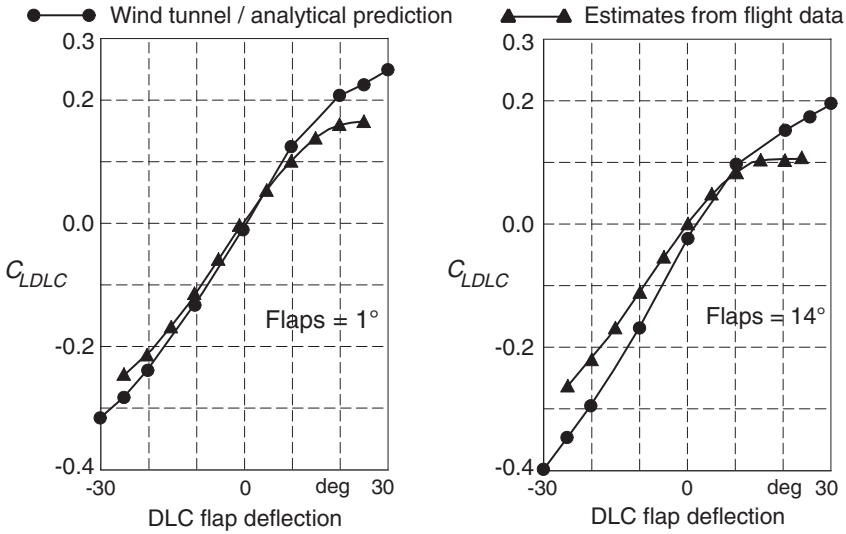

Fig. 12.8 DLC flaps effectiveness—static characteristics.

12.2.3 MODELING OF SPEED BRAKE EFFECT

The "Transall" C-160 is a military transport aircraft capable of carrying troops, casualties, freight, supplies, and vehicles, and serves the needs of the German Air Force; see Fig. 12.9. It is powered by twin turboprop Rolls Royce Tyne engines with four-bladed propellers. Each engine is capable of generating 70,000 N thrust. The maximum takeoff and landing weight is 51,000 kg, and

Fig. 12.9 Transall C-160.

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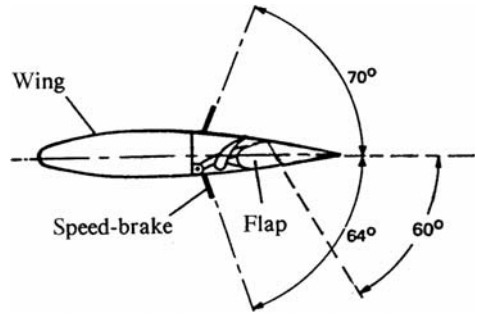

Fig. 12.10 Speed-brake and flap configuration.

the maximum cruising speed at an altitude of 16,000 ft is 277 kt (513 km/h). The cantilever high-wing has an area of 160.1 m^2 with a span of 40 m. The overall length is 32.4 m. The horizontal tail area is 43.8 m^2. Apart from the primary control surfaces, that is, elevator, rudder, and ailerons, the C-160 is equipped with hydraulically operated air-brakes mounted inboard on the wings and spoilers mounted outboard forward of the flaps on each wing; see Fig. 12.10. The spoilers are designed to improve the aileron effectiveness and are deflected proportionally for aileron deflections greater than 3 deg and less than 10 deg. For aileron deflections greater than 10 deg, the spoilers are deflected to the maximum limit of 45 deg. The underside of the upswept rear fuselage lowers to form a loading ramp. The tricycle landing gear is retractable. Each main unit comprises two pairs of wheels in tandem, and the twin-wheel nose unit is steerable.

As a part of a comprehensive data gathering and aerodynamic database development program for a training simulator [16–18], which will be addressed in Sec. 12.8, static and dynamic effects for all operational configurations including all primary and secondary control surfaces were modeled and estimated from flight data. In the present section, we concentrate on a single specific aspect, namely, speed brakes. The aerodynamic effects due to speed brakes are modeled as incremental changes in the lift, drag, and pitching-moment coefficients, that is, as ΔC_{LSB}, ΔC_{DSB}, ΔC_{mSB} where the subscript "SB" denotes speed brakes. Flight tests during which, starting from trimmed level flight, 40% speed brakes were applied are taken as typical cases for evaluation. Based on the aforementioned comprehensive aerodynamic model, the preceding incremental effects are identified separately for three landing flap positions, namely, $\delta_F = 0$, 30, and 60 deg. This identification yields the estimated responses shown by dashed lines in Fig. 12.11, which are compared with the flight measurements shown by solid lines, where a_x is the longitudinal acceleration, q the pitch rate, θ the pitch angle, α the angle of attack, and δ_{SB} the speed-brake deflection [18]. The braking effect is clearly observed from the reduction in the longitudinal acceleration a_x for $\delta_F = 0$ deg; see Fig. 12.11a. Variations in the pitching motion as seen from pitch rate q are negligible. In the other two cases of 30- and 60-deg flaps, Figs. 12.11b and 12.11c respectively, the braking effect is, of course, present, but not directly visible in a_x due to pitch-up. The speed brakes, as seen in Fig. 12.10, consist of lower and upper flaps on each wing and are deflected symmetrically. As such, they primarily work in a classical sense as a drag-inducing device.

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A closer look at the response match for $\delta_F = 30$ and 60 deg in Fig. 12.11, however, shows some discrepancies. A time lag between the flight measurements and the estimated responses is clearly to be seen, this time lag increasing with the increasing flap deflection. As in the preceding two cases presented in Secs. 12.2.1 and 12.2.2, these discrepancies are modeled as a downwash generated by the speed brakes. Accounting for the transit time effect, the angle of attack α_H at the tail is now given by [18]

$$\alpha_H = \alpha + i_H - \frac{\partial \varepsilon_H}{\partial \alpha}\,\alpha(t-\tau_\alpha) + \frac{\partial \varepsilon_H}{\partial C_S}\,C_S(t-\tau_{CS}) + \frac{\partial \varepsilon_H}{\partial \delta_{SB}}\,\delta_{SB}(t-\tau_\alpha) + \alpha_{dyn} \tag{12.12}$$

where α is the angle of attack at the wing, C_S the thrust coefficient, ε_H the downwash angle, and i_H and δ_{SB} have been defined earlier. τ_α and τ_{CS} represent the transit time for flow changes generated at the wing and at the propeller respectively to reach the tail. The downwash parameters $\partial\varepsilon_H/\partial\alpha$ and $\partial\varepsilon_H/\partial C_S$ are known from the foregoing step of basic aerodynamic model identification without speed brakes. The downwash parameter $\partial\varepsilon_H/\partial\delta_{SB}$ is required to be estimated from flight tests with speed brakes.

The model postulated in Eq. (12.12) yields the responses shown in Fig. 12.12. Compared to Fig. 12.11, the significantly improved match for the pitch rate q, and

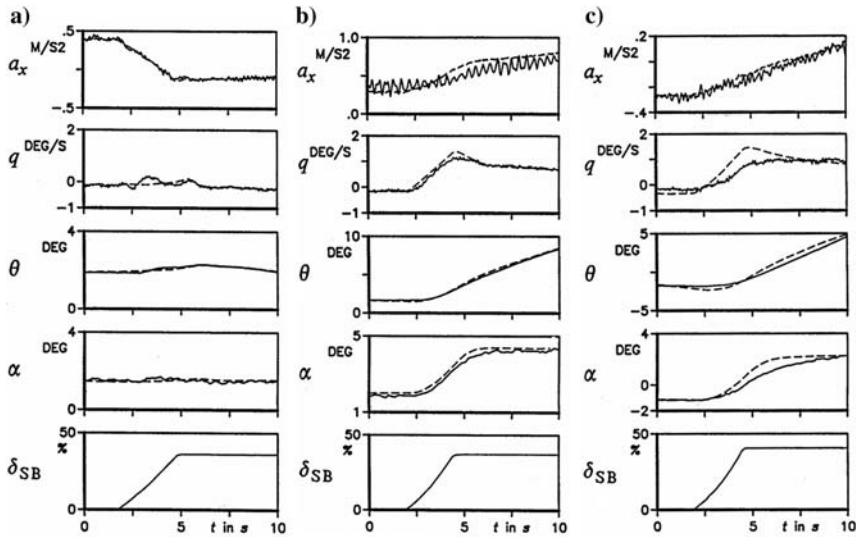

Fig. 12.11 Identification of aerodynamic effects caused by speed brakes: ——, flight measured; -------, estimated. Landing flap = a) 0, b) 30, and c) 60 deg.

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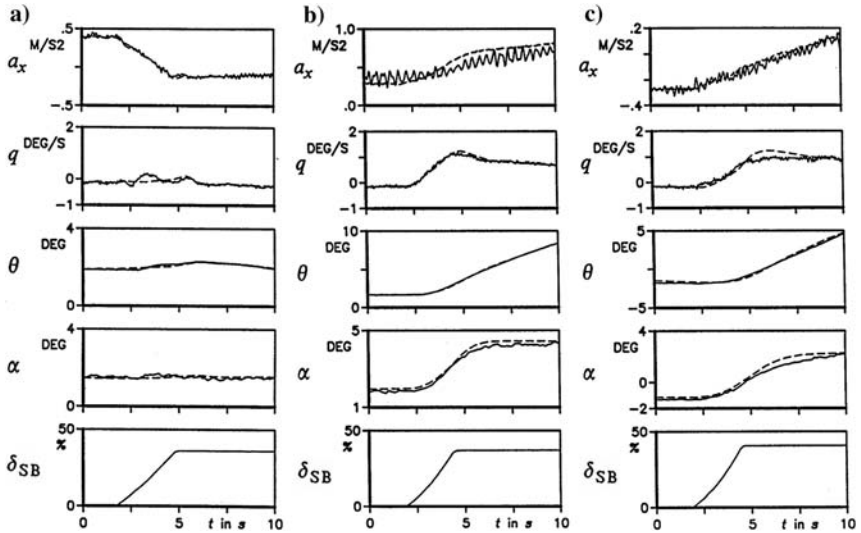

Fig. 12.12 Influence of downwash lag caused by speed brakes: ——, flight measured; -------, estimated. Landing flap = a) 0, b) 30, and c) 60 deg.

pitch attitude θ, particularly for the 30- and 60-deg landing flap, clearly demonstrates the appropriateness of the formulated model. The parameter $\partial\varepsilon_H/\partial\delta_{SB}$ was found to be zero for $\delta_F = 0$ deg, as expected from Fig. 12.11. The estimates for the other two flap positions were found to be near linearly dependent on the flap deflection.

To summarize, in this Sec. 12.2 we addressed a specific aerodynamic effect of downwash lag. The phenomenon of downwash is typical for conventional aircraft configurations with wing/body and horizontal tail. For high-fidelity databases, it is important to account for these effects induced through dynamic motion as well as through control deflections. Commonality of the chosen modeling technique to account for downwash resulting under different operating conditions and configurations has been presented.

12.3 AERODYNAMIC EFFECTS OF LANDING GEAR

In general, determination of primary derivatives from flight-test data does not pose any serious difficulties. For high-fidelity databases for flight simulators, however, it is necessary to properly model the aerodynamic effects of secondary order as well. This is particularly true when the model fidelity has to meet the Level D requirements [19, 20]. The ramp door and landing-gear effects, or the downwash due to speed brakes, are some typical examples of this class, the last one we have already addressed in the preceding section.

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To enable estimation of aerodynamic effects due to landing gear, flight maneuvers exciting the longitudinal and lateral-directional motion were carried out with extended landing gear [18]. The investigations covered the takeoff and landing configurations. As a typical example, Fig. 12.13 shows two flight maneuvers flown with extended landing gear; the first maneuver is a short-period motion excited by a multistep 3-2-1-1 input, and the second is an aileron and rudder input maneuver resulting in rolling and yawing motion. Once again following the general approach adopted in the preceding sections, modeling and identification of aerodynamic effects due to the landing gear is carried out stepwise. Proceeding from the basic aerodynamic model identified for the clean configuration, that is, neglecting the landing-gear effects, the flight maneuvers with extended landing gear are analyzed. This yields the response match shown in Fig. 12.13a, where a_x is the longitudinal acceleration, r is the yaw rate, θ is the pitch attitude, α is the angle of attack, and β is the angle of sideslip. The discrepancies in the longitudinal acceleration are demonstrative of additional drag being introduced. Note that for $t = 0$, the initial pitch attitude matches with the measurement, but not the longitudinal acceleration, which is an indicator that the mismatch in a_x is not due deviations in pitch attitude, but result directly from the increased drag. As a consequence of additional drag not yet accounted for in the model, this leads to deviation in the match for other variables including that for the pitch attitude. The lateral-directional

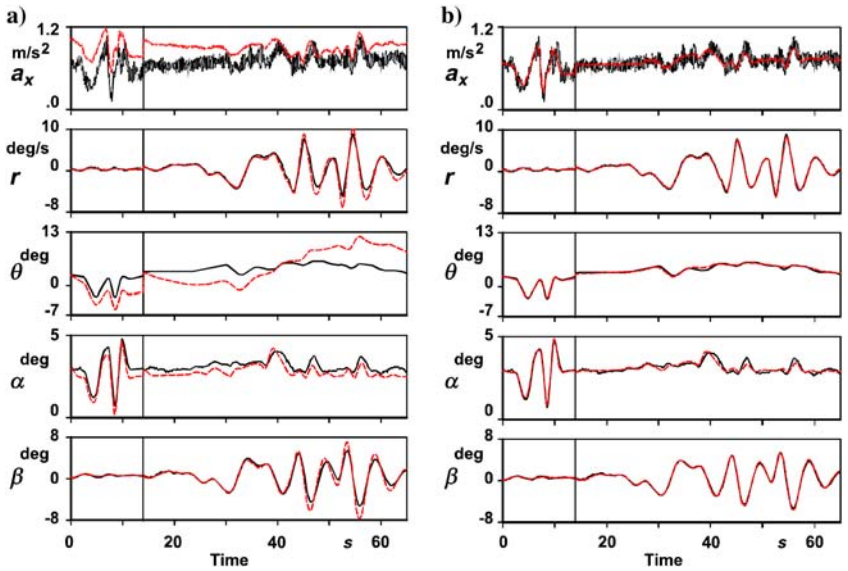

Fig. 12.13 Identification of landing-gear effects: ——, flight measured; -------, estimated; a) neglecting and b) accounting for landing-gear effects.

variables, angle of sideslip β and yawing rate r, also show some discernible differences.

Through a systematic approach and keeping in mind the principle of parsimony, additional parameters were identified to model the incremental effects due to the landing gear. The basic aerodynamic model is augmented with additional parameters, namely, incremental changes in the lift, drag, and pitching-moment coefficients, that is, ΔC_{LLG}, ΔC_{DLG}, and ΔC_{mLG}, with the subscript "LG" denoting the landing gear. Furthermore, the derivatives $C_{n\beta\mathrm{LG}}$ and $C_{Y\beta\mathrm{LG}}$ were introduced to model the influence on the lateral-directional motion. As expected, the identified value of ΔC_{DLG} suggests increased drag. The positive value $C_{\mathrm{n}\beta\mathrm{LG}}$ indicates increased weathercock stability. The two derivatives $C_{n\beta\mathrm{LG}}$ and $C_{Y\beta\mathrm{LG}}$ are adequate to model the influence of the landing gear on the lateral-directional motion. Attempts to estimate increased yaw damping due to landing gear did not lead to any improvements in the response match whereas a destabilizing effect of the landing gear on the pitching motion, as indicated by a positive value of the pitch damping derivative C_{mqLG}, is identified. Estimation of the aforementioned aerodynamic derivatives from the maneuvers with extended landing gear posed no difficulties. They were estimated with very low standard deviations. The resulting model yields the response match shown in Fig. 12.13b, which is significantly better compared to that in Fig. 12.13a. The landing-gear effects are typically in the range of 5 to 10% of the total, but still important for simulator fidelity. Proper test techniques and a systematic modeling approach enable accurate identification of such aerodynamic effects of secondary order [21].

12.4 CONTROL SURFACE MALFUNCTION EFFECTS

Pilot training under abnormal conditions is one important application area of flight simulators. Extreme flight conditions can arise due to several reasons, for example, flow separation at high angles of attack leading to stall (which we will address in the next section), due to asymmetric flight with large sideslip angles or due to control surface malfunctions leading to partial loss of control power. Modeling of aircraft behavior under such extreme conditions is complex, and validation from flight tests is particularly difficult because such configurations are difficult to fly and are safety critical involving some risk. Here, we consider the specific case of roll spoiler hardover, tested on a Dornier 328 for the landing configuration of 32-deg flaps. Starting from a horizontal level flight at an altitude of 8700 ft and 115 kt true airspeed (roughly 100 kt CAS) and trim angle of attack of about 2.5 deg with extended landing gear, the roll spoilers are drawn out on the left wing to the hardover limit of 75 deg and mechanically held fixed. Under normal operating conditions the roll spoilers are coupled to the ailerons; they are deflected from 0 to 45 deg linearly for aileron deflections between 3 and 25 deg; for aileron deflections less than 3 deg, the roll spoilers are not active.

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As common to the general procedure followed in the aerodynamic model development, first we simulate the aircraft response to flight maneuver inputs using a database that was developed from flight maneuvers under normal operating conditions, that is, in which aileron and roll spoilers are coupled and deflected proportionately as just described. The comprehensive aerodynamic model included angle of attack and thrust dependencies, as well as nonlinearities in the aileron and roll spoiler effectiveness for deflections up to 25 and 45 deg, respectively. For example, the contributions to the rolling- and yawing-moment coefficients due to roll spoiler are given by

$$\begin{aligned}\Delta C_{\ell_\mathrm{RS}} &= \left(C_{\ell\mathrm{RS}} + C_{\ell\mathrm{RS3}}\ \delta_{\mathrm{RS}}^2 + C_{\ell\mathrm{RS}\alpha}\ \alpha + C_{\ell\mathrm{RSCS}}\ C_S\right)\delta_{\mathrm{RS}} \\ \Delta C_{n_\mathrm{RS}} &= \left(C_{n\mathrm{RS}} + C_{n\mathrm{RS}\alpha}\ \alpha + C_{n\mathrm{RSCS}}\ C_S\right)\delta_{\mathrm{RS}}\end{aligned} \tag{12.13}$$

where δ_{RS} is the roll spoiler deflection, C_S the thrust coefficient, and $C_{\ell()}$ and $C_{n()}$ the respective derivatives. The left and right spoilers are treated separately.

Applying such a model to the roll spoiler hardover maneuver leads to the simulated responses shown by dashed lines in Fig. 12.14, showing the translational and rotational accelerations on the left-hand side and the flow angles, bank angle, aileron, spoiler, and rudder deflections on the right-hand side. From the second plot from bottom on the right-hand side showing the roll spoiler deflection, we

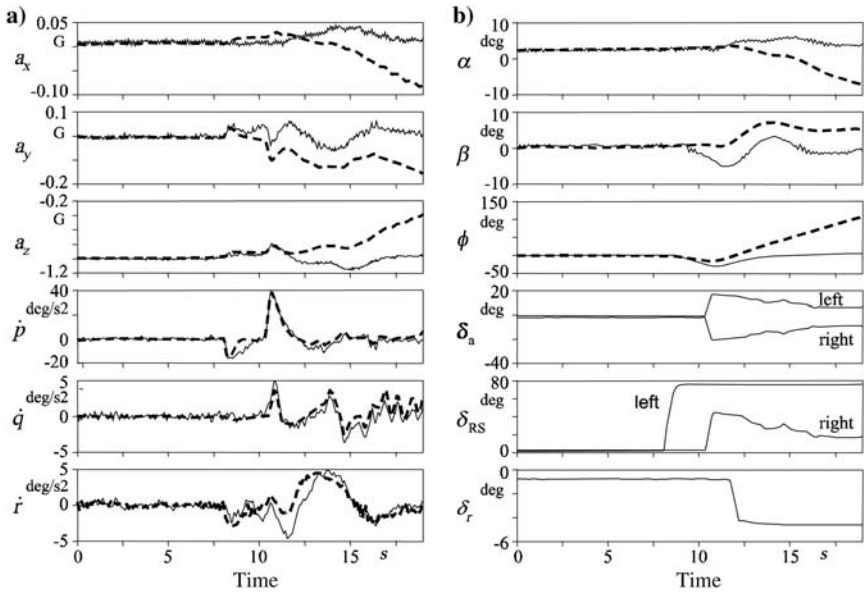

Fig. 12.14 Simulation of roll spoiler hardover using basic model: ——, flight measured; -------, model predicted.

notice that after 8 s the left roll spoiler is jammed at the maximum limit of 75 deg. We observe that both the left and right roll spoilers are deflected, which is in contrast to the normal conditions under which depending upon the aileron deflection only one of the roll spoilers either on the left or right is deflected. The simulated aircraft response in Fig. 12.14 shows drastic deviations compared to the flight measured data shown by continuous lines. The trim flight prior to the control surface jam is well simulated without showing any tendency to drift or misbehavior otherwise, but after the malfunction starts at 8 s the simulated responses are unable to predict the aircraft behavior.

The roll spoiler hardover affects primarily the rolling and yawing motion and leads to increased drag. Through a systematic approach, the discrepancies observed in Fig. 12.14 are traced to strong nonlinearities in the roll spoiler effectiveness. The comprehensive model is now updated through simple correction terms for roll spoiler deflections greater than 45 deg. The contributions to the rolling- and yawing-moment coefficients due to roll spoiler in Eq. (12.13) are now augmented with corrections given by

$$\Delta C_{\ell_\text{RSNL}} = \begin{cases} C_{\ell\text{RSNL}}\,(\delta_{\text{RS}} - \delta_{\text{RSBP}}), & \text{for } \delta_{\text{RS}} > \delta_{\text{RSBP}} \\ 0, & \text{for } \delta_{\text{RS}} \leq \delta_{\text{RSBP}} \end{cases}$$
$$\Delta C_{n_\text{RSNL}} = \begin{cases} C_{n\text{RSNL}}\,(\delta_{\text{RS}} - \delta_{\text{RSBP}}), & \text{for } \delta_{\text{RS}} > \delta_{\text{RSBP}} \\ 0, & \text{for } \delta_{\text{RS}} \leq \delta_{\text{RSBP}} \end{cases} \tag{12.14}$$

In addition, it was found that the drag and pitching-moment coefficients had to be modified as follows to include the effect due to roll spoiler hardover.

$$\Delta C_{D_\text{RSNL}} = \begin{cases} C_{\text{DRSNL}}\,(\delta_{\text{RS}} - \delta_{\text{RSBP}}), & \text{for } \delta_{\text{RS}} > \delta_{\text{RSBP}} \\ 0, & \text{for } \delta_{\text{RS}} \leq \delta_{\text{RSBP}} \end{cases}$$
$$\Delta C_{m_\text{RSNL}} = \begin{cases} C_{m\text{RSNL}}\,(\delta_{\text{RS}} - \delta_{\text{RSBP}}), & \text{for } \delta_{\text{RS}} > \delta_{\text{RSBP}} \\ 0, & \text{for } \delta_{\text{RS}} \leq \delta_{\text{RSBP}} \end{cases} \tag{12.15}$$

where the equivalent roll spoiler deflection $\delta_{\text{RS}} = \delta_{\text{RS}}^{R} + \delta_{\text{RS}}^{L}$ is the sum of the right and left roll spoiler deflections (they come out on one side only, deflected upwards and measured positive); δ_{RSBP} is the so-called breakpoint; and $C_{\ell\text{RSNL}}$, $C_{n\text{RSNL}}$, C_{DRSNL}, $C_{m\text{RSNL}}$ are the linear derivatives representing corrections for roll spoiler deflections greater δ_{RSBP}, taken to be 45 deg. The choice of 45 deg as breakpoint is based on the fact that the basic model was estimated and validated from flight maneuvers covering deflections up to this range.

Estimation of the four additional parameters without affecting the performance of the basic model for deflections less than δ_{RSBP} (that is, under normal operating conditions) leads to the results shown in Fig. 12.15. Compared to Fig. 12.14, it is now observed that the model-computed responses for all of the variables match the flight measured data very well. Some minor deviations in the match

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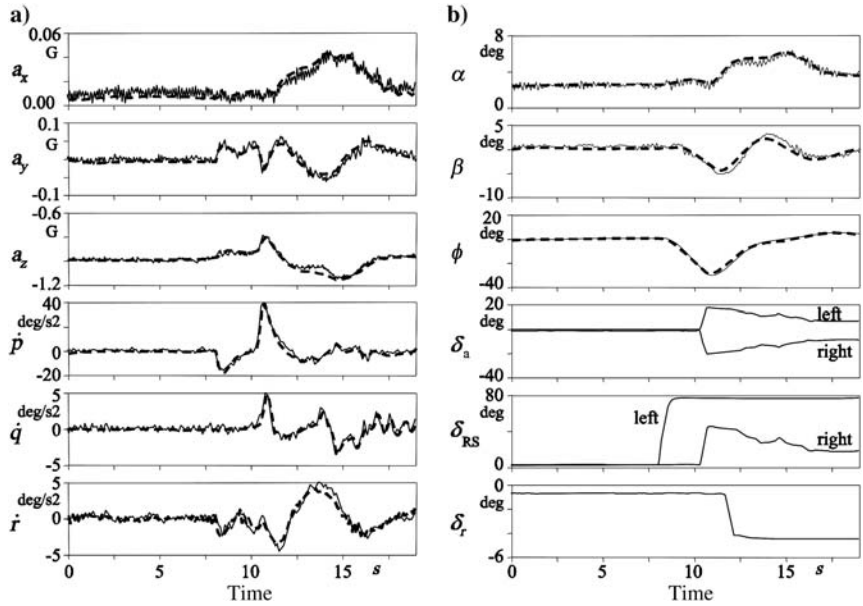

Fig. 12.15 Identification of roll spoiler hardover: ——, flight measured; --------, model predicted.

for angle of sideslip β and yaw acceleration $\dot{r}$ are observed, which are attributed to the simplified linear corrections for the highly complex flow phenomenon under such extreme flight conditions. A careful look at the match for the yaw acceleration $\dot{r}$ suggests that other effects, possibly sidewash leading to a delayed response, need to be accounted for. Nevertheless, simple model extensions in the rolling, yawing, and drag coefficients and parameter estimation from flight data were adequate to capture the basic aerodynamic phenomenon of roll spoiler hardover. In the majority of the cases, the correct trends in motion in all directions and a reasonable match between model prediction and flight measurements is usually sufficient for simulator pilot training under such abnormal conditions.

12.5 UNSTEADY AERODYNAMICS MODELING

Under stationary attached flow conditions, aerodynamic effects can be adequately described using time-invariant parameters and linear models. At higher angles of attack and aircraft undergoing stall, models are highly nonlinear because flow separation and unsteady effects are dominant. Flow separation results primarily in a reduction of the lift-curve slope before the maximum lift point whereas a significant loss of lift occurs in the poststall region. At such flight conditions the generated lift is influenced strongly by unsteady effects. Different types of flow

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separation occur depending upon the airfoil shape or wing configuration. The most common type is the trailing-edge separation, which is typical for most conventional aircraft having a turbulent boundary layer.

Unsteady aerodynamics has been a subject of extensive investigations using computational-fluid-dynamic methods, wind-tunnel tests, and semi-empirical models. The common approach has been based on indicial functions [22–26]. Although such models provide a basis for analytical investigations of the complex flow phenomena, postulating them in analytical form suitable for parameter estimation is difficult. Therefore, we follow the alternative approach of [27, 28] to describe analytically the flow separation including stall hysteresis as a function of an internal state variable. Because the approach retains the state-space formulation, it is directly amenable to identification and validation from flight data, as has been validated on flight data in [29]. For a review of unsteady aerodynamic modeling, the reader is referred to [30].

In this section we draw heavily on the mathematical developments presented in [28] and on the applications to flight data in [29, 31]. Trailing-edge stall is a dominating phenomenon on wings having sufficient thickness ($t/\bar{c} > 0.15$, where t represents wing thickness and $\bar{c}$ the wing mean aerodynamic chord), and undergoing a relatively slow variation in the angle of attack ($\dot{\alpha} < 0.02V/\bar{c}$, where $\dot{\alpha}$ is the rate of change of angle of attack and V is the freestream velocity). For such a case, a nondimensional state X is introduced, describing the instantaneous location of an idealized flow separation point along the chord on the upper surface of the wing ($0 \leq X \leq 1$) (see Fig. 12.16); $X = 1$, and $X = 0$ correspond to attached and fully separated flow, respectively. Based on Kirchhoff's theory of flow separation, the wing lift can be modeled as a function of angle of attack α and flow separation point X:

$$C_L(\alpha, X) = C_{L\alpha}\left\{\frac{1+\sqrt{X}}{2}\right\}^2 \alpha \tag{12.16}$$

where $C_{L\alpha}$ is the lift-curve slope.

The position of the flow separation point can be conveniently described using a single ordinary differential equation:

$$\tau_1 \frac{dX}{dt} + X = X_0(\alpha, \tau_2\dot{\alpha}) \tag{12.17}$$

where τ_1 and τ_2 are the time constants representing the transient and the quasi-steady aerodynamic

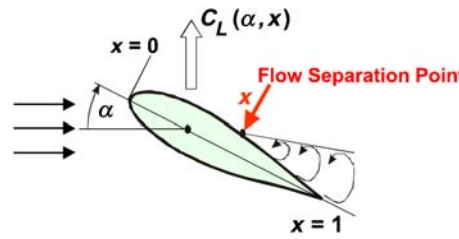

Fig. 12.16 Trailing-edge flow separation phenomenon.

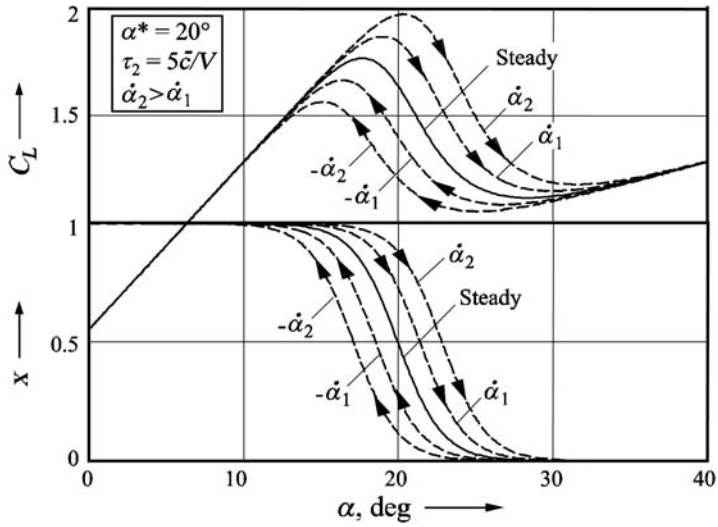

Fig. 12.17 Quasi-steady lift coefficient and flow separation point.

effects, respectively, and X_0 the separation point under steady conditions, which is, as shown on the right-hand side, a nonlinear function of α, $\dot{\alpha}$, and τ_2. To bring out the hysteresis effect resulting from τ_2, for typical values, we plot in Fig. 12.17 the flow separation point $X(\alpha, \tau_2\dot{\alpha})$ and the corresponding lift coefficient $C_L(\alpha, X(\alpha, \tau_2\dot{\alpha}))$. The solid lines correspond to steady conditions of $\dot{\alpha} = 0$, and the dashed lines show X and C_L for different but constant rates $\dot{\alpha}$. For a positive $\dot{\alpha}$, flow separation occurs at higher α whereas for negative $\dot{\alpha}$ the flow reattachment is delayed to α smaller than that of the steady case. The resulting delay (hysteresis width) is proportional to $\dot{\alpha}$ and $\tau_2\bar{c}/V$. We obtain an appreciation of the influence of these parameters by varying them and running the program /FVSysID2/chapter12/HystTau2.m.

The steady flow-separation point X_0 depends upon the airfoil and wing configuration. Using Eq. (12.16) with $X = X_0$, the function can be determined statically in a wind tunnel. An approximation based on two exponential functions, which can be adapted with several parameters, is used in [32]. We use here an alternative approximation based on hyperbolic-tangent investigated in [29, 31].

$$X_0 = \frac{1}{2}\{1 - \tanh(a_1\ (\alpha - \alpha^*))\} \tag{12.18}$$

where a_1 defines the static stall characteristics of the airfoil and α^* the breakpoint corresponding to $X_0 = 0.5$. This approximation is better suited for parameter estimation purposes because it is a continuous function in its entire range and has just two unknown parameters, namely, a_1 and α^*. The ease of adapting Eq. (12.18) to different airfoils and configurations is apparent from Fig. 12.18, showing X for

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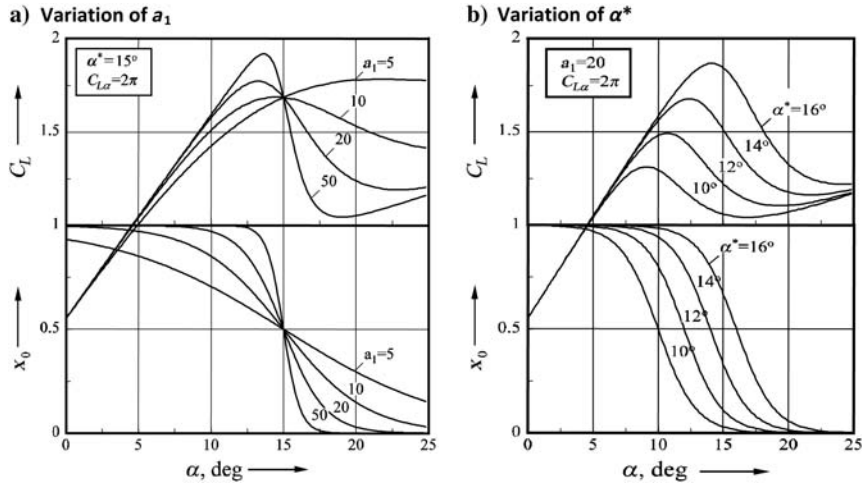

Fig. 12.18 Behavior of approximation function.

various values of a_1 for a fixed value of $\alpha^* = 15$ deg in Fig. 12.18a and for various values of α^* for a fixed value of $a_1 = 20$ in Fig. 12.18b. As seen from this figure, it is obvious that the function represented in Eq. (12.18) has its turning point at α^*. Varying α^* would move the function, that is, X_0, horizontally. The lift computed from Eq. (12.16) using $X = X_0$ is also shown in the same figure.

The applicability of the proposed hyperbolic-tangent function to approximate the steady flow separation is demonstrated on wind-tunnel measurements on an OA213 rotor airfoil. Because the flow separation point was not measured in the wind-tunnel tests, it was computed point wise using the inverted Kirchoff's formulation of flow separated lift, Eq. (12.16), extended by C_{L0} for a nonsymmetrical profile, and solving for X, that is, $X_0 = \{2\sqrt{[(C_L - C_{L0})/(C_{L\alpha}\alpha)]} - 1\}^2$. Figure 12.19 shows the wind-tunnel derived separation points and the lift curve. It also shows the approximation of the wind-tunnel data using Eqs. (12.19) and (12.18). The two unknown parameters in Eq. (12.18) turn out to be $\alpha^* = 15.2$ deg and $a_1 = 17.8$. From the plot of X_0, it is observed that flow separation starts at about 10-deg angle of attack, and it is fully developed at about 20 deg. The stall characteristic is quite symmetrical with $X_0 = 0.5$ to the turning point α^*, which is represented by the hyperbolic-tangent function in Eq. (12.18).

Now, combining Eqs. (12.17) and (12.18), we arrive at the general representation of unsteady flow, characterizing the transient and quasi-steady effects:

$$\tau_1 \frac{dX}{dt} + X = \frac{1}{2}\{1 - \tanh(a_1\,(\alpha - \tau_2\dot{\alpha} - \alpha^*))\} \tag{12.19}$$

Successful estimation of the four parameters a_1, α^*, τ_1, and τ_2 appearing in Eq. (12.19) from dynamic stall maneuvers with a Transall C-160 has been

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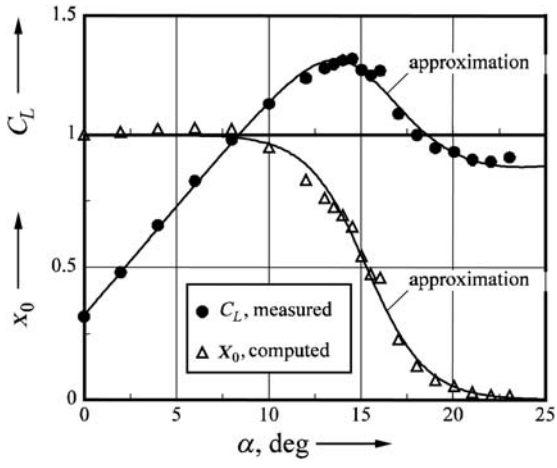

Fig. 12.19 Approximation of steady flow separation point.

reported in [29]. A sample result in Fig. 12.20, taken from the same reference, shows a cross plot of C_L and X vs wing angle of attack during the dynamic stall portion. The plot shows the identified, steady $C_{L,st}(\alpha)$ and $X_0(\alpha)$, as well as the unsteady model outputs $C_L(t, \alpha, \dot{\alpha})$ and $X(t, \alpha, \dot{\alpha})$. We reach the following two conclusions from this figure: 1) there is a dynamic lift overshoot of about 10% above $C_{L\,\max}$, and 2) the angle of attack reaches into the poststall region. The delaying effect of τ_1 can be inferred from the fact that at the maximum angle of

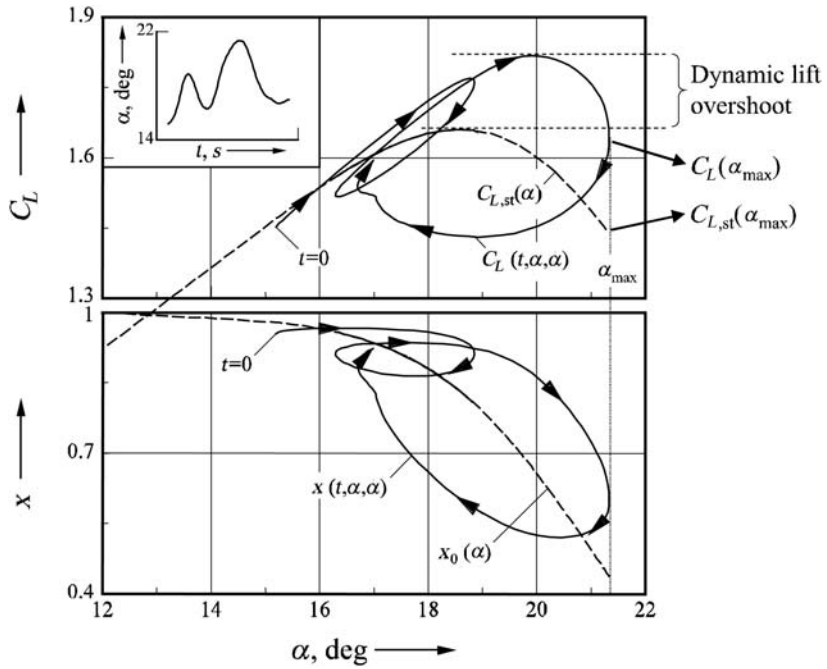

Fig. 12.20 Estimated lift coefficient and flow separation point during C-160 dynamic stall.

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attack of about 21.4 deg, the rate of change of angle of attack is zero, but $C_L(\alpha_{\max})$ is not the same as $C_{L,st}(\alpha_{\max})$ under static conditions.

12.6 QUASI-STEADY STALL MODELING

Although Eq. (12.19) provides a model characterizing both the transient and quasi-steady stall characteristics in terms of four parameters, namely, a_1, α^*, τ_1, and τ_2, as common to any exercise, estimation of these parameters requires appropriate flight maneuvers containing adequate information necessary to estimate each parameter separately. Determination of both the time constants τ_1 and τ_2 requires highly dynamic stall maneuvers.

In many cases, however, a simplified approach accounting for quasi-steady stall characteristics is quite adequate and recommended. For example, in the case of real-time flight simulators, the primary goal is usually to train pilots for correct recovery procedures from quasi-steady stall conditions. Elimination of an additional state equation also leads to minor reduction in overall computational burden [31]. Furthermore, flight data with dynamic stall are more difficult and risky to gather whereas quasi-steady stalls can be performed more readily. Flight data with quasi-steady stall would enable estimation of the hysteresis time constant τ_2 only. Accordingly, in the following investigation, we neglect the transient effects by setting τ_1 to zero, which eliminates the need for differential equation and simplifies Eq. (12.19) to

$$X = \frac{1}{2}\{1 - \tanh[a_1\ (\alpha - \tau_2\,\dot{\alpha}\ - \alpha^*)]\} \tag{12.20}$$

The three parameters a_1 (airfoil static stall characteristics), τ_2 (time constant), and α^* (break point) are completely adequate to model the stall hysteresis.

Thus, the total lift given by Eq. (12.16), which is repeated here, and considering simplified expressions for the purpose of demonstration, the total drag and pitching-moment coefficients are modeled as

$$C_L(\alpha, X) = C_{L0} + C_{L\alpha}\left\{\frac{1+\sqrt{X}}{2}\right\}^2 \alpha \tag{12.21}$$

$$C_D = C_{D0} + \frac{1}{e\pi\, AR}\ C_L^2(\alpha, X) + \frac{\partial C_D}{\partial X}(1 - X) \tag{12.22}$$

$$C_m = C_{m0} + C_{m\alpha}\ \alpha + C_{mq}\ \frac{q\bar{c}}{V} + C_{m\delta e}\ \delta_e + \frac{\partial C_m}{\partial X}(1 - X) \tag{12.23}$$

where AR is the wing aspect ratio, e the Oswald factor, and δ_e the elevator deflection. The major contribution to the unsteady drag comes from the drag polar using $C_L\ (\alpha,\ X)$, that is, from the lift modified due to flow separation. Any

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additional effects are accounted for through an empirical correction term $\partial C_D/\partial X$. The parameter $\partial C_m/\partial X$ models the hysteresis effect in the pitching moment. It is once again pointed out that Eqs. (12.21–12.23) are simplified expressions, which neglect additional terms accounting for contributions due to other variables and control deflections. They have been omitted in these expressions for the sake of brevity, but included in the model used for actual parameter estimation from flight data.

The parameters of the unsteady aerodynamic model for flow separation and stall were identified from flight data for four different aircraft, namely, 1) ATTAS, 2) Dornier 328, 3) AT-26 Xavante, and 4) Transall C-160. We present here the application of the software developed in Chapter 4 to the first case of ATTAS aircraft. This test case can be run and results verified using the models and recorded data during a quasi-stall maneuver supplemented to the book. For the second and third example, we just provide typical results. In the fourth case of Transall C-160, the basic quasi-steady longitudinal stall model has been extended to asymmetric stall behavior observed in flight.

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12.6.1 ATTAS STALL MODELING

In this section we return to the example considered in Sec. 8.7.3, namely, modeling of quasi-steady stall. Here, we apply the advanced modeling concepts based on the physics of the unsteady aerodynamics as elaborated in the foregoing section. The case considered pertains to flight data gathered with the test aircraft VFW-614 ATTAS (see Fig. 12.3) at an altitude of 16,000 ft and for clean configuration undergoing quasi-steady stall. Both output-error method and regression analysis are applied to estimate parameters characterizing the stall characteristics. The results obtained applying the output-error method are found in [29], in which the six-degrees-of-freedom (DOF) equations of motion combined with the aerodynamic model including stall model as postulated in Eqs. (12.20–12.23) has been used to match the recorded motion variables such as acceleration, angular rates, and attitude angles. We present here the application of the regression analysis, that is, least-squares method, to this case and restrict ourselves to longitudinal motion only.

Because the postulated model, Eqs. (12.20–12.23), is nonlinear in parameters, it is imperative that we can obtain the LS estimates only iteratively, that is, by applying suitably the OEM software/FVSysID2/chapter04/ml_oem2. We designate this case as test_case = 27. As in the case of Secs. 6.9.2 and 8.7.3, the first step consists of data preprocessing to compute the aerodynamic force and moment coefficients. These precomputations are performed in the function "umr_reg_attas.m." Because it is the same function that was used in the other two cases, we do not repeat the details here, but can be easily traced from the said function. The recorded flight data from two stall maneuvers (data files \FVSysID2\flt_data\ fAttas_qst01.asc and fAttas_qst02.asc) are analyzed. The time-segment-related information Nzi and izhf is defined by following a procedure similar to that described in Sec. 8.7.3.

Having derived the aerodynamic force and moment coefficients from the relevant measured data, we now define the input–output variables for parameter estimation based on the postulated model of Eqs. (12.20–12.23) as follows:

	No. of Variables		Function Name
States	0	—	—
Outputs	3	C_D, C_L, C_m	"obs_TC27_attas_regStall"
Inputs	14	δ_e, δ_a, δ_r, p, q, r, α, β, V, ρ, F_{eL}, F_{eR}, $\dot{\alpha}$, Ma	

Not all signals from the input array Uinp are required in the present case, but it is made oversized for the sake of similarity with other cases.

The 14 unknown derivatives C_{D0}, e, C_{L0}, $C_{L\alpha}$, $C_{L\alpha Ma}$, C_{m0}, $C_{m\alpha}$, C_{mq}, $C_{m\delta e}$, a_1, $\alpha*$, τ_2, $\partial C_D/\partial X$, and $\partial C_m/\partial X$ appearing in Eqs. (12.20–12.23) are to be estimated. The contribution of the elevator deflection to the lift is indirectly accounted for through the term $(-C_{m\delta e}{}^{*}\bar{c}/r_H)$. Because we apply the LS method, there are no state equations, but for the reasons elaborated in Sec. 6.9.2, we either use the same dummy function "xdot_attas_reg" or define the function name for state equations as an empty string. The right-hand sides of the model, that is, Eqs. (12.20–12.23), in terms of the just-mentioned 14 parameters are programmed in the function "obs_TC27_attas_regStall." Thus, the model definition provided by the function "mDefCase27.m" for the software program "/FVSysID2/chapter04/ml_oem2" is as follows:

```
test_case = 27;                          % Test case number
state_eq  = [ ];                         % Empty string for state equation
obser_eq  = "obs_TC27_attas_regStall";   % Function for observation
                                           equations
Nx        = 0;                           % Number of states
Ny        = 3;                           % Number of observation variables
Nu        = 14;                          % Number of input variables
Nparam    = 14;                          % Total number of parameters
dt        = 0.04;                        % Sampling time
```

For the reasons mentioned in Sec. 6.9.3 dealing with the estimation of a nonlinear lift and drag model, we specify some reasonable starting values for the parameters, which are found in the model definition "mDefCase27.m" for this example. The estimation is started by calling "/FVSysID2/chapter04/ml_oem2." The final estimation results are provided in Figs. 12.21 and 12.22. From the crossplot of lift coefficient vs angle of attack in Fig. 12.21, it is observed that the estimated hysteresis loop (shown in dashed lines) matches fairly well with the flight data (shown in continuous lines). The data contain a considerable amount of noise resulting from buffet, but the estimates are still reliable in the presence of noise. In Fig. 12.22, the three plots from the top show the match

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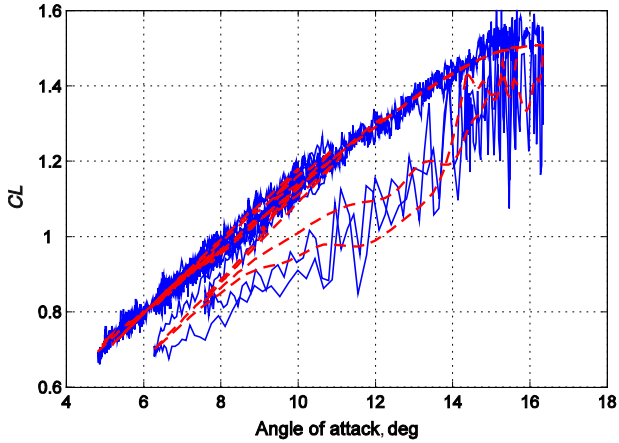

Fig. 12.21 ATTAS stall hysteresis: ——, flight derived; - - - -, model estimated.

between the flight measured and model estimated aerodynamic coefficients. The match for the pitching-moment coefficient shows some discrepancies in Fig. 12.22, which are attributed to the simplified linear model used in the regression. A more complex model incorporating angle-of-attack dependent elevator effectiveness provided a significantly improved match than that observed here. The three parameters modeling the stall characteristics are estimated to be $a_1 = 23.72$, $\alpha^* = 0.3087$ rad ($\approx$17.7 deg), and $\tau_2 = 24.02\ \bar{c}/V$.

12.6.2 DORNIER 328 STALL MODELING

The Dornier 328 is a high-wing, high-tail, twin turboprop regional transport aircraft designed to carry up to 33 passengers; see Fig. 12.23. It is powered by two Pratt & Whitney Canada PW 119A turboprop engines with six-bladed constant-speed propellers, each capable of generating up to 2180 SHP. The aircraft has an overall length of 69 ft 10 in., wingspan of 68 ft 10 in., mean aerodynamic chord of 6 ft 8 in., wing area of 430.4 ft^2, and horizontal tail area of 97.2 ft^2. The maximum takeoff and landing weight is 27,558 lb, maximum rate of climb is 2420 ft/min, and the maximum cruising speed is 335 KTAS. It has a maximum range of 700 n miles with 30 passengers onboard. In addition to high-speed capability at all altitudes, the Dornier 328 has a low final approach and landing speed providing short takeoff and landing capability. As will be discussed in Sec. 12.8, a comprehensive database for a training simulator was developed from flight-test data applying system identification. Modeling of quasi-steady stall characteristics from flight data was a part of this comprehensive database development program [33].

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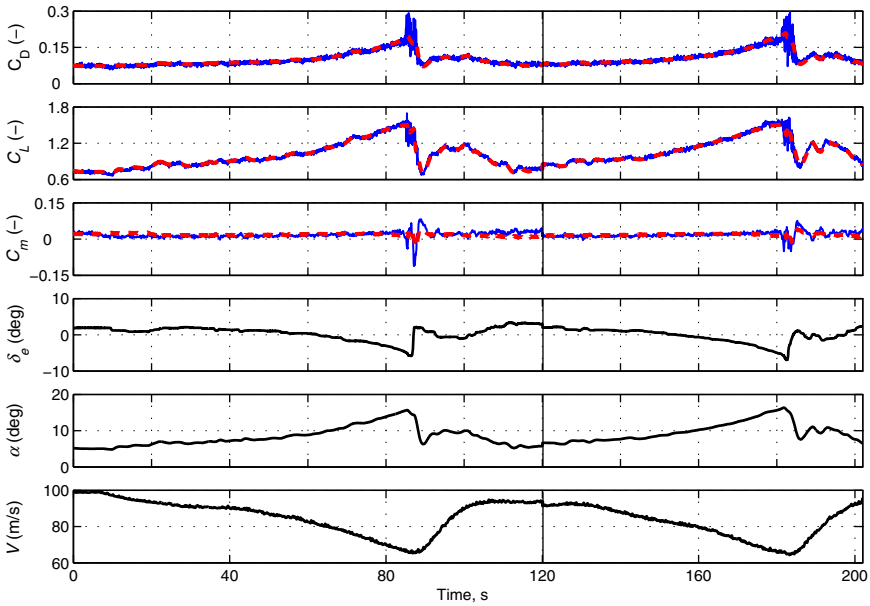

Fig. 12.22 Time histories of quasi-steady stall: ——, flight measured; - - - - -, model estimated.

A model similar to that in Eqs. (12.20–12.23) has been applied to identify the stall characteristics of the Dornier 328 aircraft flying a stall maneuver with quasi-steady approach [4, 33]. All parameters of the basic aerodynamic model estimated in a foregoing step were kept fixed, estimating only the three parameters

Fig. 12.23 Dornier 328 regional transport aircraft.

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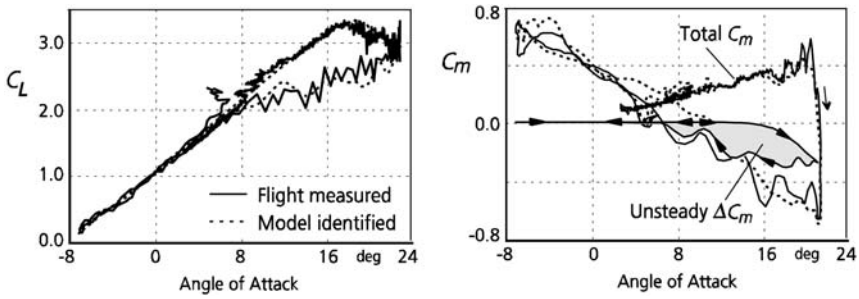

Fig. 12.24 Modeling of Dornier 328 stall hysteresis.

of the stall model. These three parameters are determined to $a_1 = 15.0$, $\alpha^* = 21.8$ deg, and $\tau_2 = 18.5\,\bar{c}/V$. Figure 12.24 shows a comparison of the total lift and pitching-moment coefficients derived from flight-measured data (shown by solid lines) with those estimated using system identification (shown by dashed lines) [4]. The increment in pitching moment due to unsteady aerodynamics, leading to hysteresis, is plotted separately in Fig. 12.24 (shaded area). The time histories of important parameters are compared in Fig. 12.25.

The efficacy of the analytical model postulate is evident from the hysteresis in the lift and pitching-moment coefficients. It is obvious from these figures that there are strong hysteresis effects during the pitch-down part of the stall. The estimated model reproduces the hysteresis quite satisfactorily; only during the recovery phase some small discrepancies are observed.

12.6.3 EMBRAER AT-26 XAVANTE STALL MODELING

The Embraer AT-26 Xavante is a fighter jet trainer, which was in service of the Brazilian Air Force for the last four decades; see Fig. 12.26. It has an overall length of 10.67 m, wingspan of 10.85 m, and wing area of 19.3 m^2. The empty weight is about 6300 lbf, the MTO weight 11,500 lbf, and the maximum operational speed of Mach 0.80 or 470 KCAS (whichever is reached first). It is powered by the Rolls-Royce Viper Mk 540 turbojet, rated at a maximum static thrust of 3400 lbf at sea level.

The AT-26 Xavante was used extensively in the past for flight-test training activities at the Brazilian Air Force Test Pilot School (Portuguese: *Divisão de Formação em Ensaios em Voo*, EFEV), which is part of the Flight Test and Research Institute (IPEV) of the Aerospace Science and Technology Department (Portuguese: *Departamento de Ciência e Tecnologia Aeroespacial*; DCTA). Several flight-test campaigns were performed with AT-26 Xavante equipped with dedicated flight-test instrumentation. We focus here only on stall maneuvers, which were performed in several configurations of flaps, engine thrust, landing gear, and aerodynamic speed brakes. As in the examples presented earlier in this section, the

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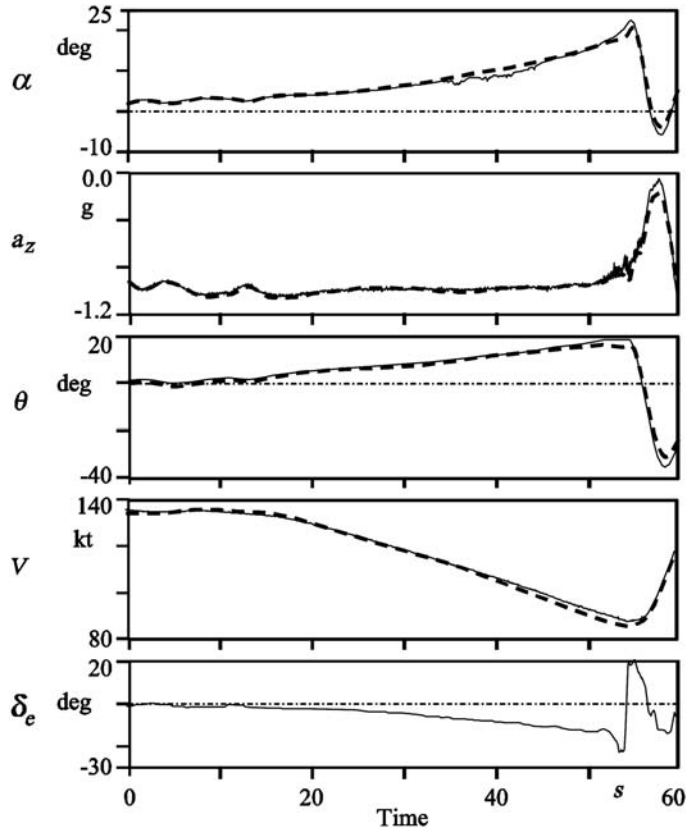

Fig. 12.25 Dornier 328 stall hysteresis time histories: ——, flight measured; - - - - -, model estimated.

stall maneuvers consisted of stabilizing the aircraft at 1.3 Vs, then reducing continuously the airspeed using the longitudinal control only and aileron inputs as required to keep wings leveled. As a typical example, we present here the results for the clean configuration (flaps up, landing gear up, aerodynamic brakes in) and engine set in idle [34].

Applying the quasi-steady model postulated in Eqs. (12.20–12.23), the aerodynamic parameters were estimated from a stall maneuver starting from reference airspeed of 80 m/s with angle of attack of about 10 deg at an altitude of 4200 m. The output variables a_x, a_z, α, q, θ were matched applying the output-error method discussed in Chapter 4. The three parameters pertaining to the stall model were determined to be $a_1 = 12.82$, $\alpha^* = 18.33$ deg, and $\tau_2 = 15.11\,\bar{c}/V$.

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Figure 12.27 shows a comparison of the time histories of flight-derived aerodynamic force coefficients C_L, C_D and pitching-moment coefficient C_m (shown by solid lines) with the model estimated force and moment coefficients (shown by dashed lines). The match between the flight-derived and model-estimated coefficients is found to be good for the complete stall maneuver. The cross plots of the lift coefficient and pitching-moment coefficients as function of angle of attack and the cross plot $C_L - C_D$ are shown in Fig. 12.28. The estimated separation point X is plotted as a function of time in the same figure. It is observed that the flow separation is more than 50% during the stall. The stall hysteresis is clearly observed from the cross plot of lift vs angle of attack in Fig. 12.28; the model estimated stall hysteresis matches fairly well the flight data.

The three examples—ATTAS, Dornier 328, and AT-26 Xavante—demonstrate wide applicability of the quasi-steady stall model discussed in this section to different aircraft configurations. This is even more interesting, given the simplicity of the model. In each case the model yields quite a good match for predicted model response with the flight-measured data, whereby just three parameters are sufficient to characterize the stall mechanism during quasi-steady stall maneuver. Further investigations on AT-26 also indicate that the same stall model structure is applicable for high-speed stall pattern up to Mach 0.74. Above this value, the shock waves appear, and it gets harder to match the flight data using this model because the stall mechanism changes.

In this section we have demonstrated the applicability of a fairly simple analytical model for quasi-steady stall characteristics. In general, the three parameters in Eq. (12.20) were estimated accurately and reliably, as suggested by the associated very low standard deviations, uncorrelated estimates, and the good match between the flight-measured and model-estimated responses. This was the case for the three examples presented here and for the case that will be discussed in the next section as well. The investigations with four different aircraft

Fig. 12.26 Embraer AT-26 Xavante.

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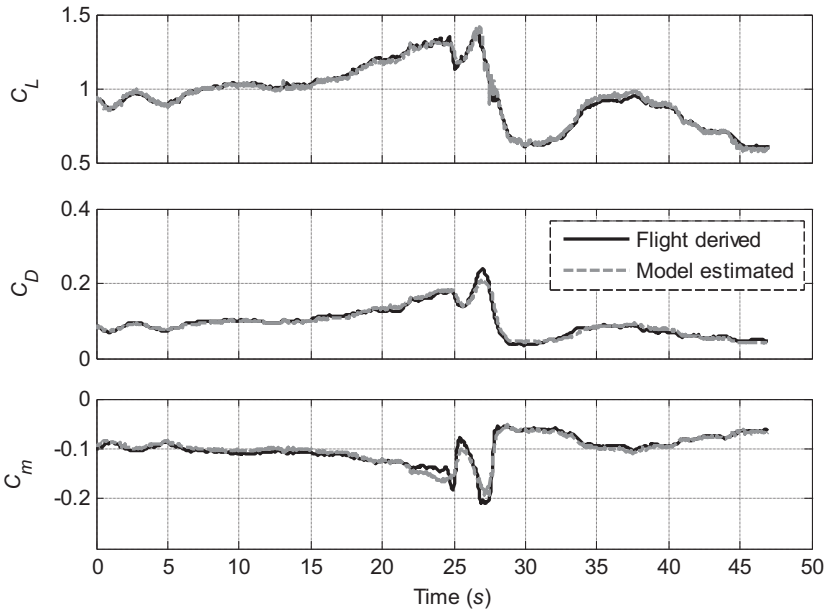

Fig. 12.27 AT-26 Xavante time histories of aerodynamic coefficients during stall maneuver: ——, flight derived; - - - - -, model estimated (Source: [34]).

indicate that the quasi-steady stall model is quite adequate for simulation of flow separated maneuvers with low dynamic effects. Such flight-validated stall models are incorporated in the aerodynamic databases for the aforementioned Level D training simulators Transall C-160 and Dornier 328 and the in-flight simulator ATTAS. The same approach was also applied to the X-31A poststall maneuvering program covering up to 70-deg angle of attack. Accurate stall modeling leads to an effective flight crew training on a simulator under extreme flight conditions with flow separation and meets the recommendations made by the NTSB (National Transportation Safety Board) after a Jetstream 4101 aircraft crash following a stall due to improper pilot reaction to stall warning [35].

12.6.4 ASYMMETRIC STALL MODELING

Analysis in the stall and poststall region is more or less confined only to the longitudinal dynamics. As demonstrated in the preceding sections, a state-space model of unsteady aerodynamics has been applied to longitudinal flight data. Flight analysts have, however, encountered cases in the past where an aircraft in stall is known to have exhibited large excursions in the lateral-directional motion. This asymmetric behavior is attributed mainly to the significant sideforce and

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yawing moments generated because of formation and shedding of the forebody vortices [36, 37]. An asymmetric behavior was observed during the quasi-steady stall with the military transport aircraft Transall C-160, exhibiting more than 20-deg bank variations. One possible cause for this phenomenon observed in flight could possibly be that the two engines equipped with four-bladed propellers rotating in the same direction influence the flow separation asymmetrically.

Applying the quasi-steady stall model discussed in Sec. 12.6 to Transall C-160 data, it was found that the longitudinal behavior in stall could be satisfactorily reproduced. However, the same cannot be stated for the lateral case where poor simulation quality, specifically in quasi-steady stall approach, for the lateral-directional time histories was observed. It was observed that the quasi-steady stall is accompanied by an uncommanded left roll-off. This is attributed to unequal lift caused by different flow separation behavior on the left and the right wing. To account for the asymmetric flow separation, as depicted schematically in Fig. 12.29, the two wing panels are treated separately [38]. The total lift

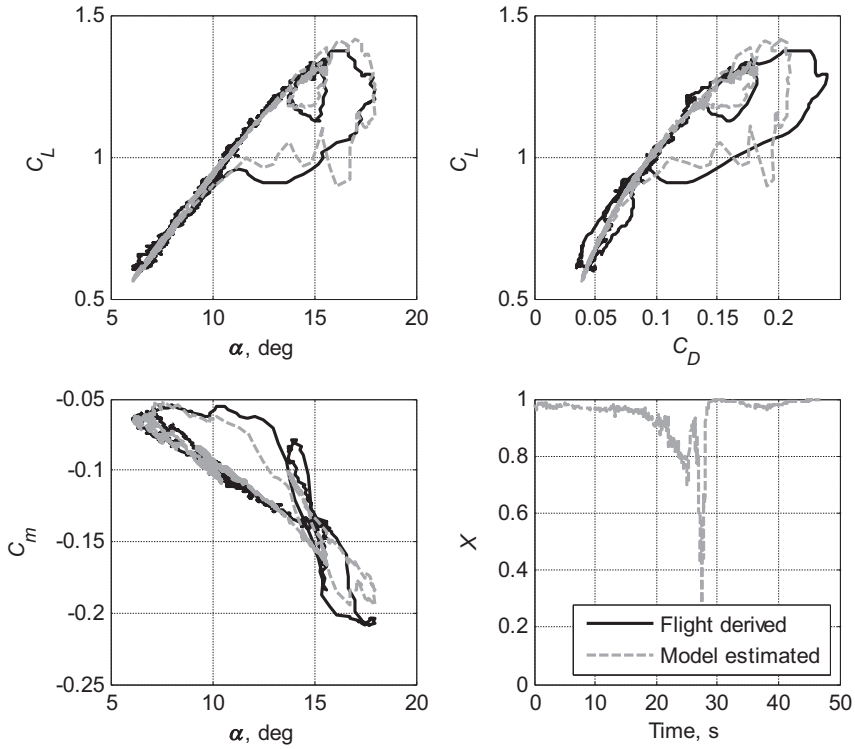

Fig. 12.28 Modeling of Embraer AT-26 Xavante stall hysteresis (Source: [35]).

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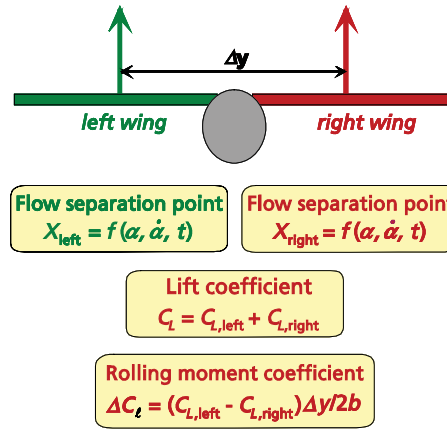

Fig. 12.29 Schematic of asymmetric stall modeling.

and drag appearing in Eqs. (12.21) and (12.22) are given by the sum of the components due to the left and right wings. Now, a simplified representation of the rolling and yawing motion resulting due to lift and drag differentials can be formulated as

$$\Delta C_\ell = (C_{L_L} - C_{L_R})\frac{\Delta y}{2b}$$
$$\Delta C_n = (C_{D_L} - C_{D_R})\frac{\Delta y}{2b} \tag{12.24}$$

where the suffixes L and R represent the entities for the left and right wing, respectively; Δy the effective lever arm; and ΔC_ℓ and ΔC_n the incremental changes to the rolling and yawing moments resulting from asymmetric flow separation. The effective lever arm that can be assumed to be half the wingspan or estimated from flight-test data [38].

Equation (12.24), incorporating the lift and drag differentials, leads not only to a parsimonious model, but is also based on the physical interpretation of the force–moment relationship. This approach appears to be preferable compared to the other possible approach of modeling the changes in rolling and yawing moments due to asymmetric unsteady flow through additional aerodynamic derivatives in the respective moment coefficients.

The stall model of Eqs. (12.20–12.23) extended by treating the left and right wings separately and through Eq. (12.24) is applied to analyze the quasi-steady stall maneuver. The stall parameters a_1, α^*, and time constant τ_2 for the left and the right wing are estimated separately. The efficacy of such a fairly simple formulation to satisfactorily capture the lateral-directional motion during stall is evident from the comparison of the estimated and flight measured responses shown in Fig. 12.30. The asymmetry in flow separation is clearly observed from the plots of the separation points X_L and X_R for the left and right wing.

12.7 GROUND-EFFECT MODELING

Ground effect is a physical phenomenon that results from the proximity of the moving flight vehicle to the ground, mainly because close to the ground the downward flow is interrupted by the ground. Loosely speaking, for example, during the

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landing phase, ground effect leads to increasing pressure below the wings as the aircraft approaches the ground and thereby reduces the sink rate. Such effects are encountered, in general, for altitudes above ground level lower than the wingspan. They have a dominant influence on the aircraft landing and takeoff performance, which are safety critical flight phases. In general, the three main effects resulting from ground effect are [8] as follows:

1. Reduced downwash angle at the tail
2. Increase in the wing-body and the tail lift-curve slopes
3. Reduction in the induced drag

The ground effect mainly influences the longitudinal motion although, in general, almost all other aerodynamic derivatives might be slightly altered. Ground-effect updates from flight data are usually necessary to accurately reproduce the landings and takeoffs, as well as for better assessment of touchdown sink rate.

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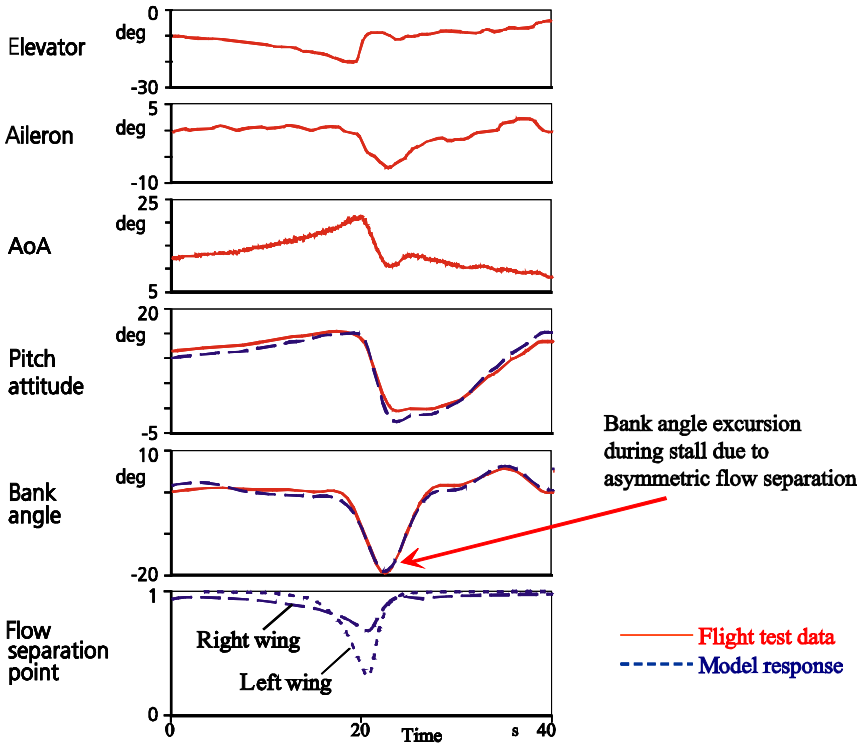

Fig. 12.30 Time histories of asymmetric stall behavior.

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The basic theory on ground effect was developed by Wieselsberger, analyzing the same through a simple model based on Prandtl's lifting-line theory [39, 40]. To date, extensive research has been devoted to this important topic. There is a vast literature, including dedicated textbooks, on ground-effect modeling based on theory, numerical investigations, wind-tunnel tests, and determination from flight tests [41–45]. A detailed discussion on the theoretical background and on the experimental investigations applying conventional methods is far beyond the scope of this book. Consistent with the overall philosophy followed in this book, we take a cursory look at the physics of the phenomenon and then address the present problem by considering analytical models amenable to parameter estimation applying the system identification techniques developed in this book. In the following, we draw heavily on the investigations reported in [46, 47].

Loosely speaking, ground effect leads to reduction of the wing-tip vortices, which is equivalent to a reduction of the induced angle of attack. This implies that for a constant geometrical angle of attack the lift coefficient will increase as the aircraft approaches ground. The reduction in the induced angle of attack results consequently in a reduction of the induced drag. Under ideal conditions (wing with elliptical lift distribution), Wieselsberger formulates the induced drag in terms of a ground-effect influence function σ as [39]

$$C_{\mathrm{Di}} = (1 - \sigma)\,\frac{C_L^2}{e\,\pi AR} \tag{12.25}$$

The various terms in Eq. (12.25) have already been defined in Sec. 6.9.3. Equation (12.25) is the same as Eq. (6.78), but for the modification factor $(1 - \sigma)$ due to ground effect. The ground-effect influence function σ is a nonlinear function of nondimensional height h/b and varies between zero and one. Based on the lifting-line theory [40], Prandtl provided values for σ, which are plotted in Fig. 12.31. Assuming a quadratic polar, it turns out that the L/D (lift-to-drag) ratio under ground effect to that for the freestream conditions is given by $1/\sqrt{1-\sigma}$.

The contributions to pitching moment due to wing as well as horizontal tail are also affected by ground effect. The change in downwash at the horizontal tail is more pronounced than the pitch-up generated by the ground effect on the wing alone. Because of the reduced downwash, the tail angle of attack increases, resulting in increased tail lift contribution and pitch-down moment of the overall aircraft.

In general, the incremental aerodynamic coefficients ΔC_L, ΔC_m, and $\Delta\varepsilon$ (lift, pitching moment, and downwash increments respectively) induced by ground effect are functions of 1) flap configuration, 2) Mach number, 3) angle of attack, 4) height above ground, 5) flight-path angle, and 6) thrust level through jet interaction. For a given flap configuration, the ground effect varies nonlinearly as the aircraft approaches the ground. To enable parameter estimation, as in

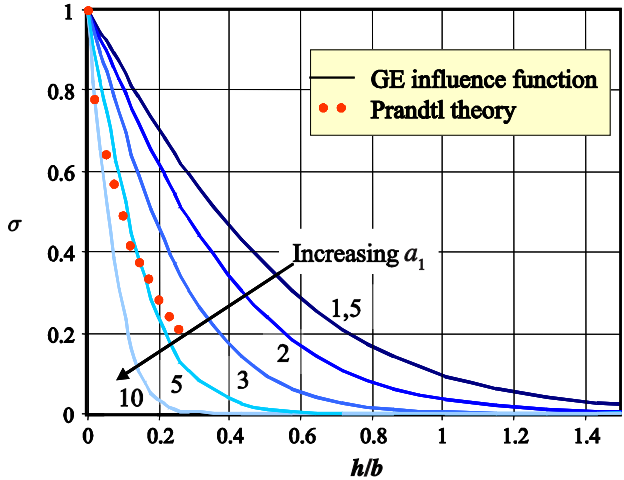

Fig. 12.31 Ground-effect influence function.

the case of stall hysteresis discussed in the preceding section, we once again use the hyperbolic-tangent function to analytically model the ground effect as follows:

$$\sigma = 1 - \tanh\left(a_1 \frac{h}{b}\right) \tag{12.26}$$

where h is the height above ground level and b the wingspan. The functional behavior Eq. (12.26) for different values of a_1 is shown in Fig. 12.31. It is apparent that the functional value is limited from 0 to 1, and the parameter a_1, which is a function of the configuration like flap setting, characterizes the curvature of the curves. The analytical model approximates roughly the theoretical values provided by Prandtl shown as solid circles; for such a case, in the limit the function has infinite slope for $h/b = 0$ and thus conforms to the theory.

Equation (12.26) is a general representation of the ground-effect influence function, whereby the altitude of the wing above ground is the critical parameter that determines the intensity of the ground effect. It approximates adequately the nonlinear behavior of aerodynamics in the proximity to ground for $h < b$ [43, 44] and hence particularly suitable for low-wing configurations, for example, the VFW-614 ATTAS aircraft (see Fig. 12.3). For many high-wing configurations, for example, Transall C-160 (see Fig. 12.9), such low altitudes are more or less not encountered by the wing. In such cases, as a first-order approximation, a linear ground-effect influence function is used instead of Eq. (12.26). Here we investigate both these cases from flight-test maneuver.

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Irrespective of the type of model (linear or nonlinear) being identified to account for ground effect, there are three important aspects critical to determination of ground effect from flight data, namely,

1. Flight-test technique
2. Accurate height above ground level signal
3. High fidelity in-air aerodynamic model including landing-gear effects

The flight-test techniques that allow estimation of ground effect consist of 1) landings, 2) takeoffs, 3) touch and go, 4) tower flyby over runway at different heights AGL (typically 100, 40, 20, and 10 ft), and 5) tower flyby at different heights AGL with small dynamic inputs in pitch, roll, and yaw. Extreme caution is called for during such flight testing to avoid any mishap. Furthermore, such maneuvers should be performed under very calm atmospheric conditions, that is, with no wind and no turbulence. Such ideal conditions exist rarely; hence, some minimum amount of horizontal wind must be tolerated, but the turbulence should be negligible. If the horizontal wind is nonnegligible, a more advanced approach incorporating GPS/DGPS measurements will be necessary. A wind gradient, if encountered during such maneuvers, is more critical than constant horizontal wind.

Besides gathering flight data with sufficient information content, reliable extraction of a ground-effect model needs accurate height AGL information because the ground effect is a strong function of the nondimensional height h/b. Some aspect of height AGL calibration using radar altimeter and independent measurements using cameras have already been discussed in Sec. 10.7. Furthermore, the ground effect is usually extracted from flight maneuvers as incremental deltas to the in-air aerodynamic model. Thus, invariably an accurate model for in-air aerodynamic, particularly in the low landing speed regime, including the landing-gear effects is a prerequisite. Aspects of high-fidelity in-air aerodynamic modeling will be covered in the next section whereas those of determining the landing-gear effects have been presented in Sec. 12.3.

The first case pertains to the Transall C-160 aircraft with a high-wing configuration (see Fig. 12.9), for which the wing is about 4 m above the ground and the mean aerodynamic chord $\bar{c}$ is 4.18 m. Thus, for the reasons already elaborated, namely, altitudes for which the ground-effect changes nonlinearly ($h < \bar{c}$) are not reached by the high-wing configuration, we consider here a linear model for the ground effect. Accordingly, the model postulate for the incremental lift, drag, and pitching-moment coefficients is formulated as

$$\left.\begin{aligned} \Delta C_{L,GE} &= \frac{\partial C_L}{\partial h}\,(h - h_{GE}) \\ \Delta C_{D,GE} &= \frac{\partial C_D}{\partial h}\,(h - h_{GE}) \\ \Delta C_{m,GE} &= \frac{\partial C_m}{\partial h}\,(h - h_{GE}) \end{aligned}\right\}, \quad h < h_{GE} \tag{12.27}$$

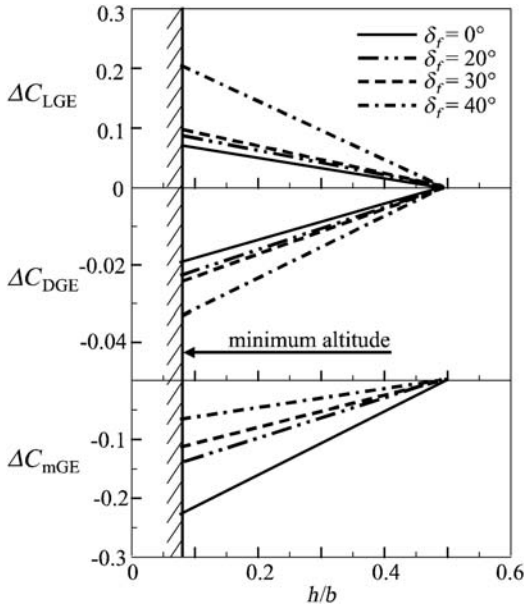

Fig. 12.32 Estimated ground effect (GE) for top-wing C-160 aircraft.

where for modeling purposes h is the altitude of the c.g. above ground and h_{GE} the maximum ground-effect altitude (that is, for altitudes h greater than h_{GE} the ground effect vanishes).

The three unknown parameters $\partial C_L/\partial h$, $\partial C_D/\partial h$, and $\partial C_m/\partial h$ are to be estimated from flight data. Figure 12.32 shows the identified coefficients plotted as function of h/b. The maximum altitude h_{GE} was determined to be 20 m, which corresponds to $h/b = 0.5$. Estimation of ground-effect parameters appearing in Eq. (12.27) was mostly based on the landings and takeoffs, mainly because turbulence was encountered during the low-altitude flyby flights. In general, variations of ± 0.5 kt indicate very low turbulence, ± 1 to 1.5 kt light turbulence, and ± 2.0 to 2.5 kt light-to-moderate turbulence. The ground effect was identified separately for each flap position. The preceding simplified ground-effect model provided good landing performance whereas the model fidelity for takeoff was just adequate. To further improve model characteristics during takeoffs, particularly after rotation, increased elevator effectiveness as a function of tail altitude above ground was identified. This is a possible empirical correction because during this phase, based on the ratio of the altitude above ground to the tail span, the tail comes into regions where the ground effects are not negligible. The linear model identified in Eq. (12.27) primarily accounts for the contributions due to the wing.

The second case pertains to the VFW-614 ATTAS aircraft with low-wing configuration; see Fig. 12.3. As already described in Secs. 12.2.2 and 12.2.3, the ATTAS aerodynamic model is a two-point model considering wing-body and horizontal tail separately. The overall model incorporates various dependencies and nonlinearities; the basis (that is, those out of ground effect) contributions are denoted appending the respective subscripts through "basic." Accordingly, the lift coefficient for the wing-body combination, including the

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ground effect, is given by

$$\begin{aligned} C_{\mathrm{L,WB}} &= C_{\mathrm{L,WB_basic}} + C_{L\alpha\mathrm{GE}}\,\alpha\,\sigma_1 \\ C_{L,H} &= C_{L,H_\mathrm{basic}} + C_{L\alpha\mathrm{HGE}}\,\alpha_H\,\sigma_2 \\ C_{m25} &= C_{m25,\mathrm{basic}} + C_{m0,\mathrm{GE}}\,\sigma_1 \end{aligned} \tag{12.28}$$

where the subscript GE indicates contributions due to ground effect. The downwash at the horizontal tail is given by

$$\varepsilon_H = \varepsilon_{H,\mathrm{basic}} + (\partial\varepsilon/\partial\alpha)_{\mathrm{GE}}\,\alpha(t-\tau)\,\sigma_2 \tag{12.29}$$

The major portion of the influence of the ground effect at the horizontal tail on the pitching moment is automatically accounted for in the two-point aerodynamic model through the term tail-lift multiplied by the lever arm; see Eq. (12.5). The additional correction in the pitching moment is introduced to account for the ground effect on the wing-body through a zero pitching-moment term.

We note that two influence functions σ_1 and σ_2 have been introduced to model the ground effect at the wing-body and horizontal tail separately. These functions are evaluated from Eq. (12.26) using two parameters a_{1W} and a_{1H}, respectively. Thus, a total of five parameters, namely (a_{1W}, a_{1H}, $C_{L\alpha\mathrm{GE}}$, $(\partial\varepsilon/\partial\alpha)_{\mathrm{GE}}$, $C_{m0\mathrm{GE}}$), characterize the ground-effect model for the VFW-614 ATTAS. They are estimated from the following flight tests: 1) flybys at 100, 50, 20, and 10 ft AGL, each at 1.3 and 1.4 V_s; 2) similar flybys with small-amplitude elevator, aileron, and rudder doublets; and 3) touch and go. All maneuvers were performed with extended landing gear. The horizontal wind was moderate and turbulence level light to moderate. The estimates of the preceding five parameters for the three flap settings are provided in Table 12.2, and the ground-effect influence functions for the wing-body and horizontal tail are plotted in Fig. 12.33. As stated at the beginning of this section, the reduction in downwash at the tail has a more significant effect than that due to changes in lift and drag. From Table 12.2, it is observed that the downwash parameter due to ground effect $(\partial\varepsilon/\partial\alpha)_{\mathrm{GE}}$

TABLE 12.2 ESTIMATES OF GE PARAMETERS (SOURCE: [47])

Parameter	Clean	Flap	
		5 deg	14 deg
a_{1W}	6.9	4.4	5.0
a_{1H}	3.7	3.8	3.7
$C_{L\alpha\mathrm{GE}}$	1.3	2.3	1.1
$(\partial\varepsilon/\partial\alpha)_{\mathrm{GE}}$	−0.43	−0.69	−1.03
$C_{m0\mathrm{GE}}$	0.11	0.085	0.111

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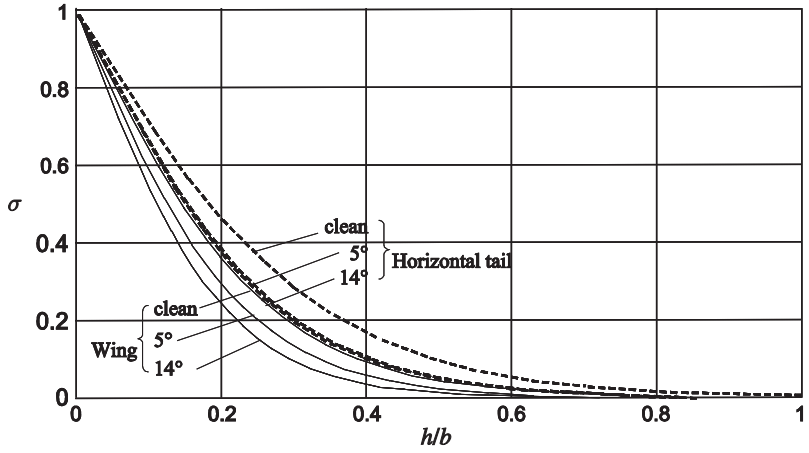

Fig. 12.33 Estimated ground-effect influence function for low-wing VFW-614 ATTAS.

increases with flap position, leading to stronger pitch-down motion. The influence function characterizing parameter for the tail a_{1H} is almost independent of the flap setting.

To better appreciate the influence of ground effect, the total longitudinal aerodynamic coefficients computed for different heights AGL are plotted in Fig. 12.34,

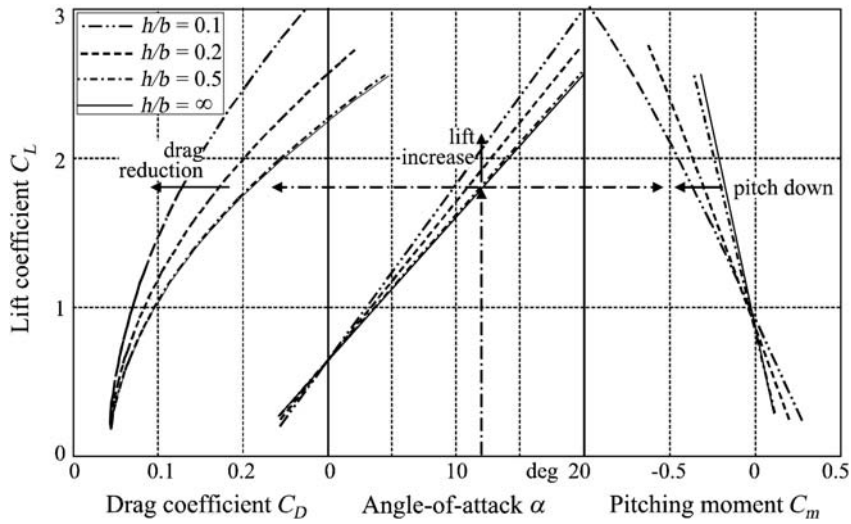

Fig. 12.34 VFW-614 ATTAS flight-identified total aerodynamic coefficients for 14-deg flap (Source: [47]).

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which confirms the three observations made at the beginning of this section. This can be inferred starting from the middle plot showing increase in C_L as h/b decreases. Now, as shown schematically in the figure, for a constant angle of attack (or lift), moving along the dotted lines with arrow to the left and right respectively, we notice reduction in the drag and the pitch-down moment. The change in the total C_m results primarily from the reduction in the downwash.

To summarize, in this section we demonstrated estimation of ground-effect parameters for high-wing and low-wing configurations from flight data using linear and nonlinear model postulates. Besides the two cases presented here, the same approach was also successfully applied in the case of X-31 VECTOR to identify ground effect from ESTOL (extremely short takeoff and landing) touch and go and landing maneuvers [48]. An analytical model in terms of influence functions and three parameters to quantify the variation in downwash, lift, and drag proved very effective. The model not only conforms to the theoretical predictions, but also leads to a smooth transition from the in-air model to one accounting for ground effect. It leads to a homogeneous overall representation.

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12.8 HIGH-FIDELITY DATABASES FOR TRAINING SIMULATORS

Prior to the advent of digital computers, the accuracy of aircraft simulation was marginal due to the inherent limitations of analog computers. The digital computers opened up new horizons not only in the field of aircraft parameter estimation but also in the field of flight simulators [49]. With the evolution of high-performance modern aircraft and with the spiraling developmental and experimental costs, the importance of ground-based flight simulators and that of in-flight simulators increased continuously over the years. Simulators are increasingly used not only for pilot training but also for other applications such as flight planning, envelope expansion, design and analysis of control laws, handling qualities investigations, and pilot-in-the-loop studies. Many of these applications demand a high-fidelity flight simulator. The fidelity of a simulation depends to a large extent on the accuracy of the mathematical model and of the aerodynamic database representing the flight vehicle.

To arrive at flight-validated high-fidelity global databases for flight simulators, there are two viable approaches, namely, the following:

1. To verify and, if necessary, update the preflight (theoretical, CFD, or wind-tunnel) predictions through incremental coefficients from flight data analysis.
2. To generate a new database from flight data through a systematic system identification procedure starting from a basic six-DOF model and extending it to include special effects like landing gear, high angle-of-attack regime, unsteady aerodynamics, ground effects, and engine-out effects.

In both approaches the task is formidable and the process iterative, requiring model structure determination, and can be equally complex and tedious.

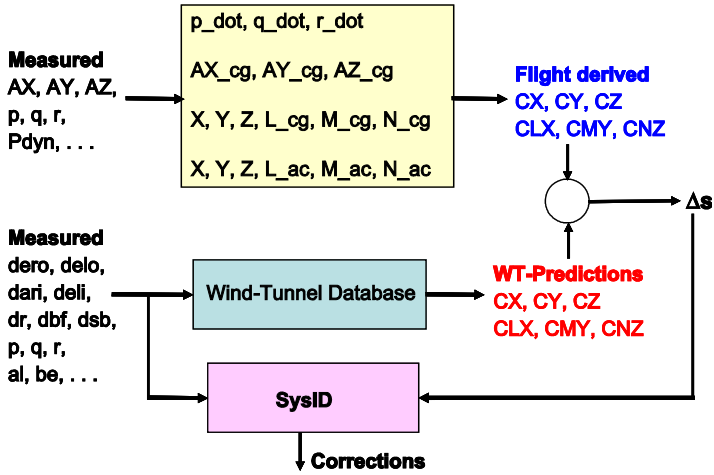

Fig. 12.35 Schematic of incremental model update procedure.

The first approach, often called incremental model update, makes maximum use of the a priori information available on aerodynamic database derived from wind-tunnel measurements, theoretical predictions, or even intermediate updates based on preliminary flight data. It starts with computation of aerodynamic force and moment coefficients using flight-measured linear accelerations and angular rates and comparing them with those computed from the so-called preflight database fed with measured aircraft motion variables and control surface deflections. The discrepancies between the two sets, called Δ, have to be eliminated or, better said, minimized to arrive at a flight-validated and updated database. Figure 12.35 depicts schematically the approach just described. In this figure we have given the notational names of the variables as they were used in a particular project, which we will address in Sec. 12.12. It leads to a two-step procedure: first, finding the discrepancies, and then identifying suitable models for these discrepancies. The first step of finding the deltas is fairly straightforward. The procedure to compute the aerodynamic forces and moments from the measurements of linear accelerations and angular rates has already been dealt with in Sec. 6.9.1.

The second step of the approach is to fit appropriate models to the increments. In most of the cases, however, it turns out that this step is not a trivial one. Such updates are possible only through a repetitive procedure because they are invariably functions of the angle of attack, Mach number, angle of sideslip, and other variables. They are also configuration dependent. For estimation purposes, mostly the regression analysis based on the least-squares technique is applied, which allows direct handling of the aerodynamic coefficients. Although some model structure determination procedures can be applied to arrive at an acceptable model, considerable engineering judgments are usually necessary. The

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approach has found successful application in practice [50–53]. It will be applied in Sec. 12.12 to verify and update the longitudinal database of a reusable launch vehicle from free flights. A highly complex procedure following this general philosophy has also been applied to identify increments to the X-31A aerodynamic database to arrive at a global model. The increments were estimated in a table-look-up form, along with the break points as well, which automatically allowed more precision in matching the nonlinear regions. Discussion of this advanced X-31A global model development is far beyond the scope of this book. An excellent account is found in [48, 54], analyzing in a single run up to 10 hours of flight-test time, separated into up to 99 time slices with a sampling rate of 12.5 Hz, covering representative flight maneuvers in cruise, high lift, and power approach configurations including their transitions and the acceleration/deceleration segments during takeoff and landing.

The second approach of generating a comprehensive homogeneous database, valid over the entire operational flight envelope is equally, if not more, elaborative. It is also an iterative process, which starts with point identification at all of the points flight tested. Point identification results in models related to specific trim conditions. To simplify parameter estimation, the point identification can be applied to the longitudinal mode and lateral-directional model separately. If dynamic maneuvers are repeated, the estimates will show scatter if they are analyzed separately. Such a scatter mainly results due to various reasons such as nonoptimal input, small variations in the nominal trim conditions, unavoidable atmospheric turbulence, and also the assumption that the noise in the measurements is white noise whereas, in practice, it is mostly colored noise. Such a scatter in the estimates from individual, repeat maneuvers is unavoidable in the task of parameter estimation. Hence, the common approach followed to overcome these practical difficulties and harmonize the estimation is to estimate a single set of aerodynamic derivatives from several repeat runs by concatenating the maneuvers and analyzing them simultaneously. This is a powerful approach, provided suitable estimation software, as provided with this book, is available.

Based on this bulk of the estimation results in the normal flight regime, the aerodynamic model postulates can be extended to include angle of attack, Mach-number and angle-of-sideslip dependencies, coupling derivatives, and nonlinearities. Through multipoint identification several flight conditions are analyzed simultaneously to arrive at an extended model. After freezing the validated model for the normal flight regime, the submodels of other aerodynamic effects resulting from landing gear, ground effect, stall approach and quasi-steady stall, takeoff and landing, single engine, etc. are estimated and appended to the basic model. Depending upon the aircraft, special configurations might have to be modeled, for example, ramp door or load drops for military transport aircraft. Figure 12.36 depicts schematically the approach just described, which was also followed in several cases to generate aerodynamic databases meeting the Level D fidelity criteria specified by the FAA for training simulators [17, 33]. Both the approaches to update or generate databases, namely, aerodynamic coefficient

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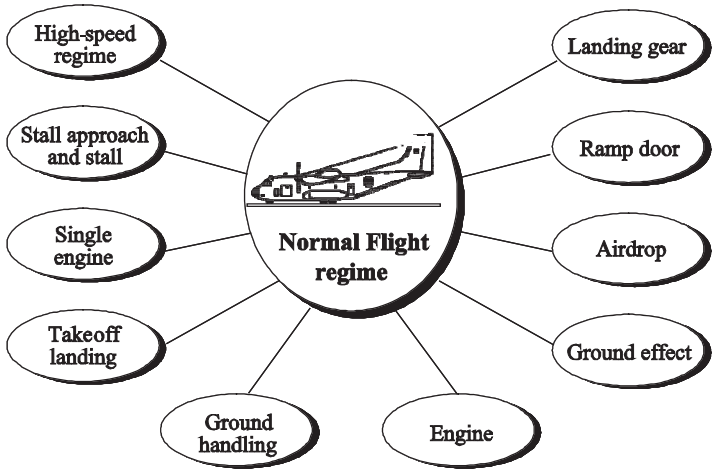

Fig. 12.36 Schematic of comprehensive model identification procedure.

matching by regression method and aircraft motion responses matching by output-error method, have found successful applications.

The two approaches of database generation, verification, and update are more or less equivalent. The choice between the two is mainly dictated by the familiarity and background of the team of engineers involved in this task. The expected quality of the a priori model will also be a decisive factor, so also the form (table look-up or derivative) because of the significant differences in the computational overhead. To some smaller extent, the quality of recorded data plays a role; as already discussed elaborately in Chapter 6, the two-step approach is primarily based on the least-squares method, which requires good quality measurements and extensive data checking and a correction step to eliminate measurement errors. The second approach is less sensitive to these errors, as they can be determined during the estimation of the aerodynamic parameters. It also incorporates in the estimation procedure the numerical integration of the six-degrees-of-freedom equations of aircraft motion, which are very similar to those finally implemented on the training simulator. It is also possible to directly transfer the complete database including the equations of motion from estimation to simulator. It might also be better suited for modeling of more complex aerodynamic phenomena such as ground effect, stall, and landing and takeoff cases, which are found to be highly sensitive to influencing parameters, for example, height above ground. Even very small errors in the measurement of such influencing variable will affect the quality of the estimated model significantly; hence, in such cases the use of integrated variables might be preferable.

Irrespective of the approach we adopt, the importance of flight validation and updates is well recognized and accepted by database generators. This necessity

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arises mainly from the fact that although the analytical predictions, wind-tunnel measurements on a scaled model, or extrapolation of existing data from similar configurations provide aerodynamic databases valid over the entire flight envelope, they are often associated with certain limitations arising from model scaling, Reynolds number, dynamic derivative, and cross-coupling effects. Moreover, in some instances, particularly for old-generation aircraft, only limited information is available. In recent years, most aircraft manufacturers consider a simulator data package as an integral part of the aircraft development program [4, 49].

Consistent with the goal of this chapter, which is to provide appreciation of various aerodynamic modeling and parameter estimation problems, we present some snapshots of results that were generated during the development of comprehensive databases for the military transport aircraft Transall C-160 (Fig. 12.9) and the commuter aircraft Dornier 328 (Fig. 12.23), meeting the FAA Level D quality criteria in both cases [17, 33]. This general approach was also followed in the generation of an aerodynamic database for the ATTAS in-flight simulator [14, 15].

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12.8.1 TRANSALL C-160 MILITARY TRANSPORT AIRCRAFT

As a part of the data gathering for the Transall C-160 (see Fig. 12.9), 30 flights each of roughly 2.5 h were carried out with the instrumented aircraft. A total of 37 configurations covering five flap settings, three c.g. locations, five different speeds at five altitude levels were tested. Roughly 1000 flight maneuvers were analyzed to derive a rigid-body aerodynamic database valid over the operational flight regime [16–18]. In Secs. 12.2 through 12.7, we have already discussed modeling of specific aerodynamic effects. Here, in this section we present two examples highlighting nonlinear effects in the basic aerodynamic model, derived from the flights in the normal flight regime.

The lift, drag, and pitching-moment coefficients are estimated from the equations of motion pertaining to the longitudinal motion, using the motion variables of the lateral-directional motion as pseudocontrol inputs. As elaborated in Sec. 3.6, introducing variables of the other modes of motion as pseudocontrols allows working with the complete nonlinear equations of motion in each case. The longitudinal derivatives are estimated for a two-point model considering, as described in Sec. 12.2, wing/body and tail separately. The underlying equations for the lift and pitching-moment coefficients are given by

$$C_L = C_{L0} + C_{L\alpha W}\,\alpha(t) + \frac{S_H}{S}\,C_{\mathrm{LH}} \tag{12.30}$$

where the lift due to the horizontal tail is given by

$$C_{\mathrm{LH}} = C_{L\alpha H}\,\alpha_H + C_{L\delta e}\,\delta_e \tag{12.31}$$

The pitching-moment coefficient is given by

$$C_m = C_{m0} - \frac{r_H^*}{\bar{c}} \frac{S_H}{S} C_{\mathrm{LH}} + C_{mqW} \frac{q\bar{c}}{V} \tag{12.32}$$

The effective angle of attack at the horizontal tail α_H, accounting explicitly for the downwash lag effect, is modeled as follows:

$$\alpha_H = \alpha + i_H - \frac{\partial \varepsilon_H}{\partial \alpha} \alpha(t - \tau_\alpha) + \frac{\partial \varepsilon_H}{\partial C_S} C_S(t - \tau_{CS}) + \alpha_{\mathrm{dyn}} \tag{12.33}$$

where the various variables have already been defined in earlier sections. Equations (12.30–12.32) are basically the same as Eqs. (12.4–12.5) that we had used in Sec. 12.2 for the downwash lag effect investigations. Similarly, Eq. (12.33) is same as Eq. (12.12), but for the term corresponding to the speed brake.

The derivatives appearing in Eqs. (12.30–12.33) are estimated from multirun evaluation at each trim point by combining a 3-2-1-1 elevator input maneuver exciting the short-period motion, an elevator pulse input maneuver exciting the phugoid motion, and a thrust input maneuver. A typical plot of the point identification for the estimated elevator control effectiveness as a function of trim angle of attack is shown in Fig. 12.37 for the landing flap of 0 deg, that is, clean configuration. Estimates obtained from eight different flights clearly show that the elevator effectiveness is a function of the angle of attack, and the dependency is nonlinear.

An extended model accounting for the dependency of elevator effectiveness on the angle of attack yields the match shown in Fig. 12.38, when applied to simulate two different maneuvers, one a doublet input with not so large elevator deflection and the other a stall approach in which large elevator doublet inputs were simultaneously applied while pulling the stick. The match between the flight

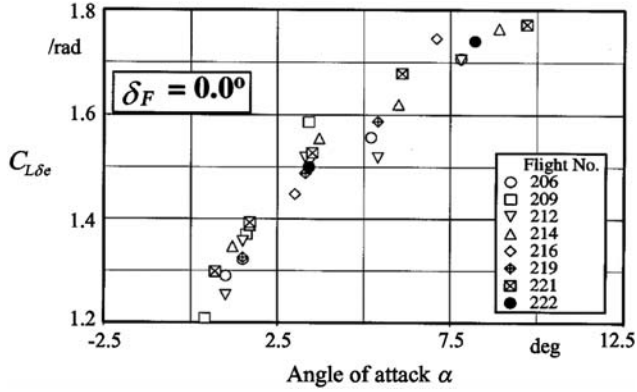

Fig. 12.37 Transall C-160 elevator control effectiveness.

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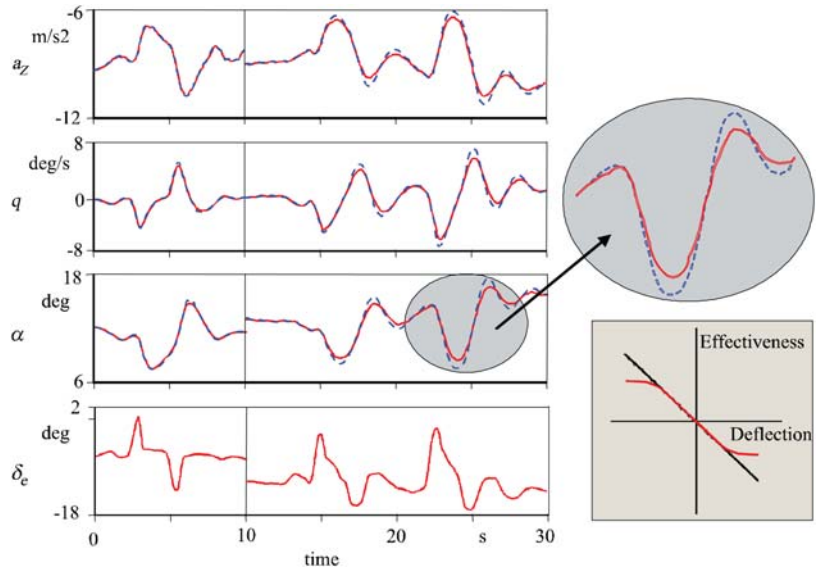

Fig. 12.38 Identification of linear model for elevator effectiveness: ——, flight measured; -------, model estimated.

measurements (shown by continuous lines) and model prediction (shown by dotted lines) is very good for the first maneuver whereas the second maneuver shows some discernible discrepancies. From the time-expanded plot of a portion of the maneuver shown in the same figure, it is apparent that the model responses overshoot, but only when the deflections are large. This is typical of nonlinear characteristics of control surfaces, as shown in an inset in the same figure, having reduced effectiveness for larger deflections, which is not yet accounted for.

Now, to account for the nonlinear control surface effectiveness, we extend further the model of Eq. (12.31) for the lift generated at the horizontal tail as

$$C_{\mathrm{LH}} = C_{L\alpha H}\ \alpha_H + [C_{L\delta e} + C_{L\delta\alpha}\,\alpha_H + C_{L\delta\alpha 2}\,\alpha_H^2 + C_{L\delta 3}\,\delta_e^2]\ \delta_e \qquad (12.34)$$

where the derivatives $C_{L\delta\alpha}$ and $C_{L\delta\alpha 2}$ account for the nonlinear dependency seen in Fig. 12.37, and the derivative $C_{L\delta 3}$ the nonlinearity in control surface effectiveness as schematically depicted in Fig. 12.38. The extended model of Eq. (12.34) yields the response match shown in Fig 12.39 for the same two flight maneuvers. Compared to Fig. 12.38, the match between model prediction and flight measurement is now excellent [17].

The second example pertains to the lateral-directional motion. Once again, following the same philosophy of point identification at a large number of trim points, strong nonlinear effects and dependency of parameters pertaining to the lateral-directional motion on the angle of attack were identified. Without going

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into any further specific details, we state here the final identified model for the yawing moment coefficient [4, 17]:

$$\begin{aligned} C_n = {} & C_{n0} + [C_{n\beta} + C_{n\beta\mathrm{NL}}\ (\alpha - \alpha_S)]\ \beta + (C_{\mathrm{np}} + C_{\mathrm{np}\alpha}\ \alpha)\ p^* \\ & + [C_{\mathrm{nr}} + C_{\mathrm{nrNL}}\ (\alpha - \alpha_S)]\ r^* + (C_{n\delta a} + C_{n\delta a\alpha}\ \alpha)\ \delta_a \\ & + (C_{n\delta r} + C_{n\delta r\alpha}\ \alpha)\ \delta_r \end{aligned} \tag{12.35}$$

where α is the angle of attack, β the angle of sideslip, p^* and r^* the normalized roll and yaw rates, δ_a the aileron deflection, δ_r the rudder deflection, and $C_{()}$ the unknown aerodynamic parameters. Nonlinearities in the weathercock stability $C_{n\beta\mathrm{NL}}$ are modeled as follows by two slopes and an angle α_S, which is also treated as unknown:

$$C_{n\beta\mathrm{NL}} = \begin{cases} C_{n\beta 1} & \text{for } \alpha > \alpha_s \\ C_{n\beta 2} & \text{for } \alpha \le \alpha_s \end{cases} \tag{12.36}$$

A similar formulation is also applied to the yaw damping parameter C_{nrNL}. Figure 12.40 shows a comparison of the analytical/wind-tunnel (WT) predictions with the flight-estimated weathercock stability [4]. Thus, in both the cases systematic approach and judicious choice of flight maneuvers led to unmasking the model deficiencies and then improvements through model extensions and parameter estimation to a model valid over the entire operational flight envelope.

The fidelity of the final model was demonstrated on roughly 110 proof-of-match tests defined by FAA [19, 20]. We have already addressed this aspect in

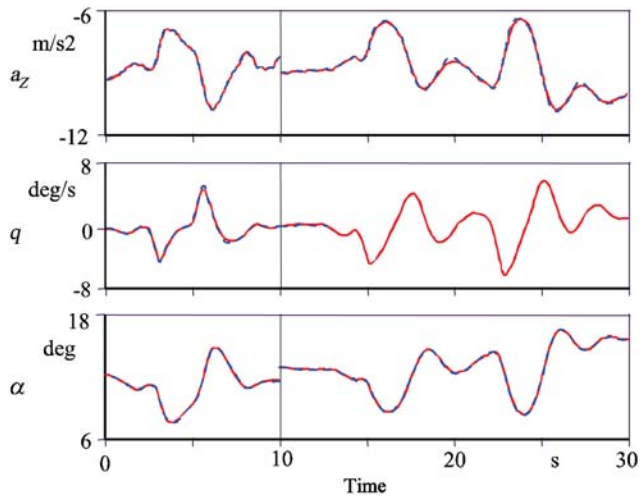

Fig. 12.39 Identification of nonlinear model for elevator effectiveness: ——, flight measured; ------, model estimated.

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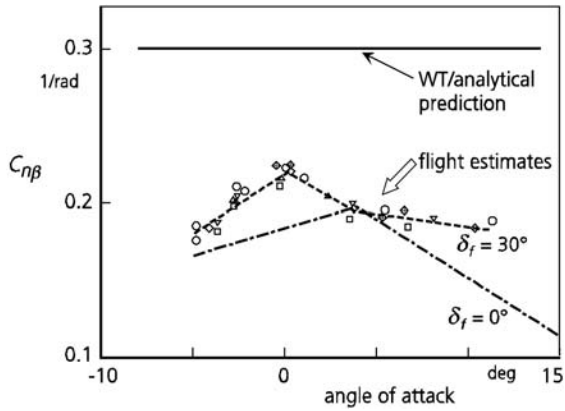

Fig. 12.40 Transall C-160 Weathercock stability.

Chapter 11, discussing the tolerances and the procedure to be followed to run such verifications. As a simple example of verification of an aerodynamic database generated from flight data, we evaluate the validation test 2d7.11, pertaining to the Dutch roll. Simulations using the predictions and estimated parameters are compared in Fig. 12.41 with the flight-recorded aircraft response. The WT predictions yield Dutch roll dynamics having a period of 4.18 s and a damping of 0.207; the flight-estimated database yields a period of 5.04 s and 0.202 damping factor. A comparison of these values with those approximated from a flight-recorded response (period of 5.12 s and damping of 0.198) clearly demonstrates that only the flight-estimated database meets the FAA tolerances of ± 0.5 s or $\pm 10\%$ of period and ± 0.02 on damping for this oscillatory mode [4].

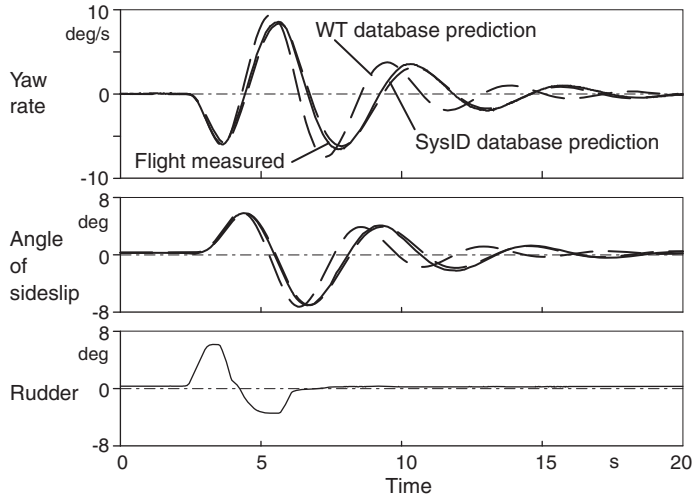

Fig. 12.41 Validation test for the Dutch roll dynamics of Transall C-160.

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12.8.2 DORNIER 328 REGIONAL TRANSPORT AIRCRAFT

For aircraft such as the Dornier 328 with reversible flight controls, in addition to rigid-body aerodynamics it becomes necessary to identify hinge moments for each of the controls. We had briefly touched upon the aspects of model validation for aircraft with mechanical controls in Sec. 11.6 and briefly elaborated the concepts of stand-alone and integrated models in Fig. 11.4. In the case of the high-performance Dornier 328 (see Fig. 12.23), a high-wing, high-tail regional transport turboprop aircraft, roughly 1200 flight maneuvers were analyzed to derive a rigid-body aerodynamic database [33]. It includes the high-angle-of-attack regime with stall hysteresis, large steady sideslip effects, ground effects, and those due to control surface malfunctions. Nonlinear influences due to the propeller slipstream were found to be very dominant. The same set of dynamic maneuvers is analyzed to derive dynamic models including friction and hinge-moment databases for the three primary flight controls [55]. Because the development of the rigid-body aerodynamic model was based on the approach that has already been covered in the preceding section, we concentrate in this section on the hinge-moment database and a few more validation tests. As a typical example, modeling of the elevator flight control is briefly presented here [4, 55].

In the elevator control system, the pilot forces are transmitted from the stick to the elevator by cables and rods; see Fig. 12.42. The system is preloaded by a downspring. For the dynamic model, the mass and moments of inertia of the connecting rods and cables are lumped together with those of the elevator surface (including trim tab) into an equivalent moment of inertia Θ_{elv}. As the pilot force is measured in the first rod, the stick inertia effects are already included in this input signal. This leads to a quasi-static transmission of the input force to the elevator axis. The elevator dynamics can, therefore, be formulated as a one-DoF system using deflection and deflection rate about the swivel axis as the system states [55]:

$$\ddot{\delta}_e = 1/2\,(ME_{\text{elv}} + MF_{\text{elv}})/\Theta_{\text{elv}} \tag{12.37}$$

where δ_e is the elevator deflection, MF_{elv} the lumped friction moment of the entire elevator control path, and ME_{elv} the sum of all external moments consisting of aerodynamic hinge moment MH_{elv}, forces transmitted to the aft part, inertia effects due to dynamic motions of the aircraft, and downspring moment.

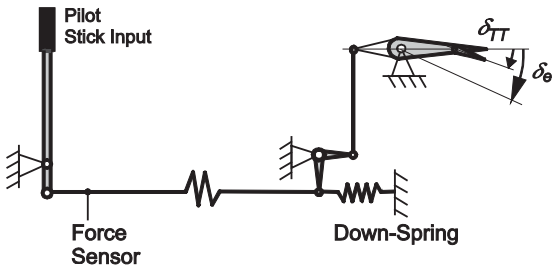

Fig. 12.42 Schematic of elevator control system.

The aerodynamic hinge moment MH_{elv} is given by

$$MH_{\text{elv}} = \frac{1}{2} CH_{\text{elv}}\, \overline{q}\, S_{\text{elv}}\, \overline{c}_{\text{elv}} \tag{12.38}$$

where $\overline{q}$ is the dynamic pressure, S_{elv} the elevator surface area, and $\overline{c}_{\text{elv}}$ the elevator reference chord. Without listing all parameters explicitly, the elevator hinge-moment coefficient was identified as

$$\begin{aligned} CH_{\text{elv}} = {} & CH_{\text{elv},0}(\alpha_{\text{elv}}, Ma, C_{\text{SP}}, \delta_F) + CH_{\text{elv},\delta e}(\alpha_{\text{elv}}, Ma, C_{\text{SP}}, \delta_e, \delta_F) \\ & + CH_{\text{elv},\delta TT}(\alpha_{\text{elv}}, Ma, \delta_{\text{TT}}, \delta_F) + CH_{\text{elv},\dot{\alpha}}(\delta_F)\, \dot{\alpha}^*_{\text{elv}} \\ & + CH_{\text{elv},\dot{\delta}e}(\delta_F)\, \dot{\delta}^*_e + CH_{\text{elv},\beta}(\beta_{\text{elv}}, \delta_F) + CH_{\text{elv,GE}}(\delta_e, \delta_F)\sigma_{\text{elv}} \end{aligned} \tag{12.39}$$

where the subscript "elv" denotes variables related to the elevator and superscript * the normalized rates.

Without going into too much details, the elevator hinge-moment coefficient is a function mainly of elevator deflection δ_e, trim tab deflection δ_{TT}, slipstream thrust coefficient C_{SP}, Mach number Ma, flap position δ_F, and local flow angles α_{elv} and β_{elv}. The force gradient $CH_{\text{elv},\delta e}$ is modeled separately for positive and negative deflections. An angle-of-attack rate influence $CH_{\text{elv},\dot{\alpha}}$ was found, and aerodynamic damping is provided by an elevator rate term $CH_{\text{elv},\dot{\delta}e}$. The elevator hinge moment for symmetrical flight was identified by evaluating dynamic maneuvers (elevator doublets and push-pulls), trims at normal, forward, and aft c.g. positions, and elevator mistrims. About 60 maneuvers at different speed and thrust levels were analyzed simultaneously for each flap position. The influence of sideslip was identified from steady sideslip maneuvers, and no additional modeling was required for single engine cases.

The identification of the rigid-body aerodynamics and of the hinge moments carried out based on the analytical model postulates yields four sets of derivatives for four flap positions, which are finally combined in a single database incorporating a linear interpolation procedure for flap position only. This so-called *derivative model* is, however, rarely used in the flight simulators mainly because the database in the so-called *table model* form was so far mostly generated from wind-tunnel measurements. Following the conventional way and in the present case as desired by the aircraft manufacturer, the flight-identified database is converted into a table model. In this form the rigid-body aerodynamics database consists of 124 tables out of which 14 are five-dimensional, 28 are four-dimensional, 45 are three-dimensional, 27 are two-dimensional, and 10 are one-dimensional functions. The elevator, aileron, and rudder hinge moments required just 8, 18, and 27 tables, respectively. Although the derivative model is better suited for parameter estimation, keeping track of a large number of derivatives being estimated is not easy, particularly those representing the secondary effects. Furthermore, overparameterization of the model, although leading to a better response

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match, is detrimental to identifiability of parameters and can lead to unrealistic estimates.

The two alternative model forms, that is, the derivative model and the table model, are equivalent in terms of accuracy, but the derivative model is significantly smaller in size and computationally faster than the table model because the linear interpolation is reduced to that for just one independent variable. The stand-alone rigid-body aerodynamics in the table model form required about eight times the computational time compared to the derivative model, the ratio becoming as large as 12–14 for the integrated model using both the rigid-body aerodynamics and the hinge moments. Although as realized in the present case, implementation of computationally complex large-size databases is feasible with modern computers, the differences in the two approaches are significant and pave way to consider the alternative way of implementing the flight-data-identified models, that is, a derivative model, in flight simulators [33].

Apart from the 170 flight maneuvers specified in the Acceptance Test Guide (ATG) covering just a part of the flight envelope, a large number (roughly 1100) of additional in-flight maneuvers, maneuvers in ground effect at various altitudes above ground for different power settings under symmetric and asymmetric (one-engine) flight conditions, were evaluated to establish global validity of the model. In each of these cases, an end-to-end simulation was performed, incorporating force-driven flight controls, providing position-driven rigid-body models, incorporating the identified rigid-body and hinge-moment databases. Two selected critical examples that bring out the different features of the database are presented here: 1) steady-state sideslip maneuver and 2) critical engine failure on takeoff [4, 33].

The first example pertains to flight maneuvers at large sideslip angles and serves the purpose of exhibiting correct reproduction of the interrelationship of steady-state lateral and directional flight characteristics. Figures 12.43 and 12.44 show typical variables pertaining to the lateral-directional motion for two time segments with negative and positive angles of sideslip for an approach configuration of 20-deg flaps with landing gear down; β is the angle of sideslip, p the roll rate, r the yaw rate, ϕ the bank angle, and δ_r the rudder deflection. Fairly large rudder deflections of about 20 deg were applied resulting in steady sideslip up to 11 deg. Each maneuver is of 25-s duration and covers both the transition phase from one level of sideslip to another as well as trim phases. Figure 12.43 shows the match between the flight measured and model predicted responses for the stand-alone rigid-body model (that is, six-DOF equations of aircraft motion excited through the flight-measured control surfaces including the identified rigid-body aerodynamics). The exceptionally good match in this case and also that in Fig. 12.25 for the stall maneuver demonstrate the excellent fidelity of the rigid-body aerodynamic database.

For the same case, Fig. 12.44 shows the end-to-end match with the integrated model (that is, six-DOF equations of aircraft motion coupled with dynamic models of the flight controls incorporating identified databases for rigid body

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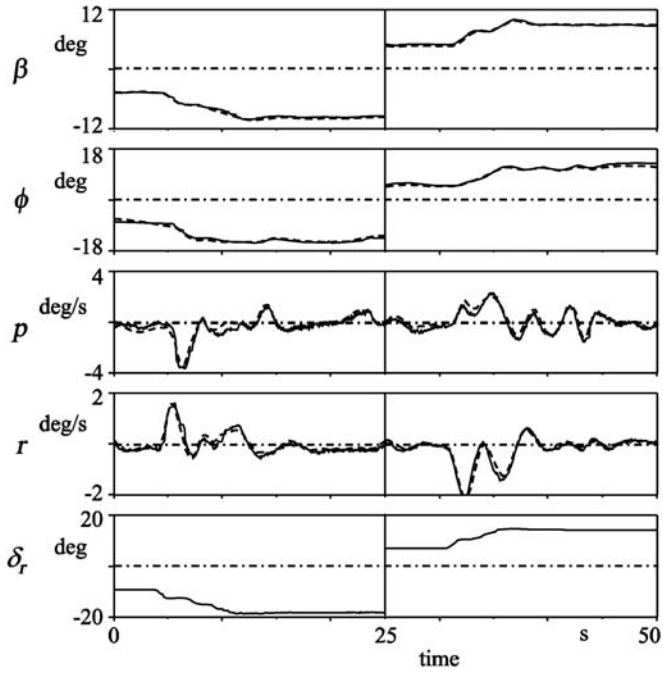

Fig. 12.43 Stand-alone rigid-body aerodynamics: ——, flight measured; ------, model estimated.

and hinge moments). The distinction between the two evaluations shown in Figs. 12.43 and 12.44 is apparent from the bottom-most plot, which shows just the measured rudder deflection in Fig. 12.43 as input to the model whereas in Fig. 12.44 the model-predicted rudder deflection is compared with the flight measurement. In Fig. 12.44 the computed control deflections are then the inputs to the rigid-body aerodynamics. Although some deterioration in the end-to-end match is observed in Fig. 12.44 compared to Fig. 12.43, the overall match between flight-measured and model-predicted variables is good and within the FAA tolerances of ± 2 deg for bank angle and ± 1 deg for sideslip; the aileron deflection not shown here was within $\pm 10\%$ or ± 2 deg, the wheel force bias within ± 1.3 daN, and the pedal force bias within ± 2.2 daN. It can be pointed out that it was necessary to account for asymmetric propeller effects, particularly on the vertical tail, at such extreme flight conditions.

The deterioration of the match in Fig. 12.44 compared to Fig. 12.43 clearly brings out the difficulties faced by the database developers in demonstrating the end-to-end match obtained by cascading two highly complex models although each might have an excellent fidelity, as shown in Fig. 12.43 for rigid body. Any

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attempts to further refine the stand-alone models appear unrealistic [33]. In the worst case and if critical, some tuning might be possible using the integrated model.

The second example is the validation test, called critical engine failure on takeoff, generally denoted as 1b5 and pertains to simulation of the response to an engine failure during the high activity task of takeoff; responses to rudder and aileron are of particular interest. Figure 12.45 shows the results for a few pertinent variables. The complete sequence from stand-still, acceleration, rotation, and climb to 200 ft above ground is simulated as a single maneuver. The results were obtained without any closed-loop controller (also termed as math pilot). The use of closed-loop controller helps in such demanding validation tests to meet the required tolerances. They should, however, be used with caution because they mask the modeling deficiencies [56].

From the bottom-most plot in Fig. 12.45 (F_L and F_R, the thrust due to the left and right engines), it is observed that the critical left engine is shut down immediately after rotation. To maintain the bank and heading, the pilot applies about 15-deg rudder, δ_r in Fig. 12.45, and about 6-deg ailerons, not shown. Except for a small lead in building up the bank angle, which is still within the tolerance, the match for all of the variables is good. Specifically, the airspeed is within the tolerance of

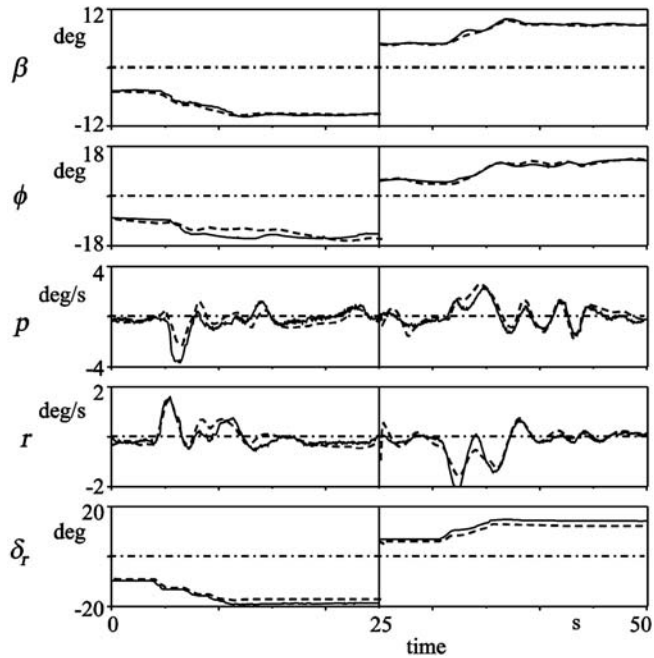

Fig. 12.44 End-to-end match with integrated model: ——, flight measured; ------, model estimated.

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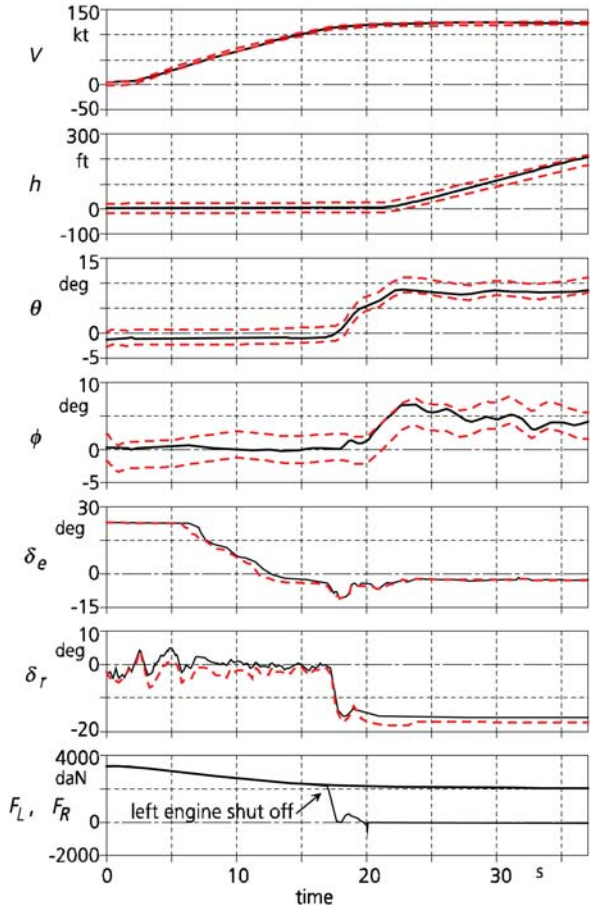

Fig. 12.45 Proof-of-match for critical engine failure on takeoff: ------, flight-measured ± tolerances; ——, model estimated.

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±3 kt, altitude within ±20 ft, pitch angle within ±1.5 deg, and bank angle within ±2 deg. Thus, the database meets the FAA fidelity requirements.

As mentioned at the outset of this section, the task of generating the databases was highly complex and iterative in nature. In general, the primary aerodynamic derivatives can be extracted well; estimation of the secondary effects needs some careful choice of the maneuvers. Particularly, in such cases care needs to be taken to avoid overparameterization, which yields better response match but not necessarily physically justifiable estimates. It needs to be ensured that the identified model is parsimonious. Analytical models for highly complex phenomena such as stall hysteresis and ground effects proved very efficient from a parameter

estimation viewpoint and also from the ability to capture the physics of the phenomenon. Based on these investigations, as a rule of thumb, it can be said that depending upon the type of aircraft (jet or propeller; with or without reversible flight controls) roughly 1000 maneuvers or 30–50 flight hours are required to generate high-fidelity databases suitable for Level D training simulators. Similar rough estimates have been reported from other investigations as well [57].

These two successful applications of high-fidelity database generation and other similar cases reported in the literature are the high-end applications of the system identification methodology. Of course, such exercises are not without intricacies. In this connection, the lessons learned from these two exercises, reported in [58], are worth repeating here:

1. The development of an integrated model is much harder than the development of stand-alone models.
2. The amount of effort required to arrive at a suitable model depends on the types of maneuvers and on the propeller slipstream interference effects.
3. The development of a FAA-adequate friction model for flight control systems seems to be impossible (for pilot force-driven end-to-end testing).
4. A gradient-based optimization method is not quite suitable to handle systems with static friction in an identification context.
5. The direct identification approach works, but there are some inherent, practical limitations.

To summarize, validation, and if necessary update of existing aerodynamic databases from flight-test data, as well as generation of completely new models, has been one of the most extensive and important applications of aircraft parameter estimation in the past. As pointed out in [1], this will continue to be one of the major applications of system identification in the future.

12.9 FLEXIBLE AIRCRAFT MODELING

The assumption of the flight vehicle being rigid, like the measurement noise being white noise, is made for convenience to simplify the modeling task. In reality, all structures undergo some deformation when excited through external inputs. The interaction between the rigid-body and the structural dynamics is dependent on the separation in their frequencies. Use of rigid-body models incorporating the classical decoupled or six-DOF equations of motion in parameter estimation from flight data yields so-called equivalent derivatives. The estimated aerodynamic derivatives are a sum total of pure rigid-body effects, flexibility effects, and other unmodeled effects. The structural modes can influence the accuracy of the flight mechanics models, that is, of the stability and control derivatives estimated from flight data. In general, there are two approaches to analyze flight data from flight vehicles, which can show effects

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due to structural modes. They are based on the different ways adopted to account for the flexibility effects. The first one is the quasi-steady approach, and the other is the generalized approach. We do not discuss the technique based on removal of structural effects through filtering of recorded data; though simplest, it has obvious limitations. The choice between the quasi-steady or generalized approach is usually dictated mainly by the separation between the rigid-body and structural modal frequencies and to some extent by the purpose for which the model is required and by the measurements available for accurate estimation of model parameters.

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12.9.1 QUASI-STEADY APPROACH

Classical point identification, as discussed in Sec. 12.8, yields stability and control derivatives related to a specific trim condition. As already pointed out, the estimated parameters are the sum of all influences and encompass the primary effect, its dependency on flight conditions, for example, Mach-number effects and structural deformation. However, it is possible to describe, even for large transport aircraft, the influence of structural deformation in a quasi-steady manner, provided the rigid-body and structural frequencies are sufficiently separated. It implies implicitly that the structure reaches its equilibrium deformation instantaneously. The influence of the structural deformation is then formulated as a function of dynamic pressure such that only the effects due to stationary deformation are accounted for. In such cases the additional degrees of freedom for the structural deformations and thereby the unsteady effects in the aerodynamics are neglected [4, 59].

In such a case, each derivative C_i is formulated as a sum of the rigid-body derivative (for dynamic pressure $\overline{q} = 0$) and a dynamic pressure dependent part, which is mostly described in a linear form through a flex factor k [8].

$$C_i(\overline{q}) = C_i\ (1 + \overline{q}\, k_i) \tag{12.40}$$

For example, assuming a linear model, the vertical force coefficient can be modeled as

$$C_z = C_{z0} + C_{z\alpha}(1 + k_{cz\alpha}\overline{q})\ \alpha + C_{zq}(1 + k_{czq}\overline{q})\ q^* + C_{z\delta e}(1 + k_{cz\delta e}\overline{q})\ \delta_e \tag{12.41}$$

where the derivatives C_{zi} pertain to the rigid-body derivatives and k_{czi} are the corresponding flex factors of the ith motion and control input variable such as angle of attack α, pitch rate q, and elevator deflection δ_e. The equations of motion remain the same; only the aerodynamic model is appropriately appended, whereby Mach number dependency must be known or is identified with a Prandtl–Glauert formulation [8, 9]:

$$C_i(Ma, \overline{q}) = \frac{C_{i_(Ma=0)}(1 + \overline{q}\, k_i)}{\sqrt{1 - a_i\, Ma^2}} \tag{12.42}$$

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where a_i is an identifiable empirical correction factor to account for differences between the ideal Prandtl–Glauert factor and the real factor and *Ma* the Mach number. The (generalized) Prandtl–Glauert factor can be applied to lift-generation-dependent derivatives. Thus, three parameters for each derivative—the derivative for the rigid-body $C_{i_(Ma=0)}$ for incompressible flow, the flex factor k_i, and the correction factor a_i—are required to be estimated. To such a model the output-error method discussed in Chapter 4 can be readily applied. Simultaneous estimation of rigid-body derivatives and flex factors requires a judicious choice of dynamic flight maneuvers and flight conditions as well as estimation software with multirun analysis capability. The preceding approach has been applied to flight data.

For example, for flight data gathered from a wide-body transport aircraft, it has been demonstrated that concatenating flight maneuvers at several selected flight conditions the dynamic pressure-dependent quasi-steady effects due to structural deformation and rigid-body aerodynamic derivatives can be separated and determined using the output-error method discussed in Chapter 4. The flight data analyzed covered a range of Mach number from 0.4 to 0.81, flight levels from 100 to 300, and dynamic pressure from 85 to 163 hPa. As a typical example, Fig. 12.46 shows the lift-curve slope as a function of Mach number showing the rigid-body part ($\overline{q} = 0$) as well as the dynamic pressure influence [59]. Prediction and flight estimates show a reasonable agreement for the primary derivatives, which were accurately estimated. Other parameters, for example, damping and control derivatives, often showed somewhat larger discrepancies between predictions and flight estimates, as structural influences on these derivatives are much more complex to predict and can be nonlinear. The investigations clearly demonstrate that separation of the Mach-number effects and flex factors was possible through judicious choice of flight data. The effects due to structural flexibility could be adequately accounted for in the lift gradient and other derivatives through linear flex factors. In the present case, the separation between the frequency of the first structural mode and that of the rigid-body mode for the investigated flight condition was higher than the usually recommended ratio of five [60].

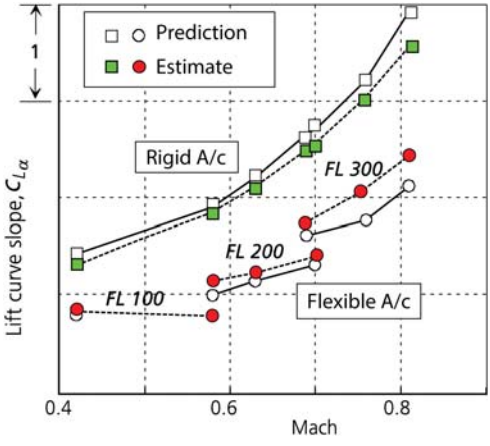

Fig. 12.46 Flight-estimated and -predicted lift-curve slopes of a typical large transport aircraft.

The preceding approach is based on the classical equations of aircraft motion incorporating conventional stability and control derivatives. The approach

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does not require any a priori information about mode shapes and modal mass; the complex generalized equations for elastic degrees of freedom as additional differential equations, discussed in the next section, are avoided. Under the assumptions made, namely, the rigid-body and structural frequencies are sufficiently separated, it is fairly straightforward to apply. The flight mechanics simulations, typically incorporated in the pilot training simulators, use such models including the flex factors. However, for certain flight vehicles such as very large transport aircraft or sailplanes, the lower structural weight and longer wingspan reduce the natural frequencies of the structural modes. In certain cases, these frequencies are in the same range or quite close to the rigid-body frequencies. In such a case the classical approach of treating flight mechanics and aeroelasticity separately as two decoupled dynamic systems is no longer valid. A simplified approach based on a first-order lag to represent aeroelastic effects is feasible to extend the scope of the rigid-body models [61]. We do not pursue this approach because it is associated with limitations and not sufficient for high-fidelity models we are trying to arrive at. A more general model for the dynamics of flexible aircraft that includes the coupling between rigid-body and structural dynamics is necessary. Such a model is generally termed as flexible aircraft model or as an integrated model.

12.9.2 GENERALIZED APPROACH

The generalized approach calls for explicit representation of rigid-body dynamics and of dynamics of structural deformation. Furthermore, to enable parameter estimation from flight data in time domain, the model must adequately represent the aerodynamic and structural characteristics, retaining the salient dynamic effects, through a finite number of parameters and also be amenable to state space representation. Such an approach and model was developed by Waszak and Schmidt [62]. The equations of motion of a flexible aircraft are derived for the mean axes reference system under the following assumptions: 1) structural deformation is assumed to be small and described by a finite set of orthogonal modes, and 2) moments and products of inertia do not vary with the deformation [60, 62, 63]. A thorough treatment including derivation, assumptions, and all other aspects related to flexible aircraft are found in [64]. The said model formulation has been widely used in investigations pertaining to flexible aircraft analysis and also parameter estimation of flexible aircraft, but mostly based hitherto on the simulated data from the aircraft model provided therein [65–67]. It has also been recently applied to flight data gathered from a highly flexible sailplane [68, 69]. Though we restrict ourselves to identification in the time domain, it is also possible to apply the frequency-domain techniques to identify generalized flexible aircraft models [70, 71].

The generalized approach incorporates the classical rigid-body dynamics and the elastic degrees of freedom explicitly. Each structural mode is modeled through additional differential equations, and the integrated model includes coupling

between rigid-body and structural dynamics. It can be applied to longitudinal and lateral-directional modes of aircraft motion. The first assumption, namely, structural deformation is assumed to be small and described by a finite set of eigenmodes, allows us to write the structural deformation $\mathbf{d}$ at a point (x, y, z) at any instant of time t as

$$\mathbf{d}(x, y, z, t) = \sum_{i=1}^{n_F} \Phi_i(x, y, z)\ \eta_i(t) \tag{12.43}$$

where n_F is the number of structural modes to be accounted for, Φ_i represents the shape of the ith structural mode, and η_i represents its generalized displacement. In practice, n_F is usually limited to a finite number of modes closest to the rigid-body modes. A generally accepted rule of thumb suggests that a structural mode needs to be explicitly incorporated into the dynamic model if its natural frequency is lower than five times the highest rigid-body mode frequency [60, 71]. To demonstrate the basic principle of an integrated model, the attention is restricted to the longitudinal dynamics accounting only for the first symmetric wing bending mode.

For the considered simplified case of accounting only for the first symmetric wing bending mode, the equations of motion can be simplified to [62, 64]

$$\begin{aligned}
\dot{u} + qw + g\sin\theta &= \frac{X}{m} \\
\dot{w} - qu - g\cos\theta &= \frac{Z}{m} \\
\dot{\theta} &= q \\
\dot{q} &= \frac{M_y}{I_{yy}} \\
\dot{h} &= u\sin\theta - w\cos\theta \\
\ddot{\eta}_1 + 2\zeta_1\omega_1\dot{\eta}_1 + \omega_1^2\eta_1 &= \frac{Q_1}{m_1}
\end{aligned} \tag{12.44}$$

We note that the first five equations are the well-known, classical flight mechanics equations, with the only difference that in the present case the total aerodynamic forces along the longitudinal and vertical axes, denoted by X and Z, and the total aerodynamic pitching moment M_y, include the effects due to structural deflections. The additional equation characterizes the structural modal deflection η_1, in the present case first symmetric wing bending mode, where m_1, ω_1, and ζ_1 are the modal mass, in vacuo natural frequency, and damping ratio respectively, and Q_1 is the generalized aerodynamic force acting on this mode. If further structural modes are to be included in the analysis, we get one additional second-order differential equation for each additional mode considered in the model. The

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second assumption, namely, moments and products of inertia do not vary with the deformation, leads to coupling resulting only through the aerodynamic forces appearing on the right-hand side of the equations.

The total aerodynamic forces are obtained by summation of forces resulting from rigid-body motion and control inputs, denoted by subscript R, and due to aeroelastic motion, denoted by subscript F.

$$\begin{aligned} X &= X_R + X_F; \qquad Z = Z_R + Z_F \\ M_y &= M_{yR} + M_{yF}; \quad Q_1 = Q_{1R} + Q_{1F} \end{aligned} \tag{12.45}$$

It is now necessary to postulate appropriate models for these forces and moments. We adopt here an approach based on stability and control derivatives, which is amenable to parameter estimation from flight data in the time domain.

12.9.2.1 RIGID-BODY AERODYNAMICS

The rigid-body aerodynamic forces and the pitching moment are given by

$$\begin{aligned} X_R &= \overline{q} S C_{XR} \\ Z_R &= \overline{q} S C_{ZR} \\ M_{yR} &= \overline{q} S \overline{c} C_{mR} \end{aligned} \tag{12.46}$$

where $\overline{q}$ is the dynamic pressure, S the wing reference area, and $\overline{c}$ the mean aerodynamic chord. The longitudinal and vertical force coefficients C_{XR} and C_{ZR} are obtained through axes transformation from experimental to body-fixed axes system:

$$\begin{aligned} C_{XR} &= C_{LR} \sin\alpha - C_D \cos\alpha \\ C_{ZR} &= -C_{LR} \cos\alpha - C_D \sin\alpha \end{aligned} \tag{12.47}$$

The lift coefficient is modeled as

$$C_{LR} = C_{L0} + C_{L\alpha}\,\alpha + C_{Lq}\,\frac{q\overline{c}}{2V} + \frac{S_H}{S}\,C_{L\delta e}\,\delta_e + C_{L\delta f}\,\delta_f \tag{12.48}$$

where S_H is the horizontal tail reference area. Total drag is given by the drag polar as a function of lift coefficient and a contribution proportional to the flap deflection:

$$C_D = C_{D_0} + \frac{C_{LR}^2}{e\,\pi\,AR} + C_{D\delta f}\,\delta_f \tag{12.49}$$

where AR is the aspect ratio and e the Oswald factor characterizing the increase in induced drag over the ideal condition caused by nonelliptical lift distribution.

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The pitching-moment coefficient referred to the aircraft center of gravity is modeled as

$$C_{mR} = C_{m0} + C_{m\alpha}\alpha + (C_{mq} + C_{mq\alpha}\alpha)\frac{q\bar{c}}{2V} + C_{m\dot{\alpha}}\frac{\dot{\alpha}\bar{c}}{2V} + \frac{S_H r_H}{S\bar{c}} C_{L\delta e}\delta_e + C_{m\delta f}\delta_f \tag{12.50}$$

where r_H is the horizontal distance between the aircraft center of gravity and the aerodynamic center of the horizontal tail.

We note that the preceding postulated model considers the unsteady aerodynamic effects in the rigid body through the rate of change of angle-of-attack $\dot{\alpha}$ derivative. For demonstration purposes, linear terms are used to model the lift and pitching-moment changes due to flaps. In practice, they are slightly more complex. Other forms of model postulates for the rigid-body aerodynamics, such as a two-point model considering the wing body and horizontal tail separately discussed in Secs. 12.2 and 12.8.1, can also be used.

12.9.2.2 AERODYNAMIC EFFECTS OF FLEXIBILITY

The generalized derivatives that represent the effects of modal displacements and rates on the aerodynamic forces, moments, and generalized loads can be derived based on a quasi-steady strip theory [62]. This, however, requires a priori knowledge of the modal shapes, which can be obtained from a finite element analysis or through a ground vibration test. In the present case such a priori knowledge about mode shapes was not available. To formulate models for these forces and moments, we use, however, the same approach based on the generalized derivatives, estimating their numerical values from flight tests.

In general, the shapes of the structural eigenmodes correspond to a combination of bending and torsion of the lifting surfaces and the fuselage. As such, the forces and moments resulting from the structural modes are functions of the modal deflections and their rates. Accordingly, in the present case, the contributions of the first aeroelastic mode (symmetrical wing bending) to the vertical force, to the pitching moment, and to the generalized load are modeled as

$$\begin{aligned} Z_F &= \bar{q}S\left(C_{Z\eta 1}\eta_1 + C_{Z\dot{\eta}1}\frac{\dot{\eta}_1\bar{c}}{2V}\right) \\ M_{yF} &= \bar{q}S\bar{c}\left(C_{m\eta 1}\eta_1 + C_{m\dot{\eta}1}\frac{\dot{\eta}_1\bar{c}}{2V}\right) \\ Q_{1F} &= \bar{q}S\bar{c}\left(C_{\eta 1\eta 1}\eta_1 + C_{\eta 1\dot{\eta}1}\frac{\dot{\eta}_1\bar{c}}{2V}\right) \end{aligned} \tag{12.51}$$

where η_1 is modal deflection and $\dot{\eta}_1$ its time rate of change. It is assumed that the longitudinal force due to the first wing bending mode can be neglected, that is, $X_F = 0$.

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In the original model formulation, the effects of the control deflections and rigid-body motion on the elastic degrees of freedom are obtained from the quasi-steady strip theory and are functions of flow angles, control deflections, angular rates, structural deformations, and their rates. In the present example, the contributions to the generalized load on the first structural mode due to rigid-body motion are modeled as

$$Q_{1R} = \overline{q} S \overline{c} \left(C_{\eta 1 \delta e} \delta_e + C_{\eta 1 \delta f} \delta_f + C_{\eta 1 \alpha} \alpha + C_{\eta 1 q} \frac{q \overline{c}}{2V} \right) \tag{12.52}$$

The preceding model does not include the effects due to unsteady aerodynamics, which would increase considerably the complexity of the formulation. Now, combining Q_{1F} from Eq. (12.51) and Q_{1R} from Eq. (12.52) in Eq. (12.45) to get Q_1, and then substituting the same in the last equation in (12.44) leads to the differential equation for the dynamics of the first structural mode given by

$$\begin{aligned} \ddot{\eta}_1 = & -2\zeta_1 \omega_1 \dot{\eta}_1 - \omega_1^2 \eta_1 + \frac{\overline{q} S \overline{c}}{m_1} \left(C_{\eta 1 \eta 1} \eta_1 + C_{\eta 1 \dot{\eta} 1} \frac{\dot{\eta}_1 \overline{c}}{2V} \right) \\ & + \frac{\overline{q} S \overline{c}}{m_1} \left(C_{\eta 1 \delta e} \delta_e + C_{\eta 1 \delta f} \delta_f + C_{\eta 1 \alpha} \alpha + C_{\eta 1 q} \frac{q \overline{c}}{2V} \right) \end{aligned} \tag{12.53}$$

From Eq. (12.53), we observe that there are two terms that depend on η_1 and two terms that depend on $\dot{\eta}_1$. If the parameter estimation is carried out for a single dynamic pressure, attempts to identify simultaneously the intrinsic modal characteristics, namely, the natural frequency ω_1 and damping ζ_1, and the aerodynamic parameters $C_{\eta 1 \eta 1}$ and $C_{\eta 1 \dot{\eta} 1}$ would lead to identifiability problems due to very high correlations.

Restricting ourselves to analysis of flight data at a single flight condition, we overcome the problem of high correlation by identifying equivalent modal frequency and damping:

$$\begin{aligned} \omega_{\text{eq}}^2 &= \omega_1^2 - \frac{\overline{q} S \overline{c}}{m_1} C_{\eta 1 \eta 1} \\ \zeta_{\text{eq}} &= \frac{1}{\omega_{\text{eq}}} \left(\zeta_1 \omega_1 - \frac{\overline{q} S \overline{c}^2}{4 V m_1} C_{\eta 1 \dot{\eta} 1} \right) \end{aligned} \tag{12.54}$$

Substituting the equivalent parameters into Eq. (12.53) leads to

$$\begin{aligned} \ddot{\eta}_1 = & -2\zeta_{\text{eq}} \omega_{\text{eq}} \dot{\eta}_1 - \omega_{\text{eq}}^2 \eta_1 + \frac{\overline{q} S \overline{c}}{m_1} \\ & \times \left(C_{\eta 1 \delta e} \delta_e + C_{\eta 1 \delta f} \delta_f + C_{\eta 1 \alpha} \alpha + C_{\eta 1 q} \frac{q \overline{c}}{2V} \right) \end{aligned} \tag{12.55}$$

From Eq. (12.55), we notice that simultaneous estimation of the modal mass m_1 and of the aerodynamic derivatives $C_{\eta 1 \delta e}$, $C_{\eta 1 \delta f}$, $C_{\eta 1 \alpha}$, and $C_{\eta 1 q}$ is not possible

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because they would be linearly correlated. As the modal mass depends on the modal shape and is not known a priori, without a loss of generality, we assume $m_1 = 1\,\text{kgm}^2$ for parameter estimation purposes. Of course, for parameter estimation purposes the preceding second-order differential equation is rewritten into two first-order differential equation as seen in Eqs. (11.14) and (11.15).

12.9.2.3 MEASUREMENTS AND FLIGHT DATA ANALYSIS

For parameter estimation purposes, it is also necessary to define the observation variables that contain adequate information to allow determination of aerodynamic derivatives pertaining to the rigid-body and the structural dynamics. As in the various applications discussed hitherto in the book, for the estimation of rigid-body aerodynamics we need the standard set of parameters, such as flow variables, angular rates, accelerations at any one location, typically close to the center of gravity, and control surface deflections. This is, however, not sufficient for identification of an integrated model incorporating the structural dynamics as discussed here. To enable estimation of the shapes of the structural modes we need additional accelerations at various suitable points along the wingspan, the length of the fuselage and the vertical tail. For the estimation of the first wing bending mode, we need vertical accelerations at multiple points along the wingspan.

The assumption that structural deformations are small and can be represented by a linear combination of orthogonal modes allows us to compute the vertical acceleration measured at a point (x_k, y_k, z_k) from the center of gravity as a sum of the accelerations at the center of gravity, the inertial effects resulting from the displacement relative to the center of gravity, and the effects of the structural motion. It is given by

$$a_{zk} = a_{z\text{CG}} + x_k\,\dot{q} - z_k\,q^2 + \Phi_z^k\,\ddot{\eta}_1 \tag{12.56}$$

where Φ_z^k is the z-component of the vector function $\Phi(x, y, z)$ appearing in Eq. (12.43) that represents the modal shape at the measurement point k and is a parameter to be estimated [64, 71]. The first three terms on the right-hand side are obtained from Eq. (10.8) for the simplified case of longitudinal motion only. The last term is the contribution due to the modal acceleration. Equation (12.56) is written by considering one single structural mode. If multiple modes are to be accounted for, the last term in the right-hand side is substituted by a summation of the contributions from each individual mode. Similar equations can be easily written for other accelerations as well, if required in the model.

To further augment the information content for better estimation of modal states, we also measure the strains at a few points along the wing. The transient stress at any point on a flexible structure is computed using the so-called "mode displacement method," given by a linear combination of the individual contributions of each normal mode [72]. The assumption of structural

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deformation being small allows linear formulation of stress-strain and strain-displacement characteristics. Accordingly, the strain at an arbitrary point k on the structure due to the first structural mode is given by

$$\varepsilon_k = \varepsilon_0^k + \varepsilon_{\eta 1}^k \, \eta_1 \tag{12.57}$$

where ε_0^k represents the zero (residual) term or bias in the measurement and $\varepsilon_{\eta 1}^k$ the sensitivity, that is, the effect of the modal deflection on the strain. The measurements of strain are sensitive to temperature variations; hence, care needs to be taken while using such data. Otherwise, a temperature-dependent correction factor can be included.

The just-discussed integrated model is applied to analyze flight data gathered from a glider SB 10; see Fig. 12.47. It is a high-performance two-seater glider designed and manufactured by Akaflieg Braunschweig. It has a wingspan of 26 m with aspect ratio of 31.0, extendable to 29 m through the attachment of wing-tip extensions resulting in an aspect ratio of 36.63. The wing reference area S is 21.81 m^2, and the mean aerodynamic chord $\bar{c}$ is 0.872 m. The very high aspect ratio, combined with the lightweight composite structure, leads to high flexibility; the first symmetric bending mode has a natural frequency of the order of 1.2 Hz, which is in the same range as some modes of the rigid-body aircraft motion. All of the flight controls (elevator, ailerons, rudder, and flaps) are mechanically actuated [68, 69].

The test vehicle was instrumented appropriately to enable parameter estimation from flight data. Pertinent to rigid-body motion, the inertial measuring unit (IMU) provided the measurement of linear accelerations, angular

Fig. 12.47 Glider SB 10.

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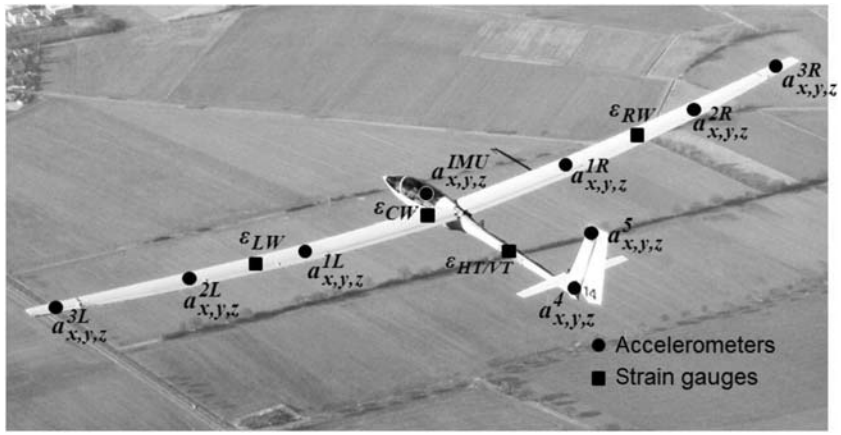

Fig. 12.48 Distribution of accelerometers and strain gauges on the flight-tested glider SB 10.

rates, and attitude angles. The flow angles are measured by vanes located at the tip of a mast mounted on the right wing at about one-third of the wingspan, and control surface deflections were recorded by potentiometers. The indicated airspeed, static pressure, and temperature were tapped from the basic aircraft system. In addition to the preceding standard set of parameters, accelerations along the three axes were recorded at several locations along the wingspan, fuselage and vertical tail; see Fig. 12.48. The locations of the three accelerometers, roughly at one-third and two-thirds of the wingspan and at the wing tip, on the right wing are denoted as 1R, 2R, and 3R, respectively; the corresponding locations on the left wing as 1L, 2L, and 3L. The locations of the accelerometers mounted on the horizontal tail and on the tip of the vertical tail are denoted as 4 and 5. Furthermore, strain gauges were mounted on the wings and the fuselage to measure the strains. The locations of these four strain gauges are denoted as CW (center wing), RW (right wing), LW (left wing), and HT/VT (horizontal/vertical tail).

Based on the measurements available from the instrumented test vehicle, the vector of observation variables in the parameter estimation consisted of

$$y = [a_{x\mathrm{IMU}}, a_{z\mathrm{IMU}}, V, \alpha, \theta, q, \dot{q}, h, p_s, T_s, a_{z1}, a_{z2}, a_{z3}, a_{x3}, a_{z4}, \varepsilon_{\mathrm{CW}}, \varepsilon_{\mathrm{RW}}, \varepsilon_{\mathrm{LW}}, \varepsilon_{\mathrm{VT}}] \tag{12.58}$$

where a_{z1}, a_{z2}, a_{z3} represent the mean values of the accelerations measured at the locations (1R, 2R, 3R) and (1L, 2L, 3L) on the right and left wings, for example,

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$a_{z1} = (a_{z1L} + a_{z1R})/2$. The parameters to be estimated from the flight data are given by

$$\Theta = [C_{L0}, C_{L\alpha}, C_{Lq}, C_{L\delta e}, C_{L\delta f}, C_{D0}, e, C_{D\delta f}, C_{m0}, C_{m\alpha}, C_{mq}, C_{mq\alpha}, C_{m\dot{\alpha}}, C_{m\delta f},$$
$$\omega_{1\text{eq}}, \zeta_{1\text{eq}}, C_{\eta 1\delta e}, C_{\eta 1\delta f}, C_{\eta 1\alpha}, C_{\eta 1q}, C_{z\eta 1}, C_{z\dot{\eta}1}, C_{m\eta 1}, C_{m\dot{\eta}1},$$
$$\Phi_z^{\text{IMU}}, \Phi_z^1, \Phi_z^2, \Phi_z^3, \Phi_z^4, \ \varepsilon_0^{\text{CW}}, \varepsilon_0^{\text{RW}}, \varepsilon_0^{\text{LW}}, \varepsilon_0^{\text{VT}}, \varepsilon_{\eta 1}^{\text{CW}}, \varepsilon_{\eta 1}^{\text{RW}}, \varepsilon_{\eta 1}^{\text{LW}}, \varepsilon_{\eta 1}^{\text{VT}}]^T \tag{12.59}$$

The parameters appearing in the first line on the right-hand side of Eq. (12.59) pertain to the rigid-body aerodynamics appearing in Eqs. (12.48–12.50). The first two parameters in the second line are ($\omega_{1\text{eq}}$, $\zeta_{1\text{eq}}$), the equivalent natural frequency and damping of the first structural mode appearing in Eq. (12.55). The next four parameters in the same line ($C_{\eta 1\delta e}$, $C_{\eta 1\delta f}$, $C_{\eta 1\alpha}$, $C_{\eta 1q}$) represent the influence of control deflections, angle of attack, and pitch rate on the generalized load acting on the first flexible mode. The last four parameters in the second line ($C_{z\eta 1}$, $C_{z\dot{\eta}1}$, $C_{m\eta 1}$, $C_{m\dot{\eta}1}$) are the vertical aerodynamic force and pitching-moment coefficients due to the modal deflection and its rate of change. The first five parameters (Φ_z^{IMU}, Φ_z^1, Φ_z^2, Φ_z^3, Φ_z^4) in the third line in Eq. (12.59) are the vertical deflections at the corresponding sensor locations indicated by the respective superscripts due to the unitary modal displacement appearing in equations similar to Eq. (12.56) for each of the accelerometers. The vertical deflection at the accelerometer near the wing tip Φ_z^3 is normalized to 1 m and kept fixed and serves as a scale factor for the other displacements. Because the sensors were symmetrically mounted on the left and right wing, it is sufficient to estimate only one set of these mode shape parameters, which are then valid for accelerometers on both wings. If antisymmetric structural modes are to be taken into account, a similar logic is valid, recalling that the displacements at corresponding positions on both sides of the aircraft have opposed senses. Figure 12.49 shows schematically the shape of the first symmetric wing bending mode. The last eight parameters ($\varepsilon_0^{\text{CW}}$, $\varepsilon_0^{\text{RW}}$, $\varepsilon_0^{\text{LW}}$, $\varepsilon_0^{\text{VT}}$) and ($\varepsilon_{\eta 1}^{\text{CW}}$, $\varepsilon_{\eta 1}^{\text{RW}}$, $\varepsilon_{\eta 1}^{\text{LW}}$, $\varepsilon_{\eta 1}^{\text{VT}}$) represent the residual/bias terms and the sensitivity to the respective modal deflections appearing in equations similar to Eq. (12.57).

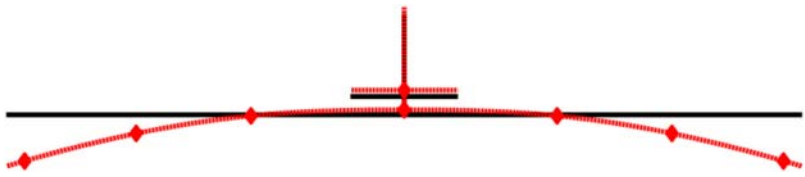

Fig. 12.49 Shape of first symmetric wing bending mode.

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The data were gathered with a sampling rate of 100 Hz. Because the flight maneuvers were performed manually by the pilot and as the maximum frequency that can be excited by a human pilot is below 20 Hz, the sampling rate of 100 Hz is quite sufficient to estimate various structural modes. As discussed in Chapter 2, a lower sampling rate, typically 25–50 Hz, is usually sufficient for identification of rigid-body aerodynamics only.

A comprehensive flight-test program, consisting of 11 flights each of a duration of 1 h and covering 72 test points, was carried out to gather the flight data for parameter estimation purposes [68, 69]. For the configuration flown, the aircraft mass was 783 kg and the center of gravity 209.7 mm behind the reference datum. All of the test points were initiated with the flaps lever set to position −1. The dynamic maneuvers were initiated from indicated airspeeds of 110 and 160 km/h to enable assessment of the effects of dynamic pressure changes on the aircraft responses. Classical maneuvers for system identification in time domain were manually performed using ailerons, elevator, rudder, and flaps inputs. A typical multistep input maneuver was 10 to 25 s long, including the initial stabilization and free response. The duration of the pulse input maneuver exciting the low frequency and weakly damped phugoid mode was roughly 60 s whereas the frequency sweeps were 50 to 140 s long. The time steps of the maneuvers were varied in order to maximize the frequency content of the inputs, enabling the excitation of the rigid-body and the structural modes. Additionally, frequency sweeps were performed to augment the data. In the present case, the flaps are used as dynamic inputs to provide a direct generation of lift on the wing, which is expected to be more efficient to excite the wing symmetric bending modes than the indirect effect obtained through elevator deflection, where the lift on the wing develops only after the angle-of-attack variations, which, due to the aircraft dynamics, are delayed and attenuated at higher frequencies with regard to the elevator inputs.

The 3-2-1-1 input maneuver, discussed in Chapter 2, was performed using all of the primary control surfaces and flaps, with targeted basic time steps of 1 and 0.3 s. The shorter time step of 0.3 s leads to higher frequency contents in the input signal, thereby exciting simultaneously both rigid-body and structural dynamics. The value of 0.3 s for the time step resulted from the rule $\Delta t_{3211} \approx 0.3/f_c$ discussed in Sec. 2.3.2; in the present case the estimated first wing bending frequency, as already mentioned, is 1.2 Hz. In practice, the pilot was able to achieve on an average a time step of 0.5 s. For demonstration purposes, we restrict our attention to a sequence containing an elevator pulse maneuver, two elevator 3-2-1-1 maneuvers (longer and shorter time steps), and two flaps 3-2-1-1 maneuvers (longer and shorter time steps) at a trim speed of 110 km/h. Prior to aerodynamic model estimation, a data compatibility check, discussed in Chapter 10, was carried out to calibrate the angle-of-attack and angle-of-sideslip measurements.

As a typical example, time histories for a 3-2-1-1 elevator input maneuver are shown in Fig. 12.50. The top-most plot shows the measured elevator deflection δ_e, followed by a comparison of measured and estimated responses for the angle of

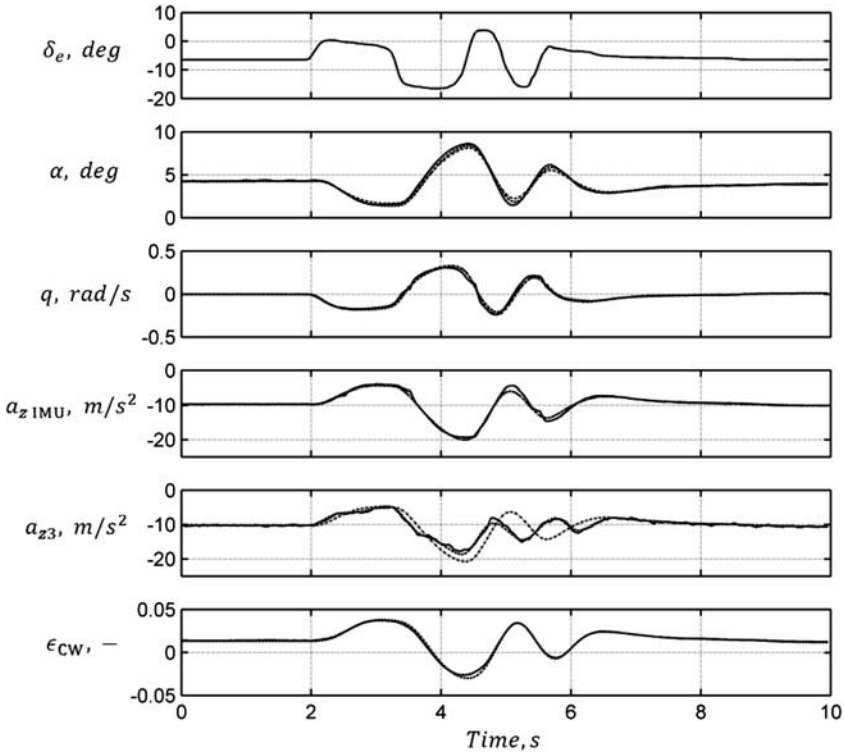

Fig. 12.50 Comparison of flight-measured and model-estimated responses for a 3-2-1-1 elevator input maneuver: ——, flight measured; ....., rigid-body model; - - - - -, flexible model.

attack α, the pitch rate q, the vertical acceleration near the center of gravity $a_{z\mathrm{IMU}}$, and finally the vertical acceleration at the wing tip a_{z3}. Each of those four plots contains three superimposed time histories. The last plot shows the match for the strain at the wing root provided by the flexible model. It is observed from the plot for a_{z3} that the match between the flight-measured data (shown by continuous lines) and those estimated by the rigid-body model, that is, neglecting the flexible modes (shown by dotted lines), for the acceleration at the wing tip a_{z3} shows discernible discrepancies whereas the extended flexible model shows a good match. This becomes all of the more evident from the enlarged plot of vertical acceleration a_{z3} shown in Fig. 12.51 for the portion of the maneuver with dynamic variations.

The integrated model accounting for the first wing bending mode yields, on the other hand, a significantly improved match (shown by dashed lines). The differences are minor for the acceleration measured near the center of gravity

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as seen from the match for the vertical acceleration in the cockpit $a_{z\mathrm{IMU}}$. It was also observed that the flexible model is found to be capable of estimating the strain on the aircraft structure with a good agreement as shown in the plot of the strain at the wing root $\varepsilon_{\mathrm{CW}}$, which makes this modeling approach useful for predicting structural loads in maneuvering flight. In general, it is noticed that the integrated model incorporating the rigid-body dynamics and the elastic degrees of freedom yields a significantly improved match between measured data and model-estimated responses for all accelerations measured off the center of gravity, for example, at the wing tip or the vertical tail. The farther the measurement point, the larger is the structural deformation and consequently the larger the effect of flexibility on the measured data. Such effects can be very dominant in the case of an aircraft with a longer and slender fuselage, large wings with thin airfoils, and high aspect ratios as in the present case of the glider SB 10. Such a flexible aircraft model including just the first symmetric wing bending mode is quite useful in practice because a major part of the improvements in the match between the measured data and the model responses results from this mode. Furthermore, this mode contributes mainly to the high dynamic loads and affects most the ride comfort; hence, accounting for this mode might be adequate in a few applications such as designing control laws.

Extending the philosophy presented so far in this section, further structural modes were added to the longitudinal model discussed in this section. Following the same procedure, the model for lateral-directional motion was extended, leading to a full 11-DOF model with a total of five structural modes modeled as

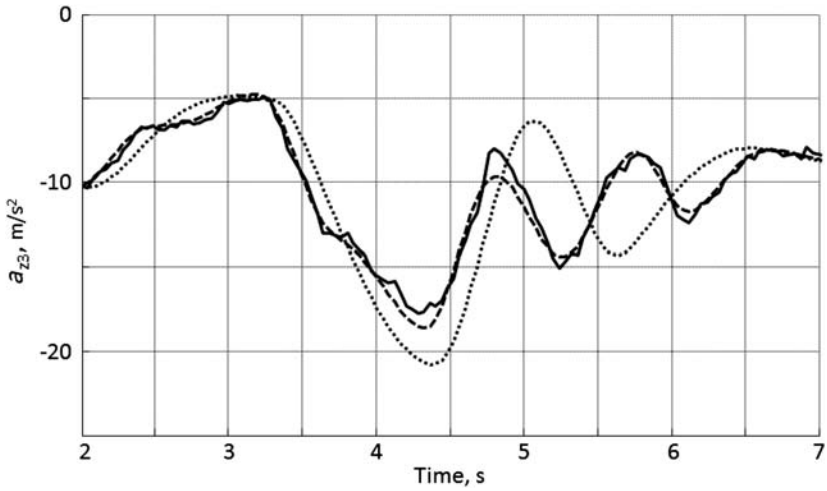

Fig. 12.51 Enlarged plot of flight measured and model estimated vertical acceleration during a 3-2-1-1 elevator input maneuver: ——, flight measured; ....., rigid-body model; - - - - , flexible model.

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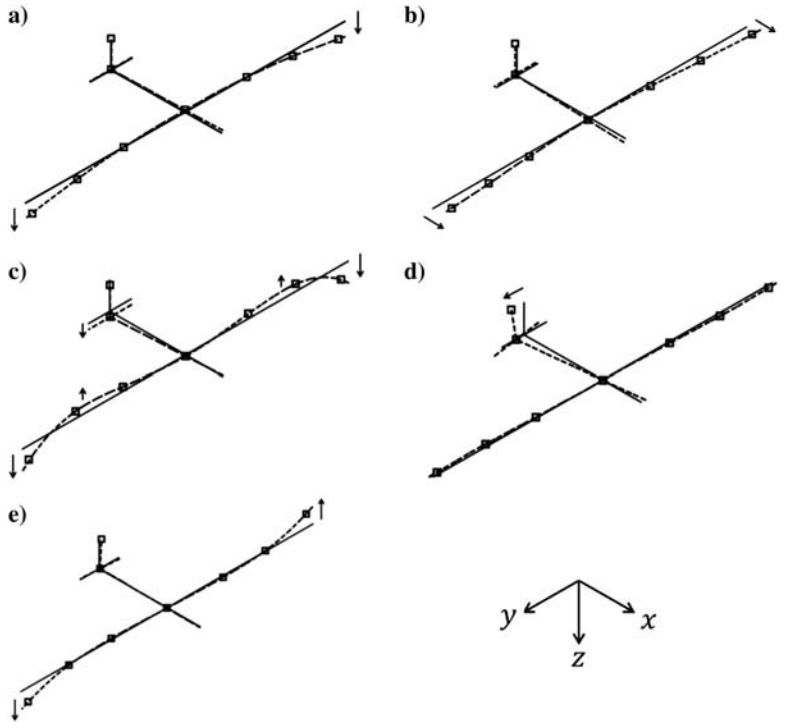

Fig. 12.52 Flight-estimated modal shapes: a) mode 1—first symmetric wing bending, b) mode 2—wing fore-aft bending, c) mode 3—second symmetric wing bending, d) mode 4—fuselage bending/torsion, and e) mode 5—first antisymmetric wing bending (Source: [69]).

dynamic systems. The five structural modes included three symmetric and two antisymmetric modal forms. They were first symmetric wing bending, wing fore-aft bending, second symmetric wing bending, fuselage bending/torsion, and first antisymmetric wing bending; see Fig. 12.52. The estimated frequencies of these structural modes were 8.38, 16.2, 25.0, 21.5, and 16.9 rad/s. The whole set of parameters in Eq. (12.59) was estimated in a single run using about 1300 s of flight data from 37 flight maneuvers. The maneuvers included elevator and flap pulse, 3-2-1-1 and frequency sweeps with all primary control and flaps, rudder doublets, and bank-to-bank maneuvers. The estimates of the parameters were found to be uncorrelated with low standard deviation indicating good identifiability. The additional accelerations, three on each wing and on horizontal and vertical tail, were found to be adequate for estimation of structural dynamics and other modal parameters. The measurements of vertical accelerations at three locations along the wing provide mostly the information to estimate the first and the second symmetric and the first antisymmetric wing bending

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modes whereas the wing fore-aft bending is mostly determined from the longitudinal accelerations close to the wing tip. The fuselage bending/torsion is determined from the lateral accelerations at the rear fuselage and the vertical tail. The wing fore-aft bending mode is also called in the literature as symmetric chord mode or as swing mode.

To include additional flexible modes, the model must be adapted accordingly. For example, the pitching moment due to structural modal displacements can be written as

$$M_{yF} = \bar{q}\, S\, \bar{c} \sum_{i=1}^{n_F} \left(C_{m\eta_i}\, \eta_i + C_{m\dot{\eta}_i} \frac{\dot{\eta}_i\, l}{2V} \right) \tag{12.60}$$

where the index i corresponds to the ith of the n_F structural modes incorporated into the dynamic model and l is a reference length, which is usually the mean aerodynamic chord length in case of symmetric modes and the wingspan in case of antisymmetric modes.

When multiple structural modes are modeled, the effects of flexibility on the accelerometers are composed by a superposition of the effects of each mode:

$$a_{zk} = a_{z\mathrm{CG}} + x_k(pr - \dot{q}) + y_k(qr + \dot{p}) - z_k\left(p^2 + q^2\right) + \sum_{i=1}^{n_F} \Phi^k_{z\eta_i}\, \ddot{\eta}_i \tag{12.61}$$

Comparing Eqs. (12.60) and (12.61) to their simplified versions for the case of symmetric motion only with one structural mode, Eqs. (12.51) and (12.56), it becomes clear that every mode incorporated into the dynamic model brings an increase in the complexity of the model pertaining to the number of parameters. Once again, the parsimony principle shall be a guide when choosing the number of structural modes to be retained in the formulation: inclusion of an additional flexible mode is justified only if the model accuracy is significantly improved, keeping in mind the desired application of the identified model.

One difficulty encountered in the preceding analysis was separate estimation of aerodynamic and intrinsic structural mode damping. The structural damping, represented in Eq. (12.53) by the variable ζ_1, is usually very small, of the order of 0.02 and seldom greater than 0.05 [72], whereas the total or equivalent damping ζ_{eq} can be much larger, typically in the range of 0.45, being mostly influenced by the aerodynamic term comprising the derivative $C_{\eta 1 \dot{\eta} 1}$, which plays a major role in matching the estimated responses to flight-measured data. Several attempts to separate them by analyzing maneuvers at two different airspeeds were not successful; the estimates were highly correlated. Therefore, only equivalent damping was estimated. A detailed discussion of all aspects of parameter estimation and of results of the current investigation based on a model pertaining to longitudinal motion including just the first wing bending mode, as well as those for extended analysis based on an 11-DOF model with identification of five structural modes, is beyond the scope of this book, which is restricted to demonstrating the concept

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and discussing the salient points from practical aspects. For more detailed information, the reader is referred to [69].

Nevertheless, it is important to point out that in this application the modal shapes were unknown a priori. They were estimated from the flight data through a systematic approach of analyzing the comparison of flight data with model responses and analyzing the residuals. Additional modal shapes were introduced only when improvements in the match were discernible, and the modal parameters were estimated with sufficient accuracy and were plausible. Estimation of aeroelastic states is not a trivial task because the displacement at any point is a sum of individual contributions from each elastic mode. It is also pertinent to point out that a direct measurement of the modal displacement was not available. The modal parameters were estimated from the indirect measurements corrupted by measurement noise, mainly from accelerations measured at several locations as discussed earlier. If mode shapes are known a priori and measurements of structural deformations are available and used as model inputs, the task would be much simpler. This would, however, call for more sophisticated sensors to measure structural deflections and strains, such as Fiber Bragg Grating. Installation and calibration of such sensors is involved and possible during the production phase and usually restricted to research aircraft but might be difficult for commercial aircraft. Thus, a much more generalized approach considering practical limitations of a priori knowledge about structural modal forms, of measurements, and of identifiability of parameters from flight maneuvers has been demonstrated and validated to estimate a flexible aircraft model in the time domain from flight data. It yields a parsimonious model that retains the salient dynamic effects, has requisite accuracy, and is valid over a broad frequency range.

In the case of the investigated flight vehicle, the improvements in the match for accelerations in the cockpit were marginal. This is attributed, as already discussed, to the fact that the offset between the cockpit and the center of gravity is not large. However, in the case of large transport aircraft, the cockpit is far ahead of the center of gravity. In such cases the flexible aircraft model yields inertial outputs with considerably improved fidelity. Such high-fidelity models are useful in flight simulators, flight dynamics, and handling-qualities investigations because the structural motion can influence the coupled motion and the flying qualities. Two other applications where comprehensive models as discussed here are required pertain to design of control laws to 1) improve passenger comfort in transport aircraft and 2) actively control the aeroelastic modes and/or alleviate loads due to gusts. In such applications, inertial measurements (accelerations, angular rates) are fed back through the control laws. These measured data can contain a significant contribution due to elastic deformation, which, if not correctly accounted for, can be further excited by the closed-loop control system, leading to instabilities. Common practice is to low-pass filter the inertial measurements above a certain frequency in order to eliminate the contributions of elastic modes; it is termed as passive approach. However, the increase of aircraft flexibility makes this approach much more difficult to adopt because of the

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closeness between rigid and flexible modes. Once again, an integrated flexible aircraft model would be advantageous in the design of better control systems. For the specific purpose of designing efficient active control laws to control the structural modes, leading to better handling qualities and passenger comfort, a different approach based on the Eigensystem Realization Algorithm is feasible and has been demonstrated on A340-600 aircraft [73]. Other applications of the flexible aircraft model are 1) assessment of stability in a broader sense, including flight mechanics and aeroelastic stability, and 2) analysis of structural loads due to maneuvers or atmospheric gusts.

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12.10 X-31A MODEL VALIDATION AND UPDATE

The U.S./German experimental aircraft X-31A is a highly control augmented fighter with enhanced maneuverability; see Fig. 12.53. Poststall maneuvering is enabled by applying advanced technologies such as high angle-of-attack aerodynamics and flight control system integrated thrust vectoring [74]. In Secs. 9.14 and 9.16.2, we have already addressed the challenges encountered in estimating the aerodynamic derivatives of the basic airframe. To recapture the essentials once again, the following was demonstrated:

1. Highly correlated control surface deflections and thrust vector vanes and states due to integrated flight control laws resulted in poor estimates with large standard deviations and large scatter.
2. Because of data collinearity, it was possible to estimate only combined derivatives from pilot input maneuvers.
3. Single surface excitation provided significantly improved estimates.
4. From a variety of algorithms in the time domain (artificially stabilized output-error method, filter-error method, equation decoupling, regression analysis, and extended and unscented Kalman filter techniques), the least-squares and filter-error methods accounting for process noise were found to be preferable.

Fig. 12.53 X-31A at 70-deg angle of attack.

A detailed description of X-31A system identification is beyond the scope of this book. A thorough treatment of various aspects is found in [48] and [75]. Principally, the results of X-31A system identification have been used to validate and in several cases update the wind-tunnel predicted database. We provide here two typical results. In Fig. 12.54a, we find the estimates of the dihedral effect obtained from pilot input maneuvers and those from the SSE inputs, which once again strongly support the conclusion drawn in Sec. 9.14, namely, SSE yields drastically improved estimates. It is also evident that the flight estimates did not confirm the wind-tunnel-predicted large value of the dihedral effect between 30–45-deg angle of attack. Similarly, noticeable discrepancies were also observed in the estimates of the directional stability (Fig. 12.54b; QT, quasi-tailless configuration).

In conclusion, diligent modeling and a suitable choice of estimation technique provided flight-validated aerodynamic characteristics, including poststall regime and thrust vector control. Furthermore, the crux of the problem pertaining to highly augmented aircraft was the data correlation and insufficient excitation, and an efficient solution to this problem was provided by separate surface excitation. System identification techniques provided improved results for flight-test

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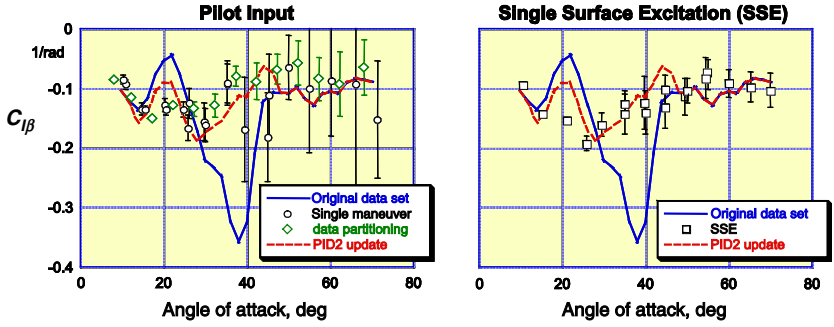
a) Dihedral effect



b) Directional stability

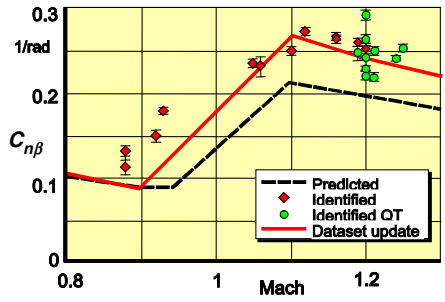


Fig. 12.54 Examples of X-31A database updates.

planning, flight envelope expansion, and a database for simulation and control law modification and validation [2, 75].

Two other advanced applications in the area of unstable aircraft and poststall maneuvering using thrust vectoring are the X-29A and F-18 HARV. The prime research objective of the X-29A program was to test the predicted aerodynamic advantages of the unique forward swept-wing configuration. The aircraft is statically unstable in the longitudinal axis, with a negative static margin of up to 35% at subsonic speeds (time to double of about 0.15 s). In the lateral-directional axes, it is statically stable [76, 77]. The F-18 High Angle of Attack Research Vehicle (HARV) with multi-axis thrust vectoring is a widely used test bed for research investigations [78, 79]. In both of these cases, the estimation of aerodynamic stability and control derivatives of the open-loop aircraft posed problems that we have talked about earlier in this section and also in Chapter 9. The uncommanded motion caused by unsteady aerodynamic called for the use of estimation methods accounting for process noise, like least squares or filter error. The high-gain feedback led to correlated motion and control variables, and under such cases the single-surface excitation provided significantly improved estimates of the aerodynamic derivatives. For exact details of these challenging projects, the reader is referred to the respective literature just cited.

12.11 WAKE-VORTEX AIRCRAFT ENCOUNTER MODEL

Wake-vortex encounters are critical situations that can result in flight conditions dominantly affecting the aircraft reaction during safety-critical flight phases. Optimization of air traffic distribution and volume in the vicinity of airports, particularly under a mix of light, medium, and heavy aircraft, calls for a better understanding of the basic physics underlying this phenomenon [4, 80]. For this purpose, validated models for aerodynamic flowfield characteristics and aircraft wake-vortex encounter reaction models are indispensable.

Aerodynamic interaction models (AIM) are an indispensable part of flight mechanic simulation models concerned with *wake vortex encounter* investigations, for example, simulator studies or encounter risk assessment. Two typical AIMs are the strip method and the lifting surface method [81, 82]. Both can be adapted to various aircraft configurations and compute the wake-vortex flowfield induced additional forces and moments that are added to the forces and moments computed by the basic aircraft aerodynamic model. For reliable simulation results, the quality of the AIM model is of prime importance. Up to now, quality has often been proven by comparing model forces and moments to static wind-tunnel measurements, in which the follower aircraft is set to distinct positions behind a wake-generating configuration.

Within the 5th European framework technology project S-WAKE [82], one objective was to validate the AIMs using data of *full-scale flight tests*, which were carried out with three aircraft: ATTAS of DLR, a Do-128 of Technical

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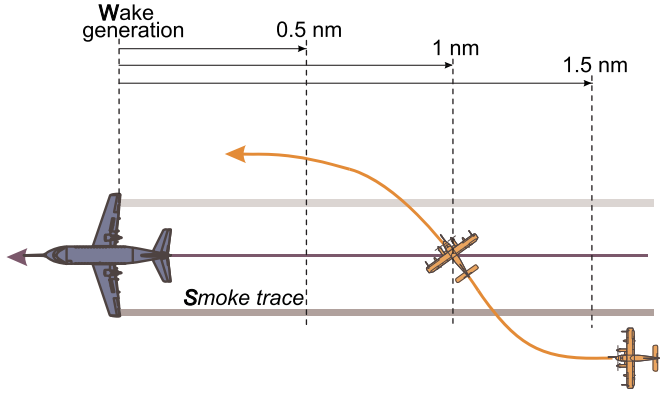

Fig. 12.55 Schematic of encounters flown.

University of Braunschweig, and a Cessna Citation of NLR, the Netherlands. The task of the leading aircraft ATTAS (ICAO separation class: medium) was to generate and visualize the wake with a wing-mounted smoke generator. The follower aircraft (separation class: light) encountered the ATTAS wake at different, distinct horizontal distances (Fig. 12.55). Both of the follower aircraft were fully instrumented with inertial sensors and several flow probes in order to measure the flowfield characteristics of the wake vortex as well as the aircraft response during each encounter. About 100 encounters were flown under steady atmospheric conditions, and bank angles up to 80 deg could be observed, with typical bank angles of 30–40 deg during a lateral encounter. Bank angle is only one aspect of wake encounter. It results in a bump and can be uncomfortable, but only for a short duration, and it does not usually lead to loss of control because the banking motion is averaged out. More important is the lateral acceleration resulting from such an encounter because it might lead to injury to crew or passengers. Similarly, vertical motion is important for separation.

The flight data analysis composed a two-step procedure depicted in Fig. 12.56 consisting of 1) flowfield characterization through an analytical wake vortex model incorporated in the flight-path reconstruction and estimating the vortex-model parameters and 2) encounter model validation through simulation, using prevalidated basic rigid-body model and identified wake-vortex characteristics in analytical form. The link between the two steps is shown by the dotted line in the figure.

The most critical part of this complex task is the determination of the wake model parameters, which comprises four subparts of a single overall computational procedure, shown in Fig. 12.57. During the first part, the classical flight-path reconstruction (FPR) technique, elaborated in Sec. 10.2 with $(u, v, w, \phi, \theta, \psi, x_g, y_g, h)$ as the state variables and the measured linear

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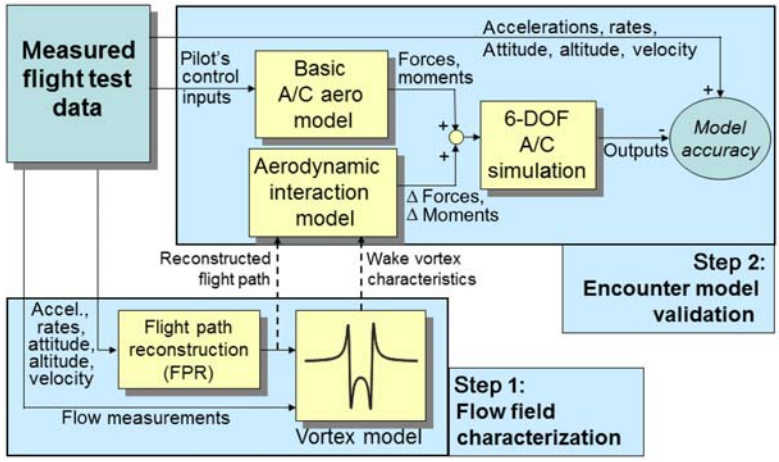

Fig. 12.56 Schematic of two-step procedure for vortex model identification and validation.

accelerations (a_x, a_y, a_z) and angular rates (p, q, r) as the inputs, is applied to obtain a c.g. flight path in the geodetic coordinate system, whereby the input sensor biases are also eliminated. In the second part, the x_g, y_g, h positions of each flow angle sensor are reconstructed. In the third part, these sensor positions are transformed into the wake axis system, which is the same as the geodetic system but twisted through ψ_W, the wake system azimuth angle. In the last part, the resulting sensor positions in the wake y–z plane are used as inputs to the analytical wake model fitting its outputs to the measured flow sensor information (e.g., angle of attack and sideslip angles) in order to identify the wake model parameters, namely, the circulation, core radius, and position of left and right vortex in the wake axis system.

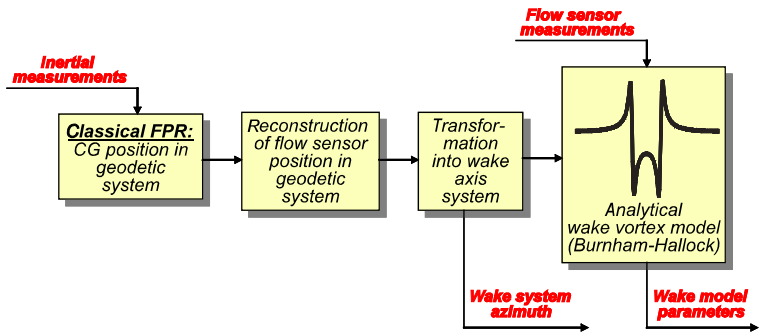

Fig. 12.57 Computational parts of wake model parameter estimation.

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There are two important issues related to the practical application of the preceding procedure. First, it requires accurately precalibrated measurements of the flow angles. Second, the modeling and identification is very sensitive to the wake system azimuth ψ_W. This azimuth angle cannot be measured, but is roughly equal to the heading of the aircraft generating the wake. Because ψ_W is a critical parameter, which has to be known accurately, it is desirable to identify the same for each individual wake fly-through. Identification of ψ_W is possible if flow angle measurements at several different locations, for example, nose and wing tips, are available. If flow angles are measured at a single location, which is mostly the case, identification is still possible, but only when the lateral distance between the two vortices is known a priori, which should be $\pi/4$ of the wingspan of aircraft generating wake, assuming elliptical wing lift distribution.

12.11.1 DETERMINATION OF FLOWFIELD CHARACTERISTICS

Precise determination of the flowfield characteristics to which the follower aircraft is subjected to during each of its individual encounters is a prerequisite and critical to aerodynamic encounter model validation. Two different analytical velocity distributions (*Burnham–Hallock* and *Lamb–Oseen*) were investigated to match the measured flowfield characteristics [83, 84]. First, the flight path was reconstructed precisely using system identification techniques. Then, the parameters of the analytical wake-vortex model were identified [85]. The model consists of two idealized, superimposed counter-rotating single vortices (see Fig. 12.58) characterizing the aircraft wake-vortex system after wake roll-up. The model parameters are vortex circulation Γ, core radius r_c, lateral vortex separation b_V, and vortex location in space. The tangential velocity of one vortex as a function of the distance from the core $V_t(r)$ is described in terms of the

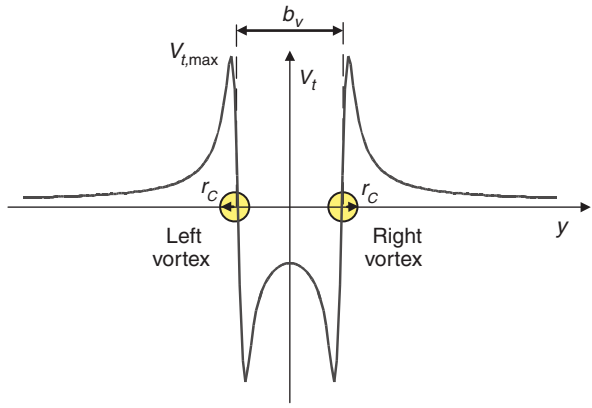

Fig. 12.58 Analytical wake-vortex model.

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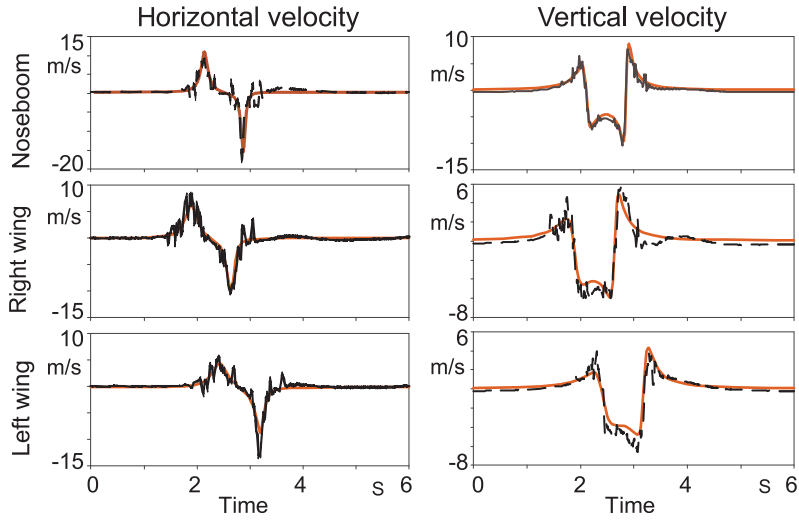

Fig. 12.59 Wake velocity components during lateral encounter: - - - - -, measured; ——, Burnham–Hallock model.

circulation Γ and the core radius r_c:

$$\text{Lamb–Oseen:} \quad V_t(r) = \frac{\Gamma}{2\pi r}\left(1 - e^{-1.2544\, r^2/r_c^2}\right) \tag{12.62}$$

$$\text{Burnham–Hallock:} \quad V_t(r) = \frac{\Gamma}{2\pi}\frac{r}{r_c^2 + r^2} \tag{12.63}$$

Both vortex models allow computation of flow measurements for most wake vortex ages. As an example, Fig. 12.59 shows the match of the measured velocity components from three flow probes during a Do-128 lateral wake encounter, 0.53 nm behind ATTAS. In this plot, it can be seen that the encounter is flown from the left to the right, as the right wing sensor enters the flowfield first, followed by the center-mounted noseboom sensor and then the left wing sensor. In general, compared to the Lamb–Oseen, the Burnham–Hallock model is slightly better suited to characterize the wake-induced velocities, especially close to the vortex core with high gradients.

Figure 12.60 summarizes the identified core radius and circulation of the Burnham–Hallock model for Do-128 encounters from three flights. The estimates are plotted as a function of normalized time $t^* = t/t_0$, which indicates the vortex age (t_0, ATTAS = 11 s). In conformance with the theory, Fig. 12.60 shows the expected decay of circulation, as well as the increase of the core radius. The initial core radius is about 0.75 m, which is 3.5% of the wake-generating wingspan; it is somewhat smaller than the commonly stated value of 5%.

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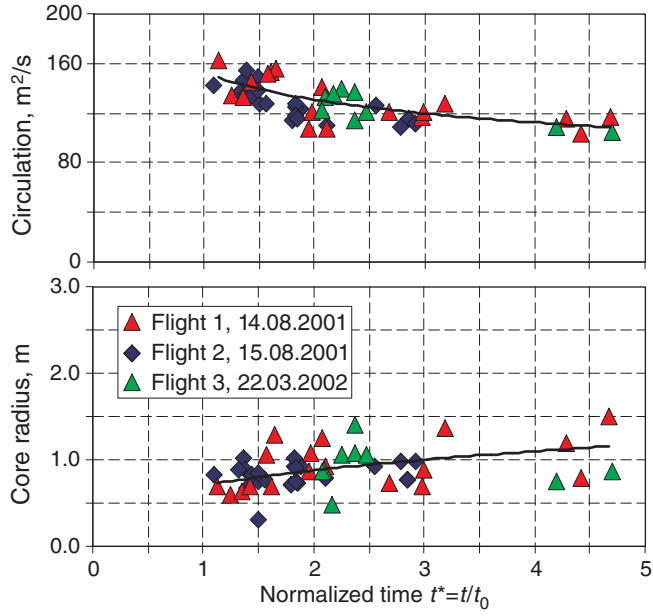

Fig. 12.60 Flight-estimated vortex model parameters for Burnham–Hallock velocity distribution.

12.11.2 VALIDATION OF AERODYNAMIC INTERACTION MODEL

Having obtained a model description of the flowfield characteristics and estimated the wake-vortex parameters, we now turn our attention to the second step of data analysis, namely, encounter model validation through simulation. This step characterizes the aircraft behavior during wake encounters, that is, computes the reaction of follower aircraft entering the wake generated ahead of its path [82, 86]. This is the step two depicted in Fig. 12.56. For this purpose, it is assumed that the high-fidelity aerodynamic model of the basic aircraft is already estimated accurately and known a priori. Such a model could be a global model or a point model valid for flight conditions similar to wake-vortex encounter, but under normal flight conditions. This is to ensure that the model deficiencies during the encounter using only the basic aerodynamic model in the simulation can be attributed to the unaccounted additional forces and moments due to aircraft entering the wake vortex and not due to the model deficiencies in the basic aerodynamics. To simulate the wake encounter, it is required to postulate model for the incremental forces and moments generated due to the interaction. As seen from Fig. 12.56, the inputs to the basic aerodynamic model are the measured control surface deflections, and those to the AIM the precomputed flight path from the foregoing flight-path reconstruction,

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stored as time histories for each encounter, and the wake-vortex model with estimated vortex parameters (strength and position) as discussed in the foregoing subsection [86]. The precomputed flight path stabilizes the wake encounter simulation and proved to be essential, as wake-induced forces and moments are sensitive to small flight-path inaccuracies. Furthermore, in this case too, the general procedure elaborated in Sec. 11.6 was applied, allowing small biases in the control deflections; the initial conditions of the attitude angles were set to the measured values, and for the other states small biases were permitted.

In the present analysis, we investigate the strip model that is easy to use and commonly applied as aerodynamic interaction model. It is based on lifting-line theory and describes the additional aerodynamic forces and moments acting on an aircraft in a nonconstant spatial wind field, for example, wake turbulence. The lift-generating surfaces of an aircraft (wing, horizontal, and vertical tail) are divided into strips; see Fig. 12.61. A well-proven number of strips in simulation are 16 for the wing, 8 for the horizontal tail, and 4 for the vertical tail. At the 25% chord location of each strip of the wing and horizontal tail, the additional angles of attack, and for each strip of vertical tail the angles of sideslip due to the local wind/wake field, are computed. Using a suitable lift gradient, an additional lift is obtained for each strip. These local lift increments are weighted in span direction elliptically and then summed up. Additionally, the corresponding moments of all strips are computed and summed up [9, 86]. To start with, we neglect the incremental drag and fuselage effects resulting from wake-vortex encounter. Accordingly, the present strip model describes wake effects in remaining five degrees of freedom.

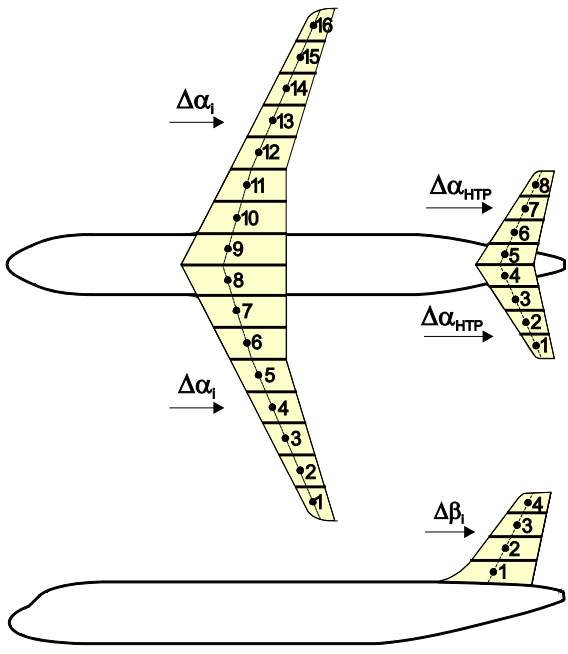

Fig. 12.61 Strip model.

The strip model is based on a priori data, which depend on aircraft geometry and aerodynamics. The geometry of wing, horizontal, and vertical tail and the lever arms for moment computation are generally known accurately from the three-dimensional

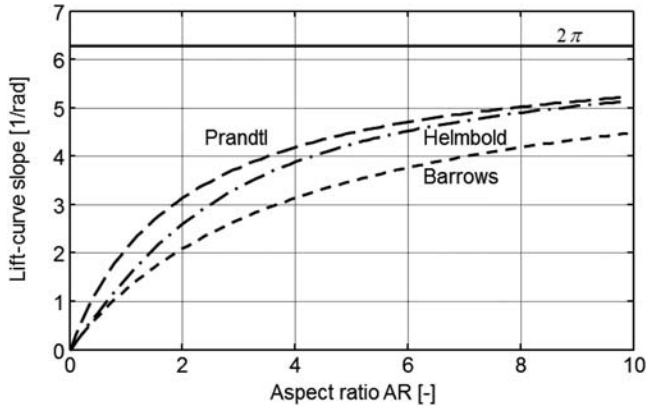

Fig. 12.62 Different approximations for lift-curve slope as a function of aspect ratio.

drawings. The aerodynamic derivatives (lift gradients for wing, horizontal and vertical tail, downwash gradient) can be known from predictions; if not, they can be estimated. The Helmbold equation, which considers the influence of the aspect ratio on the lift gradients, leads to good a priori values [9].

$$C_{L\alpha} = C_{L\alpha\infty} \frac{AR}{2 + \sqrt{4 + AR^2}} \tag{12.64}$$

where AR is the aspect ratio and $C_{L\alpha\infty}$ $(=2\pi)$ the theoretical lift-curve slope for infinite aspect ratio. The other formulas for the lift gradient as a function of the aspect ratio are by Prandtl and Barrows [9]. We observe that the Helmbold equation is a good approximation for high-aspect-ratio wings as well as low-aspect-ratio tails and is a compromise between the Prandtl and Barrows formulations; see Fig. 12.62. Lift-curve slope is also a function of wing sweep Λ; a simple approximation to account for this is given by [9]

$$C_{L\alpha} = C_{L\alpha\infty} \cos\Lambda \tag{12.65}$$

Dependency of the lift-curve slope on the Mach number can be modeled using the Prandtl–Glauert formulation:

$$C_{L\alpha} = C_{L\alpha,M=0} \frac{1}{\sqrt{1 - M^2}} \tag{12.66}$$

Thus, the total lift-curve slope is a function of the aspect ratio, wing sweep, and Mach number. Depending upon the aircraft configuration and flight condition, the preceding three equations have to be combined together and used to compute the lift contribution due to each strip. If flight data-derived values for these derivatives are available, they can be used in place of the predictions.

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Following the simulation procedure described at the beginning of this section and depicted as step 2 in Fig. 12.56, we apply the postulated strip model to two encounters in different directions. This typical example of lateral fly-through pertains to flight test with Do128 aircraft (4 tons) into the wake of the VFW-614 aircraft (20 tons), for which the simulation results are shown in Fig. 12.63. Typical model outputs (shown by dashed lines) in all six degrees freedom are compared to the corresponding flight test data (shown by solid lines). Looking at responses for each degree of freedom separately and keeping in mind that this is a typical encounter out of more than 50 other, we assess the model quality as follows: the rolling motion (see roll rate p) and the vertical motion (see vertical acceleration a_z) during a wake fly-through is simulated accurately. This is an outstanding result of the applied strip model with its widely linear structure, particularly in light of the complexity of validation procedure including sensor calibration, wake identification, and basic model inaccuracies. Both of these DOF (rolling and vertical motion) nowadays are the most important inputs into wake hazard assessment tools. The match for pitching motion (see pitch rate q) is also reasonably good, despite some minor deficiencies at the beginning of the wake encounter. The lateral motion (see acceleration a_y) has some minor discrepancies, which are considered to be tolerable. The longitudinal motion (see acceleration a_x) shows

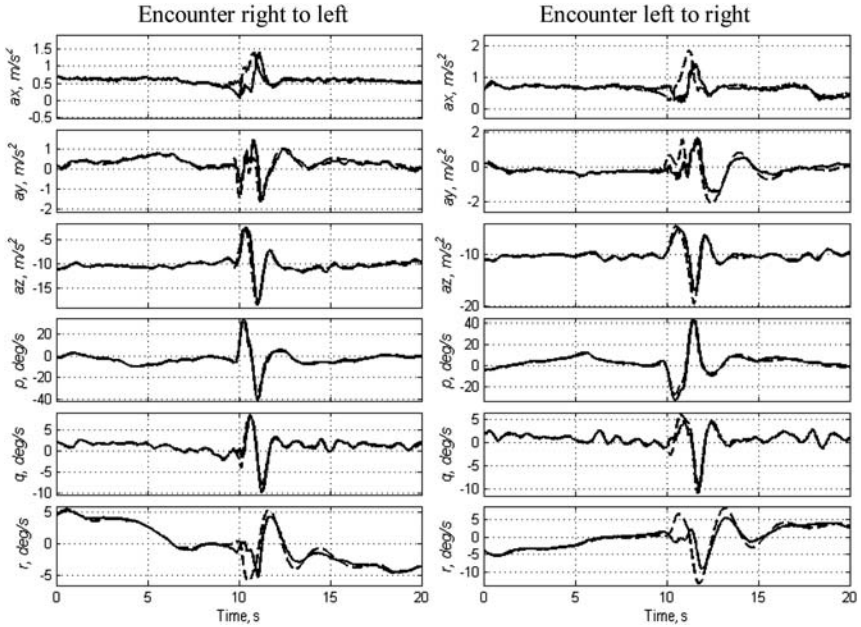

Fig. 12.63 Do128 lateral wake fly-through simulation with five-DOF model: ——, flight measured; - - - -, simulation model.

some discernible discrepancies; they are attributed to the fact that the investigated strip model, as pointed out earlier, accounts for wake effects in five DOF, whereby, to start with, the drag effects are not accounted for as this degree of freedom was considered not to be very important and also due to problems associated with the formulation, discussed next.

The simulation quality in the yawing motion (see yaw rate r) is observed to be poor; this discrepancy is considered to be more critical. At the wake entry, the simulated model response is contrary to that observed in the flight. This anomalous, adverse behavior has been observed in many Do128 encounter validations through simulations. Use of such a model with inconsistencies in yawing motion in the pilot training simulators can lead to wrong pilot reactions to counteract the yawing motion encountered while entering the wake. As a consequence, this adverse reaction can lead to undesirable outcome, which might turn out to be safety critical. Accordingly, we now focus first on investigating the cause for the discrepancies in the match for the yaw rate and then extend the strip model to account for the unaccounted aerodynamic effects.

In many validation cases, the residual analysis, that is, the analysis of the differences between the flight-measured and model-estimated responses, showed clearly a correlation between the model deficiencies in the longitudinal axis and the yawing motion. Recall that the incremental drag effects due to wake encounter were neglected in the hitherto applied strip model. Obviously, drag effects at the wing affects the longitudinal motion and also have impact on the yawing motion of the total aircraft. Based on this consideration, we now extend the strip model to account for drag effects. However, the problem we encounter hereby is to find a suitable formulation within the framework of strip model. We know that drag is a quadratic function of the lift, and such nonlinearities cannot be implemented in the strip model independent of the basic aerodynamic model. The very basis of strip model is the independence of the two models. Accordingly, to retain the linear formulation required in the strip model, we introduce one linear drag derivative and apply it to each single strip to compute the drag due to wing and tail. Having computed the drag contributions due to each strip, the increments to the yawing motion can be computed easily by multiplying them by the corresponding lever arms that are geometrically known.

Having accounted for drag effects resulting from wing and tail, we also recall that the strip model does not consider any fuselage effects. To account for fuselage effects, we implement an empirical model. We divide the fuselage in 20 strips (see Fig. 12.64) and compute a wake-induced local flow angle at each strip. Using a suitable fuselage strip derivative, the summation of the strip increments gives a fuselage force in the lateral direction, and multiplying them with the corresponding lever arms yields incremental contributions to yawing moment due to fuselage interaction with the wake encountered.

The preceding two model extensions introduce two unknown parameters, namely, the wing drag derivative and fuselage derivative. They are determined

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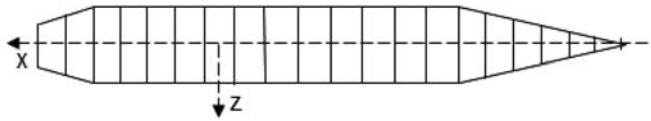

Fig. 12.64 Strip model extension for fuselage effects.

during the overall validation procedure, which is step 2 of Fig. 12.56, by minimizing the deviations between model output and flight-test data. This identification process was carried out for 23 wake encounters of the Do128 aircraft into the ATTAS wake. The results of this analysis are summarized in Fig. 12.65, which shows the standard deviations of the response errors in the three linear accelerations and three angular rates for two cases, first based on the strip model neglecting the drag effects (points marked as "x") and then those including the two model extensions discussed in the foregoing (points marked as "o"). In each case the arithmetic mean of the 23 values is plotted. It is obvious, the lower the standard deviation of the response errors is, the better is the match between the model response and the flight-measured data. From these results, we observe a considerable model improvement in the yaw and longitudinal motion, namely, 51% reduction in the mean values of the standard deviations for the yaw rate and 48% for the longitudinal acceleration. These primary improvements through the two model extension also lead to some minor improvements in the roll axis (16%), in the lateral axis (10%), and in the vertical

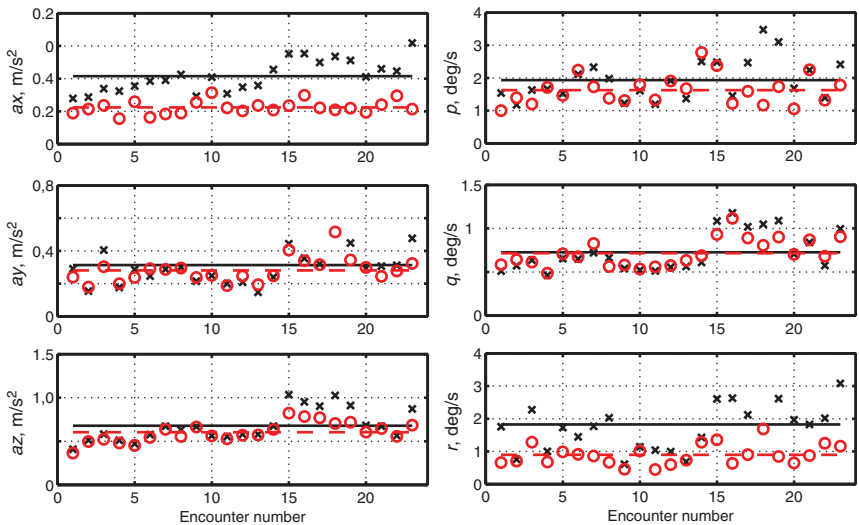

Fig. 12.65 Standard deviations of response errors and mean for 23 wake encounter simulations; x, without, and o, with wing and fuselage drag effects.

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axis (11%), resulting from the kinematic and aerodynamic coupling existing in the basic aerodynamic model. The two parameters, namely, the wing drag and fuselage derivative, were determined to be roughly 0.8, which is valid for the flight condition analyzed.

Having extended the strip model to account for the wing and fuselage drag effects on the longitudinal and yawing motion, we now return to the validation example presented in Fig. 12.63. For the two encounters, one from right to left and other from left to right, the extended model now yields the match between the model-simulated responses and flight-measured data shown in Fig. 12.66. Despite some discrepancies in the lateral motion, a considerable improvement in the longitudinal axis (see acceleration a_x) and the yawing motion (see yaw rate r) is seen. In particular, the initial contrary model reaction in the yawing motion is now largely eliminated. The response match for all of the primary parameters, that is, for the three accelerations and three angular rates, is found to be quite good and considered acceptable.

Although we have demonstrated here a feasible approach to model extension to account for wing and fuselage drag effects, it is, however, an empirical approach. The empirical approach is amenable to flight data analysis, allowing suitable values for these derivatives to be determined from flight-test data applying the procedure described in this section. If flight-test data are not available, the

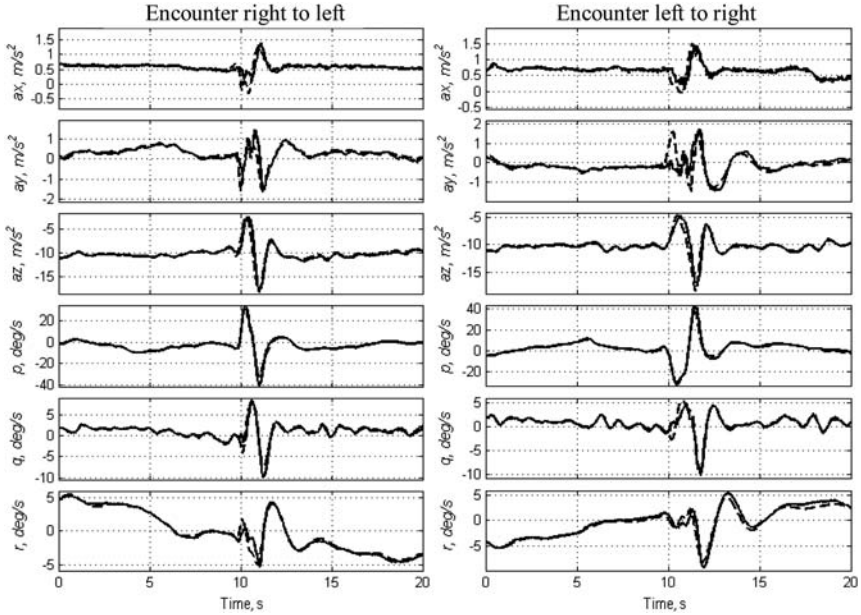

Fig. 12.66 Do128 lateral wake fly-through simulation with wing and fuselage drag modeling: ——, flight measured; ------, simulation model.

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value of 0.8 can be taken as an empirical value, which will have to be adapted for other flight conditions. Summarizing, the strip model presented in this section is capable of reproducing the primary flight mechanic forces and moments acting on an aircraft during a wake encounter quite accurately and in all important axes. It is considered to be near the maximum of what is achievable using the straightforward, linear model structure, which is moreover independent of the aircraft's basic aerodynamic model. The same models have also been successfully applied to validate AIM of the larger aircraft with swept-wing configuration and at high Mach numbers, such as Airbus A318 for cruise condition and A320 in approach entering the wakes generated by aircraft such as Airbus A340, Boeing 747, and Airbus A380.

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12.12 PHOENIX RLV DEMONSTRATOR

The Phoenix flight-test vehicle was developed and tested within the framework of the German ASTRA (Advanced Systems & Technologies for RLV Applications) Program, which is a precursor to the European Future Launcher Preparatory Program (FLPP) [87, 88]. The primary objective of the Phoenix project was to demonstrate the unpowered automatic landing of a representative reusable-launch-vehicle (RLV) configuration. The secondary goal was to generate a flight-validated database incorporating representative models of such vehicles and its systems as well as development tools for future applications. The project had its kick-off in April 2001 and reached its climax with the flight-test campaign in May 2004. Although a detailed discussion of several interdisciplinary topics is not within the scope of this book, for the specific aspect related to flight data analysis presented in this example, we draw heavily upon the more detailed discussion in [89, 90].

RLV configurations compromise between partly contradicting demands derived from various system optimization aspects as well as the different flight phases covering ascent, orbital phase, reentry, hypersonic flight, and landing. The resultant vehicle typically is characterized by a compact shape with small span wings and stabilizers and a rearward c.g. location. Such properties challenge an automatic horizontal landing on a runway, which, in addition, must be performed without engine thrust. The important characteristics and their impact on the landing phase can be summarized as follows:

- Low L/D (5.5 for Phoenix) causing steep approach path
- Low achievable C_L causing high landing velocity (71 m/s for Phoenix)
- Small span causing high roll sensitivity
- Aft c.g. position causing a statically unstable configuration (time to double amplitude <0.5 s for Phoenix)

The test vehicle shape was derived from the suborbital RLV concept Hopper, one of the three RLV concepts currently pursued in Europe. Besides GNC and avionics

systems for autonomous approach and landing, it is equipped with flight-test instrumentation to measure local aerodynamic flow and structural stress.

Two other projects, namely, X-40A and ALFLEX, have also demonstrated successful free-flight and automatic landing experiments in the past [91, 92]. Compared to Phoenix, the ALFLEX vehicle was smaller and lighter, with not so high wing loading resulting in lower landing speed and did not have an extendable landing-gear system. Moreover, it was equipped with a ground-supported microwave based landing system. The Phoenix configuration is more demanding and incorporates a fully autonomous DGPS-based autoland system, which is more advanced. Three free flights, launched from a carrier helicopter at a 2.4 km altitude, were successfully completed in May 2004; see Fig. 12.67. The data obtained from these tests established the source for the verification of the vehicle model. In particular, methods of system identification were applied to upgrade the aerodynamic and the wheel-ground contact and ground roll model.

12.12.1 FINAL CONFIGURATION AND SENSORS

The Phoenix vehicle originated from the FESTIP (Future European Space Transportation Investigation Program) Hopper shape, but was downscaled by a factor of 7. Wing loading and c.g. location were retained to keep the same absolute speed and aerodynamics as well as the unstable configuration of the full-scale vehicle. Thus, all technological challenges to master the automatic landing of such shapes are preserved. Iterative wind-tunnel test campaigns, including full-scale vehicle tests in DNW-LLF (German-Dutch Wind Tunnel—Large Low-Speed Facility), resulted in the final airframe design and aerodynamic database [90, 93]. The final airframe consisted of a conventional aluminum structure with several composite parts, such as covers for the fuselage, the elevons, and the body flap. The basic dimensions are as follows:

- Overall length including noseboom: 7.8 m
- Span: 3.84 m
- Overall height with landing gear retracted: 2.56 m
- Mass: 1200 kg
- Center of mass at 70% of the bare fuselage length from the nose

Electromechanical rotary actuators drive the aerodynamic control surfaces and the nose-gear steering. The landing gear is a tricycle configuration with telescopic deployment mechanism and oleo shock absorbers. The main gear wheels are equipped with brakes.

Phoenix is equipped with an extensive flight-test instrumentation system. Besides measuring all of the control deflections (inboard elevons, outboard elevons, rudder, body flap, speed brakes), landing-gear and nose-gear positions, variables pertaining to aircraft motion, like accelerations, angular rates, velocity,

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a) Phoenix release from carrier helicopter

b) Phoenix in free flight

c) Phoenix touchdown

Fig. 12.67 Phoenix RLV Demonstrator free flight.

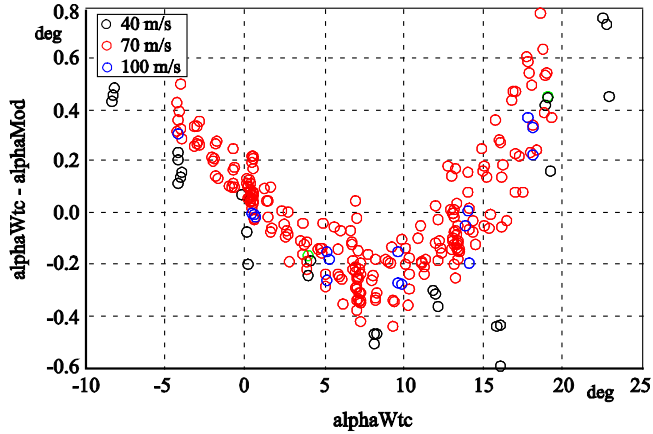

Fig. 12.68 Errors in linear calibration of angle of attack.

attitude, and position were measured using an integrated GPS/INS platform. A five-hole probe mounted on a noseboom provided the measurements of flow variables (angle of attack, angle of sideslip and dynamic pressure). The static pressure and total air temperature sensors were also mounted on the boom. Differential GPS, radar altimeter, and laser ranger were used to improve the achievable height accuracy at different segments of altitudes and height above runway.

The five-hole probe providing the measurements of feedback variables for the flight control was calibrated during the full-scale wind-tunnel test campaign prior to free flights. Three nominal speeds covering all configurations were investigated. Applying estimation algorithms, calibration factors, and bias parameters, including time delays, were determined. A linear calibration factor for angle of attack yields the error between the probe measured and the nominal angle shown in Fig. 12.68. The nonlinear nature of the error necessitated a nonlinear calibration factor to achieve the desired accuracy of better than 0.5 deg throughout the range; moreover, the measurement of the angle of attack was also found to be a function of the sideslip angle. Correction factors were also necessary for the angle of sideslip. Likewise, measured dynamic pressure had to be corrected for errors in the static-pressure probe.

12.12.2 BASICS OF AERODYNAMIC MODEL VALIDATION AND UPDATE

A two-step procedure illustrated in Fig. 12.35 is adopted to verify and update the aerodynamic model from free-flight data. The computational steps have already been elaborated in Sec. 12.8. Besides the two characteristics mentioned there, namely, 1) direct comparison of flight-determined force and moment coefficients with the wind-tunnel predictions, and 2) correction terms can be identified as

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deltas; in the present case the use of the regression method is preferable for the additional reasons of its applicability to 1) unstable configurations and 2) account for atmospheric turbulence in the estimation.

Following the procedure described in Sec. 6.9.1, the aerodynamic force and moment coefficients are computed from measured body-fixed accelerations and the angular accelerations. The angular accelerations ($\dot{p}, \dot{q}, \dot{r}$) were obtained by numerical differentiation of the measured angular rates. Implicit to this data preprocessing step, sensor positions and c.g. positions have to be accounted for while computing the force and moment coefficient refered to as the aerodynamic reference point. This data preprocessing step, depicted in the upper part of Fig. 12.35, yields flight-derived force and moment coefficients (*CX, CY, CZ, CLX, CMY, CNZ*). Special emphasis was also placed on reverification of the wind-tunnel calibration of the airflow data, namely, angle of attack, angle of sideslip, and dynamic pressure measured by the five-hole probe, and also of the static-pressure probe on the noseboom. The flow variables were verified through flight-path reconstruction applied to free-flight data. The wind-tunnel calibration of the five-hole probe measured dynamic pressure required about 10.2% correction, which was found to result from the erroneous measurement of the static pressure measured by a probe on the noseboom. Similar corrections have also been reported in other investigations [94]. Plausibility of this correction in the static pressure was verified by comparison with balloon measurements taken just prior to the free flight.

12.12.3 COMPARISON OF PREFLIGHT ADB AND FLIGHT-DERIVED COEFFICIENTS

Three free flights performed on 8, 13, and 16 May 2004 are analyzed here. The attention is restricted first to the in-air aerodynamic. To facilitate verification of landing-gear effects separately, the first part of data analysis was based on 50-s long time segments corresponding to the portions starting just after the drop up to just before the landing gear is extended. The guidance maneuver in the longitudinal axis results in a fair amount of pitching motion. The altitude above ground varies from 2400 m to about 120 m.

Applying the procedure depicted in Fig. 12.35, aerodynamic force and moment coefficients according to preflight Aerodynamic Database (ADB) predictions (version V3.1) [93] and those from the flight-measured accelerations and angular rates are computed and compared in Fig. 12.69. The table-lookup database and application rules as derived from wind-tunnel and other computational investigations were taken over without any modifications [95]. The vertical lines at 50 and 100 s demarcate the three flights. The total coefficients *CX* and *CZ* are plotted in Fig. 12.70 as functions of angle of attack. From Figs. 12.69 and 12.70, it is evident that there are some discernible differences between the flight-derived and wind-tunnel predicted forces and moments pertaining to the longitudinal motion. Such a comparison was not possible for (*CY, CLX, CNZ*) because there was no dynamic motion of the lateral-directional mode.

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The rough order of magnitude of discrepancy in the vertical force coefficient (lift-curve slope) was observed to be of the order of 9–10%. Briefly, lift generated in flight is higher than predicted, and the component due to pitch rate q is not adequately accounted for in the preflight ADB. The discrepancies are less than roughly 3% in the pitching-moment curve slope. Because of the nonlinear relationship, the error in the axial-force coefficient *CX* could not be quantified as in the other cases. From Figs. 12.69 and 12.70, the error in *CX* is evident for the clean configuration (initial portion of each segment); with increasing speed brakes the mismatch between flight test and preflight database decreases implying that the basic longitudinal-force coefficient was underestimated in the wind tunnel whereas the impact of the speed brakes was overestimated. The preflight ADB does not account for drag due to pitch rate, which seems not to be negligible. The aerodynamic zero terms *CX*0, *CZ*0, and *CMY*0 also show deviations. The magnitudes of the deviations are typical of the two approaches, namely, wind-tunnel and flight-test derived information [96].

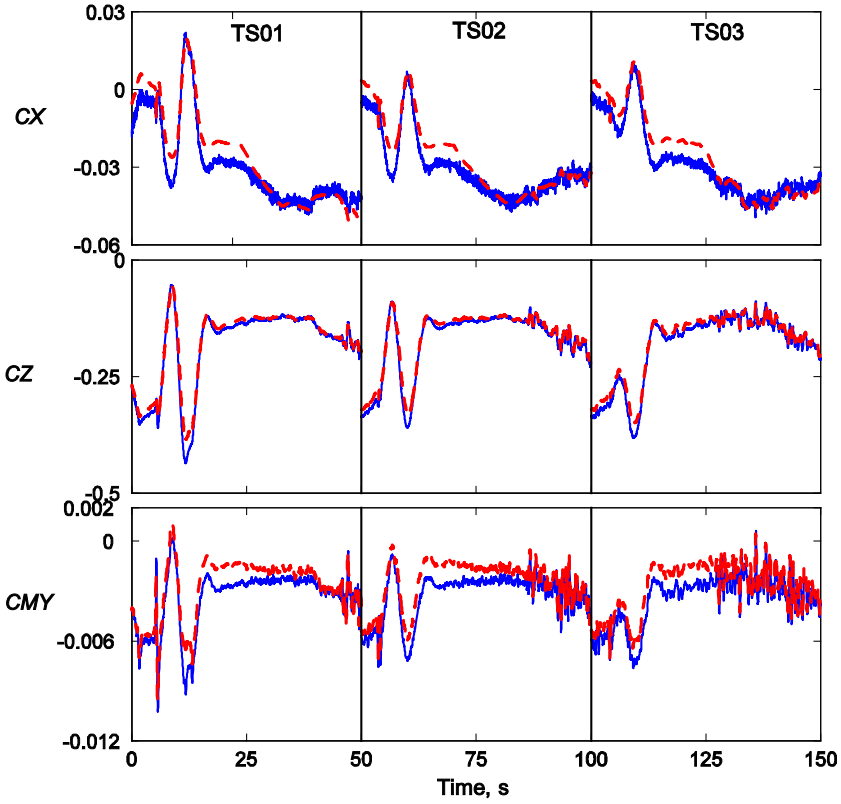

Fig. 12.69 Comparison of aerodynamic force and moment coefficients: - - - -, wind-tunnel predicted; ——, flight derived.

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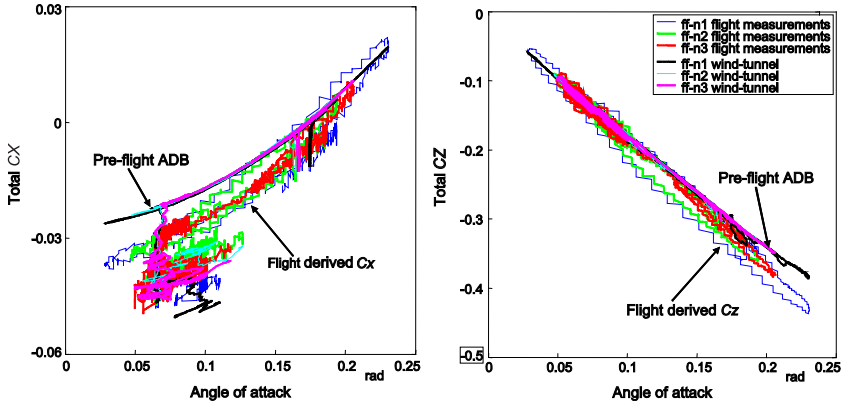

Fig. 12.70 Comparison of wind-tunnel predicted and flight-derived aerodynamic force and moment coefficients.

Through a systematic approach of evaluating multiple cross-plots, a better appreciation of a possible dependence of these deviations, the so-called deltas (that is, the deviations between the flight-derived values and preflight predictions), on different motion variables was acquired. Thus, an engineering approach was adopted to determine the primary influencing parameters, rather than that of the rigorous stepwise regression procedure. The deviations in *CX* were found to be functions of angle of attack, pitch rate, and speed brake; those in *CZ* were found to be dependent on angle of attack, pitch rate, and body-flap deflection, and those in *CMY* were observed to be dependent on angle of attack and body flap.

12.12.4 IDENTIFICATION OF DELTAS FOR FORCE AND MOMENT COEFFICIENTS

Based on the inferences drawn from the comparison, the updates for *CX, CZ*, and *CMY* are identified applying a stepwise procedure. The resulting model in terms of derivatives turned out to be as follows:

$$\begin{aligned} \Delta CX &= CX_0 + CX_\alpha\,\alpha + CX_q\,qV/L_{\text{ref}} + CX_{\delta\text{sb}}\,\delta_{\text{sb}} \\ \Delta CZ &= CZ_0 + CZ_\alpha\,\alpha + CZ_q\,qV/L_{\text{ref}} + CZ_{\delta\text{bf}}\,\delta_{\text{bf}} \\ \Delta CMY &= CM_0 + CM_\alpha\,\alpha + CM_{\delta e}\,\delta_e + CM_{\delta\text{bf}}\,\delta_{\text{bf}} \end{aligned} \tag{12.67}$$

where δ_{bf} is the body flap deflection, δ_{sb} the speed brakes, L_{ref} the reference length. The parameters CZ_α and CM_α are modeled nonlinear, having one value for angle of attack up to 0.175 rad and another value for higher angle of attack. Parameters (CX_0, CX_α, CX_q, $CX_{\delta sb}$), (CZ_0, CZ_α, CZ_q, $CZ_{\delta\text{bf}}$), and (CM_0, CM_α, $CM_{\delta e}$, $CM_{\delta\text{bf}}$) are estimated.

The results of parameter estimation are plotted in Fig. 12.71 showing the force and moment coefficients [90]. A comparison of Figs. 12.69 and 12.71 clearly

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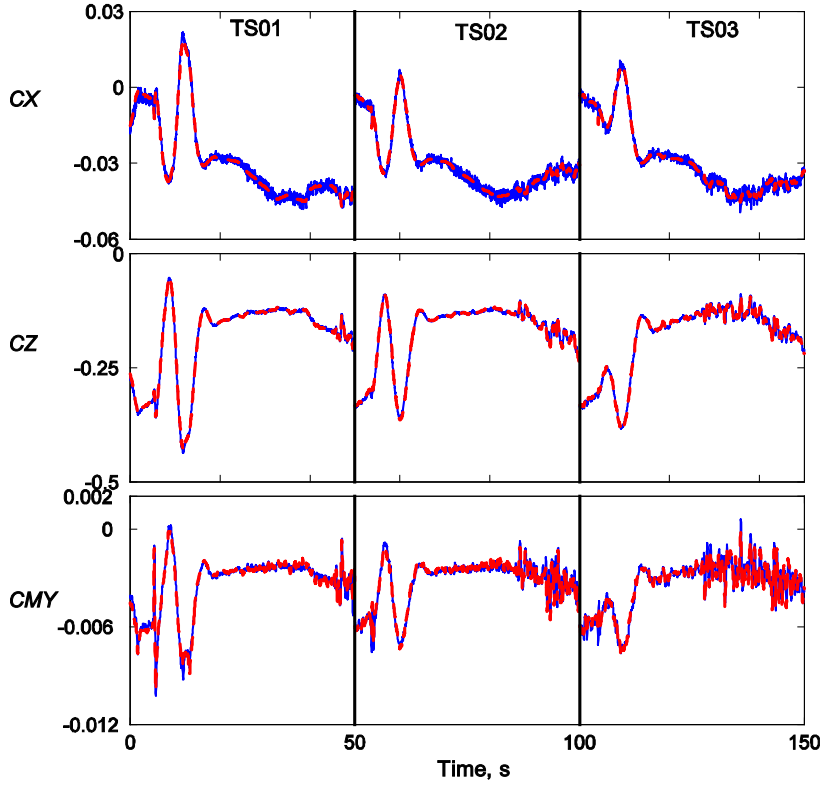

Fig. 12.71 Comparison of aerodynamic force and moment coefficients: - - - -, model estimated (wind-tunnel + updates); ——, flight derived.

shows the improvements obtained through the updates given in Eq. (12.67). It is observed that the residuals, that is, the deviations between the flight derived coefficients and those computed with updates are not anymore deterministic in nature and now represent the measurement noise. The cross-plots of total coefficients *CX* and *CZ* as a function of angle of attack showed a good agreement. It is necessary to mention here that during this phase of flight the speed brakes and body flap are deflected simultaneously, and accordingly they are correlated. As such, from the available flight data it is not possible to identify the components in *CX* due to speed brakes and body flap separately. The correction term is modeled with respect to speed brake only.

12.12.5 LANDING-GEAR EFFECTS

As the next step of data analysis, the in-air aerodynamic database, updated from flight segments in which the landing gear was not extended, is now used

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to simulate the complete time segment including the landing-gear extension phase, down to roughly 3–4 m above ground level. The final 3–4 m before touchdown is not included in the analysis to avoid ground effect, which is mostly dominant for heights up to about span width. Under this consideration any deviations that might be observed during the segment with landing gear can be attributed to landing-gear effects. Each time segment is now 64 s long, the first 50 s being the same portion that was analyzed to update the preflight database.

The input to the aerodynamic model is the measured signal for landing-gear command, which is a discrete signal, changing from 0 to 1 for retracted and fully deployed positions. The actual deployment is modeled as a simple first-order process. The gear extension time is of the order of 1 s. Now keeping the in-air aerodynamics consisting of the preflight ADB plus the updates given in Eq. (12.67) fixed, the three flights with longer time segments including the landing gear are simulated. From the time history plots (not shown here) for the three flights with the updated database, but including wind-tunnel predicted landing gear effect, it was observed that for each flight the initial part of 50 s is matched nicely, but the last part in each case showed deviation in *CX, CZ,* and *CMY*. The cross-plots of total *CX, CZ*, and *CMY* as functions of angle of attack clearly showed some mismatches during the phase with extended landing gear. As a typical example, Fig. 12.72 shows the deviation in the *CX* and *CZ* coefficients as a function of angle of attack. From Fig. 12.72 it becomes apparent that the portion of flight during which the landing gear is extended (marked as LG down) shows discernible deviations whereas for the initial 50-s segments in each flight (marked as clean configuration) the error is randomly distributed around zero.

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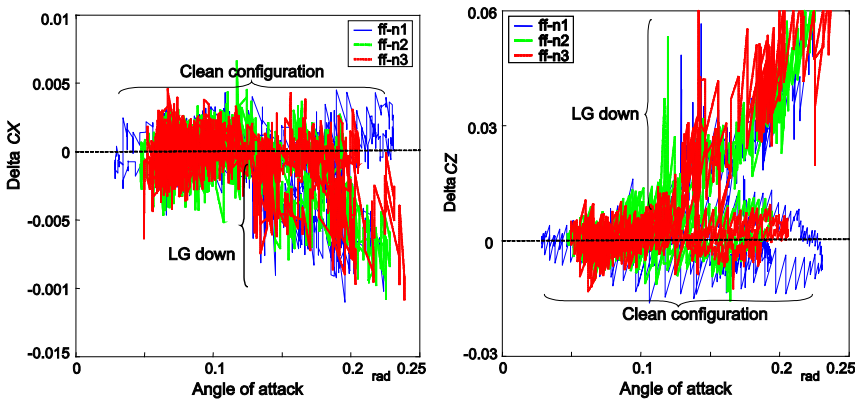

Fig. 12.72 Simulation of free flight including extended landing-gear phase.

Based on this information, the correction model presented in Eq. (12.67) is now expanded to include corrections for landing gear as well.

$$\begin{aligned}\Delta CX &= CX_0 + CX_\alpha \alpha + CX_q\, qV/L_{\text{ref}} + CX_{\delta\text{sb}}\delta_{\text{sb}} + CX_{\alpha\text{LG}}\alpha\delta_{\text{LG}}\\ \Delta CZ &= CZ_0 + CZ_\alpha \alpha + CZ_q qV/L_{\text{ref}} + CZ_{\delta\text{bf}}\,\delta_{\text{bf}} + CZ_{\alpha\text{LG}}\alpha\delta_{\text{LG}}\\ \Delta CMY &= CM_0 + CM_\alpha \alpha + CM_{\delta e}\delta_e + CM_{\delta\text{bf}}\delta_{\text{bf}} + CM_{\alpha\text{LG}}\alpha\delta_{\text{LG}}\end{aligned} \tag{12.68}$$

where δ_{LG} is the landing-gear position. The additional three parameters, namely, $CX_{\alpha\text{LG}}$, $CZ_{\alpha\text{LG}}$, and $CM_{\alpha\text{LG}}$ are now estimated from the extended time segments, keeping the parameters (CX_0, CX_α, CX_q, $CX_{\delta\text{sb}}$), (CZ_0, CZ_α, CZ_q, $CZ_{\delta\text{bf}}$), and (CM_0, CM_α, $CM_{\delta e}$, $CM_{\delta\text{bf}}$) fixed.

The correction terms based on Eq. (12.68) yielded an improved match for the force and moment coefficients (not shown here). The corresponding errors are shown in Fig. 12.73. The appropriateness of the additional correction terms for the landing-gear effect is evident from the comparison of Fig. 12.72 with Fig. 12.73. From Fig. 12.73, it is now observed that the errors observed in Fig. 12.72 are mostly eliminated. There were some minor deviations to be seen in *CZ* and *CMY* at higher angles of attack with landing gear down. However, due to a limited amount of data, any further modeling is not attempted. Nevertheless, it is an indicator that there are some nonlinear effects at angles of attack higher than 0.175 radians. From the analysis pertaining to the landing-gear system, it can be concluded that the wind-tunnel measurements underpredicted the change in drag and overpredicted the change in lift due to the landing gear. The landing-gear effects as modeled here are valid for the configuration as tested on the present vehicle. They result mainly from the door flap, which was not retracted after the gear is fully down. In future applications of Phoenix, it is

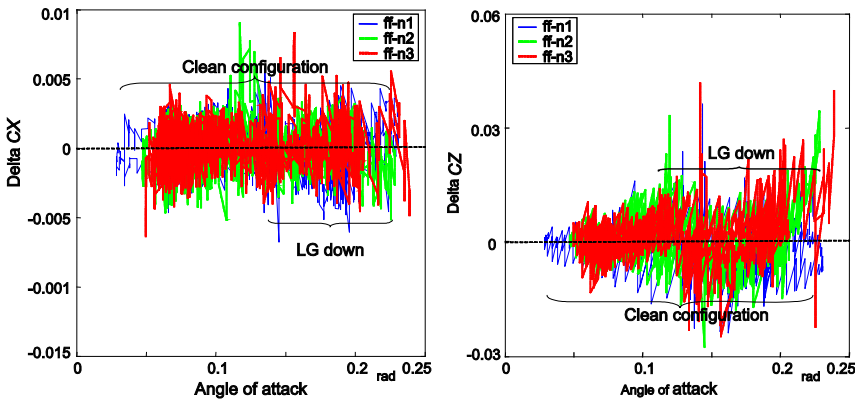

Fig. 12.73 Identification of landing-gear effects.

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expected that the landing-gear and gear-door mechanisms will be modified. In such a case the landing-gear updates might not be appropriate after configuration changes.

To summarize, wind-tunnel calibration of the air-data sensors was performed at three typical speeds, and nonlinear sensitivity factors for the flow angles were estimated to meet the accuracy requirements. The flight-recorded data from the three free flights and ground roll have been analyzed applying system identification methods. After verification of the wind-tunnel predicted aerodynamic characteristics, updates for the forces and moments pertaining to the longitudinal motion were modeled and estimated. The rough orders of corrections required to the prediction were 3 to 10%. It was found that the lift generated in flight is higher than predicted and the component due to pitch rate is not adequately accounted for in the preflight aerodynamic database. The results from system identification emphasize that the basic longitudinal force coefficient for clean configuration was underestimated in the wind tunnel whereas the impact of the speed brakes was overestimated. An update of side-force, rolling-, and yawing-moment coefficients pertaining to lateral-directional motion was not possible because there was no dynamic motion of the lateral-directional modes. The analysis of the landing-gear system showed some discrepancies, which could not be directly interpreted as deviations in the wind-tunnel predictions because of the nonretracted door-flap configuration, which was not tested in wind tunnel. The landing-gear effects as modeled here are valid for the configuration tested, which might not be applicable to the future vehicles.

12.13 ROTORCRAFT MODELING AND SIMULATION

Having dealt with several applications of fixed-wing (both jet and propeller type, as well as both stable and unstable) aircraft and a RLV demonstrator, we now turn our attention to the last case study related to rotorcraft. From an analytical modeling standpoint, rotorcraft are considered to be more complex and from system identification viewpoint more difficult due to high aerodynamic cross-coupling and rotor-coupling effects, and due to poor quality of airflow measurements at low speed and hover conditions. Rotorcraft system identification is a subject in itself; an excellent repertoire on this topic is found in [97–99]. We consider here briefly three applications to demonstrate the possibilities of rotorcraft modeling and estimation applying time-domain methods.

12.13.1 MODELING APPROACHES

Various applications demand analytical rotorcraft models and databases of different complexities and fidelities. Three approaches, namely, system identification (SID), simulation (SIM), and integrated SIM & SID, depicted in Fig. 12.74 are possible to arrive at a suitable model description for a specific application [21,

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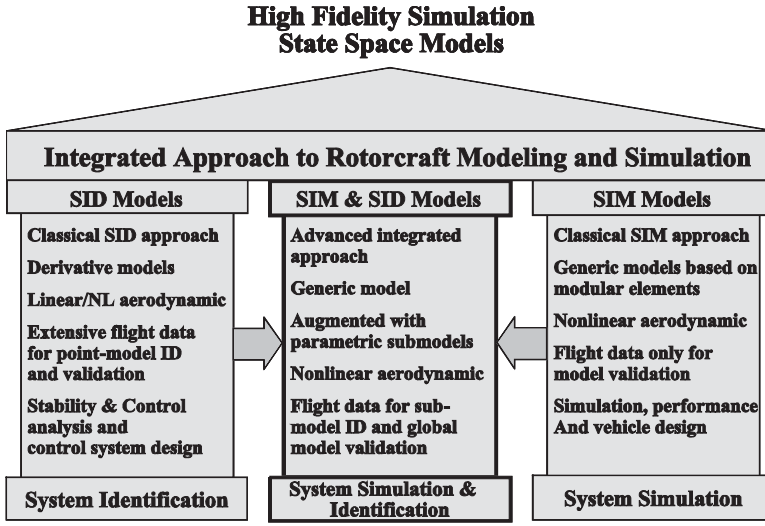

Fig. 12.74 Three-columns approach to rotorcraft modeling and simulation.

100, 101]. Typical characteristics of the three approaches are indicated in the same figure.

In general, helicopter simulation models (SIM models, right column in Fig. 12.74) are generic models, incorporating a component-by-component description of 1) main rotor, 2) tail rotor, 3) fuselage, 4) empennage-horizontal stabilizer, and 5) fin-vertical stabilizer; they can also include other configuration specific and other subsystems. The aerodynamic forces and moments due to each are computed using different procedures [102]. For example, the blade segmental aerodynamic forces and moments due to main rotor are determined from a fully nonlinear transformation of inflow velocity components with respect to the blade element profile. The aerodynamic forces and moments are then determined by two- or three-dimensional profile aerodynamics; the blade profile coefficients are normally taken from tables dependent on the local inflow angle and Mach number. Both quasi-steady and unsteady effects are accounted for. The inflow angle is also a function of induced velocities, which are computed based on the momentum theory accounting for downwash distribution and dynamic wake distortion. In contrast to the main rotor, due to the high rotation speed, the dynamic inertia forces from the tail rotor do not influence the flight dynamic response. So the tail-rotor model description used is normally a pure aerodynamic description using either the just-mentioned analytical approach or a blade element approach. The aerodynamics forces and moment due to fuselage and horizontal and vertical stabilizer modules are normally determined by using aerodynamic coefficient tables depending on angle of attack and sideslip and normalized for dynamic

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pressure. In each case the forces and moments are derived from extensive theoretical formulation and wind-tunnel tests; flight data are usually used only for model validation purposes. In general, SIM models are highly complex, but they cover the complete flight envelope.

The approach called SID, left column of Fig. 12.74, is based on the classical six-DOF equations of motion, incorporating usually linear derivatives estimated from small excursion maneuvers. It is also possible to expand on the aerodynamic model to include nonlinearities and dependencies on flight conditions leading to model valid over larger operating range. As a typical example, we consider here identification of rigid-body aerodynamics of EC-135 helicopter (Fig. 12.75). For this purpose a comprehensive flight-test program was carried out applying multistep and sweep control inputs at different forward speeds and in hover [4]. We consider here the flight at 60 kt forward speed. The state, input, and output vectors consisted of the following:

State vector:

$$x = [u \;\; v \;\; w \;\; p \;\; q \;\; r \;\; \phi \;\; \theta]^T$$

Control input vector:

$$u = [\delta_{\text{lon}} \;\; \delta_{\text{lat}} \;\; \delta_{\text{ped}} \;\; \delta_{\text{col}}]^T$$

Output vector:

$$y = [a_x \;\; a_y \;\; a_z \;\; u \;\; v \;\; w \;\; p \;\; q \;\; r \;\; \phi \;\; \theta \;\; \dot{p} \;\; \dot{q} \;\; \dot{r}]^T$$

where δ_{lon}, δ_{lat}, δ_{ped}, and δ_{col} denote the longitudinal, lateral, pedal, and collective inputs, respectively; all other variables have been used and defined earlier in the book.

Fig. 12.75 EC-135 FHS—flying helicopter simulator.

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Identification of aerodynamic database at a chosen flight condition is based on a multiple-run analysis applying the Gauss–Newton method in the time domain, concatenating several dynamic multistep inputs as well as frequency sweeps with longitudinal, lateral, collective, and pedal inputs. Besides the linear derivatives, the aerodynamic model incorporated angle-of-attack dependent parameters pertaining to the lateral directional motion and also nonlinear effects, for example, the weathercock stability $C_{n\beta}$ (equivalently, the dimensional derivative N_v) being different for positive or negative angles of sideslip. Exemplarily, we show here the equation for the specific moment $\widetilde{N}$ of the yawing motion:

$$\begin{aligned}\widetilde{N} = {} & N_0 + N_u\, u + N_v\, v + N_w\, w + N_p\, p + N_q\, q + N_r\, r \\ & + N_{\delta\text{lon}}\, \delta_{\text{lon}} + N_{\delta\text{lat}}\, \delta_{\text{lat}} + N_{\delta\text{ped}}\, \delta_{\text{ped}} + N_{\delta\text{col}}\, \delta_{\text{col}} \\ & + N_{v\alpha}\, \alpha\, v + N_{r\alpha}\, \alpha\, r + N_{\delta\text{ped}\alpha}\, \alpha\, \delta_{\text{ped}} + N_{\delta\text{ped}\beta}\, \beta\, \delta_{\text{ped}} \\ & + N_{\delta\text{lon}\alpha}\, \alpha\, \delta_{\text{lon}} + N_{\delta\text{lon}\beta}\, \beta\, \delta_{\text{lon}}\end{aligned} \tag{12.69}$$

The first two rows in Eq. (12.69) are the linear derivatives and those in the third and fourth row their α and β dependencies. Equation (12.69) also includes, as already mentioned, nonlinearity in the derivative N_v for directional stability. Starting from a linear model, the additional parameters in the third and fourth rows of Eq. (12.69) were determined in a step-by-step procedure from the error analysis. They were necessary because maneuvers analyzed were with larger excursions around the trim point.

The model predictive capability of the identified rigid-body model is demonstrated in Fig. 12.76 for two flight maneuvers, namely, lateral and pedal input respectively [4]. As already pointed in Sec. 11.6, for such validations only the trim conditions are matched, keeping the aerodynamic model fixed. As seen from Fig. 12.76, the predictions for complementary data, that is, flight maneuvers not used in aerodynamic identification, is fairly good. Such so-called low-frequency rigid-body six-DOF models are generally adequate for flying qualities investigations and autopilot functional design [101].

High-bandwidth models are necessary for high-fidelity simulation and for high precision fly-by-wire/light flight control law design. They are also necessary for in-flight simulator incorporating model-following control based on feedforward regulation providing a more accurate mode control. Extension of the six-DOF rigid-body (SID) model through additional time delays is one possibility. Such an extension is easier to deal with and face no problems in parameter estimation. They are, however, not suitable for model-following control, which involves inversion of the mathematical model of the host aircraft. Inversion of the model with time delays amounts to model with time lead. For a causal system this is not realizable. From nonlinear rotor dynamics investigations, it is known that the response behavior of the rotor contributes significantly to the time delays. Hence, we have to adopt a much more complex approach based on augmenting the rigid-body model with rotor degrees of freedom to meet the high bandwidth requirements [101, 103,

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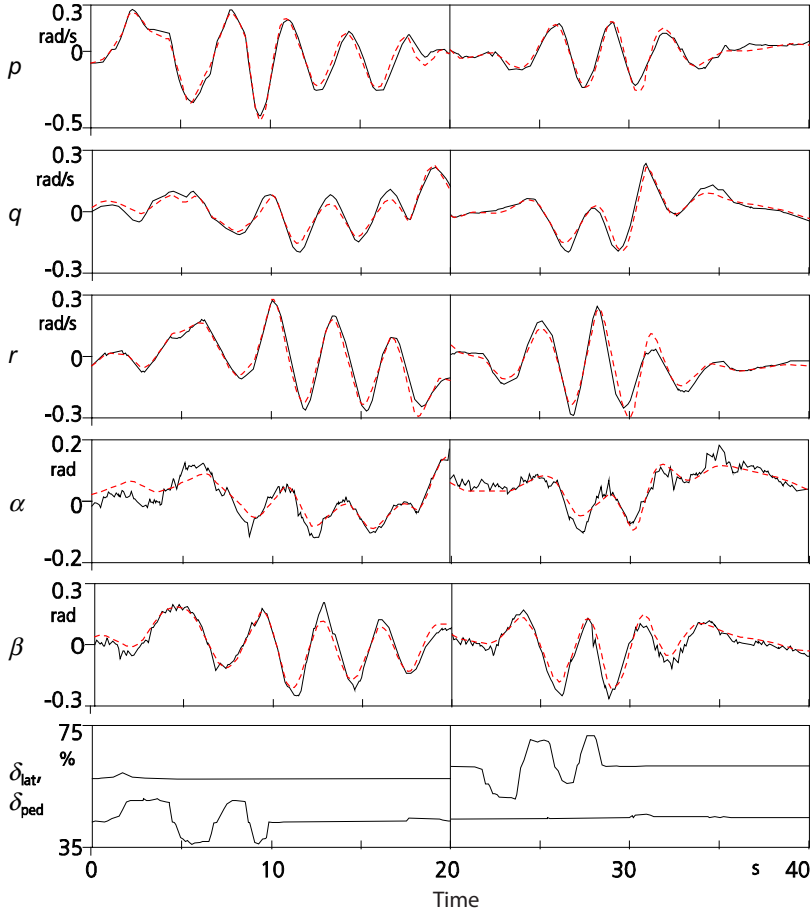

Fig. 12.76 EC-135 model predictive capability: ——, flight data; - - - -, model estimated (Source: [4]).

104]. Two approaches are possible to include the rotor degrees of freedom in the model, namely, 1) implicit first-order approximation of the main rotor and 2) explicit second-order flapping equations for the main rotor.

Looking at the roll response described within a classical six-DOF model, the basic equation for roll response can be written as

$$\dot{p} = -L_p p + L_{\delta p}\ \delta_{\text{control}} \tag{12.70}$$

which indicates that a step input into the control results in a step response in the roll acceleration.

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The approach of first-order approximation of the main rotor can be demonstrated by the high correlation between the flapping motions of the rotor-tip path plane (lateral and longitudinal flapping) and the body-fixed rotational (roll and pitch) accelerations [105]. For example, the correlation between the roll acceleration and lateral flapping, which mainly exists for helicopters with rigid rotors and high hinge offsets, can be expressed as

$$\dot{p} = L_{b1}\, b_1 \tag{12.71}$$

and

$$\dot{b}_1 = -\frac{1}{\tau_b}\, b_1 - p + \frac{L_{\delta b}}{\tau_b}\, \delta_{\text{control}} \tag{12.72}$$

where b_1 is the lateral flapping, δ_{control} the control input at the blade root, and τ_b a flapping time constant. This coupled differential equation indicates that a step into the control input leads to a first-order response of the rotor itself coupled with the body response driven by the rotor flapping. Differentiating Eq. (12.71) and substituting Eq. (12.72) and then again making use of Eq. (12.71) leads to a second-order differential equation for the rotor/body motion:

$$\ddot{p} = L_{b1}\, \dot{b}_1 = -\frac{1}{\tau_b}\, \dot{p} - L_{b1} p + \frac{L_{b1}\, L_{\delta b}}{\tau_b}\, \delta_{\text{control}} \tag{12.73}$$

which can be equivalently reformulated as

$$(\dot{\dot{p}}) = \widetilde{L}_{\dot{p}}\, \dot{p} + \widetilde{L}_p\, p + \cdots + \widetilde{L}_{\delta p}\, \delta_{\text{control}} \tag{12.74}$$

where $\widetilde{L}_{()}$ denote the new set of lateral system parameters. Thus, either the two first-order equations (12.71) and (12.72) are now used to model the roll motion and lateral flapping, or depending on the application in the control system design the equivalent second-order equation (in roll rate p), Eq. (12.74), can be used for system identification.

The derivatives appearing in Eq. (12.74) are not the same as those appearing in the classical rolling motion equation of the rigid-body motion. To obtain a relationship between the two, an analytical estimation of the new parameters is made. For this purpose, we extend the roll response of the classical six-DOF model given by Eq. (12.70) through a short-term behavior typical of helicopters, which leads to [103]

$$\tau_b\, \ddot{p} = -\dot{p} - L_p p + L_{\delta p}\, \delta_{\text{control}} \tag{12.75}$$

where the flapping time constant τ_b is estimated to be roughly one-fourth of the rotor speed. Now, dividing both sides of Eqs. (12.75) by τ_b and comparing

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it with Eqs. (12.73) and (12.74), we obtain

$$\widetilde{L}_{\dot{p}} = -\frac{1}{\tau_b}, \quad \widetilde{L}_{p} = -\frac{L_p}{\tau_b} = -L_{b1}, \quad \widetilde{L}_{\delta p} = \frac{L_{\delta p}}{\tau_b} = \frac{L_{b1}\, L_{\delta b}}{\tau_b} \tag{12.76}$$

Thus, when we use Eq. (12.74) for parameter estimation purposes, the classical parameters gets scaled through the flapping time constant τ_b, and the flapping effects appear indirectly through these scaled parameters [103].

Extending the same procedure to the longitudinal motion requires two additional equations $\dot{q}$ and $(\dot{\dot{q}})$ for the rotor/body motion in terms of derivatives M_{a1} and $\widetilde{M}_{()}$. Incorporation of these second-order models in the parameter estimation is little tricky because the estimation programs require models postulated as first-order differential equations. This is elegantly done by treating $\dot{p}$ and $\dot{q}$ as the state variables, leading to a state vector given by

State vector:

$$x = [u \;\; v \;\; w \;\; p \;\; \dot{p} \;\; q \;\; \dot{q} \;\; r \;\; \phi \;\; \theta]^T$$

compared to the state vector $x = [u\, v\, w\, p\, q\, r\, \phi\, \theta]^T$ for the pure rigid-body motion. The control input and output vectors remain the same as before.

The main advantage of this approach is that the extended model implicitly represents dynamics of the rotor degrees of freedom as a first-order system. Together with the eighth-order state vector of the rigid-body motion defined earlier, the preceding approximation of the rotor coupling in the pitch and roll leads to a tenth-order model, covering a wider frequency range. As rotor dynamics are implicitly modeled, the time delays are significantly reduced to the pure influence of the actuator dynamics.

The second approach, based on extending the six-DOF (SID) model through explicit rotor degrees of freedom, is illustrated in Fig. 12.77. Considering the blade flapping motions in terms of the tip-path plane variables, the states of the rigid-body motion are extended by

$$x_a = [a_1 \;\; b_1 \;\; a_0 \;\; \dot{a}_1 \;\; \dot{b}_1 \;\; \dot{a}_0]^T$$

where a_1 and b_1 denote the longitudinal and lateral flapping respectively, and a_0 the coning motion, each of which is modeled as a second-order system, leading to an extended model with nine degrees of freedom. An additional second-order filter function applied to the lateral control input is also included to model approximately the rotor-blade lead/lag motion influence. Thus, together with the eighth-order state vector of the rigid-body motion defined earlier, it results in a 16^{th}-order model. The SID model structure consists of two sets of equations, representing the fuselage and rotor characteristics. The individual submatrices describe rigid-body and rotor behavior and the corresponding coupling. This model structure provides a detailed representation of the rotorcraft and is closer to the reality. The limitation of such an explicit model is that parameter estimation purpose measurements of the blade motions are necessary. In the absence

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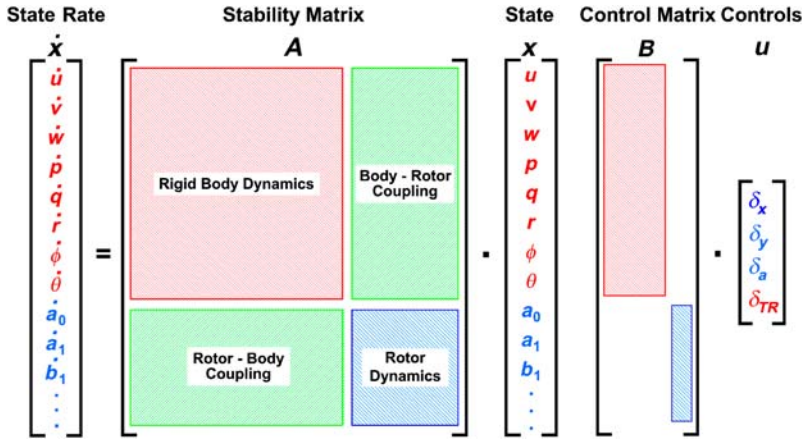

Fig. 12.77 Extended rigid-body and rotor model structure.

of such measurements, the other approach based on an implicit first-order approximation is to be used, which does not require the blade measurements.

In the case of BO-105 (see Fig. 12.78), the measurements of blade motions were available, and hence we can apply the more detailed nine-DOF SID model. The results of analyzing a maneuver with lateral input are summarized in Fig. 12.79. The high-frequency modeling resulting from the nine-DOF SID model can be judged best from the roll response. From the time-expanded plot

Fig. 12.78 Flying rotorcraft test bed BO-105 S123.

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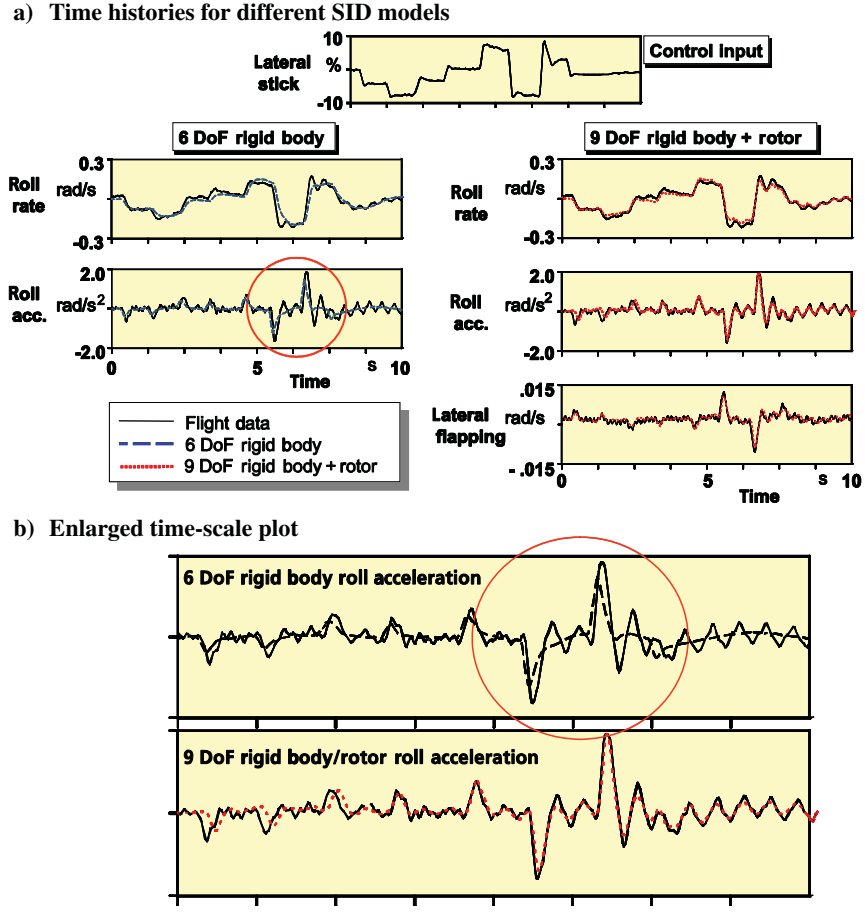

Fig. 12.79 Comparison of six- and nine-DOF models.

of roll acceleration in Fig. 12.79b, we observe that only nine-DOF model can match the amplitude peaks and the almost undamped high frequency of rotor lead/lag mode, although low amplitude.

These differences in terms of the model fidelity become more evident from the frequency responses of the error function we had discussed in Sec. 11.7. For the present case, Fig. 12.80 provides the magnitude and phase plots of the error function p_m/p, that is, ratio of measured roll rate to the roll rate predicted by the model, for six-DOF, nine-DOF, and the generic models. We recollect from Sec. 11.7 that a 0-dB magnitude and 0-deg phase angle of the preceding ratio throughout the frequency range indicate a perfect model. Accordingly, we observe that the

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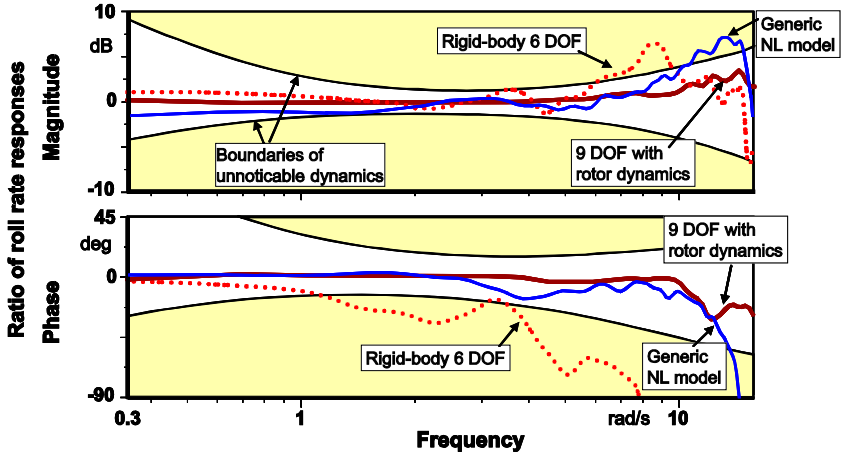

Fig. 12.80 BO-105 model validation in frequency domain (Source: [101]).

controller bandwidth based on the nine-DOF model with rotor dynamics is close to the ideal solution. We also notice that the bandwidth of the nonlinear (NL) generic SIM model is also good whereas that of the rigid-body model limited in range as determined from the suggested boundaries of the unnoticeable dynamics discussed in Sec. 11.7.

12.13.2 ROTOR WAKE MODELING

During roll and pitch maneuvers in hover and at low speeds, unsymmetrical vortex compression and dilatation act on the induced velocity field in the proximity of the main rotor. These act on the effective angle of attack at the blade sections, which directly affect the aerodynamic rotor loads. The gyroscopic behavior of the rotor due to the blade flapping dynamics forced by these loads leads to strong cross-coupling effects of the helicopter due to the wake distortion. Development of suitable flight dynamic models describing this phenomenon is a research topic in rotorcraft modeling to obtain improved dynamic fidelity in off-axis response. A combined SIM & SID approach (middle column of Fig. 12.74) allows extensions of the generic nonlinear simulation models with simpler parametric terms for complex phenomenon that cannot be readily brought in the format of analytical models [101].

The helical vortex sheet and blade-tip vortex propagation can be described either by geometrically prescribed or free wake formulation with discrete vortices or with an equivalent vortex ring/sheet formulation. Figure 12.81 shows the principle of a pitching helicopter in hover. The just-mentioned models require an extensive and complex formulation for the aerodynamics within the flight

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dynamics model. For various applications a less complex model is more appropriate. By analysis of the local phenomena with respect to their global effect on the helicopter dynamic response, it is possible to parameterize them and formulate a global model associated with the dynamics of the inflow. One possible approach is a parametric extension of the Pitt & Peters dynamic wake model, which can be represented as [101, 106–108]

$$\underbrace{\underbrace{\underline{M}\,\dot{\underline{\lambda}} + \hat{\underline{L}}^{-1}\,\underline{\lambda} = \underline{c}}_{\textit{Pitt \& Peters Inflow dynamics}} \quad + \frac{1}{\Omega}\,\hat{\underline{L}}^{-1}\begin{bmatrix} 0 \\ K_p\,(p - \dot{\beta}_s) \\ K_q\,(q - \dot{\beta}_c) \end{bmatrix}}_{\textit{Inflow + Wake distortion dynamics}} \tag{12.77}$$

where M is the apparent mass matrix associated with the acceleration term from momentum theory, $\underline{\hat{L}}$ the gain matrix, $\lambda(=[\lambda_0, \lambda_s, \lambda_c]^T)$ the inflow ratio describing the first harmonic terms, $c(=[c_T, c_l, c_m]^T)$ the rotor load coefficients with respect to rotor thrust and aerodynamic pitch and roll moment, and Ω the main rotor rotation speed. The part denoted as Pitt & Peters inflow dynamics describes the dynamic behavior of the three inflow components due to changes in the corresponding global aerodynamic rotor loads. K_p and K_q are the wake distortion parameters for lateral and longitudinal distribution of the induced velocity, and $\dot{\beta}_s$ and $\dot{\beta}_c$ are the lateral and longitudinal flapping rates. The last term on the right-hand side of Eq. (12.77) is the parametric term that feeds back the total roll and pitch rates of the rotor-tip path plane with respect to the surrounding

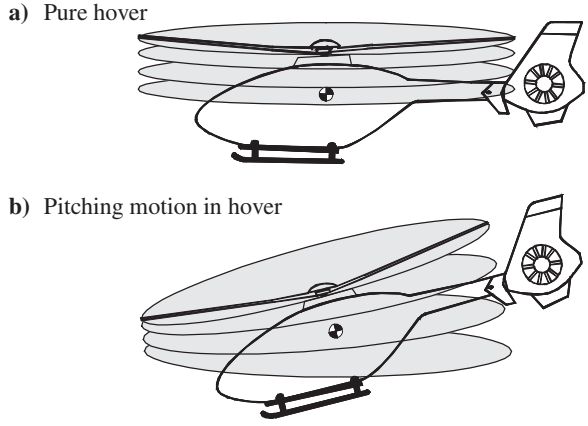

Fig. 12.81 Schematic of wake propagation.

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air to the induced velocity distribution over the rotor disk. In the extension to the classical Pitt & Peters model, the dynamics of the inflow is now additionally driven by the pitch and roll motion of the rotor itself. The wake distortion parameters can either be estimated by linearization of the nonlinear formulation or determined by system identification techniques. Several studies based on different wake prediction theories yield values for wake distortion parameter in hover ranging from 0.5 to 1.75. At hover, neglecting the fuselage aerodynamic, the two parameters in roll and pitch axis are the same. With increasing forward speed, the wake is dragged (washed) out of the influencing zone of the main rotor, reducing the distortion effect. Numerical studies yield K_p and K_q shown in Fig. 12.82 as a function of the advance ratio $\mu(=V_H/\Omega R)$, where V_H is the forward speed in m/s, Ω the main rotor rotation speed in rad/s, and R the rotor radius in meters [109].

Investigations were carried out for hover and forward flight using flight data from BO-105-S123 helicopter; see Fig. 12.78. Upgrading the Pitt & Peters dynamic flow model to parametric wake distortion according to Eq. (12.77) and using the nominal values of $K_p = K_q = 1.5$, simulation yields the pitch and roll rate responses to longitudinal and lateral inputs shown in Fig. 12.83. Two flight maneuvers, demarcated by the vertical lines, are shown: the first with a longitudinal and the second with a lateral pilot input. As seen from the match for the pitch rate for the second maneuver, the updated PWD model with theoretical values for the two parameters is seen to capture the basic phenomenon although some discrepancies in matching the peak are still noticeable.

Applying the Gauss–Newton method, we now estimate the two wake distortion parameters from the same set of data. No convergence problems were encountered, and the two parameters were estimated without correlation and with low standard deviations, yielding the response match shown for comparison purposes in the same figure (see Fig. 12.83). The match for the off-axis response is

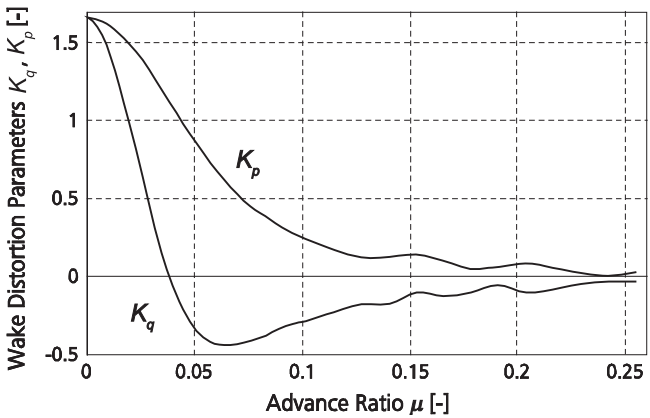

Fig. 12.82 Theoretical estimates of wake distortion parameters.

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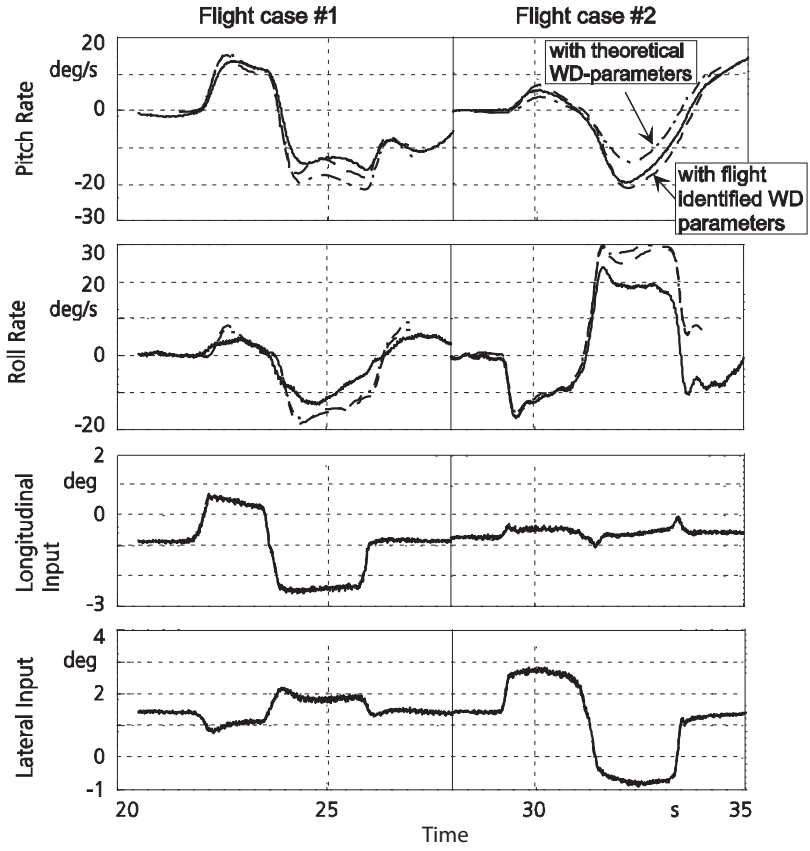

Fig. 12.83 Improvement of simulation fidelity at hover by parametric wake distortion model: ——, flight measured; - - - -, model estimated.

now much better. The estimates of two parameters turn out to be $K_p = 2.5$ and $K_q = 1.6$. The estimate of K_q matches well with the theoretical estimate as seen in Fig. 12.82 for $\mu = 0$ whereas that of K_p is on a higher side than the expected value. As already mentioned, results plotted in Fig. 12.83 have been generated using the PWD model with two different values for wake distortion parameters. Accounting for the wake distortion dynamics is necessary in such case, as evident from Fig. 12.84 showing on the left side the match for pitch rate by considering on the inflow dynamics only [see Eq. (12.77)]; the plot on the right-hand side is the same as that shown in Fig. 12.83 for the optimized wake distortion parameters.

In the next step, we carry out the same investigations for a forward speed of 40 m/s. In this case, based on the theory we first neglect the PWD because, as

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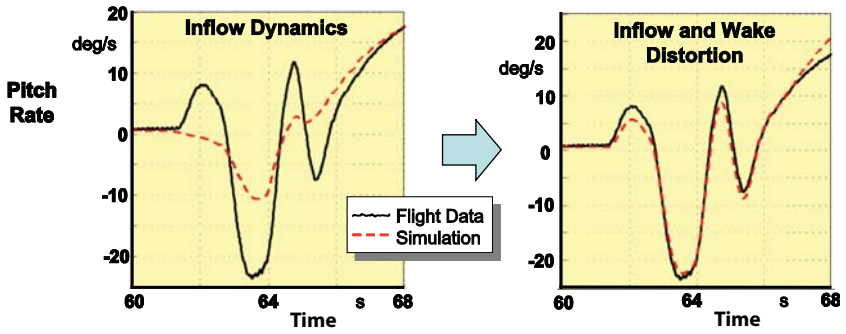

Fig. 12.84 Off-axis response at hover: ——, flight measured; - - - - -, model estimated.

seen from Fig. 12.82, these parameters are almost zero for the advance ratio of roughly 0.18 for this case ($V_H = 40$ m/s, and for BO-105 the main rotor radius and rotation speed are given by $R = 4.91$ m and $\Omega = 44.4$ rad/s). The results of simulation without PWD and those from estimation of these two parameters are shown in Fig. 12.85 for two maneuvers, the first with a longitudinal input and the second with a lateral pilot input; once again the corresponding blade root measurements are plotted. The model response without wake distortion shows considerable discrepancies compared to the measured data, particularly in the off-axis response of pitch rate resulting from lateral input. Estimation of wake distortion parameters leads to significantly improved match for the off-axis pitch rate response due to lateral input, as seen from the second maneuver of Fig. 12.85. The two parameters turn out to be $K_p = 1.1$ and $K_q = 1.6$. These estimates do not conform to the wake prediction theory; as just pointed out, they should be roughly zero.

The anomaly in the estimates from flight data and theoretical predictions for the case of forward speed represents a classical issue often faced in modeling applying system identification methods. In the present case, the estimated parameters do not represent the wake distortion phenomenon, which mainly occurs at hover and at low speeds; rather, they account for some other unmodeled effects, for example, those resulting from rigid blade formulation as assumed in the present investigations. The estimation method optimizes these parameters, which we could call *equivalent parameters*, by minimizing the error between the measured and the estimated responses. The interpretation is not part of the estimation procedure, but the task of the analyst. Further studies revealed that a model based on an elastic blade formulation significantly reduced the aforementioned discrepancies in theory and experimental results.

Although the preceding investigations have been carried out with a BO-105 helicopter, the modeling aspects are general enough to be valid for other rotorcraft as well. The results shown in Figs. 12.83 and 12.85 are derived based on a rotor model with a single flapping mode and show the optimum achievable with the

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model complexity used. Although the basic response characteristics are fairly well captured by the preceding model extension, some deviations in the amplitude resulting from dynamic input are observed. Recent investigations have demonstrated that higher-order free wake models and more blade deflection modes lead to further improvements in the on- as well as off-axis response predictions [110].

To summarize, in this section we addressed the various approaches to rotorcraft modeling and presented extensions of rigid-body model to account for rotor degrees of freedom. Two alternative forms were presented; the exact choice depends on the availability of blade measurements. Wake distortion phenomenon, which is dominant in hover, is briefly studied, and estimation

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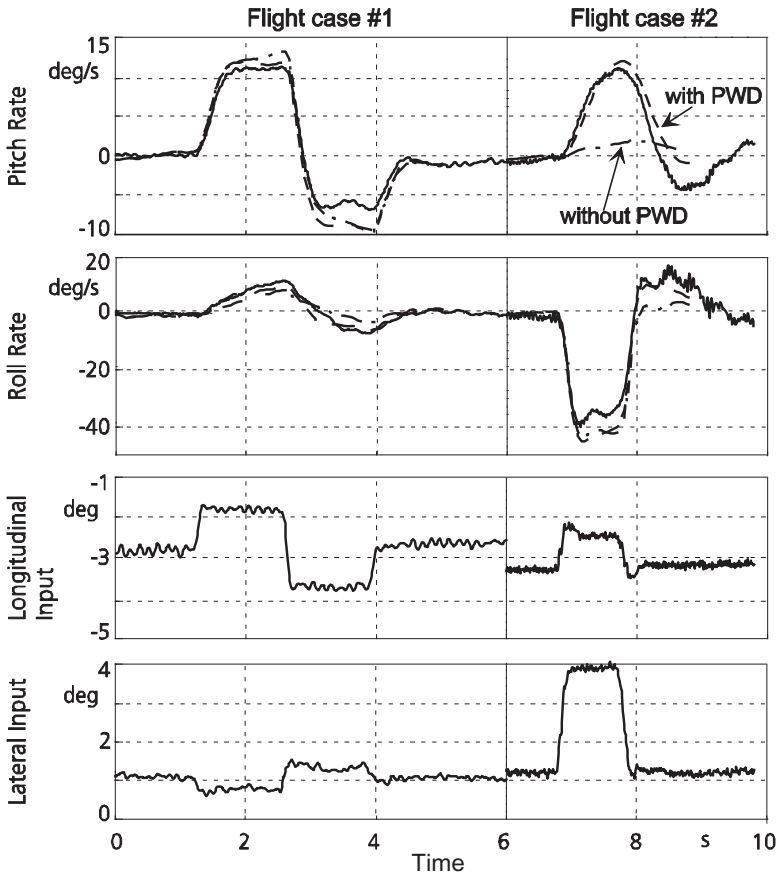

Fig. 12.85 Flight at forward speed of 40 m/s: ——, flight measured; - - - -, model estimated.

method in the time domain is applied to a parametric wake distortion model. The subtle issues related to use of such a model to flight at higher forward speed have been brought out.

12.14 CONCLUDING REMARKS

In this chapter we have provided an account of different types of aerodynamic modeling and parameter estimation problems, namely, 1) downwash and other transit time lag effects; 2) aerodynamic effects of secondary order; 3) modeling of nonlinear effects resulting from large control deflections, including those under abnormal conditions; 4) unsteady aerodynamics and quasi-steady stall; 5) high-precision aerodynamic databases for Level-D flight simulators; 6) database validation and update of an unstable aircraft; 7) integrated flexible aircraft model with rigid-body and structural dynamics; 8) wake vortex encounter and validation of aircraft interaction model; 9) reusable launch vehicle demonstrator; and 10) high-bandwidth rotorcraft and parametric wake distortion modeling. In each case, results obtained from flight data analysis and applying time-domain parameter-estimation methods have been discussed. It is evident from these advanced applications that a judicious combination of conventional and specially designed flight maneuvers and analytical model formulations supported by a suitable choice of the estimation algorithm is the key to the successful generation of flight-validated submodels or complete databases. The major thrust has been to derive aerodynamic models to explain physics of the underlying phenomenon. In some cases, to capture the observed response characteristics, we have resorted to simplified or empirical model extensions. Much more complex model formulations including those based on computation fluid dynamics are, of course, possible, for example, for the unsteady aerodynamics or ground effect; however, they are not directly amenable to system identification purposes from flight data based on state-space models as addressed here. The models identified using the approach pursued here are suitable for several applications including those of flight simulators, design of control laws, and handling qualities or other flight mechanical investigations.

A brief summary of the main outcome has been presented at the end of each subtopic. Here, we would like to repeat the conclusions drawn in [1, 2], namely, because the quest for better understanding of aerodynamic phenomena will continue in the future, it can be said that today the scope of flight vehicle modeling will mainly be limited by the ability and skill of the analyst to interpret the modeling discrepancies, to look beyond the obvious, and to formulate mathematical models purported to underlie the physical phenomena, and little by the methods of parameter estimation. This has been demonstrated in this chapter through a number of applications of widely diversified nature, which was the main goal of this chapter. It is also substantiated by the numerous successful applications reported in the literature

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REFERENCES

[1] Jategaonkar, R. V., "Flight Vehicle System Identification – Engineering Utility," (Guest Editorial), *Journal of Aircraft*, Vol. 42, No. 1, 2005, p. 11.

[2] Hamel, P. G., and Jategaonkar, R. V., "Evolution of Flight Vehicle System Identification," *Journal of Aircraft*, Vol. 33, No. 1, 1996, pp. 9–28.

[3] Iliff, K. W., "Parameter Estimation for Flight Vehicles," *Journal of Guidance, Control, and Dynamics*, Vol. 12, No. 5, 1989, pp. 609–622.

[4] Jategaonkar, R. V., Fischenberg, D., and von Gruenhagen, W., "Aerodynamic Modeling and System Identification from Flight Data – Recent Applications at DLR," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 681–691.

[5] Wang, K. C., and Iliff, K. W., "Retrospective and Recent Examples of Aircraft Parameter Identification at NASA Dryden Flight Research Center," *Journal of Aircraft*, Vol. 41, No. 4, 2004, pp. 752–764.

[6] Morelli, E. A., and Klein, V., "Application of System Identification to Aircraft at NASA Langley Research Center," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 12–24.

[7] Cowley, W. L., and Glauert, H., "The Effect of the Lag of the Downwash on the Longitudinal Stability of an Airplane and on the Rotary Derivative Mq," *British Aeronautical Research Council*, R.&M. 718, London, 1921.

[8] Etkin, B., *Dynamics of Atmospheric Flight*, Wiley, New York, 1972.

[9] McCormick, B. W., *Aerodynamics, Aeronautics, and Flight Mechanics*, Wiley, New York, 1995.

[10] Maine, R. E., and Iliff, K. W., "Maximum Likelihood Estimation of Translational Acceleration Derivatives from Flight Data," *Journal of Aircraft*, Vol. 16, No. 10, 1979, pp. 674–679.

[11] Jategaonkar, R. V., and Gopalratnam, G., "Two Complementary Approaches to Estimate Downwash Lag Effects from Flight Data," *Journal of Aircraft*, Vol. 28, No. 8, 1991, pp. 540–542.

[12] Mönnich, W., "Ein 2-Punkt-Aerodynamikmodell für die Identifizierung," *Proceedings of the Symposium on "Systemidentifikation in der Fahrzeugdynamik,"* DFVLR-Mitt. 87-22, Paper No. 3.1, Germany, Nov. 1987.

[13] Hanke, D., Wilhelm, K., and Meyer, H.-L., "Development and Application of In-Flight Simulator Aircraft for Flying Qualities Research at DFVLR," *Proceedings of the IEEE NAECON '86 Symposium on "Developing Technologies for Revolutionary Applications,"* Paper 117, IEEE, Dayton, OH, May 1986.

[14] Jategaonkar, R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DFVLR-FB 90-40, Germany, July 1990.

[15] Jategaonkar, R. V., "Identification of Actuation System and Aerodynamic Effects of Direct-Lift-Control Flaps," *Journal of Aircraft*, Vol. 30, No. 5, 1993, pp. 636–643.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[16] Krag, B., Jategaonkar, R. V., Mönnich, W., and Fischenberg, D., "Estimation of an Aerodynamic Data Base for a New C-160 Transall Flight Simulator from Flight Data," *Proceedings of the RAeS Symposium on 'Data Issues for Flight Simulators – An On-going Problem,'* RAeS, London, 1993, pp. 7.1–7.12.

[17] Jategaonkar, R. V., Mönnich, W., Fischenberg, D., and Krag, B., "Identification of C-160 Simulator Data Base from Flight Data," *Proceedings of the 10th IFAC Symposium on System Identification and Parameter Estimation*, IFAC, Copenhagen, Denmark, July 1994, pp. 1031–1038.

[18] Jategaonkar, R. V., Mönnich, W., Fischenberg, D., and Krag, B., "Identification of Speed Brake, Air-Drop, and Landing Gear Effects from Flight Data," *Journal of Aircraft*, Vol. 34, No. 2, 1997, pp. 174–180.

[19] "Airplane Simulator Qualification," Federal Aviation Administration, FAA Advisory Circular, AC 120-40C, Draft Version, July 1995.

[20] "Joint Aviation Requirements – Aeroplane Flight Simulators," JAR-STD 1A, Westward Digital, Ltd., Cheltenham, England, U.K., April 1997.

[21] Hamel, P. G., and Jategaonkar, R. V., "The Role of System Identification for Flight Vehicle Applications: Revisited," RTO-MP-11, Paper 2, March 1999.

[22] Tobak, M., "On the Use of the Indicial Function Concept in the Analysis of Unsteady Motions of Wings and Wing-Tail Combinations," NACA Report 1188, 1954.

[23] Tobak, M., and Schiff, L. B., "On the Formulation of the Aerodynamic Characteristics in Aircraft Dynamics," NASA TR R-456, 1976.

[24] Klein, V., and Noderer, K. D., "Modeling of Aircraft Unsteady Aerodynamic Characteristics, Part 1 – Postulated Models," NASA TM-109120, May 1994.

[25] Klein, V., "Modeling of Longitudinal Unsteady Aerodynamics of a Wing-Tail Combination," NASA CR-1999-209547, Sept. 1999.

[26] Abramov, N., Goman, M., and Khrabrov, A., "Aircraft Dynamics at High Incidence Flight with Account of Unsteady Aerodynamic Effects," AIAA Paper 2004-5274, Aug. 2004.

[27] Leishman, J. G., and Nguyen, K. Q., "State Space Representation of Unsteady Airfoil Behavior," *AIAA Journal*, Vol. 28, No. 5, 1990, pp. 836–844.

[28] Goman, M., and Khrabrov, A., "State-Space Representation of Aerodynamic Characteristics of an Aircraft at High Angles of Attack," *Journal of Aircraft*, Vol. 31, No. 5, 1994, pp. 1109–1115.

[29] Fischenberg, D., "Identification of an Unsteady Aerodynamic Stall Model from Flight Test Data," AIAA Paper 95-3438, Aug. 1995.

[30] Greenwell, D. I., "A Review of Unsteady Aerodynamic Modelling for Flight Dynamics of Manoeuvrable Aircraft," AIAA Paper 2004-5276, Aug. 2004.

[31] Fischenberg, D., and Jategaonkar, R. V., "Identification of Aircraft Stall Behavior from Flight Test Data," RTO-MP-11, Paper No. 17, March 1999.

[32] Leishman, J. G., and Beddoes, T. S., "A Semi-Empirical Model for Dynamic Stall," *Journal of the American Helicopter Society*, Vol. 34, No. 3, 1989, pp. 3–17.

[33] Jategaonkar, R. V., and Mönnich, W., "Identification of DO-328 Aerodynamic Database for a Level D Flight Simulator," AIAA Paper 97-3729, Aug. 1997.

[34] Dias, J. N., "Post-Stall Model Identification from Flight Test Data," Society of Flight Test Engineers, SFTE 43rd Annual International Symposium, Paper No. 43-06, Oct. 2012.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[35] Ray, P. A., "Federal Aviation Administration Flight Simulation Standards – A Regulatory Update," AIAA Paper 95-3389, Aug. 1995.

[36] Skow, A. M., Titiriga, A., Jr., and Moore, W. A., "Forebody/Wing Vortex Interactions and Their Influence on Departure and Spin Resistance," AGARD, CP-247, Oct. 1978, pp. 6-1–6-26.

[37] Orlik-Rückemann, K. J., "Aerodynamic Aspects of Aircraft Dynamics at High Angles of Attack," *Journal of Aircraft*, Vol. 20, No. 9, 1983, pp. 737–751.

[38] Singh, J., and Jategaonkar, R. V., "Identification of Lateral-Directional Behavior in Stall from Flight Data," *Journal of Aircraft*, Vol. 33, No. 3, 1996, pp. 627–630.

[39] Wieselsberger, C., "Wing Resistance near the Ground," NACA TM-77, 1922.

[40] Prandtl, L., "Der Induzierte Widerstand von Mehrdeckern," Ergebnisse der Aerodynamischen Versuchsanstalt Göttingen, II. Lieferung, Germany, 1923.

[41] Schweikhard, W., "A Method for In-Flight Measurement of Ground Effect on Fixed-Wing Aircraft," *Journal of Aircraft*, Vol. 4, No. 2, 1967, pp. 101–104.

[42] Campbell, J. P., Hassel, J. L., Jr., and Thomas, J. L., "Ground Effects on Lift for Turbofan Powered-Lift STOL Aircraft," *Journal of Aircraft*, Vol. 15, No. 2, 1978, pp. 78–84.

[43] Staufenbiel, R., "Some Nonlinear Effects in Stability and Control of Wing-in-Ground Effect Vehicles," *Journal of Aircraft*, Vol. 15, No. 8, 1978, pp. 541–544.

[44] Staufenbiel, R. W., and Schlichting, U.-J., "Stability of Airplane in Ground Effect," *Journal of Aircraft*, Vol. 25, No. 4, 1988, pp. 289–294.

[45] Curry, R. E., "Dynamic Ground Effect for a Cranked Arrow Wing Airplane," NASA TM-4799, Aug. 1997.

[46] Fischenberg, D., Mönnich, W., Krag, B., and Jategaonkar, R. V., "Aspects of C-160 Simulator Model Determination and Validation on and Close to the Ground," AIAA Paper 94-3404, Aug. 1994.

[47] Fischenberg, D., "Ground Effect Modeling Using a Hybrid Approach of Inverse Simulation and System Identification," AIAA Paper 99-4324, Aug. 1999.

[48] Rohlf, D., "Global Model Approach for X-31 VECTOR System Identification," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 54–62.

[49] Allen, L. D., "Evolution of Flight Simulators," AIAA Paper 93-3545, Aug. 1993.

[50] Mulder, J. A., Baarspul, M., Breeman, J. H., Nieuwpoort, A. M., Verbraak, J. P. T., and Steeman, P. S. J. M., "Determination of the Mathematical Model for the New Dutch Government Civil Aviation Flying School Flight Simulator," *Proceeding of the 18th Annual Symposium of the Society of Flight Test Engineers*, Paper 15, Society for Flight Test Engineers, Amsterdam, The Netherlands, 1987.

[51] Baillie, S. W., Hui, K., and DeLeeuw, J., "The Flight Test and Data Analysis Program for the Development of a Boeing/deHavilland Dash 8 Simulator Model," AGARD CP-519, Paper 30, Oct. 1992.

[52] Neville, K. W., and Stephens, A. T., "Flight Update of Aerodynamic Math Model," AIAA Paper 93-3596, Aug. 1993.

[53] Trankle, T. L., and Bachner, S. D., "Identification of a Nonlinear Aerodynamic Model of the F-14 Aircraft," *Journal of Guidance, Control, and Dynamics*, Vol. 18, No. 6, 1995, pp. 1292–1297.

[54] Rohlf, D., "Direct Update of a Global Simulation Model with Increments via System Identification," RTO-11, Paper 28, March 1999.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[55] Weiss, S., Gockel, W., Mönnich, W., and Rohlf, D., "Identification of Dornier-328 Reversible Flight Control Systems," AIAA Paper 98-4163, Aug. 1998.

[56] Shikany, D. A., "Math Pilots in Validation Data Packages – Appropriate Use or Potential Abuse?" *Proceedings of the RAeS Symposium on "Data for Simulation – The Weakest Link?"* Paper 12, London, Nov. 2002.

[57] Hall, F. C., "Flight Testing for Simulator Data," AIAA Paper 91-2930, Aug. 1991.

[58] Mönnich, W., and Jategaonkar, R. V., "Data Base Development for Level D Simulators – Lessons Learned," RTO-MP, Paper 14, March 1999.

[59] Fischenberg, D., and Rohlf, D., "Identifizierung der Einflüsse von Kompressibilität und Strukturelastizität auf die Aerodynamischen Derivative eines Widebody-Verkehrfluzeugs," DLR, Rept. IB 111-98/03, Braunschweig, Germany, Jan. 1998 (in German).

[60] McLean, D., *Automatic Flight Control Systems*, Prentice–Hall International, Hertfordshire, England, U.K., 1990, pp. 102–126.

[61] Zerweckh, S. H., von Flotow, A. H., and Murray, J. E., "Flight Testing a Highly Flexible Aircraft: Case Study on the MIT Light Eagle," *Journal of Aircraft*, Vol. 27, No. 4, April 1990, pp. 342–349

[62] Waszak, M. R., and Schmidt, D. K., "Flight Dynamics of Aeroelastic Vehicle," *Journal of Aircraft*, Vol. 25, No. 6, 1988, pp. 563–571.

[63] Waszak, M. R., Buttrill, C. S., and Schmidt, D. K., "Modeling and Model Simplification of Aeroelastic Vehicles: An Overview," NASA-TM-107691, Sept. 1992.

[64] Schmidt, D. K., *Modern Flight Dynamics*, McGraw–Hill, New York, 2012.

[65] Ghosh, A. K., and Raisinghani, S. C., "Parameter Estimates of an Aeroelastic Aircraft as Affected by Model Simplifications," *Journal of Aircraft*, Vol. 31, No. 2, 1994, pp. 452–454.

[66] Singh, J., Basappa, K., and Kumar, R., "Identification of Aeroelastic Quasi-Steady Effects Through Flex Factors," *Journal of Aerospace Sciences and Technologies*, Vol. 60, No. 4, Sept 2008, pp. 1–8.

[67] Majeed, M., Singh, J., and Kar, I. N., "Identification of Aerodynamic Derivatives of a Flexible Aircraft," *Journal of Aircraft*, Vol. 49, No. 2, 2012, pp. 654–658.

[68] de Oliveira Silva, B. G., "Data Gathering and Preliminary Results of the System Identification of a Flexible Aircraft Model," AIAA Paper 2011-6355, Aug. 2011.

[69] de Oliveira Silva, B. G., and Mönnich, W., "System Identification of Flexible Aircraft in Time Domain," AIAA Paper 2012-4412, Aug. 2012.

[70] Theodore, C. R., Ivler, C. M., Tischler, M. B., Field, E. J., Neville, R. L., and Ross, H. P., "System Identification of Large Flexible Transport Aircraft," AIAA Paper 2008-6894, Aug. 2008.

[71] Tischler, M. B., and Remple, R. K., *Aircraft and Rotorcraft System Identification: Engineering Methods with Flight Test Examples*, 2nd ed., AIAA, Reston, VA, 2012, pp. 575–643.

[72] Bisplinghoff, R. L., Ashley, H., and Halfman, R. L., *Aeroelasticity*, Addison-Wesley, Cambridge, MA, 1955, pp. 632–650, 753–758.

[73] Le Garrec, C., and Kubica, F., "In-Flight Structural Modes Identification for Comfort Improvements by Flight Control Laws," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 90–93.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[74] Ross, H., "X-31 Enhancement of Aerodynamics for Maneuvering Beyond Stall," AGARD CP-497, Paper 2, Nov. 1991.

[75] Weiss, S., Friehmelt, H., Plaetschke, E., and Rohlf, D., "X-31A System Identification Using Single-Surface Excitation at High Angles of Attack," *Journal of Aircraft*, Vol. 33, No. 3, 1996, pp. 485–490.

[76] Smith, R. E., "Flying Qualities Development for the X-29A Demonstrator," International Symposium on In-Flight Simulation for the 90's, DLR Braunschweig, Germany, July 1991.

[77] Iliff, K. W., "X-29A Lateral-Directional Stability and Control Derivatives Extracted from High-Angle-of-Attack Flight Data," NASA TP-3664, Dec. 1996.

[78] Iliff, K. W., and Wang, K. C., "Flight-Determined Subsonic Longitudinal Stability and Control Derivatives of the F-18 High Angle of Attack Research Vehicle (HARV) with Thrust Vectoring," NASA/TP-97-206539, Dec. 1997.

[79] Iliff, K. W., and Wang, K. C., "Flight-Determined, Subsonic, Lateral-Directional Stability and Control Derivatives of the Thrust-Vectoring F-18 High Angle of Attack Research Vehicle (HARV), and Comparison to the Basic F-18 and Predicted Derivatives," NASA/TP-1999-206573, Jan. 1999.

[80] Vicroy, D. D., Brandon, J., Greene, G., Rivers, R., Shah, G., Stewart, E., and Stuever, R., "Characterizing the Hazard of a Wake Vortex Encounter," AIAA Paper 97-0055, Jan. 1997.

[81] Pete, K. R., Smith, S. T., and Vicroy, D., "Model Validation of Wake-Vortex/Aircraft Encounters," AIAA Paper 2000-3979, Aug. 2000.

[82] Fischenberg, D., "S-WAKE: Results of Flight Test Data Analysis," DLR-IB 111-2002/11, Germany, Sept. 2002.

[83] Burnham, D. C., and Hallock, J. N., "Chicago Monoacoustic Vortex Sensing System, Volume IV: Wake Vortex Decay," Federal Aviation Administration, DOT/FAA/RD-79-103 IV, Vol. 4, Springfield, VA, 1982.

[84] Lamb, H., *Hydrodynamics*, Cambridge Univ. Press, Cambridge, MA, 1932, pp. 590–592.

[85] Fischenberg, D., "Bestimmung der Wirbelschleppencharakteristik aus Flugmessdaten," *Proceedings of the DGLR Annual Meeting*, Paper DGLR-2002-170, Stuttgart, Germany, Sept. 2002 (in German).

[86] Fischenberg, D., "A Method to Validate Wake Vortex Encounter Models from Flight Test Data," 27th International Congress of the Aeronautical Sciences, Paper ICAS2010-6.9.1, Sept. 2010.

[87] Spies, J., "HOPPER – ein ASTRA Systemkonzept," *Proceedings of the DGLR Annual Meeting*, Paper DGLR-JT2000-070, Leipzig, Germany, Sept. 2000 (in German).

[88] Dittmann, R., Brücker, E., and Noack, E., "ASTRA – das Deutsche Technologieprojekt für Zukünftige Raumtransportsysteme," *Proceedings of the DGLR Annual Meeting*, Paper DGLR-JT2000-117, Germany, Sept. 2000 (in German).

[89] Gockel, W., Kyr, P., Janovsky, R., and Roenneke, A., "Reusable RLV Demonstrator Vehicles – Phoenix Flight Test Results and Perspectives," *Proceedings of the 55th International Astronautical Congress*, International Astronautical Federation, Paris, Oct. 2004.

[90] Jategaonkar, R. V., Behr, R., Gockel, W., and Zorn, C., "Data Analysis of Phoenix Reusable Launch Vehicle Demonstrator Flight Test," *Journal of Aircraft*, Vol. 43, No. 6, 2006, pp. 1732–1737.

Downloaded by 80.82.77.83 on February 20, 2018 | http://arc.aiaa.org | DOI: 10.2514/4.102790

[91] Janicik, J. L., "The Phase I Space Maneuver Vehicle Test Program – Leading the United States into 21st Century Space Test and Evaluation," AIAA Paper 1999-4539, Sept. 1999.

[92] Yanagihara, M., Shigemi, M., and Suito, T., "Estimating Aerodynamic Characteristics of Automatic Landing Flight Experiment Vehicle Using Flight Data," *Journal of Aircraft*, Vol. 36, No. 6, 1999, pp. 926–933.

[93] Behr, R., "Phoenix: Aerodynamic Data Base, Version 3.1," EADS-ST Rept. PHX-DP-01, Issue 1, Jan. 2004.

[94] Parameswaran, V., Jategaonkar, R. V., and Press, M., "Five-Hole Flow Angle Probe Calibration from Dynamic and Tower Flyby Maneuvers," *Journal of Aircraft*, Vol. 42, No. 1, 2005, pp. 80–86.

[95] Gockel, W., "Phoenix Definition & Justification File – Plant Model," EADS-ST Rept. PHX-DDD-008, Issue 3, Jan. 2004.

[96] Cobleigh, B. R., "Development of the X-33 Aerodynamic Uncertainty Model," NASA/TP-1998-206544, April 1998.

[97] Hamel, P. G. (ed.), "Rotorcraft System Identification," AGARD, AR-280, Sept. 1991.

[98] Hamel, P. G., and Kaletka, J., "Rotorcraft System Identification – An Overview of AGARD FVP Working Group 18," AGARD, CP-552, Paper 18, 1995.

[99] Tischler, M. B., "System Identification Methods for Aircraft Flight Control Development and Validation," RTO MP-11, Paper 3, March 1999.

[100] Rohlfs, M., von Grünhagen, W., and Kaletka, J., "Nonlinear Rotorcraft Modeling and Identification," RTO-11, Paper 23, March 1999.

[101] Hamel, P. G., "Advances in Aerodynamic Modeling for Flight Simulation and Control Design," Vol. 23, No. 1-2, International Association of Applied Mathematics and Mechanics: GAMM-Mitteilungen, Wiley-VCH Verlag, Berlin, 2000, pp. 7–50.

[102] Bramwell, A. R. S., Done, G., and Balmford, D., *Bramwell's Helicopter Dynamics*, AIAA, Reston, VA, 2001.

[103] Kaletka, J., and von Grünhagen, W., "System Identification of Mathematical Models for the Design of a Model Following Control System," *Vertica*, Vol. 13, No. 2, 1989, pp. 213–228.

[104] Tischler, M. B., "Identification Requirements for High-Bandwidth Rotorcraft Flight Control System Design," *Journal of Guidance, Control, and Dynamics*, Vol. 13, No. 5, 1990, pp. 835–841.

[105] von Grünhagen, W., Bouwer, G., Pausder, H.-J., Henschel, F., and Kaletka, J., "A High Bandwidth Control System for Helicopter In-Flight Simulator ATTHeS - Modeling, Performance and Applications," *Advances in Aircraft Flight Control*, edited by M. B. Tischler, Taylor & Francis, London, 1996, pp. 73–101.

[106] Pitt, D. M., and Peters, D. A., "Theoretical Prediction of Dynamic Inflow Derivatives," *Vertica*, Vol. 5, No. 1, 1981, pp. 21–34.

[107] Hamers, M., and von Gruenhagen, W., "Nonlinear Helicopter Model Validation Applied to Real Time Simulations," *Proceedings of the 53rd Annual AHS Forum*, Vol. 2, American Helicopter Society, Alexandria, VA, 1997, pp. 958–972.

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[108] Kraemer, P., Gimonet, B., and von Gruenhagen, W., "A Systematic Approach to Nonlinear Rotorcraft Model Identification," *Journal of Aerospace Science and Technology*, Vol. 6, No. 8, 2002, pp. 579–590.

[109] Basset, P.-M., and Tchen-Fo, F., "Study of the Rotor Wake Distortion Effects on the Helicopter Pitch-Roll Cross-Couplings," *Proceedings of the 24th European Rotorcraft Forum*, Royal Aeronautical Society, London, Sept. 1998.

[110] Theodore, C., and Celi, R., "Helicopter Flight Dynamic Simulation with Refined Aerodynamics and Flexible Blade Modeling," *Journal of Aircraft*, Vol. 39, No. 4, 2002, pp. 577–586.

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EPILOGUE

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> "Science is not formal logic – it needs the free play of the mind in as great a degree as any other creative art. It is true that this is a gift which can hardly be taught, but its growth can be encouraged in those who already posses it" [1].
>
> Max Born (1882–1970)
> German physicist and Nobel Laureate

The leading idea of this book has been to provide an overview of the system identification methodology as applied to flight vehicles, focusing on the key methods of parameter estimation in the time domain and emphasizing nonlinear modeling. Although a certain degree of mathematical treatment is unavoidable, we have attempted to keep the theoretical development simple and understandable, including that for the methods in the stochastic framework. The major thrust has been on practical issues and on explaining in simple terms the need, purpose, background theory, and scope of the various topics covered in this book. The book is not intended as a cookbook, but certain guidelines demonstrated and recommended here help the reader to arrive at an adequate model. In a few cases the discussion has been on a philosophical level, but this was intentional, with the goal of sharing the process learned and experience gathered during the exercises.

It is our strong belief that system identification should be a goal-oriented task. Hence, at the outset we need to define the specific purpose for which the model is required because this determines to a large extent the amount of effort that will be spent on model building. A better understanding of the coordinated Quad-M approach helps one to arrive at an adequate model of minimum complexity, yet retaining the salient dynamic effects and having adequate accuracy for the intended purpose with a minimum amount of effort. We recapture here the principle outcomes of our discussions:

1. It has been demonstrated that, without getting bogged down in theoretical details, through simple basic understanding and the existing a priori

knowledge of the modes of aircraft motion, adequate inputs can be designed in most of the cases. Multistep inputs are necessary to excite the dynamic motion; however, their exact shape and time steps, although desirable, are not critical; empirical rules are quite adequate to determine the time steps. Besides the classical dynamic maneuvers, other system identification maneuvers, for example, acceleration-deceleration, pushdown-pullups, steady heading steady sideslips, and so on, are useful in system identification.

2. Data compatibility checks help to verify and improve the quality of the recorded data. In most of the cases, the standard output-error method accounting for measurement noise only is adequate for this purpose. The data gathering consisting of performing maneuvers, and recording, checking, and improving data mainly limits the scope of the aerodynamic modeling.
3. The output-error method is most widely used to estimate the aerodynamic parameters. Extensions and practical issues discussed here are helpful in catering for specific problems one may encounter in practice. It can be applied to large-scale problems without difficulties. If problems are encountered, in most of the cases they are due to some other cause, like poor information content in the data being analyzed or a wrong model postulate. The more advanced filter-error method accounting for state noise is required in a few cases only, for example, for data with atmospheric turbulence or stochastic vortex shedding from aircraft nose at high angles of attack. The least-squares method provides an alternative approach, but it needs high-precision, low-noise measurements, which are free of systematic bias and scale-factor errors. Furthermore, state-space representation and output-error method incorporating simulation is more efficient and closer to many other applications of the model. In the specific case of highly unstable aircraft, however, techniques other than the output-error method are preferable, such as filter error, least squares, or extended Kalman filter.
4. Basically, there are no restrictions on postulating a model to be identified. Highly complex and nonlinear models can be identified from data, provided adequate information is available in the flight data.

It is apparent that the time-domain methodology, as advocated in this book, provides for a large number of parameter estimation methods. The choice of a particular method mostly depends on the purpose and type of model one is estimating. If one is doing postflight data analysis, it makes little sense to go in for recursive parameter estimation because off-line methods are much more powerful and accurate. If one is interested in phenomenological models, that is, those based on physics of the problem with parameters having physical meaning such as aerodynamic derivatives, then the choice of output-error or least-squares method is obvious because the artificial neural networks yield black-box models belonging to the category of behavioral models. As already pointed out in this section, the filter-error method might be suitable only in very special cases when the

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turbulence level is considerable. As such, the best choice to start with appears to be to apply least-squares or output-error method. If we have flight data of reasonable quality and the recorded data have been calibrated reasonably well, both methods would give comparable estimates. The least-squares (equation-error) method is somewhat simpler to apply and provides direct matching of aerodynamic coefficients. It can be applied to each aerodynamic force and moment coefficient separately, which helps in model building. But it yields biased estimates in the presence of noise and systematic errors in the independent variables, also called regressors. Furthermore, the data preprocessing step would include computing aerodynamic force and moment coefficients from the flight measurements of accelerations and angular accelerations/rates. If we do not want to go through these steps of precomputation and precise calibration and elimination of measurement errors, then output error would be the choice. This, in turn, would need equations of motion to be coded in the form of state equations. It is more complex than the least-squares method, but then it is also more accurate because it accounts for noise in the measured variables. Furthermore, simulation, being a part of the output-error method, yields model that can be used directly in many applications. Simulation incorporating numerical integration of state equations is also sensitive to small changes in the parameters and initial conditions; as such, parameter estimation based on a model that incorporates simulation, which, in turn, implies the output-error method, is preferable.

In short, we have a fairly good understanding of the basic theory and parameter estimation algorithms; the challenge is in the applications. There is no doubt that in the majority of the cases flight vehicle system identification can be performed accurately and reliably. We have covered numerous examples, providing an appreciation of a wide variety of aerodynamic modeling and parameter estimation problems. The choice of the examples presented might appear biased, but it is unavoidable because we wish to report only on those cases with which we have worked. As a consequence, the views represented in this book mostly reflect the philosophy practiced by the authors and at their organization. Others might not totally agree with some of the issues discussed, but the reported schemes have worked in practice, providing acceptable solutions in some of the most demanding tasks. In this sense it seems to be a reasonable philosophy. In some of the more advanced and emerging applications, the ability to formulate mathematical models purported to underlie the physical phenomenon and the skill to interpret the modeling discrepancies looking beyond the obvious will mainly limit the scope of modeling. In other words, experience and engineering judgment will be called for, and in this sense system identification remains a scientific art, but an art supported by well-established methodology.

We would venture a guess about the likely direction of future system identification activities although forecasting is rather risky [2]. Three classical topics—1) generation of flight-validated high-fidelity databases for simulators and other applications, 2) modeling of nonlinear aerodynamics, and 3) modeling of unstable aircraft—will continue, as before, to be the prime areas of applications.

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Furthermore, better updating techniques will be needed to accommodate the slope type, that is, derivative, estimates on nonslope databases in a table look-up form for force and moment coefficients. The recent past has shown successful application of system identification and validation of flexible aircraft, integrating the rigid body and structural dynamics from flight data. This will pave way to more such applications because very large aircraft configurations and the use of lighter materials as well as overall weight reduction through structural optimization lead to a strong coupling between the rigid body and structural dynamics. Real-time parameter estimation is reemerging as yet another area of current and future research. Some theoretical aspects will focus on estimation of modeling uncertainties. Application of system identification methodology to low-cost unmanned aerial vehicles, both fixed wing and rotary wing, has now emerged as a topic of wide interest. Such vehicles pose a whole lot of new problems, such as poor signal quality due to high noise level, highly coupled dynamics and nonlinear aerodynamics, as well as adverse atmospheric conditions at low altitudes and vehicle speed. Reliable application to this class of vehicle needs to be established and will be one major focus in the future. Other advances will result from the actual needs, for example, better air data measurements. Although in the distant future, another thrust will be possibly on integrating system identification and computational-fluid-dynamics methodologies. These can call for new modeling concepts and associated measurement and experimental techniques to enable parameter estimation. In short, we will face some most demanding and fascinating modeling problems in the future, and we are convinced that in these cases system identification will provide answers to unmasking the modeling deficiencies, reducing developmental risk and improving flight safety issues.

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REFERENCES

[1] Born, M., "The Teaching of Theoretical Physics in Universities," *Reports on Progress in Physics*, Vol. 8, 1941, pp. 1–10.

[2] Jategaonkar, R. V., "Flight Vehicle System Identification – Engineering Utility," (Guest Editorial), *Journal of Aircraft*, Vol. 42, No. 1, 2005, p. 11.

Power Spectrum of a Multistep Input Signal

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A multistep input signal of arbitrary shape can be synthesized by a sequential combination of equidistant pulse inputs (see Fig. A1a) and each pulse of a fixed duration as a difference of two time displaced step signals (see Fig. A1b). Accordingly, a mutltistep input signal $u(t)$ in time domain can be expressed as [1, 2] [also Koehler, R., personal correspondence (nondocumented), DLR Inst. of Flight Mechanics, Braunschweig, Germany, 1977]

$$u(t) = \sum_{k=1}^{n_P} V_k[\sigma(t-(k-1)\Delta t) - \sigma(t-k\Delta t)] \tag{A1}$$

where n_P the number of pulses (for example, seven for the case depicted in Fig. A1a), V_k the amplitudes of the individual pulses, $\sigma(t)$ the unit step at time t, and Δt the time step (width) of the each pulse.

The frequency-domain transformation of Eq. (A1) is given by [3]

$$u(\omega) = \sum_{k=1}^{n_p} V_k \left[\frac{1}{j\omega} e^{-j(k-1)\omega\Delta t} - \frac{1}{j\omega} e^{-jk\omega\Delta t}\right] \tag{A2}$$

where ω is the angular frequency, e the base of the natural logarithm (a constant), and j the imaginary number. Equation (A2) can be rewritten as

$$u(\omega) = \Delta t \frac{1-e^{-j\omega\Delta t}}{j\omega\Delta t} \sum_{k=1}^{n_P} V_k e^{-j(k-1)\omega\Delta t} \tag{A3}$$

In terms of the normalized frequency $\Omega = \omega\Delta t$, Eq. (A3) leads to

$$u(\omega) = \Delta t \frac{e^{j\Omega/2} - e^{-j\Omega/2}}{j\Omega} e^{-j\Omega/2} \sum_{k=1}^{n_P} V_k e^{-j(k-1)\Omega} \tag{A4}$$

Using Euler's formula for the complex exponential function in terms of trigonometric functions, namely, $e^{jx} = \cos x + j\sin x$, where x is a real number, it turns out that $e^{j\Omega/2} - e^{-j\Omega/2} = 2j\sin(\Omega/2)$, which when substituted in Eq. (A4), after

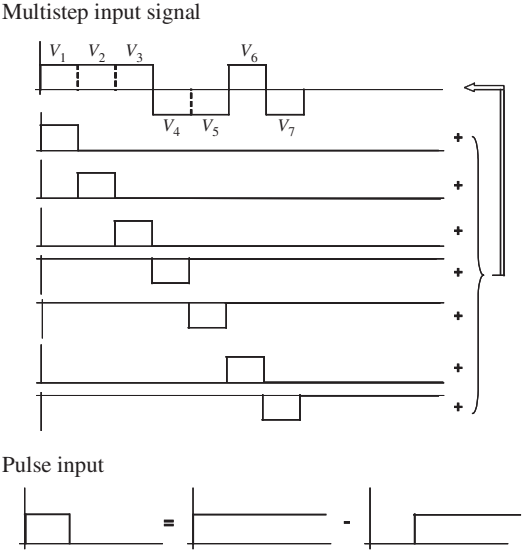

Fig. A1 Schematic decomposition of multistep input.

minor simplification, yields

$$u(\omega) = \Delta t \frac{\sin(\Omega/2)}{(\Omega/2)} \sum_{k=1}^{n_P} V_k e^{-j(2k-1)\Omega/2} \tag{A5}$$

Now, applying the same Euler's formula once again, we can expand the exponential term within the summation on the right-hand side of Eq. (A5), which leads to

$$u(\omega) = \Delta t \frac{\sin(\Omega/2)}{(\Omega/2)} \sum_{k=1}^{n_p} V_k \left(\cos \frac{(2k-1)\Omega}{2} + j \sin \frac{(2k-1)\Omega}{2} \right) \tag{A6}$$

The power (or energy) spectrum $E(\omega)$ of an arbitrary multistep signal represented in the time domain by Eq. (A1) can be computed easily using its frequency-domain representation of Eq. (A6) as follows:

$$\begin{aligned} E(\omega) &= |u(\omega)|^2 = \mathrm{Re}^2[u(\omega)] + \mathrm{Im}^2[u(\omega)] \\ &= \Delta t^2 \left(\frac{\sin(\Omega/2)}{(\Omega/2)} \right)^2 \left\{ \left[\sum_{k=1}^{n_P} V_k \cos\left(\frac{2k-1}{2} \Omega \right) \right]^2 \right. \\ &\qquad \left. + \left[\sum_{k=1}^{n_P} V_k \sin\left(\frac{2k-1}{2} \Omega \right) \right]^2 \right\} \end{aligned} \tag{A7}$$

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Now, applying the trigonometric identity, known as the Carnot formula, namely, $\sin^2 u = (1 - \cos 2u)/2$, to the term $\sin^2(\Omega/2)$, and expanding on the terms of square of the sums within flower bracket, after some minor simplification, Eq. (A7) leads to the desired form:

$$E(\omega) = 2\Delta t^2 \frac{1 - \cos\Omega}{\Omega^2} \left[\sum_{k=1}^{n_P} V_k^2 + 2 \sum_{i=1}^{n_P - 1} \cos i\Omega \sum_{l=1}^{n_P - i} V_l V_{l+i} \right] \tag{A8}$$

REFERENCES

[1] Koehler, R., and Wilhelm, K., "Auslegung von Eingangssignalen für die Kennwertermittlung," DFVLR-IB 154-77/40, Germany, Dec. 1977 (in German).

[2] Proskawetz, K.-O., "Optimierung Stufenförmiger Eingangssignale im Frequenzbereich für die Parameter Identifizierung," *ZFW*, Vol. 9, No. 6, Nov.–Dec. 1985, pp. 362–370 (in German).

[3] Backman, R. B., and Tukey, J. W., *The Measurement of Power Spectra*, Dover, New York, 1958.

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Identifiability of Initial Conditions and Bias Parameters

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Given a linear model postulate of the form

$$\dot{x} = Ax + B[u - \Delta u], \quad x(t_0) = x_0 \tag{B1}$$

$$y = Cx + D[u - \Delta u] + \Delta z \tag{B2}$$

we would like to investigate, in general, the identifiability of parameters appearing in matrices A, B, C, and D, of the initial conditions x_0, and of the zero shifts Δu and Δz. But, as discussed in Sec. 3.5.1, proper flight maneuvers ensure identifiability of aerodynamic derivatives, that is, of A, B, C, and D. Hence, we consider here the issue related to other constant terms x_0, Δu, and Δz only. They are totally $(n_x + n_u + n_y)$ in number, where n_x is the number of states, n_u the number of control inputs, and n_y the number of outputs. The state, control, and output variables are functions of time; however, we drop the time dependence, that is, (t), in the notation throughout this section.

Identifiability of any parameter, in general, is determined by the observability of the system. A system is said to be observable if the initial state x_0 can be determined uniquely by examining the system output $y(t)$ for $t > t_0$ in a finite interval over some period of time 0 to T. If the initial conditions are determined, the system behavior at any time t in future is uniquely determined. To investigate identifiability, we apply the conventional observability analysis. We follow here the commonly applied Kalman criterion [1]. Drawing upon the discussion presented in [2], the identifiability investigations for a system represented in Eqs. (B1) and (B2) are performed in two steps; the first one deals with initial conditions only and the other with zero shifts and initial conditions together.

Accordingly, we start by considering the following system without bias terms:

$$\dot{x} = A\,x; \quad x(t_0) = x_0 \tag{B3}$$

$$y = C\,x \tag{B4}$$

where the control input u has also been dropped without loss of generality because it does not affect these investigations. The general procedure of checking

observability, leading to the Kalman criterion, consists of deriving a system equations in terms of A and C, which can be solved for x. This is possible by starting from the observation equation (B4) and successively differentiating and substituting the right-hand side of Eq. (B3) for $\dot{x}$. This leads to

$$\begin{aligned} y &= Cx \\ \dot{y} &= C\dot{x} = CAx \\ \ddot{y} &= CA\dot{x} = CA^2x \\ &\dots \end{aligned} \tag{B5}$$

Now, y are known from the measurements; hence, their derivatives $\dot{y}, \ddot{y} \dots$ are also known. We can then rewrite Eq. (B5) in a familiar matrix form $Ax = b$ as

$$\begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix} x = \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \\ \vdots \end{bmatrix} \tag{B6}$$

Equation (B6) can be solved for x uniquely provided the matrix on the left-hand side

$$Q = \begin{bmatrix} C^T & (CA)^T & (CA^2)^T & \dots & (CA^{n-1})^T \end{bmatrix}^T \tag{B7}$$

has rank n_x. The matrix Q defined in Eq. (B7) is commonly called the observability matrix. Unique solution for x, ensured by rank n_x of observability matrix, implies we can estimate initial conditions x_0 completely.

Having investigated the necessary conditions for identifiability of x_0, in the second step we get back to the complete system model including initial conditions and zero shifts. By multiplying out the terms in Eqs. (B1) and (B2), and, as discussed earlier, dropping out the terms involving control inputs u, we get

$$\dot{x} = Ax - B\Delta u; \quad x(t_0) = x_0 \tag{B8}$$

$$y = Cx - D\Delta u + \Delta z \tag{B9}$$

We once again follow a similar procedure of successive differentiation and substitution. Starting from Eq. (B9), differentiating the same, substituting Eq. (B8) for $\dot{x}$, and using the knowledge that $\Delta\dot{u}$ and $\Delta\dot{z}$ are zero because the bias parameters

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are assumed to be constant over the period of observation, we obtain

$$\begin{aligned} y &= Cx - D\Delta u + \Delta z \\ \dot{y} &= C\dot{x} - 0 + 0 \\ &= CAx - CB\Delta u \\ \ddot{y} &= CA\dot{x} - 0 \\ &= CA^2x - CAB\,\Delta u \\ &\dots \end{aligned} \tag{B10}$$

which can be written in matrix form as

$$\underbrace{\begin{bmatrix} C & -D & I \\ CA & -CB & 0 \\ CA^2 & -CAB & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} x \\ \Delta u \\ \Delta z \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \\ \vdots \end{bmatrix} \tag{B11}$$

where I is the $n_y \times n_y$ identity matrix. To allow estimation of all of the initial conditions and bias parameters, the system matrix $\mathcal{A}$ in Eq. (B11) must have rank $(n_x + n_u + n_y)$.

Because of the presence of the identity matrix, the first row of $\mathcal{A}$ yields rank n_y. The rest of the matrix consists of

$$\begin{bmatrix} CA & -CB \\ CA^2 & -CAB \\ CA^3 & -CA^2B \\ \vdots & \vdots \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix} \begin{bmatrix} A & -B \end{bmatrix} \tag{B12}$$

It now remains to find the rank of the matrix product on the right-hand side of Eq. (B12). The first term on the right side of Eq. (B12) is the observability matrix Q appearing in Eq. (B7), and we already know it has rank n_x, provided the system is fully observable. The second term on the right-hand side has a maximum rank of n_x. Now, from matrix algebra results, we know that the rank of a matrix product cannot be greater than the smallest rank of the multiplied matrices, which implies that the rest matrix of Eq. (B12) has a maximum rank of n_x. Accordingly, the system matrix $\mathcal{A}$ of Eq. (B11) has a maximum rank of $(n_y + n_x)$. This implies that a maximum of $(n_y + n_x)$ terms can be determined independently and not $(n_x + n_u + n_y)$ as ideally desired.

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REFERENCES

[1] Gelb, A., *Applied Optimal Estimation*, MIT Press, Cambridge, MA, 1974.

[2] Plaetschke, E., "Maximum-Likelihood-Verfahren," *Lecture Notes: Praktische Einführung in die Systemidentifizierung*, Carl-Cranz-Gesellschaft, Oberpfaffenhofen, Germany, June 1985, Sec. 5 (in German).

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Derivation of the Likelihood Function

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The principle of likelihood is based on a probability density function. Although the method is applicable to any form of the density function distribution, for mathematical tractability we consider the Gaussian (normal) distribution, which is completely determined by the mean and covariance matrix. It is the most widely used assumption in practical cases.

The probability density function of a Gaussian (normally) distributed, scalar real random variable x is given [1]

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-m)^2}{2\sigma^2}\right] \tag{C1}$$

where $m = E\{x\}$ and $\sigma^2 = E\{(x-m)^2\}$ denote the mean value and variance respectively and E the expected value.

Consider two statistically independent, Gaussian variables x_1 and x_2 with zero mean and variances σ_1 and σ_2, respectively. Then the joint probability density function is given [2]

$$\begin{aligned} p(x_1, x_2) &= p(x_1) \cdot p(x_2) \\ &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{x_1^2}{2\sigma_1^2} - \frac{x_2^2}{2\sigma_2^2}\right] \end{aligned} \tag{C2}$$

Define two random variables y_1 and y_2, which are obtained from x_1 and x_2 through the rotational transformation

$$\begin{aligned} y_1 &= x_1 \cos\varphi - x_2 \sin\varphi \\ y_2 &= x_1 \sin\varphi + x_2 \cos\varphi \end{aligned} \tag{C3}$$

satisfying the condition $(x_1^2 + x_2^2 = y_1^2 + y_2^2)$. Because we assumed that the random variables x_1 and x_2 have zero mean, the mean values of the random variables y_1 and y_2 are also zero. Their variances are given by the expected

values of Eq. (C3):

$$\mu_{11} = E\{y_1^2\} = \sigma_1^2 \cos^2\varphi + \sigma_2^2 \sin^2\varphi$$
$$\mu_{22} = E\{y_2^2\} = \sigma_1^2 \sin^2\varphi + \sigma_2^2 \cos^2\varphi \tag{C4}$$

and the nonzero covariance is given by

$$\mu_{12} = E\{y_1 y_2\} = (\sigma_1^2 - \sigma_2^2) \sin\varphi \cos\varphi \tag{C5}$$

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The joint probability density function of y_1 and y_2 is now obtained as

$$p_2(y_1, y_2) = p_1\{x_1(y_1, y_2), x_2(y_1, y_2)\} \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_2}{\partial y_1} \\ \frac{\partial x_1}{\partial y_2} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$
$$= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{(y_1 \cos\varphi + y_2 \sin\varphi)^2}{2\sigma_1^2} - \frac{(-y_1 \sin\varphi + y_2 \cos\varphi)^2}{2\sigma_2^2} \right] \tag{C6}$$

In terms of the second moments of y_1 and y_2, the preceding expression can be expressed as

$$p(y_1, y_2) = \frac{1}{2\pi(\mu_{11}\mu_{22} - \mu_{12}^2)^{\frac{1}{2}}} \exp\left[\frac{-\mu_{22} y_1^2 + 2\mu_{12} y_1 y_2 - \mu_{11} y_2^2}{2(\mu_{11}\mu_{22} - \mu_{12}^2)} \right] \tag{C7}$$

Combining the two variables y_1 and y_2 into a vector y and the second moments into covariance matrix R, that is,

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}; \quad R = \begin{bmatrix} \mu_{11} & \mu_{12} \\ \mu_{12} & \mu_{22} \end{bmatrix} \tag{C8}$$

the probability density function can be expressed in the following concise form:

$$p(y_1, y_2) = \frac{1}{2\pi\sqrt{|R|}} \exp\left[-\frac{1}{2} y^T R^{-1} y \right] \tag{C9}$$

This result can be readily extended to the multidimensional distribution. The joint probability distribution function of n Gaussian distributed random

variables x_i with mean m_i is given by

$$p(x_1, \ldots, x_n) = \frac{1}{(2\pi)^{n/2}\sqrt{|R|}} \exp\left[-\frac{1}{2}(x-m)^T R^{-1}(x-m)\right] \tag{C10}$$

where $x^T = (x_1, \ldots, x_n)$ and $m^T = (m_1, \ldots, m_n)$ denote the vectors of random variables and of the mean values respectively, and

$$R = \begin{bmatrix} r_{11} & \cdot & \cdot & \cdot & r_{1n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ r_{1n} & \cdot & \cdot & \cdot & r_{nn} \end{bmatrix} \tag{C11}$$

the covariance matrix. The elements of R are the second moments

$$r_{ij} = E\{(x_i - m_i)(x_j - m_j)\} = \sigma_i \sigma_j \rho_{ij} \tag{C12}$$

where ρ_{ij} are called the correlation coefficients, which automatically implies that $\rho_{ii} = 1$.

The likelihood function is now defined as the conditional probability density $p(z(t_1), \ldots, z(t_N)|\Theta, R)$ of the n_y-dimensional measurement vector at N discrete time points for a given parameter vector Θ and given measurement error covariance matrix R. The conditional probability density of the measurements at a particular measurement time point t_k is given by [using Eq. (C10) just derived]

$$p(z(t_k)|\Theta, R) = \frac{1}{(2\pi)^{n_y/2}\sqrt{|R|}} \exp\left[-\frac{1}{2}\{z(t_k) - y(t_k)\}^T R^{-1}\{z(t_k) - y(t_k)\}\right] \tag{C13}$$

where y is the vector of observation variables (model output).

Assuming the error $v = z - y$ at different time points to be statistically independent, that is,

$$E\{v(t_k)v^T(t_\ell)\} = R\delta_{k\ell} \tag{C14}$$

results in the following likelihood function:

$$\begin{aligned} p(z(t_1), \ldots, z(t_N)|\Theta, R) &= \prod_{k=1}^{N} p(z(t_k)|\Theta, R) \\ &= \{(2\pi)^{n_y}|R|\}^{-N/2} \\ &\quad \times \exp\left[-\frac{1}{2}\sum_{k=1}^{N}[z(t_k) - y(t_k)]^T R^{-1}[z(t_k) - y(t_k)]\right] \end{aligned}$$

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REFERENCES

[1] Davenport, W. B., and Root, W. L., *Introduction to the Theory of Random Signals and Noise*, McGraw–Hill, New York, 1958.

[2] Plaetschke, E., "Maximum-Likelihood-Verfahren," *Lecture Notes: Praktische Einführung in die Systemidentifizierung*, Carl-Cranz-Gesellschaft, Oberpfaffenhofen, Germany, June 1985, Sec. 5 (in German).

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Statistical Properties of Maximum Likelihood Estimates

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D.1 ASYMPTOTIC CONSISTENCY

The maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ are asymptotically consistent, that is, $\widehat{\Theta}_{\mathrm{ML}}$ converges in probability to the true values Θ. In the following we investigate this property.

We know from the properties of the probability functions

$$\int p(z|\Theta)\mathrm{d}z = 1 \tag{D1}$$

Partial differentiation of Eq. (D1) with respect to Θ, and interchanging the order of integration and differentiation assuming sufficient regularity conditions [1], yields

$$\int \frac{\partial p(z|\Theta)}{\partial \Theta}\mathrm{d}z = 0 \tag{D2}$$

Equation (D2) can be rewritten as

$$\int \left(\frac{1}{p(z|\Theta)}\frac{\partial p(z|\Theta)}{\partial \Theta}\right)p(z|\Theta)\mathrm{d}z = \int \frac{\partial \ell np(z|\Theta)}{\partial \Theta}p(z|\Theta)\mathrm{d}z = 0 \tag{D3}$$

or equivalently,

$$E\left\{\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right\} = 0 \tag{D4}$$

Differentiating Eq. (D3) and rewriting yield

$$\int \left\{\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)^{T} + \frac{\partial^2 \ell np(z|\Theta)}{\partial \Theta^2}\right\}p(z|\Theta)\mathrm{d}z = 0 \tag{D5}$$

or

$$J = E\left\{\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)^T\right\} = E\left\{-\frac{\partial^2 \ell np(z|\Theta)}{\partial \Theta^2}\right\} \tag{D6}$$

The Fisher information matrix J, defined in Eq. (D6), is generally positive definite. However, if the observations are independent of Θ, that is, $p(z|\Theta)$ is not a function of Θ as assumed, then J in this case will reduce to zero. In practice, this implies that it would not be possible to estimate Θ from sample observations that do not contain information about Θ.

The Taylor-series expansion of the term $[\partial \ell np(z|\Theta)/\partial \Theta]$ in the preceding equation about the true values Θ evaluated at $\widehat{\Theta}_{\mathrm{ML}}$ leads to

$$\frac{\partial \ell np(z|\widehat{\Theta}_{\mathrm{ML}})}{\partial \Theta} = \frac{\partial \ell np(z|\Theta)}{\partial \Theta} + \frac{\partial^2 \ell np(z|\Theta^*)}{\partial \Theta^2}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \tag{D7}$$

where $\Theta^* = \lambda\Theta + (1-\lambda)\widehat{\Theta}_{\mathrm{ML}}$; $0 \le \lambda \le 1$.

Because $\widehat{\Theta}_{\mathrm{ML}}$ is the solution of the likelihood equation, equating Eq. (D7) to zero yields

$$\frac{\partial \ell np(z|\Theta)}{\partial \Theta} = -\frac{\partial^2 \ell np(z|\Theta^*)}{\partial \Theta^2}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \tag{D8}$$

We recall our assumption that the measurements at different time points are assumed to be statistically independent. This provides

$$\begin{aligned}\frac{\partial \ell np(z|\Theta)}{\partial \Theta} &= \frac{\partial}{\partial \Theta}\left(\ell n \prod_{k=1}^{N} p(z_k|\Theta)\right) \\ &= \frac{\partial}{\partial \Theta}\left(\sum_{k=1}^{N} \ell np(z_k|\Theta)\right) \\ &= \sum_{k=1}^{N} \frac{\partial}{\partial \Theta} \ell np(z_k|\Theta)\end{aligned} \tag{D9}$$

Similarly,

$$\frac{\partial^2 \ell np(z|\Theta)}{\partial \Theta^2} = \sum_{k=1}^{N} \frac{\partial^2 \ell np(z_k|\Theta)}{\partial \Theta^2} \tag{D10}$$

Now, from Eqs. (D9) and (D10), the strong law of large numbers, which indicates that the sample average converges in probability to the ensemble average, yields

$$\frac{1}{N}\sum_{k=1}^{N} \frac{\partial \ell np(z_k|\Theta)}{\partial \Theta} \;\rightarrow\; E\left\{\frac{\partial \ell np(z_k|\Theta)}{\partial \Theta}\right\} \tag{D11}$$

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and

$$\frac{1}{N}\sum_{k=1}^{N}\frac{\partial^2 \ell np(z_k|\Theta)}{\partial\Theta^2} \;\rightarrow\; E\left\{\frac{\partial^2 \ell np(z_k|\Theta)}{\partial\Theta^2}\right\} \tag{D12}$$

Further, it can be shown that the likelihood function is concave in Θ [2, 3]. Hence, it is appropriate to assume that the matrix $E\{\partial^2 \ell np(z_k|\Theta)/\partial\Theta^2\}$ is positive definite. From Eqs. (D8), (D11), and (D12), it directly follows that

$$\widehat{\Theta}_{\mathrm{ML}} \;\rightarrow\; \Theta \tag{D13}$$

with probability of one. Thus, $\widehat{\Theta}_{\mathrm{ML}}$ is consistent.

D.2 ASYMPTOTIC NORMALITY

The asymptotic normality implies that for a large number of data points N, the maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ converge to normal distribution, that is,

$$\sqrt{N}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \;\rightarrow\; r_1 \quad \sim\mathcal{N}(0, \bar{J}^{-1}) \tag{D14}$$

where $\bar{J}$ is the average Fisher information matrix per sample and Θ the true parameter values. In Eq. (D14), we have used the standard notation used in statistics, which implies that the term on the left-hand side tends to random variables denoted by r_1, having $\mathcal{N}(0, \bar{J}^{-1})$ distribution, that is, normal (Gaussian) distribution with zero mean and variance $\bar{J}^{-1}$. In the following, we investigate this property.

The Taylor-series expansion of $[\partial \ell np(z|\widehat{\Theta}_{\mathrm{ML}})/\partial\Theta]$ about the true parameter value Θ yields

$$\frac{\partial \ell np(z|\widehat{\Theta}_{\mathrm{ML}})}{\partial\Theta} = \frac{\partial \ell np(z|\Theta)}{\partial\Theta} + \frac{\partial^2 \ell np(z|\Theta)}{\partial\Theta^2}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) + \text{higher-order terms} \tag{D15}$$

Using the property that $\widehat{\Theta}_{\mathrm{ML}}$ are asymptotically consistent, higher-order terms can be neglected. Because $\widehat{\Theta}_{\mathrm{ML}}$ satisfies the likelihood equation, equating Eq. (D15) to zero gives

$$\frac{1}{\sqrt{N}}\frac{\partial \ell np(z|\Theta)}{\partial\Theta} = \sqrt{N}\left\{-\frac{1}{N}\frac{\partial^2 \ell np(z|\Theta)}{\partial\Theta^2}\right\}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \tag{D16}$$

Once again, using Eqs. (D6) and (D12), the strong law of large numbers leads to

$$-\frac{1}{N}\frac{\partial^2 \ell np(z|\Theta)}{\partial\Theta^2} \;\rightarrow\; E\left\{-\frac{\partial^2 \ell np(z_k|\Theta)}{\partial\Theta^2}\right\} = \bar{J} \tag{D17}$$

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where $\bar{J} = J/N$ represents the average information matrix per sample, and J is the Fisher information matrix.

The assumption that the samples z_k are independent allows the left-hand side of Eq. (D16) to be written as

$$\frac{1}{\sqrt{N}}\frac{\partial \ell np(z|\Theta)}{\partial \Theta} = \frac{1}{\sqrt{N}}\sum_{k=1}^{N}\frac{\partial \ell np(z_k|\Theta)}{\partial \Theta} \tag{D18}$$

Now, it is already known from Eqs. (D4) and (D6) that

$$\mathrm{E}\left\{\frac{\partial \ell np(z_k|\Theta)}{\partial \Theta}\right\} = 0 \tag{D19}$$

and

$$\mathrm{E}\left\{\left(\frac{\partial \ell np(z_k|\Theta)}{\partial \Theta}\right)\left(\frac{\partial \ell np(z_k|\Theta)}{\partial \Theta}\right)^{T}\right\} = \bar{J} \tag{D20}$$

Using Eqs. (D19) and (D20) and the central limit theorem, it follows from Eq. (D18) that

$$\frac{1}{\sqrt{N}}\frac{\partial \ell np(z|\Theta)}{\partial \Theta} \;\rightarrow\; r_2 \quad \sim \mathcal{N}(0, \bar{J}) \tag{D21}$$

Now, Eqs. (D16) and (D17) yield

$$\sqrt{N}\bar{J}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \;\rightarrow\; r_2 \quad \sim \mathcal{N}(0, \bar{J}) \tag{D22}$$

which can be equivalently expressed as

$$\sqrt{N}(\widehat{\Theta}_{\mathrm{ML}} - \Theta) \;\rightarrow\; r_1 \quad \sim \mathcal{N}(0, \bar{J}^{-1}) \tag{D23}$$

The property of asymptotic normality implies that the estimates $\widehat{\Theta}$ obtained from different sets of data samples corresponding to different experiments are clustered around Θ with a normal distribution.

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D.3 ASYMPTOTIC EFFICIENCY

The maximum likelihood estimates $\widehat{\Theta}_{\mathrm{ML}}$ are asymptotically efficient in the sense that they attain Cramér–Rao lower bounds. In the following, we study the property of asymptotic efficiency.

To establish the asymptotic efficiency of $\widehat{\Theta}_{\mathrm{ML}}$, consider a general estimator $\widehat{\Theta}(z)$, not necessarily the maximum likelihood. This gives

$$E\{\widehat{\Theta}\} = \int \widehat{\Theta} p(z|\Theta)\mathrm{d}z = \Theta + b(\Theta) \tag{D24}$$

where Θ represents the true values of the parameters and $b(\Theta)$ the bias that can result in the estimates.

Differentiation of Eq. (D24) yields

$$\int \widehat{\Theta}\left(\frac{\partial p(z|\Theta)}{\partial \Theta}\right)^T \mathrm{d}z = I + \frac{\partial b(\Theta)}{\partial \Theta} \tag{D25}$$

which can be rewritten as

$$\int \widehat{\Theta}\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)^T p(z|\Theta)\mathrm{d}z = I + \frac{\partial b(\Theta)}{\partial \Theta} \tag{D26}$$

Once again starting from the basic relation,

$$\int p(z|\Theta)\mathrm{d}z = 1 \tag{D27}$$

differentiation of Eq. (D27) yields

$$\int \frac{\partial p(z|\Theta)}{\partial \Theta}\mathrm{d}z = 0 \tag{D28}$$

which can be rewritten as

$$\int \left(\frac{1}{p(z|\Theta)}\frac{\partial p(z|\Theta)}{\partial \Theta}\right)p(z|\Theta)\,\mathrm{d}z = \int \frac{\partial \ell np(z|\Theta)}{\partial \Theta}p(z|\Theta)\mathrm{d}z = 0 \tag{D29}$$

Multiplying Eq. (D29) with $\bar{\Theta}$, the expected value of $\widehat{\Theta}$, that is, $\bar{\Theta} = E\{\widehat{\Theta}\}$, and subtracting from Eq. (D26) gives

$$\int (\widehat{\Theta} - \bar{\Theta})\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)^T p(z|\Theta)\mathrm{d}z = I + \frac{\partial b(\Theta)}{\partial \Theta} \tag{D30}$$

Using the Cauchy–Schwarz inequality for two integrable quadratic functions $f(x)$ and $g(x)$, namely,

$$\left(\int f(x)g(x)\,\mathrm{d}x\right)^2 - \int f^2(x)\,\mathrm{d}x \int g^2(x)\,\mathrm{d}x \leq 0 \tag{D31}$$

it follows from Eq. (D30) that

$$\int (\widehat{\Theta} - \bar{\Theta})(\widehat{\Theta} - \bar{\Theta})^T p(z|\Theta)\,\mathrm{d}z \times \left(\int \left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)\left(\frac{\partial \ell np(z|\Theta)}{\partial \Theta}\right)^T p(z|\Theta)\,\mathrm{d}z\right) \geq \left(I + \frac{\partial b(\Theta)}{\partial \Theta}\right)^2 \tag{D32}$$

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However, the second term on the left-hand side of the inequality is defined as the information matrix J.

Using Eqs. (D6) and (D24), Eq. (D32) leads to

$$\operatorname{cov}(\bar{\Theta}) = E\left\{(\hat{\Theta} - \bar{\Theta})(\hat{\Theta} - \bar{\Theta})^T\right\} \geq \left(I + \frac{\partial b(\Theta)}{\partial \Theta}\right) J^{-1} \tag{D33}$$

The expression in Eq. (D33) is called the Cramer–Rao inequality. Because the maximum likelihood estimates are asymptotically bias free, it follows that

$$E\left\{\hat{\Theta}_{\mathrm{ML}}\right\} = \Theta \quad \text{and} \quad b(\hat{\Theta}_{\mathrm{ML}}) = 0 \tag{D34}$$

Therefore, Eq. (D33) yields

$$\operatorname{cov}\left(\hat{\Theta}_{\mathrm{ML}}\right) \geq J^{-1} \tag{D35}$$

It has already been shown in Secs. D.1 and D.2 that $\hat{\Theta}_{\mathrm{ML}}$ are asymptotically consistent and normal. Combining these results with Eq. (D35), it follows that $\hat{\Theta}_{\mathrm{ML}}$ is asymptotically efficient in the sense of achieving the Cramér–Rao lower bound.

The property of asymptotic efficiency is of practical significance. It implies that the maximum likelihood estimator makes efficient use of the available data. The Cramér–Rao lower bound indicates the theoretically maximum achievable accuracy of the estimates.

REFERENCES

[1] Cramér, H., *Mathematical Methods of Statistics*, Princeton Univ. Press, Princeton, NJ, 1946.

[2] Kashyap, R. L., "Maximum Likelihood Identification of Stochastic Linear Systems," *IEEE Transactions on Automatic Control*, Vol. AC-15, No. 1, 1970, pp. 25–34.

[3] Wilks, S. S., *Mathematical Statistics*, Wiley, New York, 1962.

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Minimization of Likelihood Function with Respect to Covariance Matrix *R*

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In Chapters 4 and 5, parameter estimation applying the maximum likelihood method required minimization of the likelihood cost function given by

$$L(z|\Theta, R) = \frac{1}{2}\sum_{k=1}^{N} \nu^T R^{-1} \nu + \frac{N}{2}\ell n(|R|) + \frac{Nn_y}{2}\ell n(2\pi) \tag{E1}$$

where $\nu(t_k) = z(t_k) - y(t_k)$ represents the residuals at the discrete time point t_k.

For a specified model structure and the data being analyzed, the number of observations n_y and the data points N to be evaluated are defined and fixed. Accordingly, the last term in the preceding equation is a constant and, hence, neglected without affecting minimization.

The first and the second terms on the right-hand side of Eq. (E1) contain R^{-1} and $|R|$, respectively. To facilitate the derivation, we rewrite the right-hand side such that both terms contain R^{-1}. Using the matrix expression

$$|R| = 1/|R^{-1}| \tag{E2}$$

and after minor simplifications, the preceding equation can be rewritten as

$$L(z|\Theta, R) = \frac{1}{2}\sum_{k} \nu^T R^{-1} \nu - \frac{N}{2}\ell n(|R^{-1}|) \tag{E3}$$

Now, writing Eq. (E3) in a component form and denoting the elements of R^{-1} as a_{ij} lead to

$$L = \frac{1}{2}\sum_{k}\sum_{i}\sum_{j} a_{ij}\nu_i\nu_j - \frac{N}{2}\ell n\left(\sum_{i} a_{ij}A_{ij}\right) \tag{E4}$$

where we had used the matrix algebra result:

$$|A| = \sum_i a_{ij} A_{ij} \tag{E5}$$

in which A_{ij} denote the cofactor of the element a_{ij} of a matrix A.

Partially differentiating Eq. (E4) and making use of the results

$$\frac{\partial(\ell n y)}{\partial x} = \frac{1}{y}\frac{\partial y}{\partial x} \tag{E6}$$

leads to

$$\frac{\partial L}{\partial a_{ij}} = \frac{1}{2}\sum_k v_i v_j - \frac{N}{2}\frac{1}{\left(\sum_i a_{ij} A_{ij}\right)} A_{ij} \tag{E7}$$

From Eq. (E5), we know that the quantity in the denominator of the second term on the right-hand side of Eq. (E7) is nothing but $|A|$. For the minimum of L with respect to the measurement error covariance matrix R, the partial derivative of L with respect to the elements a_{ij} of R^{-1} must be zero. Accordingly, setting the right-hand side of Eq. (E7) to zero, we get

$$\frac{1}{2}\sum_k v_i v_j - \frac{N}{2}\frac{1}{|A|} A_{ij} = 0 \tag{E8}$$

Now, making use of another matrix algebra result,

$$(A^{-1})_{ij} = A_{ji}/|A| \tag{E9}$$

Eq. (E8) leads to

$$\frac{1}{2}\sum_k v_i v_j = \frac{N}{2} r_{ji} \tag{E10}$$

Thus, for each element r_{ij}, we get

$$r_{ij} = \frac{1}{N}\sum_k v_j v_i \tag{E11}$$

which can be rewritten in the matrix notation as

$$R = \frac{1}{N}\sum_{k=1}^{N} v(k) v^T(k) \tag{E12}$$

which is the solution that we were interested in [1, 2]. For any fixed value of the parameter vector Θ, Eq. (E12) maximizes the likelihood function with respect to R.

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It is also possible to derive the same expression using matrix differentiation results; see [3].

REFERENCES

[1] Plaetschke, E., "Maximum-Likelihood-Verfahren," *Lecture Notes: Praktische Einführung in die Systemidentifizierung*, Carl-Cranz-Gesellschaft, Oberpfaffenhofen, Germany, June 1985, Sec. 5 (in German).

[2] Groves, R. D., Bowles, R. L., and Mayhew, S. C., "A Procedure for Estimating Stability and Control Parameters from Flight Test Data by Using Maximum Likelihood Method Employing Real-Time Digital System," NASA TN D-6735, May 1972.

[3] Goodwin, G. C., and Payne, R. L., *Dynamic System Identification*, Academic Press, New York, 1977.

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Derivation of Kalman Filter and Extended Kalman Filter

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F.1 KALMAN FILTER

When the process under investigation contains stochastic nonmeasurable inputs, we cannot propagate the states by solving an initial-value problem through simple numerical integration as in the case of deterministic systems. Instead, we have to estimate the states using a suitable state estimator. The Kalman filter provides an efficient computational procedure to estimate the states of a linear system [1]. It has two main advantages, namely, 1) it is optimal in the least-mean-squares sense yielding a minimum variance estimator and 2) it operates recursively, making it suitable for computer implementations. For a dynamic system represented in state space, the Kalman filter consists of two steps:

1. Prediction, also called extrapolation, propagation or time update
2. Correction, also called measurement update

The first step deals with propagating the states and the errors in the state prediction between the two discrete time points k to $k+1$. The second step corrects a posteriori these predictions once the new measurements at sample time point $k+1$ become available (see Fig. F1), where x is the state vector, y the model outputs, and z the corresponding measurements. The predicted states are denoted by the superscript $\sim$ (tilde) and the corrected states by $\hat{}$ (hat). There are different notations used in the literature on estimation theory to denote the predicted and corrected states, as well as different derivations of the Kalman filter considering purely continuous-time or discrete-time or mixed continuous-discrete representations [2–6]. In our specific case, we are faced with the last type, namely continuous-discrete representations

Extending our brief discussion in Sec. 3.8.2 and using the discrete-time theory of linear dynamic systems, it can be shown that the solution to a linear stochastic system is represented by

$$\dot{x}(t) = Ax(t) + Bu(t) + Fw(t) \tag{F1}$$

$$y(t) = C(t)x(t) + Du(t) \tag{F2}$$

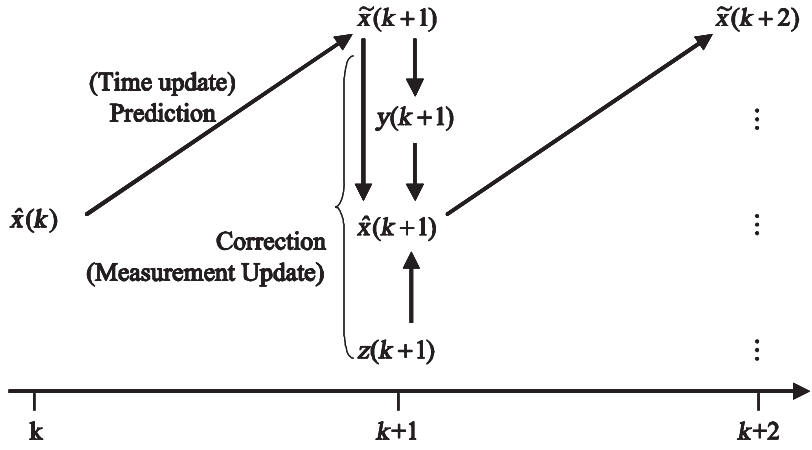

Fig. F1 Schematic of Kalman filter computations.

is given by

$$x(k+1) = \Phi x(k) + \Psi B u(k) + \Psi(\Delta t)^{-1/2} F\, w(k) \tag{F3}$$

where the process noise $w(t)$ is assumed to be zero-mean white Gaussian noise with an identity power spectral density matrix and F the process noise distribution matrix. The corresponding discrete-time white-noise process $w(k)$ has to have the covariance $1/\Delta t$ times the identity matrix. This is reflected in Eq. (F3) through the factor $1/\sqrt{\Delta t}$ associated with F, which is necessary to account for the transition between the continuous-time and discrete-time representations [3]. It is also critical to get the limit of discrete-time system into continuous time; otherwise, in the limit case the system will not react to noise, which is physically not realistic. A more detailed treatment of this subtle issue is provided by Maine and Iliff [6]. We specifically stress that $w(t)$ and $w(k)$ are not the same although we use the same symbol w for the two processes.

As already defined in Sec. 3.8.2, the state transition matrices Φ and its integral Ψ are given by

$$\Phi = e^{A\Delta t} \approx I + A\Delta t + A^2 \frac{\Delta t^2}{2!} + \cdots \tag{F4}$$

$$\Psi = \int_0^{\Delta t} e^{A\tau}\, \mathrm{d}\tau \approx I\Delta t + A\frac{\Delta t^2}{2!} + A^2 \frac{\Delta t^3}{3!} + \cdots \tag{F5}$$

We draw the attention of the readers to the definition of the matrix Ψ in Eq. (F5), which defers slightly from that used in the literature on estimation theory.

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Specifically, we do not include the input matrix B in the definition of Ψ. This allows us to use the matrix Ψ of Eq. (F5) in the last term of Eq. (F3) as well, but now multiplied by the process noise distribution matrix F. Here, we are mainly concerned with time-invariant systems, for which the system matrix A is not a function of time. Accordingly, the transition matrix Φ is also time invariant. Otherwise, in a general case, the exact notation of Eq. (F3) should include $\Phi(k)$ and also for its integral $\Psi(k)$.

Because we cannot measure the process noise $w(t)$, it is not possible to apply directly the solution given by Eq. (F3) to compute the true states x (true state is denoted without $\sim$ and without ^). Instead, the solution is obtained applying the Kalman filter by predicting the states and errors in their predictions across the time points and correcting them at each time point.

For the prediction, the states are propagated by considering the deterministic part of Eq. (F1). Accordingly, the time propagation of states is given by

$$\tilde{x}(k+1) = \Phi\hat{x}(k) + \Psi Bu(k) \tag{F6}$$

where $\tilde{x}$ is the predicted state and $\hat{x}$ the corrected state.

Now, because of the presence of process noise $w(t)$, the state estimates $\tilde{x}$ are not precise; the uncertainty in their estimates is given by the covariance matrix of the state predictions $\tilde{P}(k+1)$ defined as

$$\tilde{P}(k+1) = E\{[\tilde{x}(k+1) - x(k+1)][\tilde{x}(k+1) - x(k+1)]^T\} \tag{F7}$$

where $x(k+1)$ is the true state and $\tilde{x}(k+1)$ the estimate at the $(k+1)$th time point. In Eq. (F7), we substitute Eq. (F6) for $\tilde{x}(k+1)$ and Eq. (F3) for the true state $x(k+1)$, where for notational simplicity we have used $F_d = (\Delta t)^{-1/2}F$. This yields

$$\tilde{P}(k+1) = E\{[\Phi\hat{x}(k) - \Phi x(k) - \Psi F_d\, w(k)][\Phi\hat{x}(k) - \Phi x(k) - \Psi F_d\, w(k)]^T\} \tag{F8}$$

where $E\{\}$ denotes the expected value; the terms $\Psi Bu(k)$ cancel out. Equation (F8) can be rewritten in terms of the error $\hat{e}(k)$ as

$$\tilde{P}(k+1) = E\{[\Phi\hat{e}(k) - \Psi F_d\, w(k)][\Phi\hat{e}(k) - \Psi F_d\, w(k)]^T\} \tag{F9}$$

where $\hat{e}(k) = \hat{x}(k) - x(k)$, the difference between the updated estimate $\hat{x}(k)$ and the true state $x(k)$. By multiplying out the terms and using linearity of

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expectations, Eq. (F9) can be simplified as follows:

$$\begin{aligned}\tilde{P}(k+1) &= E\{\Phi\hat{e}(k)\hat{e}^T(k)\Phi^T - \Phi\hat{e}(k)w^T(k)F_d^T\Psi^T \\ &\quad -\Psi F_d w(k)\hat{e}^T(k)\Phi^T + \Psi F_d\, w(k)w^T(k)F_d^T\Psi^T\} \\ &= \Phi E\{\hat{e}(k)\hat{e}^T(k)\}\Phi^T - \Phi E\{\hat{e}(k)w^T(k)\}F_d^T\Psi^T \\ &\quad - \Psi F_d E\{w(k)\hat{e}^T(k)\}\Phi^T + \Psi F_d\, E\{w(k)w^T(k)\}F_d^T\Psi^T \qquad \text{(F10)}\end{aligned}$$

Now making use of the assumption that the noise $w(k)$ and errors $\hat{e}(k)$ are uncorrelated, the second and the third terms on the right-hand side in the preceding equation vanish, and Eq. (F10) simplifies to

$$\tilde{P}(k+1) = \Phi E\{\hat{e}(k)\hat{e}^T(k)\}\Phi^T + \Psi F_d\, E\{w(k)w^T(k)\}F_d^T\Psi^T \qquad \text{(F11)}$$

The expected value of $\hat{e}(k)\hat{e}^T(k)$ is nothing but the covariance matrix $\hat{P}(k)$, and by the definition we had made at the beginning we know that the expected value of $w(k)w^T(k)$ is the identity matrix. Thus, Eq. (F11) can be rewritten as

$$\tilde{P}(k+1) = \Phi\hat{P}(k)\Phi^T + \Psi F_d\, F_d^T\,\Psi^T \qquad \text{(F12)}$$

For small sampling time Δt, Eq. (F5) can be approximated as $\Psi \approx I\Delta t$ and consequently Eq. (F12) through

$$\tilde{P}(k+1) = \Phi\hat{P}(k)\Phi^T + (\Delta t)^2 F_d F_d^T \qquad \text{(F13)}$$

Now backsubstituting $F_d = (\Delta t)^{-1/2}F$, Eq. (F13) reduces to a form in terms of the original process noise distribution matrix F appearing in Eq. (F1) in which we are really interested, giving

$$\tilde{P}(k+1) = \Phi\hat{P}(k)\Phi^T + \Delta t FF^T \qquad \text{(F14)}$$

Thus, Eqs. (F6) and (F14) provide procedures to propagate the states and state estimation errors across the data points.

Making use of Eqs. (F3) and (F6), it is fairly straightforward to show that under the assumptions of error and noise being of zero mean the error in the state estimates is also zero:

$$\begin{aligned}E\{[\tilde{x}(k+1) - x(k+1)]\} &= E\{\Phi\hat{x}(k) - \Phi x(k) - \Psi F\, w(k)\} \\ &= E\{\Phi\hat{e}(k) - \Psi F\, w(k)\} \\ &= \Phi E\{\hat{e}(k)\} - \Psi F\, E\{w(k)\} \\ &= 0 \qquad \text{(F15)}\end{aligned}$$

Equation (F.15) implies that the estimates of the states are unbiased.

Having derived the procedure to propagate the states and prediction errors across the time points, we now need to derive a procedure for the measurement updates at discrete time points. Such a procedure, developed by Kalman,

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is based on finding an equation to compute updated estimates $\hat{x}(k+1)$ as a linear combination of predicted estimates $\tilde{x}(k+1)$ and the measurements $z(k+1)$. For notational simplicity, in the following we agree to denote the discrete time point at which the measurements become available simply as k instead of $k+1$; this is without loss of generality because the computations are anyway carried out recursively, that is, from time point to time point. Accordingly, the linear correction equation is formulated as

$$\hat{x}(k) = K^*(k)\tilde{x}(k) + K(k)z(k) \tag{F16}$$

where K^* and K are weighting matrices that are to be determined such that the estimates $\hat{x}(k)$ are unbiased and efficient.

We know that the measurement equation is given by

$$z(k) = C\,x(k) + v(k) \tag{F17}$$

where the control observation matrix D has been dropped from Eq. (F2) for convenience without loss of generality. Substituting Eq. (F17) in Eq. (F16) yields

$$\hat{x}(k) = K^*(k)\tilde{x}(k) + K(k)[C\,x(k) + v(k)] \tag{F18}$$

Now, denoting the errors in the predicted and the corrected states as $\tilde{e}(k)$ and $\hat{e}(k)$ respectively, in Eq. (F18) we replace $\tilde{x}(k)$ and $\hat{x}(k)$ in terms of the true states $x(k)$ and corresponding errors, that is, substitute $\tilde{x}(k) = x(k) + \tilde{e}(k)$ and $\hat{x}(k) = x(k) + \hat{e}(k)$, respectively. This leads to

$$x(k) + \hat{e}(k) = K^*(k)[x(k) + \tilde{e}(k)] + K(k)[C\,x(k) + v(k)] \tag{F19}$$

which can be simplified to

$$\hat{e}(k) = [K^*(k) + K(k)C - I]x(k) + K^*(k)\tilde{e}(k) + K(k)v(k) \tag{F20}$$

The estimate $\hat{x}(k)$ will be unbiased provided the expected value $E\{\hat{e}(k)\}$ is zero. From the assumptions made, we know that the measurement noise is zero mean, and from Eq. (F15) we also know that the errors $\tilde{e}(k)$ are zero mean. Therefore, the last two terms on the right-hand side of Eq. (F20) vanish. For any arbitrary nonzero states x, the first term would vanish only when the term in the square bracket is zero, which leads to the necessary condition for unbiased estimates given by

$$K^*(k) = I - K(k)C \tag{F21}$$

Now substituting Eq. (F21) in Eq. (F16) yields

$$\begin{aligned}\hat{x}(k) &= [I - K(k)C]\tilde{x}(k) + K(k)z(k)\\ &= \tilde{x}(k) + K(k)[z(k) - C\tilde{x}(k)]\end{aligned} \tag{F22}$$

Thus, if we can determine a suitable gain matrix $K(k)$, then Eq. (F22) provides a simple procedure to correct the predicted states using the measurements $z(k)$.

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The Kalman gain matrix $K(k)$ is now determined such that the estimates $\hat{x}(k)$ are efficient. The estimates are efficient when the variance of the estimation errors $\hat{e}(k)$, which is the same as the trace of the estimation error covariance matrix $\hat{P}(k)$, is minimized. Substitution of Eq. (F21) into Eq. (F20) leads to

$$\hat{e}(k) = [I - K(k)C]\tilde{e}(k) + K(k)v(k) \tag{F23}$$

Thus, the covariance matrix $\hat{P}(k)$ is given by

$$\begin{aligned}\hat{P}(k) &= E\{\hat{e}(k)\hat{e}^T(k)\} \\ &= E\{([I - K(k)C]\tilde{e}(k) + K(k)v(k))([I - K(k)C]\tilde{e}(k) + K(k)v(k))^T\} \\ &= E\{[I - K(k)C]\tilde{e}(k)\tilde{e}^T(k)[I - K(k)C]^T + [I - K(k)C]\tilde{e}(k)v^T(k)K^T(k) \\ &\quad + K(k)v(k)\tilde{e}^T(k)[I - K(k)C]^T + K(k)v(k)v^T(k)K^T(k)\}\end{aligned} \tag{F24}$$

The second and the third terms on the right-hand side of Eq. (F24) vanish because the measurement noise $v(k)$ and the state errors $\tilde{e}(k)$ are assumed to be uncorrelated. Furthermore, by definition we know that $E\{v(k)v^T(k)\} = R(k)$ and $E\{\tilde{e}(k)\tilde{e}^T(k)\} = \tilde{P}(k)$. Thus, Eq. (F24) leads to

$$\hat{P}(k) = [I - K(k)C]\tilde{P}(k)[I - K(k)C]^T + K(k)R(k)K^T(k) \tag{F25}$$

The cost function to be minimized is then given by

$$J(k) = \text{trace}[\hat{P}(k)] \tag{F26}$$

Here, we use the following result from matrix algebra for the partial differentiation of a trace:

$$\frac{\partial}{\partial A}\{\text{trace}(ABA^T)\} = 2AB \tag{F27}$$

where A and B are square and square symmetric matrices, respectively. Substituting Eq. (F25) in Eq. (F26), partially differentiating with respect to $K(k)$, and applying the preceding rule on the two terms on the right-hand side lead to

$$\frac{\partial J(k)}{\partial K(k)} = -2[I - K(k)C]\tilde{P}(k)C^T + 2K(k)R(k) \tag{F28}$$

Equating Eq. (F28) to zero for a minimum and solving for $K(k)$ yield the Kalman gain matrix given by

$$K(k) = \tilde{P}(k)C^T[C\tilde{P}(k)C^T + R(k)]^{-1} \tag{F29}$$

Let us take a look at the physical significance of the Kalman gain matrix K. The gain matrix given by Eq. (F29) optimally weighs the influence due to the

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measurement noise covariance matrix $R(k)$ and that due to the a priori estimate of the error covariance matrix $\tilde{P}(k)$. It is inversely proportional to $R(k)$, the measurement noise, and proportional to $\tilde{P}(k)$, the uncertainty in the estimates. The two limiting cases of Eq. (F29) are interesting. Although the following discussion is valid for a general case, to better appreciate the influence of the two terms, for convenience we assume that only states are measured, which implies that C is an identity square matrix. For $R(k) \to 0$, Eq. (F29) leads to $K(k) = C^{-1}(k)$, implying that the residuals are more heavily weighted. In other words, actual measurements are trusted more than the predicted outputs. In the other case, as $\tilde{P}(k) \to 0$ it leads to $K(k) = 0$, implying residuals are less weighted, which is to say that predicted states and as consequence predicted outputs are associated with very low uncertainty. The estimates are trusted more than the actual measurements in such a case.

Having determined the optimal Kalman gain matrix $K(k)$, it now becomes possible to derive an alternative form for the covariance matrix given by Eq. (F25):

$$\begin{aligned}
\hat{P}(k) &= [I - K(k)C]\tilde{P}(k)[I - K(k)C]^T + K(k)R(k)K^T(k) \\
&= [\tilde{P}(k) - K(k)C\tilde{P}(k)][I - K(k)C]^T + K(k)R(k)K^T(k) \\
&= \tilde{P}(k) - K(k)C\tilde{P}(k) - \tilde{P}(k)C^TK^T(k) \\
&\quad + K(k)C\tilde{P}(k)C^TK^T(k) + K(k)R(k)K^T(k) \\
&= [I - K(k)C]\tilde{P}(k) - \tilde{P}(k)C^TK^T(k) \\
&\quad + \underbrace{K(k)\{C\tilde{P}(k)C^T + R(k)\}}_{\tilde{P}(k)C^T} K^T(k) \\
&= [I - K(k)C]\tilde{P}(k)
\end{aligned} \tag{F30}$$

where Eq. (F29) was used in the second last step of the preceding equation. The two forms given by Eqs. (F25) and (F30) are equivalent. The form given by Eq. (F30) is less complex and computationally faster. Hence, it is often used in practice. However, the differencing in Eq. (F30) and roundoff errors can lead to nonpositive diagonal elements, which is physically unrealistic. On the other hand, the longer form given by Eq. (F25) involves on the right-hand side the summation of two symmetric matrices. First, one is positive definite, and the second is nonnegative definite. Thus, the resulting matrix is always positive definite. Hence, we recommend the longer form of Eq. (F25), which is numerically more stable. In the literature this longer form is called as the "Joseph" form [7].

In some specific cases, due to computer roundoff errors and ill-conditioning of the intermediate result $C\tilde{P}(k)C^T + R(k)$ for inversion in the gain formula, the Kalman filter can fail [5]. This might happen more often on machines with smaller word length, for example, 16-bit computers used in the past. In such

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cases the factorization methods will perform better. The Kalman filter implementations using factorization methods are, in general, numerically most stable. Details of such techniques are found in books exclusively addressing the subject of Kalman filtering, for example, [5]. For the purpose of off-line parameter estimation from flight data, performed on a 32-bit machine, the Kalman filter using the Joseph form for the covariance matrix update has proved sufficient in the cases analyzed.

The computational steps of the Kalman filter for a linear dynamic system represented in Eqs. (F1–F3) are summarized here:

Prediction (extrapolation/time update):

$$\tilde{x}(k+1) = \Phi\hat{x}(k) + \Psi Bu(k)$$

$$\tilde{P}(k+1) \approx \Phi\hat{P}(k)\Phi^T + \Delta t\, F F^T$$

where

$$\Phi = e^{A\Delta t} \approx I + A\Delta t + A^2\frac{\Delta t^2}{2!} + \cdots$$

$$\Psi = \int_0^{\Delta t} e^{A\tau}\,\mathrm{d}\tau \approx I\,\Delta t + A\frac{\Delta t^2}{2!} + A^2\frac{\Delta t^3}{3!} + \cdots$$

Correction (measurement update):

$$\tilde{y}(k) = C\tilde{x}(k) + Du(k)$$

$$K(k) = \tilde{P}(k)C^T[C\,\tilde{P}(k)C^T + R(k)]^{-1}$$

$$\hat{x}(k) = \tilde{x}(k) + K(k)[z(k) - \tilde{y}(k)]$$

$$\hat{P}(k) = [I - K(k)C]\tilde{P}(k)[I - K(k)C]^T + K(k)R(k)K^T(k)$$

F.2 EXTENDED KALMAN FILTER

In contrast to the linear filtering technique based on an optimal Kalman filter, there is no unique solution to nonlinear filtering, that is, state estimation based on a nonlinear model. Optimal nonlinear state estimators involve propagation of non-Gaussian functions. They are significantly more complex than the linear estimators, and in many cases computationally not realizable. A nonoptimal approach using the framework of linear filters leads to nonlinear filters useful in practice. Different approximation techniques lead to different forms, which can be broadly classified into three groups: 1) linearized Kalman filter, 2) extended

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Kalman filter, and 3) iterated Kalman filter [8]. The extended Kalman filter (EKF) based on linearization of a nonlinear model about the most recent state estimate and covariance is widely used, particularly in applications related to aircraft. Linearization about the most recent estimates helps to minimize the errors. EKF is an approximation using a linear state estimator; basically it represents a procedural approach. We have already covered the linear state estimator in necessary details and summarized it in Sec. F.1. These computational steps will be adapted in the EKF to nonlinear models.

Let us consider a nonlinear system represented by the following state and observation equations:

$$\dot{x}(t) = f[x(t),\, u(t),\, \beta] + F\, w(t), \quad x(t_0) = x_0 \tag{F31}$$

$$y(t) = g[x(t),\, u(t),\, \beta] \tag{F32}$$

where β denotes the system parameters appearing in f and g. As in the preceding case of linear systems, w represents the process noise that is assumed to be additive.

To apply the computational steps at the end of Sec. F.1, we need to linearize the nonlinear model of Eqs. (F31) and (F32). Linearized system matrices are defined as

$$A = \frac{\partial f[x,\, u,\, \beta]}{\partial x}, \quad B = \frac{\partial f[x,\, u,\, \beta]}{\partial u}$$
$$C = \frac{\partial g[x,\, u,\, \beta]}{\partial x}, \quad D = \frac{\partial g[x,\, u,\, \beta]}{\partial u} \tag{F33}$$

At any discrete time point k, linearized system matrices defined in Eq. (F33) can be approximated about the current state estimate using the central difference formulas defined in Sec. 3.5.3.

Thus, the computational steps of the EKF filter for a nonlinear dynamic system represented in Eqs. (F31) to (F32) are the same as those in the computational steps listed at the end of Sec. F.1, incorporating the system matrices A, B, C , and D defined in Eq. (F33) and approximated through the central difference formula; as before, the state transition matrix Φ and its integral Ψ are give by Eqs. (F4) and (F5), respectively.

The procedure just elaborated incorporates linearized matrices in the prediction (time propagation) step and correction step (measurement update) for the states as well as for errors in the states (that is, for covariance matrices). To retain the system nonlinearities in the state estimation as far as possible, a mixed version is used that incorporates state and model output prediction with a nonlinear model whereas the error propagation is based on a linear approximation of the state and measurement equations. Any suitable numerical integration procedure, such as Runge–Kutta methods discussed in Sec. 3.8.1, can be

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used to propagate the states across the two discrete-time points. Computational steps of such a nonlinear filter are summarized next. The various steps of the EKF are very similar to that shown in Fig. F1, except for the computational details:

Prediction (extrapolation/time update):

$$\tilde{x}(k+1) = \hat{x}(k) + \int_{t_k}^{t_{k+1}} f[x(t),\, u(t_k),\, \beta]\, \mathrm{d}t$$

$$\tilde{P}(k+1) \approx \Phi(k+1)\hat{P}(k)\Phi^T(k+1) + \Delta t\, F F^T$$

where

$$\Phi(k+1) = e^{A(k)\Delta t} \approx I + A(k)\Delta t + A^2(k)\frac{\Delta t^2}{2!} + \cdots$$

$$A(k) = \left.\frac{\partial f[x(t),\, u(t),\, \beta]}{\partial x}\right|_{x=\hat{x}(k)}$$

Correction (measurement update):

$$\tilde{y}(k) = g[\tilde{x}(k),\, u(k),\, \beta]$$

$$K(k) = \tilde{P}(k)C^T[C\tilde{P}(k)C^T + R(k)]^{-1}$$

$$\hat{x}(k) = \tilde{x}(k) + K(k)[z(k) - \tilde{y}(k)]$$

$$\hat{P}(k) = [I - K(k)C]\tilde{P}(k)[I - K(k)C]^T + K(k)R(k)K^T(k)$$

where

$$C(k) = \left.\frac{\partial g[x(t), u(t), \beta]}{\partial x}\right|_{x=\tilde{x}(k)}$$

REFERENCES

[1] Kalman, R. E., "A New Approach to Linear Filtering and Prediction Problems," *Transactions of the ASME – Journal of Basic Engineering*, Series D, March Vol. 82, 1960, pp. 35–45.

[2] Sorenson, H. W., "Least-Squares Estimation from Gauss to Kalman," *IEEE Spectrum*, Vol. 7, July 1970, pp. 63–68.

[3] Gelb, A., *Applied Optimal Control*, MIT Press, Cambridge, MA, 1974.

[4] Sorenson, H. W., "Kalman Filtering Techniques," *Advances in Control Systems: Theory and Applications*, Vol. 3, edited by C. T. Leondes, Academic Press, New York, 1966, pp. 219–292.

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[5] Grewal, M. S., and Andrews, A. P., *Kalman Filtering Theory and Practice*, Prentice–Hall, Upper Saddle River, NJ, 1993.

[6] Maine, R. E., and Iliff, K. W., "Identification of Dynamic Systems," AGARD, AG-300, Vol. 2, Jan. 1985, Chaps. 6 and 7.

[7] Bucy, R. S., and Joseph, P. D., *Filtering for Stochastic Processes with Applications to Guidance*, Wiley, New York, 1968.

[8] Schmidt, G. S., "Linear and Nonlinear Filtering Techniques," *Control and Dynamic Systems – Advances in Theory and Applications*, Vol. 12, edited by C. T. Leondes, Academic Press, New York, 1976, pp. 63–95.

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